

KATANGA MINING LIMITED

NI 43-101 TECHNICAL REPORT

ON THE MATERIAL ASSETS OF KATANGA MINING LIMITED, LUALABA PROVINCE, DEMOCRATIC REPUBLIC OF CONGO

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Kamoto Copper Company S.A. (KCC) is owner of the Technical Report (TR) and the work is prepared by or under the supervision of the Qualified Persons (QPs) named as Authors

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Appendices

APPENDIX A Certificates of QPs

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1.0 Summary

1.1 Introduction

Kamoto Copper Company S.A. (KCC) commissioned an internal team of professionals at the request of its 75% parent Katanga Mining Limited (KML or the “Company”) to prepare this Technical Report (TR), which complies with the Canadian Securities Administrators’ National Instrument 43-101 - Standards of Disclosure for Mineral Projects (NI 43-101), in respect of the Material Assets (as defined below) owned and operated by KML.

KML and KCC engaged Golder Associates Africa (Pty) Ltd (Golder) and SENET to supervise, review and sign-off on the Mineral Resource and Processing sections of this TR respectively. In this capacity all QPs have signed-off their respective sections and verified that, to the best of their knowledge, all items required by NI 43-101 have been prepared in a format that conforms to the requirements of Form 43-101F1, and that each item includes an answer addressing each question required by the NI 43-101.

This report is effective from November 7, 2019. Mineral Resources and Mineral Reserves are current as at December 31, 2018. Mining activities continued during 2019 and KCC has experienced ordinary course depletion reasonably predictable based on KML’s continuous disclosure record. Updated Mineral Resources and Mineral Reserves will be released as part of 2019 year end update, which will reflect such normal course depletion. No other material changes have occurred between the effective date and the date of signature of this TR.

1.2 Property Description and Location

This TR covers the following operations, projects and infrastructure of KML and its subsidiaries located in the Kolwezi District in the Lualaba Province of the DRC, which are collectively referred to herein as the Material Assets:

Mining Assets, namely:

- Kamoto underground (UG) Mine (KTO UG) and Kamoto East UG Mine (KTE UG);
- T17 open pit (OP) Mine (T17 OP) and UG extension (T17 UG Mine) (T17 UG);
- KOV OP Mine (KOV OP) and UG extension (KOV UG Mine) (KOV UG);
- Kamoto Interim Tailings Dam (KITD);
- Mashamba East OP Mine (Mashamba East OP);
- Tilwezembe OP Mine (Tilwezembe OP); and
- Kananga OP Mine (Kananga OP) and extension (Kananga OP Extension).

Processing Assets, namely:

- The Kamoto Concentrator (KTC); and
- The Luilu Metallurgical Plant (Luilu).

The listed infrastructure necessary for the production of saleable metals.

1.3 Ownership

KCC owns the Material Assets, including the mining and exploitation rights related to the Mining Assets. KML holds a 75% stake in KCC. La Generale des Carrieres et des Mines (“GCM”) and La Société Immobilière du Congo (SIMCO), which are state-owned mining companies in the DRC, own the other 25% of KCC. The KITD was a new project included in the 2018 TR. The KITD is partially located inside the KCC lease boundary PE 525 (60% of the total), and partially inside the GCM lease boundary PE8841 (remaining 40%) defined as Block 14. Block 14 is defined in the amended joint venture agreement (“AJVA”) as “Surfaces Necessaires”, being an

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area necessary for the proper functioning of the project, in close proximity to mining operations, in order to effectively carry out these operations without prejudice to the rights of third parties, including space for dikes and storage sites for the tailings generated by KCC (see also AJVA: “definition of the “Surfaces Necessaires” and Appendix A.5 “list and description of the “Surfaces Necessaires”).

The tailings deposited in the KITD, including the portion of the KITD situated on PE8841 belonging to GCM, are from KCC’s KTC operations. As such they are KCC’s property and can be processed utilising KCC’s facilities without the need for further approval from GCM. KCC has the legal right to mine the tailings discharged from its KTC operations as per the provisions of the AJVA with GCM. KCC has been hydraulic mining the KITD since January 2017 and will continue hydraulic mining until the depletion of the deposit.

1.4 Geology and Mineralization

1.4.1 Geology

The mineralized zones are at the western end of the Katangan copper belt.

These deposits are hosted mainly by metasedimentary rocks of the late Proterozoic Katangan system, a 7 km thick succession of sediments with minor volcanics, volcanoclastics and intrusive rocks. Geochronological data indicates an age of deposition of the Katangan sediments of about 880 million years and deformation during the Katangan Orogeny at less than 650 million years. This deformation resulted in the northwest-southeast trending Lufilian Arc, which extends from Namibia on the west coast of Africa through to Zambia, lying to the south of the DRC. Within the DRC, the zone of deformation extends for more than 300 km from Kolwezi in the north-west to Lubumbashi in the south-east.

Regional and local geological settings are not relevant to the KITD project as these “deposits” are technogenetic in nature as they are the product of previous mining activities.

The discharge of fine waste material from the KTC between 2008 and 2013 created the KITD deposit by conventional upstream discharge methods.

1.4.2 Mineralization

Primary mineralization, in the form of sulphides, within the Lower Roan is associated with the Dolomie Stratifié or Stratified Dolomite (DSTRAT) and Roches Siliceuses Feuilletées Foliated (Laminated) and Silicified Rocks (RSF) for the Ore-body Inferior (OBI) and the Schistes De Base or Basal Schists and Shales Dolomitiques Supérieurs (SDB) or Upper Dolomitic Shales for the Ore-body Superior (OBS) and is thought to be syn-sedimentary in origin. Typical primary copper sulphide minerals are bornite, chalcopryrite, chalcocite and occasional native copper while cobalt is in the form of carrolite. The mineralization occurs as disseminations or in association with hydrothermal carbonate alteration and silicification.

The “mineralization” contained in the KITD is the end product of the KTC, which processed ore from the various KCC operations. The major minerals which occur in the KITD are primarily oxides of copper and cobalt, the most common being malachite and heterogenite, together with copper and cobalt sulphides including, but not limited to, bornite, chalcopryrite, chalcocite and carrolite.

1.5 Status of KCC Material Assets

Table 1 and Table 2 provide certain details on the status of the Material Assets of KCC.

Table 1: KCC Mining Assets

Property	Type	Permit Number	Status	Licence		Comments
				Expiry Date	Area	
KTO	UG	PE525	Active	April 3, 2024	11.04 km ²	Operational
Mashamba East	OP	PE525	Active	April 3, 2024	11.04 km ²	Operational

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Property	Type	Permit Number	Status	Licence		Comments
				Expiry Date	Area	
KITD	Tailings	PE525	Active	April 3, 2024	11.04 km ²	Hydraulic mining
		GCM PE8841 (Surfaces Necessaires)	Active	GCM PE8841	Block14	Hydraulic mining within Surfaces Necessaires
T17	OP	PE11602	Depleted	April 3, 2024	1.699 km ²	Depleted
T17	UG	PE11602	Care and maintenance	April 3, 2024	1.699 km ²	Development is planned to commence in 2024
KOV and KTE	OP	PE4961	Active	April 3, 2024	8.49 km ²	Operational
KOV	UG	PE4961	Development project	April 3, 2024	8.49 km ²	Development expected to start in 2026
KTE	UG	PE4961	Development project	April 3, 2024	8.49 km ²	Development expected to commence in 2026
Tilwezembe	OP	PE4963	Dormant	April 3, 2024	7.64 km ²	Operations ceased in November 2008 due to lower copper/cobalt prices
Kananga	OP	PE4960	Dormant	April 3, 2024	11.04 km ²	Operations ceased due to pending relocation of railway
Kananga extension	OP	PE11601	Dormant	May 7, 2022	0.849 km ²	Operations ceased due to pending relocation of railway

Table 2: KCC Processing Assets

Property	Status
KTC	Active
Luilu	Active

On September 11, 2015, KCC suspended operations at KTO, KOV, Mashamba East, KTE, T17, KTC and Luilu pending construction of the Whole Ore Leach Project (WOL Project). The core of the first leach train of the WOL Project was completed and commissioned in November 2017 and first copper cathode was produced on December 11, 2017. The second leach train was commissioned in October 2018, bringing KCC up to full leach capacity.

Mining operations continued during 2018 and 2019 at KOV OP, Mashamba East OP, KTO UG and hydraulic mining of the KITD

1.6 Mineral Resources and Mineral Reserves

Mineral Resources for KOV OP/UG, KTE, KTO UG, KITD and Mashamba East OP/UG have been reported in accordance with the definitions and classification standards adopted in NI 43-101.

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Mineral Reserves for KOV OP, KTO UG, KITD and Mashamba East OP have been reported in accordance with the definitions and classification standards adopted in NI 43-101.

The T17 UG estimates of Mineral Resources and Ore Reserves have been prepared in accordance with the definitions and classification standards adopted in the JORC Code (considered an “acceptable foreign code” under Part 1 – Definitions and Interpretation of NI 43-101) and can be considered as equivalent Mineral Resources and Mineral Reserves.

The Tilwezembe OP and Kananga OP estimates of Mineral Resources have been prepared in accordance with the definitions and classification standards adopted in the SAMREC Code (considered an “acceptable foreign code” under Part 1 – Definitions and Interpretation of NI 43-101) and can be considered as equivalent Mineral Resources.

1.6.1 Mineral Resource Estimate

The consolidated Mineral Resources of the various Mining Assets of KCC as at December 31, 2018 are summarized in Table 3. Mining activities continued during 2019 and depleted Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update.

Table 3: KCC Mineral Resources estimate as at December 31, 2018¹⁻¹⁶

Mineral Resource Classification	Project Area	Million Tonnes (Mt)	%TCu	%TCo
Measured Resource	KTO UG	12.1	3.90	0.59
	Mashamba East OP UG	-	-	-
	T17 UG	4.2	2.66	0.51
	KOV OP KOV UG KTE	-	-	-
	Kananga Mine	-	-	-
	Tilwezembe OP	-	-	-
	KITD	-	-	-
	Subtotal	16.3	3.58	0.57
Indicated Resource	KTO UG	65.6	3.92	0.46
	Mashamba East OP UG	57.8	1.72	0.63
	T17 UG	9.4	4.44	0.65
	KOV OP KOV UG KTE	106.8	4.80	0.53
	Kananga Mine	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60
	KITD	6.1	1.44	0.17
	Subtotal	259.3	3.64	0.54
Measured and Indicated Resource	KTO UG	77.7	3.92	0.48
	Mashamba East OP UG	57.8	1.72	0.63
	T17 UG	13.6	3.89	0.61
	KOV OP KOV UG KTE	106.8	4.80	0.53
	Kananga Mine	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60
	KITD	6.1	1.44	0.17
	Subtotal	275.6	3.64	0.54
Inferred Resource	KTO UG	48.4	3.83	0.38

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	Mashamba East OP UG	15.9	2.72	0.51
	T17 UG	5.2	4.21	0.98
	KOV OP KOV UG KTE	77.6	4.40	0.38
	Kananga Mine	4.0	2.00	0.98
	Tilwezembe OP	13.8	1.75	0.60
	KITD	0.0	0.00	0.00
	Subtotal	164.9	3.78	0.44

- 1) Mineral Resources for the KITD, KOV OP, KTE, KTO UG and Mashamba East OP have been reported in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) Mineral Resources for T17 UG have been reported in accordance with the definitions and classification criteria standards adopted of in the JORC Code for the Reporting of Mineral Resources and Mineral Reserves. If the definitions and classification standards adopted in NI 43-101 had been used instead of those adopted in the JORC Code, the estimates of Mineral Resources would be substantially similar;
- 3) Mineral Resources for Tilwezembe OP and Kananga OP have been reported in accordance with the definitions and classification standards adopted in the SAMREC Code. If the definitions and classification standards adopted in NI 43-101 had been used as opposed to those adopted in the SAMREC Code, the estimates of Mineral Resources would be substantially similar;
- 4) Mineral Resources for the KITD are reported using a cut-off grade of 0.20% TCu and 0.07% TCo;
- 5) Mineral Resources for KOV OP and KTE are reported using a cut-off grade of 0.46% TCu or 0.12% TCo;
- 6) Mineral Resources for KTO UG are reported using a cut-off grade of 1.00% TCu;
- 7) OP Mineral Resources for Mashamba East are reported inside pit shell PB5 using a cut-off grade of 0.46% TCu for the copper only grade shell and 0.12% TCo for the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at their respective cut-offs;
- 8) UG Mineral Resources for Mashamba East are reported outside pit shell PB5 using a cut-off grade of 1.00% TCu for the copper only grade shell and 0.52% TCo for the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at their respective cut-offs;
- 9) Mineral Resources for Kananga and Tilwezembe are reported using a cut-off grade of 0.50% TCu;
- 10) Mineral Resources for T17 UG are reported using a cut-off grade of 0.40% TCu;
- 11) Grade measurements are reported as a percent (%), tonnage measurements are in metric units;
- 12) Tonnes are reported as million tonnes (Mt) rounded to one decimal place; grades are rounded to two decimal places;
- 13) Mineral Resources are inclusive of Mineral Reserves;
- 14) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 15) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes (t), grade and contained metal content; and
- 16) The Mineral Resource estimates are for KCC's entire interest, where KML owns 75% of KCC.

1.6.2 Mineral Reserve Estimate

The consolidated Mineral Reserve estimate at KCC that sits within its lease boundaries is 129.8 Mt at 3.22% TCu and 0.52% TCo, including 8.7 Mt at 3.56% TCu and 0.55% TCo of Proven Mineral Reserves. In addition, there are 2.9 Mt at 1.45% TCu and 0.17% TCo of Probable Mineral Reserve from the KITD, which are situated within GCM lease area PE8841 as shown in Table 4. As per the provisions of the AJVA concluded with GCM, KCC has the legal right to mine these reserves, as they consist of tailings discharged from its KTC operation.

The Mineral Reserve estimate for KOV OP, Mashamba East OP and KTO UG changed as a result of mining activities in 2018. Mining activities continued during 2019 and depleted Mineral Reserves will be published as a part of the 2019 year end Mineral Resources and Mineral Reserves update. The Mineral Reserve estimate for T17 is unchanged.

Table 4: KCC Mineral Reserve estimate as at December 31, 2018¹⁻¹⁰

Project Area	Proved Reserve			Probable Reserve			Total Reserve		
	Mt	%TCu	%TCo	Mt	%TCu	%TCo	Mt	%TCu	%TCo
KTO UG	6.5	3.60	0.56	20.1	3.34	0.53	26.6	3.40	0.54
T17 UG	2.2	3.42	0.54	9.1	3.71	0.64	11.3	3.65	0.62
Mashamba East OP	-	-	-	31.6	2.15	0.60	31.6	2.15	0.60

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Project Area	Proved Reserve			Probable Reserve			Total Reserve		
	Mt	%TCu	%TCu	Mt	%TCu	%TCu	Mt	%TCu	%TCu
KOV OP	-	-	-	57.5	3.72	0.48	57.5	3.72	0.48
KITD (KCC PE525)	-	-	-	2.8	1.42	0.16	2.8	1.42	0.16
KITD (GCM PE8841)	-	-	-	2.9	1.45	0.17	2.9	1.45	0.17
Total	8.7	3.56	0.55	124.0	3.15	0.51	132.7	3.18	0.52

- 1) The Mineral Reserve estimates have been prepared in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) The Mineral Reserve for T17 UG is the same as at December 31, 2017;
- 3) The KITD Probable Mineral Reserve within KCC lease PE525 are reported as 2.8 Mt and within GCM lease PE8841 are reported as 2.9 Mt
- 4) The KITD portion situated on PE8841 belonging to GCM for which KCC has the legal right to mine the tailings discharged from its KTC operation as per the provisions of the AJVA concluded with GCM;
- 5) Grade measurements are reported as a percent (%), tonnage measurements are in metric units;
- 6) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 7) Projects included were developed to a minimum PFS level of accuracy;
- 8) Life of mine (LOM) plans of existing operations were developed to a minimum PFS level of accuracy;
- 9) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes (t), grade and contained metal content; and
- 10) The Mineral Reserve estimates are for KCC's entire interest, where KML owns 75% of KCC.

The changes in the current Mineral Reserve estimate relative to the KCC 2018 NI 43-101 Technical Report released on March 31, 2018 are:

- **2018:** Mineral Reserves are depleted due to mining activities during 2018 in KOV OP, Mashamba East OP, KITD and KTO UG. Mineral Reserves are current as at December 31, 2018.; and
- **2019:** Mineral Reserves are not depleted for the mining activities that continued during 2019 in KOV OP, Mashamba East OP, KITD and KTO UG. Depleted Mineral Reserves will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update.

1.7 Development and Operations

The primary developments within the Material Assets include the following.

1.7.1 Mining

- Exploration drilling campaign for 2014 and 2015 has been completed for Mashamba East;
- Exploration drilling campaign for 2014, 2015 and 2017 has been completed for the KITD;
- Mineral Resource estimates have been updated for the KOV OP/UG, KTO UG and Mashamba East OP/UG in 2016 and for the KITD in 2017;
- A Mashamba East PFS and a KITD PFS were completed in 2016 and 2017 respectively. The studies included exploration, Mineral Resource update and OP optimization in order to define the Mineral Reserves. During 2017 KCC stripped 13.7 Mt of waste from the Mashamba East OP in preparation to deliver ore to the WOL plant. In 2018 KCC produced 3.3 Mt of ore and stripped 21.2 Mt of waste from Mashamba East OP;
- Construction and development commenced at T17 UG and KTE UG in 2013. Both projects are on hold with planned start after 2024;
- In September 2015 the KOV pit was put into care and maintenance due to the suspension of copper and cobalt production. During 2016, the North Wall of KOV pit collapsed and KCC mined 6.9 Mt of waste from KOV OP as part of the collapse remediation project. In 2017 KCC resumed stripping activities in preparation to deliver ore to the WOL plant, stripping 31.6 Mt of waste and stockpiling 0.4

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Mt of ore. In 2018 KCC ramped up ore production to 6.5 Mt and stripped 29.8 Mt of waste. Ramp up continued during 2019;

- KOV commissioned several items in its primary mobile mining fleet since acquisition of the Material Assets. Current primary mobile mining fleet includes: 35 haul trucks (CAT 793); 7 shovels (2 Terex RH-340 + 3 CAT 6060 + 2 CAT 6030); 13 track dozers (4 CAT D11 + 9 CAT D10); and 6 drills (CAT MD6240 + MD5125CL + Atlas Copco PV-271 + 3 Epiroc D65).
- KTO UG increased primary development by 10% and secondary development by 41% since 2012. The current KTO UG production plan is designed to deliver an optimal feed of sulphide ore to the Lulu roasters;
- KTO UG commissioned several items in its primary mobile mining fleet since acquisition of the Material Assets. Current primary mobile mining fleet includes: 9 haul trucks (CAT AD45B + Atlas Copco MT5020); 7 loaders (CAT R2900 + Atlas Copco 1030); 10 jumbo drills (Atlas Copco 282); 3 long-hole drills (Simba SD7); and 1 D7 dozer.; and
- There have been several improvements in the UG operations, including: installation of 2 GBTs (Grizzlies); second back fill line installed; development of a new access to Zone 3 and Zone 4, which was lost when a portion of the mine collapsed under previous ownership; development and construction of the new sump at the 505 level; concreting of Koumal 5 and Acc 20; connection of Zone 8 and 9 with Zone 3 and 4 which reduced the hauling distance to the primary crusher and improved ventilation; development of an alternative haulage route for Etang South and Etang North outside of the stressed area; started the T17 development and completed the lower portal; the upper portal was also started; Shaft 1 and Shaft 2 were rehabilitated and the winders were upgraded. Installation of new main ventilation exhaust fans at Shaft 5 to improve ventilation.

1.7.2 Processing

Current milling capacity at KTC has been increased to 12 million tonnes per annum (Mtpa) from 7.68 Mtpa.

Improvements in the plant and processing infrastructure and circuits as well as the WOL Project have been undertaken to improve throughput and increase production capacity up to 300,000 tonnes per annum (tpa) of finished copper.

Table 5 summarizes the capacity associated with Phase 1 to Phase 5 (including WOL Project).

Table 5: Phased production capacity increases

Phase	Finished Copper Capacity (tpa)
1	35,000
2	70,000
3	150,000
4	270,000
5 (including WOL)	300,000

The Phase 4 improvements included:

- Installation of new crusher;
- Expansion of flotation circuit;
- X4 and Y4 sulphide receiving thickeners;
- Three of 100 ktpa solvent extraction trains;
- 200 ktpa conversion of electrolyrefinery to electrowinning (EW2);

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- Sulphide concentrate roaster number 5;
- New lime plant;
- New oxide concentrate receiving thickener (Z12);
- Two of 10 MVA electricity co-generation plants; and
- Upgrading of existing water treatment plant.

The Phase 5 improvements included:

- Construction of Cascade Mill 5 (CM5);
- Two 30 ktpa electrowinning tankhouses (EW3);
- Construction of the Sulphuric Acid Plant (SAP);
- Commissioning of the final cobalt thickeners; and
- Commissioning of the cobalt dryers and additional cobalt filters.

The WOL Project includes:

- Conversion from oxide concentrate leaching to whole ore leaching to improve overall copper recovery;
- Conversion of oxide concentrate copper leach plant to a cobalt recovery plant;
- One 40 ktpa electrowinning tankhouse (EW4 TH3);
- Using existing KTC infrastructure to mill and Sulphide float;
- Addition of new Pre-Leach Thickeners at Luilu;
- Addition of new Leach Tanks at Luilu;
- Addition of new Post-Leach Clarifiers and Counter Current Decantation (CCD) Thickeners;
- Addition of new solution storage ponds;
- Flow modifications to current Solvent Extraction (SX) plant to allow for a High Grade (HG)/ Low Grade (LG) SX configuration;
- Modification to existing copper and cobalt areas for cobalt production with the addition of new pressure plate and frame filter cobalt driers;
- Sulphuric acid storage tank farm for the acid used in the new leach facility; and
- Addition of new neutralization.

The WOL Project is expected to achieve an overall copper recovery of 85% on mixed ore and an overall cobalt recovery of 65% on mixed and sulphide ore.

1.7.3 Safety, Health, Environmental and Community

All KCC employees and contractors continued to receive introductory and refresher training for the SafeWork Program, which incorporates 20 fatal hazard protocols and associated lifesaving behaviours. The rolling 12 month LTIFR increased from 0.11 in December 2017 to 0.30 in December 2018.

The approach to environmental management at KCC is aligned with good international industry practice and development of an improved Environmental Management System is ongoing. The Environmental and Social Impact Assessment for the mine, hereafter referred to as the Environmental Impact Study (EIS), was updated in May 2019 to incorporate new infrastructure, namely the sulphuric acid plant (SAP) and ion exchange plant (IX Plant). This 2018 update to the EIS was approved by the Direction de la Protection de l'Environnement Minier (DPEM) in August 2019 and by CAMI in September 2019.

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1.8 Economic Analysis

The economic assessment has been excluded as per instruction 1 of item 22 of Form 43-101F1.

1.9 Interpretations and Conclusions

Mining in the Kolwezi district has a long history, stretching back to the early 1900s. At its peak, in the late 1980s, the district accounted for roughly 7% of the world's copper production and 62% of the world's cobalt production. In the mid- to late-90s, production declined to a virtual standstill as a result of various political and technical factors. In 2002, the 2002 Mining Code was established and since 2008, after the KCC and DCP merger, significant growth in mining production has occurred. Due to a slump in metal prices in 2015, operations were put on care and maintenance. During this time, construction of the WOL facility was undertaken in order to achieve the planned recoveries and reduce operating costs. The commissioning of the primary components of the first leach train for the WOL facility was completed in November 2017 and first copper was produced on December 11, 2017. During October 2018 the second leach train was commissioned, bring KCC to full leaching capacity.

Since 2008, extensive drilling has been focused on defining KOV OP fragments, Mashamba East, KTO UG developments, and upgrading the Mineral Resource classification. Geological mapping in the pits and UG developments is on-going. Exploration potential exists inside the KCC concession for both the extension of known ore bodies and the discovery of new mineralized zones.

A QA/QC program was initiated on data collected since 2009, and improvements are notable year to year. A twin drilling program was carried out in 2007 to verify the Gecamines drilling database. The results of these programs are considered acceptable for Mineral Resource estimation purposes.

Geological modelling, exploratory data analysis, and geostatistics work conducted are considered acceptable for Mineral Resource estimation purposes. Mineral Resource classification is based on several factors, including: number of samples, number of drill holes, distance to a sample, drill hole spacing, and time period of drilling. The Mineral Resource estimation and classification methods are considered acceptable.

Metallurgical testwork has been conducted on KCC core since 2006, and the WOL Project testwork was initiated in 2015. This work included comminution, flotation, and hydrometallurgical studies. The results of this testwork confirm the process design parameters

This TR has taken into consideration the following:

- Updated Mineral Resources and Mineral Reserves as at Dec 31, 2018;
- The KOV OP Mineral Reserve as at December 31, 2018 is 57.5 Mt of Probable Mineral Reserve at an average TCu of 3.72%;
- The KTO UG Mineral Reserve as at December 31, 2018 is 26.6 Mt of Proved and Probable Mineral Reserve at an average TCu of 3.40%;
- The Mashamba East OP Mineral Reserve as at December 31, 2018 is 31.6 Mt of Probable Mineral Reserve at an average TCu of 2.15%;
- The KITD Mineral Reserve as at December 31, 2018 is 5.7 Mt of Probable Mineral Reserve at an average TCu of 1.44%;
- There is no change in Mineral Reserves for T17 UG;
- Cobalt debottlenecking – upgrades to the MgO make up facility and filtration systems will allow KCC to increase its cobalt production capacity from 30 ktpa to a maximum of 40 ktpa of contained cobalt in cobalt hydroxide salt;
- Cobalt dryers – the utilization of the cobalt dryers will allow KCC to produce a final cobalt hydroxide product containing 15% moisture as compared with the original 65% moisture, resulting in lower transportation costs;

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- Acid plant – construction of the acid plant, which is capable of producing 2,200 tpd of sulphuric acid or 1,900 tpd of sulphuric acid and 200 tpd of liquid SO₂, will be completed in 2019;
- Process improvements to the plant and processing infrastructure, including the WOL project, are intended to improve throughput to attain the planned maximum production rate of 300 ktpa of saleable copper; and
- Uranium removal from cobalt hydroxide product by ion exchange using an anionic exchange resin or/and by precipitation using the phosphate ion.

1.10 Recommendations

The following work programs are recommended:

- Acquire and construction of a tailings facility with sufficient capacity for the life-of-mine;
- Upgrading of Inferred Mineral Resources to Indicated Mineral Resources, specifically for KOV OP, KTE UG, KOV UG, Mashamba East OP and KTO UG; and
- Upgrading of Mineral Reserves for KTE UG and KOV UG after upgrading Inferred Mineral Resource to Indicated Mineral Resource.
- Develop a life of mine optimization, based on further drilling and upgrading of the inferred mineral resources to indicated mineral resources and then mineral reserves, which will include a thorough review and technical analysis of the surfaces required for the sustainable mining of the KOV OP and Mashamba East OP, especially in terms of dumping space requirements.

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2.0 Introduction

2.1 Issuer

This TR has been compiled for KML by KCC.

This TR has been prepared to provide an insight into any key changes in the scientific and technical information concerning the Material Assets.

This TR was prepared in compliance with the Canadian Securities Administrators' National Instrument 43-101 - Standards of Disclosure for Mineral Projects (NI 43-101), in respect of the Material Assets owned and operated by KML who owns 75% share in KCC.

2.2 Terms of Reference and Basis of Technical Report

The KCC Terms of Reference (TOR) and basis of this TR were to prepare the NI 43-101 TR to provide a summary of the key scientific and technical information concerning mineral exploration, development and production activities in respect of the Material Assets of KML in accordance with the Canadian Securities Administrators' NI 43-101, Companion Policy 43-101CP and Form 43-101F1 requirements.

This TR is based on (i) information available at the time of preparation, (ii) previous Mineral Resource reports for Tilwezembe OP, Karanga mine (SRK 2009) and T17 UG (Golder 2012), (iii) data supplied by KCC and (iv) the assumptions, conditions, and qualifications set forth in this report. Accordingly, in the preparation of this TR Mr. Usmani has relied entirely on information provided by different departments of KCC in regards to its mining permits, surface rights, socio-political conditions, financial information, environment, community relations, tax and other matters. Furthermore, Mr. Usmani has relied on other experts, as shown in Table 6, where the responsible QPs for the relevant chapters are indicated.

The effective date for this TR is November 7, 2019 and Mineral Resources and Mineral Reserves are current as at December 31, 2018. Mining activities continued during 2019 and KCC has experienced ordinary course depletion reasonably predictable based on KML's continuous disclosure record. Updated Mineral Resources and Mineral Reserves will be released as part of the 2019 year end update, which will reflect such normal course depletion. No other material changes have occurred between this date and the date of signature of this TR.

2.3 Qualified Persons and Personal Inspection

This TR was prepared by or under the supervision of:

- **Tahir Saleem Usmani**, P.Eng (Registration Number 93038), a registered Professional Engineer with the Association of Professional Engineers and Geoscientists of Alberta (APEGA);
- **Christiano Santos Goncalves**, CP (Registration Number 306079) with the Australian Institute of Mining and Metallurgy (MAusIMM); and
- **Nicholas Dempers**, Pr.Eng (Registration Number 20150196) with the Engineering Council of South Africa (ECSA) and Fellow with South African Institute of Mining and Metallurgy (SAIMM).

These individuals are defined as QPs in accordance with the definition for QPs presented in Part 1.1 of NI 43-101 and certify that, by reason of their education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, they fulfil the requirements to be a QP for the purposes of NI 43-101. Each aspect of this TR was prepared by or under the supervision of a QP. Each QP whose certificate appears in APPENDIX A is responsible for the specific chapters listed in Table 6, which they have authored, supervised or reviewed.

As far as others who have served as contributing authors, this has been explicitly indicated and they are not considered QPs. The qualifications and role of the contributing authors are listed in Table 6.

Tahir Usmani, who is a KCC employee, visited the Material Assets of KML several times before and during the study period in 2019. Christiano Santos from Golder visited the Material Assets of KML in December 2018, and Nicholas Dempers from SENET (Pty) Ltd (SENET) visited the Material Assets of KML in December 2017.

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This TR is based on (i) information available at the time of preparation, (ii) data supplied by KCC and (iii) the assumptions, conditions, and qualifications set forth in this report. Accordingly, in the preparation of this report QPs have relied entirely on information provided by different departments of KCC in regards to its mining permits, surface rights, socio-political conditions, financial information, environment, community relations, tax and other matters. Furthermore, Mr. Usmani who compiled the TR has relied on other experts, as shown in Table 6, where the responsible QPs for the relevant chapters are indicated.

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Table 6: List of responsible and contributing authors

Author	Role	Chapter Number	Company	Qualifications	Supervised / Reviewed by QP
Tahir Usmani	Mine Technical Services Manager	1, 2, 3, 5, 6, 15, 16, 21, 22, 23, 25, 26, 27 and 28	KCC	MEng Mine optimization BSc Mining Engineering P.Eng (APEGA Alberta Canada)	Tahir Usmani
Brad Rathman	General Manager Technical Services	15, 16.3, 16.5 and 21	KCC	BSc Mining Engineering (University of Arizona)	
Didier Mavungu	Legal Manager	4	KCC	Lawyer registered at the DRC bar association	
Jurgens Hamman	Chief Geotechnical Engineer	16.1 to 16.4	KCC	MSc (Geotechnical Engineering) Member of South African National Institute of Rock Engineers (SANIRE) Registered as a member (313715) of the AusIMM	
Peter Milmo	Superintendent Hydrologist	16.9	KCC	BSc Geological Sciences MSc Hydrogeology	
Jon Shenk	Chief Resource and Exploration Geologist	7, 8, 9, 10, 11, 12 and 14	KCC	MSc Geosciences BSc Geosciences	Christiano Santos
Douglas Reid	Principal Geological Engineer	14	Amec Foster Wheeler	BE Geological Engineering P.Eng (APEGBC, BC Canada)	
Yolande Jurrius	Chief Processing Officer	1.7.2, 13, 17, 21 and 24.1	KCC	BSc Chemical Engineering	Nicholas Dempers

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Author	Role	Chapter Number	Company	Qualifications	Supervised / Reviewed by QP
Alan Keeley	Project Manager (WOL)	18.1 and 18.2	KCC	National Diploma (Analytical Chemistry): Higher National Diploma (Chemical Engineering) BSc (Hons) (Pretoria University) MSc Technology Management	
Peter Beykirch	Electrical Manager	18.3	KCC	NHD Electrical BSc Electrical SA Government Certificate Electrical VDI Registration	Tahir Usmani
Tjaart Kruger	Finance Manager	4.6, 19, 21, 22 and 25.3	KCC	Bachelor of Commerce in Accounting (B-COM), (CA) SA	
Francesca Santinelli	Regional CSR Advisor	20.2 and 20.5	KCC	BSc International Relations MSc International Relations MSc Development Studies	
Becky Humphrey	Senior Environmental Manager	1.7.3, 20.1, 20.3, 20.4, 20.6 and 24.2	KCC	BSc Geography MSc Water Resources	Tahir Usmani

3.0 Reliance on Other Experts

This TR was prepared in accordance with the requirements of a NI 43-101 for KML by or under the direct supervision of following QPs:

- **Tahir Saleem Usmani**, who is an employee of KCC;
- **Christiano Santos Goncalves**, who is an employee of Golder; and
- **Nicholas Dempers**, who is an employee of SENET.

These individuals are defined as QPs in accordance with the definition for QPs presented in Part 1.1 of NI 43-101 and certify that, by reason of their education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, they fulfil the requirements to be a QP for the purposes of NI 43-101. Each aspect of this TR was prepared by or under the supervision of a QP. Each QP whose certificate appears in APPENDIX A is responsible for the specific chapters listed in Table 6, which they have authored, supervised or reviewed.

As far as others who have served as contributing authors, this has been explicitly indicated and they are not considered QPs. The qualifications and role of the contributing authors are listed in Table 6.

This TR is based on (i) information available at the time of preparation, (ii) data supplied by KCC and (iii) the assumptions, conditions, and qualifications set forth in this report. Accordingly, in the preparation of this report QPs have relied entirely on information provided by different departments of KCC regarding its mining permits, surface rights, socio-political conditions, financial information, environment, community relations, tax and other matters. Furthermore, Mr. Usmani who compiled the TR has relied on other experts, as shown in Table 6, where the responsible QPs for the relevant chapters are indicated.

The document also contains the expression of professional opinion of individuals based on the information available at the time of preparation.

The quality of information, conclusions and estimates contained within are consistent with the intended level of accuracy as well as the circumstances and constraints under which the mandate was performed.

This TR is intended to be used by KML to file as a TR with Canadian Securities Regulatory Authorities pursuant to NI 43-101. Except for the purposes legislated under provincial securities law, any use of this report by any third party is at that party's sole risk.

4.0 Property Description and Location

4.1 Location

KCC holds a valid mining concession proximal to the city of Kolwezi, in the Lualaba Province of the DRC. The location is illustrated in Figure 1.

4.2 KCC Rights: Summary of Permits

Table 7 presents a summary of KCC's permits and current legal tenure in regard to concession areas:

Table 7: Legal tenure of KCC¹⁻²

Property	Exploitation Permit Number	Rights Granted	Area of Title	Valid Until
KTO UG and Mashamba East OP	PE525	Cu, Co and associated minerals and metals	13 blocks, 11.04 km ²	April 3, 2024 Renewable
T17 OP and T17 UG	PE11602	Cu, Co, Ni and Au	2 blocks, 1.699 km ²	April 3, 2024 Renewable
Extension of Kananga	PE11601	Cu, Co, Ni and Au	1 block, 0.849 km ²	May 7, 2022 Renewable
KOV OP and KOV UG	PE4961	Cu, Co	10 blocks, 8.49 km ²	April 3, 2024 Renewable
Tilwezembe OP	PE4963	Cu, Co	9 blocks, 7.64 km ²	April 3, 2024 Renewable
Kananga mine	PE4960	Cu, Co	13 blocks, 11.04 km ²	April 3, 2024 Renewable

- 1) KITD is partially located within Surfaces Necessaires, which is Block 14 of GCM PE8841 (Figure 2) for which KCC has mining rights and KCC is currently hydraulic mining since January 2017; and
- 2) Cu - Copper, Co - Cobalt, Ni - Nickel and Au – Gold.

On January 27, 2018, the DRC parliament adopted the 2018 Mining Code (the “2018 Mining Code”), replacing the previous mining code adopted by the DRC in 2002 (the “2002 Mining Code”) (refer to Section 4.6.1.1 for additional details). Permis d’Exploitation or Exploitation Permits (PEs) under the 2018 Mining Code are renewable in accordance with the terms of the 2018 Mining Code for periods of 15 years. In addition to other conditions, Article 80 of the 2018 Mining Code provides for a 5% free carried interest in operating companies upon each renewal of the relevant PEs. KCC and its shareholders reserve entirely the right to the continuing application of the provisions of the 2002 Mining Code for a period of 10 years pursuant to the stability guarantee provided at Article 276 of the 2002 Mining Code.

A PE grants its holder the exclusive right to carry out exploration and exploitation works for the minerals for which it has been granted. This right covers the construction of necessary facilities for mining exploration, the use of water and wood resources, and the free commercialisation of products for sale, in compliance with corresponding legislation.

All figures included as part of this TR that include locations, utilise Universal Transverse Mercator (UTM) reference co-ordinate system, located in Zone 35L.

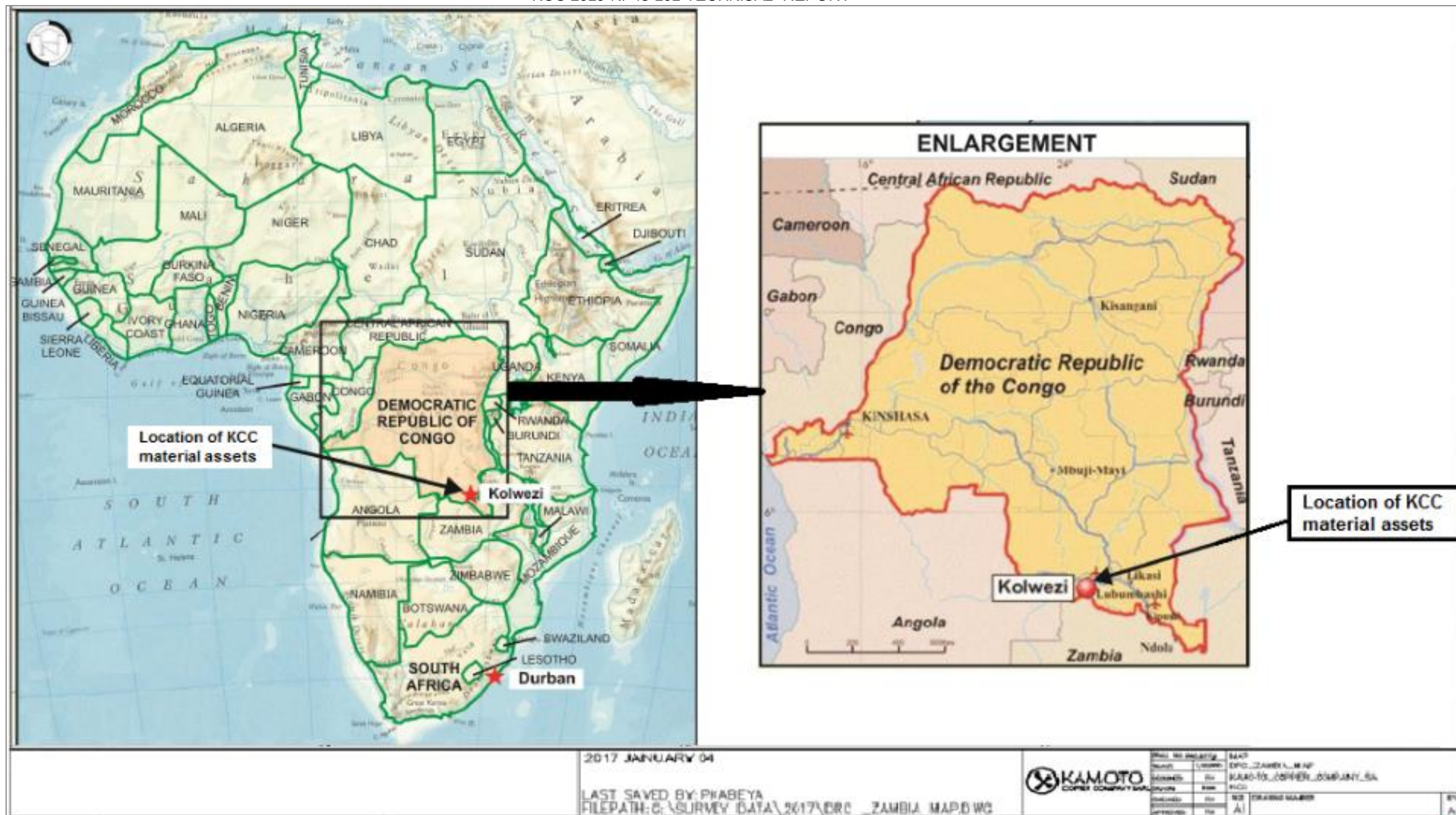


Figure 1: Location of KCC Material Assets

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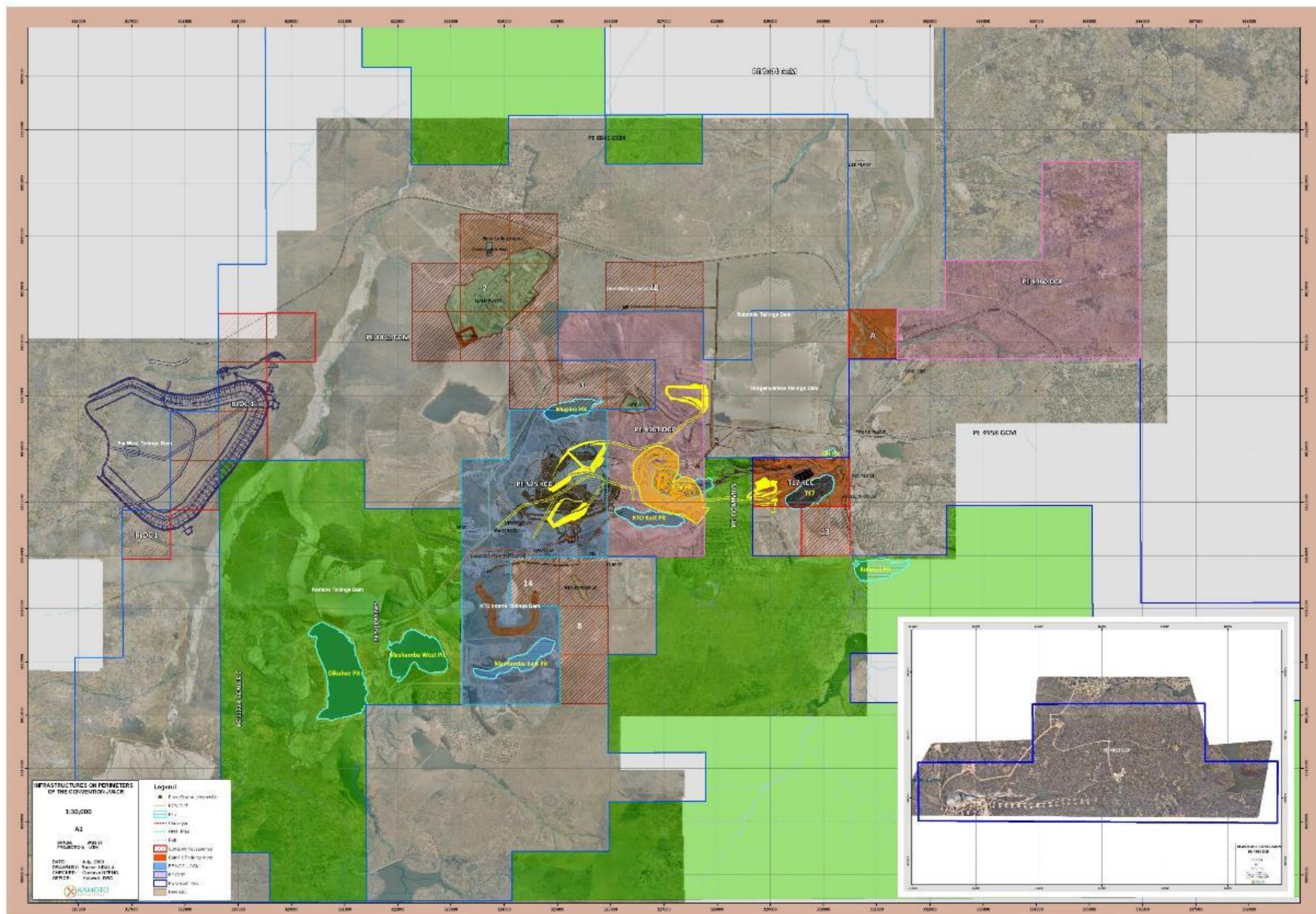


Figure 2: KCC mining permits and Surfaces Necessaires

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4.3 KCC Rights: Joint Venture Agreement

GCM, KFL Limited (KFL) and Global Enterprises Corporate Limited (GEC), in the presence of KCC, DRC Copper and Cobalt Project SARL (DCP), SIMCO, Katanga Mining Holdings Limited (KMHL), Katanga Mining Finance Limited (KMFL) and KML (BVI) Holdco Limited, entered into the AJVA on July 25, 2009 (which was also the effective date) which resulted in the merger of KCC and DCP, as ratified by Presidential Decree on April 27, 2010.

As a result of the AJVA:

- Certain exploitation permits (each a PE) were transferred from GCM to KCC, namely the entirety of PE525 (comprising 13 blocks), part of PE4958 (i.e. the new PE11602 comprising two blocks containing the T17 deposit), and part of PE7044 (i.e. the new PE11601 comprising one block containing the extension of Kananga). These transfers were completed pursuant to transfer deeds dated July 27, 2009 and evidenced by the CAMI in the related certificates;
- DCP PEs (i.e. PEs 4960, 4961 and 4963) were transferred to KCC following completion of the merger with DCP. These transfers were evidenced by CAMI in the related certificates; and
- The perimeter of the merged KCC/DCP concession area may contain some “Surfaces Necessaires” (as defined in the AJVA) which may be useful to KCC project and are subject to certain conditions.

A portion of the KITD extends beyond KCC perimeter PE525 and is located on GCM’s PE8841 Block 14. Block 14 is considered in the AJVA as “Surfaces Necessaires”, an area necessary for the proper functioning of the project, in close proximity to mining operations, in order to effectively carry out the latter, without prejudice to the rights of third parties, including space for dikes and storage sites for the tailings generated by KCC. See also AJVA: definition of the “Surfaces Necessaires” and Appendix A.5 “list and description of the “Surfaces Necessaires””.

The KITD material located on GCM Block 14 and is as a result of KCC’s KTC operations. As per the provisions of the AJVA with GCM, it is therefore KCC’s property and can be processed utilising KCC’s facilities without seeking further approvals from GCM. KCC has been hydraulic mining the KITD since January 2017 and will continue hydraulic mining until depletion of the deposit.

Table 7 presents a summary of KCC’s permits and current legal tenure regarding concession areas:

- Area of each property;
- The location of each property;
- The PE corresponding to each property;
- The type of mineral tenure of each property; and
- The expiration date of each PE.

Other provisions of the AJVA include:

- On the winding up or liquidation of KCC, the mining rights and titles of KCC shall revert to GCM without further consideration.
- KCC is required to pay GCM rent for the leased Equipment and Installations and the Surfaces Nécessaires of USD 1.8 million per year net of the applicable rental income tax.
- KCC shall retrocede the Equipment and Installations free of charge to GCM upon lawful termination or final expiry of the AJVA.
- GCM grants and/or makes available to KCC, subject to payment of the reasonable maintenance costs, the following rights:
 - The right to use roads, railways, rail routes, waterways;
 - The right to avail itself of rights of way, easements, rights to water;

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- All the supplementary rights that can facilitate access to or use of the lands involved and the facilities located thereon, which GCM enjoys outside of the perimeter of the KCC project insofar as they are necessary or desirable to carry out the project in the most cost effective manner;
- The requirement that: (i) all future intercompany loans to KCC, if any, be provided by KMFL at an interest rate equal to the lesser of (a) USD 6-month LIBOR + 3%, and (b) 6%; (ii) a tender process be run by KCC on any commercial agreement to be entered into by KCC with a value in excess of USD 5 million; (iii) the prior approval of GCM for any capital expenditure of KCC that would allow the expansion of capacity of KCC's production facilities to over 300,000 tons per annum of copper, if such expenditure exceeds USD 500 million; and (iv) the provision by KCC to GCM of the proposed annual terms of the copper and cobalt offtake arrangements between KCC and affiliates of Glencore International AG ("Glencore") and other further documentation in KCC's possession which GCM may reasonably request; and
- The requirement to pay royalties and other fees as described in Section 4.6.

In connection with an agreement announced on June 12, 2018 (the "GCM Settlement"), with (amongst others) GCM, it was agreed that a certain portion of KCC's debt (the "Residual Debt") held by KMFL is required to be amortized over a period of eight years in accordance with an agreed amortization schedule. The amortization schedule provides that, during 2018 and 2019, only interest payments on the Residual Debt are required to be made to the extent KCC has available free cash flow. Thereafter, the amortization schedule provides for the repayment of principal and interest of the Residual Debt until its expected completion in 2025.

In any fiscal year of KCC, to the extent KCC has profit available for distribution after the required principal and interest payments in respect of the Residual Debt have been made, such profits will be paid out as dividends to shareholders of KCC in proportion to their respective shareholdings. If, however, after the Residual Debt principal and interest payments have been made, there is no profit available for distribution but KCC is in possession of cash or a there is portion of cash in excess of profit available for distribution, KCC's shareholders may cause KCC to distribute such cash in proportion to their respective shareholdings by way of shareholder loans on reasonable commercial terms. These shareholder loans would be required to be repaid via set-off against future dividends payable by KCC.

4.4 Replacement Reserves

In February 2008, KCC renounced certain mineral reserves within its mineral concession in favour of GCM pursuant to a concession release agreement amongst the parties to the JVA. In connection with the concession release agreement, GCM agreed that it would procure or provide to KCC, no later than December 31, 2015, a replacement for the renounced reserves, either in certified copper and cobalt reserves to be classified following the completion of a drilling program financed by GCM, or by way of a payment of the agreed equivalent financial value of USD 285 million. In circumstances where GCM failed to provide the replacement reserves and make payment of the financial compensation, KCC had the right to offset dividends and royalties due to GCM against the amount of the financial compensation. An amendment to the JVA was concluded in January 2015 to preserve the above contractual set-off rights with GCM following the conclusion of the tripartite royalty agreement between GCM, Africa Horizon Investments Limited ("AHIL") and KCC. The drilling program to locate the replacement reserves was commenced in 2009, and notwithstanding the agreement of GCM to finance the project, the Company agreed to fund the exploration and drilling costs on the condition that all such expenditures would be reimbursed by GCM at the conclusion of the program. Over the course of the drilling program between 2009 and 2014, the Company funded approximately USD 57 million in exploration expenditures on behalf of GCM. The deadline to procure or provide the replacement reserves to KCC was extended until March 2019 when KCC suspended the processing of copper and cobalt in September 2015.

Per the GCM Settlement, KCC agreed to waive its contractual right to receive the replacement reserves or an equivalent cash payment of USD 285 million and waive its contractual right to be reimbursed for the approximately USD 57 million in exploration and drilling expenditures incurred on behalf of GCM in connection with the replacement reserves program. As a consequence, KCC does not have the right to offset dividends due to GCM and royalties due to AHIL against the replacement reserves or equivalent cash payment of USD 285 million owing by GCM.

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Additionally, the Company agreed to fund the payment of approximately USD 41 million in outstanding unpaid invoices for contractors in charge of the replacement reserves exploration program, resulting in an aggregate financial impact on the Company of USD 383 million.

4.5 Property Boundaries

The property boundaries of the PEs and the Necessary Surfaces of KCC are presented in Figure 2 and Figure 3.

4.6 Royalties, Duties and Other Fees

4.6.1 Royalties Payable to the State

The holder of a mining exploitation title is subject to mining royalties payable to the State. The royalties are due upon the sale of the product. A detail on the mining royalties is included in the following section.

4.6.1.1 2018 Mining Code

On January 27, 2018, the DRC parliament adopted the 2018 Mining Code, replacing the previous 2002 Mining Code, which provided for a 10-year stability clause by which existing operations would, after introduction of any new regulations amending the 2002 Mining Code, continue to benefit under the terms of the 2002 Mining Code for 10 years.

On March 7, 2018, representatives of the Company, along with fellow international miners with operations in the DRC, met with the President of the DRC and outlined concerns, the cornerstone being the need to preserve the stability arrangement provided in the 2002 Mining Code. Additional concerns included the increase of mining royalties, conditions regarding renewal of mining permits, tax on "super-profits", the increase of import duties, repatriation of funds and strategic substances qualification. It was agreed that the miners' concerns would be addressed and a working group hosted by the Ministry of Mines, which would include the miners and civil society was to have commenced a 30-day review of the concerns. Despite repeated attempts, no such formal discussions have taken place.

The 2018 Mining Code was published in the Official Gazette on March 28, 2018. The position of the DRC government and the application of the new measures provided by the 2018 Mining Code and subsequent mining regulations published on June 8, 2018 as well as an instruction issued by the Secretary General of the Ministry of Mines on 16 June 2018, confirmed that the 2018 Mining Code is intended to supersede the 10-year stability clause provided by the 2002 Mining Code and that all new obligations will be immediately applicable.

The 2018 Mining Code substantially amended the 2002 Mining Code, introducing a number of new obligations, taxes and charges, and increasing existing ones. KML and KCC consider that they are entitled to the continuing application of all relevant provisions of the 2002 Mining Code, including, but not limited to, the tax, custom and exchange rate regimes existing thereunder, for a further ten years starting from the date of entry into force of the 2018 Mining Code, pursuant to Article 276, paragraph 2 of the 2002 Mining Code (the "2002 Stability Clause"). KML and KCC have consistently contested the applicability of the 2018 Mining Code and reserved all rights in this regard. Where in this section there is reference to agreement or compliance with the 2018 Mining Code, KCC has stated that it was under no obligation to do so and reserved entirely its rights pursuant to the 2002 Stability Clause. Further detail on the 2018 Mining Code is included in Sections 25.3.7.3.

The royalties on copper and cobalt under the 2018 Mining Code were increased from 2% of the net revenue basis to 3.5% of the gross revenue calculated on the basis of the average price of the concerned commodity on international markets. Additionally, the DRC customs authorities have established cobalt tables to predetermine the quality of cobalt products for exports. The tables consist of multiple quality brackets, within which the cobalt product for export is automatically deemed to be the top end of the bracket for the purpose of the royalty calculation. The new rates and basis have been implemented in June 2018.

As allowed by the 2018 Mining Code, on November 24, 2018, the Prime Minister of DRC published a decree qualifying cobalt as a strategic mineral substances, resulting in a royalty payable of 10% instead of 3.5% on KCC contained cobalt sales. The effect of such a decree on KCC operations to date is currently relatively limited due to the fact that cobalt quantities exported since the issuance of the decree have been impacted by the presence of uranium in the cobalt produced. The 2018 Mining Code also provides for the possible adoption

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by the Order of the Ministers responsible for Mines and Finance of an entirely different regulatory regime for "strategic mineral substances" including cobalt, concerning the conditions for access, research, exploitation and marketing thereof. The non-deductibility of freight costs lead KCC to add back freight costs related to the lots exported after June 8, 2018. As at September 30, 2019, the loss to be carried forward indefinitely for tax purposes has been reduced by USD 107.4 million.

The 2018 Mining Code creates a 50% tax on 'exceptional' profits, also called the "Super Profit Tax" or "SPT", made on a given mining project when the prices of the metals and minerals produced exceed by more than 25% the prices used in the bankable feasibility study ("BFS") for that project. Specifically, the 2018 Mining Code introduced a new SPT equating to 50% of the portion of the effective EBITDA higher than 125% of the EBITDA presented by the BFS, when the concerned commodity price exceeds 125% of the price used in the bankable feasibility study. In July 2019, KCC settled its 2018 SPT liability with the authorities. As part of a memorandum of understanding entered into in connection with the settlement of its 2018 SPT liability, KCC agreed to produce a BFS for future SPT calculation by the end of 2019.

The copper and cobalt price assumptions used in the BFS are key judgments applied in estimating the impact resulting from the changes in the 2018 Mining Code on Katanga's estimated fair value. The 2018 Mining Code provides for the application of substantive rates of imports duties (essentially 5% or 10% instead of 3%) when the mining license has been issued for more than 6 years. KCC has been importing under substantive regime since July 2018, resulting in an additional monthly cost of \$3 million (duties and taxes collected by DGDA).

The non-deductibility of interest payable to group companies (such as interest payable on intragroup loans) remains unclear and will depend on the interpretation of the new regulations by the DRC central bank. Following the enactment and entry into force of the 2018 Mining Code and Ministerial Decree n°020/CAB/MIN/EDD/AAN/TNT/SAA/2019, the Ministry of Environment has updated the list of classified installations to be declared by KCC to serve as basis for the calculation of various environmental taxes. KCC indicated that it will proceed with making the requisite declarations on the basis of the updated schedule.

The application of the 2018 Mining Code to KCC also impacts the approval process of some of the future waste dumps to be developed for the continuity of its operations. Article 279 of the 2018 Mining Code provides for an approval procedure for certain installations such as tailings and dumps when located within a certain distance from residential dwellings. KCC chose to comply with this requirement

As part of the approval procedures for the Ion Exchange Plant (the "IX Plant"), the Ministry of Mines has requested that KCC submit a complete BFS for the entire KCC project, pursuant to the 2018 mining regulations made under the 2018 Mining Code. Given the effectiveness of the interim solutions using phosphoric acid over the past two quarters, KCC undertook instead to provide a BFS for the entire KCC project by 2019 year-end which will cover the IX Plant long term option.

Renewal of the concession permits also have been amended by the 2018 Mining Code adding several requirements which have increased the risk of non-renewal of the permits held by KCC. In particular, the renewal of the permits are subject to the submission of a bankable feasibility study and the assignment of 5% of the shares to the DRC State. The renewal of the permits is also subject to the submission of a new social plan (which needs to be complied with) and the uninterrupted operations of the mining permits. Although KCC's mining permits were not issued under the 2018 Mining Code and some conditions appear not be applicable, in virtue of the Stability Clause, the 2018 Mining Code additional requirements to renew the permits are considered as a high risk for KCC.

The rules governing foreign exchange under the 2018 Mining Code could also have a significant impact on the Company. Specifically, the 2018 Mining Code provides that KCC is required to repatriate to the DRC at least 60% of the proceeds from exports of mineral products, which proceeds may only be used for domestic expenses of KCC. The remaining 40% of export proceeds may be used for foreign expenses, including payment of foreign suppliers, servicing of foreign debt and the payment of dividends to non-DRC resident shareholders. Under the 2002 Mining Code, only 40% of export proceeds were required to be repatriated. This restriction may affect the ability of KCC to fully maintain its supply relationships with certain key non-DRC suppliers, service the Residual Debt and pay dividends to KMFL and ultimately to the Company.

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KCC is considering certain commercial and structural options to mitigate the negative impact of such restrictions.

4.6.2 Surface Rights Fees Payable to the State

Under Article 199 of the 2002 Mining Code, KCC is required to pay surface rights fees of USD 5 per hectare (ha) per year for exploitation permits, i.e. USD 424.78 per block.

Additional surface rights fees are payable by KCC as the holder of exploitation mining rights pursuant to Article 238 of the 2002 Mining Code at the rate of USD 0.08 per ha.

4.6.3 Royalties Payable under the AJVA

Under the AJVA, KCC is also required to pay a royalty equivalent to 2.5% of net revenues of the project, for the use of GCM's equipment and facilities and the depletion of the deposits. Initially, pursuant to the AJVA, the royalty was payable directly to GCM. Following the acquisition of rights from GCM by companies affiliated with Mr. Gertler, GCM directed KCC to make payment of the royalty to Africa Horizons Investment Limited ("AHIL"). The direction to pay the royalty to AHIL in lieu of GCM was formalized on January 22, 2015 pursuant to a tripartite royalty agreement between GCM, AHIL and KCC. In March and July 2015 respectively, AHIL and KCC entered into two prepayment agreements pursuant to which KCC made advance payments of royalties to AHIL for a total amount of USD 54.7 million, which resulted, after accounting for time value of money, in an overall prepayment amount of USD 57.5 million to be set-off against royalties which become due or payable to AHIL. Subsequent royalty liabilities to AHIL have been, and future royalty liabilities to AHIL will be, offset against the royalty advances until the royalty advances are recovered by KCC in full. The remaining balance of royalty advances in the amount of USD 57.1M results from the payment of invoices by KCC on behalf of GCM to contractors in charge of the Exploration Program conducted under the supervision of GCM, as described in Section 4.4.

In December 2017, the United States government designated Mr. Gertler and affiliated companies, including AHIL as specially designated nationals ("SDNs"). Additional companies affiliated with Mr. Gertler, including Ventora Development Sasu ("Ventora") were designated SDNs on June 15, 2018.

On June 15, 2018, the Company, KCC, AHIL and Ventora entered into an assignment and assumption agreement whereby all of the rights of AHIL pursuant to the tripartite royalty agreement were assigned to Ventora. Future royalty liabilities will be offset against the Exploration Program payments until the Exploration Program payments are recovered by KCC in full. Royalty liabilities arising from sales in the period since the prepayments were made have been offset against the prepayments made.

The Company believes that payment in non-US dollars of royalties to Ventora without the involvement of US persons would appropriately address all applicable sanctions obligations. For further information, see Section **Error! Reference source not found.**

4.6.4 Pas de Porte Payable under the AJVA

Pursuant to the AJVA, an entry premium ("pas de porte") equal to an aggregate of \$140 million was previously payable by KFL Limited ("KFL") and Global Enterprises Corporate Limited ("GEC") (both KFL and GEC are subsidiaries of the Company) to GCM for the access to the mineral concession area, payable in instalments on an agreed schedule. In 2014, GCM directed KFL to pay the last three installments of the entry premium to AHIL. One installment was paid to AHIL in 2014 and two installments were prepaid to AHIL in 2015, for a total discounted value of \$43.5 million, after taking time value of money into consideration. The undiscounted prepayment amounted to \$45.5 million, as per the AJVA provisions. The entry premium is amortized on a straight line basis over the life-of-mine.

As part of the GCM Settlement, the Company has agreed that additional entry premiums shall be paid by the Company to GCM for certain reserves that may be identified in the future subject to certain conditions and the outcome of the additional studies that are to be conducted.

GCM has agreed to provide to KCC all studies, drilling data, cores, surveys, assays and other information held by GCM and its contractors in respect of the areas that were explored as part of the Exploration Program conducted by GCM between 2009 and 2014 within the concession areas covered by KCC's mining permits.

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KCC has agreed to commission additional studies (the "KCC New Studies") on the concession areas covered by such mining permits and provide the results to GCM prior to June 12, 2023 in order to determine whether there are any new Australasian Joint Ore Reserves Committee ("JORC") compliant reserves. The payment and price of the entry premium per tonne of copper and copper equivalent reserves (classifying cobalt to copper using industry standards and long term consensus pricing prevailing at the time of payment) will depend on the result of the New Studies. If KCC fails to conduct the New Studies prior to June 12, 2023, GCM may conduct the New Studies and provide the results to KCC (the "GCM New Studies"), which in turn may result in an obligation on the Company to pay additional entry premiums to GCM.

The tables below provide a summary of the additional entry premiums to be paid by the Company depending on the outcome of the New Studies or GCM New Studies, as applicable.

	STUDIES CONDUCTED BY KCC ⁽¹⁾	
	JORC-compliant reserves Price / ton ⁽²⁾	Non JORC ore Price / ton ⁽²⁾ extracted by KCC
Extension of an ore body defined in the JVA	US\$ 85 ⁽³⁾	US\$ 115 ⁽⁵⁾
Distinct ore body and not defined in the JVA	US\$110 ⁽⁴⁾	US\$ 130 ⁽⁶⁾

(1) KCC New Studies must be conducted within 5 years of the GCM Settlement

(2) Ton of copper and copper equivalent (classifying cobalt to copper using industry standards and long term consensus pricing prevailing at the time of payment)

(3) If the KCC New Studies in the areas demonstrate that there are JORC-compliant reserves that are not geologically and lithologically distinctly separate from the KCC ore bodies identified in the JVA (the "Existing Reserves' Extension") and which do not correspond to and are in excess to the resources or reserves that have been previously disclosed by the Company in its Ore Reserves and Mineral Resources statement as at 31 December 31, 2017 or in its Technical Report as at December 31, 2017, the Company shall pay Gécamines an entry premium of USD 85 per ton of copper and copper equivalent reserves (classifying cobalt to copper using industry standards and long term consensus pricing prevailing at the time of payment) contained in the Existing Reserves' Extension. In such case, the entry premium shall only be payable in excess of the tonnage corresponding to the Total Replacement Reserves as defined in the JVA (i.e. 3,992,185 tonnes of copper and 205,629 tonnes of cobalt JORC-compliant reserves).

(4) If the KCC New Studies demonstrate that there are new JORC-compliant reserves geologically and lithologically distinctly separate from the existing KCC ore bodies identified in the JVA (the "New Identified Reserves"), KCC shall, in its sole discretion, elect, to:

- transfer for no consideration to GCM the right to mine the New Identified Reserves for its benefit, at its own cost using its own resources. The transfer or lease of the relevant exploitation rights shall be done at GCM's costs and KCC shall be reimbursed of the reasonable costs of the Studies and any related tax liability shall be fully assumed by GCM; or
- mine the New Identified Reserves itself, in which case the Company shall pay GCM an entry premium of USD 110 per ton of copper contained in such New Identified Reserves.

(5) If the KCC New Studies identify additional ore which is an extension of a defined ore body under the JVA, but such ore is a non JORC-compliant reserve, and KCC decides to extract such ore, the Company shall pay an entry premium of US\$ 115 per ton of copper or copper equivalent contained in ore effectively extracted by KCC within 30 days of the end of the year during which such ore was extracted.

(6) If the KCC New Studies identify additional ore which is distinct of a defined ore body under the JVA, but such ore is a non JORC-compliant reserve, and KCC decides to extract such ore, the Company shall pay an entry premium of US\$

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130 per ton of copper or copper equivalent contained in ore effectively extracted by KCC within 30 days of the end of the year during which such ore was extracted.

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	STUDIES CONDUCTED BY GCM ⁽¹⁾	
	JORC-compliant reserves Price / ton ⁽²⁾	Non JORC ore Price / ton ⁽²⁾ extracted by KCC
Extension of an ore body defined in the JVA	US\$ 130 ⁽³⁾	US\$ 140 ⁽⁵⁾
Distinct ore body and not defined in the JVA ⁽¹⁾	US\$160 ⁽⁴⁾	US\$ 170 ⁽⁶⁾

- (1) The GCM New Studies will only be conducted if KCC fails to conduct KCC New Studies within 5 years of the GCM Settlement
- (2) Ton of copper and copper equivalent reserves (classify cobalt to copper using industry standards and long term consensus pricing prevailing at the time of payment)
- (3) If the GCM New Studies in the areas demonstrate that there are JORC-compliant reserves that are not geologically and lithologically distinctly separate from the KCC ore bodies identified in the JVA (the "Existing Reserves' Extension") and which do not correspond to and are in excess to the resources or reserves that have been previously disclosed by the Company in its Ore Reserves and Mineral Resources statement as at 31 December 31, 2017 or in its Technical Report as at December 31, 2017, the Company shall pay Gécamines an entry premium of USD 130 per ton of copper and copper equivalent reserves (classify cobalt to copper using industry standards and long term consensus pricing prevailing at the time of payment) contained in the Existing Reserves' Extension. In such case, the entry premium shall only be payable in excess of the tonnage corresponding to the Total Replacement Reserves as defined in the JVA (i.e. 3,992,185 tonnes of copper and 205,629 tonnes of cobalt JORC-compliant reserves).
- (4) If the GCM New Studies demonstrate that there are new JORC-compliant reserves geologically and lithologically distinctly separate from the existing KCC ore bodies identified in the JVA (the "New Identified Reserves"), KCC shall, in its sole discretion, elect, to:
- transfer for no consideration to Gécamines the right to mine the New Identified Reserves for its benefit, at its own cost using its own resources. The transfer or lease of the relevant exploitation rights shall be done at GCM's costs and any related tax liability shall be fully assumed by Gécamines; or
 - mine the New Identified Reserves itself, in which case the Company shall pay GCM an entry premium of USD 160 per ton of copper contained in such New Identified Reserves.
- (5) If the GCM New Studies identify additional ore which is an extension of a defined ore body under the JVA, but such ore is a non JORC-compliant reserve, and KCC decides to extract such ore, the Company shall pay an entry premium of US\$ 140 per ton of copper or copper equivalent contained in ore effectively extracted by KCC within 30 days of the end of the year during which such ore was extracted.
- (6) If the GCM New Studies identify additional ore which is distinct of a defined ore body under the JVA, but such ore is a non JORC-compliant reserve, and KCC decides to extract such ore, the Company shall pay an entry premium of US\$ 170 per ton of copper or copper equivalent contained in ore effectively extracted by KCC within 30 days of the end of the year during which such ore was extracted.

4.6.5 Customs Duties and Taxes Payable

There is a requirement to pay customs duties and taxes in accordance with DRC Legislation, including the 2018 Mining Code.

4.7 Environmental Liabilities

Environmental liabilities are discussed in Chapter 20. In general, KCC is subject to the following environmental permitting.

- Environmental Impact Study (EIS) every five years. Current EIS is valid until 2024;

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- Periodic inspections by DRC regulatory authorities and third party environmental audits; and
- The 2018 EIS for the mine (approved by the DRC authorities in June 2018) was updated in December 2018 to incorporate the development of new infrastructure, specifically a planned sulphuric acid plant and the IX Plant. As part of the EIS process, independent consultants assessed the environmental and social impacts of KCC operations and developed an Environmental Management Plan and a Sustainable Development Plan. The EIS was re-submitted in May 2019 following the comments received from the DRC authorities and was subsequently approved by DPEM in August 2019 and CAMI in September 2019.

4.8 Other Factors

Section 25.3 discusses the risk associated with the Material Assets of KML. There are no other known significant factors or risks that may affect access, title, or the right of KML/KCC to perform work on the property.

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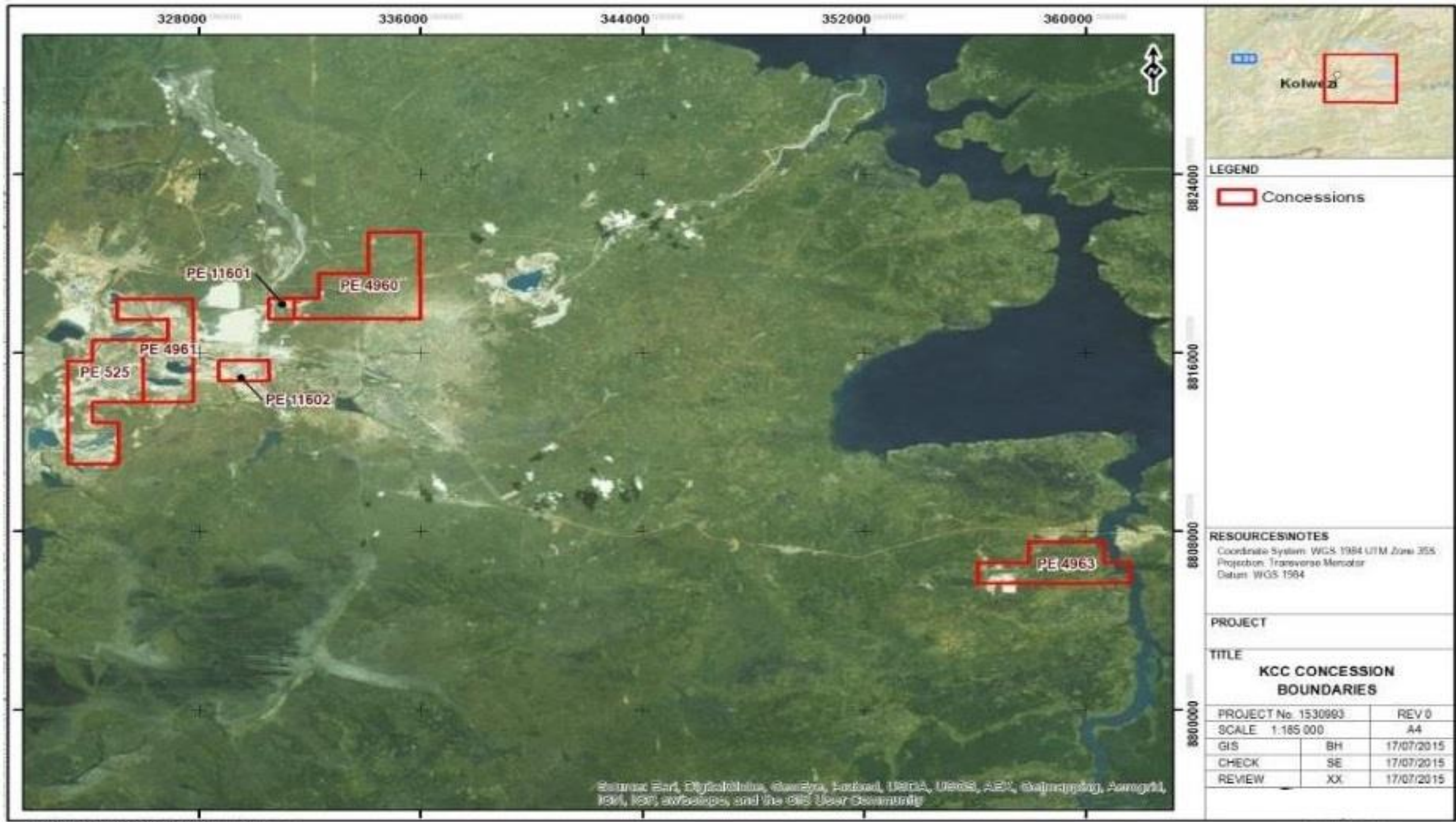


Figure 3: Property boundaries of KCC's exploitation permits

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5.0 Accessibility, Climate, Local Resources, Infrastructure and Physiography

5.1 Topography, Elevation and Vegetation

The Material Assets of KCC are located in the Congolese portion of the Central African Copper belt within the Lualaba Province of the DRC, the capital of which is Kolwezi. The DRC is located in south-central Africa.

Topographically the Material Assets are situated on rocks of the Roan Group, which is generally fairly flat, although there are several small ranges of low hills in the surrounding area. In this area the Roan Group is comprised of inter-bedded siliclastic and dolomitic units, with occasional occurrences of chert. In places the dolomitic units have been leached and silicified and where this has happened these silicified dolomites are relatively resistant to weathering, forming prominent ridges, such as is apparent around the KOV OP, which lies to the north-east of the KTO UG Mine. Where less resistant lithologies are encountered the dolomites and siltstones have been eroded to form gently sloping hillsides and shallow valleys draining to the Luilu, Musonoie, Kalemba and Potopoto rivers.

Due to the dolomitic nature of the host karstic topography is prevalent in the area, with weathering profiles known to be between 50 m to 200 m deep dependant on the orientation and position of structural controls. To the south lies the Manika Plateau, this has sandy soils with high permeability, resulting in a high ground water and surface water flow into the Kolwezi area. This flow drains through the Luilu and Musonoie rivers, which in turn converge into the Lualaba River. The Musonoie River flows towards the east.

The town of Kolwezi is situated at a latitude of 11 degrees south and longitude of 25 degrees east at an altitude of approximately 1400 meters above mean sea level (mamsl).

The Material Assets have an average surface elevation of 1400 m to 1450 mamsl.

At Kolwezi the vegetation is semi-tropical grassland with scattered trees and bushes. The surface soils are relatively well-drained and are extensively cultivated with cassava and peanut fields. Slopes at lower elevations on the Material Assets are typically covered with small brush and grasses with the largest concentration of vegetation in swampy areas in the valleys. Slopes at higher elevations are generally bare with very little vegetation and are predominantly covered by talus.

5.2 Accessibility

All the Material Assets of KCC are located in the Lualaba Province of the DRC, the capital of which is Kolwezi.

The Katanga Province was divided into four provinces subsequently to the central government law dated 9 January 2015 restructuring the provinces in DRC. Lubumbashi is now located in the Haut-Katanga Province, through which the import and export cargos for the operations also have to pass (see paragraph below relating to Road Transport). Some uncertainty as to the split of administrative responsibilities and certain tax duties between the provincial authorities remain to date and are addressed on a case-by-case basis when relevant to KCC.

A province assembly, as well as a new province Governor and government of the Lualaba, were elected in early 2016. Numerous infrastructure and community development programs are currently being conducted

The best option for travelling to KCC mines is flying by charter airplane to Kolwezi or by commercial airplane via Lubumbashi to Kolwezi. KCC has arrangements with a charter company that flies on a daily working day basis, from O R Tambo in Johannesburg, South Africa, to Kolwezi. Access to the mine is made from Kolwezi on a half gravel and half tar road. It is approximately 25 minutes driving time from the city to the mine.

Ground transport infrastructure in the DRC is limited. The terrain and climate of the Congo Basin present serious barriers to road and rail construction, and the distances are enormous. The DRC has thousands of kilometres (km) of navigable waterways, and water transport has been the traditional, dominant means of moving around approximately two-thirds of the country.

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Two civil wars have resulted in the destruction of transportation infrastructure from which the country has not yet recovered. Many vehicles were destroyed or commandeered by militias, especially in the north and east of the country, and the fuel-supply system was also badly affected.

Consequently, outside of Kinshasa, Matadi and Lubumbashi, private and commercial road transport is almost non-existent, and traffic is scarce even where roads are in good condition. The few vehicles in use outside these cities are run by the United Nations, aid agencies, the DRC government, and a few larger companies such as those in the mining and energy sectors. Air transport is the only effective means of moving between many places within the country. The government, the United Nations, aid organizations and large companies use air rather than ground transport to move personnel and freight. Compared to other African countries the DRC has a large number of small domestic airlines and air charter companies.

Specific transportation relevant to the operations:

- Rail Systems

The rail route between Kolwezi and Lubumbashi is electrified; the non-electrified rail section from Lubumbashi to the Zambian border is generally serviced by diesel electric locomotives. World Bank donated 38 diesel locomotives to Congo Railroad Company (Société Nationale des Chemins de Fer du Congo).

The rail system operates at very low capacity, as the condition of the railway network is extremely poor and not reliable. There are restrictions on this line with the maximum speed limited to about 40 kilometres per hour (km/h) while the load is limited to 600 t or the train length is restricted from 14 to 16 wagons. There are also width and load constraints where the railway line crosses the Lualaba River (width approximately 3 m and load capacity of 65 t). Delivery time for rail transport from Durban to Kolwezi historically has been on the order of three weeks.

The Luilu railway station is equipped with sidings that will assist in the logistics of the large volumes of imported reagents as well as the export of product. However, it is not connected with Kolwezi railway station currently and needs a rehabilitation of 20 km of rail tracks. KCC has not used train transportation since 2008 for import and since 2010 for export.

- Road Transport

Road access for equipment and material to be imported into the DRC from the south is via Zimbabwe and Zambia, crosses the border into the DRC at the Kasumbalesa border post. The road between Zambia and Likasi is in fair condition. The 196 km road between Likasi and Kolwezi has been improved and is now in fair to good condition. Travel time for this distance averages four to five hours. There are no current restrictions on the route. The Kapolowe Bridge, 25 km from Likasi towards Lubumbashi, was recently built and has two lanes. There is also a bypass on side of the bridge in case of any emergencies. The Lualaba Bridge, 27 km from Kolwezi towards Likasi, was also built recently and all truck loads are crossing without any restrictions.

The below changes at the Kasumbalesa border post increased the transit time recently:

- Introduction of manual payment system for Seguce fee;
- A compulsory scanner check (capacity of 10 trucks per hour and operating 8 hours a day) for all import trucks except acid and fuel;
- Change of management from a private company to the Haut Katanga Province; and
- Increase of goods traffic (import and export) due to resumption of the mining activity in the area.

Transportation time increased from 4 to 7 weeks for trucks from South Africa and from 10 to 19 days for trucks from Zambia. This caused a truck shortage in South Africa and Zambia, which affected supply of lime and other materials recently. These issues could impact the production targets for recovered copper

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and cobalt. KCC is taking necessary arrangements to mitigate these issues by requesting exemption or upgrade of scanner process at the border.

- Air Transport

Lubumbashi

Lubumbashi is the main airport for the Haut-Katanga province and caters to international flights. The airport has refuelling facilities, but there are occasional problems obtaining fuel supplies. Maintenance facilities are available. If charter flights are proceeding to other destinations in the province, customs and immigration must be cleared at Lubumbashi.

Kolwezi

The airfield at Kolwezi is an asphalt topped airstrip 2.5 km long and 30 m wide. Site altitude is 1,500 metres above sea level (masl). The condition of the strip is good, and the airfield is suitable for medium sized aircraft. KCC has arrangements with a charter company that flies on a daily working day basis, from O R Tambo in Johannesburg, South Africa, to Kolwezi.

KCC has contributed, and continues to contribute to transport infrastructure projects in the DRC. Local initiatives have included the rehabilitation of the airport of Kolwezi, the rehabilitation and maintenance of roads in and around Kolwezi, and participation in the construction of Lualaba Bridge.

5.3 Climate

Two broad climatic areas can be distinguished in the DRC. The Congo River Basin, which lies on the equator and forms around one-half of the country's area, consists of low-lying rain forest, which receives rainfall all year round. Temperatures are not as high as might be expected at the equator, however, humidity is generally high. The remainder of the country, comprising the area around Kinshasa, Kivu, Kasai, Katanga and Lualaba Provinces, experiences distinct rainy and dry seasons. Lualaba Province, lying largely at an elevation of 1,000 m or greater, experiences a climate with cooler, drier air than the majority of the country.

At only 10° latitude, daylight and night hours are almost equal, daylight lasting broadly from 06:00 to 18:00. Rapid temperature drops occur after sunset during the dry season as a result of lack of cloud cover.

Five distinct seasons can be readily distinguished, namely:

- Cool dry season: May – July;
- Hot dry season: August – September;
- Early rainy season: October – November;
- Full rainy season: December – February; and
- Late rainy season: March - April

Due to the elevation, the climate is rather uniform all year round, with a dry season between April and September, and a wet season for the remaining months. The average rainfall for the area is approximately 1,200 millimetres (mm) per year with periods of extreme precipitation and extreme aridity. Heavy rainfall occurs during the months of November to the end of March, as shown in Table 8 below. There is generally very little or no rainfall during the months of May to September.

Table 8: Kolwezi monthly average rainfall (mm)

Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
4	22	73	195	209	203	187	202	92	13	2	0

The mine is currently operating 12 months a year.

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5.4 Local Resources

The DRC has considerable hydroelectric power generating capacity, which is controlled and distributed by the national power utility, Societe Nationale de Electricite (SNEL). Kolwezi lies along the transcontinental railroad system and has access to both east and west coast ports of Tanzania and Angola, as well as South Africa. Lubumbashi, located some 300 km southeast of Kolwezi, is the commercial and industrial centre of the Haut-Katanga Province and hosts an international airport.

The existing infrastructure around Kolwezi (e.g. buildings, water lines, workshops and roads) is in a very bad state of repair. Cellular phones work in the area.

KCC has contributed and continues to contribute to electricity infrastructure projects in the DRC. In collaboration with SNEL, KCC provides financial and technical assistance for the improvement of existing infrastructure and facilitates communities' access to electricity supply through the construction of new facilities and distribution systems. KCC has supported SNEL with the donation of cables and transformers and the punctual rehabilitation of undamaged facilities. At national level, in partnership with the government, SNEL and other mining companies, KCC is participating in the INGA project to increase electricity production and supply within the DRC. The INGA project was commenced in 2012 and is expected to be completed in Q1 2020. KCC will fund USD 392.6 M of the INGA project, of which USD 261.7 M is to be reimbursed by Mutanda Mining SARL (a related party of KCC and part of the Glencore Group). Accordingly, KCC's net funding contribution will be USD 130.9 M, of which USD 119.3 M has been funded as at September 30, 2019. SNEL will retain ownership of the generation and transmission infrastructures throughout the duration of the project and thereafter.

5.5 Infrastructure

As a result of the mining activities (mainly under the control of the state-owned mining company GCM) the area around Kolwezi has numerous open pits, waste rock, ore and slag dumps, tailings dams, concentrators and other mining-related infrastructure. A substantial urban infrastructure has also developed, including housing, roads, water supply and social facilities. The processing plant sites are well established.

In terms of the AJVA with GCM, KCC has an operating lease for the land containing the production areas and surrounds used for waste dumping, stockpiles and also the processing plant as stated in Section 4.0.

5.6 Physiography

Poor living conditions and civil war have endangered much of the biodiversity of the DRC. The rising population has forced many people to become dependent on wild animals for either their livelihood or their food. Extensive logging throughout the country by corporate loggers and land clearing by farmers for agriculture or charcoal production have caused destruction of the forests throughout the country and resulted in significant degradation of habitat. Miombo woodland is the predominant vegetation type in the Lualaba Province. Such woodlands extend over about 2.8 million square kilometres (km²) of the southern sub-humid tropical zone from Tanzania and DRC in the north, through Zambia, Malawi and eastern Angola to Zimbabwe and Mozambique in the south. Their distribution largely coincides with the flat to gently undulating African (early Tertiary) and post African I (Miocene) planation surfaces that form the Central African plateau. These woodlands constitute the largest more-or-less contiguous block of deciduous tropical woodlands and dry forests in the world.

Miombo woodlands supply many goods and services that are essential to the well-being of rural communities; some products act as subsidies to agriculture while others provide for basic needs, such as food, shelter and health (Clarke et al., 1996). Cavendish (2002) records over 100 different types of resource utilizations in a single Miombo study area, with many types having multiple species (e.g. 47 wild fruits, over 40 medicinal species, 40 wild vegetables). Wood alone provides subsistence farmers and households with numerous products, including poles and construction products, timber, materials for tool handles and household utensils, foods, medicines, leaf litter, grazing and browse (Clarke et al., 1996). In the Lualaba Province, the population is heavily dependent on their supply of wood as a source of energy through wood-fuel and charcoal production.

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6.0 History

6.1 Introduction

This section gives a short historical overview of the mining history at Lualaba Province, Material Assets of KCC including historical development, infrastructure and operating results.

The mining industry of Lualaba, formerly part of Katanga province until 2015, began in 1911, with the first mine owned by the Belgian company Union Minière du Haut Katanga (UMHK). In the same year, the Benguela Railway to the Angolan port of Lobito opened the area for significant freight transportation. Shortly afterwards, the railway from the south through Zimbabwe (Rhodesia at that time) gave access to South Africa and Mozambique. UMHK prospered and in the 1950s was one of the four largest copper producers in the world and by far the largest producer of cobalt.

The Congo's independence from Belgium in 1960 led that same year to unrest and disruption. The province of Katanga declared independence, and a short war followed, ending with the failure of the independence attempt. UMHK resumed control and, on nationalization of the industry as GCM, continued through a management contract. Throughout the years of the Mobutu presidency, GCM declined due to starvation of funds, resulting from various causes such as misplacement of assets and profits. Civil war followed, and by 2002, GCM had been brought to a state of ruin. It was, in effect, bankrupt and able to sustain a production rate of only about 20 Ktpa of copper compared to its peak production of almost 500 Ktpa. It was unable to pay its employees. The major production units were flooded.

The situation was stabilized with the World Bank's intervention, and in 2002, the 2002 Mining Code was promulgated, providing for privatization of the Mining Assets by negotiated agreements with GCM. Since then a number of joint-venture agreements have been reached and confirmed by the DRC's Council of Ministers.

6.2 Prior Ownership of the Material Assets

From commencement of operations until 1967, all mining activities on the KCC property were operated by UMHK. However, because of the independence in 1960, the mines were nationalized in 1967 and all assets of the UMHK were transferred to GCM.

6.2.1 KCC Assets

The joint venture was governed by the Kamoto joint venture agreement (JVA). The parties to the agreement were GCM, a DRC public enterprise incorporated under the laws of the DRC and KFL, a private company incorporated under the laws of the British Virgin Islands. Negotiations between GCM and KFL started in June 2001. The JVA between GCM and KFL was approved by Presidential Decree dated August 2005 after all regulatory approvals were obtained.

6.2.2 DCP Assets

The joint venture was governed by the DCP SARL JVA. The parties to the agreement were GCM and Global Enterprises Corporate Limited (GECL), a private company incorporated under the laws of the British Virgin Islands. Negotiations between GCM and GECL started in May 2004. The JVA between GCM and GECL was approved by Presidential Decree in October 2005 after all regulatory approvals were obtained.

6.2.3 The Merger

On January 11, 2008, KML announced that its shareholders approved the merger with Nikanor PLC (Nikanor). The merger brought together KCC and DCP assets to create a single operation governed by the AJVA.

6.3 Historical Development

Corporate mining activity in the province of Katanga began in 1906 with the formation of the UMHK. In 1967, following national independence, the operations of UMHK were nationalized and transferred to GCM. At its peak in the late 1980s GCM produced about 7% of the world's copper and 62% of its cobalt. In 1986, GCM produced 476 thousand tonnes (Kt) of copper and 14.5 Kt of cobalt, 63.9 Kt of zinc, 34.3 t of silver, and cadmium and other minor metals (Source: GCM data).

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The majority of this production came from the Kolwezi district. In 1995, production fell to 32.5 Kt of copper, 4 Kt of cobalt, and 4.5 Kt of zinc. The decline in metal production continued to the point that primary production in the Kolwezi area virtually stopped.

GCM overall decline was due to a number of factors including:

- The political isolation of the DRC (then Zaire) in 1991;
- The loss of financial credit lines;
- The lack of sustaining capital and maintenance improvements;
- The social and political environment within the country during this period; and
- The collapse of the Plateau in the central UG portion of KTO.

The Dikuluwe-Mashamba (DIMA) pit group operated from 1975 through 1998 during which time a total of 57.7 Mt of ore grading 4.96% copper and 0.16% cobalt was mined (Source: GCM). There was no significant production from T17.

Growth in mining production since 2008 has been significant. Ore production from the KTO UG mine has increased 352%, from 0.55 Mt in 2008 to 1.94 Mt in 2014 and 1.48 Mt in 2015. Additionally, the KOV OP, which was dewatered in 2010, has increased from no ore production in 2008 to 5.4 Mt in 2014, 4.6 Mt in 2015 and 6.5 Mt in 2018 after the care and maintenance period.

Similarly, there have been significant increases in processing volumes since 2008. Ore milled at KTC increased from 1.00 Mtpa in 2008 to 6.31 Mtpa in 2014 (531% increase), 5.46 Mt in 2015 and 7.5 Mt in 2018 resulting in concentrate being produced at KTC increasing from 100 Ktpa in 2008 to 910 Ktpa in 2014 (810% increase) and 860 Ktpa in 2015. At the same time copper production at Luilu increased from 22.12 Ktpa in 2008 to 157.02 Ktpa in 2014 (610% increase), 106.8 Ktpa in 2015 and 152.3 Ktpa in 2018 while cobalt production increased from 0.75 Ktpa in 2008 to 2.9 Ktpa in 2015 (387% increase) and 11.1 Ktpa in 2018 (1480% increase).

Due to a slump in metal prices, KCC put its UG operations on care and maintenance on September 11, 2015 and reduced its OP operations to waste stripping for the remainder of 2015 and 2016. Similarly, KTC and Luilu facilities were put on care and maintenance at the same time. During the care and maintenance period, KCC also continued with the construction of its WOL facility, which will enhance copper recovery and reduce overall operating costs per tonne of metal produced. Ore mining at KOV OP recommenced in 2017 ahead of the anticipated commissioning of the WOL facility.

The drill whole logs indicate that exploration drilling commenced in the early 1940s initially targeting the KTE fragment and initial drill holes were prefixed as KTO. Although the target was to drill into the KTE fragment, a substantial number of KTO drill holes were later drilled into the present day KOV OP.

6.4 Historical Exploration

Exploration work was undertaken by UMHK and GCM. The oldest drill hole on record related to the Material Assets still retained by GCM is KTO1 (dated 23/03/1942), one of the original deep drill holes drilled for the evaluation of the geology underlying the Kamoto North (KTN) OP and KTO UG. Numerous surface and underground exploration drill holes were drilled in the 1950s and 1960s for the UG operations of KTO, with development beginning in the mid-1960s.

Prior to the exploitation of the DIMA pits, the surface area was drilled on a systematic grid (initially 400 x 400 m but typically 100 m x 100 m or 50 m x 50 m).

Prospecting, geological mapping, trenching and shallow drilling by GCM on permits has discovered the most obvious surface and near-surface copper-cobalt mineralization. Drilling by GCM proved that mineralization persists to depths of 150 m below surface.

The drill hole logs indicate that surface exploration drilling commenced in the early 1940s initially targeting the KTE fragment and the southern flank of Kamoto Principal and initial holes were prefixed as KTO. Although the

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target was to drill into the KTE fragment, a substantial number of KTO holes were later drilled into the present day KOV OP.

Later in the 1980s, another surface drilling campaign commenced to define the KOV fragments and this campaign continued into the early 1990s. The drilling was carried out along section lines spaced about 100 m apart. Where feasible, drill holes were spaced about 100 m along these section lines. The holes drilled in this campaign were prefixed KOV.

No systematic exploration of the area has been carried out since the 1990s and due to the fact that a 20-40 year resource was believed to already exist (GCM internal reports), evaluation of existing data was the focus of the project, and no new exploration work was undertaken by, or on behalf of, the issuer.

There is no further information available regarding historical exploration. Refer to Section 9.0 for further details of exploration activities in recent years.

The following sections have been extracted from SRK 2007 TR titled "Independent Technical Report on the Assets of Nikanor in the Katanga Province, Democratic Republic of Congo".

6.4.1 KOV OP

As part of the planning for mining the north and northwest side of KOV in the first 3-4 years, 8 holes (KV1-KV8) were drilled in the northwest quadrant of the pit between October and end November 2007, totalling approximately 1,100 m. This was mainly designed to check the shallower ore; particularly the northwest edge of the Virgule fragment to greater than 200 m. Along this margin, the latter is interrupted by sub-horizontal faults and is not as continuous as believed from the 100 m grid drilled by GCM. However, it is useful in showing that the fragment is almost completely oxidized with minor mineralisation extending into the base of the SDS. More than 70% of the samples from the drill holes have been dispatched for assaying.

6.4.2 Kananga

The Kananga limb covers 4 km of which more than 80% has been covered by drilling.

The eastern limb amounting to over 3 km was explored by a 13 fence lines, 200 m apart, with 2 holes each that penetrated the northern, sub-vertically dipping limb of an asymmetrical syncline to maximum depths of approximately 240 m. While sporadic significant cobalt mineralisation was noted close to the Upper Orebody position, there was very little copper, as proved by assaying, and even the heterogenite occurrences lend themselves largely to artisanal mining only.

At Kananga West at the upper portion both drilling and mining have shown that economic mineralisation is limited to ~500 m strike in an east-northeast direction, commencing 50 m southwest of the railway crossing. Drilling on a 50 m x 50 m grid has clearly demarcated the western faulted cut-off to the northern limb of the syncline which, near vertical close to surface, dips 70-50° southwards at 100-150 m depth.

At the lower portion the down-dip extension of the western segment between 150 m and >350 m depth to the southern limit of the concession boundary has been partly explored by holes on a 100 m grid. This confirms the steady increase in mineralisation westwards and also in the Upper orebody which is poorly mineralized near surface and similar or better grades are expected. There is significant upside potential to the southeast.

6.4.3 Tilwezembe

The favourable Mwashya dolomite contact zone has been covered by drilling on a 100 m spacing over 3 km, with a further 1.5 km of known strike remaining.

Significant intercepts towards the base of the weathered zone such as 6% Cu over 21 m, as well as deeper zones with 5-15 m widths indicate greater potential. In addition, the first phase reconnaissance drilling immediately east of the pit area indicates further potential above the 180 m depth.

Drilling in the far eastern half of the strike has revealed only spotty mineralisation, although more detailed drilling around 2 known artisanal and CCM pits will probably define small additional resources.

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6.5 Historical Drilling

6.5.1 KOV OP and KTE

The drill hole logs indicate that exploration drilling commenced in the early 1940s on the KTE fragment, and initial holes were prefixed as KTO. Although the target was the KTE fragment, a substantial number of KTO holes were later drilled into the present day KOV OP. During the 1950s further drilling took place systematically over the surface area of present day KOV OP on section lines spaced approximately 300 m apart. Surface drilling continued during the early to mid 1960s on a smaller 100 m grid mostly targeting the KTE, Oliveira and Virgule fragments. All these holes were prefixed KTO.

In the 1980s, another major drilling campaign was aimed at defining the KOV mineralized zones, and this campaign continued into the early 1990s. The drilling was carried out along section lines spaced about 100 m apart. Where feasible, drill holes were spaced at approximately 100 m along these section lines. The drill holes were prefixed KOV.

Most of the drill holes within the KTE and the KOV OP areas were drilled vertically, with only a few being inclined. The KTE drilling was problematic due to the steepness in the dip of the strata. As a result, the majority of the drill holes intersect the near surface expression of the KTE fragment, and only the inclined drill holes provide intersections at depth.

6.5.2 KTO UG

GCM carried out both extensive surface and underground drilling to delineate the KTO ore bodies. Drill hole logs indicate that exploration drilling commenced from surface in 1942, targeting the southern flank of the KTO Principal fragment and were systematically drilled on 50 m northwest trending section lines. Drilling continued northwards during the 1950s and 1960s, intersecting the KTO Principal "plateau", Etang South and Etang North fragments. During the 1970s, all drilling targeted the Etang South fragment. The surface drilling remains as the only drilling that covers the up-dip portion of this fragment (above 400 level). Limited surface infill drilling took place in 1990 and 1991 focused on the Etang North fragment.

Drill hole logs indicate that exploration commenced from underground development in 1959, targeting the southern flank of the KTO Principal fragment. Drilling continued northwards during the 1960s, 1970s and 1980s on north trending section lines roughly spaced at 25 to 50 m apart to cover the KTO Principal "plateau". During the 1990s up until 2002, drilling primarily focused on the Etang South and Etang North fragments, with a lesser focus on Zone 8 and Zone 9 within the KTO Principal fragment. Underground drill holes were generally drilled as fans of three or more drill holes from specifically mined cobbles.

The majority of the surface drill holes within the KTO UG were drilled vertically, with only a few being inclined. Drill hole downhole surveys were carried out at regular intervals for $\pm 80\%$ of the total drill holes and the results show minimal deviation. For the underground collared drill holes, less than 2% of the total drill holes were surveyed.

6.5.3 Mashamba East

For Mashamba East, drilling was undertaken from the surface on a 100 m x 100 m spaced grid parallel to the strike of the mineralized zone. These drill holes were drilled vertically from the surface, and show minimal deviation in the available downhole surveys.

6.5.4 T17 OP and T17 UG

Caracle Creek International Consulting Inc (CCIC) indicate that the T17 West deposit has been the subject of two diamond drilling programs by GCM, with 3,287.6 m drilled between 1938 and 1954, and 8,011.3 m from 1986 to January, 1988. Drill holes were drilled generally to a nominal 100 m x 100 m grid, with certain areas being on 50 m x 50 m spacing.

6.5.5 Tilwezembe OP

The historical drilling was undertaken by GCM, and the information was the basis for the first Mineral Resources estimates for Tilwezembe Mine undertaken by SRK. The drilling information included drill holes on a 25 m spacing within the operational Tilwezembe pit and 100 m x 50 m on Tilwezembe East.

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6.5.6 Kananga OP

The historical drilling at Kananga OP was limited to approximately eight drill holes drilled by GCM. Details of the drilling, sample collection and sample analyses for the drill holes are not available.

Refer to Section 10.0 for further details of drilling activities in recent years.

6.6 Historical Technical Reports

In 2007, Snowden Mining Industry Consultants (Snowden) completed the first phase of a Mineral Resource evaluation for the Tilwezembe and Kananga OP in accordance with SAMREC (SRK, 2007). There are no other available public domain historical Mineral Resource statements that are relevant to this TR.

In November 2007, SRK completed an Independent Technical Report (SRK, 2007) which was updated in 2009 and 2010.

KML published Independent Technical Reports, produced by Golder in regard to its Material Assets on March 22, 2011 and in March 30, 2012.

KML published the most recent TR's, produced by KCC and independently reviewed by Golder in regard to its Material Assets on March 31, 2017 and March 31, 2018.

6.7 Historical Mineral Resource and Mineral Reserve Estimates

Following the collapse in the central portion (the Plateure) in 1990 GCM produced "reserve" figures for the remaining UG Mineral Resource, which they subdivided into nine (9) zones. GCM reported "reserve" values in 1999 for KTO Principal and Etang, as well as the DIMA pits and Musonoie-T17 West are documented in Table 9 below:

Table 9: GCM reported historical "Mineral Reserve" values in 1999¹⁻⁵

Pits	Total Tonnes	%Cu/T	Tonnes Cu	%Cu/S	T-Cu/S	%Co	T-Co
Dikuluwe Total	54,921,722	4.10	2,247,723	2.70	1,495,927	0.10	42,037
Junction DIMA Total	11,373,615	2.90	334,408	0.20	22,878	0.20	19,776
Cuvette DIMA Total	46,192,860	4.20	1,933,160	2.61	1,205,285	0.10	49,883
Mashamba West Total	52,327,069	3.05	1,593,512	1.60	854,381	0.31	163,592
Mashamba East Total	29,417,528	3.60	1,060,730	0.10	15,283	0.70	196,688
Cuvette Mashamba Total	25,067,380	2.80	703,617	1.33	334,296	0.60	151,331
Musonoie T-17 Total	3,752,258	4.00	149,219			0.70	27,261
Underground	Total Tonnes	%Cu/T	Tonnes Cu	%Cu/S	T-Cu/S	%Co	T-Co
Kamoto North Total	4,852,317	3.70	181,906			0.20	8,111
Kamoto Etang Total	22,989,044	3.30	761,121			0.80	183,707
Kamoto Principal Total	6,103,067	4.90	3,236,328			0.40	251,877

- 1) %Cu/T – total copper grade in percent
- 2) %Cu/S – non-oxidized portion of the copper grade in percent
- 3) T-Cu/S – non-oxidized portion of the copper metal
- 4) %Co – total cobalt grade in percent
- 5) T-Co – total cobalt metal

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GCM personnel from the Central Geology and Engineering Office developed the “reserve” values. CCIC were unable to ascertain the exact methodology used to derive these “reserves” and as such are unable to comment on the reliability of these figures. These figures should therefore only be considered as historic estimates (i.e. an estimate prepared prior to February 1, 2001). As such, the quoted “reserves” are not in accordance with the categories set out in sections 1.2 and 1.3 of the Instrument and should be considered only as a rough indication of mineralization, grade and tonnage.

An in-house GCM Technical Report (No. 73, 1988) by Kvapil and Hustrulid provided an “ore reserve” calculation for the proposed DIMA UG area of 138,336,000 t containing 4,993,464 t of contained copper metal.

Exploration drilling undertaken on Musonoie-T17 between the 1420 and 1300 Levels in 1986 delineated a Mineral Resource of 3,752,257 t of ore grading at 4% Cu and 0.7% Co (for 149,219 t of copper and 27 261 t of cobalt) (Katekesha, 1989). Once again, these figures should be considered only as historic estimates, and are not in accordance with the categories set out in Sections 1.2 and 1.3 of the Instrument. As such they should be considered only as an indication of mineralization, grade and tonnage.

There is no information available on the reporting code GCM used to classify the Mineral Resource and Mineral Reserve estimates presented in Table 9. KCC is unable to compare these estimates with the most current Mineral Resource and Mineral Reserve estimates presented in Chapter 14 and Chapter 15 of this TR respectively.

6.8 Historical Production

Historical production and exploration prior to ownership of the Material Assets by KML has included the following:

6.8.1 KOV OP

KOV OP comprises the five fragments namely, KTE, Oliveira and Virgule (hence the name KOV), Variante and FNSR.

The KOV OP and KTE OP filled with water during the period of GCM ownership. Under KCC ownership, a program to dewater the pits and the surrounding slopes was developed and the pits were dewatered which enabled OP mining activities to resume. The KOV OP is in an area that is highly disturbed by past mining activities. The old Musonoie pit, immediately east of KOV OP, has been partly back-filled. Extensive accumulations of waste materials are present to the north and south of KOV OP.

The KTE OP started in 1959 and continued operations to 1985. Metallurgical records indicate that the ore from the KTE fragment was delivered to the plant from 1960 to 1985, while ore production from the KOV OP commenced in early 1985 and carried on, at declining production rates, through to 2000. During that time, some 38 Mt of ore was delivered to the plant at an average grade of 5.8% copper and 0.5% cobalt. Mining at the Musonoie pit commenced in 1943 and ceased production by 1984.

6.8.2 Mashamba East OP

Mashamba East OP operated from 1985 through 1988 and the pit produced a total of 9.8 Mt of ore at an average grade of 4.96% copper and 0.35% cobalt.

Production ceased in 1988 and by 1998, due to the lack of funds and increasing costs, the pit was allowed to flood. The pit was dewatered prior to the recommencement of stripping in 2015 and 2017. Ore production started in 2018 and produced 3.3 Mt of ore as feed for KTC.

6.8.3 KTO

The official opening of KTO is given by GCM as 1942, with the beginning of exploitation of the opencast Mineral Resource in 1948 and the opening of the KTO UG mineshaft in 1959.

UG operations at KTO are accessed by twin declines, two primary shafts and three secondary shafts. Primary access is through the declines and ore handling is through the primary shafts from which crushed ore is transferred directly onto a conveyor to the KTC.

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UG shaft was opened in 1959 and production began in 1969, which used a variety of large-scale techniques including cut and fill, room-and-pillar and sub-level caving. Production steadily increased to reach the rate of 3 Mtpa by mid-1970. Production reached a peak in 1989, when the mine produced 3.3 Mt of ore. In 1990, a major collapse in the central portion (the Plateure) of the UG deposit resulted in the loss of approximately 15 Mt of Mineral Resource. Since that time production from KTO has steadily decreased to the point that primary production essentially stopped. From 1969 through 2005 a total of 59.4 Mt of ore grading 4.21% Cu and 0.37% Co have been mined from KTO.

6.8.4 T17

Pre-stripping for T17 OP began in May 2007 and the first blast was at the end of September 2007. OP production from 2008 to 2014 is given in Section 6.8.9. T17 UG is a new site, with no significant production history from UG to date.

6.8.5 Tilwezembe Mine

Mining has taken place intermittently since 1999. A rail siding and contractors yard was established close to the site. GCM conducted OP mining at Tilwezembe using contract mining on and off for a period of about 7 years.

Copper and cobalt head grades are reasonably well defined using both current and historical records from the GCM geology database and from head grades of ore processed in the Kolwezi concentrator. The resulting oxide concentrate was leached and refined at the 'Old' Luilu and Shituru refineries, located at Kolwezi and Likasi respectively, producing a "B" grade copper cathode and cobalt metal. The mineralized zones in Tilwezembe also contain a high proportion of manganese (which requires an adjustment to the processing for this ore).

Latter, organised mining by GCM was replaced by artisanal mining within the existing pit and along strike, until DCP recommenced mining in 2007. The existing OP is located at the western extremity of the orebody.

6.8.6 Kananga Mine

GCM conducted OP mining at Kananga using contract mining from around mid-2004. The ore was processed in the Kolwezi concentrator. The resulting oxide concentrate was leached and refined at the 'Old' Luilu and Shituru refineries, at Kolwezi and Likasi respectively, producing a "B" grade copper cathode and cobalt metal.

Mining by GCM has been replaced by predominantly artisanal mining within the existing pit and along strike. The existing OP is at the western extremity of the orebody and had been mined to a depth of approximately 20 m.

6.8.7 KTC

KTC consists of four sections: Kamoto 1 and 2 built in 1968 and 1972 respectively, and DIMA 1 and 2 built in 1981 and 1982. The Kamoto 1 and DIMA circuits were designed to process mixed ore types, and Kamoto 2 was designed for sulphide ore. From 1969 through 2000, KTC processed over 126 Mt of ore at an average grade of 4.33% copper and 0.28% cobalt. In its current configuration, KTC is capable of processing 12 Mtpa of ore.

6.8.8 Luilu Metallurgical Plant

The Luilu metallurgical plant is located approximately 6 km north of KTC. It was constructed in 1960. In 1972, it was expanded to an annual capacity of 175 Kt of copper and 8 Kt of cobalt. The plant had three roasters, a leaching circuit and electrowinning cells for copper and cobalt production. From 1984 through 1989, annual production at Luilu averaged 173 Kt of copper and 5.9 Kt of cobalt. Production peaked in 1986 at 177.5 Kt of copper and 7.8 Kt of cobalt. By 1996, production had fallen to an estimated 27 Kt of copper and 1.2 Kt of cobalt.

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6.8.9 Historical Production since Acquisition

The chronological developments in regards to Material Assets since acquisition are presented on Table 10. There was no ore or metal production during 2016 due to the operations being on care and maintenance and construction of the WOL Project.

Table 10: Historical production since acquisition

	Unit	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Mashamba East OP												
Mined ore	Mt	-	-	-	-	-	-	-	0.004	-	-	3.32
Cu grade	%	-	-	-	-	-	-	-	3.42	-	-	1.55
Co grade	%	-	-	-	-	-	-	-	0.42	-	-	0.33
KTO UG												
Mined ore	Mt	0.55	1.1	1.31	1.61	1.84	1.74	1.94	1.48	-	-	0.37
Cu grade	%	3.92	3.85	3.82	3.71	3.6	3.35	3.42	3.22	-	-	3.35
Co grade	%	0.43	0.49	0.56	0.53	0.57	0.5	0.46	0.36	-	-	0.60
KOV OP												
Mined ore	Mt	-	-	0.72	2.52	3.71	4.39	5.38	4.55	-	0.43	6.48
Cu grade	%	-	-	4.43	4.98	4.38	4.35	4.1	3.93	-	2.18	2.38
Co grade	%	-	-	0.3	0.37	0.38	0.37	0.41	0.47	-	0.48	0.48
T17 OP												
Mined ore	Mt	0.4	1.6	1.9	0.4	-	0.11	0.11	-	-	-	-
Cu grade	%	1.72	1.3	2.35	3.32	-	3.48	3.3	-	-	-	-
Co grade	%	0.89	0.85	0.93	1.01	-	0.72	0.88	-	-	-	-
Tilwezembe OP												
Mined ore	Mt	0.6	-	-	-	-	-	-	-	-	-	-
Cu grade	%	1.39	-	-	-	-	-	-	-	-	-	-
Co grade	%	1.17	-	-	-	-	-	-	-	-	-	-
KTC												
Milled ore	Mt	1.0	1.99	2.93	4.1	4.61	5.6	6.31	5.46	-	1.92	7.53
Concentrate produced	Kt	100	190	300	480	500	710	910	860	-	95.81	205.6
Luilu												
Concentrate fed	Kt	970	180	260	300	360	490	1,110	840	-	13.76	198.3
Ore fed	Kt	-	-	-	-	-	-	-	-	-	126.47	466.7
Copper produced	Kt	22.12	41.96	52.18	57.61	61.44	87.48	157.02	106.8	-	2.2	152.4
Cobalt produced	Kt	0.75	2.53	3.44	2.43	2.13	2.3	2.78	2.9	-	-	11.11

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7.0 Geological Setting and Mineralization

7.1 Regional Geology

The mineralized zones are located at the western end of the Katangan Copper belt, one of the great metallogenic provinces of the world, and which contains some of the world's richest copper, cobalt and uranium deposits.

These deposits are hosted mainly by metasedimentary rocks of the Late Proterozoic Katangan System, a 7 km thick succession of sediments with minor volcanics, volcanoclastics and intrusives. Geochronological data indicate an age of deposition of the Katangan sediments of ~880 Ma and deformation during the Katangan Orogeny at <650 Ma. This deformation led to the formation of the northwest-southeast trending Lufilian Arc, which extends from Namibia on the west coast of Africa through to Zambia, lying to the south of the DRC. Within the DRC, the zone extends for more than 300 km from Kolwezi in the northwest to Lubumbashi in the southeast.

Stratigraphically, the rich copper and cobalt deposits found in Zambia and the DRC occur localised in the Roan Supergroup (Roan). The Roan occurs at the base of the Katanga succession, overlying the basement rock of Kibaran age (Mid-Proterozoic). The Roan is separated from the overlying rocks of the Kundelungu and the Nguba Groups by a conglomerate, the Grand Conglomerat. The Nguba is composed of sandstones and shales with a basal conglomerate while the Kundelungu consists essentially of sediments and is separated from the Nguba by a conglomerate, the Petit Conglomerat.

Within the Lufilian Arc are large scale east-west to northwest-southeast trending folds with wavelengths extending for kilometres. The folds are faulted along the crests of the anticlines through which rocks of the Roan Group have been diapirically injected into the fault zones, squeezed up fault planes and over thrust to lie above rocks of the younger Kundelungu. The overthrust Roan lithologies occur as segments or fragments on surface which are intact units preserving the original geological succession within each of the fragments. A fragment could be of hundreds of metres aligned across the fault plane.

In the Katangan Copper belt, the mining for copper and cobalt occurs in these outcropping to sub-outcropping fragments.

7.2 General Stratigraphy

The generalized stratigraphy of the Katangan System is shown in Figure 4. The Roan has been correlated across the Katangan copper belt into four main formations or groupings, R1 to R4. The divisions between each of the R series are often marked by an unconformity. The main ore-body lithologies belong to the R2 Formation, but R3 and R4 Formations are also known to contain mineralization. Within each of the R series are subdivisions identifying the different lithological units. Rocks belonging to the Roan Group are described briefly below from the oldest to the youngest:

Breche Heterogene or Heterogeneous Breccia (BH): This breccia is composed of angular and sometimes well rounded fragments of all the various rock types of the Roan. The fragments vary in size from a few millimetres to several tens of millimetres in diameter, while the matrix is made up of finer-grained sandy particles of the same material as the fragments.

Breche RAT or Brecciated RAT (B RAT): A reddish-pink brecciated rock with calcite and silica veinlets and is at times well mineralized with specular haematite, occurring as veinlets.

Roches Argilleuses Talceuse (RAT): The RAT is considered the boundary between the R2 and R1 units and consists of a Roches Argilleuses Talceuse Upper (RAT Grises) (R2) and a Roches Argilleuses Talceuse Lower (RAT Lilas) (R1). Both are massive but sheared in places, silty or sandy, dolomitic rocks. Mineralization in the form of malachite and black oxides occurs associated with the upper RAT.

Dolomie Stratifie or Stratified Dolomite (DSTRAT): This is a well-bedded, argillaceous dolomite, which forms the base of the traditional "Lower Ore Zone" in GCM nomenclature. The mineralization consists of copper and cobalt oxides.

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Roches Siliceuses Feuilletées Foliated (Laminated) and Silicified Rocks (RSF): These are grey to light-brown, thinly bedded laminated and highly silicified dolomites. The unit is generally well mineralized with copper and cobalt oxides. Together with the DSTRAT, the RSF comprise the OBI.

Roches Siliceuses Cellulaires or Siliceous Rocks with Cavities (RSC): Vuggy and in-filled massive to stromatolitic silicified dolomites. Copper mineralization is almost absent in these rocks, which were therefore regarded as barren. However, the infillings are enriched in wad (manganese oxide) and heterogenite (cobalt oxide), and RSC is the target of artisanal activity.

Schistes De Base or Basal Schists (SDB): Reddish-brown to grey silty and nodular dolomite to siltstone. This unit is well mineralized with copper and cobalt in varying amounts and forms the OBS.

Shales Dolomitiques Supérieurs or Upper Dolomitic Shales (SDS): Yellowish, cream-toned, bedded laminated dolomitic siltstones and fine-grained sandstones. The rock is sparsely mineralized with malachite.

Calcaire a Minerais Noirs or Calcareous Unit with Black Minerals (CMN): A slightly banded and laminated light-grey to grey, silicified dolomite mineralized with black oxide of iron, manganese and cobalt. The unit bears some similarities with the RSC.

Dipeta (R3): Greyish to dark red or brown stratified shales and micaceous schist.

Mwashya (R4): Altered stratified greyish siliceous dolomitic rock with oolitic horizons and a few bands of light-yellow, talcose schist. Nodules of haematite often occur.

KATANGA SYSTEM					STRATIGRAPHY	
System	Series	Formation	Local Name	Orebody	Description	Thickness (m)
KATANGA	KUNDELUNGU SUPERIEUR - UPPER KUNDELUNGU				Sediments, at base limestones, calcschists grading into shales, sandstones at base Petit Conglomerat	30 - 50
	KUNDELUNGU INFERIEUR - LOWER KUNDELUNGU				Sediments, sandstones and shales, at base Grand Conglomerat	200 - 500
	ROAN	R4-1and 2	MWASHYA		Shales, siltstone, sandstone to dolomites, limestones, oolitic, jasper and frequently siliceous beds and cherts	50 -100
		R3-2	DIPETA		Shales and sandy schists with intercalated dolomites and limestones	1000?
		R3-1	RGS		Roches Greuseuse Supérieur, dolomitic shales and sandy schists, towards top beds of silicified dolomites	100 -200
		R2-3	CMN		Calcaire a Minerais Noirs, light coloured dolomites with thin beds of white sandstone argillitic dolomites alternating with schists, sometimes graphitic and two collenia (stromatolite) horizons locally, at base dolomitic sandstones	130
		R2-2	SDS		Schistes Dolomitic Supérieur, upper dolomitic shales, subdivided as follows:-	50 -80
					SD 3b finely laminated black, graphitic shales, highly micaceous, sandy dolomites	
					SD 3a micaceous, sandy shales, dolomitic, grey-green, well bedded	
					SD 2d shale, dolomitic and graphitic	
					SD 2c shale, dolomitic and sandy, grey-green	
		R2-1	SDB		SD 2b dolomite, massive, light coloured, collenia (stromatolitic) dolomite	10 -15
					SD 2a dolomitic, graphitic black shale, micaceous	
					SD 1b BOMZ (Black Ore Main Zone) blue-grey, crystalline dolomite	
				OBS	SD 1a Schistes De Base, shale dolomitic, well bedded, micaceous, light grey nodule horizons at base	
					Roches Siliceuses Cellulaire, massive, crystalline light coloured dolomite, black crystals present (heterogenite ?), locally developed collenia horizons, occasionally shales at bottom and top, extreme cellular weathering	12 -25
			RSC		Roches Siliceuses Feuilletées, siliceous, dolomitic shales, micaceous, light-grey weathering whitish, extremely finely laminated	5
			RSF		Dolomie Stratifiée, argillitic dolomite, well bedded to laminated, light-grey in colour light-brown, yellow weathering, chert horizons, nodule layer towards top	3
			D STRAT	OBI	Roches Argilleuses Talceuse, argillitic sandstones, massive and dolomitic, light-grey talc present in varying quantities	2 -5
			RAT grises			
		R1	RAT 2		Roches Argilleuses Talceuse, argillitic siltstones, dolomitic, banded to well bedded containing beds of dolomitic sandstone and occasionally beds of collenia (stromatolitic)	190
			RAT 1		Roches Argilleuses Talceuse, upper portion silty argillite, red and bedded. Lower portion unknown	40
			POUDINGUE		Unknown formation, transgression conglomerate and arkoses in other localities	?

Figure 4: General stratigraphy of the Katanga system

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7.3 Local Geology

With the exception of Tilwezembe OP, all of the mineralized properties of KCC are localized within the Kolwezi Nappe, a northeast striking synclinal basin with major and minor axes of approximately 20 km and 10 km respectively. Tilwezembe OP is located about 20 km to the east of Kolwezi. Figure 5 shows the location of the deposits.

Within the Kolwezi Nappe, each of the project areas, T17 Mine, Kananga OP, KTO UG, KOV OP and Mashamba East OP contain fragments with intact successions of Series Des Mines lithologies, which host the copper and cobalt mineralization. The fragments are often structurally complex, being tightly folded and exhibiting variable strikes and dips both within individual rafts and between neighbouring rafts.

Within the KITD project, regional, local and property geological settings are not relevant as the “deposits” are technogenetic in nature, being the product of previous mining activities.

7.3.1 T17 Mine

The T17 West is described as dismembered structurally complex packages, which belong to the southern flank of a synclinal fold that extends 2.6 km and is overturned towards the north.

Faulting is assumed to be the predominant process in the deformation and dismemberment of the deposit.

7.3.2 Tilwezembe OP

The mineralized zone of Tilwezembe is located in a northeast-southwest anticlinal structural lineament, which extends further to the east where known copper and cobalt deposits exist (Kisanfu, Myunga, Kalumbwe and Deziwa). Strongly brecciated siliceous dolomites and shales of the Mwashya Formation (or R4) dominate with interstitial bands of haematite and oolites. The strata strike almost east-west and dips at about 45° to the south.

7.3.3 Kananga OP

The Kananga orebody outcrops and forms a ridge with a north-northeast strike. The ridge falls quite rapidly towards the south and has been cut to form part of the embankment for the Lobito railway line, which runs parallel to the ridge and 10 m to 20 m away from it for most of the strike length of the orebody. GCM interpretations indicate that Kananga is the northern limb of the Kananga-Dilala syncline, which plunges to the south.

7.3.4 KOV OP

The KOV OP deposit can be sub-divided into five main fragments within the KCC concession. These fragments can be classified as rafts of the Lower Roan lithologies which are separated by faults and/or RAT Lilas waste intrusions. The KOV fragments as follows:

KTE: Can be divided into three sub-fragments; KTEA forms a broad syncline dipping at approximately 40° north, KTEB lies below KTEA and dips approximately 30° north, whilst KTEC lies to the east of the KTEB and is sub-vertical.

Oliveira: A broad open syncline with the northern limb dipping 5° to the southwest and the southern limb dipping 25° to the northwest.

Virgule: A rectangular fragment which is faulted in the west, appearing as a ‘trouser leg’ shape. It is dipping 20° to the southwest.

Variante: Forms a rough L-shape plane, with the eastern portion steeply dipping 70° to the north, shallow dipping 20° to the west.

FNSR: An over turned fragment, rectangular fragment which is dipping 25° to the southwest.

7.3.5 Mashamba East Mine

The Mashamba East has an intact succession of Series de Mines lithologies, which host the copper and cobalt mineralization.

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Structurally, the lithologies of the Mashamba East strike to the northeast and dip 20° to the northwest in the west and wraps around to strike almost north-south and dip 35° to the east in the eastern portion of the property.

7.3.6 KTO UG

KTO UG operations extract mineralized copper ore from the Kamoto deposit, which is differentiated from KTE, predominantly mined in the KOV OP, but contains the same lithologies. The KTO UG is subdivided into five fragments as follows:

Principal: forms a rough L-shape syncline, flat lying to gently dipping (up to 8° to the north) in the central and northern areas and becoming steeper (15° to 25°) towards the east and west flanks. The steeply dipping area to the south is sub-vertical in places and is separated from the flatter dipping area by structurally complex zones and RAT Lilas waste intrusions.

Etang South: forms a broad open synform dipping 25° to 35° to the east on an approximate north-northwest strike;

Etang North: forms a roughly rectangular slab dipping 35° to 55° to the east on an approximate north-northeast strike. It extends below the Principal fragment in the north;

Ecaille Renversee: a small reversed fragment between Principal and Etang North; and

Pringle: forms a moderately open synform dipping 35° to 55° to the southeast on an approximate north strike.

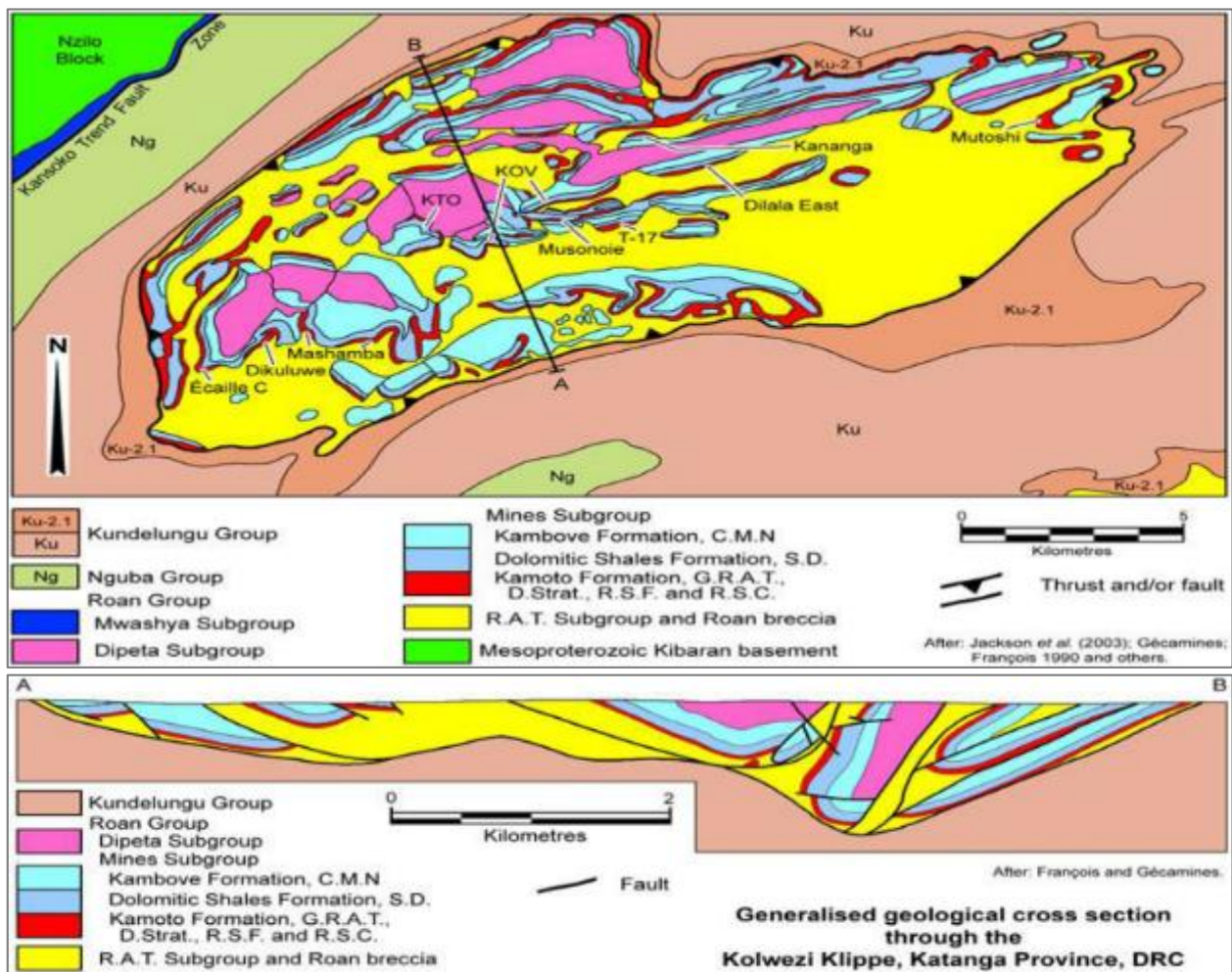


Figure 5: Kolwezi Nappe plan and section G-G showing the location of Kamoto East and Oliveira Virgule fragments

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7.4 Mineralization

Primary mineralization, in the form of sulphides, within the Lower Roan is associated with the DSTRAT and RSF for the OBI and the SDB and SDS for the OBS and is thought to be syn-sedimentary in origin. Typical primary copper sulphide minerals are bornite, chalcopyrite, chalcocite and occasional native copper while cobalt is in the form of carrollite. The mineralization occurs as disseminations or in association with hydrothermal carbonate alteration and silicification.

Supergene mineralization is generally associated with the levels of oxidation in the subsurface sometimes deeper than 100 m below surface. The most common secondary supergene minerals for copper and cobalt are malachite and heterogenite.

The RSC, a lithological unit stratigraphically intermediate between the OBS and OBI host rocks, contains relatively less copper mineralization. The RSC contains appreciable copper mineralization near the contacts with the overlying SDB formation and the underlying RSF formations. The middle portion of the RSC, considered to be “sterile” by GCM, normally contains relatively less copper mineralization and was historically not sampled. KCC currently samples the entire RSC. The mineral potential of the RSC is less well known than that of other formations.

The RSC has been observed to be well mineralized in supergene cobalt hydroxide, heterogenite, which occurs as vug infillings, especially near the surface.

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8.0 Deposit Types

The copper-cobalt minerals hosted in rocks of the Neoproterozoic Katangan Basin in the Central African copper belt metallogenic province of the DRC are a classic example of (low energy) sediment-hosted stratiform copper (SSC) ore system deposits. These deposits are economically significant, as they account for approximately 25% of the world's copper production and known Mineral Reserves being second only to porphyry copper deposits in terms of copper production and the most important global cobalt resource.

The copper-cobalt deposits contained in a sedimentary series of rocks are known as the Mines Group in the Katanga sequence and the Roan Group in Zambia. The sediments are shallow-water shales, dolomitic shales, reefal dolomites and possible evaporitic lagoonal mudstones, formed on a platform marginal to a subsiding basin. The rocks are exposed in a series of tightly folded and thrust anticlines and synclines, which generally trend east-west or southeast and are often overturned to the north. In spite of this deformation overprint, the mineralized zones, although sometimes lenticular along strike and down dip, as well as showing local diapiric forms, display remarkable large-scale continuity within the Mines Group.

Artisanal and other near-surface workings occur scattered throughout the permits, and at Kakanda mining has proved mineralization over a 1,000 m strike, similarly at Mukondo and Petit Gis, whereas at Disele, mine workings and GCM drilling show a 500 m long mineralized zone exists.

The primary mineralization, in the form of copper sulphides, is thought to be syn-sedimentary in origin. Typical primary copper sulphide minerals are bornite, chalcopyrite, chalcocite and occasional native copper while cobalt is in the form of carrollite. The mineralization occurs as disseminations or in association with hydrothermal carbonate alteration and silicification.

Supergene mineralization is generally associated with the levels of oxidation in the sub-surface sometimes deeper than 100 m below surface. The most common secondary supergene minerals for copper and cobalt are malachite, spherocobaltite and heterogenite. Malachite is the main mineral mined within the confines of the current KOV OP and Mashamba East OP.

The mineralization at Tilwezembe OP is atypical being hosted by the Mwashya or R4 Formation. The mineralization generally occurs as infilling of fissures and open fractures associated with the brecciation. The typical mineralization consists mainly of copper minerals (chalcopyrite, malachite and pseudomalachite), cobalt minerals (heterogenite, carrollite and spherocobaltite) and manganese minerals (psilomelane and manganite).

Despite the large number of variables in the basinal settings of these deposits, most SSC deposits are remarkably similar in terms of their mineralization style, morphology and mineralogy.

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9.0 Exploration

Refer to Section 6.4 for additional information on historical exploration. A brief summary of historical drilling is shown in Table 11, with details on drilling since acquisition presented in Section 10.0.

Table 11: Summary of historical exploration diamond drill holes per project area

Drill hole Age	Drilling Target	Completed By
1942 - 1969	Kamoto surface – KTO UG / KTN, KTE, KOV OP and Mashamba East	UMHK / GCM
1970 - 1979	KTO UG and Mashamba East	GCM
1980 - 1989	KOV OP and Mashamba East	GCM
1959 - 2002	KTO UG	GCM
2006 – 2008	KOV OP	DCP

During 2016, 2017 and 2018 a total of 8,662, 14,224, and 13,064 meters of diamond drilling was completed, respectively. The focus of this drilling was to provide improved definition in regard to the geotechnical, hydrological, environmental and geological aspects of KOV OP, Mashamba East OP and KITD. The results from drilling undertaken in 2018 are continuing to be processed

In addition to diamond drilling, comprehensive geological mapping was undertaken at KOV OP, T17 OP and Mashamba East OP together with continued geological mapping in KTO UG within recent development fronts. No further exploration work was undertaken.

9.1 KOV OP

Geotechnical and hydrological drilling commenced during 2016 for 29 diamond drill holes was totalling 8,662 m and continued in 2017 for 36 diamond drill holes was totalling 10,495 m. During 2017, geological exploration at KOV OP also focused on surface mapping and re-sampling/re-logging of historical drill cores. In 2018, 32 core holes were drilled totalling 9,115 m. The results from the drilling are continuing to be processed.

9.2 KTO UG

No exploration drilling at KTO UG took place during 2017. Exploration was concentrated on reviewing and validating the historical GCM drill hole database, and updating underground mapping sections whilst the mine is on care and maintenance. In 2018, 46 core holes were drilled underground for a total of 3,949 m. The results from the drilling are continuing to be processed.

9.3 Mashamba East

Geotechnical, hydrological and environmental drilling commenced during 2017 for 12 diamond drill holes was totalling 2,432m. This drilling was utilised for cross-departmental logging and sampling purposes. Exploration activities were focused on geological mapping of the ore and waste lithologies.

9.4 KITD

Geological drilling was completed in 2017 for 89 diamond drill holes totalling 1,297 m. Exploration was focused on covering the accessible surface area of the KITD, and completing nine twin verification diamond drill holes from previous drilling activities.

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10.0 Drilling

KCC exploration activities have ramped up significantly since 2009 to provide better definition of the deposits. In total 155,741 m of diamond drill core and 868 m of auger drill core has been drilled since 2009 of which approximately 32% were in KOV OP or KTE, 29% KTO UG, 15% in Mashamba East OP, 19% in KITD and 5% on other projects. A summary of diamond drill holes per project area since acquisition is shown in Table 12. Reverse Circulation (RC) drilling was not utilized for Mineral Resource estimation.

Table 12: Summary of exploration diamond drill holes per project area since acquisition

Drill hole Age	KOV OP	KTE	KTO UG	Mashamba East OP	KITD
Historical (<Pre-2009)	103	27	972	127	-
2009	8	4	37	-	-
2010	13	6	43	-	-
2011	2		-	-	-
2012	51		1	-	-
2013	25		26	-	-
2014	16	19	105	2	4
2015	82	45	93	145	79
2016	29		-	-	-
2017	36		-	12	89
2018	32		49		
TOTAL	397	101	1326	286	172

Details on historical drilling that occurred prior to 2009 can be found in Section 6.5. Details on drilling that took place in 2017 can be found in Chapter 9.0. The results from drilling undertaken in 2018 are continuing to be processed

Historically, diamond core recoveries were generally lower than desirable throughout the various project areas. This is mainly due to the fractured, weathered nature of the ore, as well as the drilling techniques used. However, improvements in core recovery are notable year on year. Further details on core recovery for each project are presented in the following sections.

The KCC Survey department demarcates the diamond drill site in the field according to the signed-off drill plan, and reconfirms the drill hole collar to the KCC Geology department once drilling has completed. A multi-shot downhole survey using Reflex™ EZ-Trac 5177 equipment is completed at 30 m intervals to the base of the drill hole, and communicated to KCC Geology staff for verification. The Reflex™ survey instrument is checked, verified and calibrated annually by Reflex™.

Under normal drilling conditions, the majority of open pit diamond drill holes are PQ (core diameter 85 mm) and HQ (core diameter 63.5 mm) in size, in contrast to HQ (core diameter 63.5mm) and NQ (core diameter 47.6 mm) in underground drilling. Orientated core has not been performed by the KCC Geology department. Once the diamond drilling contractor has removed the core from the barrel and placed it in the relevant size steel core tray, KCC staff transport the core trays to the Loco shed for core logging and sampling.

Refer to Chapter 11 for details on core logging and sampling procedures.

10.1 KOV OP

After KTO UG, KOV OP has seen the most extensive drilling campaigns since 2009. The vast majority of the diamond drilling has been to provide confirmation of the orebody ahead of the various pushbacks in the pit. Average diamond drill core recovery stands at 70% whereby the core recovery is the lowest in the highly

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weathered, completely leached portion of the Variante fragment and within the highly silicified RSC lithology unit.

The 2009-2010 drilling campaigns were focused in the pit bottom area which is comprised primarily of the Virgule fragment. A total of 21 holes were drilled for 3,840m.

The 2011-2013 drilling campaigns were focused on the second pushback, in northeast portion of KOV OP, which is in the Variant and Oliveira fragments. A total of 78 drill holes were drilled for 12,410m.

The 2014 drilling campaigns were dominantly drilling in the southwall in southwest portion of the KOV OP which is primarily in the Virgule fragment. A total of 16 drill holes were drilled for 3,722m.

The 2015 drilling campaign was focused around the pit bottom area for the fragments of Virgule, FNSR and Variante. A total of 82 drill holes were drilled for 12,996m.

The 2016 drilling campaign was focused around KOV OP according to Geotechnical and Hydrological requirements. A total of 29 drill holes were drilled for 725.5m.

10.2 KTO UG

KTO UG has seen the most extensive drilling campaigns since 2009. Most of the diamond drilling has been to provide further definition of the deposit in advance of development and to upgrade the Mineral Resource. Average diamond drill core recovery stands at 82% whereby core recovery is lowest in the highly faulted, argillitic portion of Zone 3 and Zone 4.

The 2009-2010 drilling campaigns were focused on the Etang North fragment, and to a lesser extent on the Etang South and Zone 5 (Principal) fragments. A total of 80 drill holes were drilled for 10,123m.

The 2012-2015 drilling campaigns were focused on the Etang South and Zones 3/4 (Principal) fragments, and to a lesser extent on the Etang North fragment. A total of 225 drill holes were drilled for more than 37,369 m.

10.3 MashambaEast OP

The historical core drilling in the Mashamba East OP was not of sufficient density to classify the deposit as an Indicated Resource. Consequently, an infill exploration drill program was planned and completed during 2014 and 2015. The drilling program included 147 drill holes with total length of 26,590m. Average diamond drill core recovery stands at 86% whereby the core recovery is the lowest within the highly silicified RSC lithology unit.

Exploration activities for Mashamba East during 2016 focused on geological mapping of the ore and waste lithologies.

10.4 KTE UG

The KTE deposit is planned to be mined from both the open pit and underground mining operations. To further define the deposit ahead of development, a surface drilling campaign was carried out in 2009-2010 with further infill drilling completed during 2014 and 2015 from both surface and underground.

The 2009-2010 drilling campaign totalled 10 drill holes for 3,662m.

The 2014-2015 drilling campaign totalled 64 drill holes for 11,251m.

Average diamond drill core recovery stands at 57% whereby the core recovery is low due to a high volume of dump material overlying the KTE fragment and the highly weathered, highly oxidised nature of the deposit.

10.5 T17 Mine

A total of 20 holes for 4,283m were drilled in 2010. The purpose of these drill holes was to confirm the deposit in both the OP and UG.

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10.6 KITD

A total of 2,087m were drilled in the KITD in 2014, 2015 and 2017 to define the Mineral Resource.

The 2014-2015 drilling campaign focused on the northwest corner of the KITD. A total of 83 drill holes were drilled for 790m.

The 2017 drilling campaign focused on covering the remaining area of the KITD. A total of 89 drill holes were drilled for 1,935m. This includes nine twin drill holes to confirm 2014 – 2015 drilling.

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11.0 Sample Preparation, Analyses and Security

11.1 Summary

All historical sampling, sample preparation, analysis and security were undertaken by GCM over a period of more than 50 years.

The core sample was cut along the longitudinal axis with one half of the core sent for laboratory analysis and the other retained in the boxes.

There was no systematic approach to sample lengths as indicated by the variations in the sample lengths in the database. The minimum sample taken was 0.5 m and the maximum sample was 2.5 m. The sample lengths were a consequence of the lithology unit thickness and of sample recovery within the run.

The bulk of the data within the project areas is historical, except for the recent drilling data reported above which has validated the GCM data.

Once core is collected from the drill rigs by KCC staff it is locked up behind closed off fences and there is no unauthorized access to this core during working and non-working periods. Security guards have been placed at the Loco, Moors and KTO core yards and safe guard the core 24/7.

An established chain of custody and Standard Operating Procedure (SOP) exists and is currently under review for improvements.

The actual operational procedures, analytical laboratory, QA/QC program utilizing certified reference material (CRM) and certified blanks is considered adequate for Mineral Resource estimation purposes.

With the exception of 2017 KITD samples, all KCC samples were submitted to SGS in Zambia, which is an internationally recognized and certified laboratory by SANAS (South African National Accreditation System on ISO/IEC 17025:2005) for chemical analysis. The 2017 KITD samples were analysed at the on-site Lulu laboratory facility.

11.2 Sampling Method and Approach

11.2.1 Historical Sampling

Details of the historical sampling undertaken within each of the project areas are scant and based on personal communications with the respective consultant in each of the project areas. Inferences have been drawn from the observations from the sample database.

SRK described the observations on the historical sampling for KOV OP (SRK 2009), and the review of the individual projects reports indicated that the process described for KOV OP was applicable to the other projects. SRK's review of the historical sampling for KOV OP is summarized below.

Cores from the orebody intersections were sampled for chemical analysis. The lengths of core sampled varied, and it is SRK's understanding that this was a consequence of the sample recovered within each run. In the GCM logging sheet, there is a column for percentage recovery where values ranging from 1% to 100% are entered to describe the amount of core recovered in the sample length. Core recoveries are recorded only for cores that were sampled.

The lithologies sampled were the Upper Ore-body host rocks (lower SDS and SDB) and the Lower Ore-body rocks (RSF, DSTRAT and the RATGR) and portions of the RSC deemed to be mineralized. It is understood that the visibility of copper mineralization in the core was used as the criterion for sampling the core. Core lengths deemed to be barren of copper were not sampled, and an entry was made in the sample log for that interval with the comment "sterile" or barren. It is possible that the unsampled cores could contain finely disseminated copper mineralization not visible to the naked eye. There is a further possibility, especially in the RSC, that the "sterile" zones contain cobalt mineralization. In drill holes KOV 426 and KOV 427, the entire RSC is mineralized and returned good copper mineralization (2 to 3%) within the mid-RSC. In drill hole KOV 428, the mid-portion of the RSC was sampled. Partial or selective sampling, although common in the RSC, was also evident in the other Roan lithologies.

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Due to this pre-selection during sampling, the assay database is incomplete, and this affects the accuracy of the Mineral Resource estimation.

The assay database describes the sample in terms of the length, depths (From and To) of intersection and the amount of core recovered in that sample length. The sample database contains assay data for the following:

- **%TCu:** the percentage total copper content of the sample;
- **%CuO:** the percentage of the copper present as oxide. In the modelling, this is reported as %AsCu. Fewer than half of the samples were analysed for %AsCu;
- **%Cu mal:** the percentage of the copper as malachite. Only a few samples contain values on this column;
- **%TCo:** the percentage total cobalt content of the sample; and
- **%CaO soluble:** the relative proportion of soluble calcium oxide in the sample. Less than 30% of the total database was assayed for calcium oxide.

11.2.2 T17 Mine, KTO UG and Mashamba East OP

CCIC undertook a re-sampling of existing cores for the T17, KTO UG and Mashamba East OP. CCIC reported that 26 historical drill holes were re-sampled (SRK 2009), and these included the following:

- DIK 171 (Mashamba East);
- F2418 (Kamoto Principal OBS);
- F2471 (Kamoto Principal OBI);
- F2391 (Kamoto Etang); and
- MU321 (Musonoi-T17).

CCIC's approach was to try as much as possible to replicate GCM sampling and assay results on cores from each of the above mentioned areas to gain confidence in the reported assay values. CCIC did not have the details of the original GCM sampling protocol and methodology, by inference, CCIC contend that samples were taken for assay based mainly on lithology, and as such were of irregular lengths. CCIC studied the sample lengths and found that the most popular sample length was between 1.5m and 2.0m, although sample lengths ranged from 0.02m to 10.0m.

CCIC replicated the sampling intervals of the original log sheets by quartering the existing half-cores. The remaining quarter core of the sample was put back in the sample box and remarked with the original sample number.

The quarter core samples were marked, cut and bagged under CCIC's supervision at the KTO Geological Department. Samples were cut by CCIC or GCM personnel on a Wendt L18A B61936 saw with a 340 mm diamond blade.

A total of 654 samples were generated and dispatched to SGS Lakefield Research Africa (Pty) Ltd (SGS Lakefield) (accredited to SANAS, Facility Accreditation Number: T0169) for analysis.

11.2.3 Kananga OP and Tilwezembe OP

The procedures adopted during the sampling of the cores from Kananga OP and Tilwezembe OP are similar.

After the drill hole core was photographed and logged, the core was split with a diamond saw and sampled at 1 m intervals within the mineralized units, honouring geological contacts.

11.2.4 KOV OP

The core sample was cut along the longitudinal axis with one half of the core sent for laboratory analysis and the other retained in the boxes.

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There was no systematic approach to sample lengths on historical drill holes as indicated by the variations in the sample lengths in the database. The minimum sample taken was 0.5 m and the maximum sample was 2.5 m. The sample lengths were also a consequence of the sample recovered within the run.

11.2.5 Recent Sampling

11.2.5.1 Sampling Procedures 2005 - 2009

Sampled zones are selected based on the visual observation of the lithological contacts. The geologist marks on the core the direction along which the core should be split, after considering the attitude of the bedding or foliation relative to the core axes. The drill lengths and the recoveries are recorded in the sampling notebooks.

Sampling is carried out at a maximum of 1 m drill length intervals and different stratigraphic units are sampled separately. The core samples are sawed into two halves. One half is broken up and bagged for assay while the other half is stored in the core tray in a dedicated on-site core shed for future reference.

Core bags for a particular batch are pre-labelled and arranged in order from the first to the last sample. A tag with an identification or sample number is added to the bag containing the sample before the bag mouth is tied.

Split core sampling is done from the drill core. Prior to taking samples, the geologist examines the core and marks off the intervals to be sampled by drawing a line along the core with a marker pen. When the intervals have been selected, the core is split in half using a diamond saw or core splitter. Once the core is split, individual sample lengths are selected considering the stratigraphical and lithological boundaries. The whole width of mineralization is covered.

The data is recorded as preliminary in the log sheets and is then logged in Lakefield's sample-tracking system and stored on a shelf.

11.2.5.2 T17 Mine, KTO UG and Mashamba East OP

CCIC indicated that the existing half samples were quartered using a diamond core-cutting blade, washed, flagged with a unique sample number, and bagged. The samples were collated into larger bags, per drill hole, locked in a metal trunk and shipped to SGS Lakefield in South Africa.

In all, 58 samples were taken from various drill holes for independent verification of the GCM copper and cobalt assay figures for T17 Mine, KTO UG and Mashamba East OP. These samples were sent to SGS Lakefield for preparation and analysis, with renumbered pulps resubmitted to both SGS Lakefield and Set Point Laboratories ("Set Point") (accredited to SANAS, Facility Accreditation Number: T0223) as checks. The cut samples were placed in metal containers and sealed under lock and key under the supervision of CCIC personnel. The samples were then trucked to the client's Lubumbashi offices before being sent to SGS Lakefield.

CCIC indicated that SGS Lakefield staff entered each of the samples into their LIMS system, which includes the client's details, the list of samples and the analyses required. Each of the quartered core samples was crushed to <2 mm, and the crushed sample was split, where necessary, to produce a portion of about 250 g. The split (or entire crushed sample if less than 250 g) was milled, bagged, labelled and stored in a box(s), logged in Lakefield's sample tracking system and stored on a shelf.

The analytical method used for the determination of total copper (%TCu) and total cobalt (%TCo) was X-ray fractionation.

Acid-soluble copper and cobalt (%AsCu and %AsCo) were determined by acid digestion (sulphuric acid) and analysis of the solution by Atomic Absorption Spectrometry (AAS), using methods described as follows:

- For analysis of copper oxides each sample was weighed and mixed with an aliquot of dilute sulphuric acid enriched with sulphur dioxide. This mixture was agitated at room temperature for a set period and the sample residue filtered out of the solution. The solution was made up to volume and analysed for copper and cobalt by AAS, yielding acid-soluble results; and

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- For analysis of copper sulphides the residue of the copper oxide preparation was placed in a beaker and mixed with multiple acids, with the residue being digested in the acid mixture. The solution was made up to volume and analysed for copper and cobalt by AAS. This yielded an assay of acid-insoluble copper (%AICu) and acid-insoluble cobalt (%AICo) present as sulphides.

11.2.5.3 Kananga OP and Tilwezembe OP

The samples from Kananga OP and Tilwezembe OP were sent to two laboratories, Alfred H. Knight (Alfred Knight) in Kitwe and SGS in Ndola, for preparation and analysis. Preparation activities consisted of:

- Drying of sample;
- Primary jaw and roll crushing of sample;
- Splitting a sub sample of 250 g using a riffle splitter;
- Pulverizing of the sub-sample to 75 micrometres and homogenizing;
- The samples were analysed for %TCu, %TCo, %Mn, %AsCu and %AsCo; and
- Both Alfred H Knight and SGS determined %TCu, %TCo, %Mn assays by multi-acid digestion (using hydrofluoric, nitric and perchloric acids) followed by dissolution in hydrochloric acid and AAS.

For %AsCu and %AsCo, both laboratories used cold leaching with 5% sulphuric acid. However, Alfred H Knight saturated with sulphur dioxide while SGS saturated with potassium sulphite before finishing with AAS. The laboratories may also have used different temperatures and digestion times.

Appropriate QA/QC checks were undertaken by CCIC on Kananga OP and Tilwezembe OP. No further information is available.

11.2.5.4 KOV OP

There have been three phases of drilling in the KOV OP since 2005. The initial drilling of eight drill holes was undertaken over six months commencing in November 2005 to collect material for metallurgical testwork. A second phase of drilling was undertaken for geotechnical purposes during 2006/07, and a third phase of confirmatory drill holes was drilled in 2007.

The phase 1 drilling was supervised by SRK. At the start of the drilling program, SRK devised a core-sampling protocol. The protocol outlined the procedures to be followed, and they included a review of the core to determine the contacts, which were to be used as a guide during the sampling program. The protocol ensured that samples were taken at regular intervals of approximately 1 m within the lithology as dictated by the core and the core recovered. It also ensured that samples did not cross lithological contacts. Additional samples were taken well outside the zones of visible mineralization.

The core from the phase 2 and 3, the geotechnical and confirmatory drilling, were sampled under the supervision of DCP, who had the exploitation rights for KOV OP before the merger with KML. Information on the sampling approach adopted by DCP was not relayed to SRK.

The recent sampling includes all the infill drilling undertaken since 2005 to 2009. The sampling undertaken during the infill drilling program supervised by SRK is described in the 2009 feasibility study (FS) report.

11.2.5.5 Sampling Procedures 2009 - 2019

The following sampling procedure applies to KOV OP, KTO UG, T17 OP/UG, Mashamba East OP and KITD (2017 and subsequent years only).

In the diamond drilling campaign from 2009, core was placed in core boxes in sequence and transported to KCC core shed securely. The core shed technician was responsible for laying down the core boxes in sequence on the logging tables. Logging and sampling was carried out by trained geologists in the core shed.

The geologists carefully examined the core for fabric lineaments such as beddings, lithology and mineralization. A cutting plane that transects the centre of the core was chosen, based on the fabric and

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mineralization distribution, and the plane was marked with a marker pen. Sampling was carried out in all of the orebody lithologies including the RSC regardless of any visible mineralization. Sampling intervals did not cross geological boundaries (i.e. lithologies or structures) and included at least 3 m either side of the OBI/OBS beyond visible mineralization.

The data is recorded primarily in the log sheets and then transferred into the geological database (GDMS Fusion v.9), with the hardcopy stored in the KCC Geology Department. All analytical pulp rejects from 2014-2017 and subsequent year's drilling campaigns are stored at the KCC Intervention core yard.

The core was split in half along the marked cutting planes, using a core splitting cutter. The split core was laid back in the original positions of the core trays.

Out of the two split core halves, the right side was sampled in intervals of 1 m for assaying, whilst the left side was stored onsite. Where a duplicate was inserted, the QA/QC sample was taken from the same half as the normal assay sample. Where the core is extremely broken (common in oxide zones), a visual estimation of 50% of the core interval was taken for sampling.

The sampled core pieces were packed in new, clean sample plastic bags (polyethylene plastic - high strength bags of 150µ, size 250 mm x 400 mm in size) and tagged with three clean laboratory dispatch labels printed from GDMS. One label was stuck to the inside of the bag and two were placed loose inside the sample bag. Each bag was also labelled with from/to and the borehole identification number. The samples were then tied with a cable tie to prevent contamination and spilling of the samples during transportation. The samples were placed in sacks each weighing approximately 20 kg. Except for 2017 KITD samples, the sacks were laid in a box weighing approximately 500 kg and transported to SGS laboratories in Kalulushi, Zambia. The normal practice for SGS laboratory was to complete bulk density measurements by immersion every 1 m on the whole core samples before samples were assayed. The 2017 KITD samples were transported to the on-site Luilu laboratory for analysis. Bulk density measurements were completed by the KCC Geology department by immersion each 1 m of the whole core samples prior to assaying.

For the initial KITD (2014 – 2015) drilling campaign, a portable powered auger was used for the drilling and sample extraction. The 50 mm nominal bore drill rod produces approximately 2 to 3 kg of sample per meter drilled. Samples were drawn in 0.5 m increments until the drilling was terminated upon reaching the red soil (terre rouge) base.

All samples were placed in durable, sequentially numbered clean sample plastic bags (polyethylene plastic - high strength bags of 150µ, size 250 mm x 400 mm in size) and taken by vehicle to the Geology Intervention Building. If a sample was particularly wet, the original sample bag was double bagged to prevent any sample loss.

Once samples were logged and analysed, the information was entered into a Microsoft Excel spreadsheet by the drilling contractor. The assay database was sent to KCC on 25 August 2016. The lithology database was not forthcoming from the Contractor but given that all sampling took place within the tailings itself, a rock code of TAILINGS was assigned by KCC. This information was imported into the geological database (GDMS Fusion v.9) for permanent storage.

The analytical method used for the determination of total copper and total cobalt was X-ray fractionation. Acid-soluble copper and cobalt were determined by acid digestion (sulphuric acid) and analysis of the solution by Atomic Absorption Spectrometry (AAS).

The methods are described as:

- For analysis of copper oxides each sample was weighed and mixed with an aliquot of dilute sulphuric acid enriched with sulphur dioxide. This mixture was agitated at room temperature for a set period and the sample residue filtered out of the solution. The solution was made up to volume and analysed for copper and cobalt by AAS. This yielded acid-soluble results;
- For analysis of copper sulphides the residue of the copper oxide preparation was placed in a beaker and mixed with multiple acids, with the residue being digested in the acid mixture. The solution was

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made up to volume and analysed for copper and cobalt by AAS. This yielded an assay of acid-insoluble copper (AICu) and acid-insoluble cobalt (AICo) present as sulphides; and

- Other elements were analysed to assist in the determination of copper and cobalt recoveries for the new processing method called the WOL. The elements included Ca, CaO, Mg, MgO, Mn, Fe, Al, Mn and Au. The method used to analyse the elements is called Atomic Absorption Spectrometry (AAS). Gold was analysed by fire assay using 30 g pulp, fusion and cupellation at 1,100 °C and 950 °C respectively with AAS finish.

11.3 QA/QC

QA/QC is the methodology by which confidence levels are measured and maintained for assaying.

The main objectives of QA/QC programs are to:

- Minimize bias from sampling and assaying;
- Ensure the accuracy and precision of assaying; and
- Measure and demonstrate data integrity and validity for Mineral Resource estimates and grade control.

A robust QA/QC program was initiated by KCC in all recent diamond drill hole data (2009 – 2017), with improvements notable year on year. Refer to Section 11.3.1 onwards. There is no QA/QC database available prior to KCC ownership (pre-2009). The integrity of the historical data was taken into account for the Mineral Resource classification scheme employed by KCC for KOV OP, KTE, KTO UG and by Amec Foster Wheeler for Mashamba East. Refer to sections 14.2.10, 14.3.10 and 14.4.10 for further details.

11.3.1 Purpose

The QA/QC procedure involves insertion of blanks, duplicates and CRM at predetermined and sequential intervals. The cycle is repeated until the end of the hole.

KCC uses six types of standard reference material for copper from 0.5 – 6.07 %TCu and six for cobalt from 0.0047 – 0.8557 %TCu. This ensures the full range of grade categories of both copper and cobalt are covered.

- The reference material assays provide a method by which an analytical accuracy is monitored and quantified. There are two parameters of interest when reporting analytical accuracy:
 - The relative assay deviation from the expected value of the reference material; and
 - The average bias over time.

The deviation of a reference material's assay is measured and expressed as a relative standard deviation, thus making it possible to directly compare reference materials with different standard deviations. Acceptable limits are 95% of samples submitted to be within $\pm$ two standard deviations.

When acceptable limits for reference materials are not achieved the following course of actions are taken:

- Cross-check KCC reference material assays with laboratory submitted reference material assays for the same period and/or batch;
- In the case of bias, determine if it is one reference material type or all reference material types; and
- After discussions with the laboratory, send the relevant batch of samples back for re-analysis ensuring that the re-assayed samples are imported using the same methodology as previously completed.

11.3.2 Procedures

The following procedure on QA/QC is extracted from KCC's Standard Operating Procedure MIN-GEO-PRO-316.

The following procedure was employed from 2015 onwards:

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- Insertion of blanks, duplicates and standard reference material at pre-determined and sequential intervals;
- Every tenth sample is a blank, every fifteenth sample is a standard reference material with known mean and standard deviations and every twentieth is a field duplicate. This cycle is repeated till the end of the hole;
- Six types of certified reference material are used as described in the previous section. This is to ensure that the entire grade categories of both copper and cobalt are all taken care of. The reference material assays provide a method by which analytical accuracy is monitored and quantified; and
- Field duplicates were taken to check the effectiveness of sampling procedures.

The QA/QC on the database was carried out at the end of 2015 for KOV OP, KTO UG, Mashamba East OP and Mashamba East UG as part of the previously reported Mineral Resource model (Dec 31, 2016). The 2017 Mineral Resource update utilises the same database as the previous year, with the only QA/QC addition being from the 2017 KITD data. The 2017 QA/QC verification was carried out by KCC using Golder's QA/QC software. The QA/QC report was reviewed and approved by Golder. QA/QC results have been summarized in the following sections.

11.3.3 Blanks

Blanks were introduced in the sample stream at a rate of 1 in 10. Since 2015, AMIS certified silica has been used as blanks. From 2009 - 2014, the following blank materials were used:

- Cement because it resembles fresh dolomitic shale and the fineness of the grains; and
- CMN cores considered to be sterile.

The database on blanks from 2009-2017 contains a total of 1,170 samples on which the following can be noted:

- 15% of KOV OP and 1.3% of Mashamba East OP %TCu total analysed samples were outside the maximum value. All KTO UG and KITD %TCu samples passed;
- 17.8% of KOV OP, 4% of Mashamba East OP and 0.6% of KTO UG %TCu total analysed samples were outside the maximum value. All KITD TCo samples passed; and
- Less than 1 % of %TCu and 1% of %TCu blank samples analysed in 2015 were outside of the maximum value. No blank samples analysed for KITD in 2014/15 and 2014 were outside of the maximum values. Anomalous results, mainly due to swapping labels, were verified and corrected when necessary. The vast majority of the anomalous values are close to the determined limit and there are no signs of systematic contamination.

Figure 6 to Figure 9 present results for AMIS 0415.

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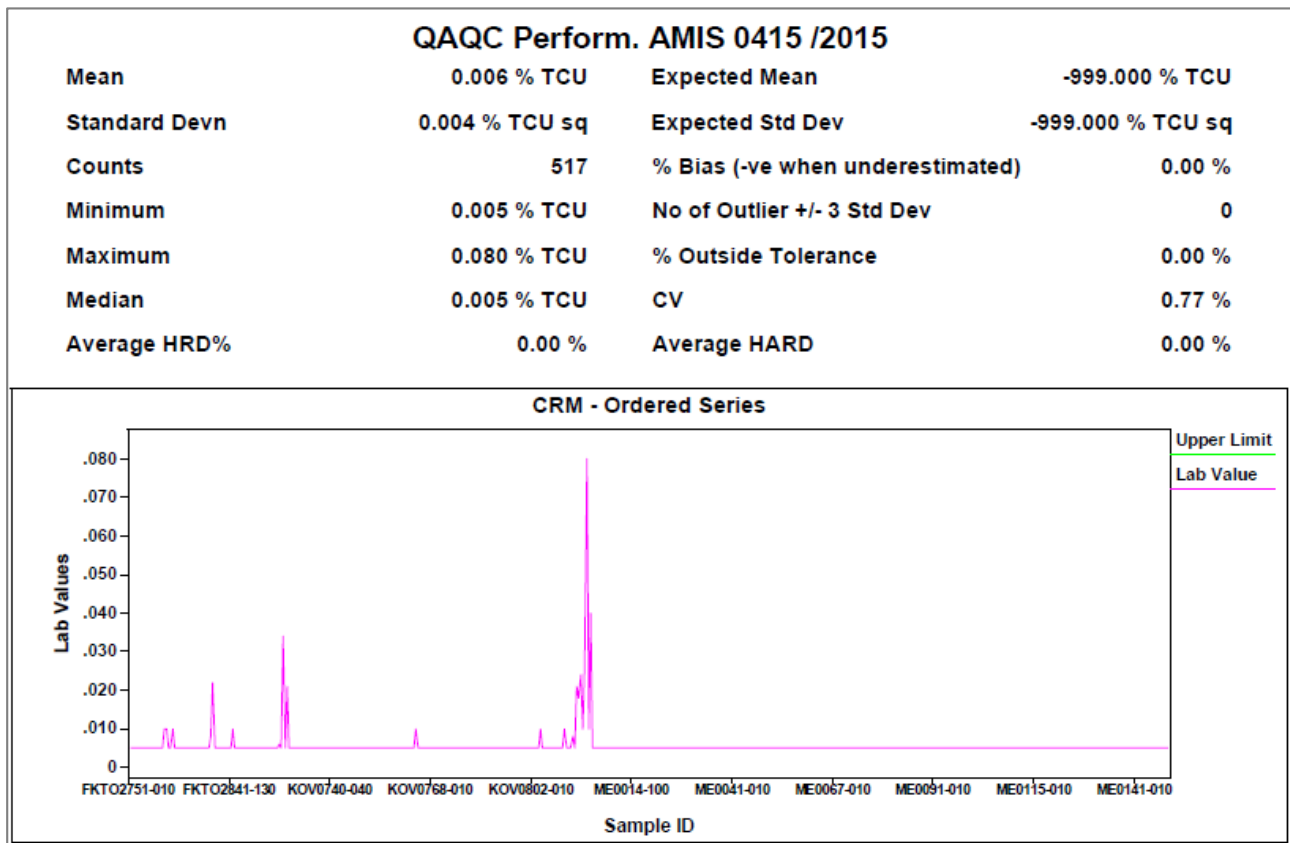


Figure 6: QA/QC performance - AMIS 0415 for %TCu (KOV, KTO and Mashamba East)

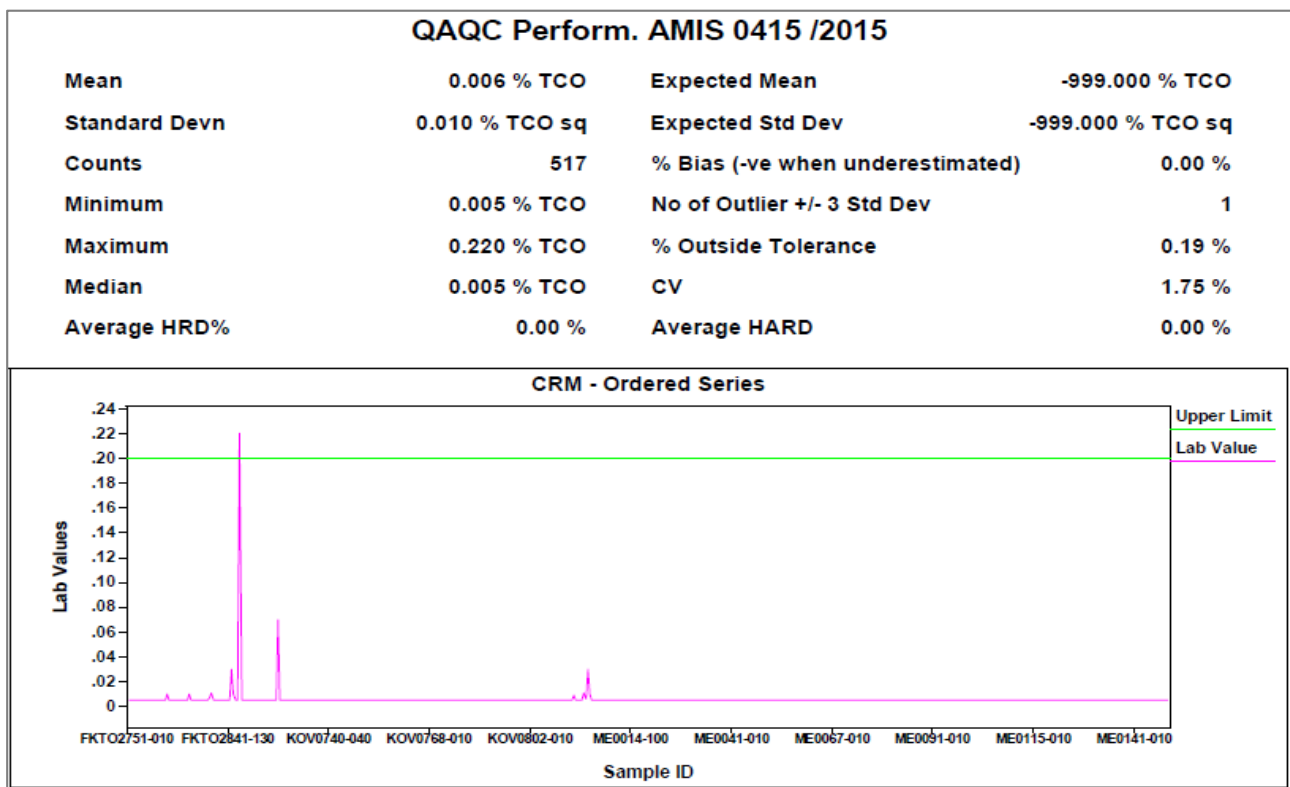


Figure 7: QA/QC performance - AMIS 0415 for %TCu (KOV, KTO and Mashamba East)

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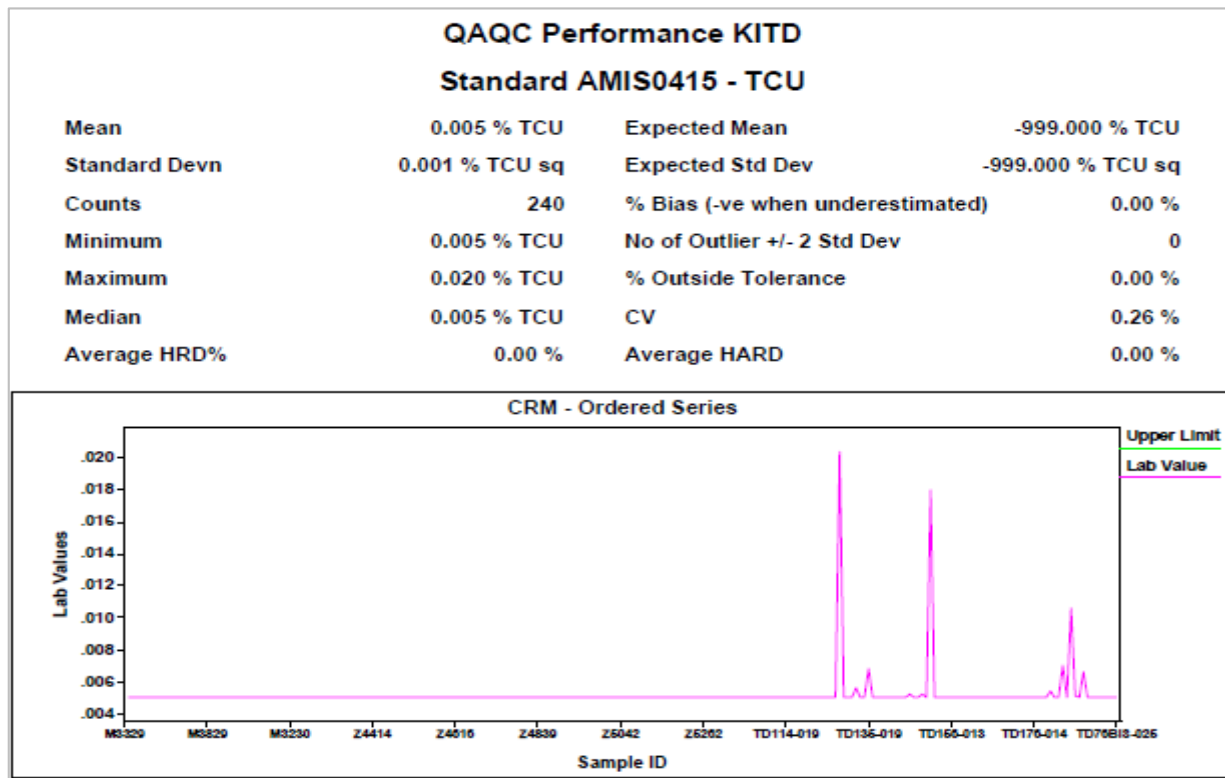


Figure 8: QA/QC performance - AMIS 0415 for %TCu (KITD)

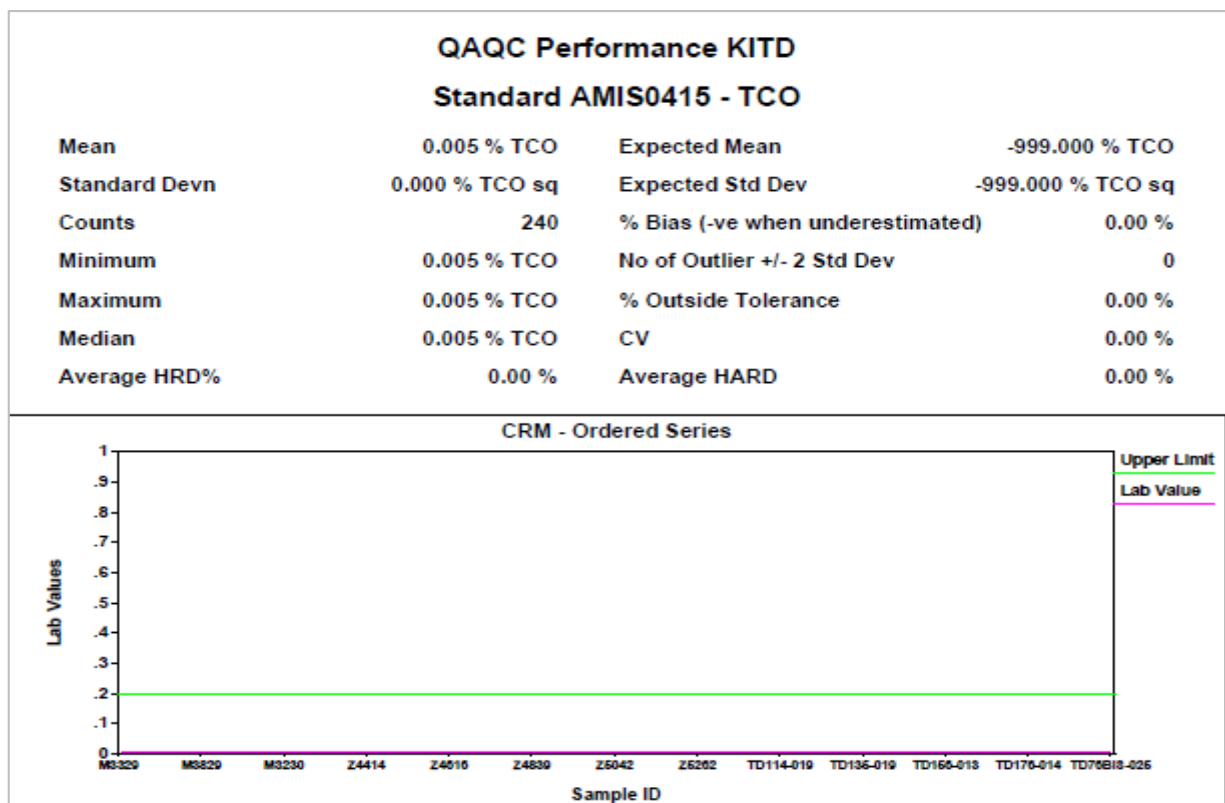


Figure 9: QA/QC performance - AMIS 0415 for %TCO (KITD)

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11.3.4 Field Duplicate Assays

For the scatter plots, good reproducibility is when the scatter points plot close to the ideal correlation line (i.e. the 45 degree line shown in pink) and within the +10 % and -10% upper and lower limits. The %TCu, %TCo and %AsCu scatter and QQ plots, as observed from Figure 10 to Figure 15, show slight positive bias. This is deemed acceptable for Mineral Resource estimation.

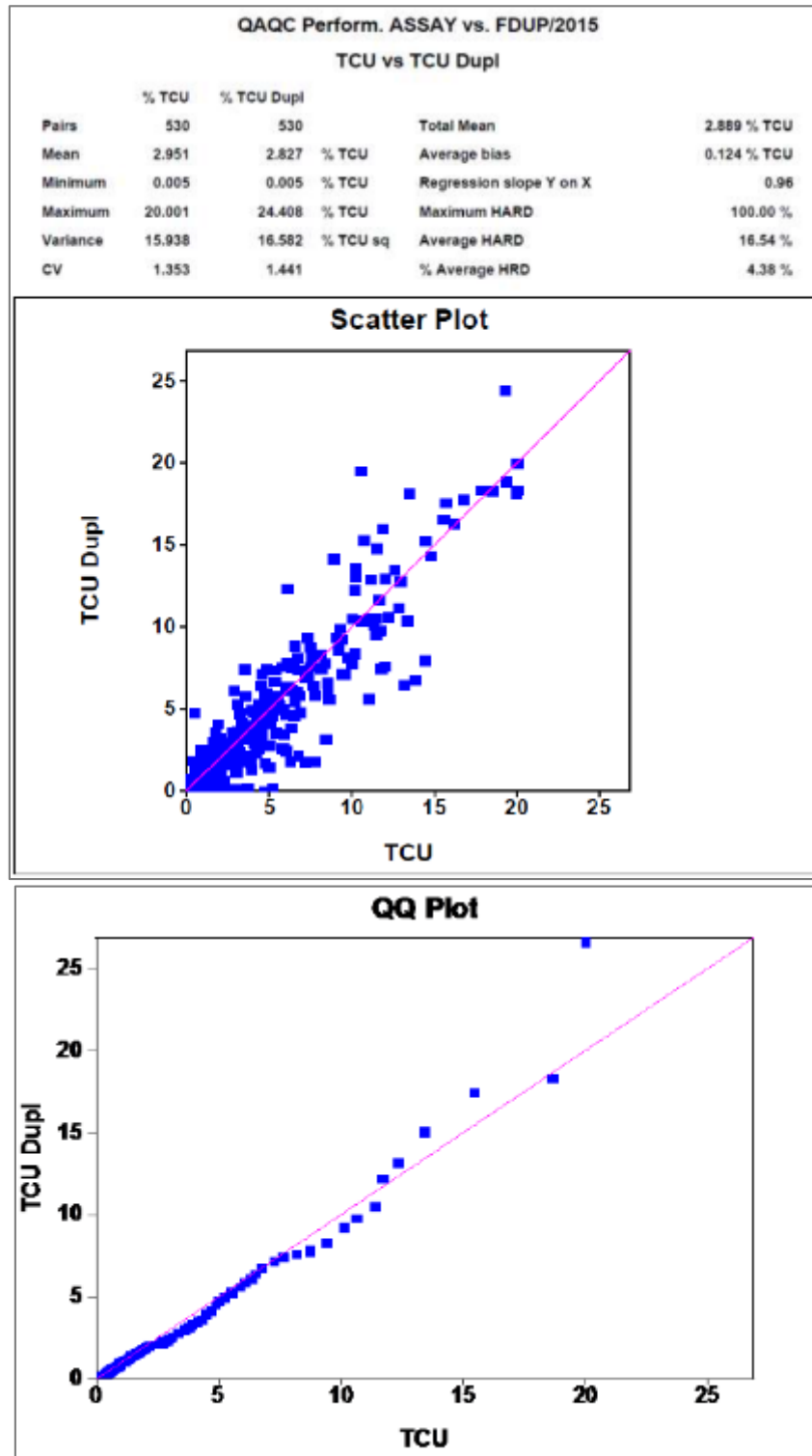


Figure 10: QA/QC performance of %TCu field duplicates (KOV, KTO and Mashamba East)

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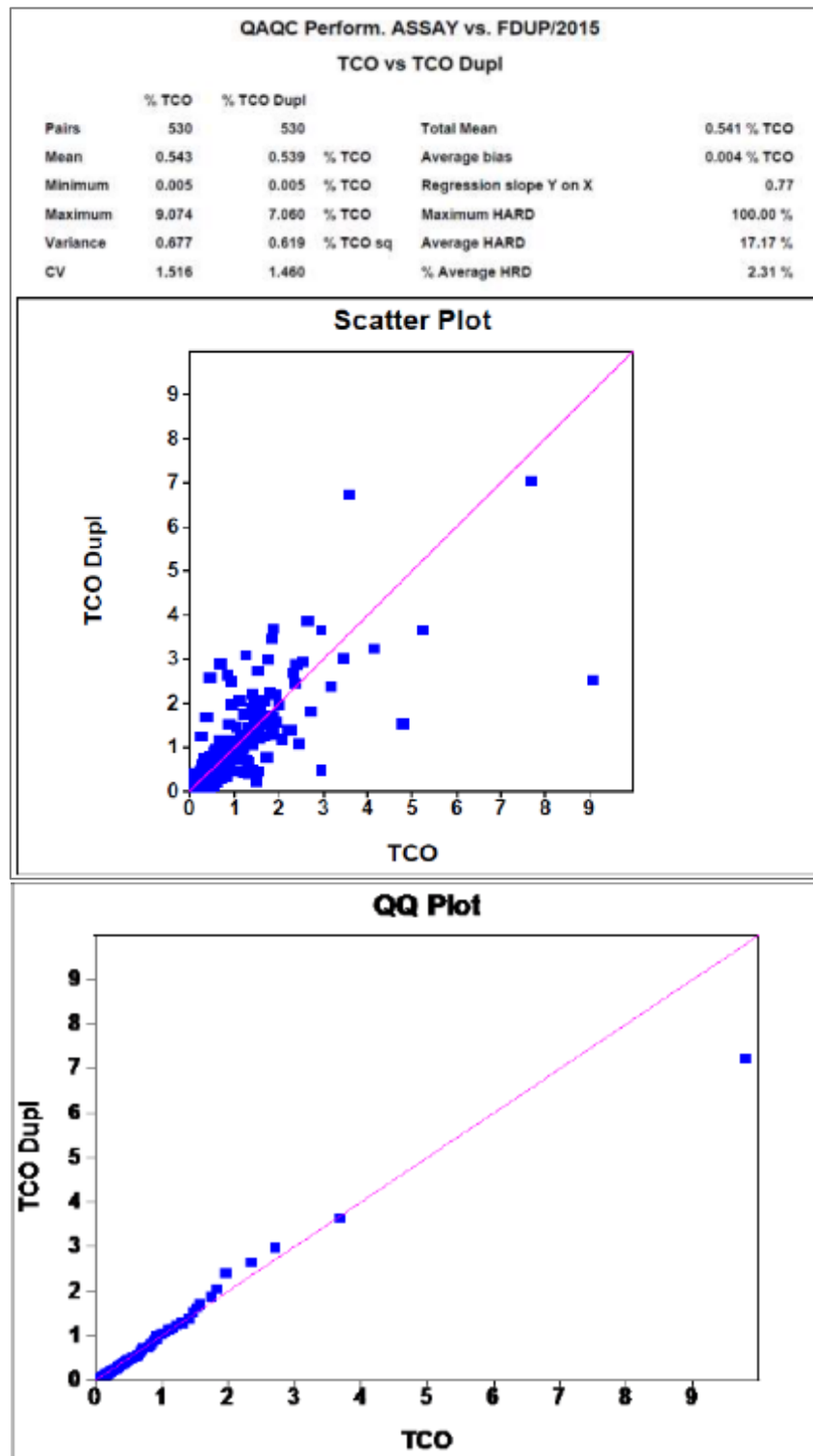


Figure 11: QA/QC performance of %TCo field duplicates (KOV, KTO and Mashamba East)

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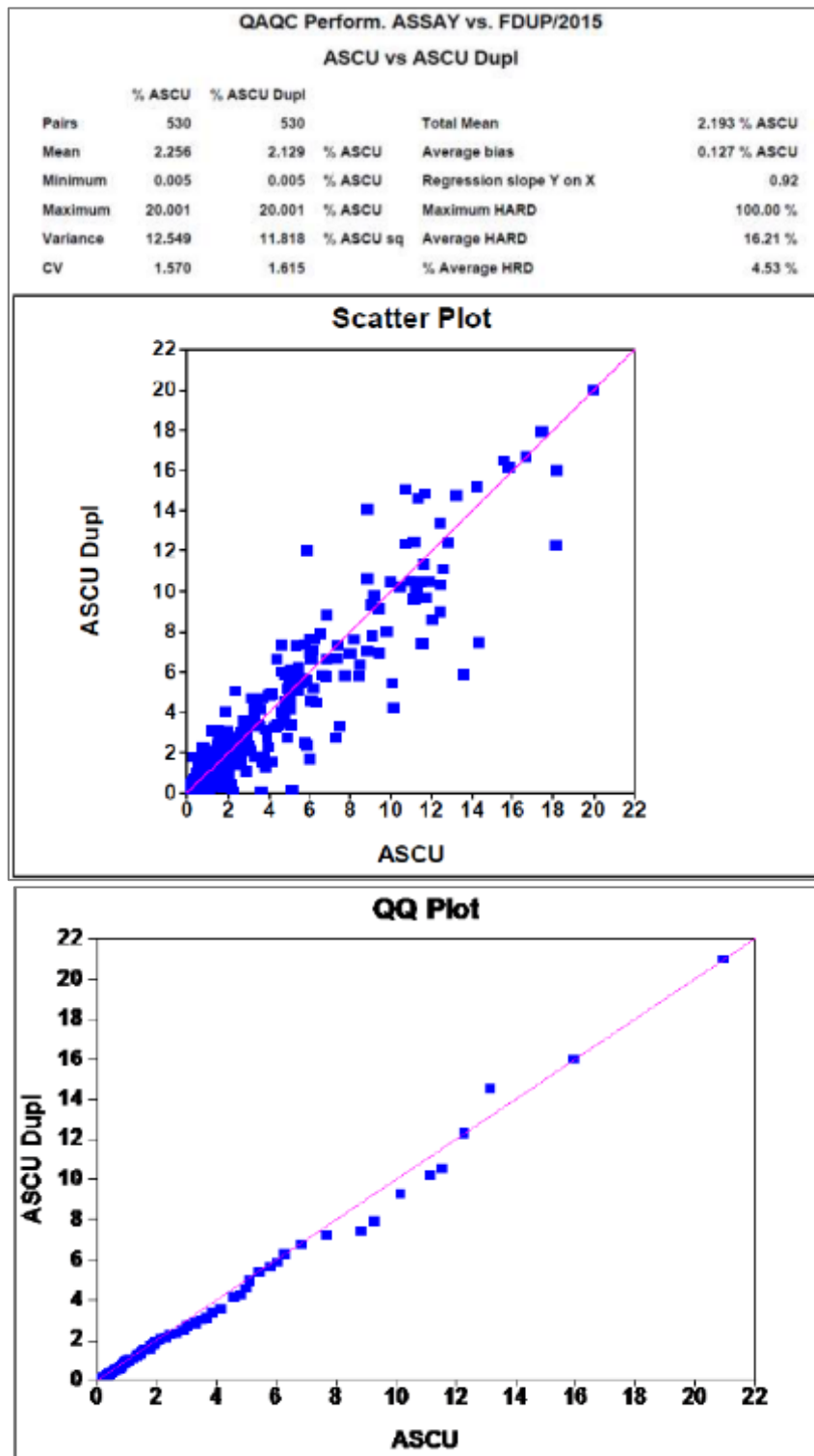


Figure 12: QA/QC performance of %AsCu field duplicates (KOV, KTO and Mashamba East)

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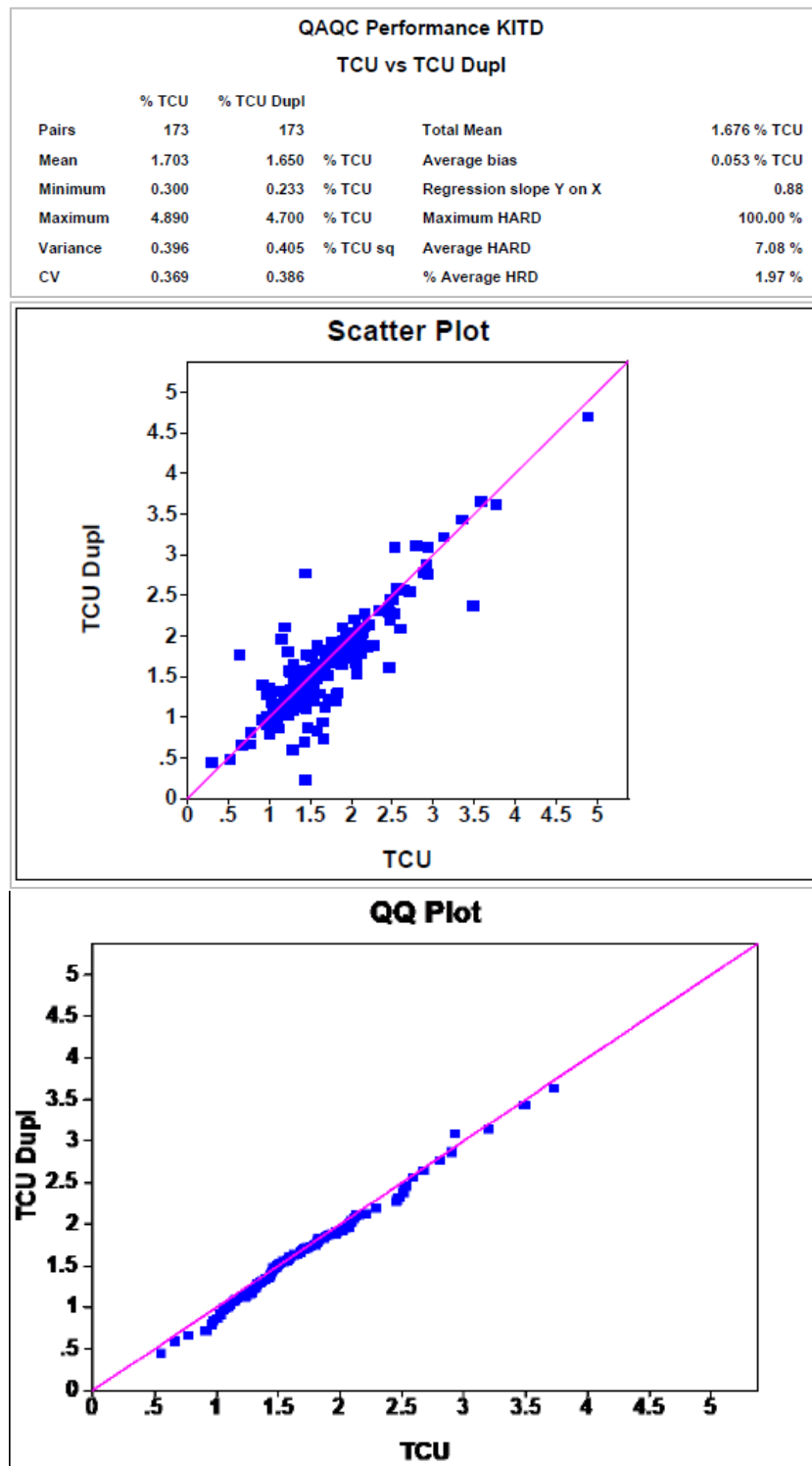


Figure 13: QA/QC performance of %TCu field duplicates (KITD)

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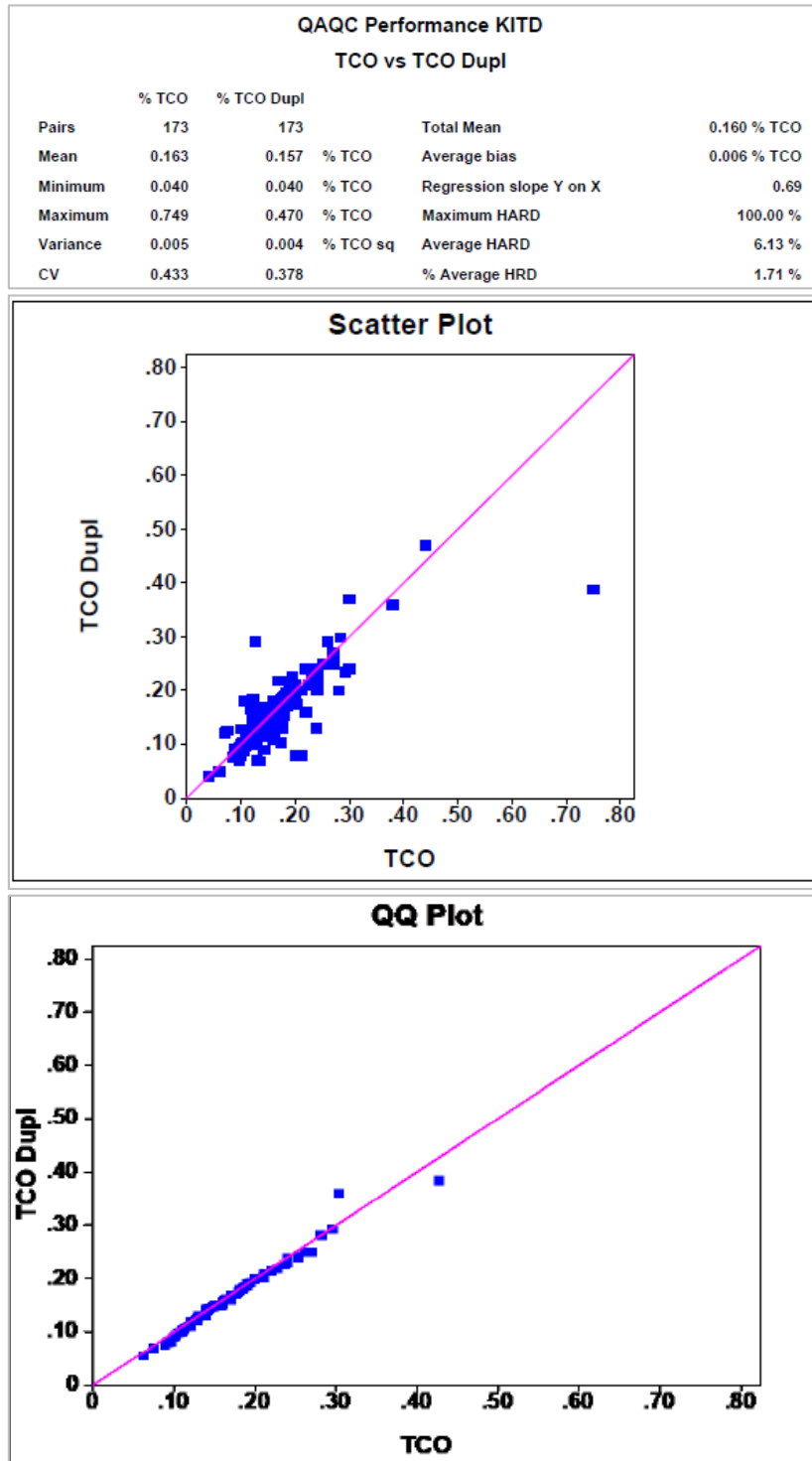


Figure 14: QA/QC performance of %TCo field duplicates (KITD)

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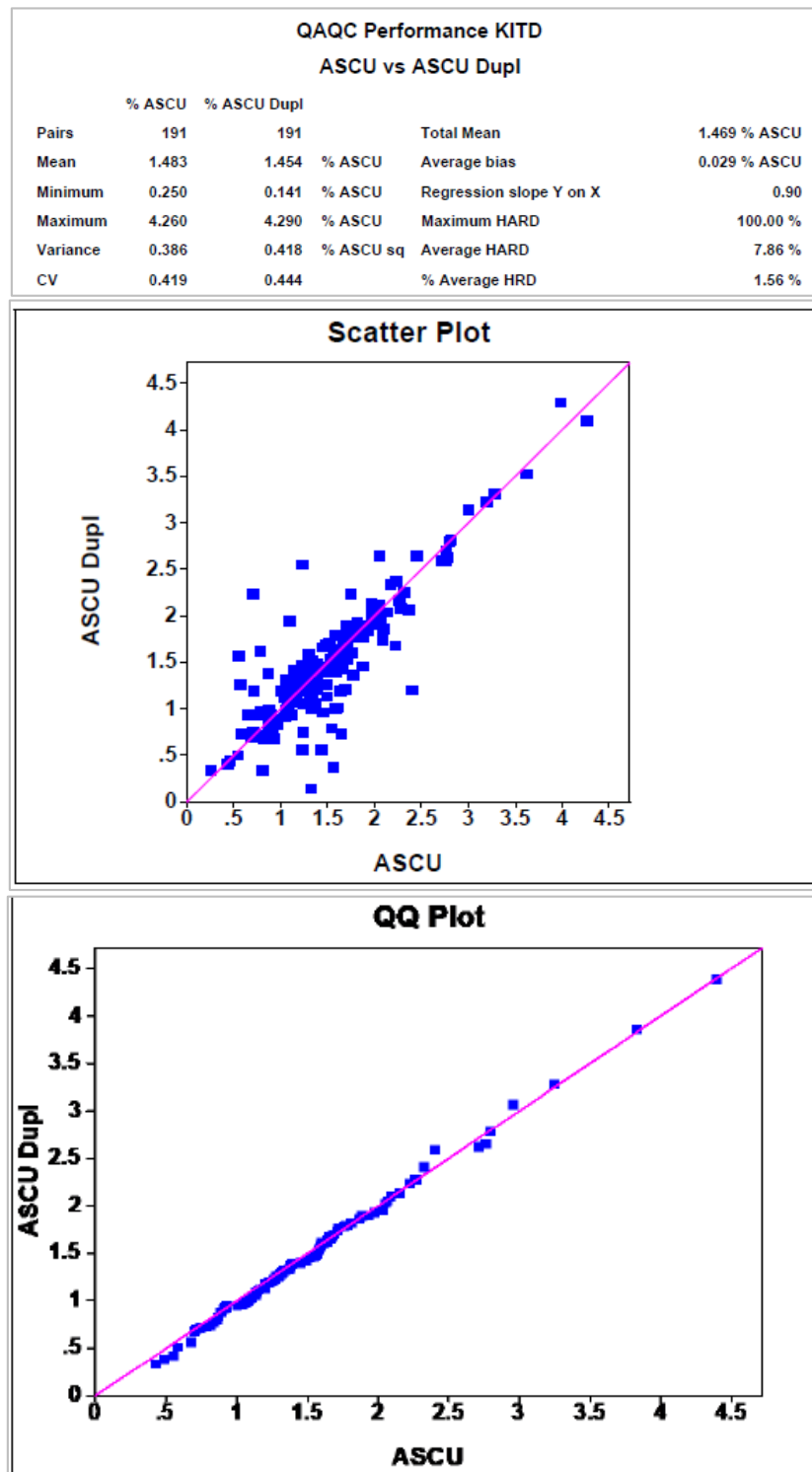


Figure 15: QA/QC performance of %AsCu field duplicates (KITD)

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11.3.4.1 Comment on the Performance of the Field Duplicates

Slight positive bias is apparent in the field duplicate performance, however, this is deemed acceptable for Mineral Resource estimation.

11.3.5 CRM

The database on CRM from 2009-2017 contains a total of 1,697 samples on which the following can be noted:

- More than 93% of KOV OP, 99% of Mashamba East OP, 97% of KTO UG and 93% of KITD (98% SGS; 87% Luilu) %TCu samples analysed passed within the confines of +/- 2 standard deviations; and
- More than 86% of KOV OP, 97% of Mashamba East OP, 94% of KTO UG and 75% of KITD (99% SGS; 51% Luilu) %TCo samples passed within the confines of +/- 2 standard deviations.

The following observations are noted for 2017:

- AMIS0398: Except for 12 %TCu and 37 %TCo samples out of a total 303 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations;
- AMIS0246: Except for 11 %TCu and 27 %TCo samples out of a total 406 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations;
- AMIS0348: Except for 1 %TCu sample out of a total 129 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations;
- AMIS0357: Except for 4 %TCu and 4 %TCo samples out of a total 224 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations;
- AMIS0417: Except for 2 %TCu and 12 %TCo samples out of a total 215 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations; and
- AMIS0201: Except for 11 %TCu and 29 %TCo samples out of a total 234 samples, the results of the remaining samples are within the confines of +/- 2 standard deviations.

Acceptable limits are 95% of samples submitted to be within +/- 2 standard deviations.

When acceptable limits for CRM are not achieved the following course of action is adhered to:

- Cross check KCC reference material assay with laboratory submitted reference material assay for the same period and/or batch;
- Any trace of biasness is ascertained by looking at all the results of all six types of reference material; and
- A follow up discussion is held with the lab for further investigation and samples re-analysed where necessary.

Figure 16 and Figure 17 provide results of CRM AMIS0246 (combined KOV, KTO and Mashamba East), and Figure 18 and Figure 19 present results of CRM AMIS 0246 for combined KITD.

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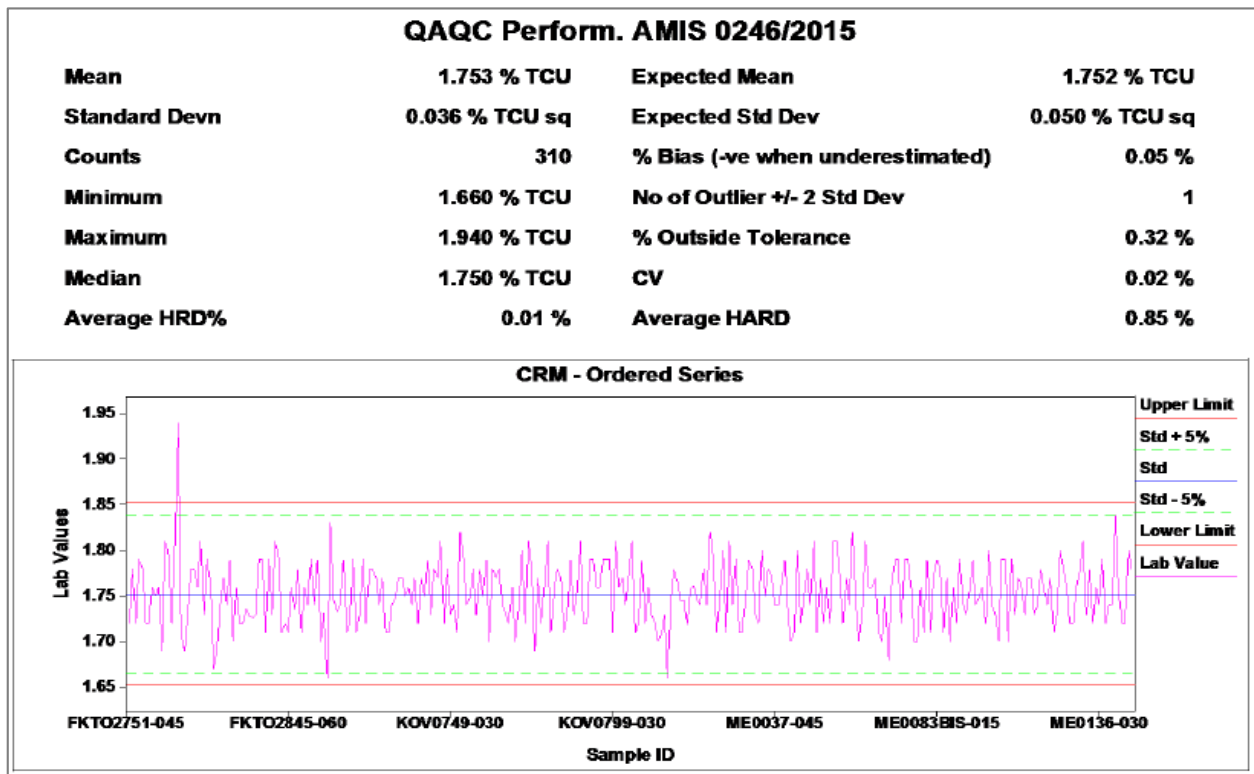


Figure 16: QA/QC performance of AMIS0246 for %TCu (KOV, KTO and Mashamba East)

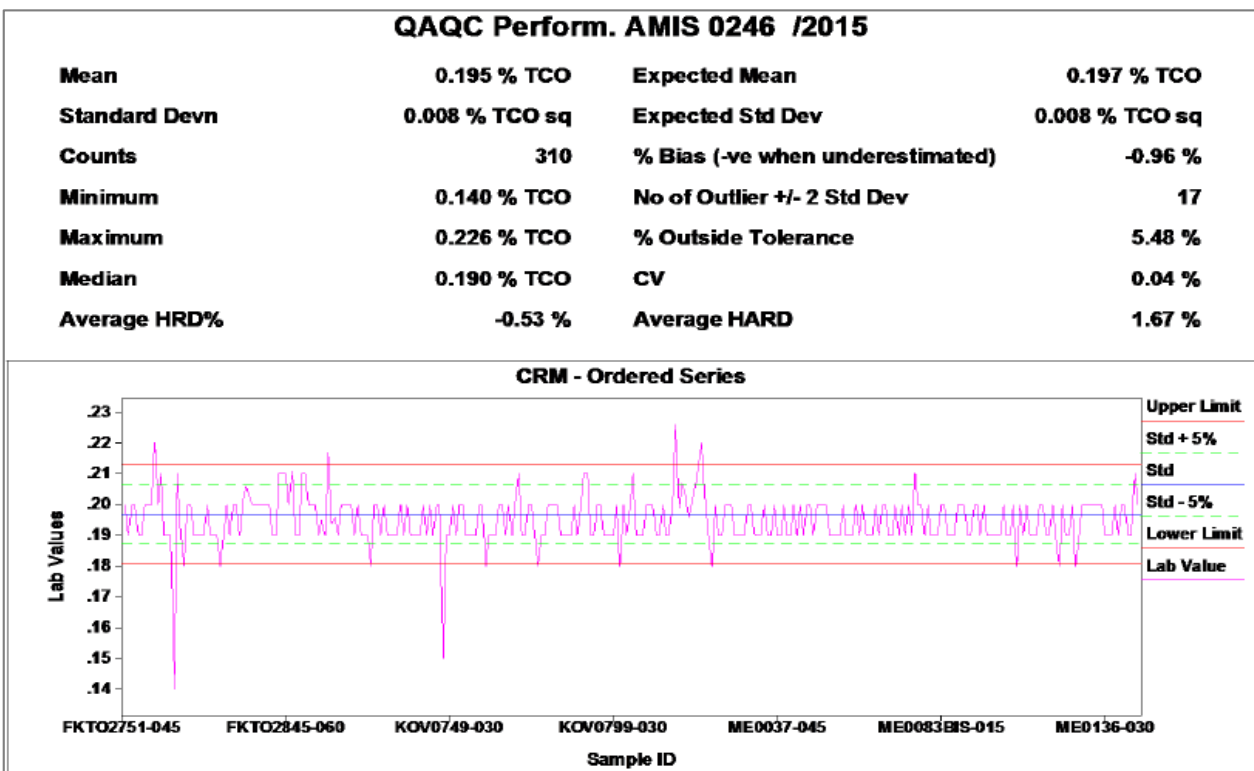


Figure 17: QA/QC performance of AMIS0246 for %TCu (KOV, KTO and Mashamba East)

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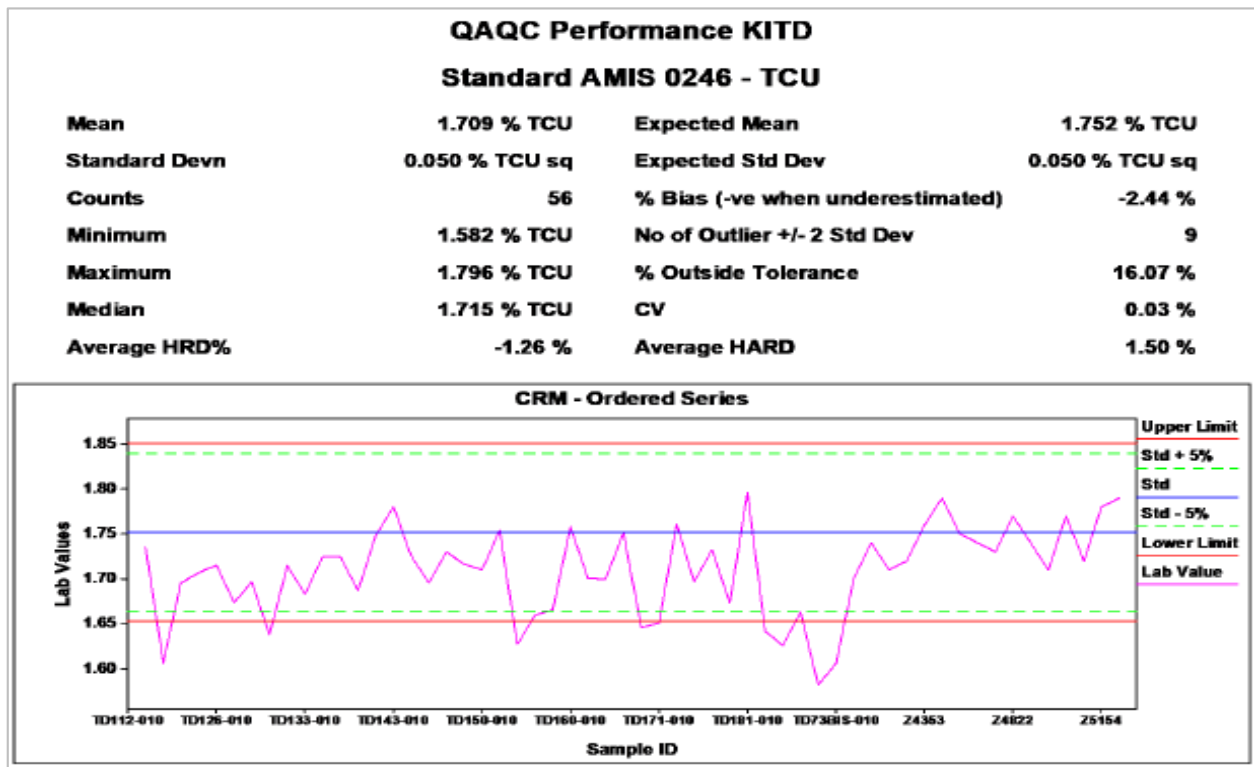


Figure 18: QA/QC performance of AMIS 0246 for %TCu (KITD)

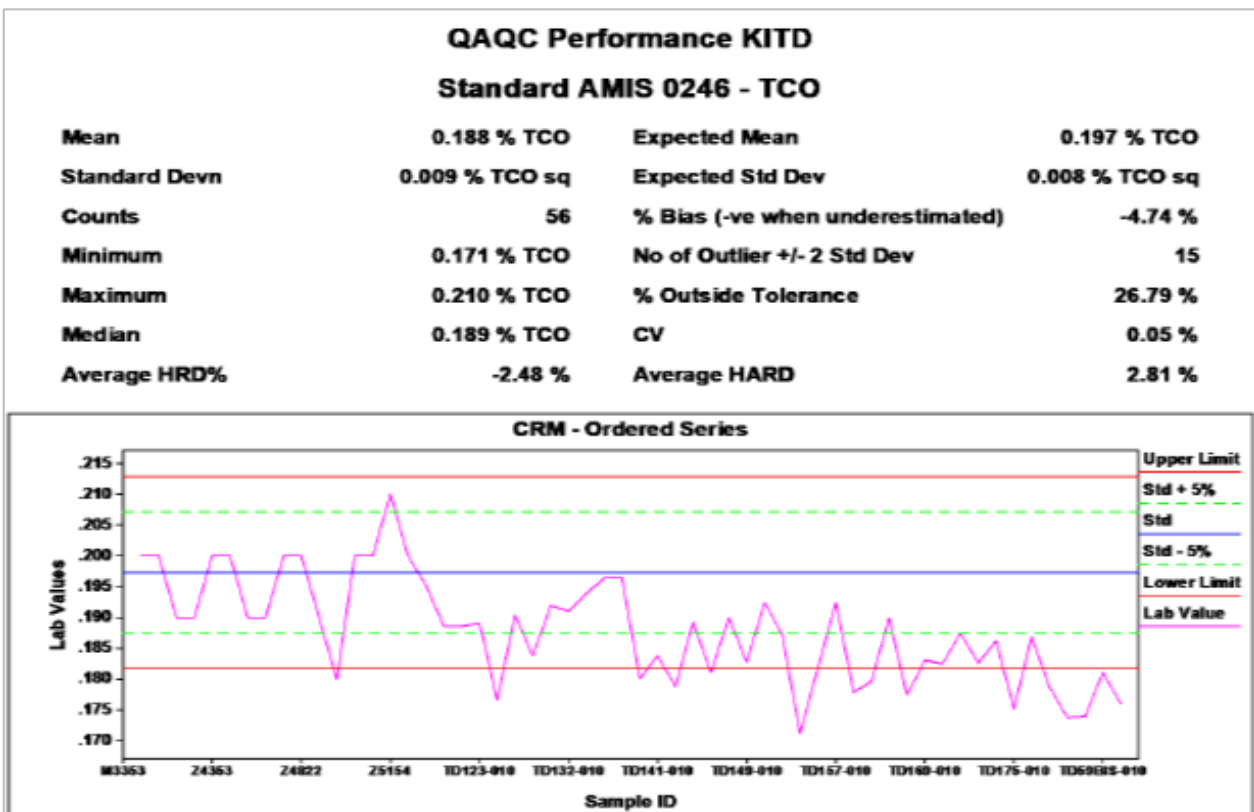


Figure 19: QA/QC performance of AMIS 0246 for %TCu (KITD)

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11.3.5.1 Comment on the Performance of the CRM

The SGS and Luilu laboratory performance on the CRM was satisfactory. Except for KITD, 95% of samples are within ± 2 standard deviations. The slight negative bias seen in KITD can be attributed to differences in laboratory technique between SGS and Luilu including, but not limited to, inaccurate calibration of carrier, differences in acid digestion methods and other variable controls. To better improve future analysis, cross-laboratory pulp reject validation will occur between accredited laboratories at Mopani Copper Mines Plc in Zambia and Mutanda Mining SARL in DRC.

11.4 Qualified Person's Opinion

It is the QP's opinion that the selection and preparation of sample intervals is acceptable for Mineral Resource estimation. As current mining and drilling continues to supplement historical sampling, the accuracy of the Mineral Resource estimate will improve. In addition, the selection of the analytical laboratory and the current QA/QC program using duplicates, CRMs, and blanks, is consistent with industry practices, and is also acceptable.

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12.0 Data Verification

A large quantity of data verification work has been completed while working on the 2017 Mineral Resource estimates for KCC. The data verification work completed included a check of the drill hole database provided against original assay records and a site visit by the QP to the site to check the materiality of the asset, drill hole collar positions, suitability of the core sheds and appropriateness of logging procedures. A 15 drill hole confirmatory drilling program was carried out in KOV OP in 2007 to verify the GCM database. A further nine drill hole confirmatory drilling program was carried out in KITD in 2017 to verify the previous drilling information.

The database used for each Mineral Resource estimate was found to be suitable for estimation purposes by the QP. Any issues with the data were considered while assigning classification to the Mineral Resource.

12.1 T17 Mine, KTO UG and Mashamba East OP - 2009

CCIC captured the data for 565 drill holes on hard-copy log sheets which were later transferred into Microsoft Excel. A parallel entry system was set up by Maxwell Geo Services (Maxwell), using a data-management system that allowed for the standardization of data capture and storage in a database. The electronic copies were verified and validated by comparing with the original hard-copy logs.

In all, 363 drill hole logs were supplied as scanned images of the original paper logs. These logs were captured in the existing Microsoft Excel database created during the Engineering Study. The verification process involved comparisons of the electronic logs against the paper logs. Every collar and survey data point was verified. The stratigraphic and sample logs were routinely verified. The validation process undertaken by CCIC involved the following:

- Checking for any zero lengths, gaps, overlaps and duplicate entries in the Microsoft Excel database and verifying 10% of the data against the hard-copy original logs;
- 3D visual validation, using Datamine™ Studio to validate collar and survey information;
- Visual flagging of the lithology using surrounding drill hole to ensure consistency in the logged stratigraphic succession across sections; and
- Histogram plots of the copper and cobalt values were generated to identify and verify any outliers.

During sampling of the 26 historical drill holes, 654 samples were generated and dispatched to SGS Lakefield for analysis. The analytical method used for the determination of total copper and cobalt was X-ray fluorescence. Copper and cobalt oxides were determined by acid digestion (sulphuric acid) and analysis of the solution by AAS.

The core verification program consisted of diamond-drill core samples selected from each Mineral Resource area and stratigraphic interval. The program was designed to check the accuracy of the recorded GCM copper and cobalt grades. Samples were prepared from the half core remaining after preparation of the original samples assayed by GCM. The half core was split in half again and one quarter of the original core was sampled and the other quarter remained in the original box at the same position. Initial QA/QC check samples were run at SGS Lakefield laboratory with additional splits and checks completed by SGS Lakefield and Set Point laboratories. Refer to Figure 20 for SGS Lakefield internal QA/QC plots for Co and Cu. The work was carried out in accordance with compliance procedures that addressed sample preparation, security, laboratory qualification, and procedures. The SGS Lakefield laboratory was accredited to SANAS, Facility Accreditation Number: T0223).

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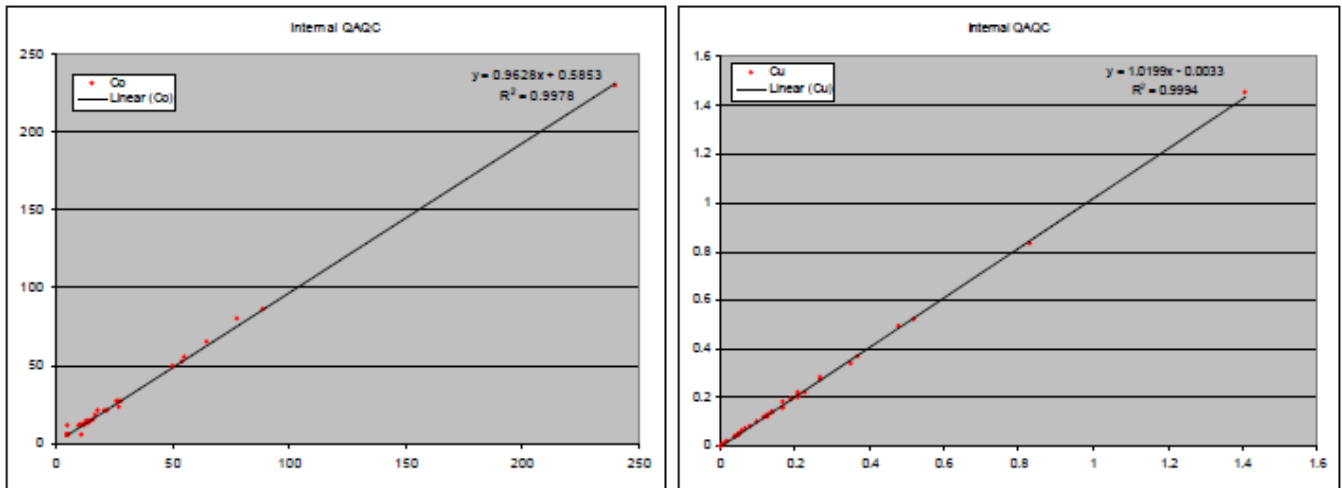


Figure 20: SGS Lakefield QA/QC plots for Cu and Co

12.2 Kananga OP and Tilwezembe OP - 2009

Snowden described a similar process for the validation of the database for both Kananga OP and Tilwezembe OP. The drill hole data were stored in Microsoft Excel spreadsheets, and Snowden imported the data into a GEMS Access database where the following validation checks were conducted:

- Missing collar coordinates;
- Interval errors (missing intervals, overlaps etc.) within drill-hole sample data;
- Duplicate sample records;
- Zero values within the sample data; and
- Collar elevation errors.

Data validation included checking for anomalous cases where the %AsCu and %AsCo values were greater than the corresponding %TCu and %TCo values. As %TCu represents the total copper in the sample, the %AsCu is expected to be lower than or equal to the %TCu in all cases. Therefore, where %AsCu was reported higher than %TCu, the sample values were reset to the %TCu value. Similarly, for the cases of %AsCo higher than %TCo, the %AsCo was reset to the %TCo value.

At Tilwezembe Mine, cases were found where the %AsCu values were greater than the %TCu and 30 cases for the %AsCo values being greater than the %TCo out of a total database of 3,000 samples. At Kananga OP, the numbers were 20 samples and 14 samples respectively out of a total database of 2,370. The %AsCu and %AsCo values were amended to the respective %TCu and %TCo values to correct this issue.

Snowden indicated that the difference in value between the acid soluble and the total grade were small with approximately 18% of the copper and 17% of the cobalt datasets reporting differences higher than 0.1% at Tilwezembe and 6% and 0% of the copper and cobalt datasets respectively reporting differences higher than 0.1% for Kananga.

12.3 KOV OP - 2009

SRK reviewed the database and the following issues with data quality were highlighted:

- Poor core recovery within the orebody varying between 65 and 75%, the worst recovery being in the RSC lithology;
- The RSC formation in contact with the SDB and the RSF was sampled, but near the middle of the formation selective sampling was undertaken on the basis of visible copper mineralization;

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- The orebody zones were sampled for total copper and total cobalt, but assays for acid soluble copper and cobalt and for calcium oxide were limited; and
- Samples with zero core recovery, mostly from the earlier drilling in KTE, are shown with high total copper values; presumably from the analyses of the sand collected in the absence of core.

SRK examined selected drill hole core but did not resample the core in any way. The KOV OP was in production for over 20 years, and head grades during that period reflect the grades in the database.

12.4 Recent Data Verification

KCC and Amec Foster Wheeler E&C Services Inc. (Amec Foster Wheeler) reviewed all the data (pre and post-2009 for KITD, KTO UG, KOV OP and Mashamba East OP). Reviews included the collar surveys, down-hole surveys, geology logs and assay data. It also included the following:

- Checking hardcopies of printed data to ensure data consistency;
- Checking for any zero lengths, gaps, overlaps and duplicate entries in the GDMS database. Validation using Datamine™ Studio for collar and survey information;
- Visual flagging of the lithology using surrounding drill holes to ensure consistency in the logged stratigraphic succession across sections; and
- Histogram plots of the copper and cobalt values were generated to identify and verify any outliers.

The audit found the database to be acceptable to support Mineral Resource estimation.

12.5 2017 and 2018 Site Visits

A site visit to the KCC office, located in the town of Kolwezi, Democratic Republic of Congo was carried out by Christiano Santos Goncalves (MAusIMMCP Geo), a QP for Mineral Resources for this TR, on November 2017 and December 2018. The visit to the property included:

- An overview tour of the exploration property; and
- Visual inspection of physiography and general conditions.

The site visit to the KCC office and core logging facilities included the following items:

- Review of the logging and sampling procedures used on the drill holes;
- Review of core logs against the core available at time of visit;
- Review of the KCC geological and mineralization characteristics with KCC staff;
- Review of QA/QC protocol;
- Review of sampling and shipping protocol; and
- Review of the Mineral Resource modelling, estimation and validation procedures.

No significant issues were identified during the review of data collection procedures, security or sample chain of custody. The core logging matched the core well and all processes were found to meet industry standards.

During the site visit no independent samples were taken. KCC has been in production for over 9 years, and head grades during that period reflect the grades in the database.

12.6 Qualified Person's Opinion

The database used for each Mineral Resource estimate was found to be suitable for Mineral Resource estimation and meets industry standards. Data provided by KCC was validated and verified during the course of the site visits, and it is the QPs opinion that the data is adequate for Mineral Resource estimation. The integrity of the historical data was considered for the Mineral Resource classification scheme employed by

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KCC and Amec Foster Wheeler for KOV OP, KTE, KTO UG and Mashamba East OP. Refer to Sections 14.2.10, 14.3.10 and 14.4.10 for further details.

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13.0 Mineral Processing and Metallurgical Testing

13.1 Introduction

Initial metallurgical testwork was conducted on the KCC cores from 2006 (van Deventer, 2006) to 2009 (Mabaso & Bryson, 2009). The testwork involved comminution, flotation and hydrometallurgical work. Additional testwork was conducted since 2010 to confirm and optimize process design parameters.

13.2 Phase 3, 4 and 5 Testwork

Testwork to generate extraction and stripping isotherms was conducted by two independent laboratories for the design of the Phase 4 SX plant. The testwork was conducted on electrolytes from the existing Luilu Plant that were adjusted to represent the pregnant leach solution (PLS) composition predicted by the METSIM® model. No testing was performed on the electrowinning process as electrowinning downstream of the SX plant is deemed to be a well-proven process.

Two previous independent testwork reports (Makhalemele, 2008) on milling were consulted as part of the Phase 4 Project development. Investigations were conducted to confirm the expectation that returning the DIMA mills to semi-autogenous operation would increase the total installed capacity to approximately 9 Mtpa. As part of the Phase 5 project, notably the construction of the semi-autogenous CM5 mill, current milling capacity has been increased to 12 Mtpa.

The results from the testwork programs were used in the design of the current circuit, which includes the Phase 3, 4 and 5 Process Improvement Projects. Data from the operating plant were also used in the development of the flotation process for Phases 3, 4 and 5.

13.3 Whole Ore Leach Testwork

A WOL Project development testwork program was launched in Q2 2015 (Slurrytec, Vietti, 2015 and Du Preez, Chagonda and Crundwell, 2015). The additional testwork was required to confirm the design of the leach circuit and the solid-liquid separation circuit for the WOL Project. The testwork program included the following:

- Hydrometallurgical testwork; and
- Solid-Liquid Separation testwork.

CM Solutions Proprietary Limited (CM Solutions) conducted the hydrometallurgical testwork while Vietti Slurrytec Proprietary Limited (Vietti Slurrytec) conducted the solid-liquid separation testwork.

The original CM Solutions testwork was performed on four cores selected from each of the FNSR, Variant, Oliveira and Virgule fragments in the KOV pit. Each fragment contained the seven relevant lithologies. No Mashamba pit cores were tested during the CM Solutions test program. Subsequently, a further 54 cores from both Mashamba and KOV pits were tested by KCC laboratory in the period from April 2016 to January 2017.

13.3.1 CM Solutions

All testwork from CM Solutions was performed on site. The following tests were conducted on selected KOV drill core samples and calcined sulphide material obtained from the existing process by CM Solutions:

- Optimization of sulphide flotation on mixed ore milled product (du Preez, Chagonda and Crundwell, 2015);
- Acid leach optimization testwork (du Preez, Chagonda and Crundwell, Preliminary Comparison of Phase 1 and Phase 2 Results, 2015);
- Bulk leach testwork using the optimum conditions established to produce a PLS (du Preez, Chagonda and Crundwell, Phase 3 Results, 2015);
- SX isotherm testwork on the PLS from the bulk leach (du Preez, Sebata and Crundwell, Phase 4 Results, 2015);

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- Precipitation testwork, namely iron/manganese removal, secondary copper precipitation and cobalt precipitation (du Preez, Sebata and Crundwell, Phase 5 Results, 2015); and
- Thickening and filtration testwork on the precipitates produced (Slurrytec, Vietti, 2015).

Figure 21 below shows a summary of the testwork conducted by CM Solutions.



Figure 21: Testwork by CM Solutions

13.3.1.1 Phase 1 Testwork Results – Ore Variability

The focus of Phase 1 was to investigate the ore variability at a P_{80} of 150 micron. During this phase milling curves were determined for each of the 28 ore samples. These milling curves were then used to generate milled samples of each ore type at the target P_{80} of 150 micron. A pre-float to remove the sulphides from the samples was performed and the flotation tails were then used for the acid leach tests. In the acid leach tests, the flotation tails of each ore sample was leached in sulphuric acid for four hours at a constant pH of 1.5 (Du Preez, Chagonda, & Crundwell, 2015).

13.3.1.1.1 Flotation Testwork Results

The weighted, overall average acid-insoluble copper recovery for the sulphide pre-float was 70%, with a weighted average copper grade in the concentrate of 11%. A wide variation in the acid-insoluble copper

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recoveries for the different ore bodies and lithologies was noted, ranging from a 1% recovery of the acid-insoluble copper up to 100% recovery of acid-insoluble copper. Similarly, the copper grades in these concentrates ranged from nil up to approximately 27%. A nil copper grade in the concentrate is possible when the ore sample contains almost no acid-insoluble copper, thus only gangue material is floated.

13.3.1.1.2 Leaching Testwork Results

For the leach tests, the weighted overall average acid-soluble (oxide) leach efficiency was 95.3%. As with the flotation, a wide variation in the oxide leach efficiency was observed across the different ore bodies and lithologies. It ranged from about 63% up to 98%.

The weighted overall average total copper leach efficiency was 93.8%. The total copper leach recovery ranged from 78% up to 98%. This lower total copper leach efficiency is due to un-floated acid-insoluble copper that was not recovered during the leach.

13.3.1.1.3 Acid Consumption Testwork Results

The weighted overall average gangue acid consumption (GAC) was determined to be 64 kg/t-ore, ranging from 0 kg/t-ore up to about 340 kg/t-ore, depending on the orebody and lithology. All non-copper minerals (including cobalt) have been considered gangue. This value represents the amount of acid required, taking into account the acid recovered from the copper removal during solvent extraction. The weighted overall average fresh acid consumption (FAC) was determined to be 85 kg/t-ore. This ranged from 0 kg/t-ore up to 340 kg/t-ore, depending on the orebody and lithology. This value is the amount of fresh acid required after taking into account the acid recovered from the copper removal during solvent extraction and lost to the cobalt bleed. It assumes that 33.3% of the leached copper is bled to the cobalt circuit from the low-grade solvent extraction raffinate stream.

13.3.1.2 Phase 2 Testwork – Particle Size Comparison

A primary outcome from Phases 1 and 2 is the selection of the blend composition and target particle size. The following preliminary results are provided to allow KCC and SENET to begin considering this topic (CM Solutions, 2015).

13.3.1.2.1 Flotation

The results from the flotation tests vary markedly across the different ore bodies and lithologies. However, it is possible to compare the overall average recoveries, grades and mass pulls for the 150- and 212-micron tests. These values were determined by averaging all the ore bodies and lithologies for both sets of flotation tests (2.5 L and 1.5 L floats) for both particle sizes. These results show, as expected, that increasing the particle size of the flotation feed reduces the recovery and grade of concentrate. The average results for the 150- and 212-micron flotation parameters are shown in Table 13. It is important to note that there is a large variance in the flotation test results for 150- and 212-micron tests.

Table 13: Average flotation parameters for 150- and 212-micron tests

Parameter	150 micron	212 micron
Recovery, %	52.8	37.0
Grade, %	7.4	6.3
Mass Pull, %	4.5	4.3

Figure 22 shows a comparison of the 150- and 212-micron flotation recovery results for all the lithologies and ore bodies. The 212-micron results have been plotted against the 150-micron results and grouped by orebody.

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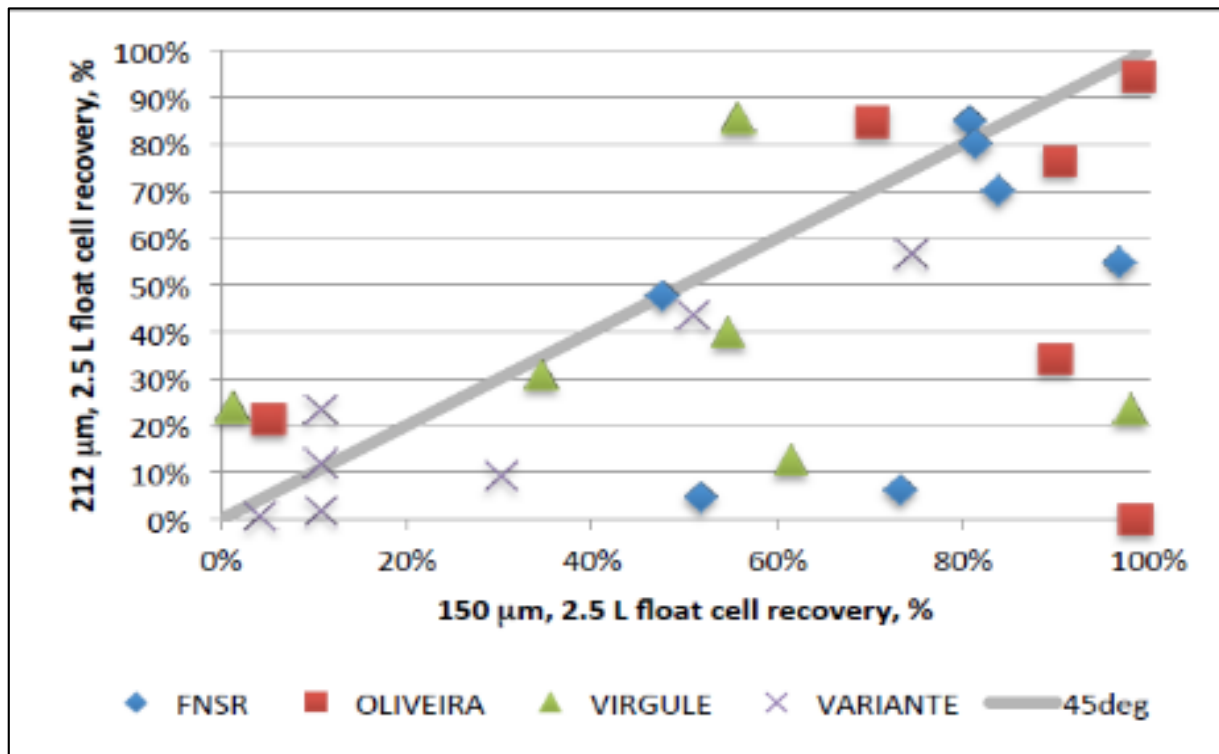


Figure 22: Comparison of 150- and 212- micron flotation recovery results

Figure 22 also shows the wide spread of recoveries achieved in the flotation tests, and a similar spread of concentrate grades was also obtained. What it does however show is that the recoveries are typically below the 45° line. This indicates that the 150-micron recoveries are typically higher than the 212-micron recoveries. This observation is in line with theoretical expectations. The results presented here indicate the reduction in the recovery and concentrate grade for a similar mass pull at a P_{80} of 212-micron compared to a P_{80} of 150-micron.

13.3.1.2.2 Acid Leach

During the leach tests the initial mass and final leach tails mass are measured, and the amount of acid added to maintain a pH of 1.5 is also measured. Based on this data it is possible to calculate the mass loss and estimate the total acid consumption (TAC). All the data for the 150-micron tests is available. Figure 23 shows the 150 and 212-micron mass losses plotted against each other.

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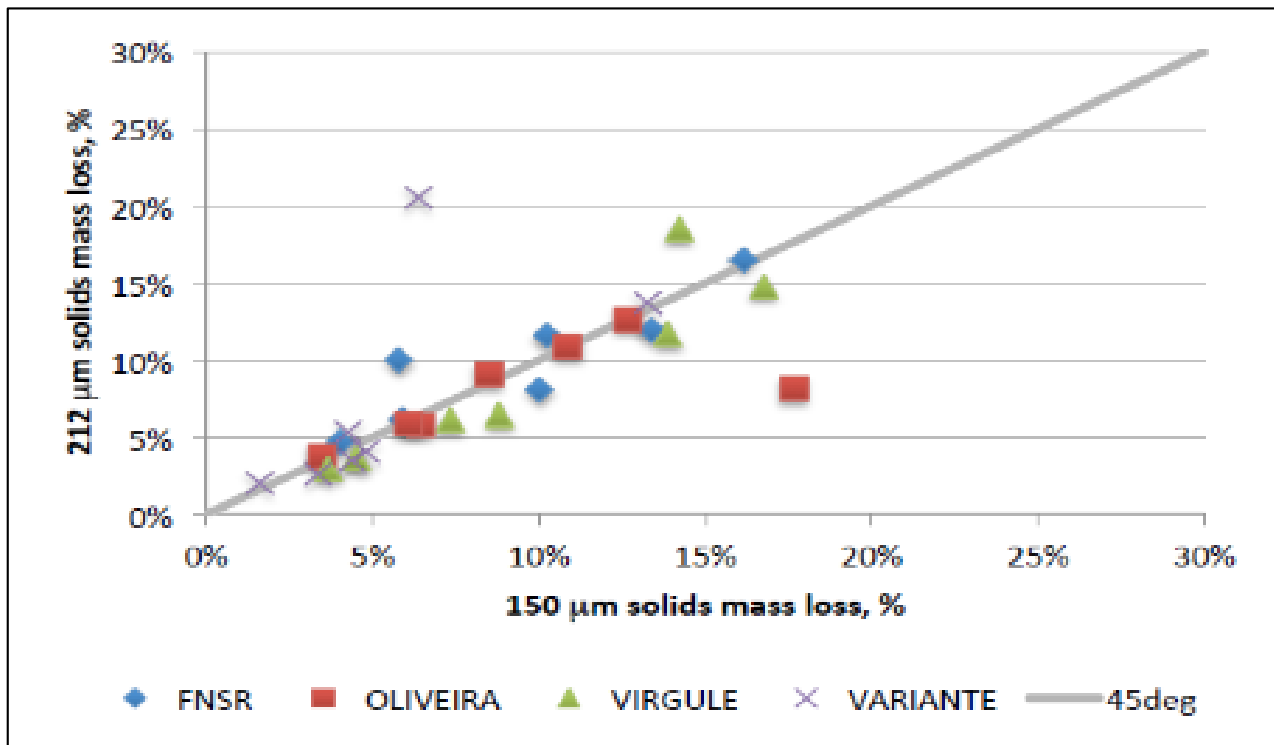


Figure 23: Comparison of 150- and 212-micron acid leach solids mass loss

These results indicate that for the majority of the lithologies across different ore bodies there is little difference in mass loss between 150- and 212-micron tests. The outliers are primarily from the DSTRAT and SDS lithologies. The overall average mass loss for both 150- and 212-micron was 9%. Figure 24 shows the 150- and 212-micron TAC plotted against each other.

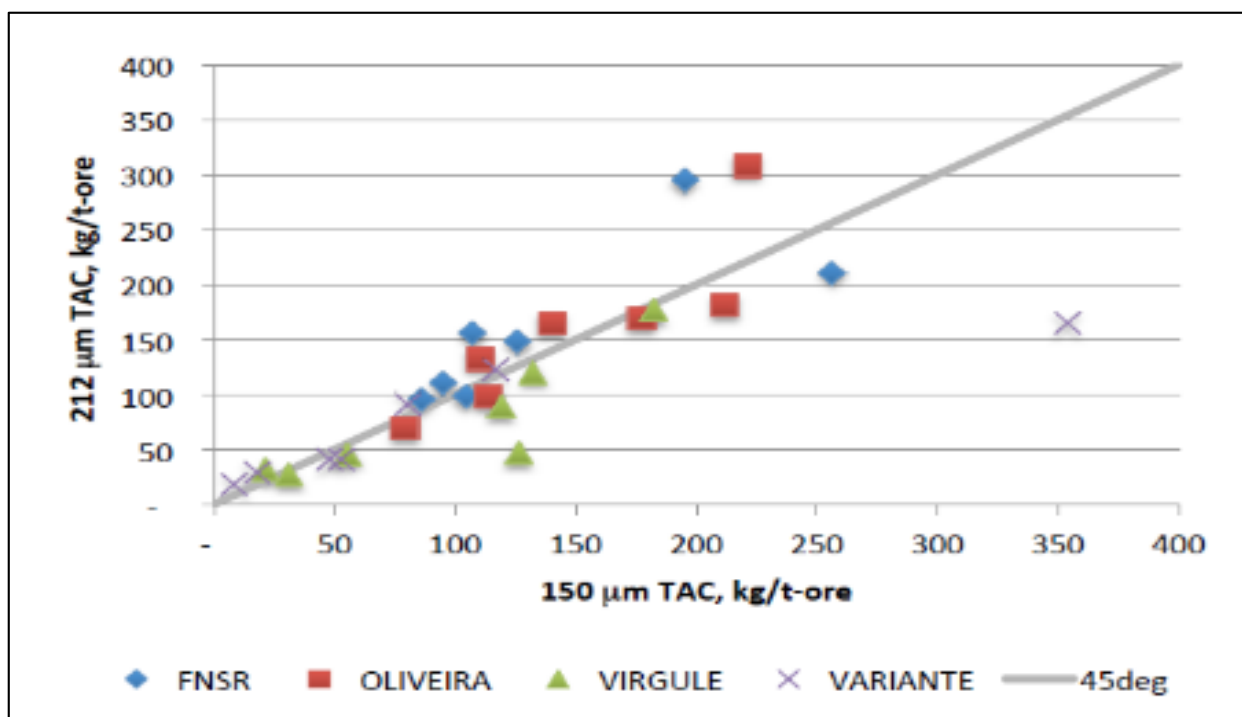


Figure 24: Comparison of 150- and 212-micron total acid consumption

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As with the solid mass loss, the TAC values are similar for the majority of lithologies and ore bodies, with the outliers primarily from the DSTRAT lithology. Based on these results, it is likely that the other leach parameters, such as leach efficiency and gangue acid consumption will be similar. The overall average TAC for 150-micron is 121 kg/t-ore and 117 kg/t-ore for 212-micron.

13.3.1.3 Phase 3 – Blended Ore Performance Characteristics

The focus of Phase 3 was to investigate the sulphide pre-float and leaching performance for a blended ore with a P_{80} of 150-micron under different conditions. The ore blend was based on KCC's current Mineral Resource estimates. The sulphide pre-float was performed under the same conditions as Phase 1 and 2. The leach tests however, were performed at different pH values, oxidation reduction potential (Eh) values, % solids, temperature, calcine addition and time of leach (du Preez, Changonda and Crundwell, Phase 3 Results, 2015).

13.3.1.3.1 Sulphide Pre-flotation Results

The acid insoluble copper recovery was determined to be 66.6%, with an acid insoluble copper grade of 6.90%. The Phase 1-based estimate of the acid insoluble copper recovery was 68.9% at a copper grade of 11.1%.

13.3.1.3.2 Acid and SO₂ Consumption Results

The gangue acid consumption (GAC) for the base case was 81 kg/t-ore and the fresh acid consumption (FAC) was 102 kg/t-ore. The SO₂ consumption was 22.6 kg/t-ore. It was found that decreasing the leach pH from 1.5 to 1.0 increases the GAC to 121.5 kg/t-ore; however increasing leach pH from 1.5 to 2.0 decreases the GAC to 65.6 kg/t-ore. Similarly the FAC increases to 142 kg/t-ore at a leach pH of 1.0 and decreases to 85.7 kg/t-ore at a leach pH of 2.0. The SO₂ consumption was also affected by the leach pH, at leach pH 1.0 the SO₂ consumption was 58.9 kg/t-ore and at a leach pH of 2.0 the SO₂ consumption was 6.4 kg/t-ore.

While the other leach parameters did affect the acid consumptions, they were all minor relative to the effect of the leach pH.

13.3.1.3.3 Leach Results

The leach efficiencies achieved in base case leach tests were 95.3% for total copper, 99.6% for acid soluble copper, and 76.0% for total cobalt. While the copper recoveries are only slightly higher than the Phase 1-based estimates, the cobalt and iron were much higher due to the use of SO₂ in the leach. As with acid consumption, the leach pH affected the leach performance, although the impact on copper recovery was relatively minor it decreased from 95.4% at a leach pH of 1.0 to 93.0% at a leach pH of 2.0. Its impact on cobalt recovery was more significant; it decreased from 82% at a leach pH of 1.0 down to 60% at a leach pH of 2.0.

It was found that substituting SMBS for SO₂ increased the acid consumptions, confirming that SO₂ should be used in preference to Sodium Metabisulphite (SMBS).

The kinetic studies indicate that the majority of the copper is leached within two hours, however a leach time of four hours is necessary to achieve the higher cobalt leach efficiencies. The tests further indicate that the acid leaching recoveries plateau after about two hours. Similarly, much of the cobalt recovery is achieved within the first two hours from when the Eh is controlled with SO₂ addition.

13.3.1.4 Phase 4 Testwork – Solvent Extraction

The focus of Phase 4 was to investigate the solvent extraction of copper. Extraction and strip isotherms were generated for each of the train organics with current plant PLS. Phase disengagement tests were also performed on each train. Thereafter these same tests were performed with fresh organic and plant PLS. Finally an extraction isotherm was generated for a synthetic PLS generated from the Phase 3 (P_{80} 150-micron, blend testwork) with fresh organic.

For all streams the phase disengagement times were found to be acceptable (less than 180 seconds). However, they did indicate that the organic from the first train is performing poorly in comparison to the other two trains. The organic will be clay treated.

The PLS from the Phase 3 testwork, specifically the blend with 2% calcine leach test, was used for the synthetic PLS. This PLS has a lower copper concentration than the current PLS. Part of the reason for this is that the

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leach tests were not performed with raffinate. For this reason both the copper and acid concentrations in the synthetic PLS are likely to be lower than the WOL process PLS (du Preez, Sebata and Crundwell, Phase 4 Results, 2015).

13.3.1.5 Phase 5 – Cobalt Bleed Precipitation Characteristics

The focus of Phase 5 was to investigate the cobalt bleed precipitation section. This included iron, aluminium and manganese precipitation, copper precipitation, cobalt precipitation and cobalt re-dissolution (du Preez, Sebata and Crundwell, Phase 5 Results, 2015). Two sets of testwork were performed for each section:

- Batch tests where a single dose of CaO/MgO was added and the pH monitored as it changed with time; and,
- Semi-batch tests where solution was fed into the reactor and the pH maintained through the addition of CaO/MgO.

The results presented indicate that further studies should be performed on the precipitation circuit.

This study provides the basis from which further controlled precipitation testwork can be done.

Batch tests indicate that the following times should be acceptable for residence times:

- 120 - 150 minutes for the iron/manganese/aluminium precipitation section;
- 20 - 60 minutes for the copper precipitation section;
- 20 - 30 minutes for the cobalt precipitation section; and,
- ~60 minutes for the cobalt re-dissolution.

The reagent consumption results presented in the Table 14 are based on the semi-batch testwork results.

Table 14: Summary of reagent consumption rates Flocculating

Section	CaO, kg/m ³ soln	MgO, kg/m ³ soln	SO ₂ , kg/m ³ soln
FAM	72.1	-	37.5
Copper	51.8	-	-
Cobalt (Part 1)	-	1.9	-
Cobalt (Part 2)	36.3	-	-

13.3.2 Vietti Slurrytec

Vietti Slurrytec performed the testwork off site in Johannesburg, South Africa. The following tests were conducted on the pre-leach slurry and post-leach slurry that was generated by the technical services department at Luilu:

- Thickening testwork;
- Filtration testwork; and
- Rheology testwork.

Figure 25 below shows a summary of the testwork conducted by Vietti Slurrytec.

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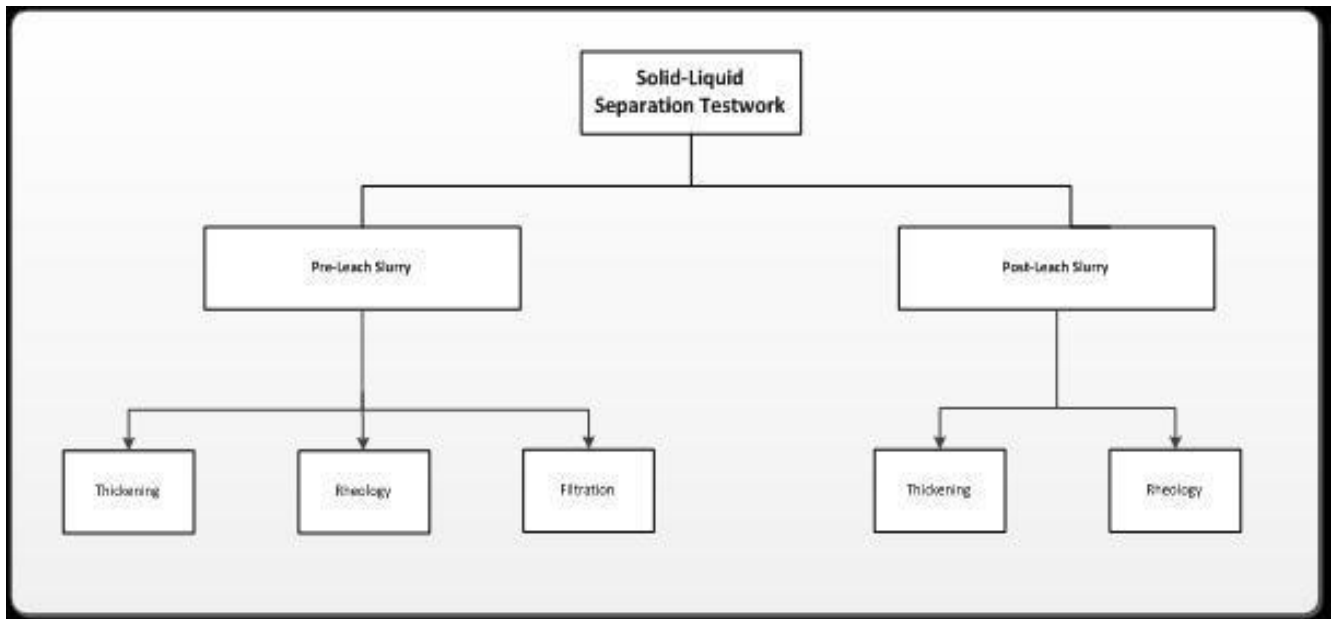


Figure 25: Testwork by Vietti Slurrytec

13.3.2.1 Thickening Testwork

Vietti Slurrytec conducted laboratory scale thickening testwork on two slurry samples from the KCC Mines in the DRC for optimization of the Current Plant Process:

- Pre-leach flotation tailings slurry (oxide receiving) – (CUR-RSL); and
- Post-leach slurry from CCD circuit (leach residue) – (CUR-PLSL).

The test results and findings of the laboratory thickening testwork are summarized below (Vietti Slurrytec, 2015).

- The CUR-RSL and CUR-PLSL slurries were found to be naturally coagulated (settling) in the unflocculated state;
- No swelling clays were detected in the ore samples. Trace amounts of Talc clays were detected;
- The conductivity of the supernatant water (process water) of the CUR-RSL slurry is moderately high at 7 mS/cm and the pH around 7 to 8;
- No conditioning of the slurry is required prior to flocculation; and
- For thickening the optimum flocculant types identified were:
 - a) CUR-RSL: Magnafloc 338 and A-120; and
 - b) CUR-PLSL: Magnafloc 10 and N-100.

The optimum thickener flocculating parameters obtained from the testwork are shown in Table 15.

Table 15: Flocculating parameters

Parameters	CUR-RSL	CUR-PLSL
	High-Rate	High-Rate
Feed solution solids concentration (% _m) (In feed well)	5	
Flocculant Dose (g/t)	70	80
Flocculant Dosing (% _m)	0.025	

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After 24 hours consolidation under bench scale High Rate thickening conditions the following mud bed solids concentrations were recorded:

- CUR-RSL: 52.8%_m
- CUR-PLSL: 46.3%_m

The un-sheared and sheared yield stress of the slurry at these solids concentrations are shown in Table 16.

Table 16: Yield stress

Parameter	CUR-RSL	CUR-PLSL
Un-sheared yield stress (Pa)	224	101
Sheared yield stress (Pa)	63	19

It was recommended that picket raking is considered to increase the underflow solids concentration of the CUR-PLSL slurry for improved liquor recovery.

13.3.3 Filtration Testwork

Dewatering Solutions conducted laboratory scale filtration testwork for KCC on a pre-leach copper slurry sample representative of the new WOL process (Slurrytec, Vietti, 2015).

13.3.3.1 Filter Press

- The material dewateres readily under pressure filtration;
- 40 mm thick cakes were considered in the final design;
- A maximum of 5 bar feed pressure was considered in the final design;
- 5 bar feed pressure produced lower cake formation times as well as lower cake discharge moistures; and
- The reduction in cake formation time is minimal (30 to 90 seconds), as is the reduction in cake moisture (17% to 16% _m).

13.3.3.2 Tower Press

- Cakes are readily formed under pressure filtration;
- 6 Bar produces marginally lower cake moisture (Low 16% _m) compared to 3 bar (High 16% _m);
- 6 Bar has lower cake formation times than 3 bar (30 seconds to 2 minutes);
- On a rapid discharge filter, this may be significant;
- 6 bar feed pressure (or membrane squeeze) should be considered; and
- Thick cakes, up to 35 mm are recommended.

13.3.3.3 Vacuum Filter

Without the use of flocculant:

- For large tonnages, thin cakes are to be avoided as they crack on large width belt filters;
- A target cake formation time is around 60 seconds;
- An 11-13 mm cake takes around 2 minutes to form;

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- Cake discharge moisture is around 20% m; and
- At high tonnages, without flocculant addition, vacuum filtration is not viable.

With flocculant addition:

- Various flocculants were tested;
- The addition of flocculant as well as the handling of flocculated filter feed will be crucial to ensure that a vacuum filter performs satisfactorily;
- SetChem SC555 proved to be the most efficient flocculant;
- Flocculation only becomes effective at high dosages, over 140 g/t; and
- The addition of flocculant increases the cake discharge moisture to 27-28% m.

13.3.3.4 Rheology Testwork

Table 17 summarizes the results of the material properties and bench top tests (Vietti Slurrytec, 2015).

Table 17: Material properties and bench top tests

Property Tested	Pre Leach Thickener Feed (WOL-RSLU)	Pre Leach Thickener Underflow (WOL-RSL)	Post Leach Thickener Underflow (WOL-(PLS))
Solids Density	2839 kg/m ³		2789 kg/m ³
D ₉₀ Particle Size	241 µm		115 µm
D ₅₀ Particle Size	23 µm		19 µm
D ₂₅ Particle Size	7 µm		5 µm
% passing 45 µm	64.0		81.7
% passing 25 µm	51.8		56.2
Average Slurry pH at 25°C	7.98	7.93	1.54
Average slurry temperature	18.0°C	17.6°C	20.5°C
Slurry conductivity	0.19 mS/cm	0.19 mS/cm	20.40 mS/cm
Free settling bed packing concentration, C _{free}	36.7% _v or 62.1% _m	36.2% _v or 61.7% _m	31.3% _v or 56% _m
Process water salinity (g _{salt} /g _{water})	N/A	N/A	0.04

Table 18 summarizes the rheological correlations used to calculate the slurry rheology as a function of mass solids concentration.

Table 18: Summary of the rheological correlations

Slurry Name	Bingham Plastic Model	
	Plastic Viscosity	Bingham Yield Stress
Pre Leach Thickener Feed	Applicable Mass Solids Concentration Range: 56% m < C < 73% m	
	$K_{BP} = \mu_w + 2.52 C^{9.8}$	$\tau_y = 42.09 \times 10^3 C^{20.3}$
Pre Leach Thickener underflow	Applicable Mass Solids Concentration Range: 51% m < C < 75% m	

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	$K_{BP} = \mu W + 23.03 C^{16.52}$	$\tau_y = 119.3 \times 10^3 C^{22.58}$
Post Leach Thickener Underflow	Applicable Mass Solids Concentration Range: 58% m < C < 71% m	
	$K_{BP} = 16.19 C^{13.20}$	$\tau_y = 6.23 \times 10^3 C^{16.76}$

μ_w = viscosity of water at 18°C (0.0010528 Pa.s)

13.3.4 Current Mashamba East Testwork

A campaign to catalogue the leach and flotation characteristics of the Mashamba East orebody is currently underway in the KCC Metallurgical Laboratory. Various core holes have been received from the KCC Geology department; the cores are divided into lithologies and tested individually to isolate the characteristics of each lithology.

Currently 15 Mashamba East cores have been received with a total of 79 lithologies tested and analysed. From the cores received, it was observed that the acid soluble copper ratio was higher than in KOV. This was expected as Mashamba East is a younger orebody that is shallower than KOV and thus the minerals are more oxidized. In this stage of the profiling campaign the overall copper grade is lower than in KOV. The results are shown in Table 19.

Table 19: Mashamba East copper grade profile

Lithology	Head Grade %TCu	Head Grade %AsCu	AsCu Ratio
SDS	1.15	1.09	94%
BOMZ	1.66	1.56	92%
SDB	2.86	3.32	98%
RSC	3.39	2.80	97%
RSF	2.86	2.84	98%
DSTRAT	3.74	3.69	98%
RATGRISE	1.72	1.67	96%
RATLILLAS	0.58	0.54	93%
Averages	2.25	2.19	96%

The leaching characteristics of the ore are exceptional in terms of GAC, pH control and leach efficiency; the results are shown in Table 20. The low recoveries on SDS, BOMZ and RATLILLAS are subject to the low head grades. There is always a fraction of oxidized ore that is not liberated, on average 0.18% TCu.

Table 20: Mashamba East leach recoveries and gangue acid consumption

Lithology	Leach AsCu Recovery	GAC (kg/DMT)
SDS	85%	1
BOMZ	75%	18
SDB	93%	7
RSC	94%	5
RSF	91%	8
DSTRAT	95%	34
RATGRISE	90%	10
RATLILLAS	82%	9
Averages	90%	12

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No flotation kinetics tests have been done on the Mashamba East ore yet, due to the high acid soluble copper ratios.

13.3.5 Current KOV OP Testwork

A campaign to catalogue the leach and flotation characteristics of the KOV OP mining operation is currently underway. KOV OP consists of 4 ore bodies namely, Oliveira, Virgule, FNSR and Variante. Various core holes have been received from the KCC Geology department; the cores are divided into lithologies and tested individually to isolate the characteristics of each lithology.

Currently 41 KOV OP cores have been received with a total of 155 lithologies tested and analysed. From the cores received, it was observed that there are localized areas of high acid consuming gangue minerals particularly in the Oliveira fragment. Consequently, there is a need for a precise blending strategy to control the calcitic minerals and the copper head grade in the feed material, so as to avoid sudden spikes in acid demand. The ore profiles are shown in Table 21 to Table 24.

Table 21: Oliveira copper grade profile

Lithology	Head Grade %TCu	Head Grade %AsCu	AsCu Ratio
SDS	1.64	1.50	91%
BOMZ	2.40	2.29	95%
SDB	4.78	3.82	80%
RSC	2.22	1.82	82%
RSF	3.42	2.79	82%
DSTRAT	4.28	3.32	78%
RATGRIS	4.33	3.38	78%
BRECHE	1.80	1.74	97%
RATL	0.27	0.25	93%
Averages	2.79	2.32	83%

Table 22: Virgule copper grade profile

Lithology	Head Grade %TCu	Head Grade %AsCu	AsCu Ratio
SDS	2.31	2.26	98%
BOMZ	5.66	5.55	98%
SDB	8.91	8.52	96%
RSC	2.41	2.36	98%
RSF	7.91	7.77	98%
DSTRAT	6.31	5.96	94%
RATGRIS	5.47	5.17	95%
BRECHE	1.6	1.6	100%
RATL	0.72	0.71	99%
Averages	4.59	4.43	97%

Table 23: FNSR copper grade profile

Lithology	Head Grade %TCu	Head Grade %AsCu	AsCu Ratio
SDS	2.30	1.95	85%

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SDB	8.89	8.22	92%
RSC	2.62	2.41	92%
RSF	4.61	4.25	92%
DSTRAT	8.51	8.03	94%
RATGRIS	3.53	3.34	95%
RATL	0.53	0.51	96%
Averages	4.43	4.10	93%

Table 24: Variante copper grade profile

Lithology	Head Grade %TCu	Head Grade %AsCu	AsCu Ratio
SDS	0.81	0.75	93%
BOMZ	0.78	0.67	86%
SDB	0.80	0.70	88%
RSC	1.25	1.21	97%
RSF	3.98	3.93	99%
DSTRAT	1.37	1.01	74%
RATL	0.57	0.55	96%
Averages	1.37	1.26	92%

From the head grade analysis it was found that the oxide to sulphide ratio is higher than initially estimated. From the cores received the ratio was been found to be 90:10. Only 13% of the 270 lithologies analysed from KOV OP and Mashamba showed higher sulphide content ratios.

The ore from all the ore bodies have high acid soluble copper recoveries; low recoveries on certain samples are due to the low head grades. There was always a fraction of copper in the oxidized ore that was not liberated, on average 0.18% TCu. Gangue acid consumption varied from sample to sample and was found to be directly correlated to the calcium content of the ore and not necessarily related to any specific lithological classification. The acid soluble copper recovery and GAC values are shown in Table 25 to Table 28.

Table 25: Oliveira leach recoveries and gangue acid consumption

Lithology	Leach AsCu Recovery	GAC (kg/DMT)
SDS	93%	131
BOMZ	85%	51
SDB	90%	112
RSC	90%	104
RSF	95%	96
DSTRAT	92%	165
RATGRIS	92%	142
BRECHE	93%	52
RATL	81%	228
Averages	91%	120

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Table 26: Virgule leach recoveries and gangue acid consumption

Lithology	Leach AsCu Recovery	GAC (kg/DMT)
SDS	86%	11
BOMZ	82%	11
SDB	97%	9
RSC	97%	11
RSF	98%	10
DSTRAT	98%	23
RATGRIS	97%	11
BRECHE	93%	6
RATL	91%	14
Averages	93%	12

Table 27: FNSR leach recoveries and gangue acid consumption

Lithology	Leach AsCu Recovery	GAC (kg/DMT)
SDS	91%	30
SDB	97%	9
RSC	85%	78
RSF	80%	58
DSTRAT	96%	42
RATGRIS	94%	118
RATL	82%	50
Averages	89%	55

Table 28: Variante leach recoveries and gangue acid consumption

Lithology	Leach AsCu Recovery	GAC (kg/DMT)
SDS	89%	27
BOMZ	88%	7
SDB	81%	21
RSC	97%	12
RSF	97%	2
DSTRAT	78%	340
RATL	92%	3
Averages	89%	59

13.3.6 Flotation Kinetics Testing

From the drill core samples, 13 suitable lithologies were isolated for flotation kinetics testing. The samples were selected based on their oxide to sulphide ratios; the samples had ratios ranging from 8:92 to 84:16 with total copper head grades ranging from 0.69% TCu to 11.13% TCu. The purpose of the flotation kinetics tests

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was to determine an achievable sulphide recovery whilst maintaining a rougher concentrate grade of more than 8% TCu. Figure 26 shows the sulphide recovery versus time trend lines.

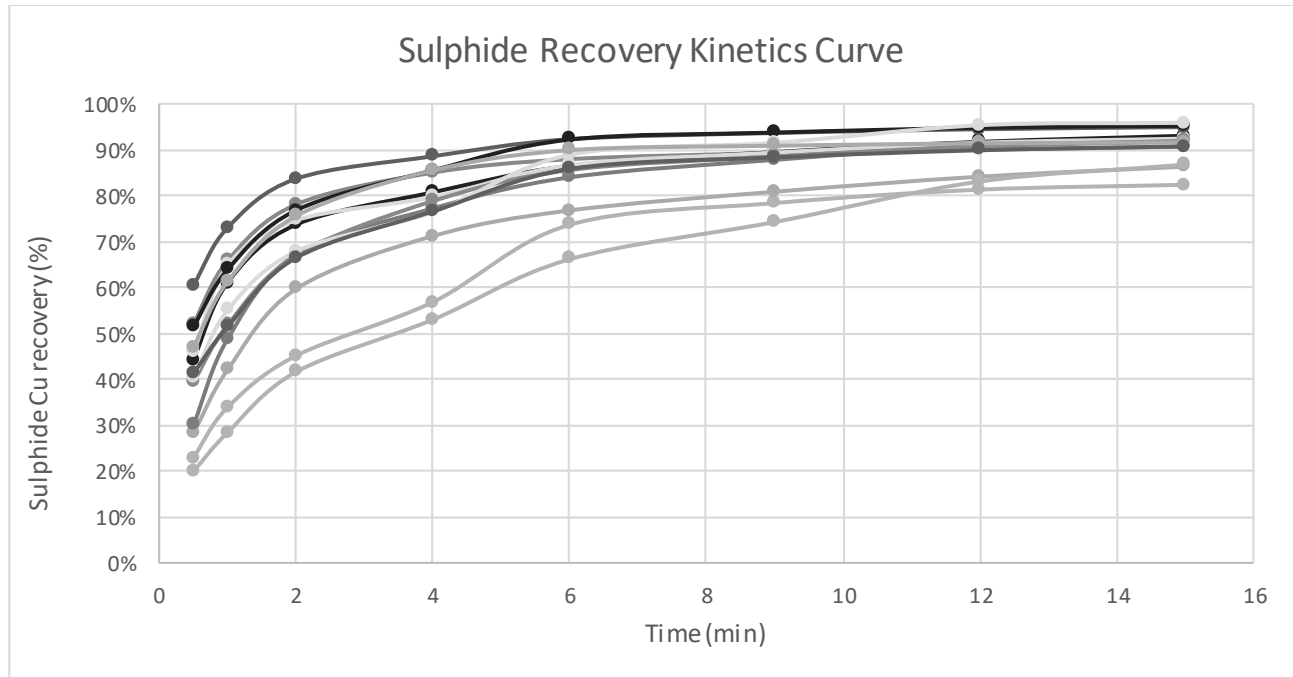


Figure 26: Sulphide flotation kinetics curve

From the kinetics trend lines it can be seen that a recovery of over 80% for copper sulphides has been achieved from all the tested samples after 12 minutes of flotation. When flotation time is increased excessively, the concentrate grade drops as a result of the gangue minerals floating along with the copper containing minerals. Figure 27 shows the phenomenon of decreasing concentrate copper grade as a function of sulphide copper recovery.

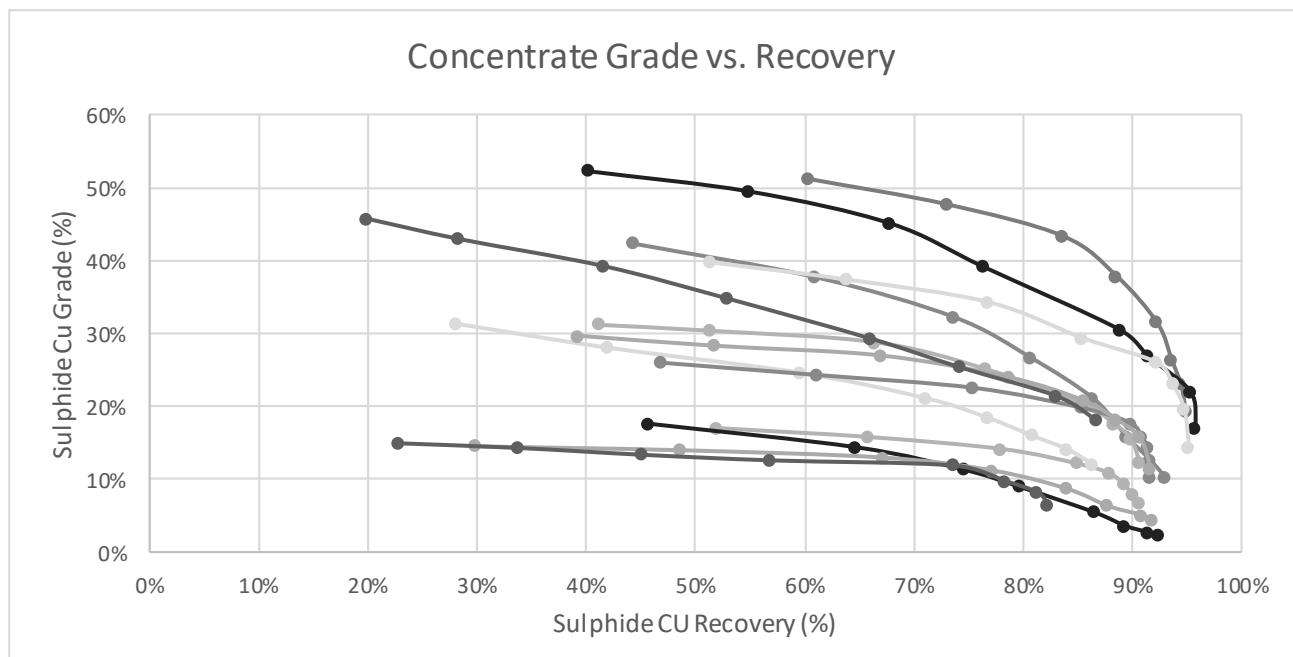


Figure 27: Concentrate Cu grade vs. recovery curve

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The design of the sulphide roasters demands a copper concentrate grade of 35% TCu. For the KTC concentrator to supply this to the roaster, a rougher grade of 8% TCu is necessary. From the test results, at a rougher grade of 8% TCu a 76.14% recovery of copper sulphides can be expected.

13.3.7 Conclusion

The testwork has confirmed the following process design parameters:

- Leach efficiencies;
- Leach reagents consumptions;
- Leach densities;
- Leach residence times;
- Precipitation efficiencies;
- Precipitation reagents consumptions;
- Precipitation densities;
- Precipitation residence times;
- Pre-leach thickener and post-leach thickener settling rates (unit area);
- Pre-leach and post-leach reagent consumptions;
- Achievable pre-leach thickener and post-leach thickener underflow densities;
- Pre-leach slurry and post-leach slurry filtration rates;
- Achievable pre-leach and post-leach filter cake moisture content;
- Pre-leach and post-leach slurry flow and pumping characteristics;
- SX isotherms for plant sizing; and
- SX reagent consumptions.

13.4 Ore Sampling and Analyses for Uranium

Although uranium (U) occurs naturally in KCC's feed ore, previous test work conducted by other experts made no mention of this element. Moreover, historically KCC produced a cobalt metal, and as uranium is not plated on the cobalt cathode it was always disposed of via the spent electrolyte and eventually through to the tailings storage facility (TSF). On November 6th 2018 KML announced the suspension of cobalt sales as a result of the uranium content in its cobalt hydroxide. The level exceeded the acceptable limit allowed for export of the product through main African ports to customers. An investigation was carried out to determine the average uranium values present in ore produced from different mines, the uranium values found from KCC's mines and KITD are currently estimated to be as follows:

- KOV open pit – 22.2 g/Tonne;
- Mashamba open pit – 8.5 g/Tonne;
- Kamoto underground – 53.6 g/Tonne; and
- KITD – 17.3 g/Tonne.

These values have been obtained within 825 blasthole samples inside of confirmed ore zones for KOV and Mashamba, 84 samples within KTO UG from diamond drill holes, and 38 samples within KITD diamond drill holes. Sampling for uranium began in late 2018 and is ongoing.

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14.0 Mineral Resource Estimates

14.1 Introduction

For this TR, Golder has the overall responsibility to sign-off on the Mineral Resources for the KOV OP, KTO UG, KITD, Mashamba East OP and Mashamba East UG, relying on other QPs as outlined in Table 29 for Tilwezembe OP, Kananga OP and T17 UG. As part of that process Golder has reviewed the methods and results adopted in the generation of the Mineral Resource models from Amec Foster Wheeler and KCC staff.

Table 29: Summary of Mineral Resources estimates completed by independent consultants and KCC

Name of Operation	Name of Company	Resource Effective Date	Responsible Person
KITD	KCC	December 31, 2018	Christiano Santos (Golder)
KOV OP / KTE	KCC	December 31, 2018	Christiano Santos (Golder)
KTO UG	KCC	December 31, 2018	Christiano Santos (Golder)
Mashamba East OP / Mashamba East UG	Amec Foster Wheeler	July 31, 2016	Christiano Santos (Golder)
KTO UG	MSA	December 31, 2015	Jeremy Witley
KOV OP / KTE	MSA	December 31, 2015	Jeremy Witley
Mashamba East OP / Mashamba East UG	Amec Foster Wheeler	December 31, 2015	Douglas Reid
KTO UG	SRK	April 30, 2015	Victor Simposya
KOV OP / KTE	SRK	April 30, 2015	Victor Simposya
T17 OP and UG	KCC	December 31, 2011	Willem van der Schyff (Golder)
T17 OP and UG	CCIC	December 31, 2008	Victor Simposya
Mashamba East	CCIC	December 31, 2008	Victor Simposya
Kananga Mine	Snowden	December 31, 2008	Victor Simposya
Tilwezembe OP	Snowden	December 31, 2008	Victor Simposya
KOV OP / KTE	SRK	December 31, 2008	Victor Simposya

The major recommendations from Golder's review of the modelling methods utilized by Amec Foster Wheeler and KCC are summarized below:

- Review the QA/QC for the drill hole sample data;
- Segment the drill hole data by each of the structural fragment areas separated by faulting and waste intrusions;
- Review the lithological intersections within each of the structural fragment areas based on the historical and new drill holes and geological mapping;
- Create a three dimensional block model with block sizes determined by the smallest mining unit for each zone and the density of drill data available;
- Select the sample data by lithological units and within each of the fragments;
- Undertake statistical and geostatistical analyses of the sample data within the defined envelopes of mineralization and derive variogram parameters;

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- Estimate grades into the zones of mineralization using Ordinary Kriging (OK) and Inverse Distance Weighting (IDW) techniques with attendant geostatistical parameters, search neighbourhood and input composite data derived from the respective lithological data sets selected;
- Classify the Mineral Resources into the various categories in compliance with CIM 2014 Definition Standards;
- Estimate grouped lithological units (OBI, RSC and OBS) into the model using the composite data; and
- Tabulate Mineral Resources based on a selected %TCu and %TCo cut-off grades where applicable.

14.2 KOV OP and KTE

The KOV OP block model and Mineral Resource estimate was completed by KCC in 2016 and includes KOV OP and KTE. The following information was summarized from the KCC internal technical report (KCC KOV, 2016). Mineral Resources are depleted as at 31 December 2018 and are effective at this date.

14.2.1 Drill Hole Data

The KOV OP Mineral Resource estimate is based on the drill hole database up to and including the 1st of August 2016. Drill holes completed since this date will be included in next resource update expected to be completed during 2020.

The validated KOV OP drill hole database contains a total of 421 holes of which 80 intersect the Virgule fragment, 173 intersect Oliveira, 61 intersect KTE, 19 intersect Variante and 88 intersect FNSR.

The average drill spacing is approximately 100 m for historical GCM drill holes and 50 m to 75 m for the KCC infill drill holes. Figure 28 shows the location of KOV OP drill holes and Figure 29 shows the cross-section AB. All of the five ore fragments with lithology codes and associated drill holes can be seen on Figure 29.

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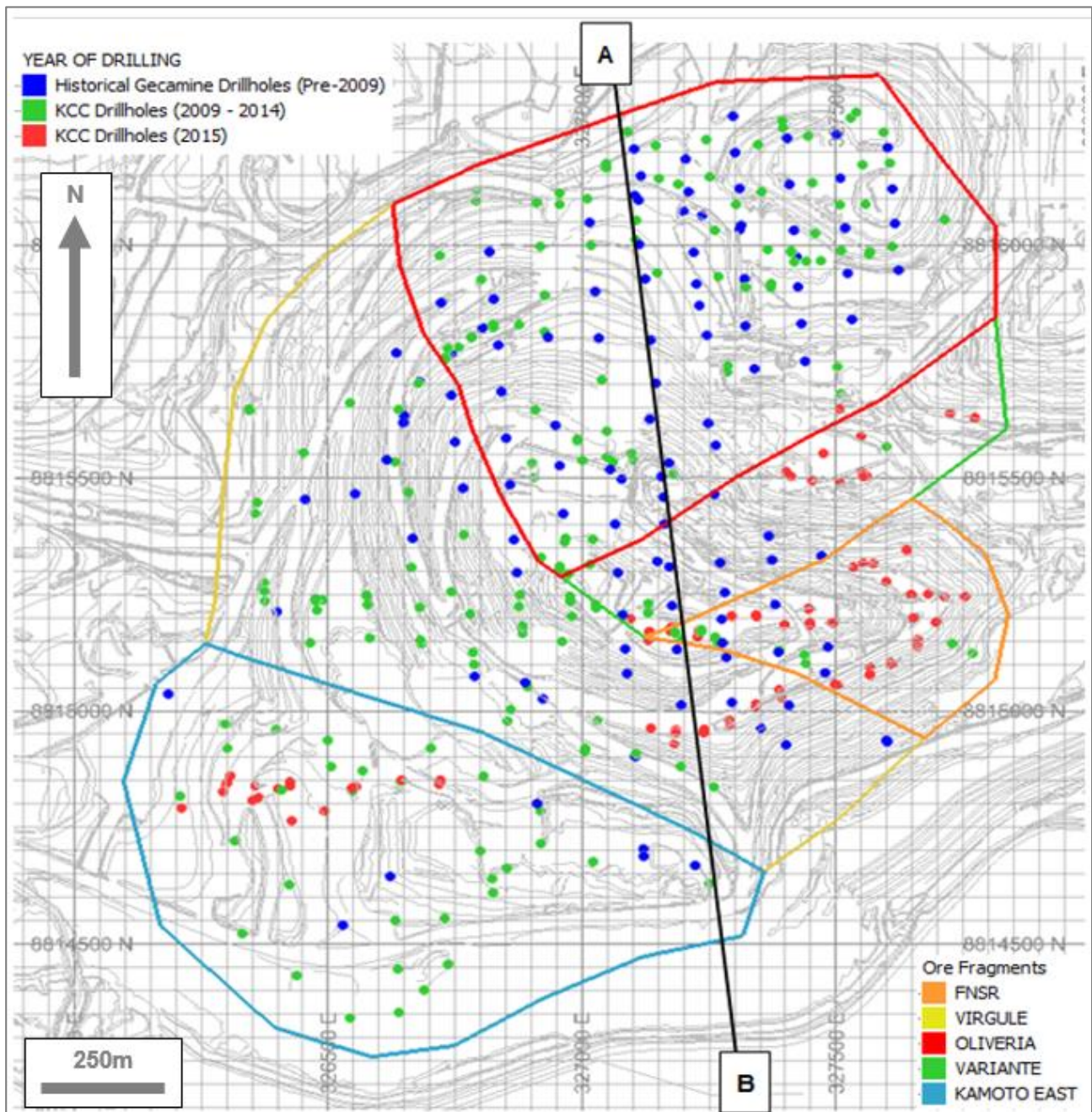


Figure 28: KOV OP drill hole locations, outline of mineralized zones and section line

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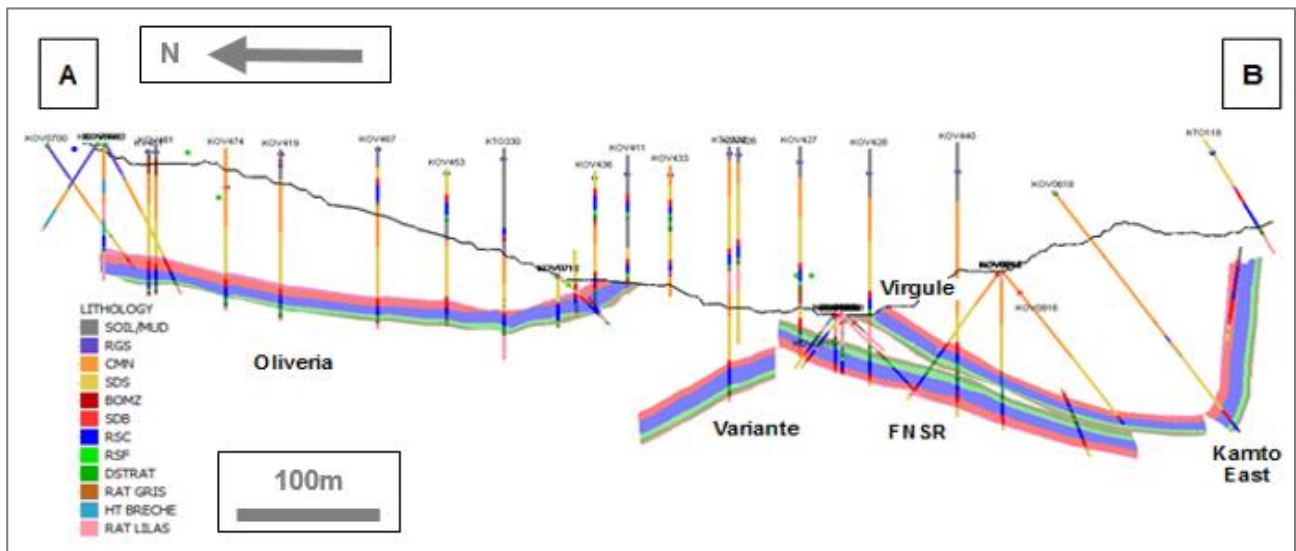


Figure 29: Cross-section AB as shown on Figure 28

The KCC drill hole data was exported from the GDMS Fusion database into a CSV (comma separated values) and imported into Datamine™ RM (version v.1.1.20.0). The drill hole file was reviewed in plan and sectional views to validate the accuracy of the drill hole collar locations, downhole survey data, lithology mistypes and assay results. The drill hole database was determined to be of suitable quality for Mineral Resource estimation purposes by the QP.

14.2.2 Geological Interpretation

The lithological models were updated in each of the five KOV OP fragments based on the combined database of the historical GCM and the recent KCC (2009-2015) data. Every structural fragment is composed of six different lithological mineralized strata and was modelled individually for domaining purposes as shown in Figure 30.

A thorough drill hole validation exercise was performed against the geological block model and as a whole the new drill hole lithological intersections are consistent with the existing historical data. In exceptions where the historical and recent data do not match, the historical data was removed from the validated drill hole database.

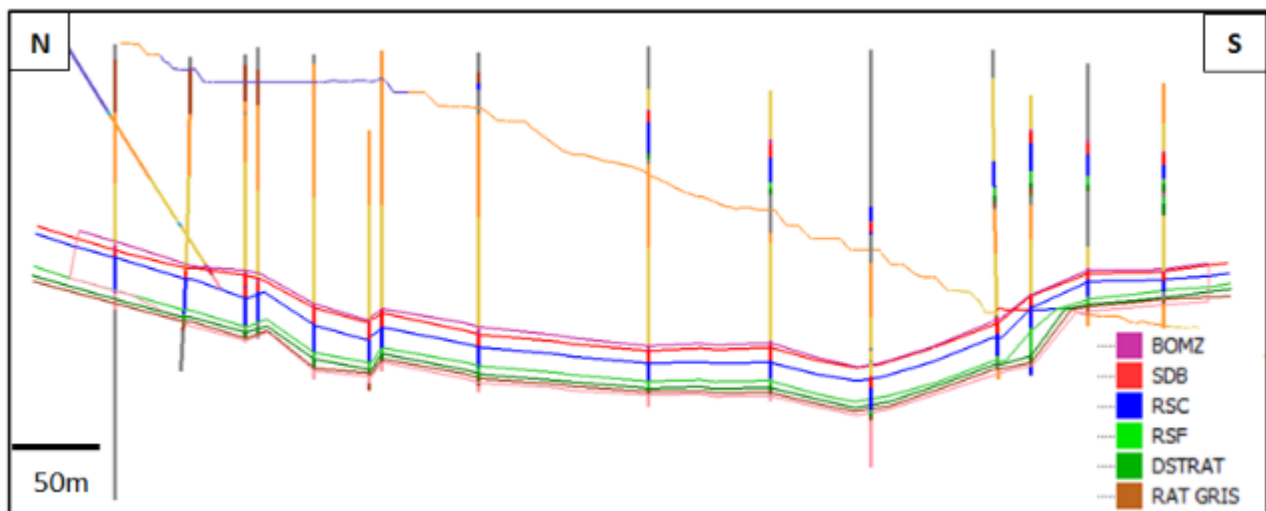


Figure 30: Interpretational vertical section, showing drill hole lithologies and line interpretation) to illustrate the modelling approach

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14.2.3 Statistics and Compositing

Exploratory data analysis was used to summarize and understand the data within each geological domain.

The validated drill holes were examined for differences in sample length. In the historical drill holes, there was a wide variation in sample length, considered to be caused by only sampling areas of visible mineralization. Any samples that were greater than 10 m in length were removed from the estimation. In the recent (2009-2015) KCC drill holes, 82% of samples occurred at a 1 m sample interval.

Assay data within each lithology was extracted from the database using the lithological wireframes. The respective lithological sample data was then composited at 5 m intervals to reflect the selective minimum mining unit and minimise grade variability.

The %AsCu and %AsCo assays data for the historical drill holes is incomplete due to the selective sampling practices. Due to this, %AsCu data is limited to 87% of the total %TCu data and %AsCo data is limited to 77% of the total %TCo data. KCC has greatly increased the number of %AsCu and %AsCo assay data since acquiring the property and is also undertaking a re-sampling campaign for the historical drill cores.

The upper AAS analytical limit for copper is 20% and 6% for cobalt. Based on these limits, any assays reported to be higher were capped at the respective upper limits. Similarly, any sample found to be below 0.01% TCu or TCo, the established lower detection limit, was set to ½ the detection limit (0.005%). Unsampling intervals were assigned a zero value (0.000%).

It was decided that within each fragment, the lithologies would be grouped together into ore zones of a similar grade and geological characteristic for estimation as per following detail:

- BOMZ and SDB lithologies were grouped to form the OBS ore zone;
- RSC was distinctively different from the other lithologies, therefore it was kept separate as the RSC ore zone; and
- RSF, DSTRAT and RATGR were grouped to form the OBI ore zone.

Original assay data and composited assay data for %TCu, %TCo, %AsCu, and %AsCo were examined with histograms and probability plots. Statistics from the respective ore zone composites within each of the five fragments were generated and reviewed for %TCu and %TCo. The normal and log histograms for original %TCu assays are shown in Figure 31 for the combined KOV fragments. Overall, the composite mean statistics show minimal variation in comparison to the raw sample data except for the RSC lithology. Historically, this lithology was only sampled where visible %TCu mineralization was present and therefore, to reduce the impact of the sampling focus in high-grade intervals, samples above the 25% quartile and below the 75% quartile were used to restrict the search ellipse for grade estimation within the RSC domain.

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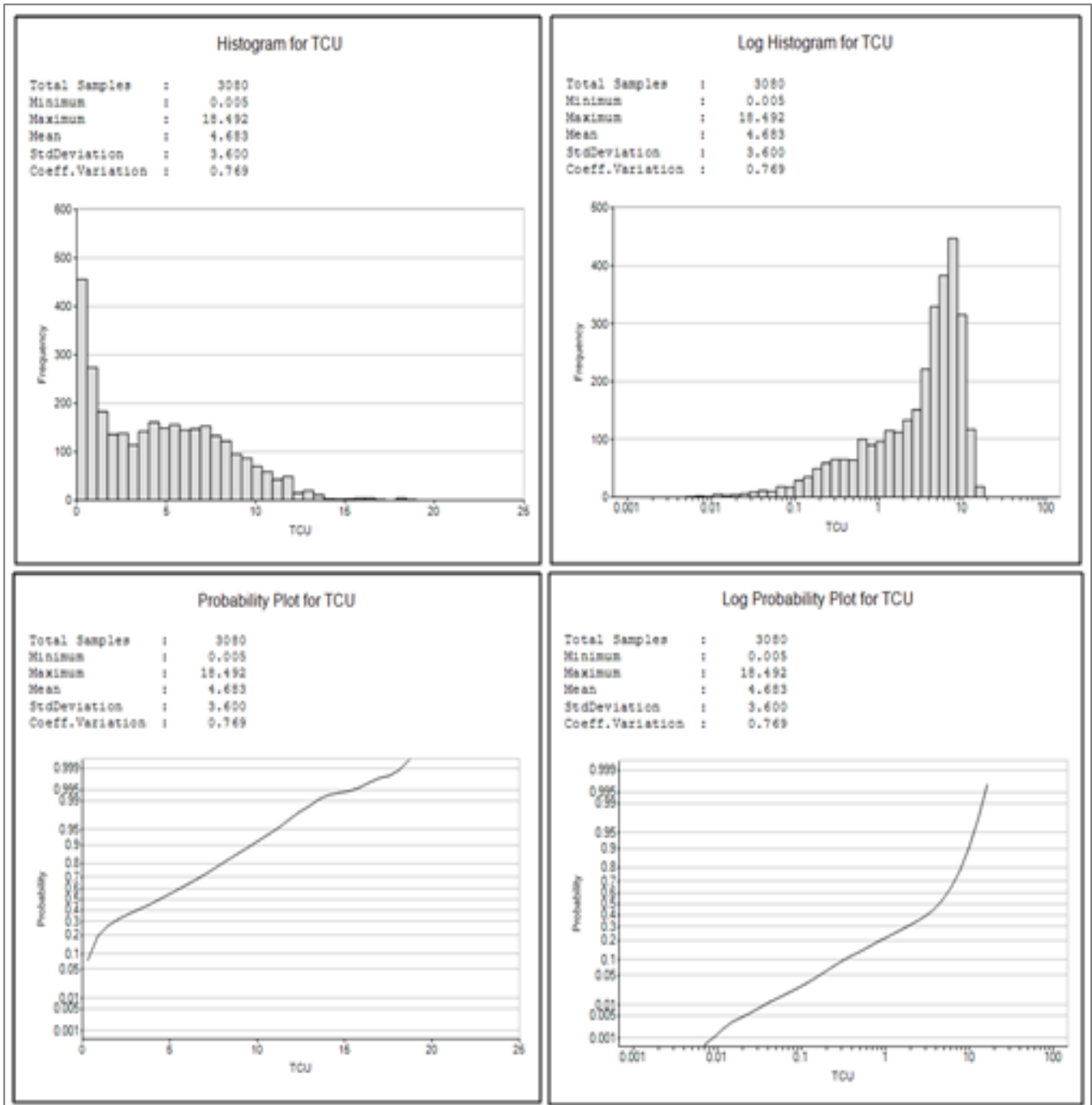


Figure 31: Normal and log histograms for original assays %TCu, all KOV fragments

14.2.4 Dry Bulk Density

Since 2012, bulk density measurements have been taken on all drill core samples using the Archimedes method. This method consists of first weighing the oven-dried sample in air, then sealing the sample in plastic cling film, immersing it in water and weighing it again; the water displacement is equal to the volume. Samples were collected ranging from deeply weathered to fresh. Due to a lack of historical density measurements, where only 156 out of 421 drill holes contain density values, instead of interpolating density, density was assigned based on average values lithology.

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No outliers were removed from the final average bulk density calculation as the dry bulk density measurements were within the accepted range for the respective lithologies. The dry bulk density assignment is summarized in Table 30.

Table 30: Dry bulk density assignment for KOV OP based on lithologies

Lithology	Assigned Dry Bulk Density	Minimum Dry Bulk Density Value	Maximum Dry Bulk Density Value	Number of Samples with Dry Bulk Density Values
BOMZ	2.05	0.93	4.97	359
SDB	2.38	1.02	4.98	1,564
RSC	2.39	1.05	9.84	2,872
RSF	2.35	1.15	3.31	932
DSTRAT	2.36	1.25	6.33	504
RATGR	2.41	1.13	3.92	407

14.2.5 Grade Variography

Experimental variograms were calculated for each ore zone and each fragments based on 5 m drill hole composites for %TCu, %TCo, %AsCu and %AsCo. Unidirectional downhole variograms were used to define the nugget variances for the grade variables based on a lag distance of 5 m reflecting the nominal (the drill hole composites lengths).

From the experimental variograms, variogram models were created for each fragment in three primary directions (x, y and z) for %TCu, %TCo, %AsCu and %AsCo. The KTE fragment was divided into three sub-fragments for variogram and estimation purposes and later combined for Mineral Resource reporting purposes.

The summary of the variograms parameters for %TCu, %TCo, %AsCu and %AsCo within the various fragments is provided in Table 31.

Table 31: KOV OP variogram parameters by fragment

Fragment	Variable (%)	Nugget (C0)	Rotation Angle			Range 1 (m)			Range 2 (m)		
			Z	X	Y	X	Y	Z	X	Y	Z
OLIVERIA	TCu	0.03	-54	11	4	124	102	10	255	213	20
	ASCu	1.18	236	0	12	112	196	10	299	499	30
	TCo	0.02	-65	10	6	84	52	9	289	239	25
	ASCo	0.01	-75	9	8	33	60	10	104	104	21
VIRGULE	TCu	0.35	-87	20	7	105	98	5	253	248	9
	ASCu	1.23	-87	20	7	110	99	5	297	404	15
	TCo	0.08	-77	21	4	34	38	10	128	159	27
	ASCo	0.04	-77	21	4	27	27	9	108	135	24
FNSR	TCu	2.37	92	43	10	22	69	10	81	156	25
	ASCu	1.06	225	11	20	110	99	5	297	404	15
	TCo	0.08	216	8	22	34	38	10	128	159	27
	ASCo	0.04	225	11	20	27	27	9	108	135	24
VARIANTE	TCu	0.33	-35	7	44	18	20	3	57	99	10
	ASCu	0.24	-1	32	32	40	33	4	113	73	11
	TCo	0.00	-42	0	44	116	25	5	198	69	20

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Fragment	Variable (%)	Nugget (C0)	Rotation Angle			Range 1 (m)			Range 2 (m)		
			Z	X	Z	X	Y	Z	X	Y	Z
	ASCo	0.00	21	41	18	10	47	12	39	92	27
KTE A	TCu	1.67	-10	0	24	68	67	10	218	217	20
	ASCu	1.67	-10	0	24	68	67	10	218	217	20
	TCO	0.02	58	22	9	119	31	13	283	103	19
	ASCo	0.00	0	-90	242	5	4	1	15	14	1
KTE B	TCu	0.96	21	0	31	89	86	5	199	201	16
	ASCu	0.96	21	0	31	89	86	5	199	201	16
	TCO	0.01	99	30	6	52	65	7	177	92	31
	ASCo	0.01	57	19	25	153	61	6	280	101	24
KTE C	TCu	2.01	16	0	44	99	99	11	202	198	20
	ASCu	2.01	16	0	44	99	99	11	202	198	20
	TCO	0.03	67	37	26	147	124	10	235	160	34
	ASCo	0.03	67	37	26	147	124	10	235	160	34

The variograms are generally of good quality within the plane of mineralization and the resulting models are broadly similar between fragments and ore zones as shown in Figure 32. For the KTE A and Variante fragments there were insufficient assay results to yield acceptable variograms for %AsCu and %AsCo therefore the corresponding variogram for %TCu and %TCO respectively was applied instead. Variogram ranges were modelled between 50 m and 300 m; in all cases longer than the average drill hole spacing in that fragment.

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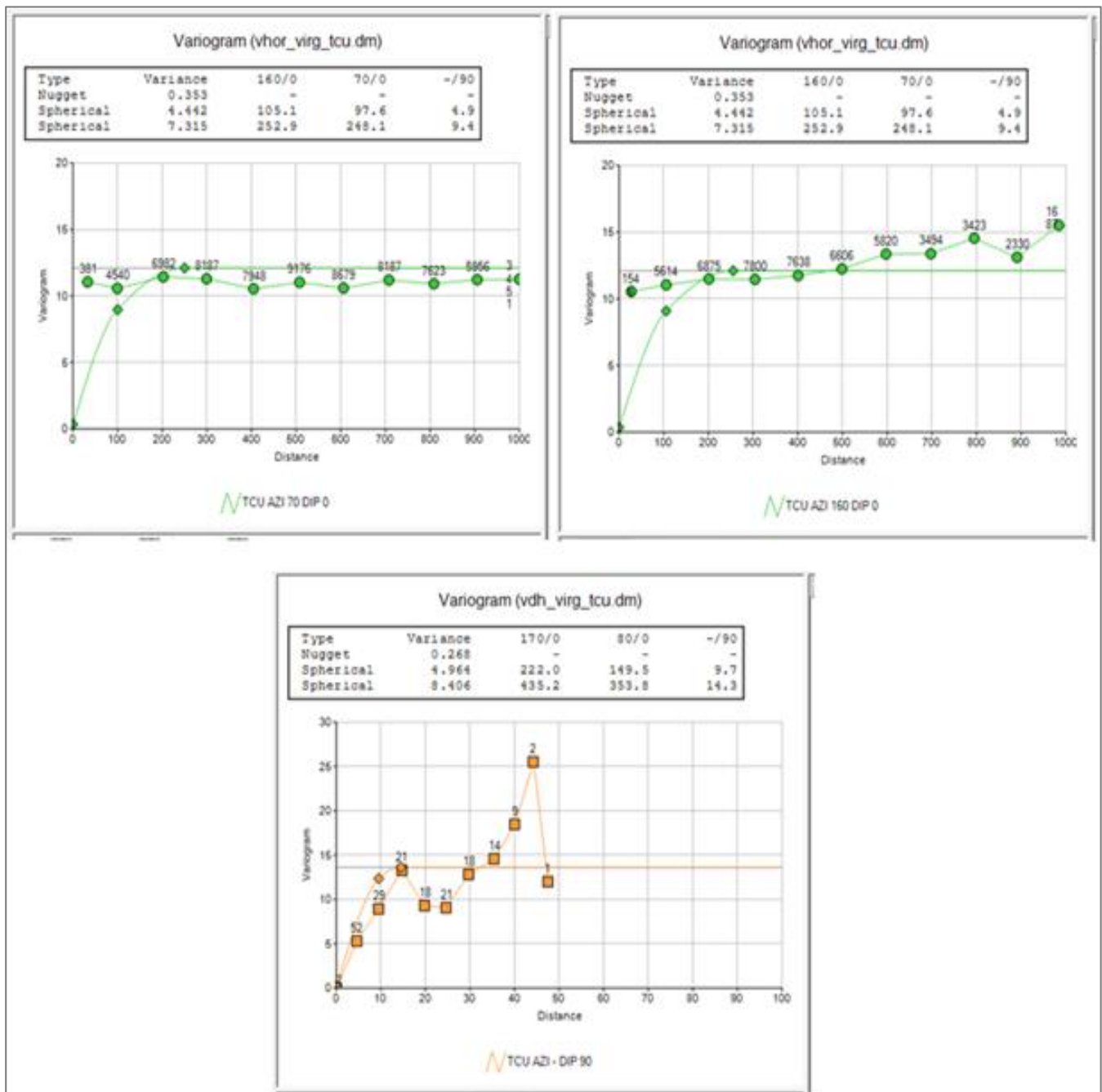


Figure 32: %TCu variogram for Virgule fragment

14.2.6 Block Model Definition

Datamine™ RM Resource modelling software was used in the creation of the KOV OP block model using extended precision format.

The KOV OP proto-type model covers a 3D block in UTM grid co-ordinates. Block shape and size is typically a function of the geometry of the deposit, density of sample data, and expected potential smallest mining unit (SMU). On this basis, a parent block size of 10 m (east-west) by 10 m (north-south) by 10 m (elevation) was chosen, however, parents blocks were divided into sub-blocks on the lithology contacts. Datamine™ uses sub-blocking methodology to improve the resolution at the lithological and mineralisation boundaries. The block model definition parameters are summarized in Table 32.

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Table 32: KOV block model origin and dimensions

Variable	Origin	Parent Block Size (m)	Number of Blocks	Minimum Block Size (m)
X	325,500	10	261	1.25
Y	8,814,000	10	261	1.25
Z	800,800	10	71	0.01

14.2.7 Lithological Modelling Codes

Each of the fragments in the model is identified by a unique code called FRAGNO as indicated in Table 33.

Table 33: Block model fragment codes

Fragment	FRAGNO
FNSR	1
Virgule	2
Oliviera	3
Variante	4
KTE	5

Within each of the fragments, the lithologies are identified by a unique code called ROCKNO as indicated in Table 34.

Table 34: Block model lithological codes within each fragment

Lithology	ROCKNO
BOMZ	5
SDB	6
RSC	7
RSF	8
DSTRAT	9
RATGRIS	10

14.2.8 Mineral Resource Estimation

Grade estimation was performed using OK method for the variables of %TCu, %TCu, %AsCu and %AsCo. IDW and Nearest Neighbour (NN) estimation methods were also used, however these results were only used for validation purposes and do not appear in the final model. The OK estimation method used the directional variogram parameters, search distance parameters and the composited data for each ore zone. It was decided to estimate grades based on ore zone (OBS, RSC and OBI) instead of separate lithologies due to narrow width of the several individual lithologies and insufficient samples for creating the individual lithology variograms.

A search ellipse controlled the search distance of estimation along the azimuth, plunge and dip of the mineralized zone. A Datamine™ macro was developed to allow the search ellipsoid to be aligned to the trend of the wireframe at any one location. The size of the ellipse was calculated from the variogram parameters, with three different sizes of ellipse being used in estimation, equating to three estimation passes. The first pass used a search distance equal to 2/3 of the range of the variogram and the minimum and maximum number of composite samples to estimate a block was set at 3 and 8 respectively. In the second pass, the search distance was expanded by a factor of 1.5 of the first pass to equal the variogram range, and the minimum and maximum number of composite samples to estimate a block remained at 3 and 8. Blocks that remained un-estimated

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after the second pass were now estimated within a search neighbourhood of 10 times the original search and the minimum and maximum number of samples expanded to 1 and 20 respectively. Due to a drill hole spacing of over 200 m in the western portion of the Variante fragment, several blocks were unpopulated with grades, therefore the third search distance was expanded to 50 times the range of the variogram during the estimation of Variante. On completion of the three passes, grades had been estimated into all blocks. Further details of these search parameters are provided in Table 35. Where a second search parameter is given (i.e. 170/120), this refers to a restriction applied in order to limit the influence of very high-grade historical values.

Table 35: Estimation passes, search volume and samples used for %TCu

Fragment	Ore Zone	Search Distance (m)	Search Volume 1			Search Volume 2			Search Volume 3		
			Factor	Min	Max	Factor	Min	Max	Factor	Min	Max
OLIVERIA	OBS	170	0.66	3	8	1.5	3	8	50	2	10
	RSC	170/120	0.66	3	8	1.5	3	8	50	2	10
	OBI	170	0.66	3	8	1.5	3	8	50	2	10
VIRGULE	OBS	169	0.66	3	8	1.5	3	8	50	2	10
	RSC	169/126	0.66	3	8	1.5	3	8	50	2	10
	OBI	169	0.66	3	8	1.5	3	8	50	2	10
FNSR	OBS	81	0.66	3	8	1.5	3	8	50	2	10
	RSC	82/60	0.66	3	8	1.5	3	8	50	2	10
	OBI	81	0.66	3	8	1.5	3	8	50	2	10
VARIANTE	OBS	38	0.66	3	8	1.5	3	8	50	2	10
	RSC	38/29	0.66	3	8	1.5	3	8	50	2	10
	OBI	38	0.66	3	8	1.5	3	8	50	2	10
KTE A	OBS	145	0.66	3	8	1.5	3	8	50	2	10
	RSC	145/109	0.66	3	8	1.5	3	8	50	2	10
	OBI	145	0.66	3	8	1.5	3	8	50	2	10
KTE B	OBS	132	0.66	3	8	1.5	3	8	50	2	10
	RSC	132	0.66	3	8	1.5	3	8	50	2	10
	OBI	132/99	0.66	3	8	1.5	3	8	50	2	10
KTE C	OBS	135	0.66	3	8	1.5	3	8	50	2	10
	RSC	135/101	0.66	3	8	1.5	3	8	50	2	10
	OBI	135	0.66	3	8	1.5	3	8	50	2	10

14.2.9 Block Model Validation

Statistical and visual validation of the block model was undertaken to assess the kriging performance and conformance to the input data. A series of checks were performed including:

- Geological model volume validation;
- Visual assessment of estimated grades versus composite grades as shown in the example in Figure 33; and
- Swath plot validations comparing block grades against composite grades.

The validity of the geological model was examined by comparing the block model volumes with the interpreted wireframe volumes. The differences between the wireframe and block model volumes are minimal at 0.05%

The block model was examined visually in sections to ensure that the composited drill hole grades were locally honoured by the model. This was carried out for %TCu, %TCu, %AsCu and %AsCo every 50 m along north-south and west-east section. Visually the model validated well against the drill hole data.

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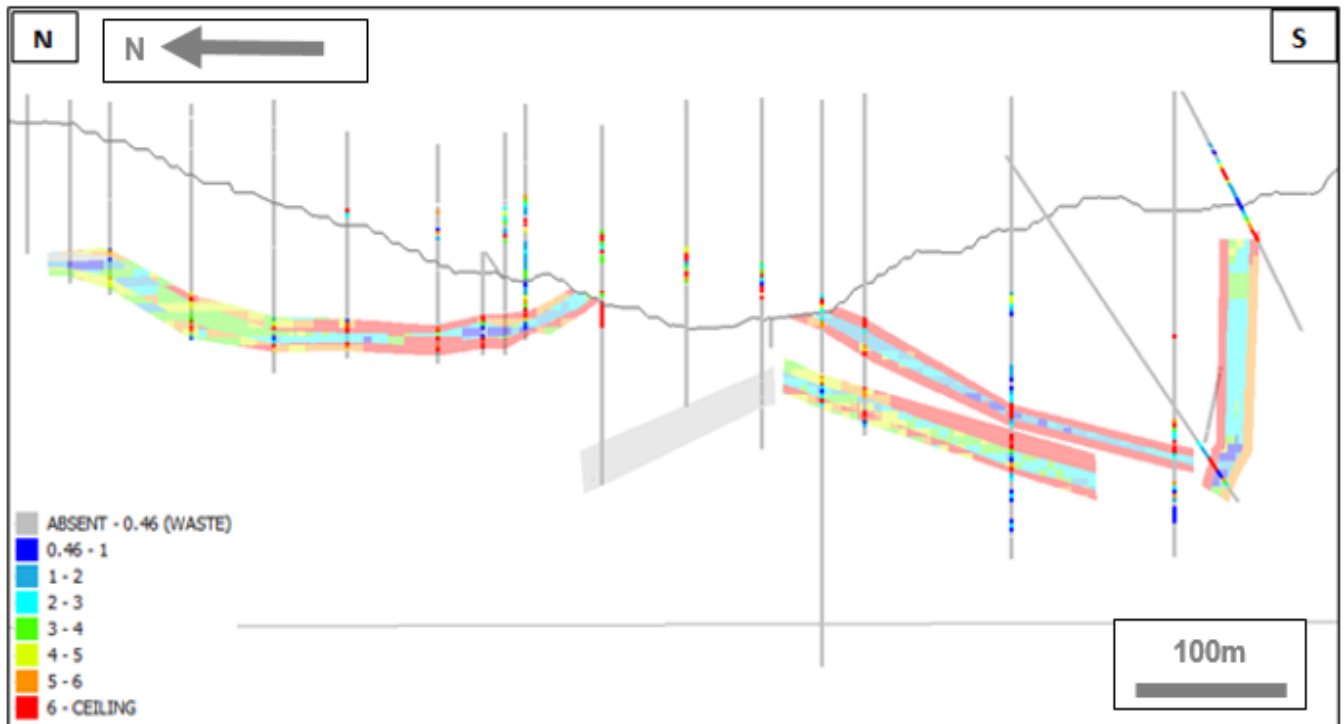


Figure 33: Visual validation of composited drill holes vs final model for %TCu

The mean grades of the drill hole composites and the block model were compared for %TCu and %TCo grades statistically and using swath plots. This validation was carried out for each fragment by ore zone, as per the estimation process and the following observations are made:

The comparable estimated %TCu grades for the OBS and OBI were within acceptable limits compared to the composite data for the majority of fragments (FNSR, Virgule, Oliviera, and KTE B) where there was a large, non-clustered sample data set, as shown in Figure 34.

Minor differences are seen in the OBS and OBI for KTE A and Variante, where there are a lower number of composites, however a large percentage of these fragments have already been mined out.

In the RSC for all fragments, there are minor differences between the mean composite grades and mean estimation grades, which is function of the estimation method used to eliminate sample bias. In the GCM historical drill holes, samples were only taken where visible copper mineralization was present. In the RSC unit this often meant only along the higher-grade contact zones with the RSC-OBS and RSC-OBI, and the lower grade middle RSC unit was not sampled. To reduce the impact of the sampling focused only in high-grade intervals, only samples above the 25% quartile and below the 75% quartile were used for grade estimation within the RSC domain.

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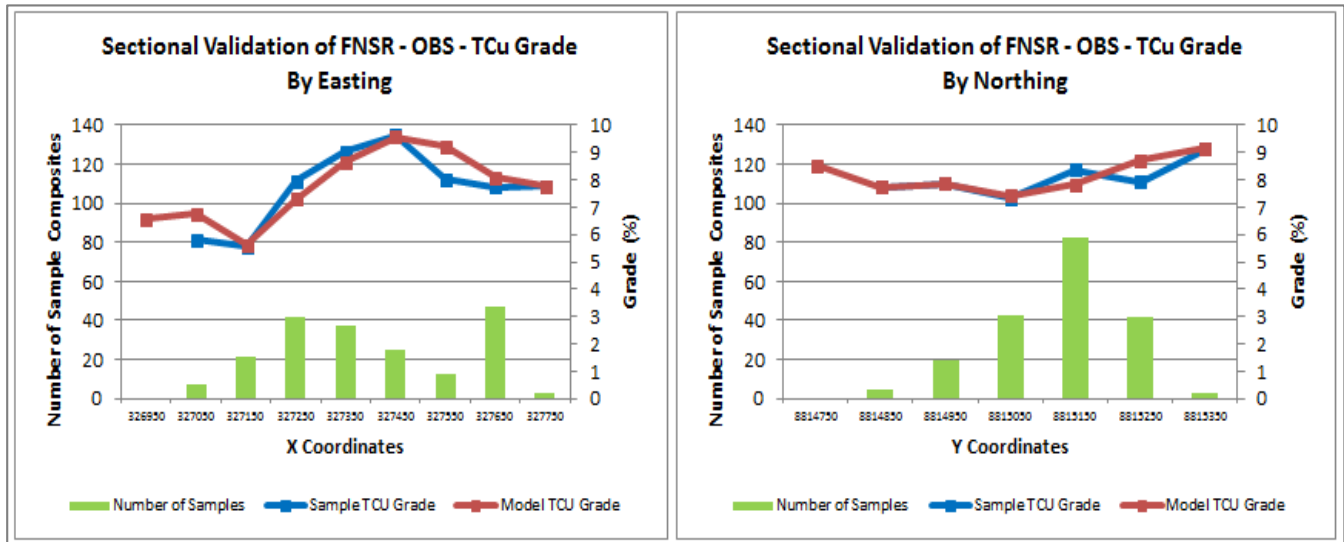


Figure 34: %TCu swath plot example for the OBS ore zone in FNSR fragment

The results of the above validation indicate that the model appears to be representative of the data and that no material bias was identified.

14.2.10 Mineral Resource Classification

Mineral Resources were classified according to CIM 2014 Definition Standards and based on drill hole spacing.

KCC classified the Mineral Resources for KOV OP into Indicated and Inferred Mineral Resources. KCC has reviewed the classification criteria and made the following observations:

- There has been a long mining history of the KOV OP deposit and therefore the general behaviour of the mineralization and grade continuity is reasonably well understood;
- Recent mapping (2016), RC and blast hole drilling have all confirmed the near-surface continuation of mineralization in the fragments;
- The variograms indicate strong lateral grade continuity over several hundred metres (well in excess of the drill hole spacing), indicating a low variance in the %TCu and %TCo grades;
- The drilling can be broadly grouped into two main phases; a historical GCM drilling phase (pre-2009) and a recent KCC drilling phase (2009-2015). Concerns have arisen with the historical drill hole database regarding the limited number of RSC samples if no visible copper mineralization was present and the absence of QA/QC; and
- There are more drill holes intersecting Oliveira (173 out of 421 drill holes) compared to other fragments. KTE (61), Virgule (80) and FNSR (88) fragments were all targets of recent drilling campaigns to increase confidence in the mineralization and grade continuity. Variante has the highest level of uncertainty surrounding it with 19 drill holes as reflected in the classification.

KCC has re-classified the KOV OP deposit based on the following classification method:

- Indicated Mineral Resources were classified in regions where the Mineral Resource was covered by the KCC recent drilling (2009-2015), with a drill hole spacing of less than 100 m and was estimated by the first pass (2/3 the variogram range) search ellipse; and
- Inferred Mineral Resources were classified in regions where the Mineral Resource was covered dominantly by the historical drill holes (pre 2009) lacking QA/QC, with a drill hole spacing of less than 200 m and within the second pass (1.5) variogram range.

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14.2.11 Mineral Resources Tabulation

The Mineral Resources for KOV OP and KTE are tabulated in Table 36. Resources are estimated as at December 31, 2018 and are effective at this date. Mining activities continued during 2019 and KCC experienced ordinary course depletion. Updated Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update, which will reflect such normal course depletion.

Table 36: KOV OP and KTE Mineral Resource tabulation as at December 31, 2018¹⁻⁹

Fragment	Classification	Mt	%TCu	%TCo
OLIVERIA	Measured	-	-	-
	Indicated	54.2	3.96	0.61
	Measured + Indicated	54.2	3.96	0.61
	Inferred	14.4	3.71	0.43
VIRGULE	Measured	-	-	-
	Indicated	29.9	5.62	0.45
	Measured + Indicated	29.9	5.62	0.45
	Inferred	31.7	4.22	0.33
FNSR	Measured	-	-	-
	Indicated	18.2	5.62	0.48
	Measured + Indicated	18.2	5.62	0.48
	Inferred	7.7	5.63	0.41
VARIANTE	Measured	-	-	-
	Indicated	-	-	-
	Measured + Indicated	-	-	-
	Inferred	1.7	0.62	0.08
KTE	Measured	-	-	-
	Indicated	4.9	6.07	0.32
	Measured + Indicated	4.9	6.07	0.32
	Inferred	22.2	4.98	0.44
Total KOV OP & KTE	Measured	-	-	-
	Indicated	106.8	4.80	0.53
	Measured + Indicated	106.8	4.80	0.53
	Inferred	77.6	4.40	0.38

- 1) Mineral Resources have been reported in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) Mineral Resources are reported using a cut-off grade of 0.46% TCu or 0.12% TCo;
- 3) Mineral Resources are estimated using an assumed long term price of \$6,500 per tonne of copper and long term price of \$60,000 per tonne of cobalt;
- 4) Grade measurements are reported as percent (%), tonnage measurements are in metric units;
- 5) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 6) Mineral Resources are inclusive of Mineral Reserves;
- 7) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 8) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 9) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

14.3 KTO UG

The KTO UG block model and Mineral Resources estimate was completed by KCC in 2016. The following information was summarized from the KCC internal technical report (KCC KTO, 2016). Mineral Resources are depleted as at 31 December 2018 and are effective at this date.

14.3.1 Drill Hole Data

The KTO UG Mineral Resource estimate is based on the drill hole data up to August 1, 2016. Drill holes completed since this date will be included in the next resource update.

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The validated KTO UG drill hole database contains a total of 624 holes of which 438 intersect the various zones within the Principal fragment, 66 for Etang North, 114 for Etang South and 6 for Pringle. The recent drilling has been in Principal Zones 3 and 4, Pringle, Etang North and Etang South, which are the subject of this Mineral Resources update.

The KTO UG database contains assay fields for %TCu, %AsCu, %TCo and %ASCo. However, the data count is disproportionate in favour of %TCu and %TCo, especially for the historical holes, as shown in Table 37. Historical sampling for acid soluble copper or cobalt was based on the visibility of oxide mineralization in the core. Samples with colored minerals were assayed for either Cu Oxide or Cu Malachite, while samples considered barren for oxides were only analysed for the sulphide content.

Table 37: Total drill hole sample intervals containing assay data (%TCu, %TCo, %AsCu, %ASCo) for KTO UG

Total Drill Hole Sample Intervals Containing Assay Data for KTO UG					
*Historical = <2009					
*Recent = 2009 - 2015					
Drill Hole Series	Assay Variables (%)	Total Assay Count	Drill Hole Series	Assay Variables (%)	Total Assay Count
Historical	TCu	5,006	Recent	TCu	8,457
	TCo	4,966		TCo	8,457
	ASCu	2,098		ASCu	6,641
	ASCo	2,011		ASCo	4,630

Down-hole survey data in the historical drill hole database is very sparse (27 holes contain a survey). Whilst drill holes up to 50 m were deliberately not surveyed, the holes that exceed this length are subject to severe deflection from the planned path.

The historical drill holes were drilled at various angles to various depths from the underground workings as well as vertically from surface. The average drill hole grid spacing is approximately 100 m with further infill drilling spaced from 25 m to 50 m apart as illustrated in Figure 35 and Figure 36. Infill drilling conducted by KCC (2009–2015) is irregularly spaced over the project area with the holes drilled at various angles from the underground workings, typically in a fan formation.

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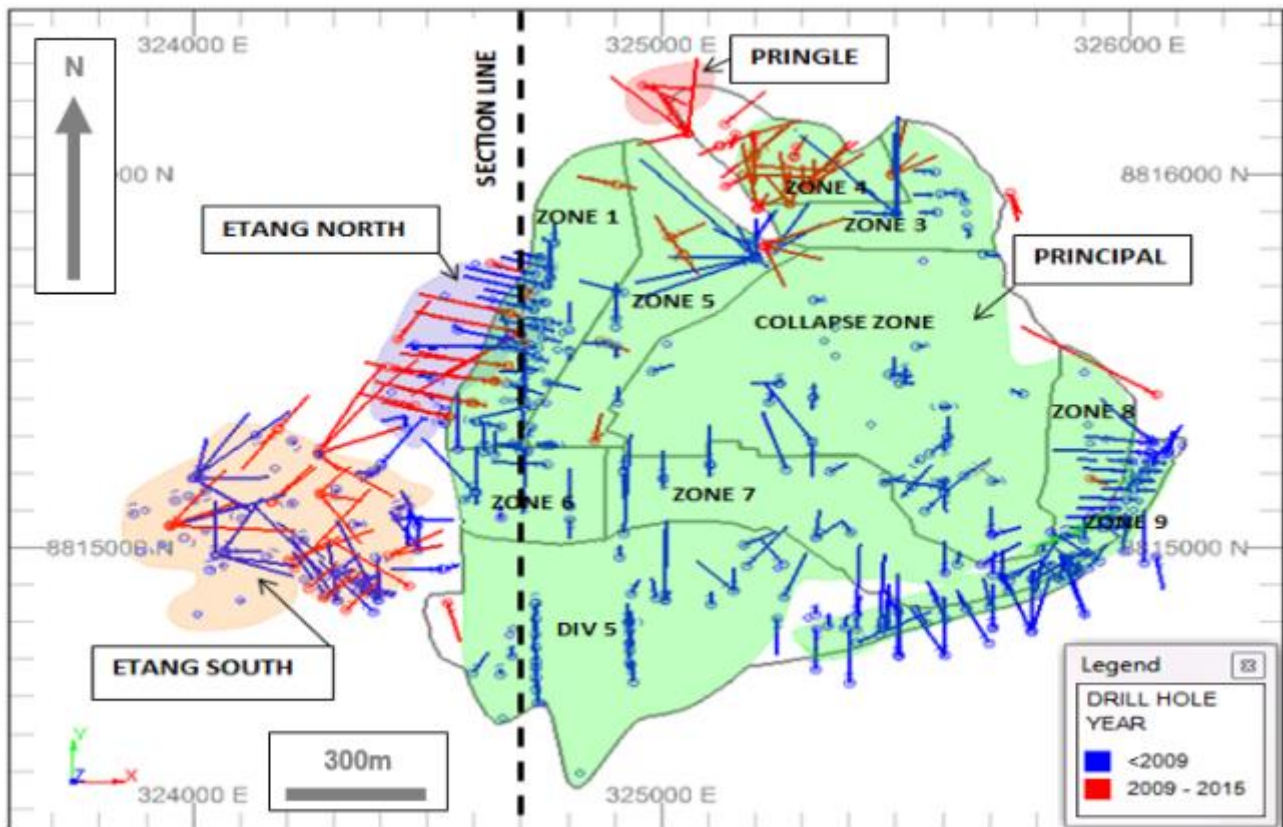
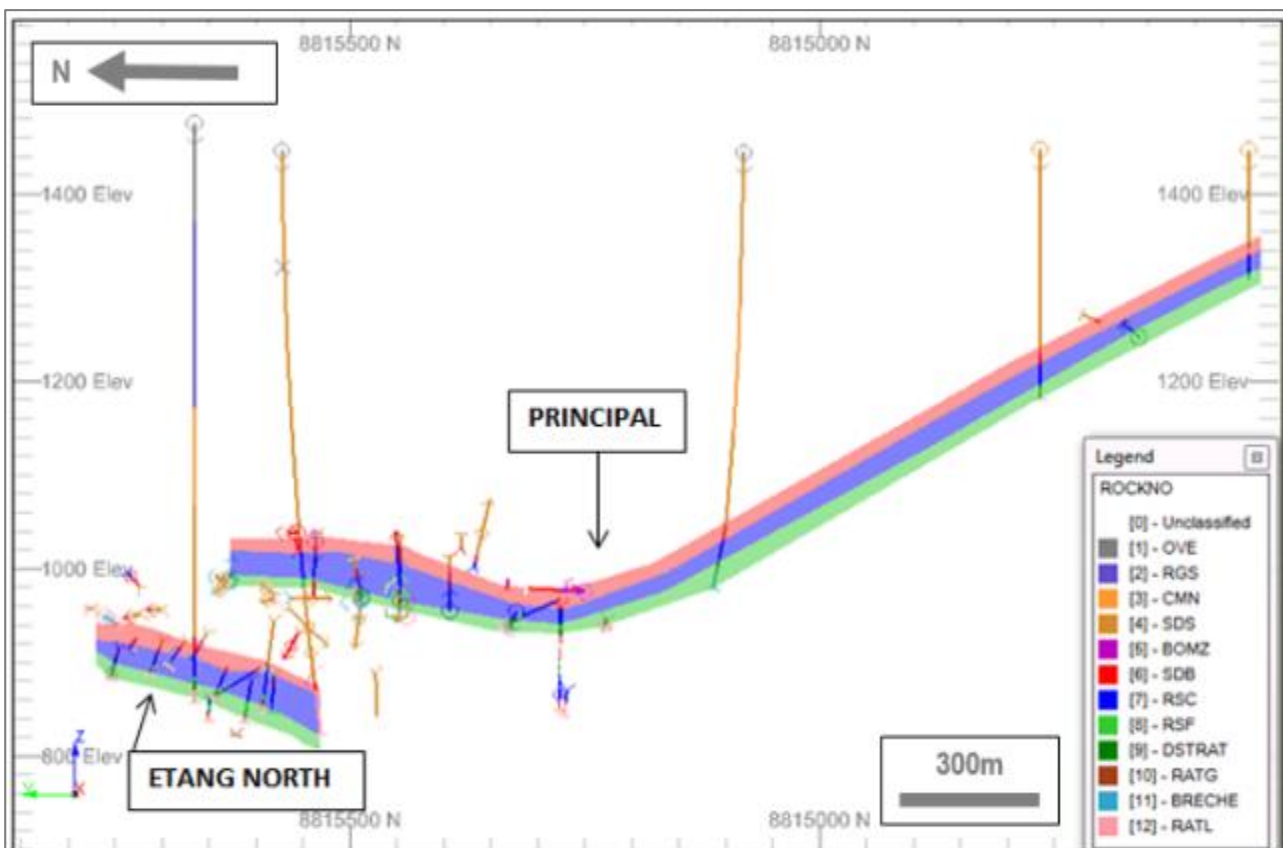


Figure 35: Plan diagram of KTO UG fragments and subdivision of zones with drill hole locations



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Figure 36: North-south section view of Etang North, Zone 1, 6 and DIV5

The KCC drill hole data was exported from GDMS Fusion into electronic CSV (comma separated values) and imported into Datamine™ RM (version v.1.1.20.0) where it was subject to data validation checks including, but not limited to, examination of missing values, determination of data completeness, checks for acid soluble values exceeding the total values and removal of duplicated drill holes.

The drill hole file was reviewed in plan and section to validate the accuracy of the drill hole collar locations from a development end or surface, drill hole orientations, down hole survey, and the assay data were checked for out of range values. The drill hole database was determined to be of suitable quality for Mineral Resources estimation purposes.

14.3.2 Geological Interpretation

The lithological models within each of the KTO UG fragments were updated based on the combined database of the historical pre 2009 (GCM) and 2009-2015 drill hole data (KCC), an example of this is shown in Figure 37. Within each fragment, four surfaces were created to separate the three ore zones (OBS-RSC-OBI) and linked across to create wireframes. These wireframes were constrained using a perimeter limit string, the lateral extents of which were defined by a lack of mineralized layers in the drill hole, geological sections and by a lack of stope and development excavations.

For grade estimation purposes, the different lithological mineralized strata within each structural fragment was grouped together into ore zones. These ore zones were defined as follows:

- OBS ore zone includes BOMZ and SDB lithologies;
- RSC ore zone was grouped individually; and
- OBI ore zone includes RSF, DSTRAT and RATGRIS lithologies.

A thorough drill hole validation exercise was performed against the geological block model and generally the new drill hole lithological intersections are consistent with the existing historical data, except for a few cases. These exceptions were removed from the validated drill hole database.

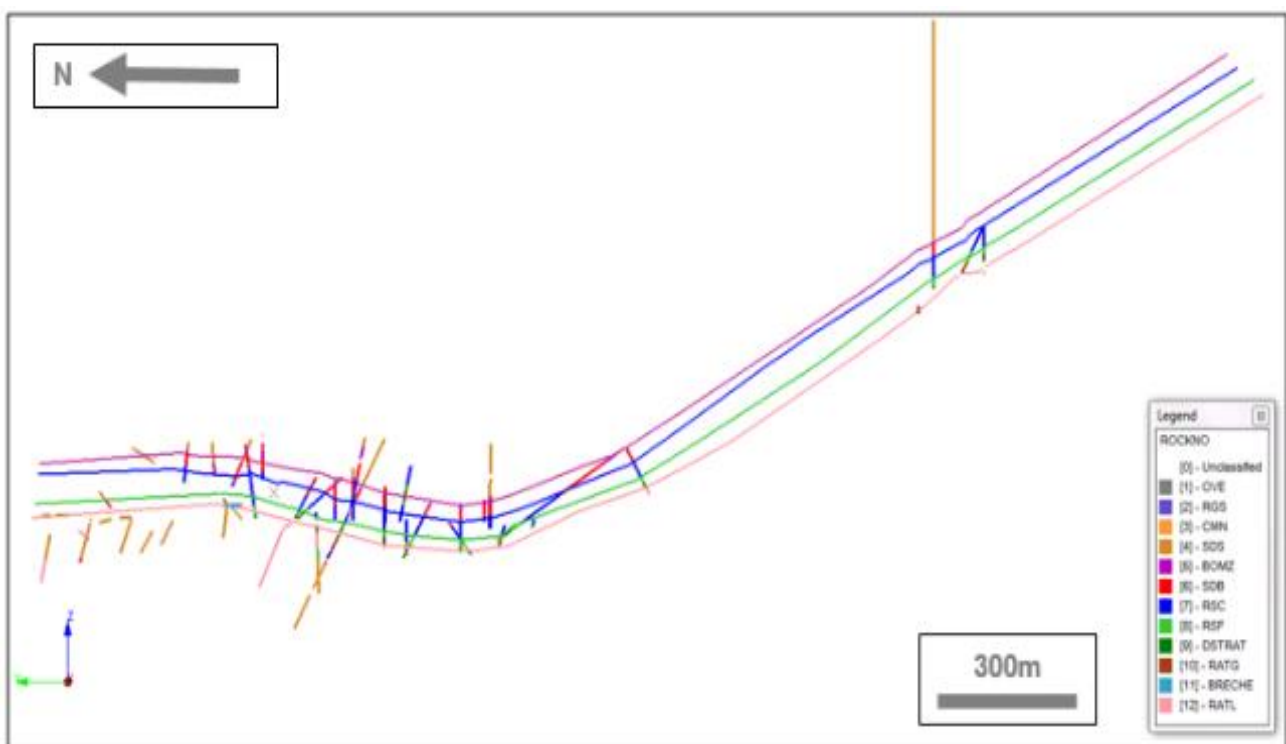


Figure 37: Vertical section interpretations, showing drill hole lithologies and line interpretations

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14.3.3 Statistics and Compositing

Descriptive statistics were used to analyse the grade distribution of each sample population and to determine the presence of outliers and correlations between the grade variables for each domain.

KCC reviewed the statistics of the sample length within each of the mineralized ore zones and within each of the fragment areas being estimated. Sample lengths were found to vary greatly in all fragments, from a minimum of 0.01 m to a maximum of 318 m. In addition, several drill holes were found with sample interval lengths of 1 m, but a further analysis suggests these were original samples of greater length subsequently divided up into equal intervals. However, preliminary examination of the data suggests the problem areas are restricted to assay intervals outside of the ore zones. These will have minimal impact on estimates inside the ore zones. Most of the sample length variations were found to be associated with the historical data.

An analysis was conducted on the drill holes using the Parrish method (1997), a process of using decile analysis to study the metal distribution as it relates to the assay frequency distribution within a sample population. It was determined that there was no compelling reason to cap the drill hole assays however, based on the upper AAS analytical limit, the drill holes were capped at 20% for %TCu and 6% for %TCo. Similarly, for samples found to be below 0.01% %TCu or %TCo, the established lower detection limit, was set to ½ the detection limit (0.005%).

Assay data within each lithology was extracted from the database using the lithological wireframes. The respective lithological sample data was then composited on 5 m intervals to reflect the selective mining unit and reduce grade variability. The minimum acceptable composite length was set at 2 m to provide for the thin nature of the BOMZ stratigraphic unit. Density weighting for compositing was not applied.

Original assay data and composited assay data for %TCu, %TCo, %AsCu, and %AsCo were examined with histograms and probability plots. Statistics from the respective ore zone composites within each of the fragments were generated and reviewed for %TCu and %TCo. The normal and log histograms for original %TCu assays versus composited %TCu assays are shown in Figure 38 and Figure 39 for the Etang South fragment. Overall, the composite mean statistics show minimal variation in comparison to the raw sample data except for the RSC ore zone which historically has only been sampled where visible %TCu mineralization was observed. The estimation method applied for this ore zone is discussed further in Section 14.3.8.

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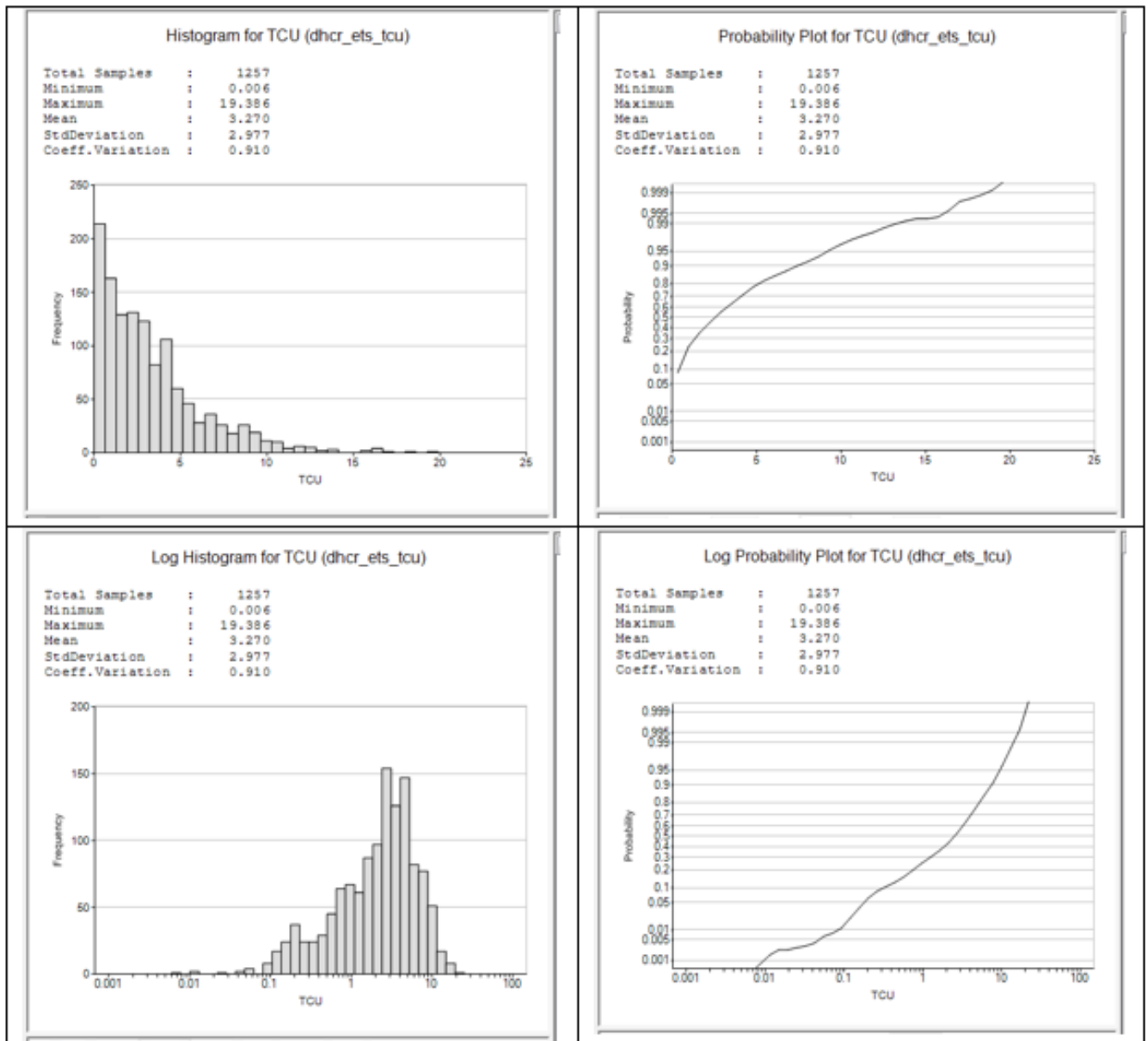


Figure 38: Normal and log based histograms and cumulative probability diagrams for composited %TCu, Etang South fragment

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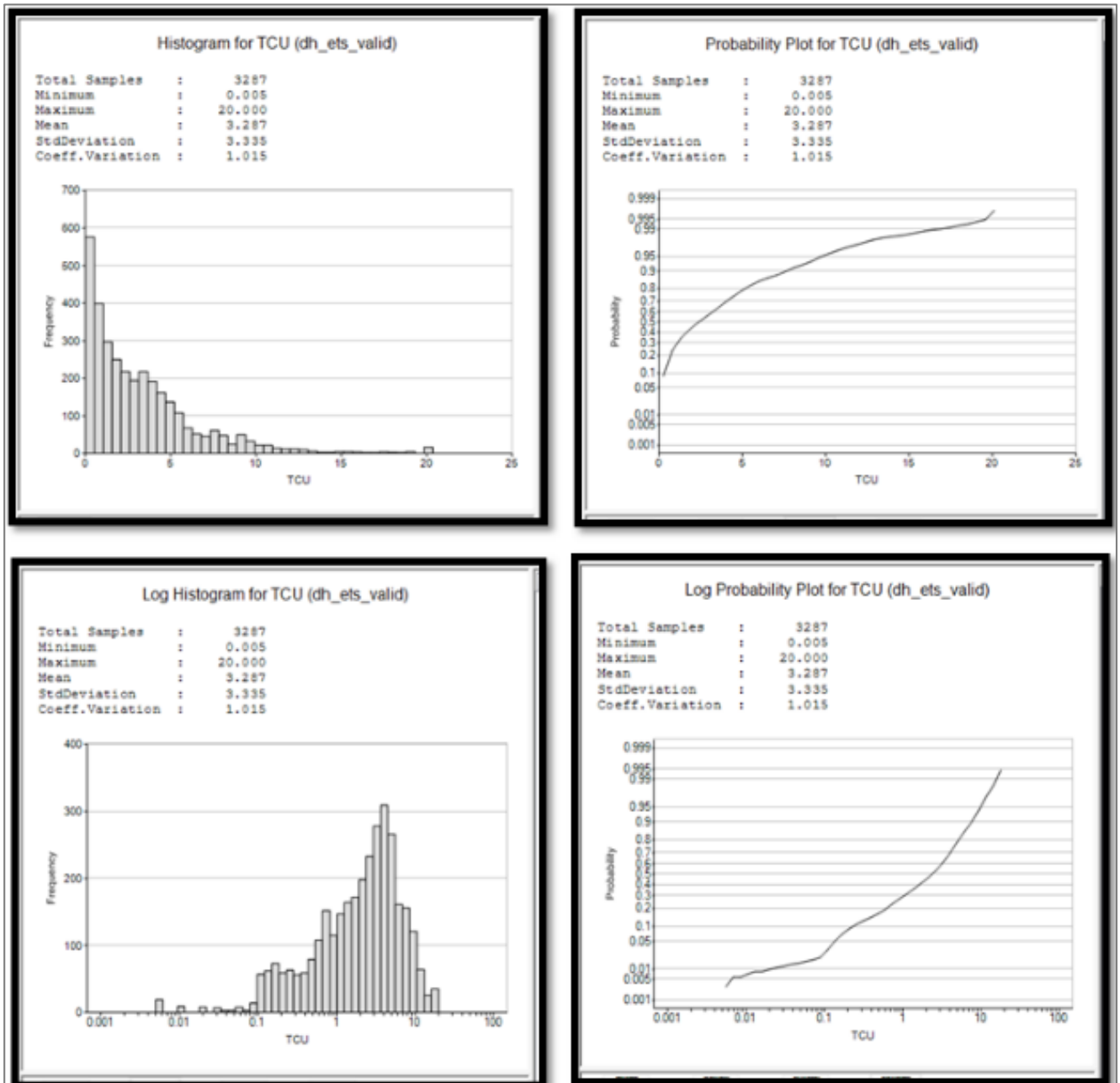


Figure 39: Normal and log based histograms and cumulative probability diagrams for composited %TCu, Etang South fragment

14.3.4 Dry Bulk Density

As there is a limited availability of the density data covering all of the KTO UG fragments, average dry bulk densities were applied to each ore zone throughout the KTO UG deposit as per the summary in Table 38. No outliers were removed from the final average calculation as the dry bulk density measurements were within the accepted range for the respective lithologies. Although the samples were limited, there was minimal variation in the average values of the various lithologies within each ore zone. To supplement the density database for the KTO UG, KCC sent 5,387 half-core samples from drill holes intersecting Zone 1, 4, 5, Etang North, Etang South and Pringle to SGS laboratory in Kalulushi for density determinations from April 2015 to March 2016. From January 2016, a further 808 half-core samples from the KTO UG drill holes had density determinations completed in-house within the KCC Geology Department. Both SGS and KCC dry bulk density

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determinations were undertaken using the Archimedes method. Samples were collected ranging from deeply weathered to fresh.

Current density determinations are consistent with the 2008 values as reported in the 2009 FS and confirm the values in Table 38.

Table 38: Dry bulk density assignment for KTO UG based on ore zones

KTO UG Dry Bulk Density Summary Table			
	OBS	RSC	OBI
Count	414	1128	447
Min	1.8	1.97	1.07
Average	2.77	2.59	2.61
Max	4	3.29	3.1
Std Dev	0.19	0.15	0.24
CV	0.07	0.06	0.09

14.3.5 Grade Variography

KCC reviewed the spatial distribution of the post 2009 drilling with respect to the data used in the MSA Group Mineral Resource estimate (MSA, 2016) and found that there were 6 additional drill holes in Etang South, Pringle and Principal Zone 4 to incorporate.

Experimental variograms were calculated for %TCu, %TCo, %AsCu and %AsCo using the 5 m composite data for selected ore zone units within each of the fragments. Omni-directional downhole variograms were used to define the nugget variance based on a lag distance of 5 m (based on the nominal drill hole composite length). Omni-directional horizontal variograms were used with a lag distance of 100 m to match the average drill hole spacing.

In zones where drilling was either limited to historical GCM data or not extensively drilled, acceptable variograms were not obtained and hence IDW estimation was applied instead of OK; this included zones 8 and 9, and the Pringle fragment. In Etang North, %AsCu and %AsCo were analysed in only a few instances and again IDW estimation was applied, but using the search ellipses obtained for %TCu and %TCo. An example of the %TCu variogram is shown in Figure 40 for Etang South fragment. The variogram parameters for different fragments are tabulated in Table 39.

Table 39: Variogram parameters by fragment

Fragment	Variable (%)	Nugget (C0)	Rotation Angle			Range 1 (m)			Range 2 (m)		
			Z	X	Z	X	Y	Z	X	Y	Z
ETANG SOUTH	TCu	0.452	90.82	21.07	27.33	30	26	5	110	104	20
	AsCu	0.465	56	0	34	88	86	10	195	188	34
	TCo	0.039	64.32	5.57	33.59	58	45	20	152	199	30
	AsCo	0.033	64.32	5.57	33.59	27	130	5	127	235	28
ETANG NORTH	TCu	0.308	170.9	36.25	15.49	29	34	5	104	113	11
	AsCu	0.308	170.9	36.25	15.49	29	34	5	104	113	11
	TCo	0.08	113.8	6.27	38.57	75	46	4	265	143	8
	AsCo	0.08	113.8	6.27	38.57	75	46	4	265	143	8
	TCu	0.169	40.36	6.46	11.3	153	155	10	293	323	20

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Fragment	Variable (%)	Nugget (C0)	Rotation Angle			Range 1 (m)			Range 2 (m)		
			Z	X	Z	X	Y	Z	X	Y	Z
PRINCIPAL FLAT	AsCu	0.016	80.52	12.2	4.51	99	101	10	203	284	25
	TCu	0.026	30.53	4.41	12.24	201	200	10	476	500	25
	AsCo	0.001	50.27	8.31	10.03	99	144	10	353	254	25

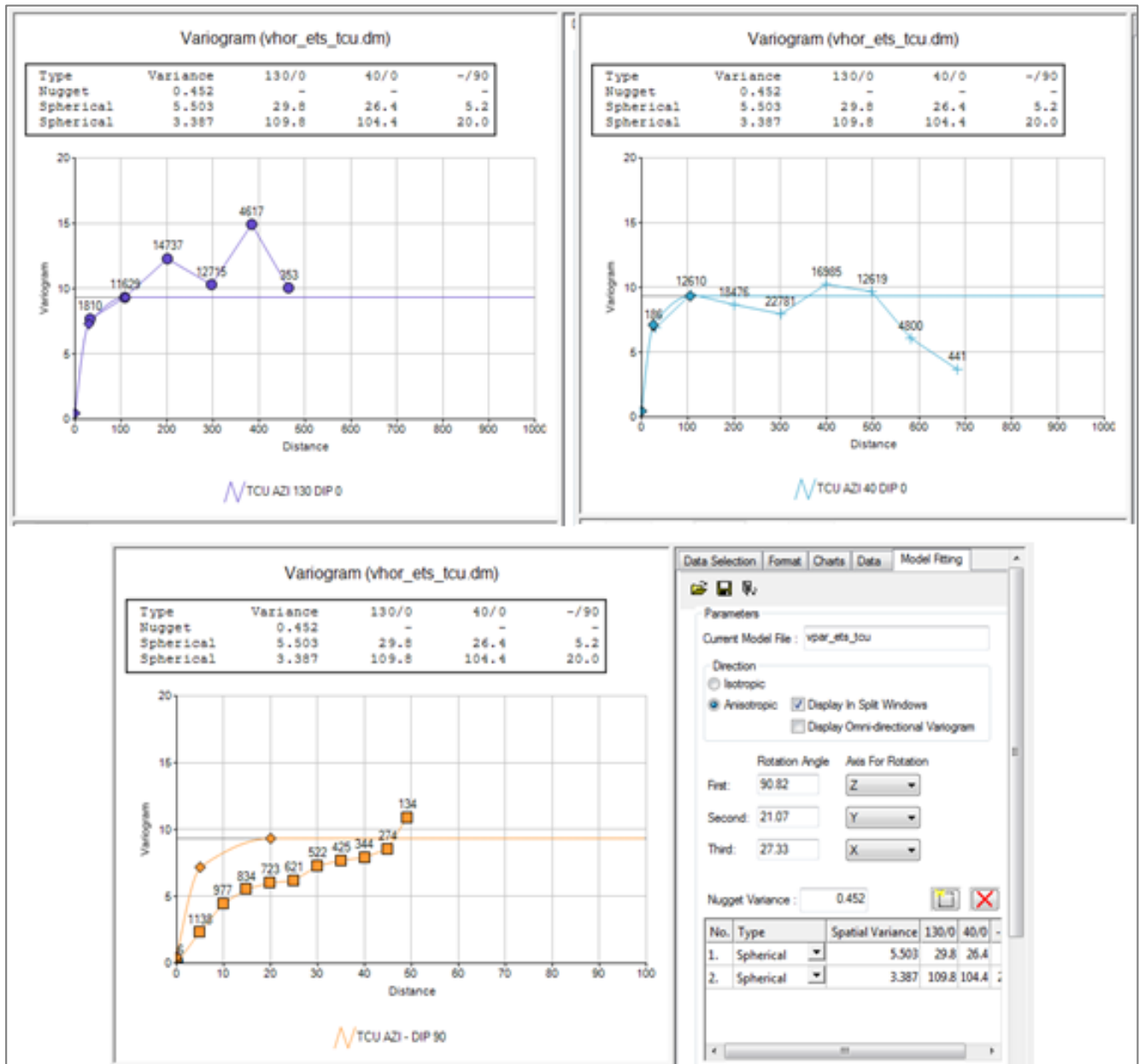


Figure 40: %TCu Variogram for Etang South Fragment

14.3.6 Block Model Definition

Historical data is provided in the local coordinate system and exported through GDMS Fusion to the UTM grid system (WGS84 L35) using an export routine that applies the appropriate conversion. The model update was therefore carried out in the UTM coordinate system with model origin, block size and dimensions as indicated

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in Table 40. A parent block size of 10 m (east-west) by 10 m (north-south) by 10 m (vertical) was chosen, however parents blocks were divided into sub-blocks on the lithology contacts.

Table 40: KTO UG block model origin and dimensions

Variable	Origin	Patent Block Size (m)	Number of Blocks	Minimum Block Size (m)
X	323,500	10	300	0.25
Y	8,814,000	10	250	0.25
Z	800	10	65	0.01

14.3.7 Lithological Modelling Codes

Five separate models for Etang North, Etang South, Pringle and Principal (subdivided into Principal Flat and Principal Steep) were generated. Within each of these models, the field FRAGNO carries a unique numeric code as an identifier for the separate fragments in the model as outlined in Table 41.

Table 41: Block model fragment codes

Fragment	FRAGNO
Principal Steep	1
Principal Flat	2
Etang North	3
Etang South	4
Pringle	5

Within each of the fragments, the lithologies are identified by a unique code in the OREZONE field as indicated in Table 42.

Table 42: Block model lithological codes within each fragment

Lithology	OREZONE
BOMZ	1
SDB	
RSC	2
RSF	3
DSTRAT	
RATGRIS	

14.3.8 Mineral Resource Estimation

Grade estimation was primarily estimated by OK using the directional variogram parameters and the composite data for each lithology. The variables estimated were %TCu, %AsCu, %TCu and %AsCo. IDW was also used to estimate these variables for validation purpose against the OK method and, as previously discussed, used as an estimate for where acceptable variograms were not obtained because of insufficient sample support (Zone 8, 9 in Principal and Pringle),

Samples were selected for estimation based on a search ellipsoid oriented along the dip of the mineralized zone with the maximum search radius being in the plane of the deposit. A dynamic search volume was used, with the first search calculated at 2/3 of the variogram range. Because of sampling bias, search distances were

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typically limited during a second estimation pass to restrict the influence of high-grade samples. This distance was calculated at 50% of the variogram range or determined based upon reasonable geological interpretations. The actual search distances are the directional variogram ranges detailed out in Table 39 in Section 14.3.5.

A Datamine™ macro was developed to allow the search ellipsoid to be aligned to the trend of the wireframe at any one location. Estimation was undertaken in three passes using the search parameters and the minimum and maximum number of samples as detailed in Table 43. Where a second search parameter is given (i.e. 150/75), this refers to a distance restriction applied in order to limit the influence of very high-grade historical assay values.

Table 43: Estimation passes, search volume and samples used for %TCu

Fragment	Ore Zone	Search Distance (m)	Search Volume 1			Search Volume 2			Search Volume 3		
			Factor	Min	Max	Factor	Min	Max	Factor	Min	Max
Etang North	OBS	150/75	0.66	3	8	1.5	3	8	50	2	10
	RSC	150/65	0.66	3	8	1.5	3	8	50	2	10
	OBI	150/75	0.66	3	8	1.5	3	8	50	2	10
Etang South	OBS	73/55	0.66	3	8	1.5	3	8	50	2	10
	RSC	73/27	0.66	3	8	1.5	3	8	50	2	10
	OBI	73/55	0.66	3	8	1.5	3	8	50	2	10
Principal Flat	OBS	200	0.66	3	8	1.5	3	8	50	2	10
	RSC	200/50	0.66	3	8	1.5	3	8	50	2	10
	OBI	200	0.66	3	8	1.5	3	8	50	2	10
Principal Steep	OBS	200/150	0.66	3	8	1.5	3	8	50	2	10
	RSC	100	0.66	3	8	1.5	3	8	50	2	10
	OBI	200/100	0.66	3	8	1.5	3	8	50	2	10
Pringle	OBS	400	0.66	3	8	1.5	3	8	50	2	10
	RSC	400	0.66	3	8	1.5	3	8	50	2	10
	OBI	400	0.66	3	8	1.5	3	8	50	2	10

For the RSC ore zone, with the lower number of analysed drill hole intersections, the data set was divided into a lower and upper 50th percentile, and for the OBS and OBI a lower 80th and upper 20th percentile. In Principal Steep, where lower grade RSC composites were unavailable for estimation, a default value of 0.5% TCu and 0.05% TCo was applied throughout the blocks on the initial pass. The second pass, with the restricted search volume, then estimated the higher portions of the ore zone.

14.3.9 Block Model Validation

Statistical and visual validation of the block model was undertaken to assess the kriging performance and conformance to the input data. A series of checks were performed including:

- Visual comparison of estimated grades versus composite grades. This was carried out for %TCu, %TCo, %AsCu and %AsCo every 50 m along north-south and west-east section. Good adherence between composite data and block estimates is observed, with only minor, local variations noted as shown in Figure 41; and
- Swath plots validations comparing block grades against composite grades, an example %TCu swath plot for Etang North is shown in Figure 42.

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Overall, the %TCu and %TCo swath plots show that the trend in the estimates is similar to that of the composites and generally the estimates are consistent with the composite data.

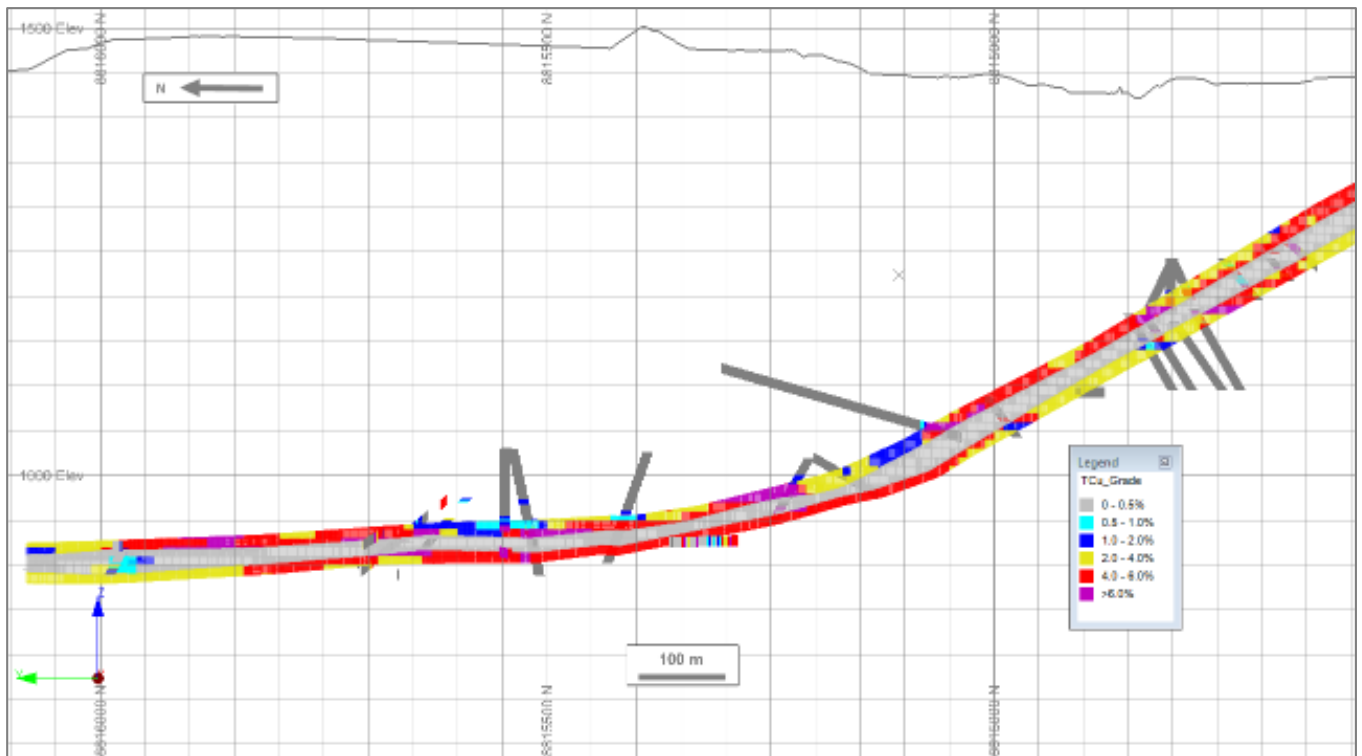


Figure 41: Visual validation of composited drill holes vs final model for %TCu

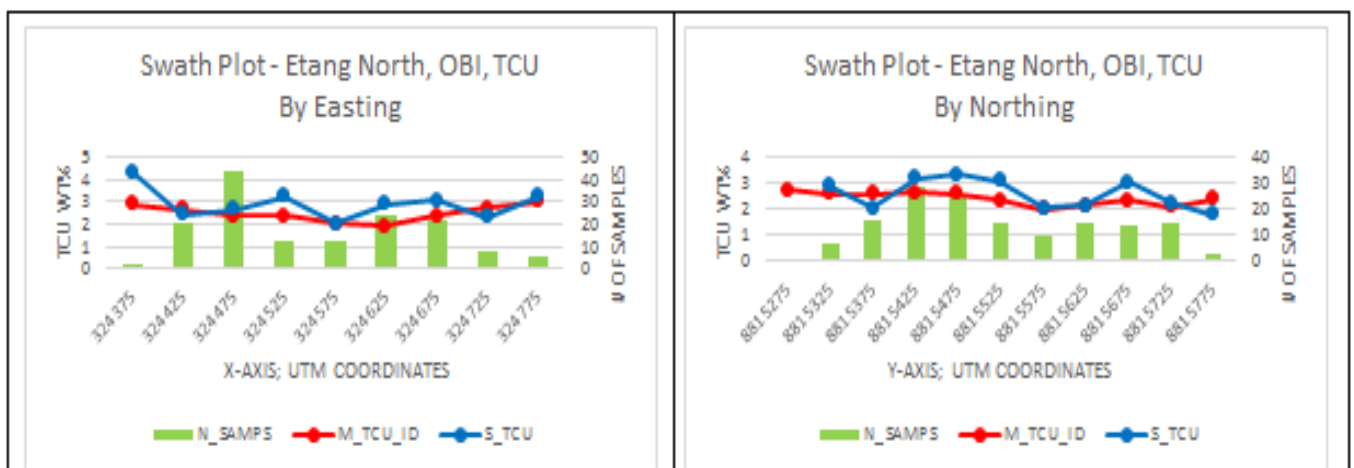


Figure 42: %TCu swath plot example for the OBI ore zone in Etang North fragment

The above validation results indicate that the model appears to be a representative of the data and that no significant bias was identified.

14.3.10 Mineral Resources Classification

Mineral Resources were classified according to CIM 2014 Definition Standards and based on drill hole spacing.

KCC classified the Mineral Resources for KTO UG into Measured, Indicated and Inferred Mineral Resources. KCC has reviewed the classification criteria and made the following observations:

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- The spatial distribution of the drill hole data from 2009 to 2015 drilling campaign is limited to Etang North, Etang South, Pringle and Principal Zones 1, 2, 3, 4 and 5;
- There has been a long mining history of the KTO UG deposit and therefore the general behaviour of the mineralization and grade continuity is reasonably well understood;
- The variograms indicate strong lateral grade continuity over several hundred metres (predominately in excess of the drill hole spacing) and show that the copper grades in the high grade layers have low variance;
- There are two main drill series in KTO UG, the historical data collected by GCM (pre 2009) and the recent data collected by KCC (2009–2015). There are prevalent data completeness and validation issues with the historical data including, but not limited to, the lack of sampling in the RSC ore zone, the absence of QA/QC, limited assaying of acid soluble copper and acid soluble cobalt, and irregular sample lengths generally biasing the data towards the high-grade mineralization. In comparison, the recent KCC drill holes have been subjected to QA/QC protocols, show an even spread of modelled assay variables (%TCu, %TCu, %AsCu, %AsCo) and, for the most part, have a well-documented history; and
- KTO UG is structurally complex, with geological sections and drill holes documenting the presence of several small-scale discontinuous folding, faulting and RAT Lilas intrusion events. Despite the challenges, the historical production illustrates that KCC is well equipped to overcome these challenges.

KCC has re-classified the KTO UG deposit based on the following classification method:

Measured Mineral Resources: where there is a high confidence in the geological interpretation due to a high density of drill holes spaced <50 m apart that include a majority of recent KCC drill holes. This is supported by a greater number of sample composites and a search ellipse of <2/3 of the variogram range;

Indicated Mineral Resources: where there is a reasonable confidence in the geological interpretation due to the drill hole spacing being between 50 m – 100 m apart but limited to the historical GCM drill holes. This is supported by a lesser number of composites and a search ellipse of >2/3 of the variogram range; and

Inferred Mineral Resources: where drill hole spacing is over 100 m apart and there is a paucity of the data to reasonably define the Mineral Resource such as the RSC ore zone and the Pringle fragment.

14.3.11 Mineral Resources Tabulation

The Mineral Resources for KTO UG are tabulated in Table 44. Resources are depleted as at 31 December 2018 and are effective at this date. Mining activities continued during 2019 and depleted Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update.

Table 44: KTO UG Mineral Resource tabulation as at December 31, 2018¹⁻⁹

Fragment	Classification	Mt	%TCu	%TCu
Etang North	Measured	1.1	3.05	0.27
	Indicated	6.0	2.52	0.51
	Measured + Indicated	7.1	2.60	0.47
	Inferred	-	-	-
Etang South	Measured	4.1	4.29	1.04
	Indicated	11.9	3.58	0.72
	Measured + Indicated	16.0	3.76	0.80
	Inferred	6.1	2.87	0.49
Principal Flat	Measured	6.9	3.81	0.38
	Indicated	36.4	4.14	0.38
	Measured + Indicated	43.2	4.14	0.38

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Fragment	Classification	Mt	%TCu	%TCo
	Inferred	29.4	4.03	0.31
Principal Steep	Measured	-	-	-
	Indicated	11.4	4.13	0.42
	Measured + Indicated	11.4	4.13	0.42
	Inferred	11.7	3.99	0.48
Pringle	Measured	-	-	-
	Indicated	-	-	-
	Measured + Indicated	-	-	-
	Inferred	1.2	2.08	0.59
Total KTO UG	Measured	12.1	3.90	0.59
	Indicated	65.6	3.92	0.46
	Measured + Indicated	77.70	3.92	0.48
	Inferred	48.4	3.83	0.38

- 1) Mineral Resources have been reported in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) Mineral Resources are reported using a cut-off grade of 1.00% TCu;
- 3) Mineral Resources are estimated using an assumed long term price of \$6,500 per tonne of copper and long term price of \$60,000 per tonne of cobalt;
- 4) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 5) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 6) Mineral Resources are inclusive of Mineral Reserves;
- 7) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 8) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 9) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

14.4 Mashamba East

The Mashamba East block model and the Mineral Resource estimate was completed by Amec Foster Wheeler in 2016 (Amec, 2016). The following sections provide a summary of the Mineral Resource estimates and modelling methodologies used as provided by Amec Foster Wheeler. Mineral Resources are depleted as at 31 December 2018 and are effective at this date. Mining activities continued during 2019 and depleted Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update.

14.4.1 Drill Hole Data

The Mashamba East Mineral Resource estimate is based on the diamond drill hole database up to June 4th, 2016 and contains 274 core holes within the Mashamba East Concession Boundary as shown on Figure 43. All drill holes intersected the fragment of Mashamba East. As of this date, assay results were available for 144 of the 147 holes completed by KCC (current drill holes). KCC collared 5 holes outside of the current concession boundary; the data from these holes were not used in construction or estimation of the current Mineral Resource.

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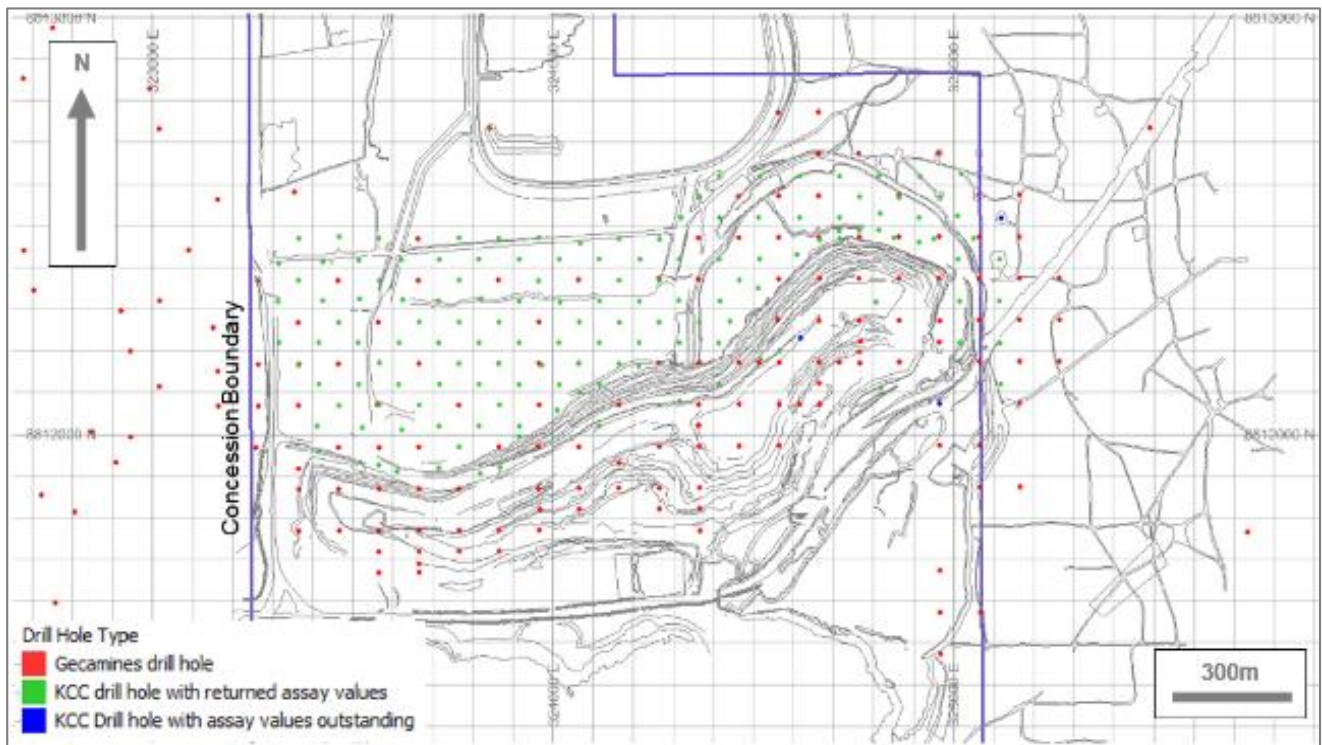


Figure 43: Plan view of Mashamba East drill hole locations

The KCC drill hole data was exported from the GDMS Fusion database into a CSV (comma separated values) and imported into Vulcan™ (GridCalc) modelling software. The drill hole file was reviewed in plan and sectional views to validate the accuracy of the drill hole collar locations, downhole survey data, lithology mistypes and assay results. The drill hole database was considered to be of suitable quality for Mineral Resource estimation purposes

14.4.2 Geological Interpretation

There is one fragment in the Mashamba East deposit which comprises six different lithological mineralized strata. Each strata was modelled individually from the drill hole database for estimation and domaining purposes, as shown in Figure 44. A thorough drill hole validation exercise was performed against the geological block model, to check the survey, collar, lithology and assay data in the historical and the recent drill holes. Occasionally where the historical and the recent data did not match, the historical data was removed from the validated drill hole database.

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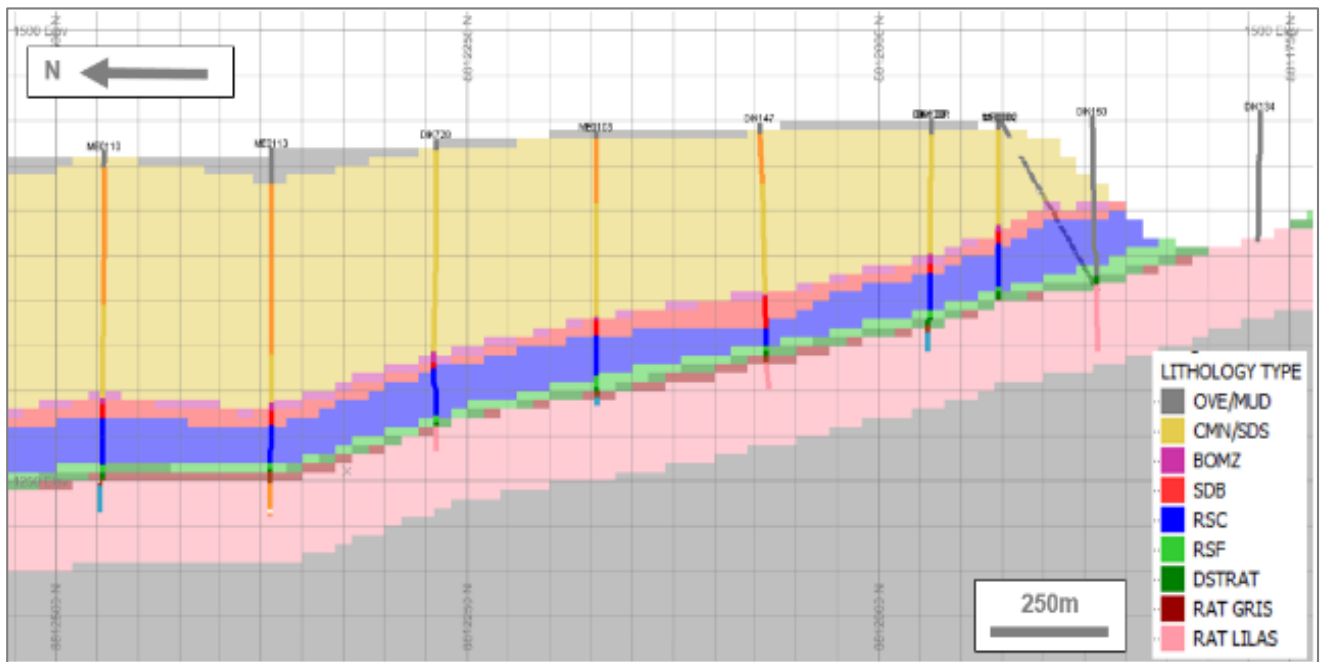


Figure 44: Mashamba East x-section showing lithologies and associated drill holes

14.4.3 Statistics and Compositing

Amec Foster Wheeler examined the validated drill holes and conducted exploratory data analysis to summarize and understand the data within each geological domain.

Assay data within each lithology was extracted from the database using the interpreted lithological wireframes. The respective lithological sample data was then composited at 2.5 m intervals to reduce the grade variability.

There are fewer %AsCu assays than %TCu assays (87%) and fewer %AsCo assays than %TCu assays (88%), due to historical selective sampling practices. This poses risk to the project economics and optimized pit shells that rely on %AsCu or %AsCo data and requires an estimation method designed to treat paired assays with unequal support.

GCM capped %TCu and %AsCu at 12% and %TCu and %AsCo at 4%, it was decided to maintain these capping grades across the entire database, including the recent 2014 and 2015 data.

Contact plots were constructed to assess the nature of the mineralization at the contacts between the stratigraphic units. If the contact is soft, then it would be reasonable to “share” all composites between the respective stratigraphic units within the individual ore zones (OBS or OBI) during the estimation process. If the contact is hard, then no composite data would be “shared” between the stratigraphic units. At Mashamba East, both assays and composites were used to assess the nature of the contact between the stratigraphic units. The following results were observed for %TCu and %TCu grades:

- OBS (SDS, BOMZ and SDB) - Grades increase progressively through the SDS, throughout the BOMZ and into the SDB units all as soft contracts;
- RSC - There is a distinct break (hard contact) at the SDB/RSC contact, and the grade decreases progressively into the RSC and increases towards the RSC/RSF hard contact; and
- OBI (RSF, DSTRAT and RGRIS) - The grade continues to increase through the RSF and DSTRAT unit and then decreases throughout the RGRIS unit. These units were treated as soft contacts.

14.4.4 Dry Bulk Density

The Mashamba East database contained 7,357 samples with dry bulk density values performed by the SGS laboratory, Zambia during 2015-16. Since these data ranged in value from 0.01 to 7.27 t/m³. These values

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were trimmed to remove anomalous values between 1.5 and 2.9 t/m³. A review of mean bulk density by stratigraphic unit compared to the measured bulk density showed that 64% of the measured values were within 10% of the mean density value. Dry bulk density was assigned based on the average for each rock type as summarized in Table 45.

Table 45: Bulk density assignment for Mashamba East based on the lithologies

Mashamba East Dry Bulk Density Summary Table								
	SDS	BOMZ	SDB	RSC	RSF	DSTRAT	RATGRIS	RAT LILAS
Count	539	503	1,190	2,701	706	412	413	610
Min	1.52	1.51	1.2	1.51	1.55	1.52	1.59	1.53
Average	2.107	2.135	2.255	2.279	2.232	2.276	2.323	2.293
Max	2.840	2.890	2.870	2.89	2.86	2.85	2.88	2.89
CV	0.176	0.139	-0.368	-0.104	0.230	0.089	-0.101	0.183

14.4.5 Grade Variography

Sage software was used to create experimental variograms for %TCu and %TCu using 2.5 m composites within the %TCu and %TCu grade shells. Down-hole variograms were used to define the nugget effect, and directional variograms were created in all directions using an azimuth increment of 30° and a dip increment of 15°. The anisotropy was then evaluated and identified as following the overall east-west strike and dip (approximately 15° to the north) of the deposit. Variogram ranges were modeled between 100 m and 800 m which is longer than the average drill hole spacing of 80 m to 100 m.

The variogram parameters for %TCu, %AsCu, %TCu and %AsCu are displayed in Table 46 and the %TCu variogram model for Mashamba East is shown in Figure 45.

Although variograms were created for %AsCu and %AsCu, these elements were estimated using the “Paired Assay Method” due to insufficient composite numbers as described below:

- Using only those samples that were assayed for both %TCu and %AsCu (“paired data”), %TCu and %AsCu were estimated (by IDW) into the block model using the search parameters in Table 46, except there was no stratigraphic unit restriction, and the search ellipses were increased to account for the decreased amount of paired samples available;
- The %AsCu: TCu Ratio was then calculated for each block by dividing paired %AsCu estimates by paired %TCu estimates;
- %TCu was estimated using the entire dataset; and
- The %AsCu grade in each block was then calculated by multiplying AsCu: TCu based on the paired data by the %TCu estimates based on the entire dataset of available %TCu samples.

Table 46: Mashamba East variogram parameters

Fragment	Variable (%)	Nugget (C0)	Rotation Angle			Range 1 (m)			Range 2 (m)		
			Z	X	Z	X	Y	Z	X	Y	Z
Mashamba East	TCu	0.05	0	15	0	125	100	20	800	600	50
	ASCu	0.05	0	15	0	125	100	20	800	600	50
	TCu	0.05	0	10	0	100	100	10	175	175	35
	ASCo	0.05	0	10	0	100	100	10	175	175	35

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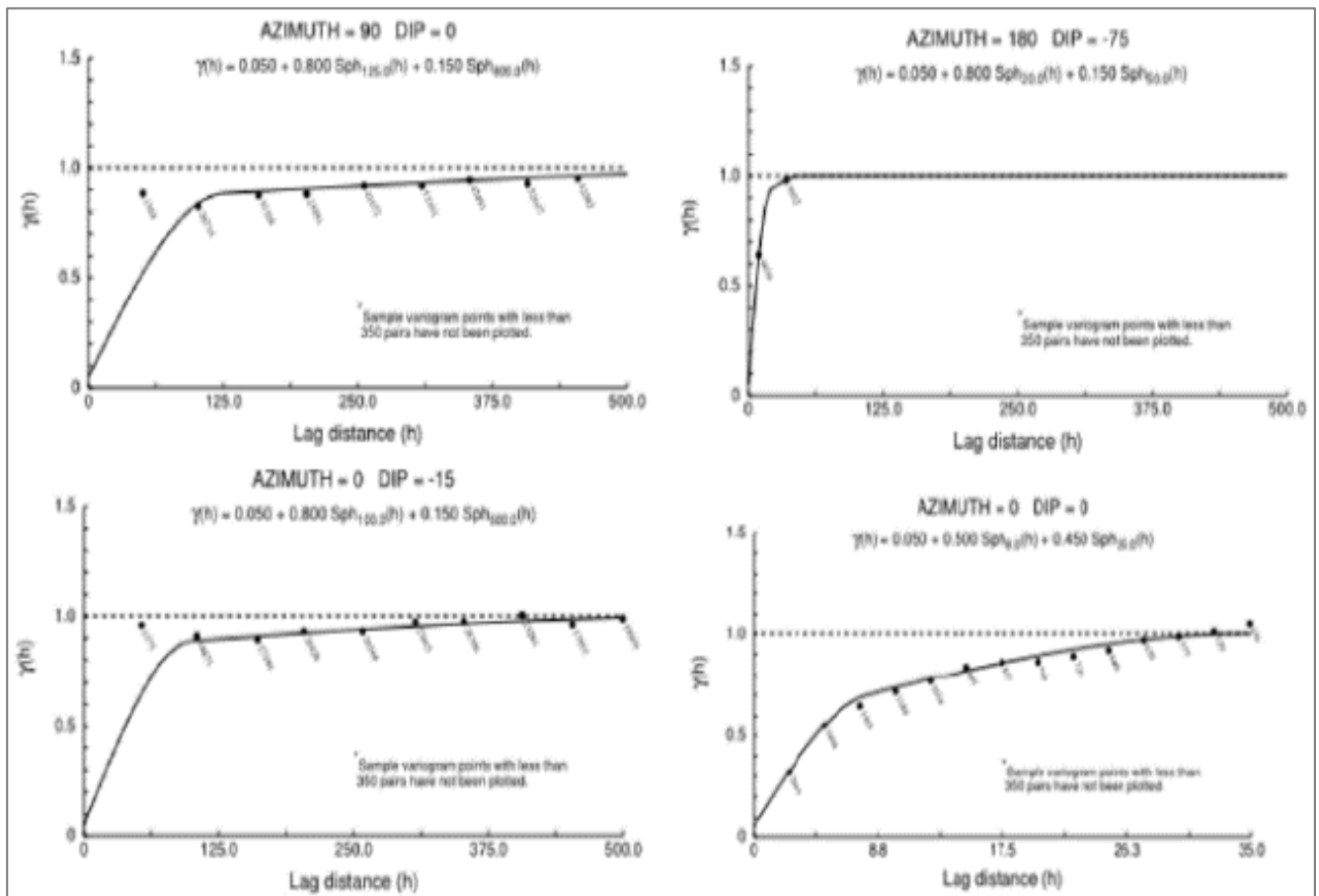


Figure 45: %TCu variograms for Mashamba East

14.4.6 Block Model Definition

The Mashamba East Mineral Resource block model was constructed over the entire project area. An appropriate block size of 10 m x 10 m x 2.5 m was selected as it is small enough to fit the geometry of the mineralized zones and can be reblocked to mining bench height of either 5 m or 10 m if required. No sub-celling was used. The block model definition parameters are summarized in Table 47.

Table 47: Mashamba East block model origin and dimensions

Variable	Origin	Block Size (m)	Number of Blocks
X	323200	10	190
Y	8811500	10	140
Z	1000	2.5	200

14.4.7 Lithological Modelling Codes

As there is only one fragment in the deposit, fragment coding was not applied. Within the single fragment, the lithologies are identified by a unique code in the ROCKNO field as indicated in Table 48.

Table 48: Block model lithological codes within the fragment

Lithology	ROCKNO
OVE (Overburden)	9

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Lithology	ROCKNO
SDS	0
BOMZ	1
SDB	2
RSC	3
RSF	4
DSTRAT	5
RATGRIS	6
RATLILAS	7

14.4.8 Mineral Resource Estimation

Grades were estimated by a variety of methods depending on which element was under assessment.

The %TCu and %TCo estimation process was as follows:

Grade Shells: First grade shells were constructed to define the geometry of the economic mineralization. This was achieved by creating variable length drill hole composites (minimum 5 m), with cut-off-grades of 0.46% for %TCu and 0.12% for %TCo. Sections were drawn and separate grade shells were created for %TCu and %TCo;

Indicator Model: An indicator model was constructed to define areas inside the grade shells as either mineralized or non-mineralized. Using the 2.5m composites, an indicator of 1 was assigned to composites greater than 0.46% and 0.12% for %TCu and %TCo respectively. These indicators were estimated into the block model using an IDW and NN and the indicators (1 or 0) were tagged to the %TCu and %TCo grades. Only the blocks above an estimated indicator threshold of 0.6 would define the economic mineralization within the grade shell be interpolated during estimation;

Search Ellipse: The search ellipse rotation angles were calculated by local varying anisotropy. This method references the dip and dip-direction of the stratigraphic surfaces and uses these orientations to control the search rotations for each block during the estimation process. The size of the ellipse was determined from the variogram parameters, with four search passes used for estimation. The first pass was a single factor of the variogram range (110 m for %TCu), increasing up to the fourth search ellipse which was 7 factors of the variogram (770 m for %TCu). The search parameters for %TCu and %TCo are displayed in Table 49; and

Table 49: Estimation passes, search volume and samples used for %TCu

Search Distance (m)	Search Volume 1			Search Volume 2			Search Volume 3			Search Volume 3		
	Factor	Min	Max	Factor	Min	Max	Factor	Min	Max	Factor	Min	Max
110	1	3	12	2	3	12	3	3	12	7	1	12

OK, IDW and NN Estimation Methods: Estimations were carried out inside the hard boundary ore zones (OBS, RSC and OBI), the mineralized indicator models and the grade shells. The grades were estimated by OK with IDW and NN being used as a validation methods.

The %AsCu and %AsCo estimation process was as follows:

Paired Estimation and IDW Methods: Since only 87% of the samples with %TCu assays were assayed for %AsCu and only 88% of the %TCo samples were assayed for %AsCo, Amec Foster Wheeler used the “paired assay” method to estimate %AsCu and %AsCo grades. The drill holes were filtered for composites, which had both %TCu and %AsCu or %TCo and %AsCo values, and these selected composites were estimated into the model by IDW. The AsCu: TCu and AsCo: TCo ratio values were calculated for each block and from these

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ratios the %AsCu and %AsCo grades were calculated by multiplying the block model ratio value with the %TCu and %TCu estimated grade.

14.4.9 Block Model Validation

Statistical and visual validation of the block model was undertaken to assess the kriging performance and conformance to the input data. A series of checks were performed including:

- Visual assessment of the estimated grades versus the composite grades;
- Global bias validation; and
- Local bias check (swath plots).

Visual comparison in cross section of the block model and the composites presents a good adherence with only minor, local variations noted, as shown in Figure 46.

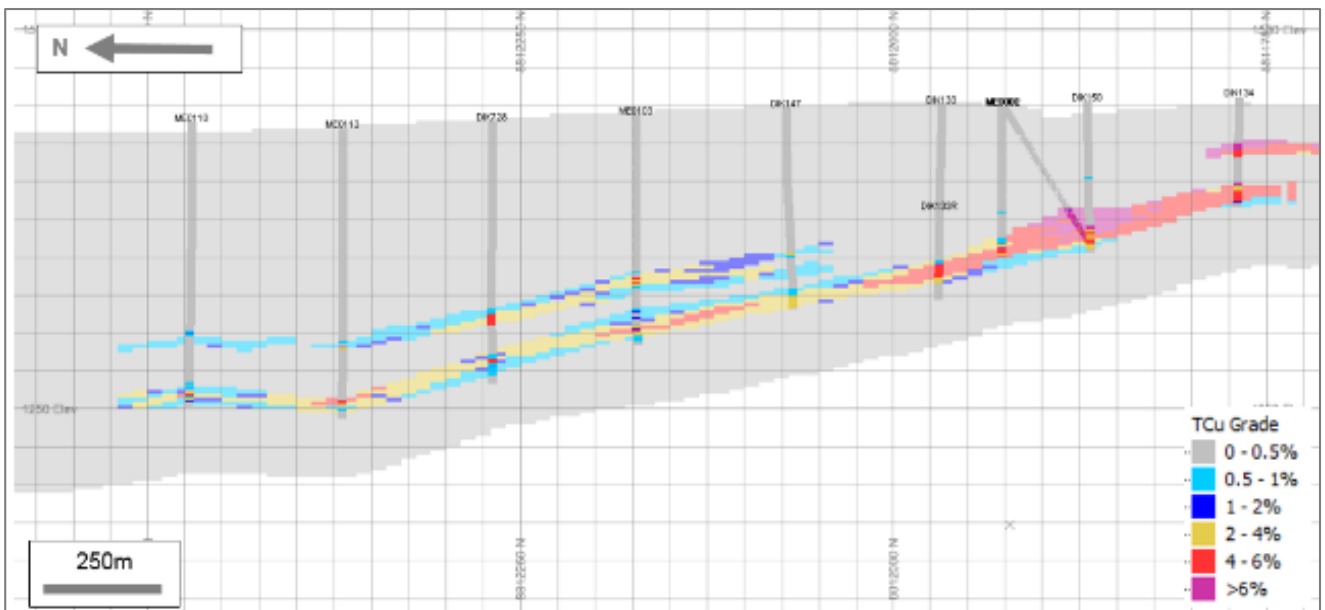


Figure 46: Visual validation of the composited drill holes vs the final block model for %TCu

Global bias validation results are provided in Table 50 showing no significant difference between OK and NN.

Table 50: Global bias check within the Indicated Resources

Stratigraphic Unit	Element	No. Blocks	NN Mean	Estimated Mean (OK)	Relative Difference (%)
					(OK-NN)/NN
OBS	TCU (%)	28,719	1.830	1.839	0.5%
	TCO (%)	31,889	0.573	0.568	-0.9%
RSC	TCU (%)	28,703	1.648	1.656	0.5%
	TCO (%)	51,191	0.536	0.524	-2.2%
OBI	TCU (%)	25,774	2.833	2.873	1.4%
	TCO (%)	26,837	0.895	0.908	1.2%
RLIL	TCU (%)	5,409	1.783	1.752	-1.7%
	TCO (%)	4,647	0.333	0.344	3.3%

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Checks for the local biases were performed by the stratigraphic units for %TCu and %TCo by creating and analysing the local trends in the grade estimates using the swath plots. Overall, the %TCu and %TCo swath plots show good agreement, especially in areas supported by a large number of blocks. See example %TCu OBI swath plot in Figure 47 where the red line represents the OK model grades, the blue line represents the IDW model grades, and the black line represents the NN model grades.

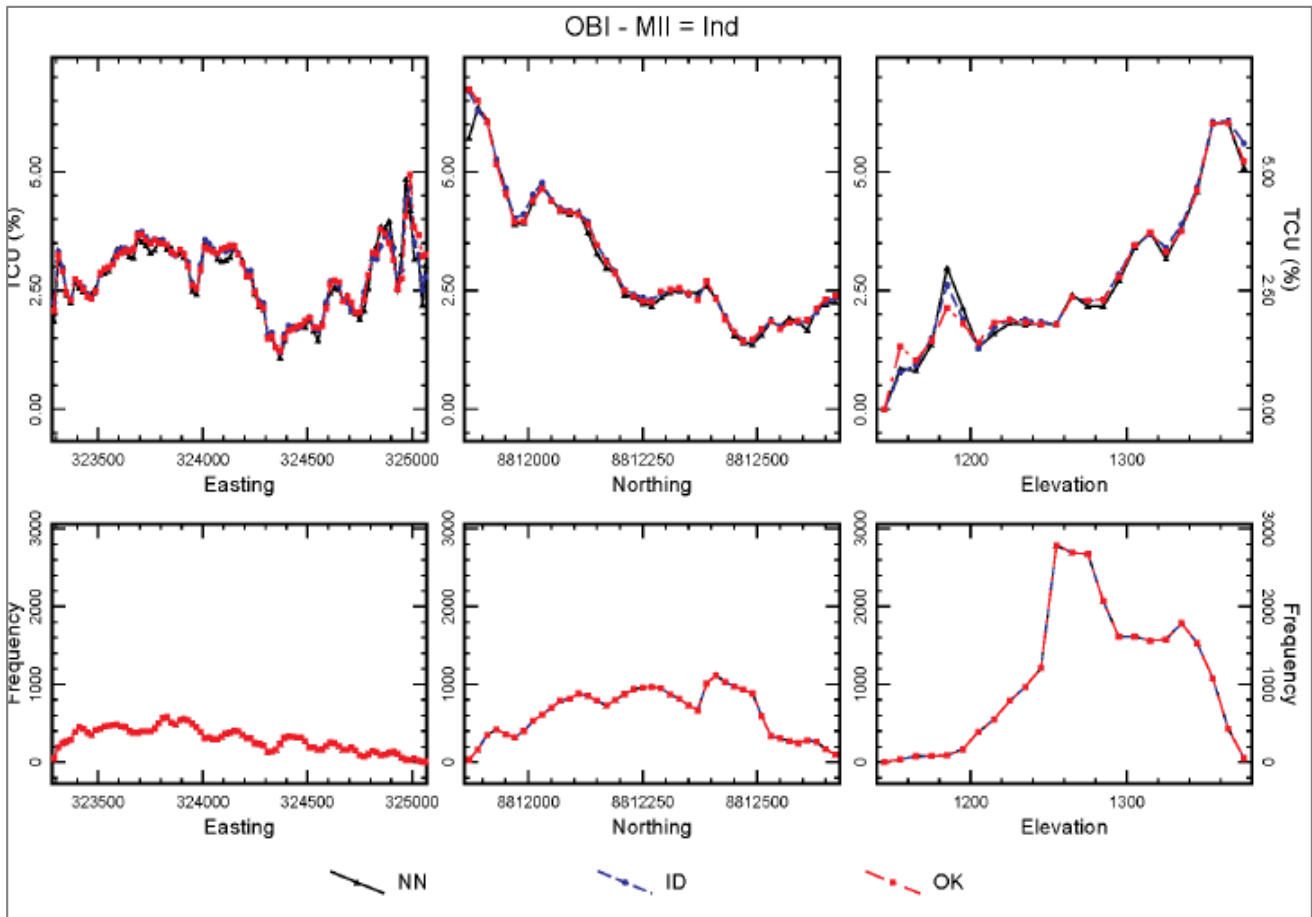


Figure 47: %TCu swath plot example for the OBI ore zone in Mashamba East fragment

The results of the above validation indicate that the model appears to be a representative of the data and that no significant bias was identified.

14.4.10 Mineral Resources Classification

Mineral Resources were classified according to CIM 2014 Definition Standards and based on the drill hole spacing.

Amec Foster Wheeler classified the Mineral Resources for Mashamba East as Indicated and Inferred Mineral Resources. Amec Foster Wheeler reviewed the classification criteria and made the following observations:

- The spatial distribution of the drill hole data from the 2014 to 2015 drilling campaign covered most of the Mashamba East fragment;
- The variograms indicate a strong lateral grade continuity over the several hundred metres (predominately in excess of the drill hole spacing) and show that the copper grades have low variance; and
- There are two main drill hole series in Mashamba East; the historical data collected by GCM (pre-2009) and the recent data collected by KCC (2014–2015). There are prevalent data completeness and validation issues with the historical data including, but not limited to, the lack of sampling in the RSC

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ore zone, the absence of QA/QC, limited assaying of the acid soluble copper and the acid soluble cobalt, and irregular sample lengths generally biasing the data towards the high-grade mineralization. In comparison, the recent KCC drill holes have been subjected to QA/QC protocols, show an even spread of the modelled assay variables (%TCu, %TCo, %AsCu, %AsCo) and, for the most part, have a well-documented history.

In view of the foregoing, Amec Foster Wheeler established the classification criteria for Mashamba East based on the following classification method:

- Indicated: where there is a high confidence in the geological interpretation within an area well informed by the KCC drilling on a <80 m drill spacing with supporting QA/QC data; and
- Inferred: where there is a reasonable confidence in the geological interpretation within an area well informed by the GCM drilling but with a lack of supporting QA/QC data and within 120 m of a single drill hole were considered Inferred.

14.4.11 Mineral Resources Tabulation

The Mineral Resources for the Mashamba East OP and UG are tabulated in Table 51: Total Mashamba East OP Mineral Resource tabulation as at December 31, 2018¹⁻¹⁰

to Table 53. Resources are estimated as at December 31, 2018 and are effective at this date. Mining activities continued during 2019 and KCC experienced ordinary course depletion. Updated Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update, which will reflect such normal course depletion.

Table 51: Total Mashamba East OP Mineral Resource tabulation as at December 31, 2018¹⁻¹⁰

Fragment	Classification	Mt	%TCu	%TCo
Mashamba East OP	Measured	-	-	-
	Indicated	35.9	1.72	0.61
	Measured + Indicated	35.9	1.72	0.61
	Inferred	8.3	2.58	0.55

Table 52: Total Mashamba East UG Mineral Resource tabulation as at December 31, 2018¹⁻¹⁰

Fragment	Classification	Mt	%TCu	%TCo (%)
Mashamba East UG	Measured	-	-	-
	Indicated	21.9	1.72	0.67
	Measured + Indicated	21.9	1.72	0.67
	Inferred	7.6	2.87	0.48

Table 53: Total Mashamba East consolidated Mineral Resource tabulation as at December 31, 2018¹⁻¹⁰

Fragment	Classification	Mt	%TCu	%TCo
Mashamba East Total	Measured	-	-	-
	Indicated	57.8	1.72	0.63
	Measured + Indicated	57.8	1.72	0.63
	Inferred	15.9	2.72	0.51

1) Mineral Resources have been reported in accordance with the definitions and classification standards adopted in NI 43-101;

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- 2) OP Mineral Resources are reported using a total copper (%TCu) cut-off grade of 0.46% for the copper only grade shell. A total cobalt (%TCo) cut-off grade of 0.12% is used for reporting Mineral Resources within the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at either cut-off;
- 3) UG Mineral Resources are reported using a total copper (%TCu) cut-off grade of 1.00% for the copper only grade shell. A total cobalt (%TCo) cut-off grade of 0.517% TCo is used for reporting Mineral Resources within the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at either cut-off;
- 4) Mineral Resources are estimated using an assumed long term price of \$6,750 per tonne of copper and long term price of \$30,000 per tonne of cobalt;
- 5) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 6) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal place;
- 7) Mineral Resources are inclusive of Mineral Reserves;
- 8) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 9) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 10) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

14.5 KITD

The KITD block model and the Mineral Resource estimate was completed by KCC in 2017. The following sections provide a summary of the Mineral Resource estimates and the modelling methodology used. Mineral Resources are depleted as at 31 December 2018 and are effective at this date.

14.5.1 Drill Hole Data

The KITD Mineral Resource estimate is based on the drill hole database up to and including the 9th of September 2017.

The validated KITD drill hole database contains a total of 171 drill holes, of which 9 are twin drill holes of the Minutor Mining drilling in 2014/15 as shown in Figure 48. There was one drill hole (TD138) that had the incorrect collar coordinates and was removed from the final database. The average drill hole spacing is approximately 50 m x 50 m for Minutor Mining drill holes (2014/15) and either 100 m x 100 m or 50 m x 50 m infill for the KCC drill holes (2017) as shown in Figure 49.

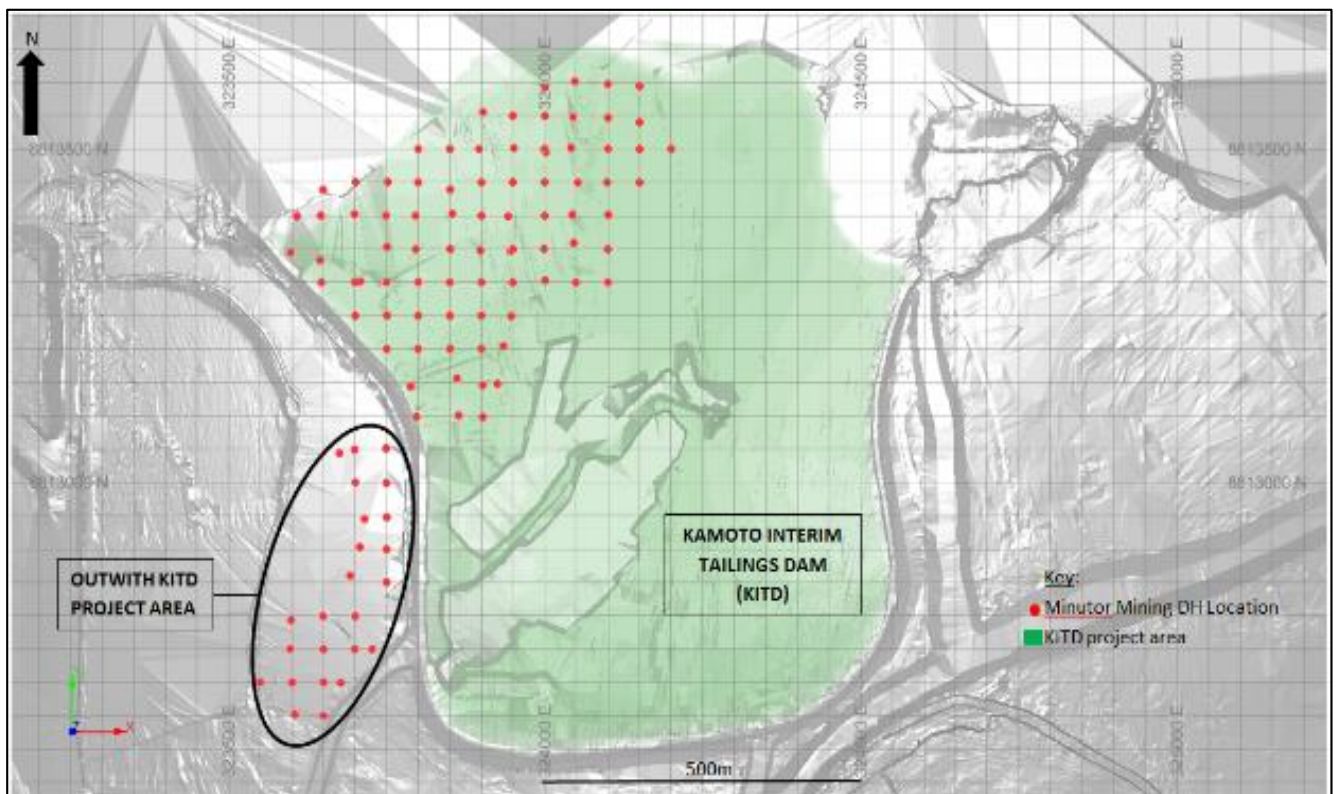


Figure 48: Plan view of the KITD drilling for the 2014/15 campaign

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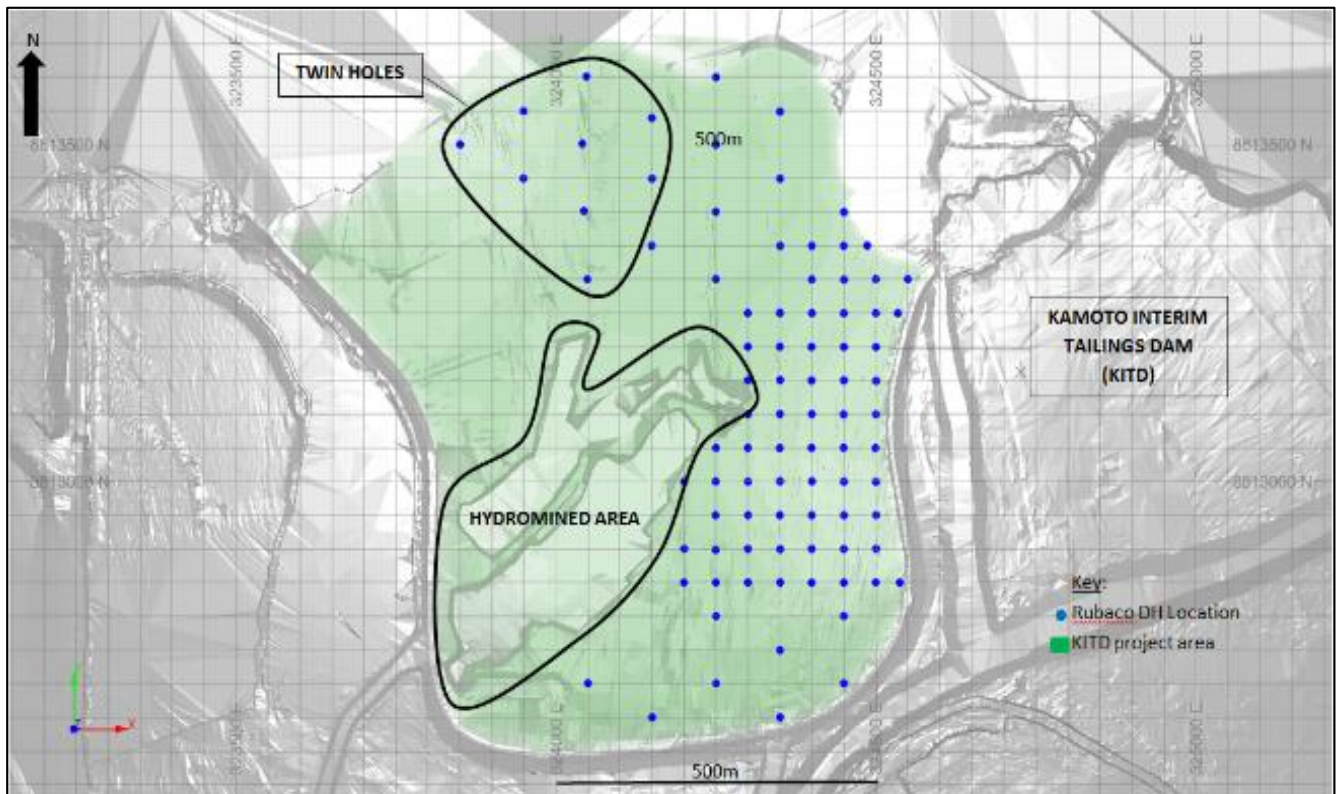


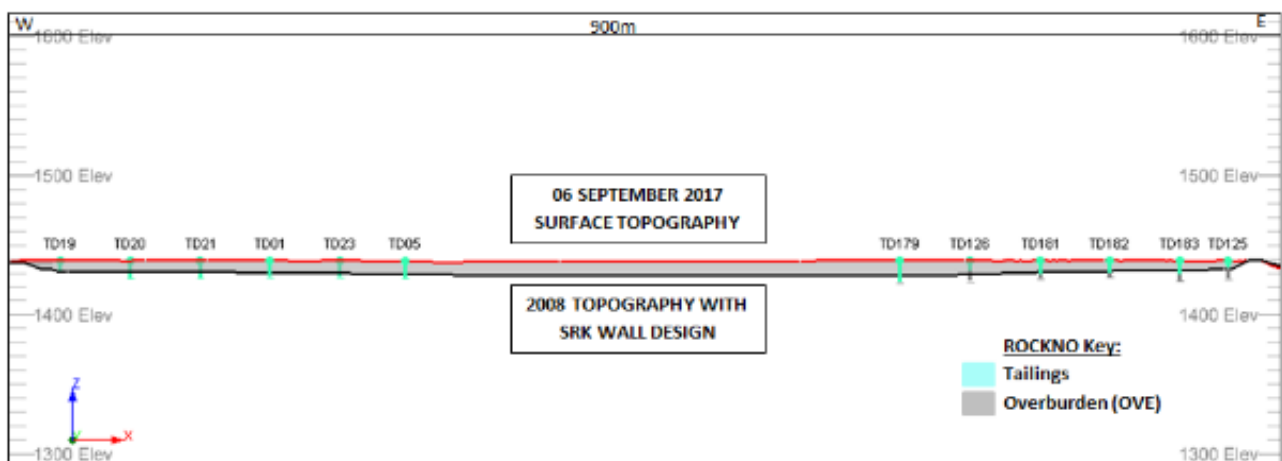
Figure 49: Plan view of the KITD drilling for the 2017 campaign

The KCC drill hole data was exported from the GDMS Fusion database into a CSV and imported into Datamine™ RM (version v.1.1.20.0). The drill hole file was reviewed in plan and sectional views to validate the accuracy of the drill hole collar locations, downhole survey data, lithology, and assay results. The drill hole database was considered to be of a suitable quality for the Mineral Resource estimation purposes

14.5.2 Geological Interpretation

The original tailings surface superimposed with the 2012 SRK designed slope was utilised as the base wireframe for the KITD, together with the 6th of September 2017 surface survey (Figure 50).

The three dimensional wireframes were used to constrain the block model estimates. The density results from the 2017 sampling were interpolated into the block model using the NN technique”.



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Figure 50: The KITD cross-section showing the lithologies and the associated drill holes

14.5.3 Statistics and Compositing

Datamine™ RM Resource software was used to generate histograms, log histograms, cumulative histograms, cumulative log histograms, cumulative probability plots, and log probability plots for %TCu, %TCo, %AsCu, and %AsCo. Results for %TCu are shown in Figure 51 to Figure 53.

The drill holes were composited by the rock type with a minimum of 0.5 m and a maximum of 2 m intervals.

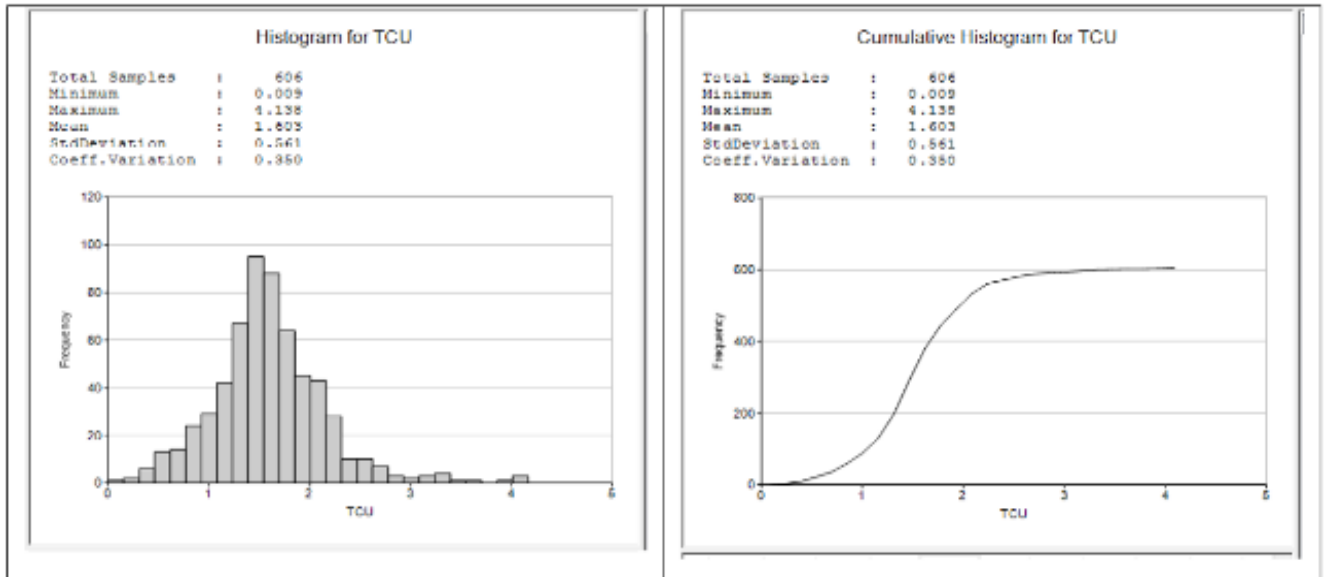


Figure 51: Frequency and cumulative distribution histogram for %TCu

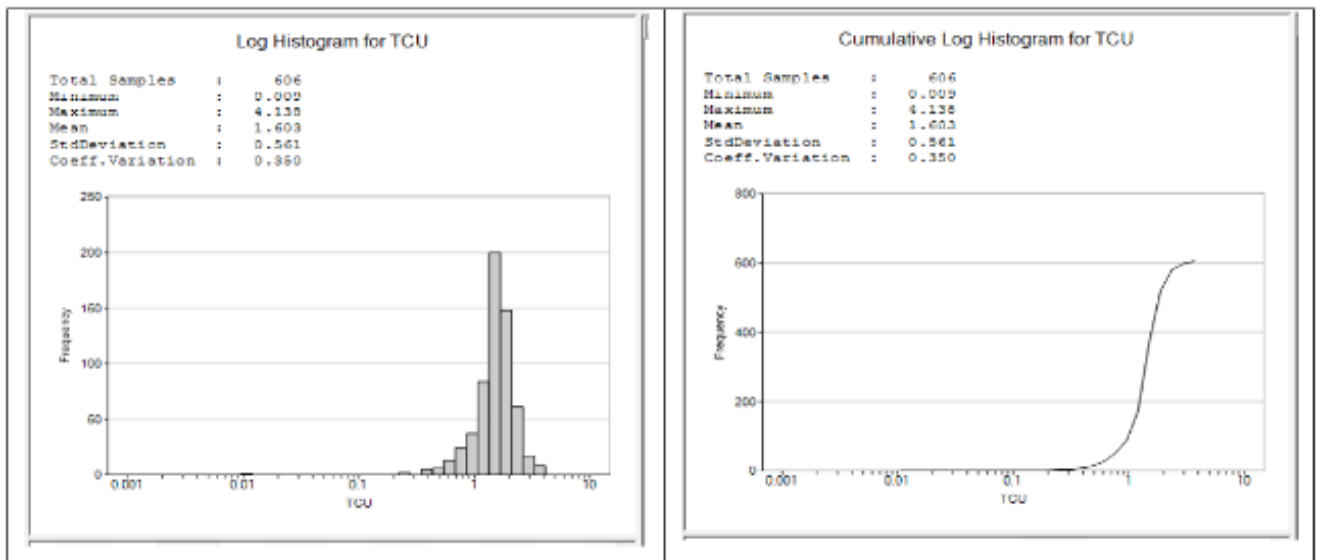


Figure 52: Log frequency and log cumulative distribution histogram for %TCu

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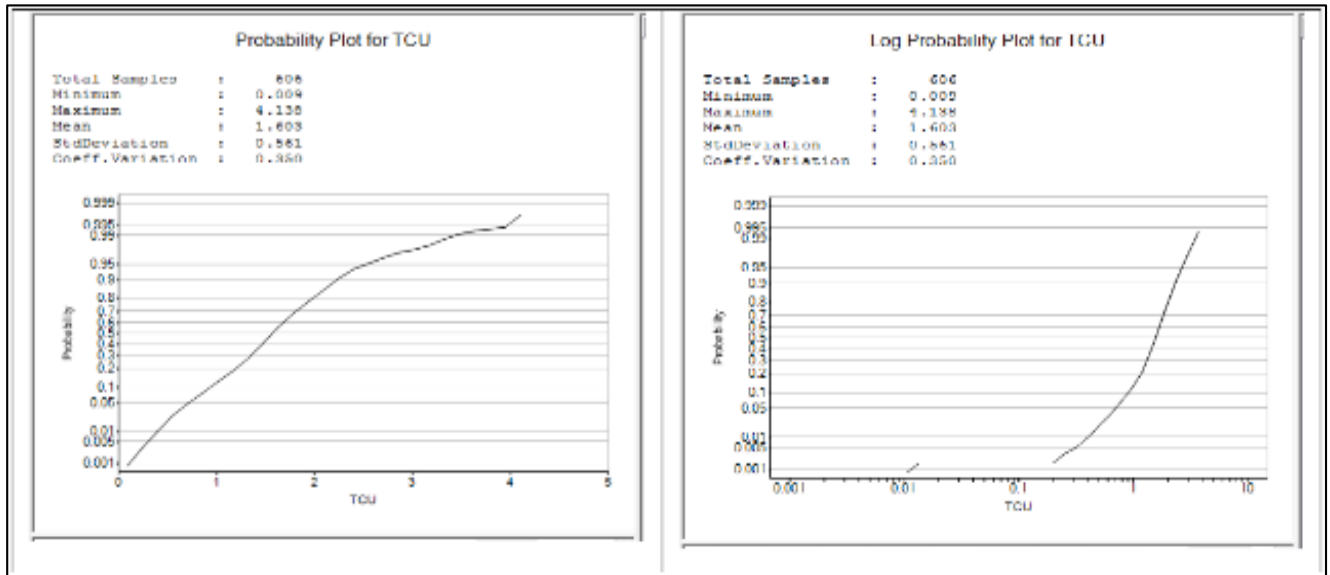


Figure 53: Cumulative probability and cumulative log probability plots for %TCu

14.5.4 Dry Bulk Density

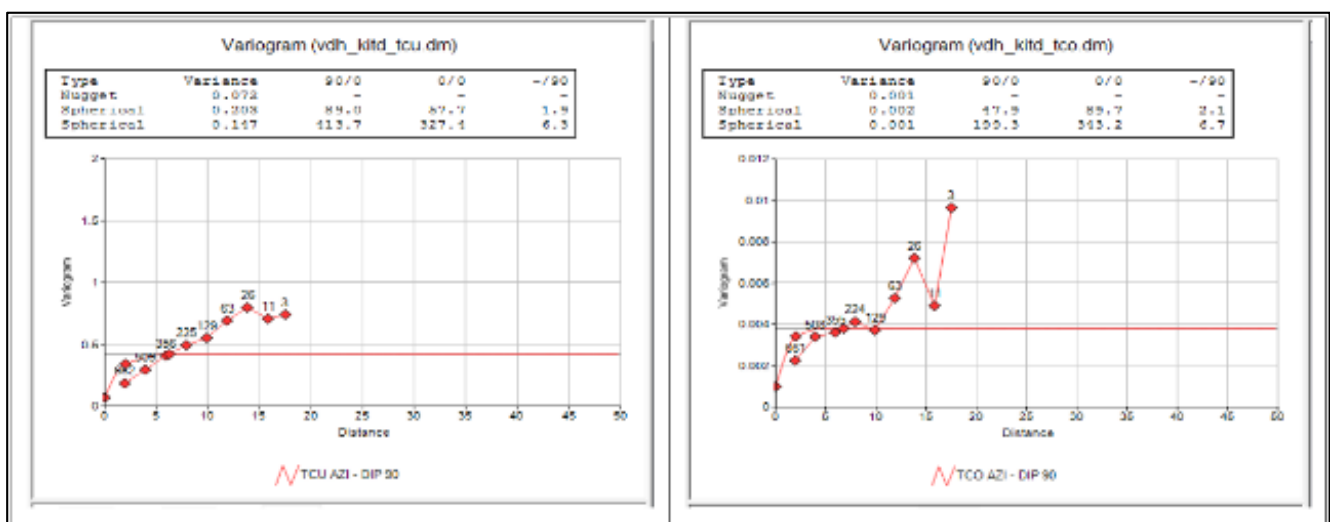
A total of 511 samples were collected from the tailings core for dry bulk density determinations using the KCC Standard Operating Procedure MIN-GEO-PRO-319. Dry bulk density was determined using the Archimedes Principle. This method consists of first weighing the oven-dried sample in air, then sealing the sample in plastic cling film, immersing it in water and weighing it again; the water displacement is equal to the volume. Statistics of the dry bulk density used as the basis for density assignment are shown in Table 54.

Table 54: Dry bulk density assignment for KITD based on the lithologies

Unit	Count	Minimum	Average	Maximum	Standard Deviation	CV
Tailings	511	1.02	1.49	2.99	0.15	0.1

14.5.5 Grade Variography

Datamine™ RM Resource modelling software was used to create the downhole and the horizontal variograms for composited %TCu, and %TCO. Results are shown in Figure 54 to Figure 56.



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Figure 54: Downhole variograms for %TCu and %TCo

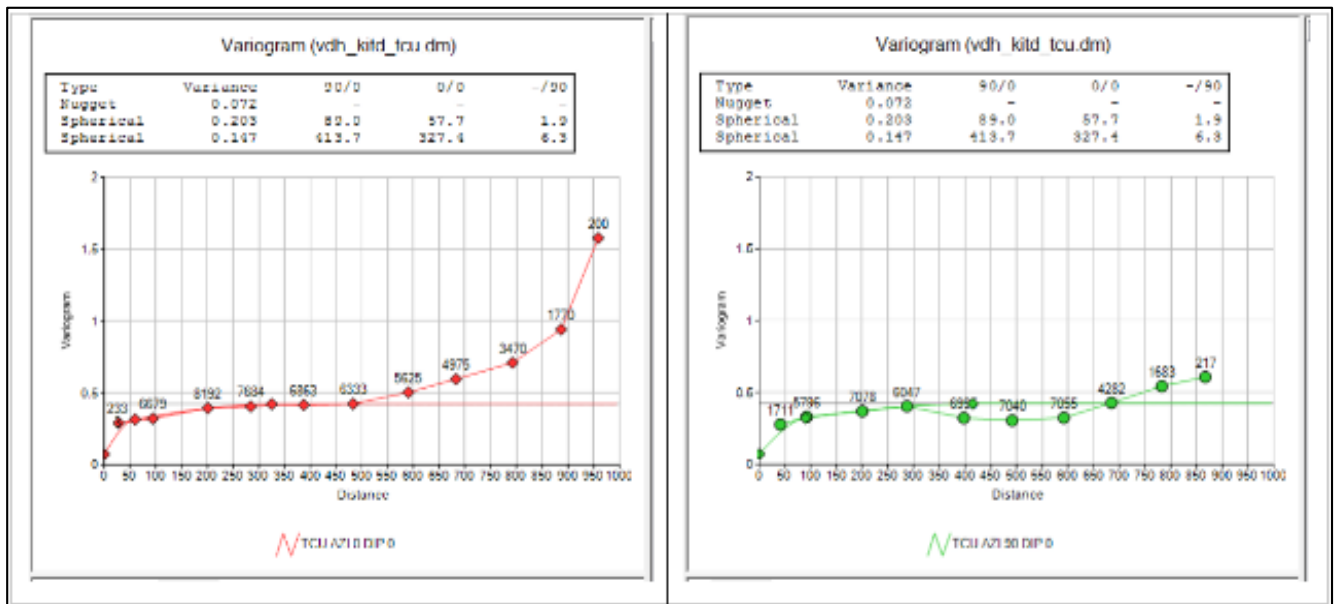


Figure 55: Horizontal variograms for %TCu

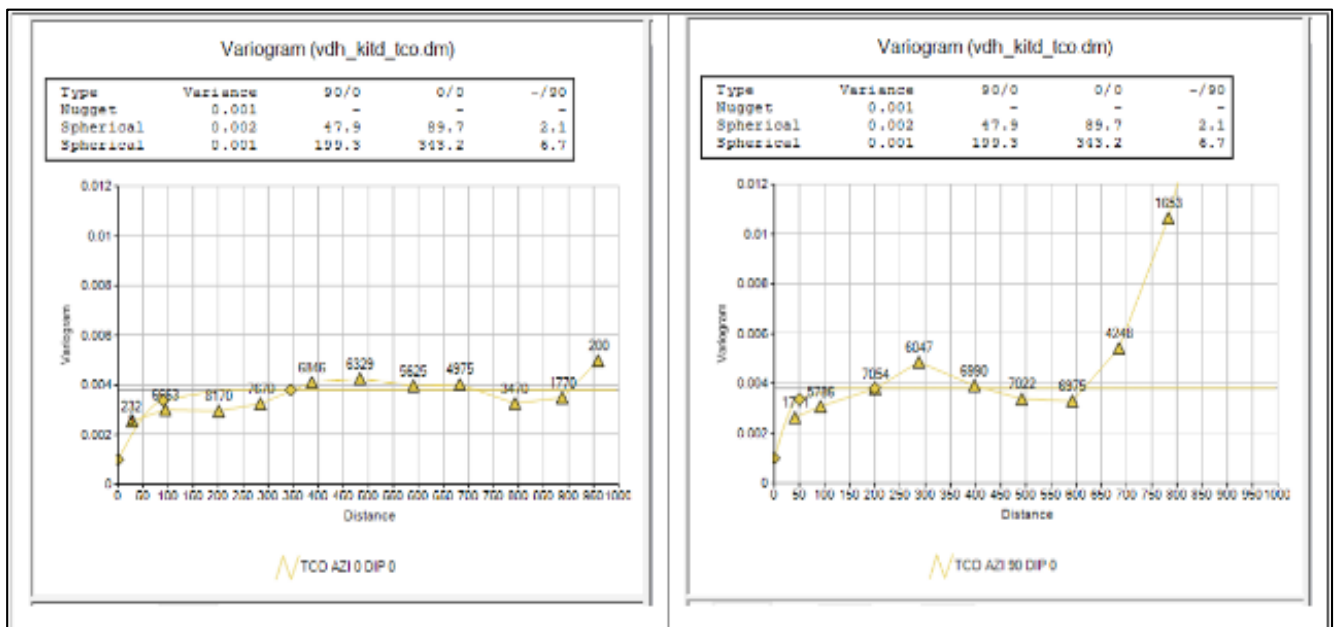


Figure 56: Horizontal variograms for %TCo

14.5.6 Block Model Definition

Datamine™ RM Resource modelling software was used to create the KITD block model using the extended precision format.

The KITD prototype model covers a three-dimensional block in the UTM grid co-ordinate system. The block configuration and the size is typically a function of the geometry of the deposit, density of the sample data, and the SMU. On this basis, a parent block size of 20 m (east-west) by 20 m (north-south) by 2 m (elevation) was chosen, with parent blocks divided into sub-blocks along the model boundaries. The block model definition parameters are summarized in Table 55.

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Table 55: KITD block model origin and dimensions

Variable	Origin	Parent Block Size (m)	Number of Blocks	Minimum Block Size (m)
X	323,200	20	80	2.5
Y	8,812,400	20	75	2.5
Z	1400	2	50	.002

14.5.7 Lithological Modelling Codes

The KITD is identified by a unique project code called FRAGNO as indicated in Table 56.

Table 56: Block model project code

Fragment	FRAGNO
KITD	17

Within the project, the different units are identified by a unique code called ROCKNO as indicated in Table 57 where overlying red soil is assigned as overburden (OVE).

Table 57: Block model “lithological” codes

Unit	ROCKNO
OVE	1
TAILINGS	100

14.5.8 Mineral Resource Estimation

The grade estimation was completed by using the following methods: NN, IDW and OK for the variables %TCu, and %TCu. To determine the acid soluble values, and to prevent exceeding AsCu/TCu ratios >1.0 from occurring, the ratio of AsCu/TCu and AsCo/TCu was first interpolated into the blocks, and the acid soluble component calculated by multiplying the interpolated ratio with the %TCu or %TCu value. Because the values for IDW were virtually identical to OK, and because of uncertainty regarding the nugget and sill values for cobalt, IDW was retained as the method for the final model for Mineral Resource reporting. The search parameters for %TCu and %TCu are displayed in Table 58.

Table 58: Estimation passes, search volume and samples used for %TCu and %TCu

Unit	Value (%)	Search Distance			Search Volume 1			Search Volume 2			Search Volume 3		
		X	Y	Z	Factor	Samples		Factor	Samples		Factor	Samples	
						Min	Max		Min	Max		Min	Max
Tails	TCu	275	220	4	1	3	8	1.5	3	8	3	1	20
	TCu	135	230	4	1	3	8	1.5	3	8	3	1	20

14.5.9 Block Model Validation

Statistical and visual validation of the block model was completed to assess the block model and its conformance to the input data. A series of checks were performed including:

- Visual assessment of the estimated grades versus the composite grades; and
- Swath plot validations comparing the block grades against the composite grades.

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The block model was examined visually in sections to ensure that the composited drill hole grades were locally well represented by the block model. This was carried out for %TCu, %TCo, %AsCu and %AsCo. Figure 57 and Figure 58 show the %TCu validation in east-west and north-south cross sections.

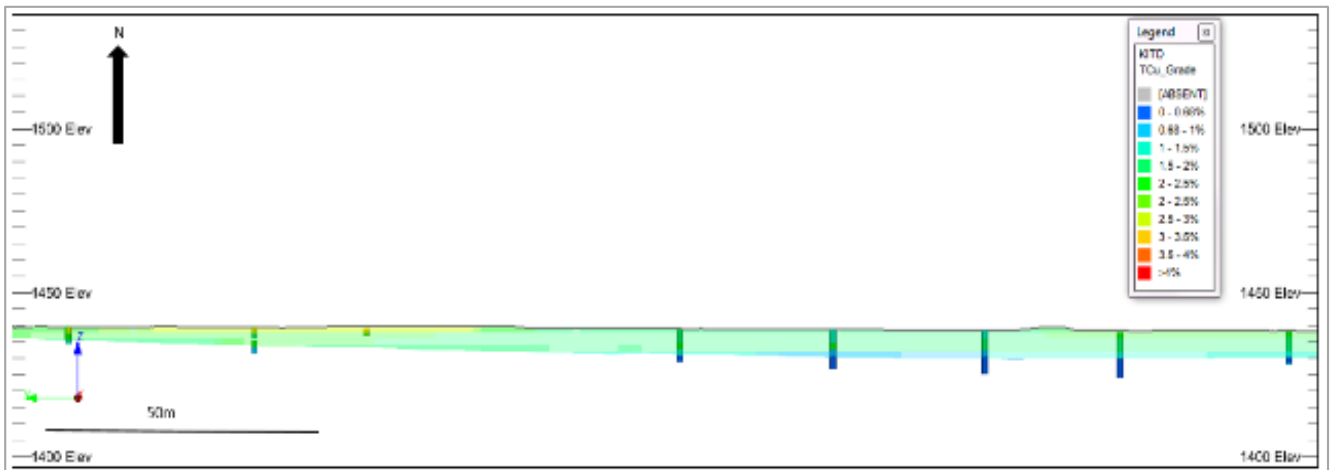


Figure 57: Visual validation of the composited drill holes vs the final block model for %TCu (NS cross-section)

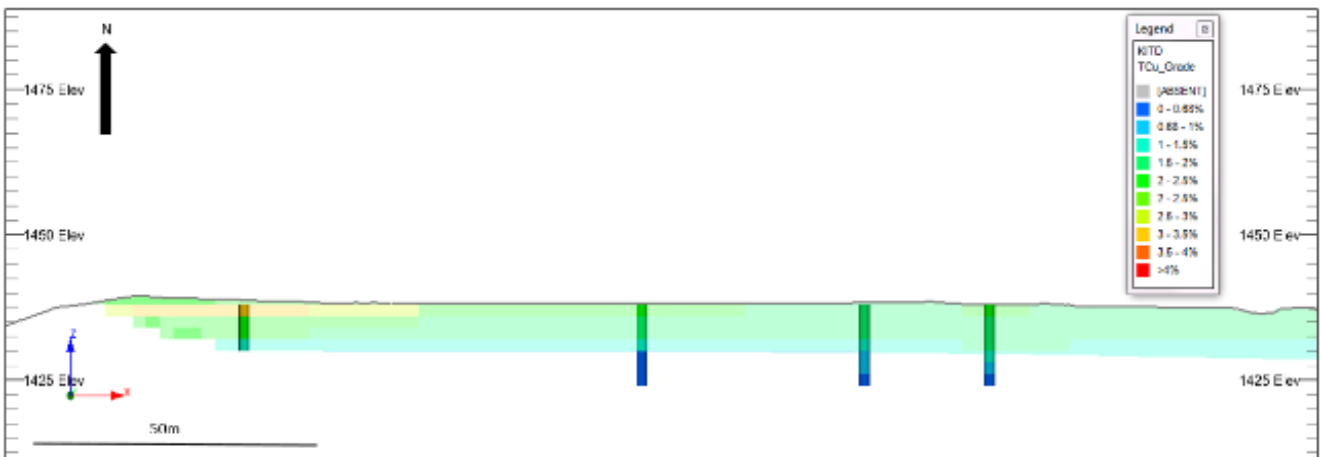


Figure 58: Visual validation of the composited drill holes vs the final block model for %TCu (EW cross-section)

The mean grades of the drill hole composites and the block model were compared for %TCu and %TCo grades by statistical validation and by swath plots (Table 59, Figure 59 and Figure 60).

Table 59: Average grade comparison of the block model versus the drill hole composites

INPUT	DENSITY	%TCu	%TCo	%AsCu	%AsCo
Block model	1.49	1.48	0.16	1.26	0.12
Composites	1.49	1.60	0.17	1.38	0.12

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Figure 59: North-south swath plots for %TCu, %TCo, %AsCu and %AsCo

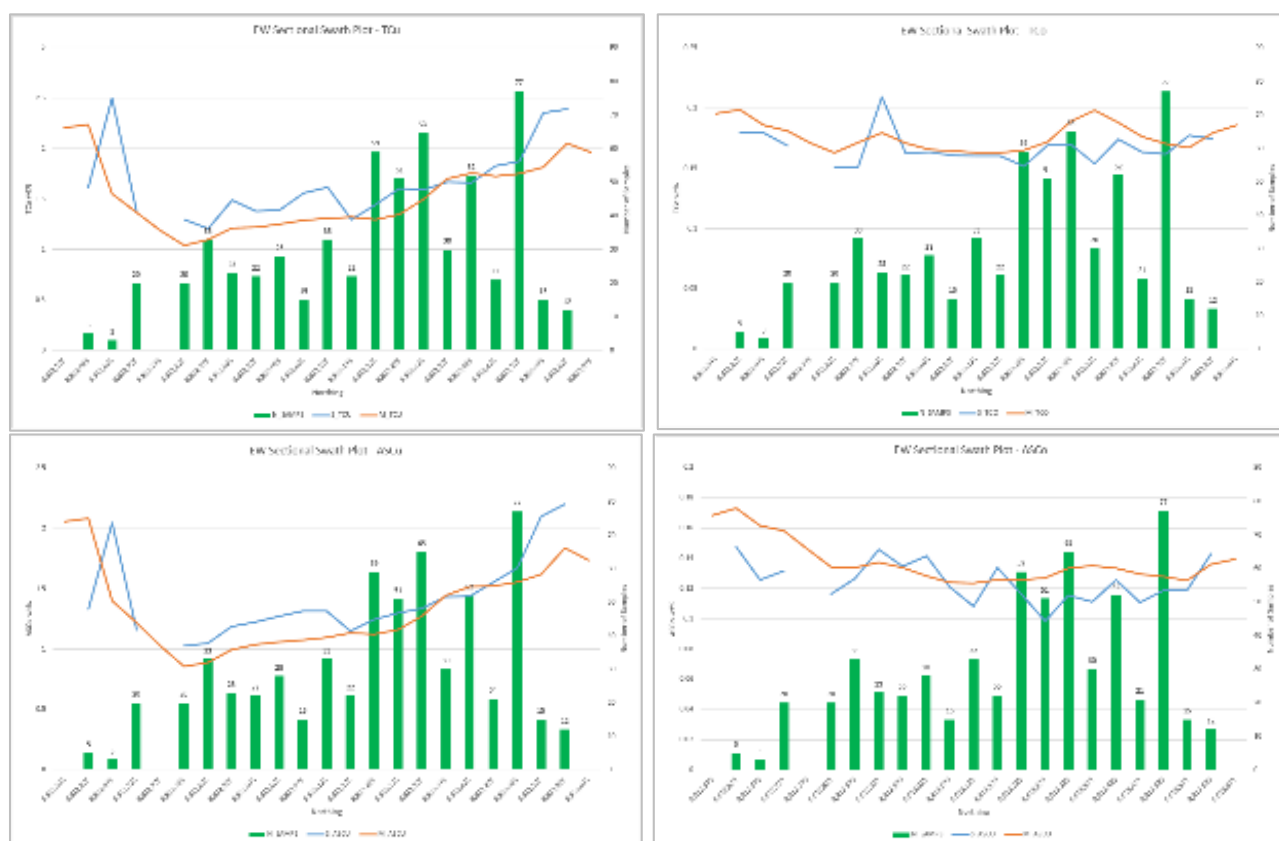


Figure 60: East-west swath plots for %TCu, %TCo, %AsCu and %AsCo

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The results of the above validation indicate that the block model appears to be a representative of the drill hole data and that no significant material bias was identified.

14.5.10 Mineral Resources Classification

Mineral Resources were classified according to CIM 2014 Definition Standards and based on the drill hole spacing.

All resources are classified as an Indicated Mineral Resource, based on the nominal drill hole spacing and as 99% of the tonnages were interpolated during the first run.

14.5.11 Mineral Resources Tabulation

The Mineral Resources for the KITD are tabulated in Table 60. Resources are estimated as at December 31, 2018 and are effective at this date. Mining activities continued during 2019 and KCC experienced ordinary course depletion. Updated Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update, which will reflect such normal course depletion.

Table 60: KITD Mineral Resource tabulation as at December 31, 2018¹⁻⁹

Classification	Mt	%TCu	%TCo
Measured	-	-	-
Indicated	6.1	1.44	0.17
Measured + Indicated	6.1	1.44	0.17
Inferred	-	-	-

- 1) The KITD Mineral Resources have been reported in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) Mineral Resources for the KITD are reported using a cut-off grade of 0.20% TCu and 0.07% TCo;
- 3) Mineral Resources are estimated using an assumed long term price of \$6,500 per tonne of copper and long term price of \$55,000 per tonne of cobalt;
- 4) Grade measurements are reported as percent (%), tonnage measurements are in metric units;
- 5) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 6) Mineral Resources are inclusive of Mineral Reserves;
- 7) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 8) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 9) The Mineral Resource estimates are for KCC's entire interest, whereas KML owns 75% of KCC.

14.6 Tilwezembe OP and Kananga OP

The Mineral Resource estimate for Kananga OP and Tilwezembe OP was reported by SRK on March 17, 2009. The following information was summarized from the 2009 and 2012 Independent Technical Reports (ITR) (SRK, 2009 and Golder, 2012).

No new Mineral Resource models have been completed for these areas since 2009. The Mineral Resources have been updated for mineral extraction and currently are dormant. The Mineral Resource models from which Mineral Resources are quoted in this section are:

- Tilwezembe Mine; and
- Kananga Mine.

No additional information is available.

14.6.1 Drill Hole Data

14.6.1.1 Kananga OP

The historical drilling at Kananga OP was limited to approximately eight drill holes drilled by GCM. Details of the drilling, sample collection and sample analyses for the drill holes are not available.

From April 2006 to December 2007, 51 drill holes were drilled in Kananga totalling 10,300 m. No new exploration has been completed after 2007.

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14.6.1.2 Tilwezembe OP

The historical drilling was undertaken by GCM, and the information was the basis for the maiden Mineral Resource estimate for Tilwezembe OP undertaken by SRK. The drilling information included holes on a 25 m spacing within the operational Tilwezembe OP and 100 m by 50 m on Tilwezembe East. A total of 157 drill holes were drilled at Tilwezembe.

14.6.2 Geological Interpretation

The lithological models within each of the areas are based on the historical and 2008 data. Every structural fragment is composed of different lithological mineralized strata and each was modelled individually for Mineral Resource estimation purposes with an individual estimation domain.

Generally the 2008 drill hole lithological intersections are consistent with the existing historical data, except for a few cases. In these cases the information from the respective drill hole in the lithological model updates was not used.

14.6.3 Statistics and Compositing

14.6.3.1 Tilwezembe OP

The summary statistics of the de-clustered 1 m composite data by the various rock types are presented in Table 61. The composite data was de-clustered using a cell size of 25 m x 25 m x 1 m (elevation) that approximates the drill hole spacing in the closer spaced areas.

Generally, coefficients of variation (CV = standard deviation/mean) for copper and cobalt (total and soluble) in the manganiferous dolomite zones are high (>1.5) and those in the other zones are lower (<1.5). For manganese, CVs in the oxidized zones were generally high (>1.5) and low in the sulphide zones (<1.5). This demonstrates the variability on the dolomite zones and the oxidized ones. All density CVs were low and displayed low variability.

Table 61: Tilwezembe OP statistics from the 1 m composites by lithology type

Domain	Variables (%)	No. Samples	Minimum	Maximum	Mean	CV
OX_MNDOL	TCu	1,054	0.01	26.50	1.27	1.91
OX_BREC	TCu	485	0.10	19.84	3.78	0.89
OX_TILAR	TCu	674	0.05	4.92	0.56	1.05
SL_MNDOL	TCu	511	0.03	37.00	1.19	2.42
SL_BRED	TCu	339	0.02	21.38	3.29	0.97
SL_TILAR	TCu	406	0.01	6.27	0.53	1.28
OX_MNDOL	AsCu	1,054	0.01	14.12	0.91	1.85
OX_BREC	AsCu	485	0.05	14.28	3.16	0.96
OX_TILAR	AsCu	674	0.01	4.90	0.44	1.17
SL_MNDOL	AsCu	511	0.01	6.88	0.24	2.97
SL_BRED	AsCu	339	0.01	3.52	0.33	1.19
SL_TILAR	AsCu	406	0.01	1.09	0.13	1.21
OX_MNDOL	TCu	1,054	0.01	11.15	0.50	1.47
OX_BREC	TCu	485	0.01	13.47	0.96	1.58
OX_TILAR	TCu	674	0.01	3.61	0.29	1.02
SL_MNDOL	TCu	511	0.01	7.94	0.38	1.52
SL_BRED	TCu	339	0.03	4.98	1.19	1.01

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SL_TILAR	TCu	406	0.02	4.00	0.34	1.15
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14.6.3.2 Kananga OP

The summary statistics of the de-clustered 1 m composite data of the various rock types are presented in Table 62. The composite data was de-clustered using a cell size of 25 m x 25 m x 1 m (elevation) that approximates the drill hole spacing in the closer spaced areas.

Table 62: Kananga OP statistics from the 1 m composites by lithology type

Domain	Variables (%)	No. Samples	Minimum	Maximum	Mean	CV
UOX_OX	TCu	250	0.08	10.05	1.13	1.10
MID_OX	TCu	528	0.02	0.64	0.16	0.60
LOB_OX	TCu	297	0.03	9.28	1.93	0.80
UOB_SL	TCu	122	0.01	6.10	1.83	0.60
MID_SL	TCu	269	0.02	1.00	0.26	0.60
LOB_SL	TCu	234	0.13	6.75	2.18	0.60
UOX_OX	AsCu	250	0.02	9.05	0.85	1.20
MID_OX	AsCu	528	0.01	0.62	0.12	0.80
LOB_OX	AsCu	297	0.01	9.26	1.26	1.10
UOB_SL	AsCu	122	0.01	1.61	0.15	1.20
MID_SL	AsCu	269	0.01	0.39	0.07	1.20
LOB_SL	AsCu	234	0.01	2.99	0.22	1.60
UOX_OX	TCu	250	0.06	4.51	0.64	1.20
MID_OX	TCu	528	0.02	2.73	0.27	0.80
LOB_OX	TCu	297	0.02	3.37	0.70	0.80
UOB_SL	TCu	122	0.01	4.79	0.94	1.10
MID_SL	TCu	269	0.06	2.03	0.53	0.60
LOB_SL	TCu	234	0.02	3.49	1.05	0.70

14.6.4 Dry Bulk Density

According to the GCM criterion, waste rock was generally assigned a density of 2.0 t/m³ if it was siliceous and 2.4 t/m³ if the rock was considered dolomitic.

SRK reviewed the historical assayed dataset for all the projects in the application of these criteria and found that there were proportionately fewer assays for %CaO than the %TCu assays available for these criteria to be applied (SRK 2009). However, SRK consider these values as the guidelines for the possible ranges of density within the respective mineralized zones.

A total of 212 density determinations were undertaken on the selected lithological cores using the Archimedes' Principle by which a sample is weighed in air and then in water using a Clover Scale. Table 63 indicates the dry bulk densities applied to the various project areas. It is unknown whether outliers were removed from the final average dry bulk density calculation.

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Table 63: Bulk density determinations on various lithologies

Project Area	Mineralized Zone	Density (t/m ³)
Tilwezembe OP	Ox_MnDol	1.96
	Ox_Brec	1.81
	Ox_TillArg	1.98
	SI_MnDol	2.26
	SI_Brec	2.24
	SI_TillArg	2.18
Kananga OP	Upper orebody oxides (UOB_OX)	1.8
	Middle low-grade oxides(MID_OX)	1.8
	Lower orebody oxides (LOB_OX)	2.0
	Upper orebody sulphides (UO_SL)	2.1
	Middle low-grade sulphides (MID_SL)	2.0
	Lower orebody sulphides (LOB_SL)	2.1

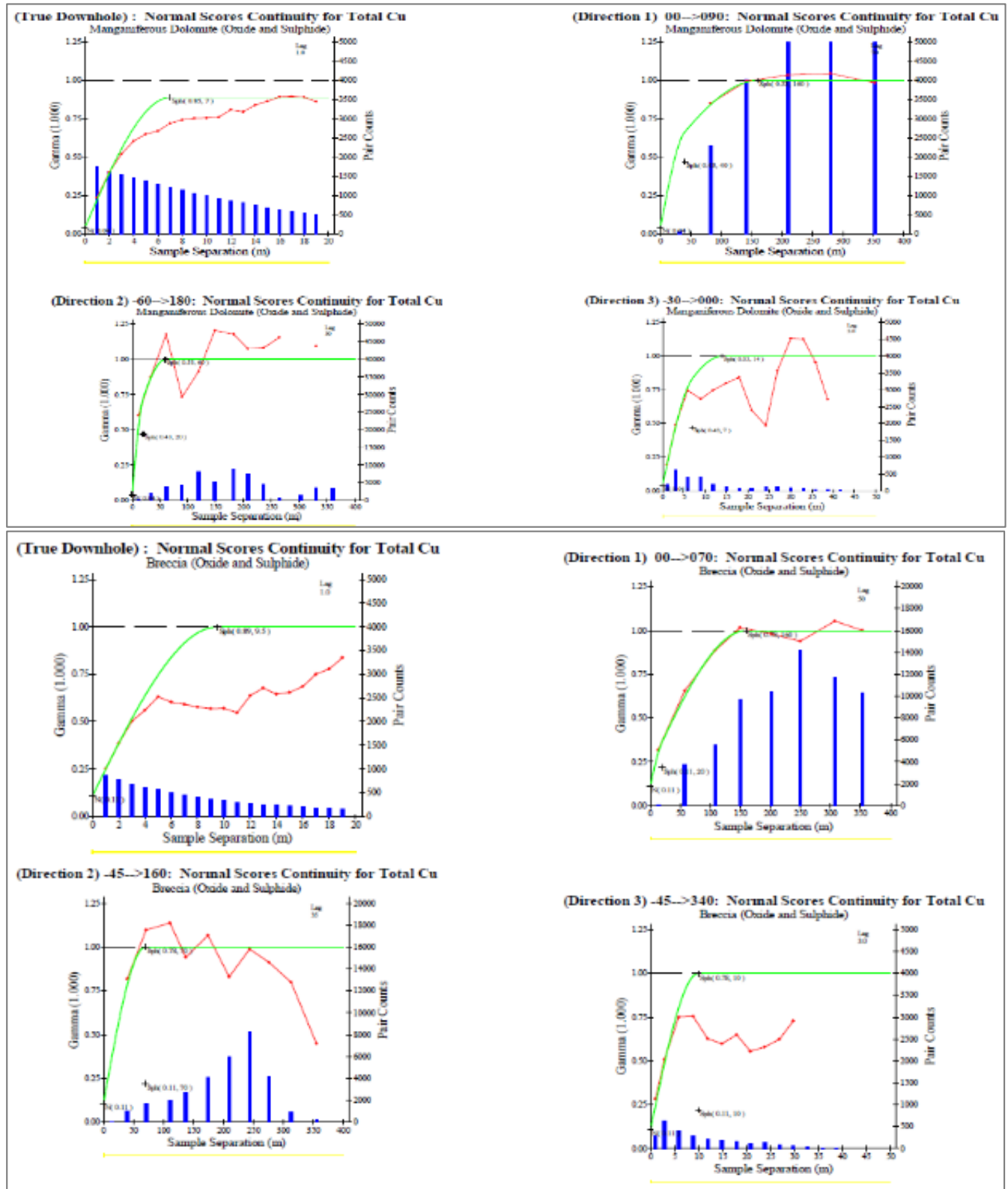
14.6.5 Variography

14.6.5.1 Tilwezembe OP

Traditional semi-variograms were calculated from the selected composite data using Supervisor software. To improve the variogram structures, a normal score transform was performed on the composites before semi-variogram calculation.

In the sulphide zone, where drilling was limited relative to the oxide zone, acceptable semi-variograms for OK were not obtained. Combining the two zones, however, did result in an adequate amount of the data to calculate reasonable and robust variograms. As a result, variograms were modelled for the entire brecciated zone, the entire manganiferous dolomite zone and the entire low-grade argillite/tillite zone. The %TCu variogram model for Tilwezembe OP is given in Figure 61.

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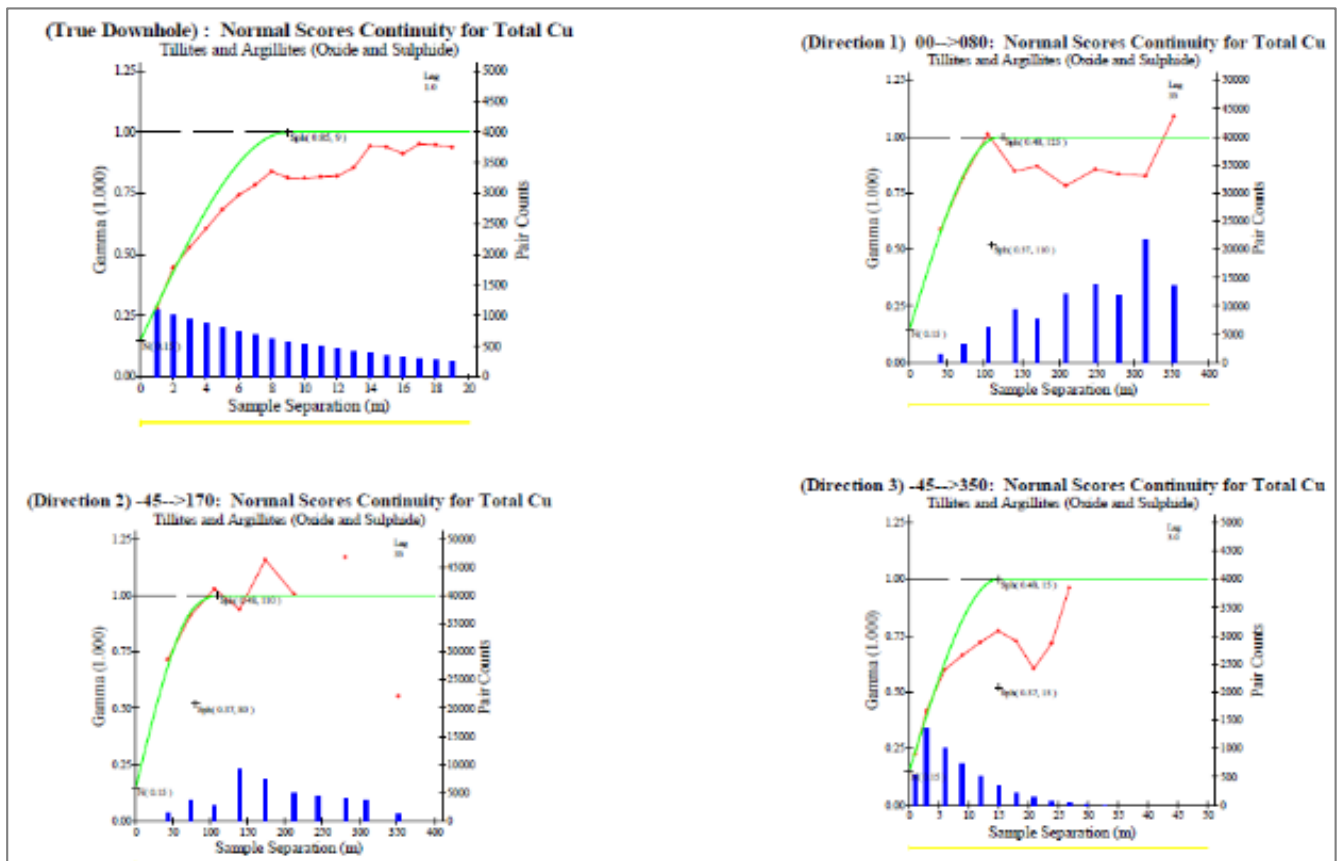


Figure 61: %TCu variograms for Tilwezembe OP fragment

Where two elements were highly correlated (correlation coefficient >0.7), the variogram model of the most significant grade variable was applied directly to the second variable. As the semi-variograms were predominantly calculated from the oxide zone data that was more extensively drilled than the sulphide zone, the oxide correlation matrices were considered.

High correlations existed between the total copper and the soluble copper and between the total cobalt and the soluble cobalt. As a result, the total semi-variograms were directly applied to the soluble elements to preserve the correlations. Therefore, semi-variograms were modelled only for the total copper, total cobalt and manganese.

The directions of continuity for these elements were evaluated by making use of the semi-variogram contours on the horizontal, across-strike and dip planes. This allowed for the determination of the strike, dip and plunge continuity. The calculated experimental semi-variograms for each of the attributes were modelled using spherical scheme models with two or three structures. All variograms were standardized to a sill of one. The nugget effect was determined by extrapolating the first two points of the downhole variogram to the Y-axis.

14.6.5.2 Kananga OP

Traditional semi-variograms were calculated from the selected composite data using Supervisor software. To improve the variogram structures, normal scores transform was performed on the composites before semi-variogram calculation.

In the sulphide zone, where drilling was limited relative to the oxide zone, acceptable semi-variograms were not obtained. Combining the two zones resulted in an adequate amount of data to calculate reasonable and robust variograms.

Where two elements were highly correlated (correlation coefficient >0.7), the variogram model of the most significant grade variable was applied directly to the second variable. As the semi-variograms were

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predominantly calculated from the oxide zone data that was more extensively drilled than the sulphide zone, the oxide correlation matrices were considered. The %TCu variogram model for Kananga is given in Figure 62.

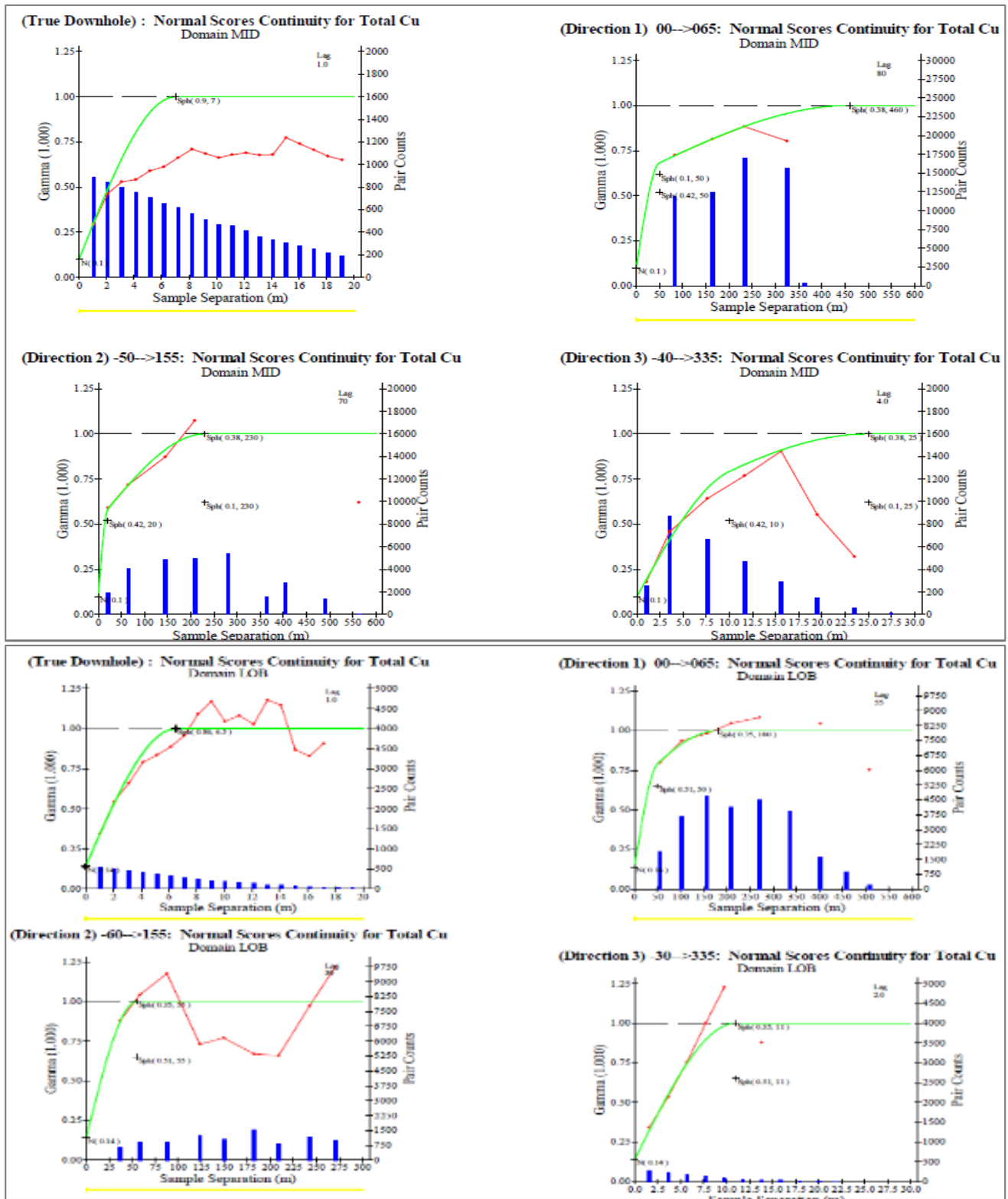


Figure 62: %TCu variograms for Kananga OP fragment

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Strong correlations exist between the total copper and the soluble copper, and between the total cobalt and the soluble cobalt. As a result, the total semi-variograms were directly applied to the soluble elements to preserve the correlations. Therefore, semi-variograms were only modelled for the total copper, cobalt and manganese.

The directions of continuity for these elements were evaluated by making use of semi-variogram contours on the horizontal, across-strike and dip planes. This allowed for the determination of the strike, dip and plunge continuity. The calculated experimental semi-variograms for each of the attributes were modelled using the spherical models with two or three structures. All variograms were standardized to a sill of one. The nugget effect was determined by extrapolating the first two points of the downhole variogram to the Y-axis.

14.6.6 Mineral Resource Estimation

14.6.6.1 Tilwezembe OP

During the grade estimation process, Snowden applied various top cuts to the data to limit the influence of high grades in the estimation into the various lithological domains and for the various grade fields, %TCu, %AsCu, %TCo, %AsCo and Mn. No details were available regarding the actual top cut values used in the Mineral Resource estimate.

OK was used for grade estimation. The defined mineralized zones were subdivided into two structural domains based on the strike orientation. Samples used to estimate the respective blocks were sourced within an ellipsoidal search oriented according to the structural domain in which a mineralization block occurred.

Composites used in the estimation process were selected with aa search distances of 130 m x 60 m x 12 m respectively along strike, across strike and perpendicular to the strike dip plane directions. The search distances are equivalent to the variogram ranges.

Where blocks remained un-estimated after the first pass, the search neighbourhood size was doubled or tripled.

14.6.6.2 Kananga OP

Similar procedures to those adopted for Tilwezembe OP were applied for Kananga OP.

- OK was used for grade estimation;
- Top cutting was applied;
- The mineralized zones were split into two structural domains;
- Composites used in the estimation process were selected within a search distances of 180 m x 60 m x 10 m equivalent to the variogram ranges; and
- Where blocks remained un-estimated after the first pass, the search neighbourhood was doubled or tripled.

14.6.7 Block Model Validation

The block models from the respective projects were validated using swath plots, comparing the mean values of the composites against the mean values of the blocks. The swath plots were specified along an easting, and composites and block model data were selected within the search limits on either side of the easting for the comparisons.

No material issues were identified during validation.

14.6.8 Mineral Resources Classification

The Mineral Resources for Kananga OP and Tilwezembe OP were originally classified in accordance with the 2007 SAMREC Code.

Mineral Resources stated in accordance with the CIM 2014 Definition Standards on Mineral Resources and Mineral Reserves are expected to comprise no significant difference.

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The following criteria was used in the classification of Tilwezembe OP and Kananga OP (SRK, 2009):

- The risk in the data informing the Mineral Resource estimate;
- The robustness of the geological models; and
- The risk in the grade estimates.

SRK established the following classification criteria for Tilwezembe OP and Kananga OP:

- **Indicated Mineral Resource:** where the estimates were within the first search, equivalent to the variogram range; and
- **Inferred Mineral Resource:** where there is a paucity of data to reasonably define the Mineral Resource.

14.6.9 Mineral Resource Tabulation

Table 64 presents KCC's consolidated Mineral Resource statement for Tilwezembe OP and Kananga OP as at 31 December 2018. The Mineral Resource estimate for Tilwezembe OP and Kananga OP is same as reported in 2012 (Golder, 2012). The Mineral Resource estimate were prepared and signed-off by SRK in 2009 (SRK, 2009) and QPs for this TR have relied on the previous work undertaken.

Table 64: Tilwezembe OP and Kananga OP Mineral Resources as at 31 December 31, 2018¹⁻⁸

Classification	Project Area	Mt	%TCu	%TCo
Indicated	Kananga OP	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60
	Total	13.6	1.81	0.66
Inferred	Kananga OP	4.0	2.00	0.98
	Tilwezembe OP	13.8	1.75	0.60
	Total	17.8	1.81	0.69

- 1) Mineral Resources have been reported in accordance with classification criteria of South African Code for the Reporting of Mineral Resources and Mineral Reserves (SAMREC Code). If the definitions and classification standards adopted in NI 43-101 had been used instead of those of the SAMREC Code, the estimates of Mineral Resources would be substantially similar;
- 2) Mineral Resources are reported using cut-off grades 0.5% TCu for Kananga OP and Tilwezembe OP;
- 3) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 4) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal place;
- 5) Mineral Resources are inclusive of Mineral Reserves;
- 6) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 7) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 8) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

14.7 T17

The T17 block model and Mineral Resource estimate was originally created by CICC in 2008 and signed-off by SRK. The model was updated by KCC in 2011 and signed-off by Golder. The following information was summarized from 2012 ITR (Golder, 2012) to present a technical overview of the T17 deposit. There has been no Mineral Resource update for the T17 deposit since 2011 and the QPs for this TR have relied on the information from the previous resource estimate.

14.7.1 Drill Hole Data

The T17 deposit was the subject of two diamond drilling programs by GCM, with 3,287 m drilled between 1938 and 1954, and 8,011 m drilled from 1986 to January 1988. The drill holes were drilled generally to a nominal 100 m by 100 m grid, with certain areas on 50 m by 50 m spacing.

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KCC drilled 20 drill holes in 2010 within the current pit perimeter to confirm the continuation of the Mineral Resource at depth. A total of 4,286 m has been drilled, and the result from this drilling indicates that mineralization continues down-dip to a depth of more than 180 m below the current pit design floor.

14.7.2 Geological Interpretation

The T17 lithological model was based on the historical and KCC drill holes. The deposit was sub-divided into three sub-fragments, separated by faults, breccia and waste intrusions as displayed in the drill hole data. Each fragment is composed of different lithological mineralized strata and each was modelled individually for the Mineral Resource estimation purposes with an individual estimation domain.

14.7.3 Statistics

The lithological wireframes were used to extract the sample data and the data was composited to 2.5 m lengths. Statistics from the composite files are given in Table 65 as produced by GAA in ITR 2012.

Table 65: Statistics from the 2.5 m composites by lithology type

Lithology	Variables (%)	No. Samples	Minimum	Maximum	Mean	CV
DSTRAT	TCu	360	0.02	12.64	4.32	3.05
RSF		601	0.07	16.79	3.20	2.85
RSC		1,059	0.02	35.03	1.44	2.91
SDB		1,389	0.01	22.69	5.12	3.96
BOMZ		211	0.10	18.71	2.53	2.84
RATGRIS		163	0.09	10.00	3.47	2.79
DSTRAT	TCu	360	0.00	4.45	0.41	0.53
RSF		601	0.03	7.60	0.88	1.05
RSC		1,050	0.00	9.21	0.67	1.02
SDB		1,389	0.00	8.55	0.80	1.17
BOMZ		211	0.05	2.23	0.47	0.48
RATGRIS		163	0.01	2.84	0.30	0.34

14.7.4 Dry Bulk Density

CCIC undertook the limited bulk density determinations of the various stratigraphic units to verify GCM empirical densities. The determinations were undertaken on the selected lithological cores using the Archimedes' Principle by which a sample is weighed in air and then in water using a Clover Scale. It is unknown whether the outliers were removed from the final average bulk density calculation.

KCC personnel undertook a new density determination study on the diamond drilling core from the recent exploration program of T17. The results from this study, as applied to the updated T17 resource block model are shown in Table 66.

Table 66: Bulk density assignment for T17

Stratigraphic Unit	Number of Samples	Minimum (t/m ³)	Maximum (t/m ³)	Average (t/m ³)
BOMZ	8	1.20	2.80	2.09
SDB	18	2.00	2.80	2.32
RSC	18	2.10	2.50	2.33
RSF	17	2.00	2.50	2.29

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Stratigraphic Unit	Number of Samples	Minimum (t/m ³)	Maximum (t/m ³)	Average (t/m ³)
DSTRAT	15	1.70	2.80	2.33
RATGRIS	11	2.10	2.70	2.36

14.7.5 Variography

Omni-directional variograms were generated in each of the lithologies within the T17 deposit. There was limited sample data precluding generation of reliable directional variograms. The variogram parameters are provided in Table 67.

Table 67: Variography parameters for T17

Zone	Variable (%)	C0 Nugget Variance	C1 Variance	Range (m)	C2 Variance	Range2 (m)
RATGRIS	TCu	0.898	5.21	75.00	0.00	0.00
DSTRAT		3.951	4.54	75.00	0.00	0.00
RSF		4.574	2.15	75.00	0.00	0.00
RSC		7.200	4.56	58.00	1.23	73.00
SDB		9.000	6.70	150.00	0.00	0.00
BOMZ		9.200	1.00	100.00	5.29	150.00
RATGRIS/DS	TC0	0.328	0.54	300.00	0.00	0.00
RSC		0.654	0.07	75.00	0.25	150.00
BOMZ		0.026	0.10	80.00	0.08	150.00
SDB		0.350	0.90	35.00	0.08	150.00

14.7.6 Mineral Resource Estimation

KCC updated the T17 Mineral Resource model in 2012. Composites for the Mineral Resource estimation were selected within each lithology to estimate the respective lithology. The criterion adopted by CCIC and SRK to account for absent data, where 0.01% TCu was assigned for unsampled sections in the drill hole, was also applied by KCC to estimate grades using the three-pass search ellipse method. The first search neighbourhood distance was set equivalent to the variogram range as indicated by the model parameters in Table 67.

14.7.7 Block Model Validation

The T17 block model was validated using swath plots, comparing the mean values of the composites against the mean values of the blocks. The swath plots were specified along an easting, and composites and block model data were selected within the search limits on either side of the easting for the comparisons.

No material issues were identified during the validation process and the average grade conformance considered acceptable.

14.7.8 Mineral Resources Classification

The Mineral Resources for T17 were originally estimated and classified in accordance with the JORC 2012 Code. However, the existing estimates are considered substantially similar stated in accordance with the CIM 2014 Definition Standards on Mineral Resources and Mineral Reserves.

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14.7.9 Mineral Resource Tabulation

Table 68 presents the T17 Mineral Resource statement as at December 31, 2018. There has been no Mineral Resource update for the T17 deposit since 2011 and QPs for this TR have relied on the prior work reported in 2012 (Golder, 2012).

Table 68: Mineral Resources of T17 as at 31 December 2018¹⁻⁸

Classification	Project Area	Mt	%TCu	%TCo
Measured	T17 UG	4.2	2.66	0.51
Indicated	T17 UG	9.4	4.44	0.65
Measured and Indicated	T17 UG	13.6	3.89	0.61
Inferred	T17 UG	5.2	4.21	0.98

- 1) Mineral Resources for T17 UG have been reported in accordance with the definitions and classification criteria standards adopted of in the JORC Code for the Reporting of Mineral Resources and Mineral Reserves. If the definitions and classification standards adopted in NI 43-101 had been used instead of those adopted in the JORC Code, the estimates of Mineral Resources would be substantially similar;
- 2) Mineral Resources are reported using a cut-off grade of 0.4% TCu;
- 3) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 4) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 5) Mineral Resources are inclusive of Mineral Reserves;
- 6) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 7) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 8) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

14.8 Consolidated Mineral Resource Statements

The Mineral Resources for KCC are reported in accordance with NI43-101 and have been estimated in conformity with generally accepted CIM 2014 Definition Standards.

Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability. There is no certainty that all or any part of this Mineral Resource will be converted into Mineral Reserves.

The consolidated Mineral Resource estimate was compiled by Christiano Santos Goncalves MAusIMM CP Geo (306079), a QP as is defined in NI 43-101. The effective date of this Mineral Resource estimate is December 31, 2018. For this TR, Golder has the overall responsibility to sign-off on the Mineral Resources for KOV OP, KTE, KTO UG, KITD and Mashamba East OP, and has relied on prior work as outlined in Table 29 for Tilwezembe OP, Kananga OP and T17 UG. Table 69 summarizes the Mineral Resources for the seven KCC areas. Details of reporting cut-off grades that have been applied are detailed in the notes below.

. Resources are estimated as at December 31, 2018 and are effective at this date. Mining activities continued during 2019 and KCC experienced ordinary course depletion. Updated Mineral Resources will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update, which will reflect such normal course depletion.

Table 69: Consolidated Mineral Resource as at December 31, 2018¹⁻¹⁶

Mineral Resource Classification	Project Area	Mt	%TCu	%TCo
Measured Resource	KTO UG	12.1	3.90	0.59
	Mashamba East OP	-	-	-
	Mashamba East UG	-	-	-
	T-17 UG	4.2	2.66	0.51

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Mineral Resource Classification	Project Area	Mt	%TCu	%TCo
	KOV OP and KTE	-	-	-
	Kananga OP	-	-	-
	Tilwezembe OP	-	-	-
	KITD Tailings	-	-	-
	Subtotal	16.3	3.58	0.57
Indicated Resource	KTO UG	65.6	3.92	0.46
	Mashamba East OP	35.9	1.72	0.61
	Mashamba East UG	21.9	1.72	0.67
	T-17 UG	9.4	4.44	0.65
	KOV OP and KTE	106.8	4.8	0.53
	Kananga OP	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60
	KITD Tailings	6.1	1.44	0.17
	Subtotal	259.3	3.64	0.54
Measured and Indicated Resource	KTO UG	77.7	3.92	0.48
	Mashamba East OP	35.9	1.72	0.61
	Mashamba East UG	21.9	1.72	0.67
	T-17 UG	13.6	3.89	0.61
	KOV OP and KTE	106.8	4.8	0.53
	Kananga OP	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60
	KITD Tailings	6.1	1.44	0.17
	Subtotal	275.6	3.64	0.54
Inferred Resource	KTO UG	48.4	3.83	0.38
	Mashamba East OP	8.3	2.58	0.55
	Mashamba East UG	7.6	2.87	0.48
	T-17 UG	5.2	4.21	0.98
	KOV OP and KTE	77.6	4.40	0.38
	Kananga OP	4.0	2.00	0.98
	Tilwezembe OP	13.8	1.75	0.60
	KITD Tailings	-	-	-
	Subtotal	164.9	3.78	0.44

- 1) Mineral Resources for KOV, KTE, KTO, KITD Tailings and Mashamba East have been reported in accordance with the definitions and classification standards adopted in NI 43-101;
- 2) Mineral Resources for T17 have been reported in accordance with the definitions and classification criteria of JORC Code for the Reporting of Mineral Resources and Mineral Reserves. If the definitions and classification standards adopted in NI 43-101 had been used instead of those adopted in the JORC Code, the estimates of Mineral Resources would be substantially similar;
- 3) Mineral Resources for Kananga and Tilwezembe have been reported in accordance with SAMREC Code. If the definitions and classification standards adopted in NI 43-101 had been used instead of those of the SAMREC Code, the estimates of Mineral Resources would be substantially similar;
- 4) Mineral Resources for KOV and KTE are reported using cut-off grades 0.46% TCu or 0.12% TCo;
- 5) Mineral Resources for KTO are reported using cut-off grades 1.00% TCu;

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- 6) OP Mineral Resources for Mashamba East are reported using a total copper (%TCu) cut-off grade of 0.46% for copper only grade shell. A total cobalt (%TCo) cut-off grade of 0.12% is used for reporting Mineral Resources within the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at either cut-off;
- 7) UG Mineral Resources for Mashamba East are reported using a total copper (%TCu) cut-off grade of 1.00% for the copper only grade shell. A total cobalt (%TCo) cut-off grade of 0.517% %TCo is used for reporting Mineral Resources within the cobalt only grade shell. Mineral Resources within the copper and cobalt grade shells are reported at either cut-off;
- 8) Mineral Resources for Kananga and Tilwezembe are reported using cut-off grade of 0.5% TCu;
- 9) Mineral Resources for T17 are reported using cut-off grades 0.4% TCu;
- 10) Mineral Resources for KITD are reported using cut-off grades 0.20% TCu and 0.07% TCo;
- 11) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 12) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 13) Mineral Resources are inclusive of Mineral Reserves;
- 14) Mineral Resources are not Mineral Reserves and do not necessarily demonstrate economic viability;
- 15) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 16) The Mineral Resource estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

A comparison of the Mineral Resource between December 31, 2017 and December 31, 2018 is shown in Table 70.

There was a decrease of 21.79% in the Indicated Mineral Resources for KITD Tailings due to hydraulic mining activity in 2018.

The decrease of 3.67% in the Indicated Mineral Resources and 13.11% in the Inferred Mineral Resources of Mashamba East OP is due to mining activities in 2018.

There was a decrease of 3.26% in the Indicated Mineral Resources and 0.51% in the Inferred Mineral Resources of KOV OP.

There was a decrease of 0.51% in the Measured and Indicated Mineral Resources and 0.21% in the Inferred Mineral Resources for KTO UG due to mining activities in 2018.

There were no changes in the Mineral Resources reported for:

- T17-UG;
- Kananga OP; and
- Tilwezembe OP.

The overall change of Mineral Resources (Measured, Indicated and Inferred) is -2.39% decrease in tonnes due to mining activity in 2018.

It is important to highlight that in KOV OP there are instances of isolated opportunistic mineralized material which were being mined during 2018. This material is not reflected in the Mineral Resource model because exploration or infill drilling of this material is ongoing. This material has not been included in the Mineral Resource reporting, and there are future plans to progress infill drilling in these areas to improve the geological confidence and thus include it in future resource model updates.

Table 70: Comparison of the December 31, 2018 Mineral Resource with the December 31, 2017

Classification	Project Area	December, 2018			December, 2017		
		Mt	%TCu	%TCo	Mt	%TCu	%TCo
Measured Resource	KTO UG	12.1	3.90	0.59	12.2	3.90	0.59
	Mashamba East OP UG	-	-	-	-	-	-
	T17 UG	4.2	2.66	0.51	4.2	2.66	0.51
	KOV OP KTE	-	-	-	-	-	-
	Kananga OP	-	-	-	-	-	-
	Tilwezembe OP	-	-	-	-	-	-

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Classification	Project Area	December, 2018			December, 2017		
		Mt	%TCu	%TCo	Mt	%TCu	%TCo
	KITD Tailings	-	-	-	-	-	-
	Subtotal	16.3	3.58	0.57	16.4	3.58	0.57
Indicated Resource	KTO UG	65.6	3.92	0.46	65.9	3.92	0.46
	Mashamba East OP UG	57.8	1.72	0.63	60.0	1.68	0.62
	T17 UG	9.4	4.44	0.65	9.4	4.44	0.65
	KOV OP KTE	106.8	4.8	0.53	110.4	4.75	0.53
	Kananga OP	4.1	1.61	0.79	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60	9.5	1.89	0.60
	KITD Tailings	6.1	1.44	0.17	7.8	1.49	0.16
	Subtotal	259.3	3.64	0.54	267.1	3.60	0.53
Measured and Indicated Resource	KTO UG	77.7	3.92	0.48	78.1	3.92	0.48
	Mashamba East OP UG	57.8	1.72	0.63	60.0	1.68	0.62
	T17 UG	13.6	3.89	0.61	13.6	3.89	0.61
	KOV OP KTE	106.8	4.8	0.53	110.4	4.75	0.53
	Kananga OP	4.1	1.61	0.79	4.1	1.61	0.79
	Tilwezembe OP	9.5	1.89	0.60	9.5	1.89	0.60
	KITD Tailings	6.1	1.44	0.17	7.8	1.49	0.16
	Total	275.6	3.64	0.54	283.5	3.60	0.54
Inferred Resource	KTO UG	48.4	3.83	0.38	48.5	3.83	0.38
	Mashamba East OP UG	15.9	2.72	0.51	18.3	2.85	0.47
	T17 UG	5.2	4.21	0.98	5.2	4.21	0.98
	KOV OP KTE	77.6	4.40	0.38	78.0	4.40	0.38
	Kananga OP	4.0	2.00	0.98	4.0	2.00	0.98
	Tilwezembe OP	13.8	1.75	0.60	13.8	1.75	0.60
	KITD Tailings	0.0	0.00	0.00	0.0	0.00	0.00
	Total	164.9	3.78	0.44	167.8	3.79	0.44

14.9 Interpretation and Conclusions

14.9.1 Mineral Resources

This report has been prepared in accordance with the definitions and standards of NI 43-101. All Mineral Resource estimates stated therein have been classified in accordance with CIM 2014 Definition Standards.

The consolidated Mineral Resources of the KCC project areas as of 31 December 2018 consists of 16.3 Mt of Measured Mineral Resource at average grades of 3.58% TCu and 0.57% TCo, and 259.3 Mt of Indicated Mineral Resource at average grades of 3.64% TCu and 0.54% TCo, with an additional 164.9 Mt of Inferred Mineral Resource at average grades of 3.78% TCu and 0.44% TCo. The cut-off grades used to estimate the Mineral Resources for each area is provided in Section 14.8.

Mr. Christiano Santos Goncalves, MAusIMM CP Geo, (306079), is the QP for the compilation of the Mineral Resources, and has visited the site in November 2017 and December 2018, reviewed the KCC data, including the geological reports, maps, technical papers, digital data including lab results, sample analyses and other relevant information. The QP believes that the data presented is an accurate and reasonable representation of the KCC operations and concludes that the data related to the 2018 Mineral Resource estimates is of suitable quality to provide the basis for the relevant conclusions and recommendations made in this report.

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14.9.2 Risks

The Mineral Resource classification provides a reasonably accurate summary of the risks associated with each mineral domain. Particular reference is noted on data completeness and validation issues associated with the historical data including, but not limited to, the lack of sampling in the RSC ore zone, the absence of QA/QC, limited assaying of acid soluble copper and acid soluble cobalt, the limited number of density samples, and irregular sample lengths generally biasing the data towards the high-grade mineralization.

There are no known environmental, permitting, legal, title, socio-economic, marketing, political, or other relevant risk factors which could materially impact the resource estimates. For a complete list of project risks refer to Section 25.3.

The 2018 Mining Code was promulgated in the in the DRC on 9 March 2018. On 16 June 2018, the Secretary General of the Ministry of Mines issued an instruction to start applying the 2018 Mining Code immediately. Further detail on the 2018 Mining Code is included in Sections 4.6.1 and 25.3.7.3.

14.9.3 Opportunities

Based on the information collected to date, there is an opportunity to increase the size and the confidence (Mineral Resource classification) of the Mineral Resource with future infill and exploration drilling. Mashamba East, KTO and KOV have a high probability of being upgraded to an Indicated or Measured Mineral Resources with the completion of adequate infill drilling and better understanding of the geology.

Re-sampling of the historical data in the inferred areas of Mashamba East, KTO and KOV is required to confirm the previous results and potentially upgrade the classification.

Development of a robust reconciliation study to support the classification of Measured Mineral Resources where appropriate.

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15.0 KCC LOM Mineral Reserve Estimates

15.1 Consolidated LOM Mineral Reserve Estimates

The KCC LOM Mineral Reserves have been prepared in accordance with the classification criteria of the Canadian Securities Administrators' National Instrument 43-101 – Standards of Disclosure for Mineral Projects (NI 43-101). The basis of the Mineral Reserve estimate such as key assumptions, parameters and methods used are described in Chapter 16.0. The total Mineral Reserve estimate at KCC is 132.7 Mt at 3.18% TCu of which 8.7 Mt is from the Proven Mineral Reserve category as shown in Table 71. Mineral Reserves are estimated as at December 31, 2018 and are effective at this date. Mining activities continued during 2019 and KCC experienced ordinary course depletion. Updated Mineral Reserves will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update, which will reflect such normal course depletion.

Table 71: KCC consolidated LOM Mineral Reserve estimate as at 31 December 2018¹⁻¹¹

Mining operations	Proved			Probable			Total		
	Mt	%TCu	%TCo	Mt	%TCu	%TCo	Mt	%TCu	%TCo
KTO underground	6.5	3.60	0.56	20.1	3.34	0.53	26.6	3.40	0.54
T-17 underground	2.2	3.42	0.54	9.1	3.71	0.64	11.3	3.65	0.62
Mashamba East open pit	-	-	-	31.6	2.15	0.60	31.6	2.15	0.60
KOV open pit	-	-	-	57.5	3.72	0.48	57.5	3.72	0.48
KITD (KCC PE525)	-	-	-	2.8	1.42	0.16	2.8	1.42	0.16
KITD (GCM PE8841)	-	-	-	2.9	1.45	0.17	2.9	1.45	0.17
Total	8.7	3.56	0.55	124.0	3.15	0.51	132.7	3.18	0.52

- 1) The Mineral Reserve estimates have been prepared in accordance with the classification criteria of the NI 43-101;
- 2) The Mineral Reserves for all areas except KTO UG, KOV OP and KITD Tailings are same as at December 31, 2016
- 3) The KITD Tailings Mineral Reserve within KCC lease PE525 are reported at 2.8 Mt and within GCM lease PE8841 are reported at 2.9 Mt
- 4) KITD Tailings portion situated on PE8841 belonging to GCM for which KCC has the legal right to mine the tailings discharged from its KTC operation as per the provisions of the AJVA concluded with GCM;
- 5) Grade measurements reported as a percent (%), tonnage measurements are in metric units;
- 6) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 7) Projects included were developed to a minimum pre-feasibility level of accuracy;
- 8) LOM plans of existing operations were developed to a minimum pre-feasibility level of accuracy;
- 9) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 10) The Mineral Reserve estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

A comparison of the Mineral Reserves between December 31, 2017 and December 31, 2018 is shown in Table 72.

Table 72: Comparison between the December 31, 2018 and December 31, 2017 LOM Mineral Reserve estimates

Mining Operation	December 31, 2018 LOM Mineral Reserve Estimate			December 31, 2017 LOM Mineral Reserve Estimate			Comments
	Mt	%TCu	%TCo	Mt	%TCu	%TCo	
KTO UG	26.6	3.40	0.54	26.9	3.40	0.54	Changed due to mining depletion in 2018
T17 UG	11.3	3.65	0.62	11.3	3.65	0.62	Unchanged – no depletion
Mashamba East OP	31.6	2.15	0.60	32.1	2.13	0.60	Changed due to mining depletion in 2018

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Mining Operation	December 31, 2018 LOM Mineral Reserve Estimate			December 31, 2017 LOM Mineral Reserve Estimate			Comments
	Mt	%TCu	%TCu	Mt	%TCu	%TCu	
KOV OP	57.5	3.72	0.48	60.5	3.70	0.48	Changed due to mining depletion in 2018
KITD (Total)	5.7	1.44	0.17	7.4	1.49	0.16	Changed due to mining depletion in 2018
Total	132.7	3.18	0.52	138.1	3.15	0.51	Net decrease in the Mineral Reserve is due to depletion of Mineral Reserve being mined in 2018

The changes in the current Mineral Reserve estimate relative to the KCC 2018 NI 43-101 Technical Report released on March 31, 2018 are:

- 2018: Mineral Reserves are depleted due to mining activities continued during 2018 in KOV OP, Mashamba East OP, KITD and KTO UG. Mineral Reserves are current as at December 31, 2018; and
- 2019: Mineral Reserves are not depleted for the mining activities that continued during 2019 in KOV OP, Mashamba East OP, KITD and KTO UG. Depleted Mineral Reserves will be released as a part of the 2019 year-end Mineral Resources and Mineral Reserves update.

15.2 Mineral Reserve Related Risks

KCC is actively engaged with GCM for land use around the Mupine pit as well as areas within or around the Near West Tailings dam footprint. Both of these combined areas would provide sufficient space for the life of mine tailings capacity. For a complete description of this issue, refer to Section 20.2.2. Failure to obtain the land could potentially impact the Mineral Reserve estimate and trigger adverse consequences for KCC's operations, up to the shutdown of the operations by the end of Q1 2021 if no agreement with third parties (including GCM) can be reached in the near future.

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16.0 Mining Methods

16.1 KOV Geotechnical Engineering

16.1.1 Introduction

Information presented in this section has been obtained from the previous investigations by SRK (2008), Golder (2012) and Knight Piesold (2017) as well as internal geotechnical work and mine planning evaluations. Geotechnical and hydrogeological properties for this design were based on the evaluations from SRK (2008), Knight Piesold (2017) and KCC (2017 and 2018).

A geotechnical exploration program, laboratory testing, data analysis and a feasibility level design was completed in 2017. Further geotechnical and hydrogeological studies are being undertaken through 2019 and 2020.

16.1.2 Design Methodology

The KOV lithology model (June 30, 2016), based on the exploration drilling and surface mapping, was used to define the geometry of the geotechnical units. A rock mass classification evaluation was completed based on 25 geotechnical holes drilled during the 2017 geotechnical drilling program as shown in Figure 63.

This geotechnical data and information from the laboratory tests were used to assign the geotechnical values to the geotechnical units that are expected to be exposed at KOV OP. Further drilling will be continued during 2019 and throughout 2020 in order to improve the geotechnical modelling and optimization.

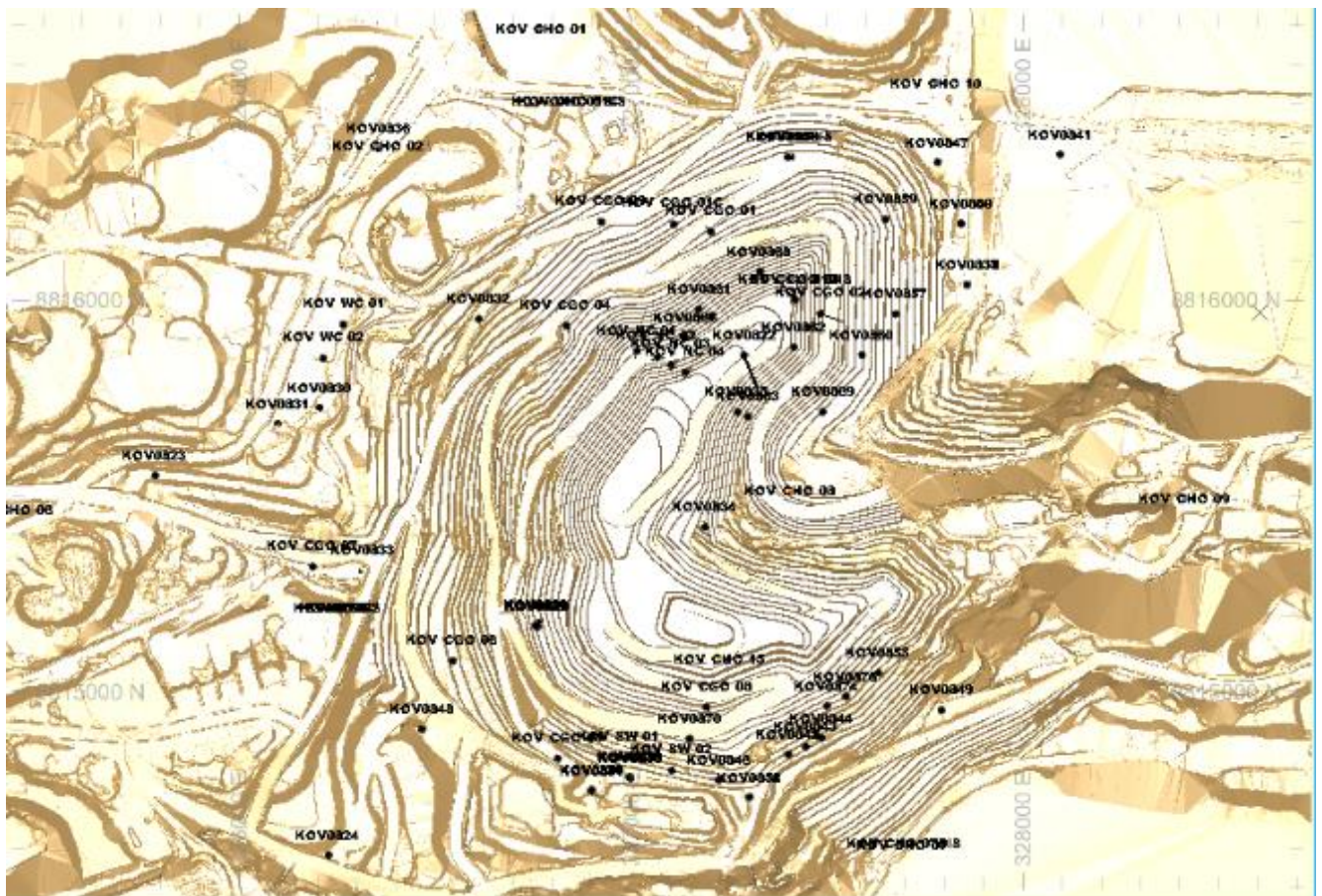


Figure 63: 2017 and 2018 drilling program

The geotechnical analysis was based on the 2017 and 2018 Mineral Reserve mine plan. The selected slope sections, representing the planned slopes, were used for the assessment of stability within SLIDE v5 and v7 (Rocscience), a software for evaluating the stability of circular and non-circular failure in rock and soil slopes.

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The factor of safety sensitivity analysis of the overall slope angles was carried out for representative groundwater levels based on the piezometer measurements in the area. The results of these analyses were used in constructing the overall slope angle versus the factor of safety for each slope region.

16.1.3 Geotechnical Properties

The lithology model was used with determined weathering boundaries to define the geotechnical units for the slope stability analysis. These are similar to the lithology units, however some simplifications were considered pertinent, as some of the lithological units included in the geological model and rock-mass database could be assumed to have similar geomechanical characteristics.

A summary of the estimated rock mass strength parameters, for each geotechnical unit, considered by KCC (2018) for the analysis of the slopes is shown in Table 74 to Table 75. These have been derived from the KP, KCC and SRK drilling programs.

Table 73: Geotechnical parameters (slightly weathered rock) for KOV OP

Unit	UCS (MPa)	GSI	mi	σ_3 max (MPa)	D	Unit Weight (kN/m ³)	Rock Mass Cohesion (kPa)	Rock Mass Friction (°)	Bedding Cohesion (kPa)	Bedding friction (°)
RGS	38	44	6	1.61	0.7	24.2	552	38	42	36
CMN	96	40	9	1.61	0.7	26.1	877	46	12	30
SDS	63	50	6	1.61	0.7	26.1	912	42	42	36
OBS	60	54	9	1.61	0.7	27.2	1048	46	Massive	
RSC	75	62	12	1.61	0.7	24.9	1540	53	Massive	
OBI	60	48	9	1.61	0.7	24.9	843	45	Massive	
RATS	47	44	9	1.61	0.7	25.5	680	42	Massive	

Table 74: Geotechnical parameters (moderately weathered rock) for KOV OP

Unit	UCS (MPa)	GSI	mi	σ_3 max (MPa)	D	Unit Weight (kN/m ³)	Rock Mass Cohesion (kPa)	Rock Mass Friction (°)	Bedding Cohesion (kPa)	Bedding friction (°)
RGS	24	39	6	1.61	0.7	22.6	377	33	40	26
CMN	38	31	9	1.61	0.7	24.5	447	37	32	26
SDS	27	36	6	1.61	0.7	23.3	376	33	40	26
OBS	38	36	9	1.61	0.7	23.8	495	39	Massive	
RSC	38	35	12	1.61	0.7	22.1	503	42	Massive	
OBI	38	41	9	1.61	0.7	19.3	494	42	Massive	
RATS	20	30	9	1.61	0.7	24.4	337	32	Massive	

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Table 75: Geotechnical parameters (highly weathered rock) for KOV OP

Unit	Unit Weight (kN/m ³)	Cohesion (kPa)	Friction (°)
RKS	20.8	24	38
RGS	20.0	101	36
CMN	17.9	53	34
SDS	20.4	56	39
ORE	21.9	107	34
RATS	20.8	24	38

16.1.4 Structural and Hydrogeological Conditions

Hydrogeological conditions were based on the observed piezometer reading as described in Section 16.8 of this report.

Failures associated with the prominent joint orientations are expected to be contained on the spill berms, whereas the bedding plane failures will be minimized due to the positioning of the proposed high wall orientations in the southwestern and the south eastern walls.

The above mentioned conditions will be verified during the 2019 and 2020 geotechnical and hydrogeological drilling project.

16.1.5 Analysis of Planned Slopes

The selected slope sections, representing the planned slopes, were used for the assessment of stability with SLIDE v5 and v7 (Rocscience) software.

The sensitivity analyses of the factor of safety for different overall slope angles were carried out for the representative groundwater levels based on piezometer measurements in the area. The analyses were performed for the northern (N1, N2, N3 and N4), south eastern (SE) and southwestern (SW) slope regions as shown in Figure 65 and results are shown in Table 76. Based on the results of the stability analysis, overall slope angles of 28° - 35° are recommended for the final pit slopes.

These recommendations take into account the lack of structural data available outside the KOV OP ore zones where the majority of the final slopes are located.

KCC completed an internal geotechnical evaluation of the KOV OP in 2017. The slope dimensions recommended by KCC for KOV OP are presented in Table 76.

Table 76: Summary of the geotechnical derived slope design parameters completed by KCC in 2017

Wall	BH	BFA	SBW	Height	BSA	OSA	FoS
N1	10 m	65 - 75°	7.5 m - 12.5 m	343 m	28° - 47°	35°	1.74
N2	10 m	65 - 75°	7.5 m - 12.5 m	200 m	28° - 46°	32°	1.35
N3	10 m	65 - 75°	7.5 m - 12.5 m	243 m	32° - 33°	28°	1.28
N4	10 m	65 - 75°	7.5 m - 12.5 m	270 m	31° - 33°	29°	3.12
SE	10 m	65 - 75°	7.5 m - 12.5 m	280 m	29° - 36°	29°	2.46
SW	10 m	65 - 75°	7.5 m - 12.5 m	310 m	23° - 32°	25°	2.15

Slope design definitions are indicated in Figure 64.

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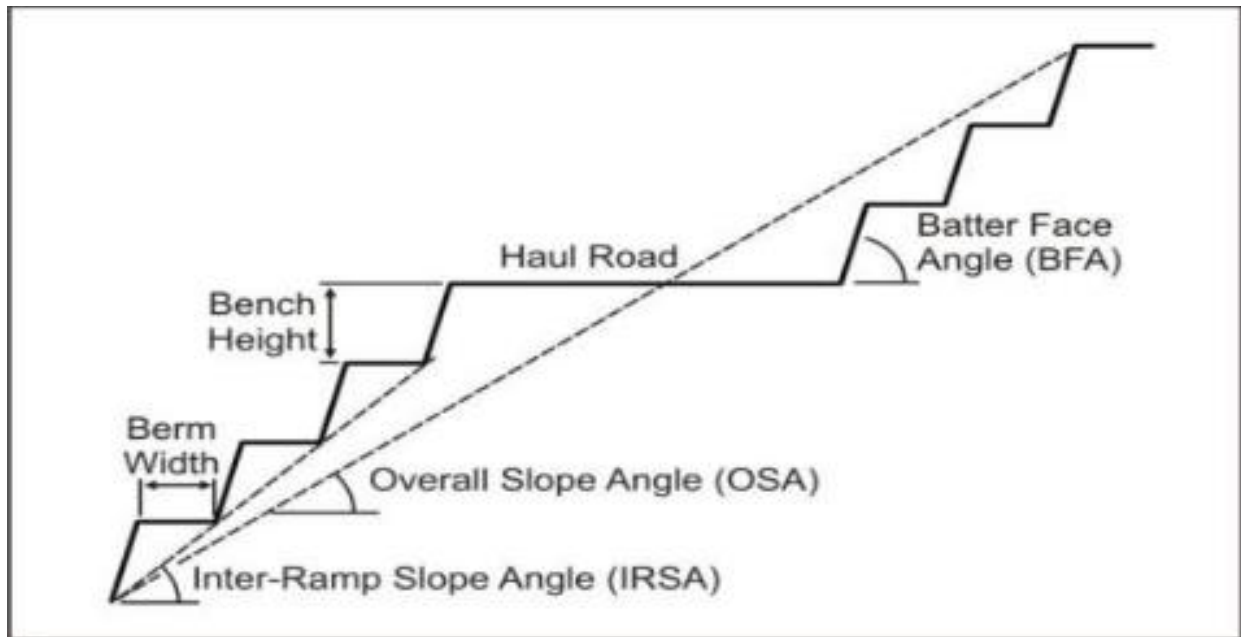


Figure 64: Slope design definitions

BH – Bench Height
 BFA – Bench Face Angle
 SBW – Spill Berm Width
 OSA – Overall Slope Angle measured from toe to crest
 BSA – Bench Stack Angle measured from toe to crest
 IRA – Inter Ramp Angle measured from toe to toe

16.1.6 Verification of the KOV OP Design

Table 77 shows the current design parameters for KOV OP. The updated geological information, internal geotechnical and mine planning evaluations were used to update the slope design parameters.

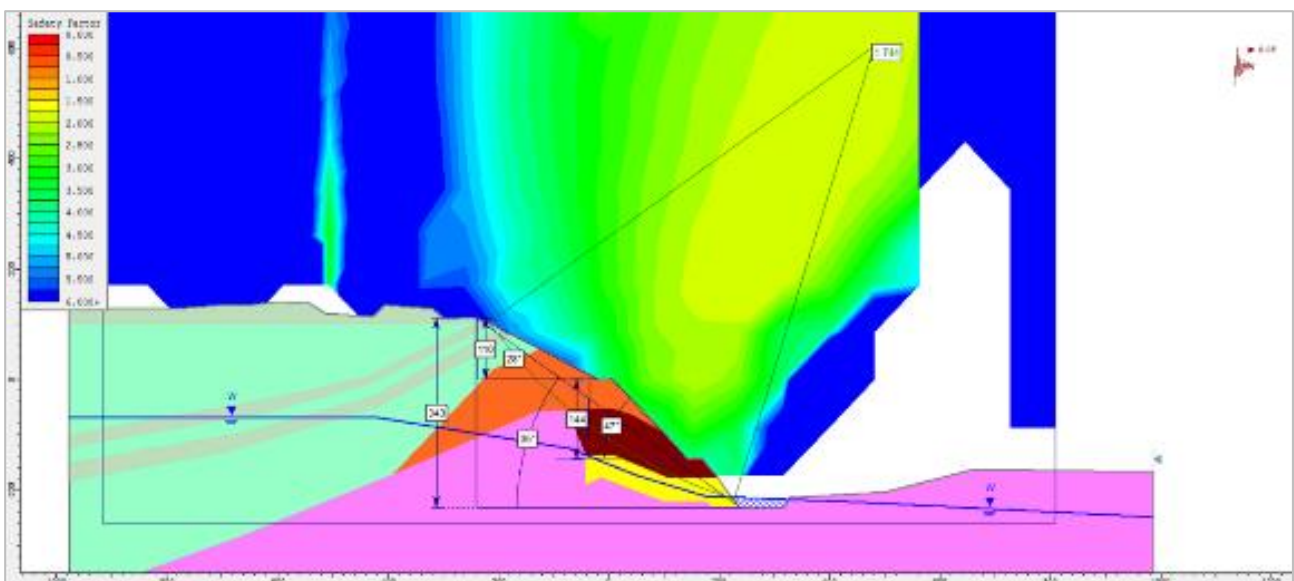
Table 77: Summary of the geotechnical derived slope design parameters (current)

Wall	BH	BFA	SBW	Ramp
All	10 m	65°	12.5 m	35 m

The intermediate pit mine design, based on the current slope design parameters was verified using Slide v5 and v7. The designs were evaluated for the representative groundwater levels based on the piezometer measurements to design the sectors as shown on Figure 65.

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Slide evaluations for the north wall of the intermediate KOV OP are shown in Figure 66 to Figure 69.



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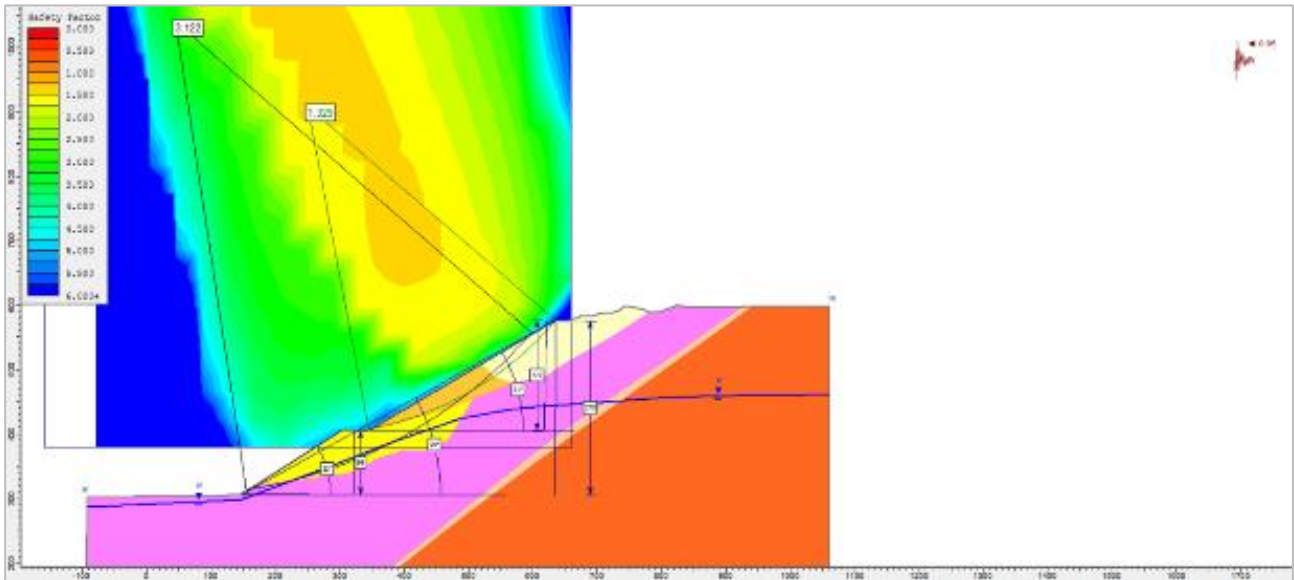


Figure 69: North wall slope design – N4 design sector

16.1.6.2 Verification of the Southwest Wall

Slide evaluations for the southwest wall of the intermediate KOV OP are shown in Figure 70.

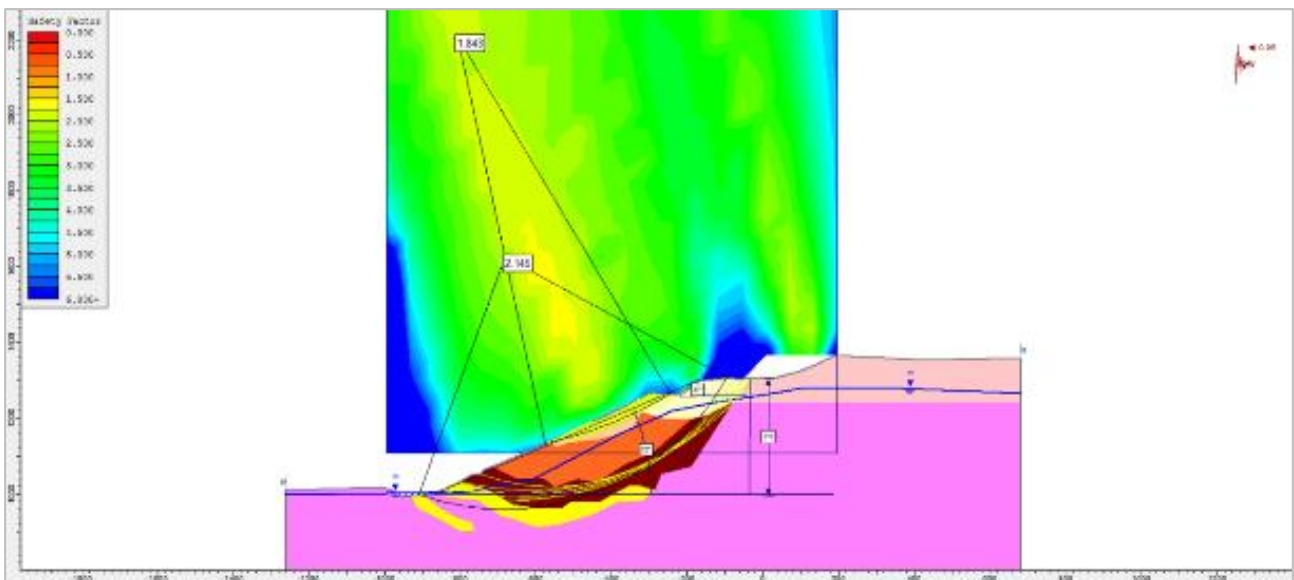


Figure 70: Southwest wall slope design

16.1.6.3 Verification of the Southeast Wall

Slide evaluations for the southeast wall of the final KOV OP are shown in Figure 71.

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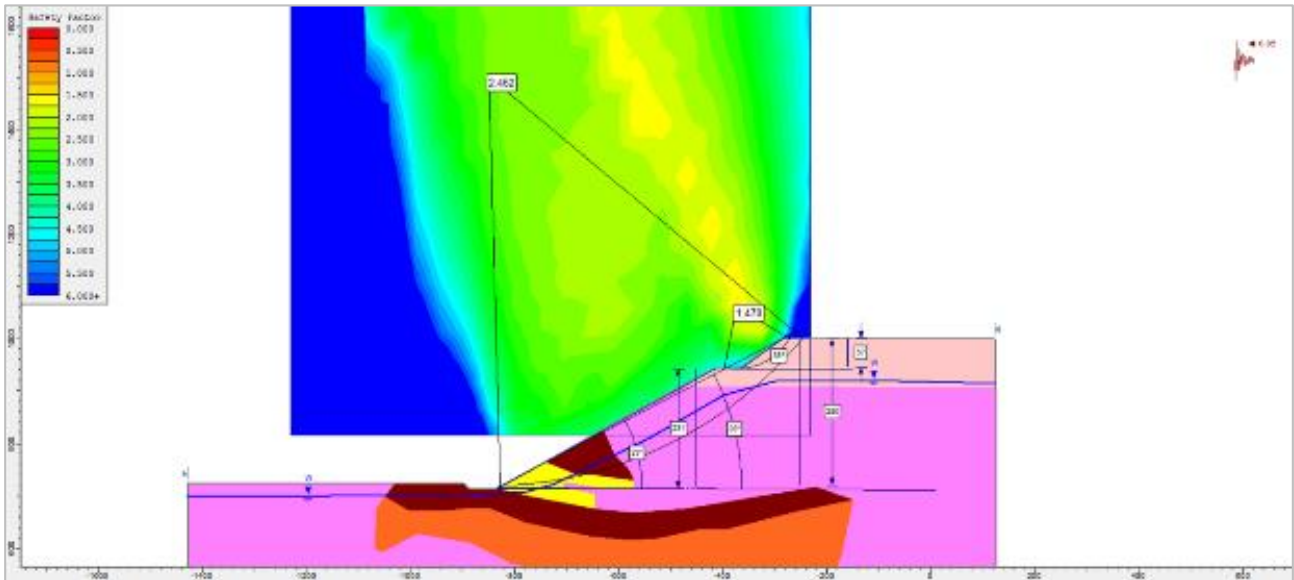


Figure 71: Southeast wall slope design

16.2 Mashamba East Geotechnical Engineering

16.2.1 Introduction

Information presented in this section was obtained from the previous investigations by SRK (October 2008), CNI (August 2016), geological exploration holes, and internal geotechnical evaluations.

Geotechnical properties for this design were based on the evaluations from SRK as well as internal geotechnical data analysis.

A geotechnical exploration program for the Mashamba East OP was completed in March 2017 and study report is not finalized at the time of this TR. This included drilling, logging, laboratory testing, and data analysis. KCC engaged SRK Consulting in 2019 to revise and define both the KOV and Mashamba East open pit geotechnical recommendations. Little or no changes are expected for the Mashamba East open pit.

16.2.2 Design Methodology

The KCC Mashamba East OP lithology model (2016), based on the exploration drilling and the surface mapping, was used to define the geometry of the geotechnical units. A rock mass classification evaluation was completed based on 83 exploration holes, drilled during the 2014-15 exploration campaign. This information was cross-referenced with data from the known geotechnical units in KOV OP and T17 OP to assign values to the similar units that are expected to be exposed at Mashamba East OP.

The geotechnical analysis was based on the 2016 mine plan included in the Mashamba East OP PFS. Selected slope sections, representing the planned slopes, were used for the assessment of stability within SLIDE v5.

The factor of safety sensitivity analyses of the overall slope angles was carried out for different slope conditions. The results of these analyses were used in constructing summary tables of overall slope angle versus factor of safety for the north slope. Scenarios analysed included semi-saturated and dry slopes.

Based on the results of the stability analysis an OSA of 40° was selected for the Mashamba East OP. This provides a factor of safety (FoS) of 1.2 with a semi-saturated slope.

16.2.3 Geotechnical Properties

A similar approach as described in Section 16.1.3 was followed to define the geotechnical units for the slope-stability analysis.

Mohr-Coulomb parameters (c and ϕ) were estimated to represent the weaker rock mass strengths such as soil and highly to completely weathered CMN and RGS.

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Hoek-Brown constants (mb, s and a) were estimated using the Hoek-Brown criterion for the moderately weathered rock mass that dips into the north high wall.

The geological strength index (GSI) was estimated using the Laubscher (1990) classification system that assumes the GSI is equivalent to RMR-5. This is consistent with the original definition published by Bieniawski (1976).

The strength parameters were estimated assuming a disturbance factor of D=0.3.

A summary of the estimated parameters for each geotechnical unit considered for the analysis of the slopes is shown in Table 78.

Table 78: Geotechnical parameters

Unit	UCS (MPa)	RMR	GSI	mi	Density (kN/m ³)	C (kPa)	φ (°)	mb	s	a
SOIL	-	-	-	-	22	15	28	-	-	-
HW CMN	-	-	-	-	25	30	30	-	-	-
MW RGS	40	41	36	7	27	-	-	0.476	0.0004	0.51
MW CMN	35	42	37	9	27	-	-	0.638	0.0004	0.51
MW SDS	40	43	38	7	27	-	-	0.517	0.0004	0.51
MW ORE	80	37	32	8	28	-	-	0.459	0.0002	0.52
MW RATL	88	48	43	16	28	-	-	1.459	0.0009	0.51

16.2.4 Structural and Hydrogeological Conditions

General conditions experienced at KOV OP and T17 OP were assumed for Mashamba East OP. Failures associated with prominent joint orientations are expected to be contained on the spill berms, whereas bedding plane failures will be minimized due to the orientation of the proposed high walls.

Hydrogeological conditions were estimated to be similar to KOV OP and groundwater levels were assumed to be at 1370 mamsl.

16.2.5 Analysis of Planned Slopes

A stability analysis along current slope sections was carried out under dry and semi-saturated conditions to verify the estimated rock-mass strengths. This analysis indicates a factor of safety (FoS) of 2.0 for the south slope (OSA <25° and excavated in RATLilas) and 1.5 for the north slope of the proposed Mashamba East OP, as illustrated in Figure 72, Figure 73 and Table 79.

These results are considered to be similar with the observed behaviour of the slopes in certain areas of the KOV and T17 pit walls. Instability has occurred in some of these slopes likely due to the unfavourable combination and orientation of a weak rock unit (siltstones and shales) with the presence of high water levels in the area. However, it is important to note that the bedded rock units of CMN and SDS dip into the northern high wall of the Mashamba East OP. This orientation will limit the potential of a bedding slip failure.

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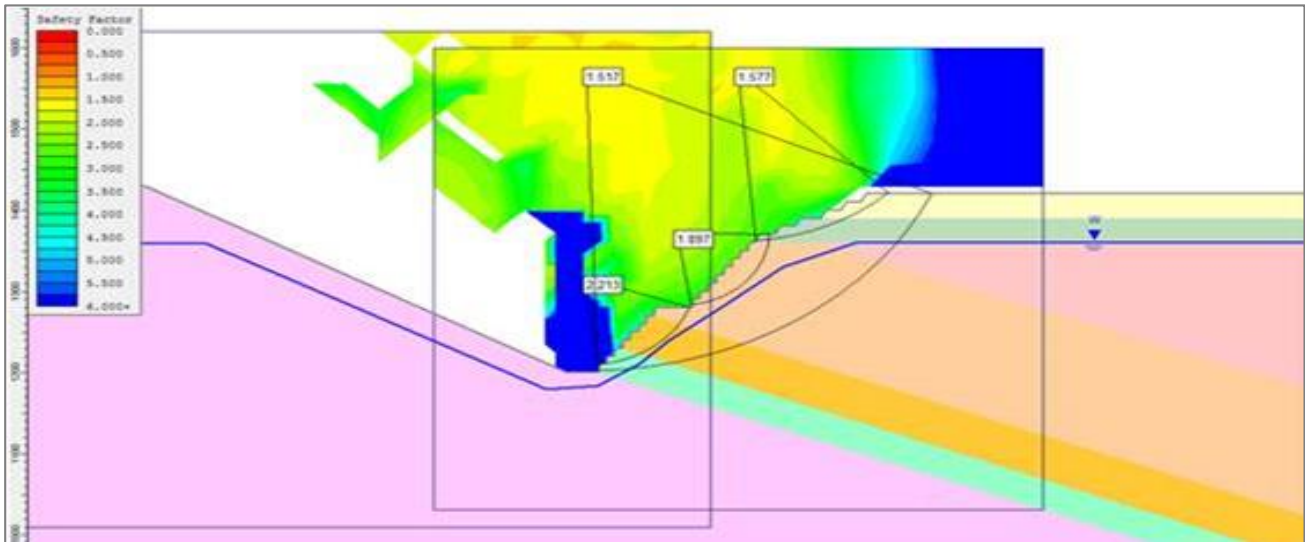


Figure 72: Slope design with dry conditions

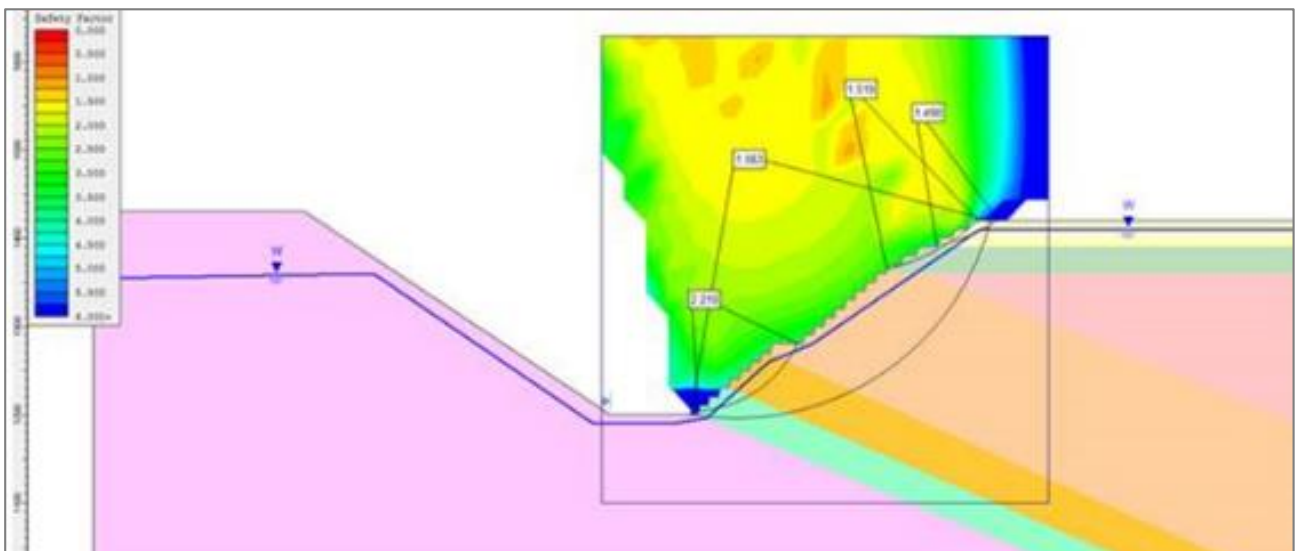


Figure 73: Slope design with semi saturated conditions

Table 79: Summary of the geotechnical derived slope design parameters

OSA 40° Dry Conditions		OSA 40° Semi Saturated Conditions	
Upper Highly Weathered Slopes and / or Soil		Upper Highly Weathered Slopes and / or Soil	
BH	10 m	BH	10 m
BA	65°	BA	65°
SBW	12.5 m	SBW	12.5 m
BSH	30 m	BSH	30 m
BSA	37.6°	BSA	37.6°
Geotechnical berm	20 m	Geotechnical Berm	20 m
FoS	1.6	FoS	1.5
Lower Moderately Weathered Slopes		Lower Moderately Weathered Slopes	
BH	10 m	BH	10 m

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OSA 40° Dry Conditions		OSA 40° Semi Saturated Conditions	
BA	80°	BA	80°
SBW	6.8 m	SBW	6.8 m
BSH	80 m	BSH	80 m
BSA	52.4°	BSA	52.4°
Geotechnical berm	35 m	Geotechnical Berm	35 m
FoS	1.9 – 2.2	FoS	1.5 – 2.2
Overall Slopes		Overall Slopes	
OSH	220 m	OSH	220 m
OSA	40°	OSA	40°
FoS	1.52	FoS	1.2

16.2.6 Influence of the KITD

The KITD hydraulic mining plan will be completed by the end of 2019. The Mashamba East OP mine plan does not intersect the KITD area until 2021, therefore, no interaction or influence from the KITD Tailings is expected on Mashamba East OP.

16.2.7 Verification of the Mashamba East OP Design

The final mine design for pushbacks 4 and 5 was verified using Slide v5 and similar parameters as indicated in Table 79. Both designs were evaluated for dry and semi-saturated conditions.

16.2.7.1 Pushback 4 Verification

The slide analysis resulted in a FoS of greater than 1.4 for all groundwater conditions in pushback 4, which creates the final wall on the west side of the Mashamba East OP. This is shown in Figure 74, Figure 75, Table 80 and Table 81.

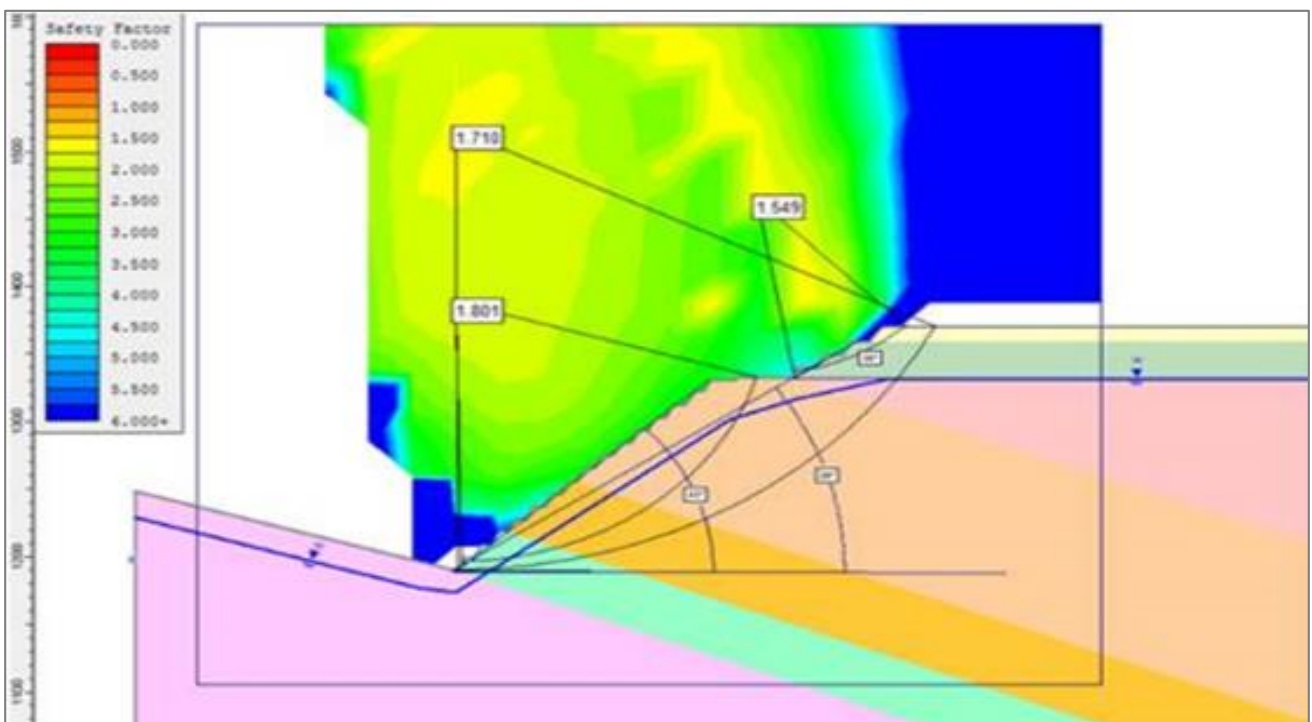


Figure 74: Slope design for pushback 4 with dry conditions

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Table 80: Factors of Safety (FoS) for pushback 4 dry conditions

Description	Height	Angle	FoS
Upper bench stack	40 m	35°	1.55
Lower bench stack	140 m	43°	1.71
Overall pit	180 m	35°	1.80

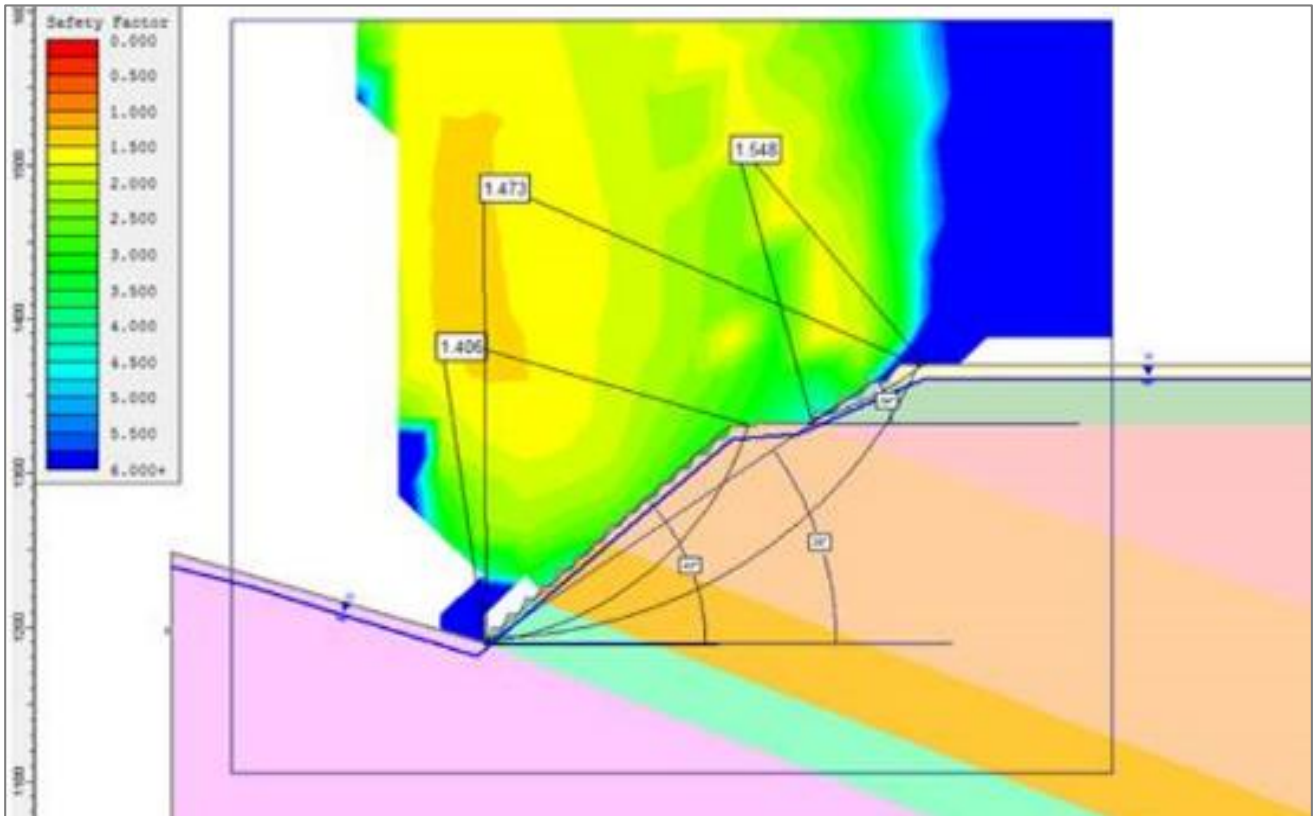


Figure 75: Slope design for pushback 4 with semi saturated conditions

Table 81: Factors of Safety (FoS) for pushback 4 semi saturated conditions

Description	Height	Angle	FoS
Upper bench stack	40 m	35°	1.55
Lower bench stack	140 m	43°	1.41
Overall pit	180 m	35°	1.47

16.2.7.2 Pushback 5 Verification

Pushback 5 creates the final wall of the Mashamba East OP on the eastern side of the pit. The slide analysis resulted in a FoS of greater than 1.4 for all groundwater conditions except for the semi-saturated conditions in the upper 40 m of the high wall as shown in Table 82, Table 83, Figure 76 and Figure 77. The FoS in the top 40 m is 1.3.

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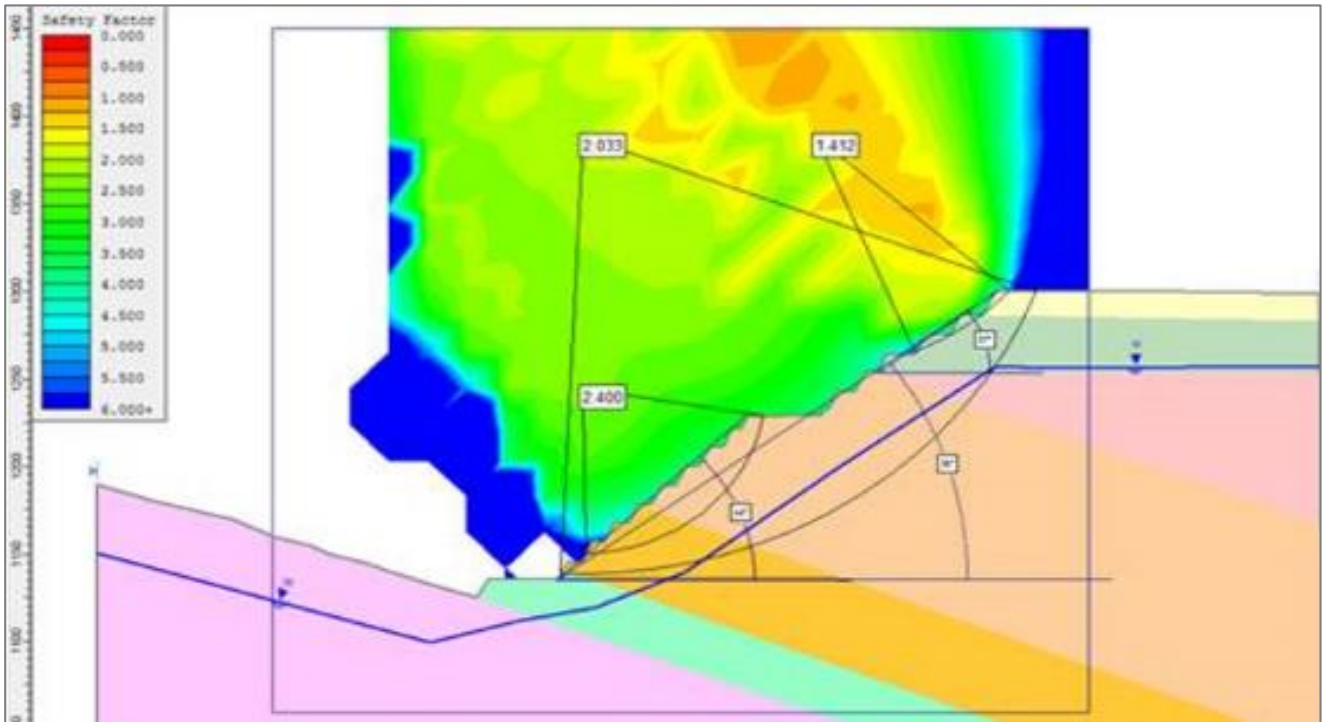


Figure 76: Slope design for pushback 5 with dry conditions

Table 82: Factors of Safety (FoS) for pushback 5 dry conditions

Description	Height	Angle	FoS
Upper bench stack	70 m	37°	1.41
Lower bench stack	95 m	44°	2.40
Overall pit	165 m	36°	2.03

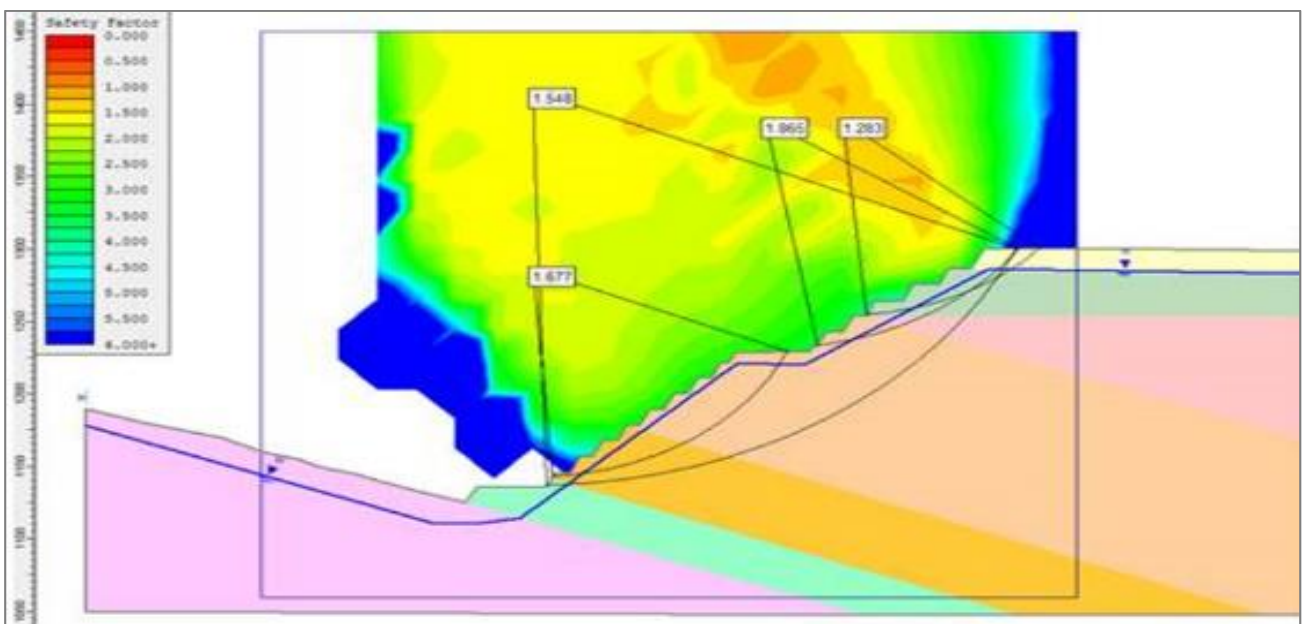


Figure 77: Slope design for pushback 5 with semi saturated conditions

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Table 83: Factors of Safety (FoS) for pushback 5 semi saturated conditions

Description	Height	Angle	FoS
Upper bench stack	70 m	37°	1.87
Lower bench stack	95 m	44°	1.67
Overall pit	165 m	36°	1.55

16.3 KTO UG Geotechnical Engineering

16.3.1 Introduction

Rock mass characterization data is limited for the KTO UG mine. Data used for this analysis have been derived from the reports listed below and UG mine design documents as well as the internal geotechnical and mine planning evaluations.

- RSV (2006);
- SRK (2008); and
- Golder NI 43-101 (2012 and 2015).

The key criteria used in developing a stope design for each of the mining methods have been derived empirically, using the industry accepted methods for stope span and stable pillar designs.

In addition, numerical models have been used which required a selection of appropriate rock-mass properties and a best estimate of an in-situ stress field.

16.3.2 Design Methodology

Geotechnical data was gathered through underground visits and back analysis of the stable and unstable excavations and pillars (SRK 2008).

Data collection methods and data analysis were performed according to industry accepted methods and norms. Rock mass classifications systems (RMR, MRMR and Q) were used to classify the various rock masses in which development declines (RATS) and stopes (RATS, OBI, RSC, OBS and SDS) will be excavated.

The stability number (N'), based on the industry standard data, was used to determine the stable hydraulic radii (HR) of the unsupported stope walls.

Industry standard pillar formulae were used to determine stable pillars sizes for different stope sizes and mining methods.

16.3.3 Geotechnical Properties

A similar approach as described in Section 16.1.3 was followed to define the geotechnical units.

Rock mass classification systems, indexes and numbers such as RMR, GSI, MRMR, Q, Q' and N' were calculated for each geotechnical unit to be utilized in the development, stope and pillar stability analysis as well as for the tunnel support strategies.

A summary of the estimated parameters, for each geotechnical unit, considered by SRK (2008) for the evaluation of the underground mine design is shown in Table 84.

Table 84: Geotechnical data parameters (averages)

Unit	UCS (MPa)	DRMS (MPa)	RMR	GSI	MRMR	Q'
SDS	100	-	55 - 65	55	45 - 55	10
OBS	190	48	65 - 75	65	50 - 60	-

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Unit	UCS (MPa)	DRMS (MPa)	RMR	GSI	MRMR	Q'
RSC	220	56	65 - 75	65	50 - 60	15
OBI	140	43	65 - 75	65	50 - 60	-
RATS	103	-	70 - 80	70	55 - 65	-

16.3.4 Mine Modelling Parameters

Numerical models have been set up by SRK (2008) using UDEC software to examine the stability of the horizontal sill pillars in open stoping, and FLAC3D software to examine the loads imposed on the pillars. In both cases rock mass properties, including rock mass strengths as listed in Table 85, are required to examine pillar failure.

Table 85: Rock mass properties used in the numerical modelling

Properties	Hangingwall	OBS	RSC
UCS (MPa)	140	220	170
GSI /RMR _{Laubscher}	60	60	60
m _i	15	10	10
E _i (GPa)	75	80	110
v	0.3	0.25	0.27
Density	2700	2700	2700
Em Young's Modulus (GPa)	39	42	57
Bulk Modulus (GPa)	33	28	41
Shear Modulus (GPa)	15	17	22
D	0	0	0
mb	2.397	2.397	2.397
s	0.0117	0.0017	0.0017

16.3.5 Current Mine Design Parameters

Current mine design parameters applied at the KTO UG mine for various mining methods are listed in Table 90.

16.3.6 Development Support Designs

The KCC geotechnical engineering department developed the support designs for various dimensions of the tunnels, declines and drives based on rock mass classification systems and industry accepted methods (Barton 1974). The Q system was used to prepare the design charts, which show the required support types including the rock bolt length, support density and shotcrete thickness. These charts are applicable for long term tunnels and chambers in reasonable rock and in-situ stress conditions. Table 86 provides an overview of the basic support regime, as determined by the geotechnical department.

Table 86: Support design for development

Areas	Roof Bolts	Shotcrete, mesh, long anchors, steel sets
All rock types	Type : Splitset	Site specific as specified by the KCC geotechnical department.
	Diameter : 46 mm	
	Length : 2.4 m	

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Areas	Roof Bolts	Shotcrete, mesh, long anchors, steel sets
	Spacing : 1.2 m x 1.2 m box pattern	
	Grout: Only in poor rock masses as specified by geotechnical department.	

16.4 T17 UG Geotechnical Engineering

16.4.1 Introduction

Information presented in this section was obtained from the T17 PFS conducted by Golder in 2013. The geotechnical engineering section of the T17 PFS was completed by Middindi. The current design criteria is based on internal geotechnical and mine planning evaluations.

16.4.2 Design Methodology

A geotechnical appraisal on the feasibility of transitioning from the open cast mining to the underground mining methods for the T17 OP was carried out by Middindi (2013).

Geotechnical data was gathered through the geotechnical logging of seven drill holes as shown in Figure 78. Underground scan line mapping and information was provided by KCC.

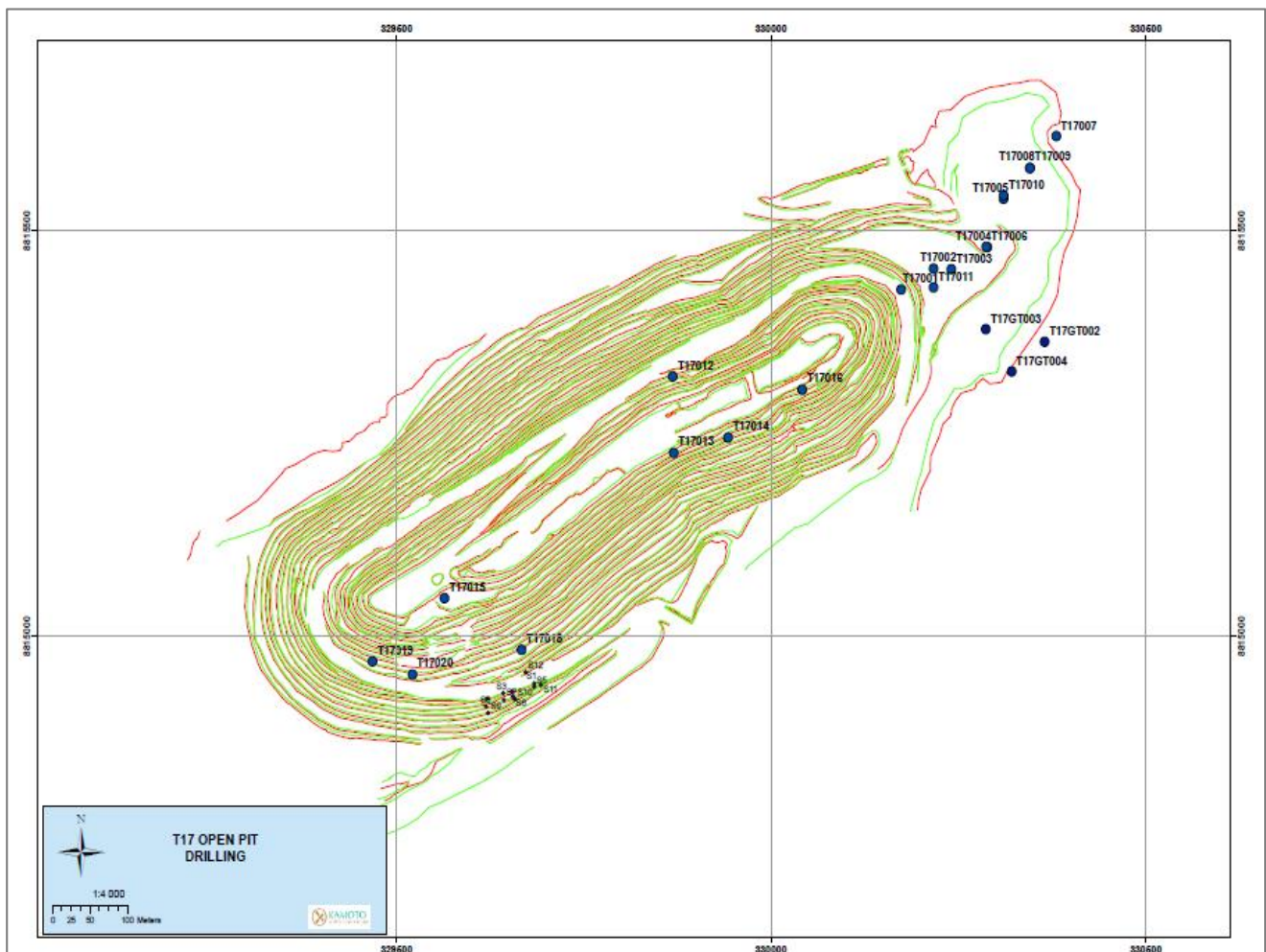


Figure 78: Location of T17 geotechnical and exploration drill holes

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Data collection methods and data analysis were performed according to industry accepted methods and standards. Rock mass classifications systems (RMR, MRMR and Q) were used to classify the various rock masses in which the development declines (RATS) and the stopes (RATS, OBI, RSC, OBS and SDS) are planned to be excavated.

The stability number (N'), based on industry data, was used to determine the stable hydraulic radii (HR) of the unsupported stope walls. Industry standard pillar formulae were used to determine stable pillars sizes for different stope sizes.

16.4.3 Geotechnical Properties

The T17 geological model was used as the basis of the geotechnical model in conjunction with the rock mass, rock strength and structural models.

Some of the lithological units in the geological model and rock mass model database could be assumed to have similar geomechanical characteristics. They were therefore grouped into a single geotechnical unit within the geotechnical model e.g. RATLILAS, OBI, RSC, OBS and SDS.

Rock mass classification systems, indexes and numbers such as RMR, GSI, MRMR, Q, Q' and N' were calculated for each geotechnical unit to be utilized in the development, stope and pillar stability analysis as well as for the tunnel support strategies.

A summary of the estimated geotechnical parameters, for each geotechnical unit (Middindi, 2013), used in the evaluation of the underground mine design is shown in Table 87.

Table 87: Geotechnical data parameters (averages)

Unit		UCS (MPa)	DRMS (MPa)	RMR	GSI	MRMR	Q	Q'	N'
RAT	RATS	74	22	58	53	37	5	11.4	31.7
Breche		24	7	49	44	32	2	4.4	12.3
SDS	SDS	100	21	49	44	31	2	4.1	11.5
SDB	OBS	100	21	47	42	31	1	3.6	10.1
RSC	RSC	79	15	41	36	27	1	1.9	5.2
RSF	OBI	92	15	40	35	26	1	1.7	4.7
DStrat		69	12	38	33	25	1	1.3	3.7

16.4.4 Mine Modelling Parameters

A combination of HR, Stability number N, Modified RMR and the Q rating index was used to determine the mine design parameters for T17 UG (Middindi, 2013).

The following parameters were determined:

- Stable stope spans;
- Rib pillar dimensions;
- Decline support; and
- Crown pillar dimensions.

The stable stope spans were determined from the Stability Number (N') as calculated from the rock mass characteristics, structural data, influence of mining stress and the equivalent HR. A HR of 10.5, based on the data from the SDS, was used for the stope stability calculations as shown in Figure 79. Middindi (2013) evaluated various stope widths to determine stable strike lengths for the stopes as shown in Figure 80.

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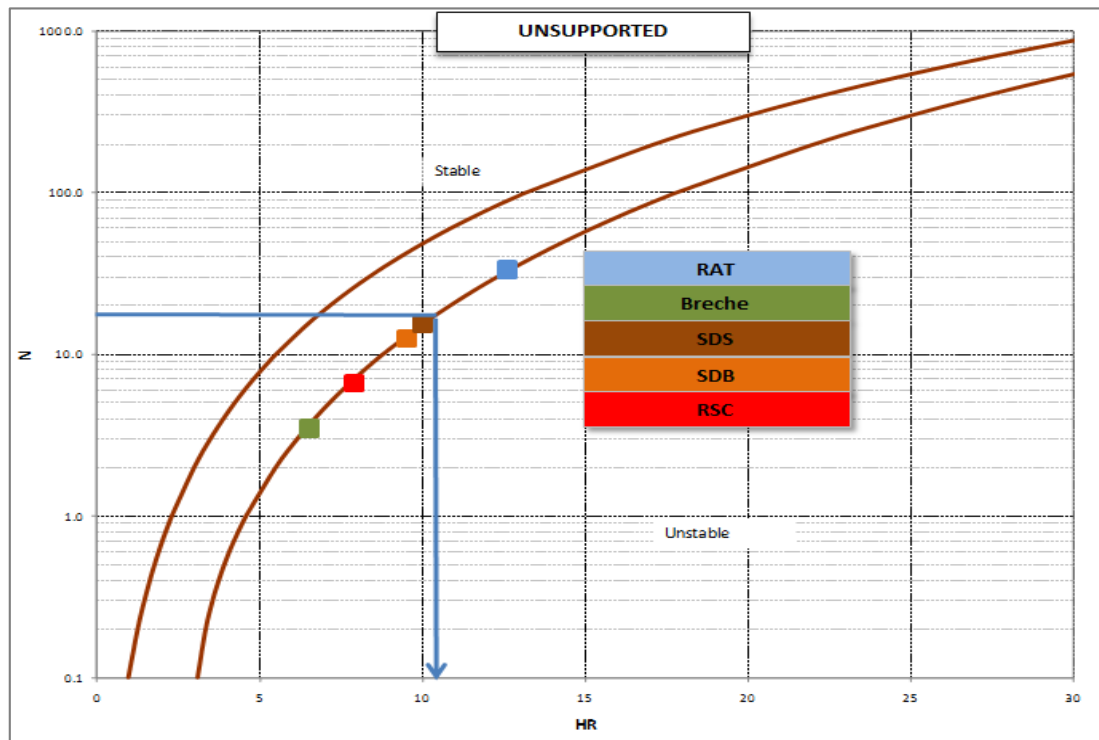


Figure 79: Hydraulic radii for various geotechnical units (Middindi 2013)

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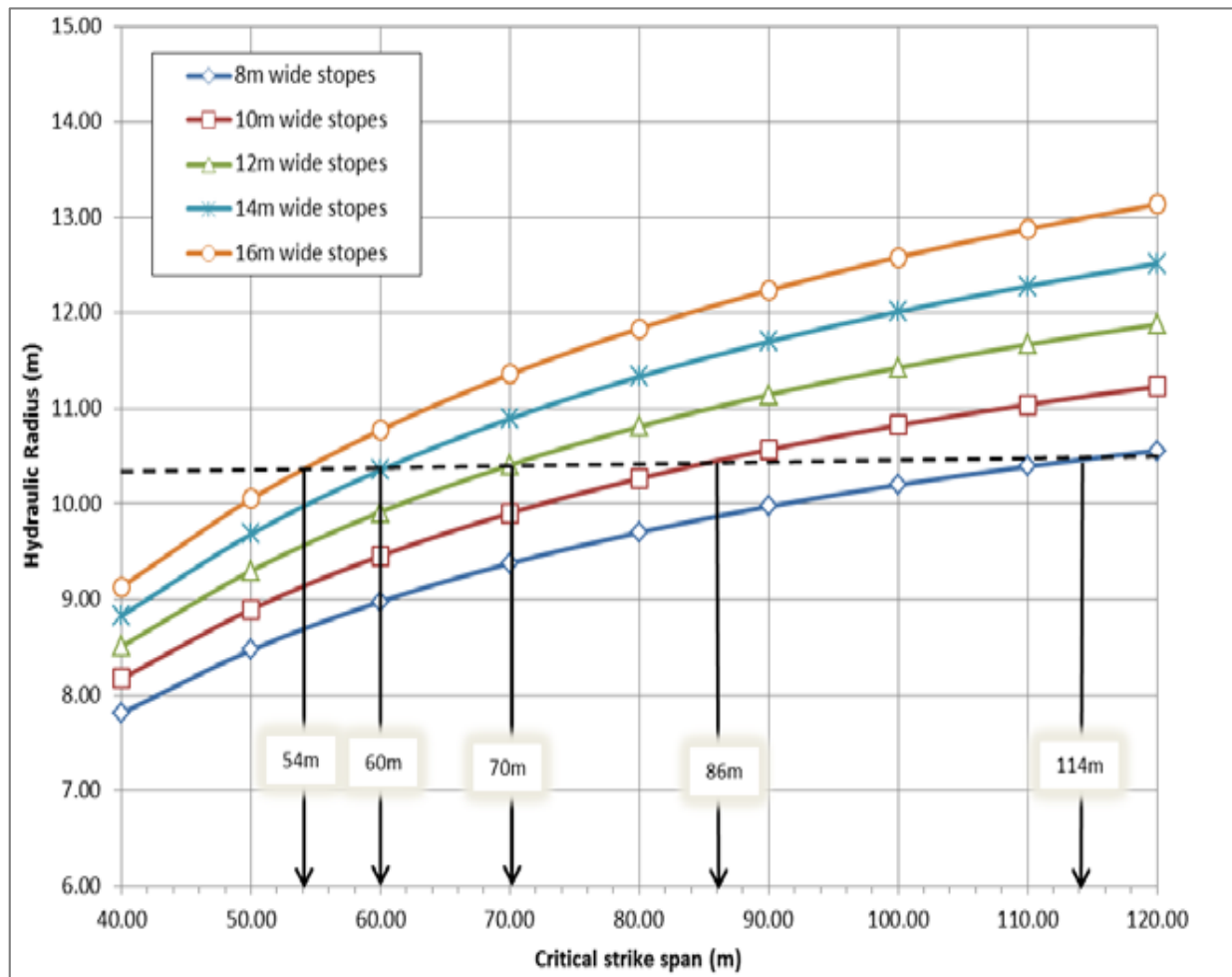


Figure 80: Stable strike lengths for various stope widths based on a HR of 10.5 (Middindi 2013)

Middindi proposed stope sizes for the longitudinal and transverse cut and fill stopes as indicated in Table 88.

Table 88: T17 UG proposed stope dimensions (Middindi 2013)

Mining Method	Orebody Width	Strike Length	Pillar Width
Longitudinal cut and fill	12 – 15 m	60 m	8 m
Transverse cut and fill	40 – 50 m	16 – 20 m	8 m

16.4.5 Rib Pillar Dimensions

The rib pillar dimensions were derived using the Hedley and Grant (1972) pillar strength formula for inclined fragments and for a level spacing of 20 m. A minimum Factor of Safety (FoS) of 1.2 was used as the design criteria before backfill is placed. The introduction of backfill provides confinement to the rib pillars and will increase the FoS.

The rib pillar widths were calculated to be 8 m for orebodies with a width of <15 m and a strike length of 60 m

16.4.6 Crown Pillar Dimensions

A crown pillar will be required for the LOM between the final pit floor and the underground workings to protect the underground workings from flooding and subsidence of surface infrastructure.

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A crown pillar design method, as developed by Barton (1974) and modified by Golder (1990), was used to estimate the minimum stable crown pillar dimensions.

A crown pillar thickness of 30 m was suggested by Middindi (2013).

16.4.7 Current Mine Design Parameters

The current mine design parameters applied for the T17 UG mine for various mining methods are listed in Table 89.

16.4.8 Development Support Design

Development support designs for 5 m x 6 m tunnels were based on the rock mass classification systems and industry accepted methods (Barton 1974). The Q system was used to prepare the design charts, which show the required support types including the rock bolt length, support density and shotcrete thickness. These charts are applicable for long term tunnels and chambers in reasonable rock and in situ stress conditions. Table 89 shows the support pattern for T17 UG development.

Table 89: Development support designs (Middindi 2013)

	Roof Bolts	Shotcrete
Declines (5 m x 6 m) All rock types	Type : Ripple bar	75 mm
	Diameter : 20 mm	
	Length : 2.4 m	
	Spacing : 1.5 m x 1.5 m box pattern	
	Grout : Cement full column	

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Table 90: KTO UG and T17 UG mine design parameters

Mine	Mining Zone	Unit	Primary Dev		Secondary Dev		Mining Method	Stope Dimensions			Pillar Dimensions			Pillar Extraction	Backfill Type Required
			H	W	H	W		L (along strike)	W (width of orebody)	H	L (along strike)	W (width of orebody)	H		
KTO UG	Zone 1	m	6	5	6	6	Transverse Stoping	15	15-30	15	8	15-30	15	Conditional	Waste/HBF
	Zone 3	m	6	5	6	6	Room & Pillar	37.5	12.5 – 12	15-20	37.5	12.5 – 13	15-20	Required	Cement Back fill
	Zone 4	m	6	5	6	6	Room & Pillar	37.5	12.5 – 12	15-20	37.5	12.5 – 13	15-20	Required	Cement Back fill
	Zone 5	m	6	5	6	6	Room & Pillar	37.5	12.5 – 12	15-20	37.5	12.5 – 13	15-20	Required	Cement Back fill
	Zone 6	m	6	5	6	6	Transverse Stoping	15	10-30	15-20	8	10-30	15-20	Conditional	Cement Back fill
	Zone 7	m	6	5	6	6	Transverse Stoping	17	25-35	15	8	25-35	15	Required	Cement Back fill
	Zone 8	m	6	5	6	6	Room & Pillar	60	12.5 – 12	10-25	60	12.5 – 13	10-25	Required	Cement Back fill
	Zone 9 below 495 Level	m	6	5	6	6	Cut & Fill	25	13-40	6	No pillar			N/A	Cement Back fill
	Etang South above 350 Level	m	6	5	6	6	Transverse Stoping	16	13-35	20	8	13-35	20	Conditional	Cement Back fill
	Etang South below 350 Level	m	6	5	6	6	Transverse Stoping	16	13-35	15	8	13-35	15	Conditional	Cement Back fill
	Etang North above 520 Level	m	6	5	6	6	Longitudinal stoping	45 – 50	10-20	15-20	20	10-20	15-20	Conditional	Cement Back fill
	Etang North below 520 Level-Primary	m	6	5	6	6	Longitudinal stoping	20 x 2	10-20	15	16	10-20	15	Conditional	Cement Back fill
	Etang North below 520 Level-Secondary	m	6	5	6	6	Longitudinal stoping	20 x 2	10-20	15	16	10-20	15	Conditional	HBF
T17 UG	All	m	5.5	5.5	5.5	5.5	Transverse Stoping	20	30-50	25	8	30-50	25	No	Waste/HBF
		m	5.5	5.5	5.5	5.5	Longitudinal Stoping	48	10-20	25	8	10-20	25	No	Waste/HBF
		m	5.5	5.5	5.5	5.5	Cut & Fill	12	20-70	5	12	20-70	5	No	Waste/HBF

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16.5 Mining Overview

16.5.1 Mine Planning

The Mineral Reserve LOM plan was prepared as a basis of the Mineral Reserve estimate and contains the Proven and Probable Mineral Reserves. The projects not developed to a minimum PFS level of detail were excluded and the Mineral Resources from the Inferred Mineral Resource category were treated as waste material in the planning and valuation processes. The Mineral Reserve estimate was based on a LOM plan completed in September 2017 to a minimum PFS level of accuracy. This plan consists of the Proven and Probable Mineral Reserves of KTO UG, KOV OP, T17 UG, KITD Tailings and Mashamba East OP. As stated in a previous Mineral Reserve estimate, the T17 OP was depleted, therefore no Mineral Reserves are stated for it in this estimate. No additional Mineral Reserves are declared for KCC other than the five stated mines.

16.5.2 Mining Operations

The active operations at KCC can be classified into surface and underground mining operations. Surface operations include KOV OP, KITD Tailings and Mashamba East OP; the T17 OP has been depleted. The underground operations include KTO UG and T17 UG, which have been developed to various stages. Figure 81 below shows the relative location of the current and planned operations of KCC.

KTO UG is an operational underground mine that produced 1.94 Mt during the full year of 2014. The current LOM plan is to ramp yearly production up to 2.1 Mt of sulphide. The previous technical reports and Mineral Reserve estimates for KTO UG from 2009 to 2012 were based on a FS conducted in 2008. Some of the proposed mining methods were not implemented as KCC's management made a strategic decision to limit the number of mining methods to those practised historically and successfully at the KTO UG.

The three mining methods practised at KTO are:

- Room and Pillar with Hydraulic Backfill (RAP) at dips of less than 12°;
- Transversal Longhole Retreat with Backfill (TLHR) between 13° and 55°; and
- Longitudinal Longhole Retreat with Backfill (LLHR) at dips of greater than 55°.

A re-design and re-scheduling of the entire KTO UG deposit in line with the KCC strategy was conducted and completed in 2013 and updated on a regular basis to date. The Mineral Reserve updated as at December 31, 2018 for KTO UG is 26.6Mt at 3.40% TCu.

A PFS was completed on the T17 UG mine in July 2013. This study indicated that the underground portion of T17 yielded a positive return. Construction of the first portal from the existing T17 OP has been completed, together with 596 m of waste access development. The T17 UG Mineral Reserve estimate is 11.3 Mt at 3.65% TCu. The first ore development tonnes are scheduled for 2025 while steady state production is planned by 2027 under the LOM plan. The two methods, TLHR and LLHR, currently practised at KTO UG will be applied to the relevant production areas of T17 UG.

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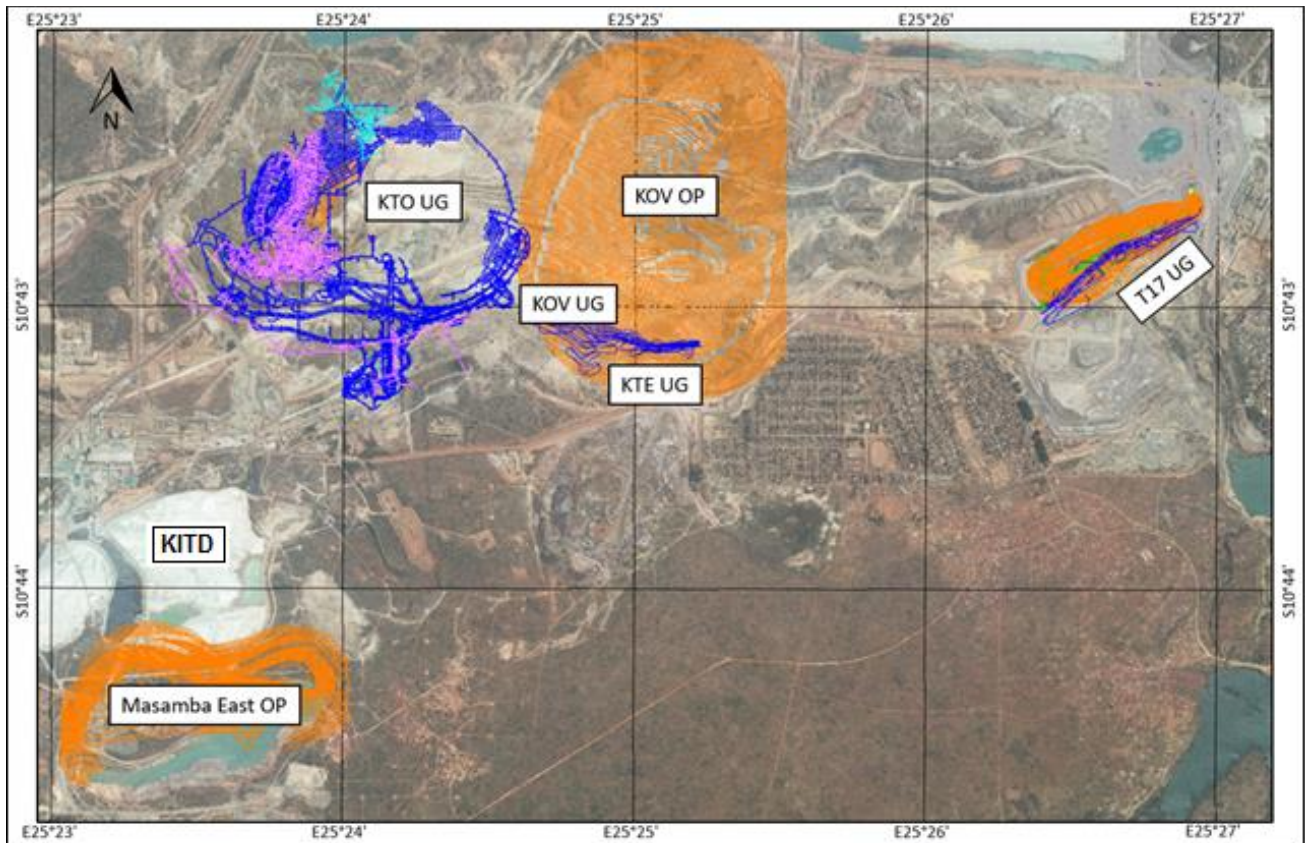


Figure 81: KCC overall mining layout

Mining related modifying factors applied across the underground mining operations are based on actual historic mining method performance:

- The dilutions and mining over breaks applied vary from 5% to 10%;
- A stope design cut-off grade of 1.5% TCu was used at KTO UG and 1.2% TCu at T17 UG;
- Pillar losses applied range from 18% to 35%; and
- Geological losses and mining losses range from 5% to 8% depending on the mining method and mining area.

KOV OP is an existing, large-scale operation. Mining related modifying factors applied to the KOV OP mine plan include a cut-off grade of 0.65% TCu and mining dilution of 20% and 5% overall losses. Ore production is primarily oxide and mixed material with only 8% sulphide ore. The Probable Mineral Reserve estimate of 57.5 Mt at 3.72% TCu is updated as at December 31, 2018 and based on the Mineral Reserve LOM plan up to 2029.

Mashamba East OP is an existing, large-scale operation. A pre-feasibility study was completed internally by KCC in November 2016. This study indicated that the Mashamba East OP yielded a positive return. Mining related modifying factors applied to the Mashamba East OP plan include a cut-off grade of 0.65% TCu, mining dilution of 7.9% and 5% overall losses. The Probable Mineral Reserve is updated as at December 31, 2018 is 31.6 Mt at 2.15% TCu. The Reserve LOM plan extends up to 2029. Mineral Resources (excluding Mineral Reserves) from the Inferred Mineral Resource category were scheduled as waste and treated as such in the Mineral Reserves only plan. Inferred Mineral Resources are considered too speculative geologically to have economic considerations applied to them and they cannot be categorised as Mineral Reserves.

The KITD Tailings area consists of a tailings dam which formed as a result of the discharge of fine tailings from KTC between 2008 and 2013. The KITD Tailings was constructed by a conventional upstream discharge

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method and is a U-shaped tailings dam, constrained to the west, east and south flank by civil engineered (SRK Consulting, 2008) containment walls, and the northern flank formed by steep natural rising ground.

The KTC processed ore from the T17 OP, KOV OP and KTO UG to produce the concentrate that was piped to Luilu Plant for solvent extraction / electro-winning. The recoveries obtained from the conventional concentrating techniques of sulphide and oxide ores left valuable amounts of copper and cobalt to discharge into the KITD Tailings. In June/July 2013, the dam was nearing full capacity, and tails were instead redirected to the Mupine OP.

KCC started remaining and reprocessing KITD Tailings in January 2017 to assess the processing viability of KITD material. Mining of KITD Tailings is still on going. With positive results achieved, KCC decided to reclaim all the KITD Tailings and completed a pre-feasibility study internally in October 2017. This study indicated that the KITD Tailings yielded a positive return. The mining related modifying factors applied to the KITD Tailings plan include a cut-off grade of 0.20% TCu and 0.07% TCo, mining dilution of 0.66% and 5% overall losses. The Probable Mineral Reserve updated as at December 31, 2018 is 5.7 Mt at 1.44% TCu.

16.5.3 LOM Plan as Basis of Mineral Reserve Estimate

A LOM plan that excludes Inferred Mineral Resources and excludes projects developed to a conceptual level of accuracy has been prepared on the basis of the Mineral Reserve estimate. The ore production schedule on the basis of the Mineral Reserve estimate is shown in Table 91 and Figure 82. The production profiles for recovered copper and cobalt are presented in Figure 83 and Figure 84.

Table 91: KCC LOM Mineral Reserve schedule

LOM Mineral Reserve plan	Unit	2019	2020	2021	2022	2023	2024	2025	2026	2027
Proven Mineral Reserve	Mt	0.1	0.1	0.3	0.6	0.5	0.3	0.5	0.4	0.4
Probable Mineral Reserve	Mt	10.1	11.4	10.2	10.4	10.7	10.7	10.6	11.0	11.0
Mineral Reserve total	Mt	10.2	11.5	10.5	10.9	11.1	11.0	11.2	11.4	11.4
Average Mineral Reserve %TCu	%	2.86	2.91	3.41	3.28	3.23	3.27	3.24	3.16	3.16
Average Mineral Reserve %TCo	%	0.42	0.43	0.58	0.57	0.58	0.63	0.58	0.51	0.49
LOM Mineral Reserve plan	Unit	2028	2029	2030	2031	2032	2033	2034	2035	Total
Proven Mineral Reserve	Mt	0.6	0.8	0.6	0.8	1.2	0.9	0.5	0.0	8.7
Probable Mineral Reserve	Mt	10.4	7.8	2.9	2.8	2.3	1.4	0.4	0.0	124.0
Mineral Reserve total	Mt	11.0	8.6	3.5	3.5	3.5	2.4	0.8	0.0	132.7
Average Mineral Reserve %TCu	%	3.29	2.55	3.46	3.52	3.69	3.60	3.71	3.52	3.18
Average Mineral Reserve %TCo	%	0.42	0.34	0.56	0.67	0.65	0.51	0.48	0.42	0.52

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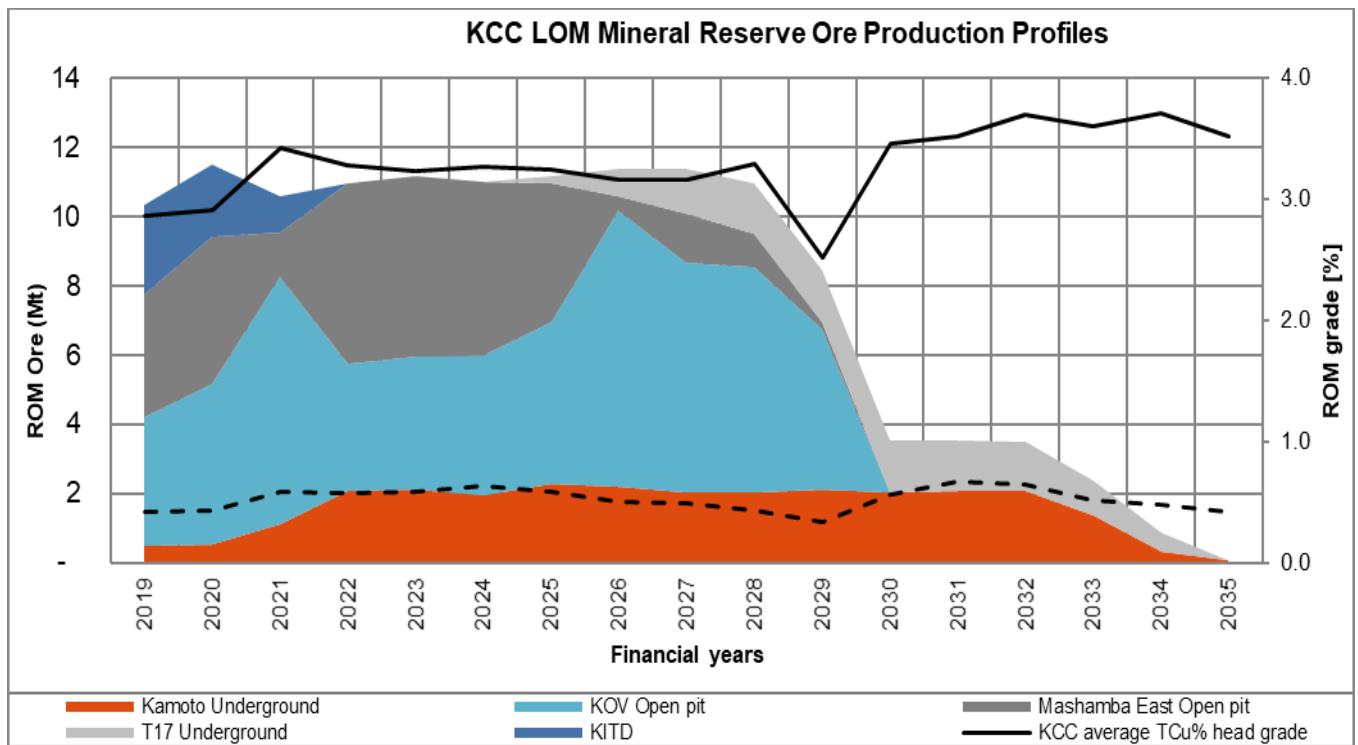


Figure 82: KCC LOM Mineral Reserve ore production profiles

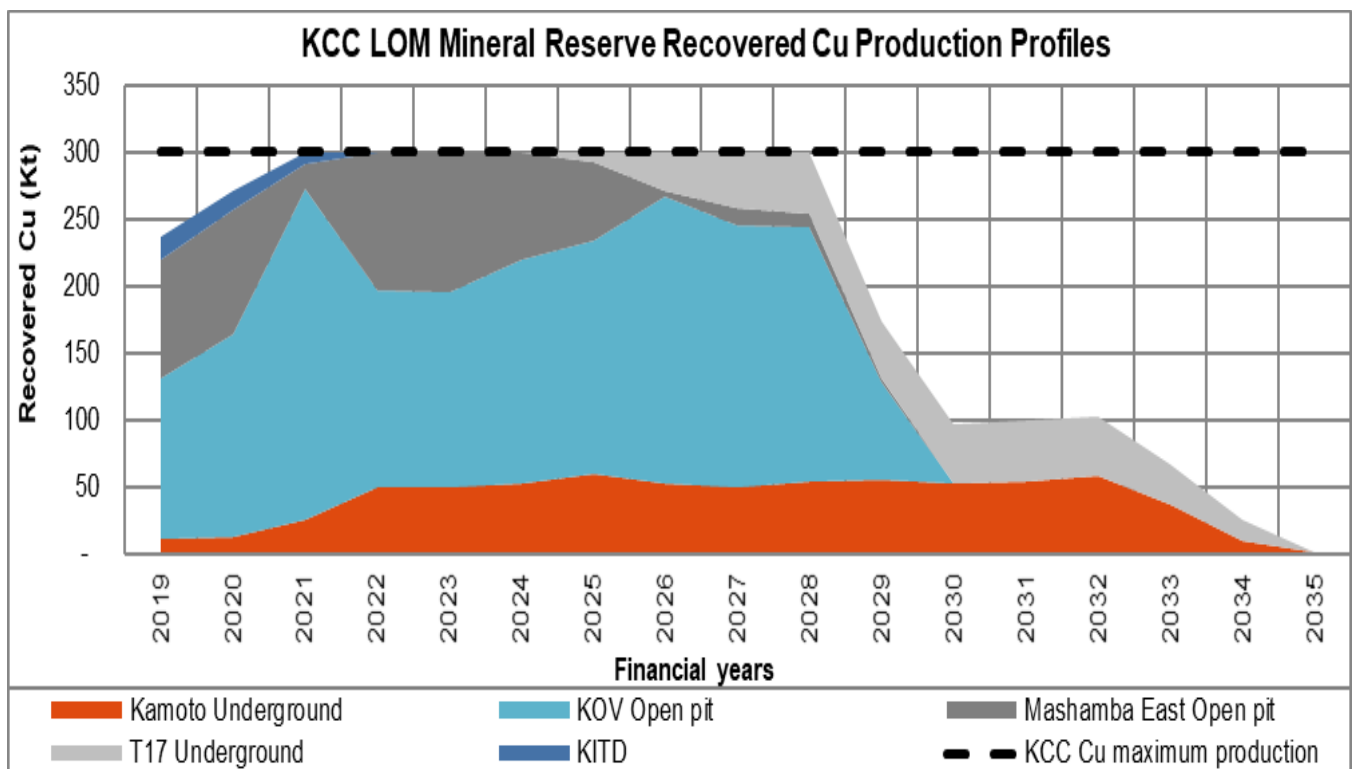


Figure 83: KCC LOM Mineral Reserve recovered Cu production profiles

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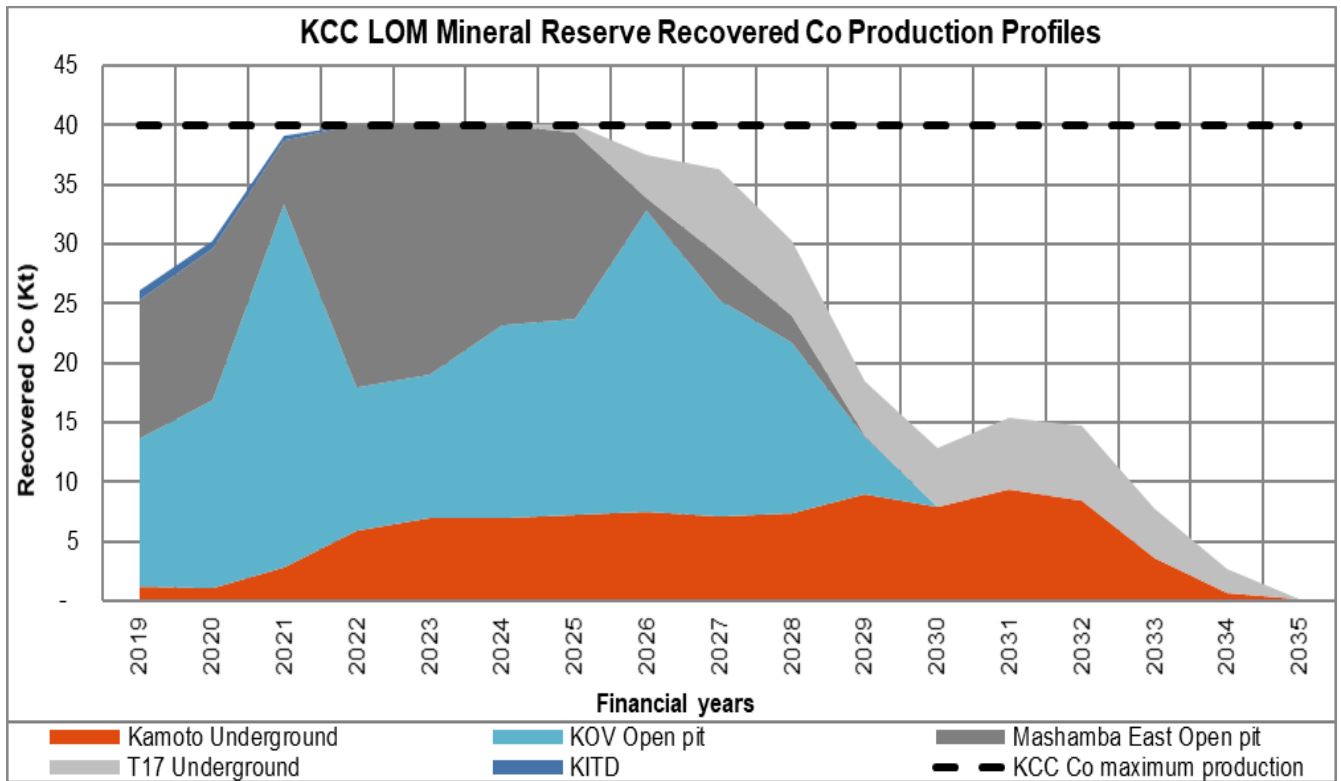


Figure 84: KCC LOM Mineral Reserve recovered Co production profiles

16.6 UG Mining

16.6.1 KTO UG

16.6.1.1 Background

In 2011, strategic technical work was commenced to simplify the mining methodologies, and to apply historical, low cost and appropriate methods in which the mining personnel are experienced and efficient. It was decided then that the mining methods and sequence, mining extraction and dilution achieved, and the cement and waste backfill requirements should be confirmed through numerous studies. Medium term technical work was conducted to reflect the change in mining methodology, but a full, appropriately detailed LOM plan that incorporates the mining methods practised at KCC was only completed in 2013. The LOM plan is now updated annually and the budget and Mineral Reserves are derived from this LOM plan.

Details of the zone geometry and mining methods that were considered during previous studies are shown in Table 92 together with the current mining methods, as designed and scheduled in 2013 (and now updated annually).

Table 92: Summary of mining methods and previous studies

Zone	Geometry	RSV 2006 Feasibility Study	RSV 2007 Detailed Design	2008 Feasibility Study	2013 LOM Re-design and
Z1 bottom and top	Flat and steeply dipping portions	Long Hole Retreat Stoping	Cut and Fill (CAF)	CAF	TLHR
Z2 OBS and OBI	Flat and steeply dipping portions	Long Hole Retreat Stoping	CAF	CAF	RAP
Z3 and Z4 OBS	Flat	RAP	RAP	RAP	RAP

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Zone	Geometry	RSV 2006 Feasibility Study	RSV 2007 Detailed Design	2008 Feasibility Study	2013 LOM Re-design and
Z3 and Z4 OBI	Flat	Long Hole Retreat Stoping	Long Hole Retreat Stoping	RAP	RAP
Z5 OBS	Flat	RAP	RAP	RAP	RAP
Z5 OBI	Flat	Long Hole Retreat Stoping	RAP	RAP	RAP
Z6 OBI	Flat and steeply dipping portions	Long Hole Retreat Stoping	CAF	Post pillar cut and fill and Cut and Fill	TLHR
Z7 OBI	Very steeply dipping	N/A	N/A	Sub level caving	TLHR
Z8 OBS	Flat	RAP	RAP	Room and Pillar	RAP
Z8 OBI	Flat	Long Hole Retreat Stoping	RAP	Room and Pillar	RAP
Z9 OBS and OBI	Very steeply dipping	CAF	CAF	Sub level caving	CAF
Etang North	Steeply dipping to vertical	Long Hole Retreat Stoping	CAF	Post pillar cut and fill and Cut and Fill	LLHR
Etang South	Moderate to steeply dipping	Long Hole Retreat Stoping	CAF	Post pillar cut and fill and Cut and Fill	TLHR

TLHR – Transverse Longhole Retreat with Backfill

LLHR – Longitudinal Longhole Retreat with Backfill

The KTO service and rock shafts are shown in Figure 85 and the access portals in Figure 86.



Figure 85: KTO surface service shaft and rock hoisting shaft

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Figure 86: KTO surface portals to underground workings

The KTO mining zones are indicated in Figure 87 below.

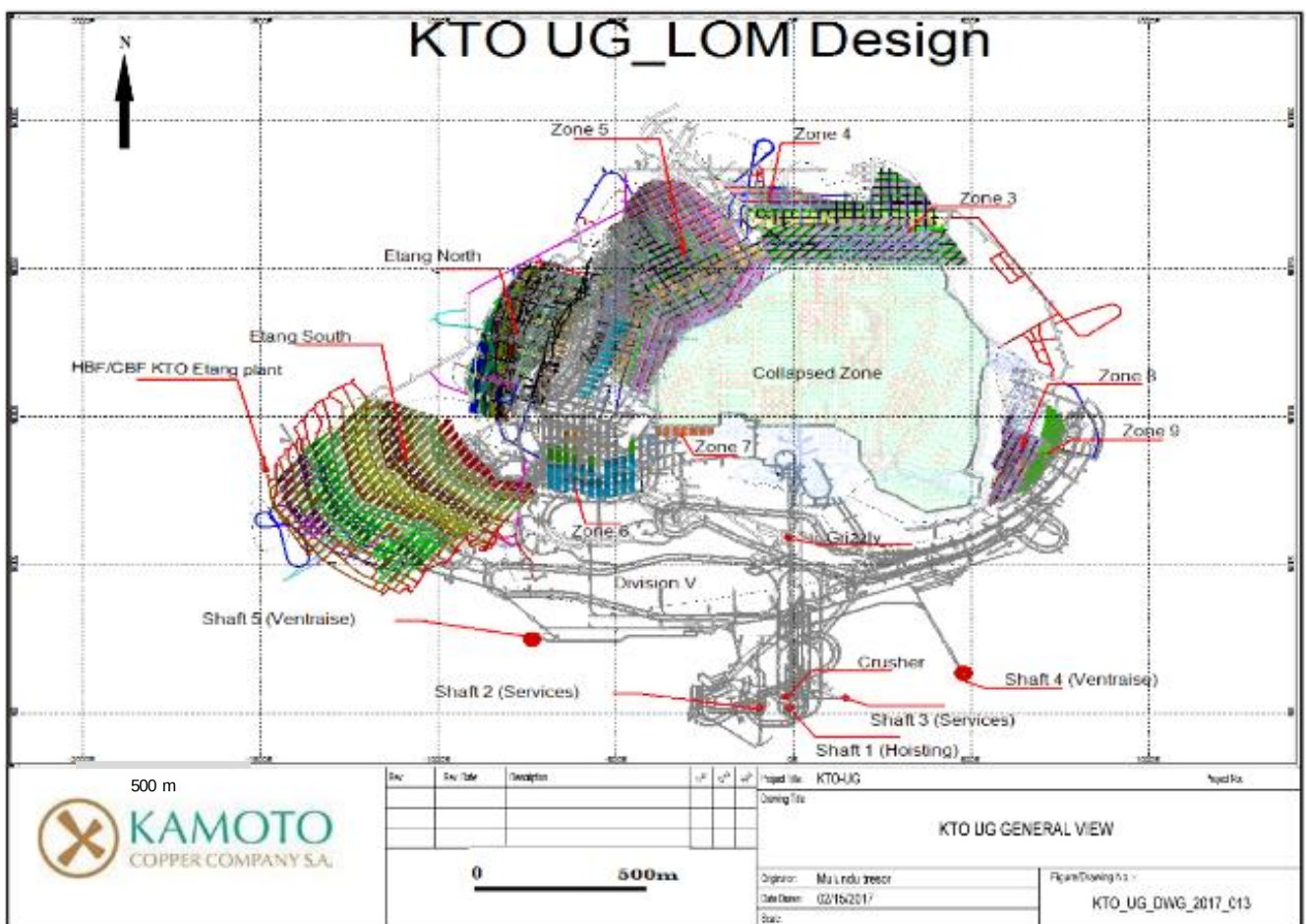


Figure 87: General configuration of the mining zones at KTO UG

16.6.1.2 KTO Modifying Factors

The modifying factors as applied for KTO UG design in the mining model for conversion to Mineral Reserve include:

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- The underground stopes have been designed at a 1.5% TCu cut-off;
- Cut-off grade is calculated using an assumed long term price of \$6,500 per tonne of copper and long term price of \$60,000 per tonne of cobalt;
- Over break in stopes are 6% at a dilution grade of 0.6% TCu and 0.05% TCo;
- Mining losses and geological losses are 5%;
- Planned dilution in stope design is 5% to 10%; and
- RAP pillar extraction is 45%.

Price assumptions listed above are different than the current long term price assumptions. However, the variances are not material for the Mineral Reserves.

16.6.1.3 KTO UG Mining Methods

The deposit is currently divided into three categories of dip/plunge, and the three mining methods applied throughout the zones are:

- RAP with Hydraulic or Cement Hydraulic Backfill at dips of less than 12°;
- TLHR with Backfill between 13° and 55°; and
- LLHR with Backfill at dips greater than 55°.

16.6.1.3.1 Room and Pillar

This method is planned for Zones 2, 5, 3, 4 and 8. A typical RAP mining layout is shown in Figure 88 and Figure 89.

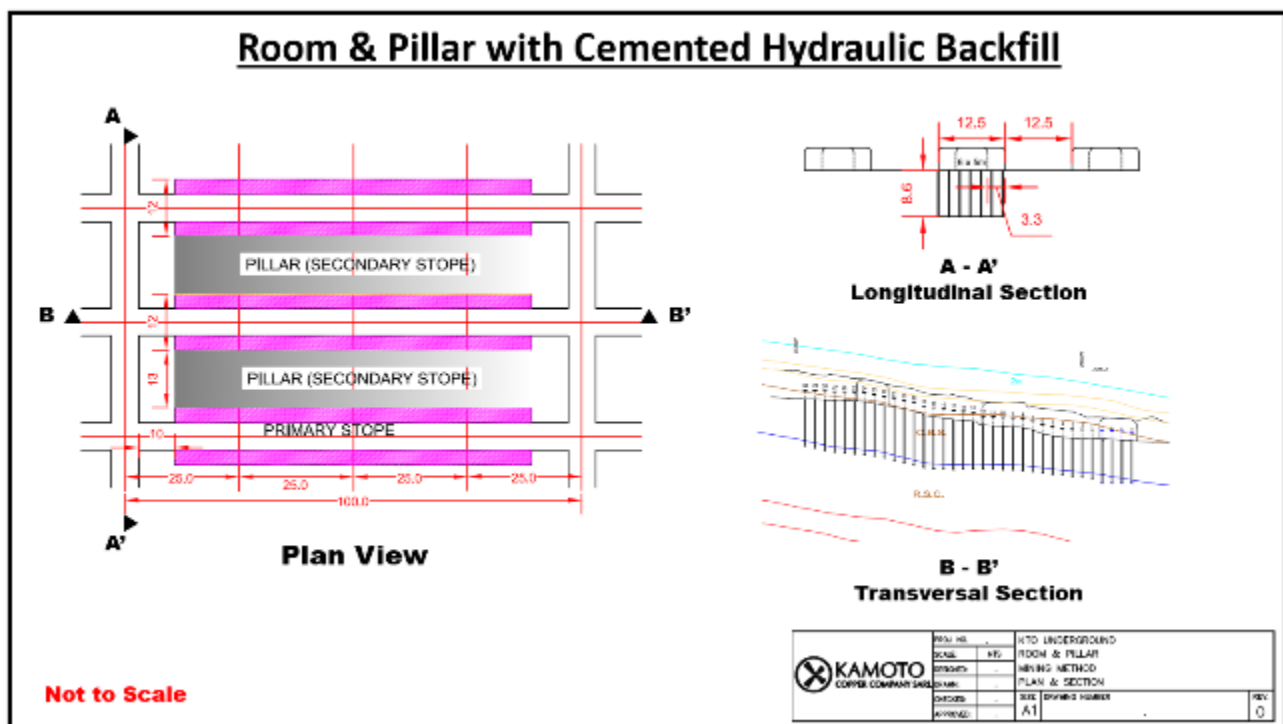


Figure 88: RAP with backfill for Zones 2,3,4,5 and 8 schematic

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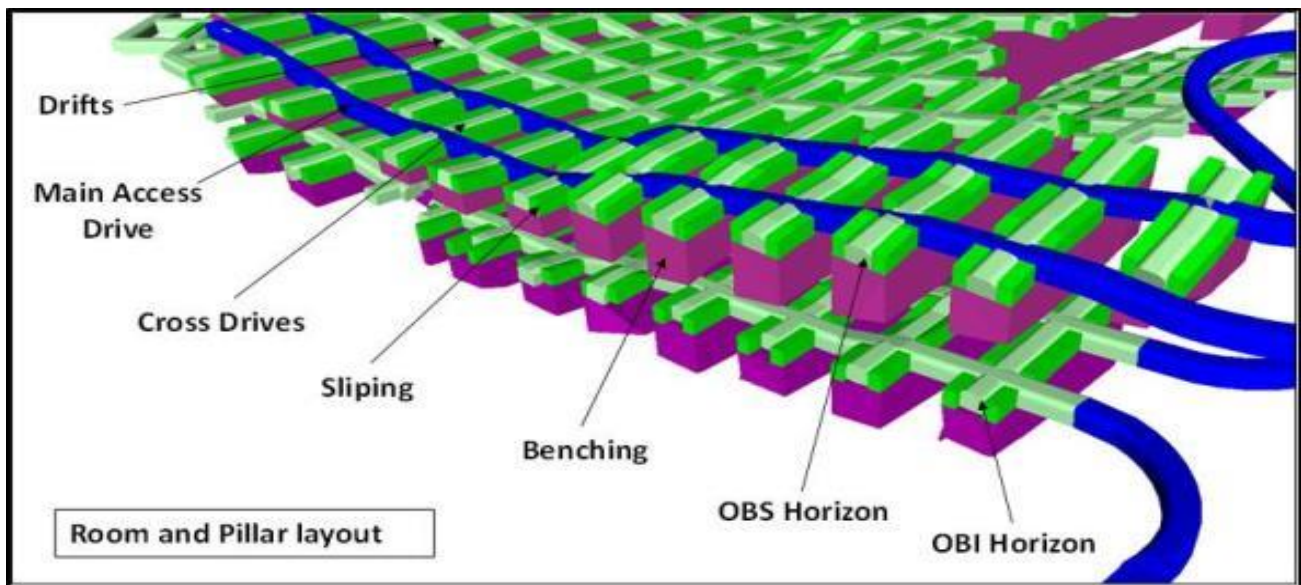


Figure 89: RAP with cemented backfill for Zones 2, 3, 4, 5 and 8

The RAP mining practised at KTO UG is applied to the OBS and OBI horizons. Access drives are developed along strike on the hanging wall contact, 6 m wide by 5 m high on 25 m centres. Cross drives (6 m x 5 m) are developed, 25 m centre to centre, along the centre of the stope for the full length thereof; connected by drifts, 25 m centre to centre, for ease of loading the broken ore and for ventilation purposes. A strike gathering drive is developed at the bottom of the stope for cleaning of ore. The cross drives are then slyped to a full width of 12.5 m and height of 5 m. Benching of drives takes place by drilling long down-holes and slots from the slyped cut to the footwall of the stope, and then blasting from the gathering drive end. Benching operations take place under a hanging wall that is supported in the development and slyping stages of mining.

The 12.5 m stopes with 12.5 m pillars are mined as primary stopes and backfilled with cemented hydraulic fill to provide support for the pillars. If fill is not used, much larger pillars would be required for a stable layout. As an indication, for a mining height of 15 m with a 12.5 m wide roadway, the required pillar width to provide a FoS of 1.2 would be close to 50 m. Overall volumetric extraction would reduce to approximately 40%. After curing, pillars can be developed and mined as secondary stopes in a similar fashion as the primary stopes. These secondary stopes are filled with waste rock or hydraulic fill for regional stability. This is a high extraction mining method with relatively minor losses and dilution. Pillar extraction or mining for the secondary stopes was not very successful historically at KTO UG due to the quality of the fill in the primary stopes. KCC has secured relevant skills to start addressing this problem. The pillar or secondary stopes account for approximately 11.8% of the total Mineral Reserve tonnages of KTO UG.

16.6.1.3.2 Transversal Longhole Retreat with Backfill

This method is practiced in Zones 1, 6, 7 and Etang South for dips between 12° and 55°. The stopes are 15 m to 17 m wide and pillars are 8 m wide along the strike. A schematic representation of a general layout is shown in Figure 90 and Figure 91.

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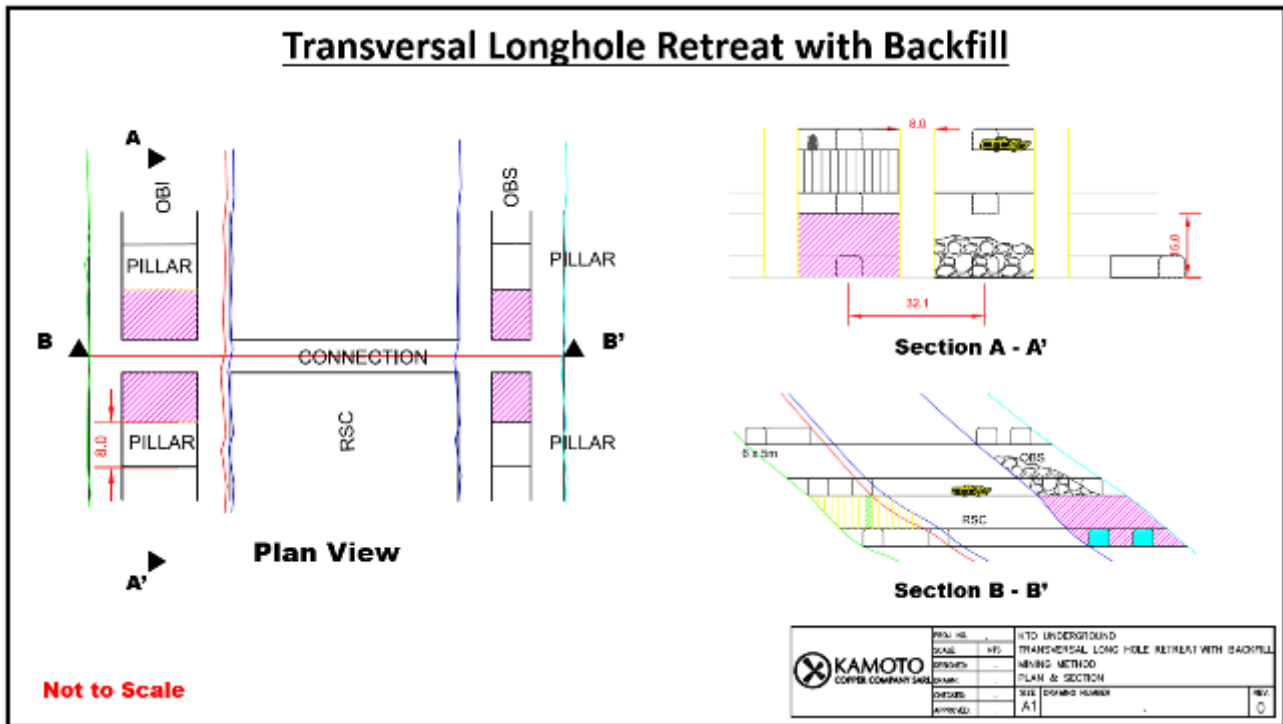


Figure 90: Schematic representation of TLHR with backfill for Zone 1, 6 and 7

The mining method requires a 6 m x 5 m drive on the hanging wall contact at 15 m to 20 m vertical intervals. Crosscuts (6 m x 5 m) are developed to the foot wall contact of the specific orebody at 23 m centres and in this position drives are again established on either side of the crosscut to eventually create one continuous drive on the footwall contact.

Crosscuts are then splayed to 16 m wide and 5 m high chambers by establishing 16 m wide stopes and pillars 8 m wide along strike. Drifts are supported with 2.4 m bolts at 1.2 m centres while the 16 m wide chambers are supported with additional 4 m long anchors at 2 m centres. A connection drive is developed to connect the chambers at the footwall contact. Slot raises and longhole blast rings are drilled from the top and bottom of the chambers to the outline of the orebody. Rings are blasted sequentially towards the slot from the top down and bottom up. Stopes are backfilled depending on the availability of waste rock from other production areas. Hydraulic fill is used if development waste is unavailable and appropriate backfill logistics exist in the area in question.

Etang South has a weak hanging wall on the OBS horizon which necessitated a design change to only use access drives with fans and not strike drives with parallel rings along the hanging wall. Pillars of 8 m wide are left as rib pillars. The Etang South mine design is shown in Figure 91. Extraction of the 8 m rib pillar may be possible for the OBI which has a stronger hanging wall than the OBS.

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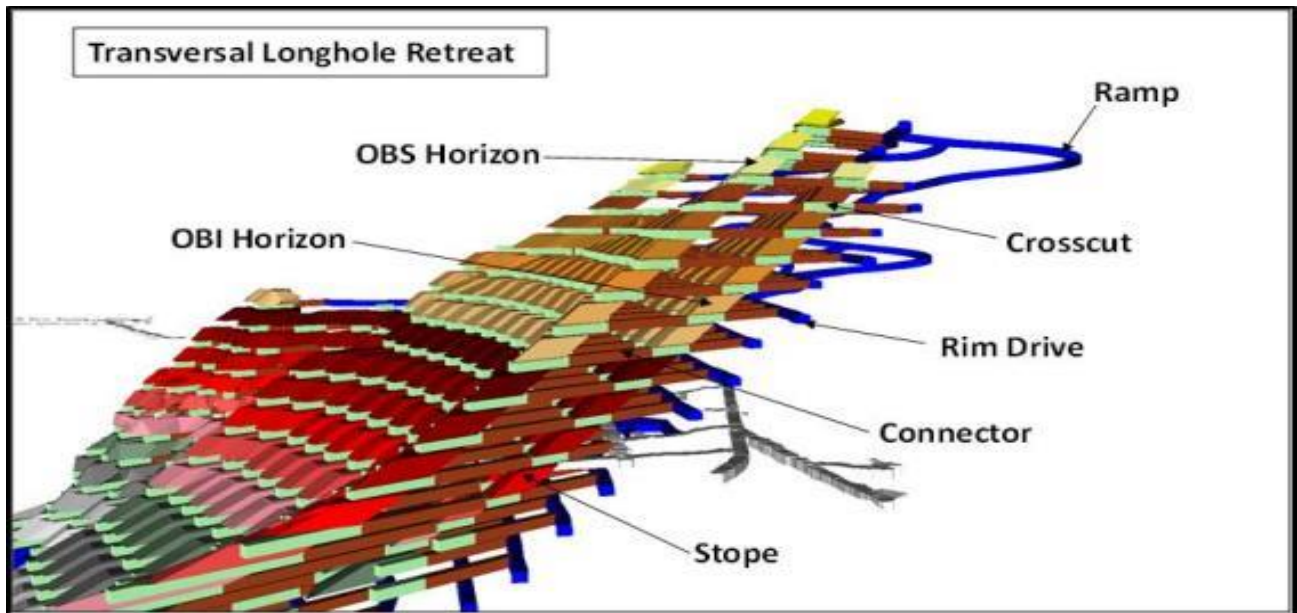


Figure 91: Isometric view of TLHR with backfill in Etang South

16.6.1.3.3 Longitudinal Longhole Retreat with Backfill

This method is used in Etang North for dips off 55° and higher. The vertical level spacing is 15 m in old areas and 15 m to 20 m in undeveloped areas. Stopes are designed at the width of the orebody, 60 m in length and with 8 m rib pillars. A schematic representation of the mining method design is shown in Figure 92 below.

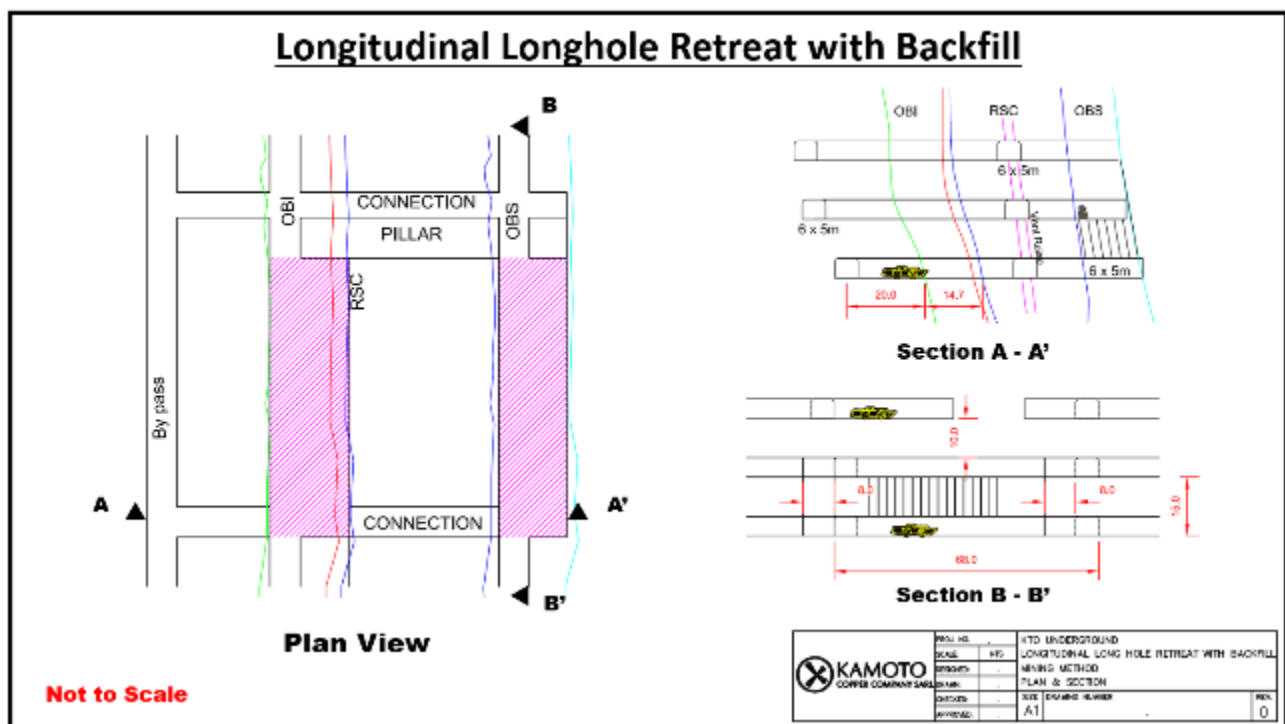


Figure 92: Schematic representation of LLHR with backfill

The generic design of the Longitudinal Longhole Retreat mining method (as applied at KTO UG) is shown in Figure 93. Stopes are designed at the width of the orebody, 60 m in length with 8 m rib pillars. These pillars serve to provide regional support and protection for the contained access ways and infrastructure.

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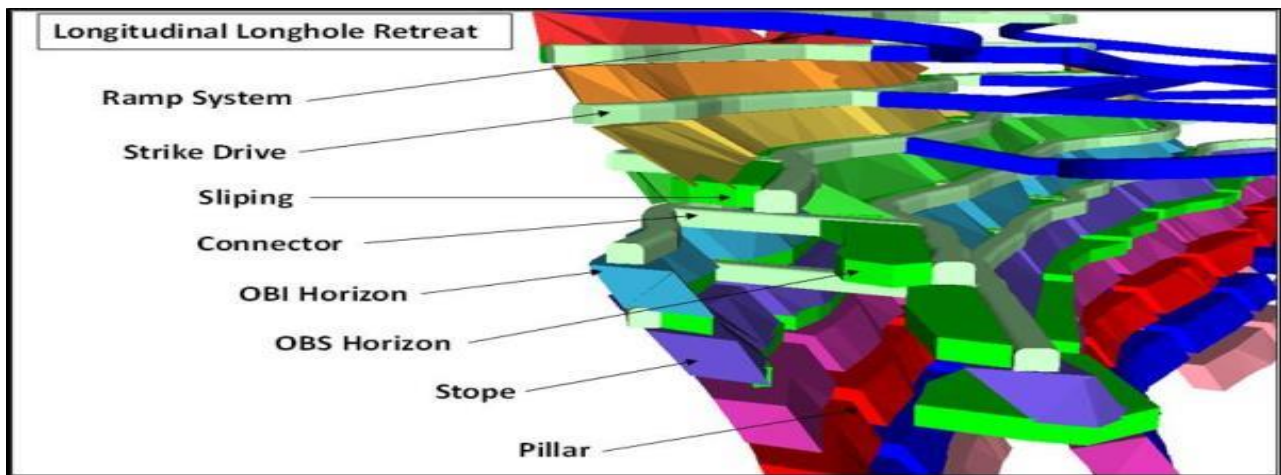


Figure 93: Three dimensional representation of LLHR with backfill

Level access is developed through a ramp system. Strike drives (6 m x 5 m) are developed and located at the safest position in the orebody, for that level, based on geotechnical and geological parameters. These drives are developed to the full width of the orebody. The initial development is 6 m x 5 m and is supported with 2.4 m bolts at 1.2 m centres. The orebody width slyped section is supported with 4 m bolts or anchors at 2 m centres. Down holes are drilled from the upper slyped drive for the slot and the stope. The slots and rings are blasted and cleaned from the lower level access on a retreat basis. Ore extraction is achieved using remotely operated loading equipment. On completion of ore extraction, barricades are constructed in the drill drives at each end of the panel and panels are backfilled with waste rock or hydraulic fill depending on availability.

16.6.1.4 Mineral Reserve LOM Production Scheduling

The Mineral Reserve LOM plan schedule for KTO UG was completed in Deswik mine design and scheduling software. The KTO UG LOM plan is shown in Table 93 below.

Table 93: KTO UG Mineral Reserve LOM schedule

KTO UG Mineral Reserve	Unit	2019	2020	2021	2022	2023	2024	2025	2026	2027
Mineral Reserve ore	Mt	0.5	0.5	1.1	2.1	2.1	1.9	2.3	2.2	2.0
ROM Cu grade	%	3.03	3.48	3.07	3.19	3.16	3.58	3.47	3.12	3.28
ROM Co grade	%	0.37	0.35	0.40	0.44	0.51	0.55	0.49	0.52	0.55
Sec. Dev.	km	4.5	5.3	7.0	8.0	8.0	8.0	7.0	7.0	3.2
Prim. Dev.	km	3.6	5.9	7.3	7.4	7.6	7.2	7.6	6.4	1.2
Recovered Cu	kt	11.3	13.3	25.5	50.1	50.3	53.0	59.9	52.3	50.1
Recovered Co	kt	1.2	1.1	2.8	5.9	7.0	7.0	7.3	7.4	7.1
KTO UG Mineral Reserve	Unit	2028	2029	2030	2031	2032	2033	2034	2035	Total
Mineral Reserve ore	Mt	2.0	2.1	2.0	2.0	2.1	1.3	0.3	0.0	26.6
ROM Cu grade	%	3.56	3.45	3.39	3.49	3.71	3.67	4.20	3.52	3.40
ROM Co grade	%	0.56	0.65	0.60	0.71	0.63	0.41	0.39	0.42	0.54
Sec. Dev.	km	1.0	0.8	0.1	0.0	0.2	0.0	0.0	0.0	60.2
Prim Dev.	km	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0	54.8
Recovered Cu	kt	54.4	55.5	52.2	54.5	58.3	37.6	9.4	1.2	688.6
Recovered Co	kt	7.4	9.0	7.9	9.4	8.4	3.6	0.7	0.1	93.4

ROM - run of mine

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Sec – secondary
 Prim – primary
 Dev – development

The Mineral Reserve LOM production profile with ROM head grade and development requirements are shown in Figure 94 and Figure 95 below. The KTO UG expansion plan considers the development required to ramp up to 4.0 Mtpa in 2028 to maintain 300 ktpa of saleable copper production.

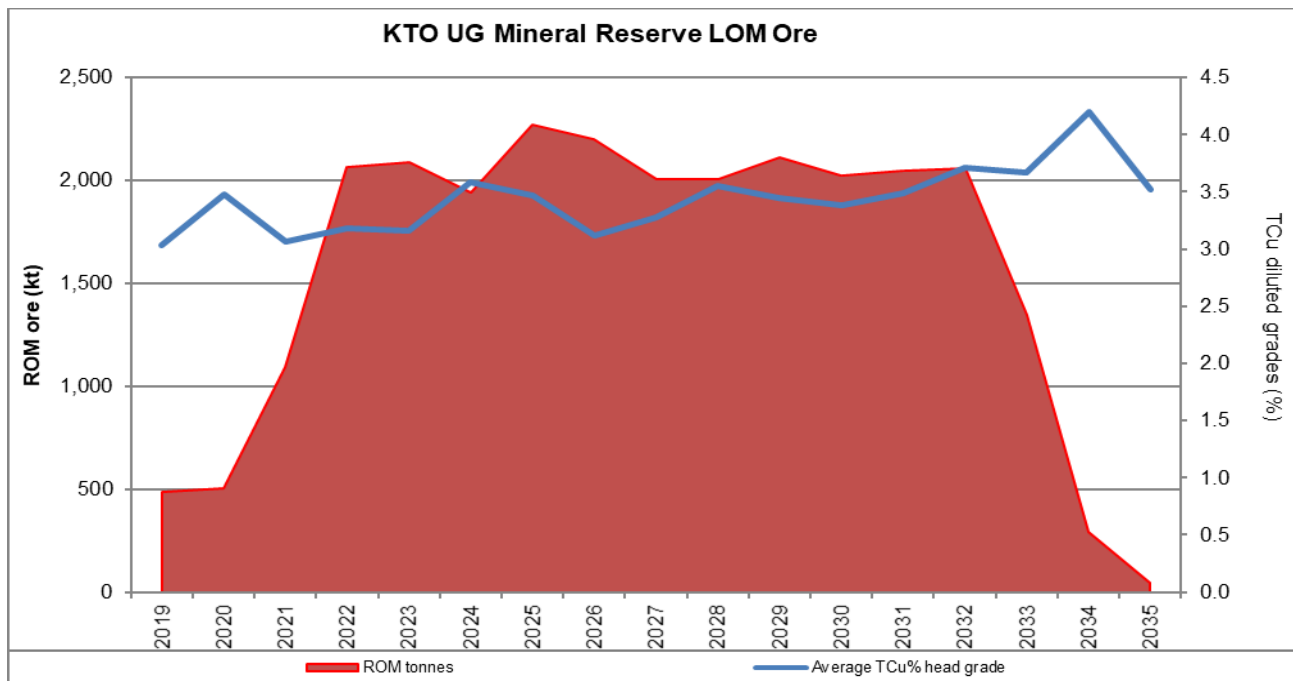


Figure 94: KTO UG Mineral Reserve LOM production profile

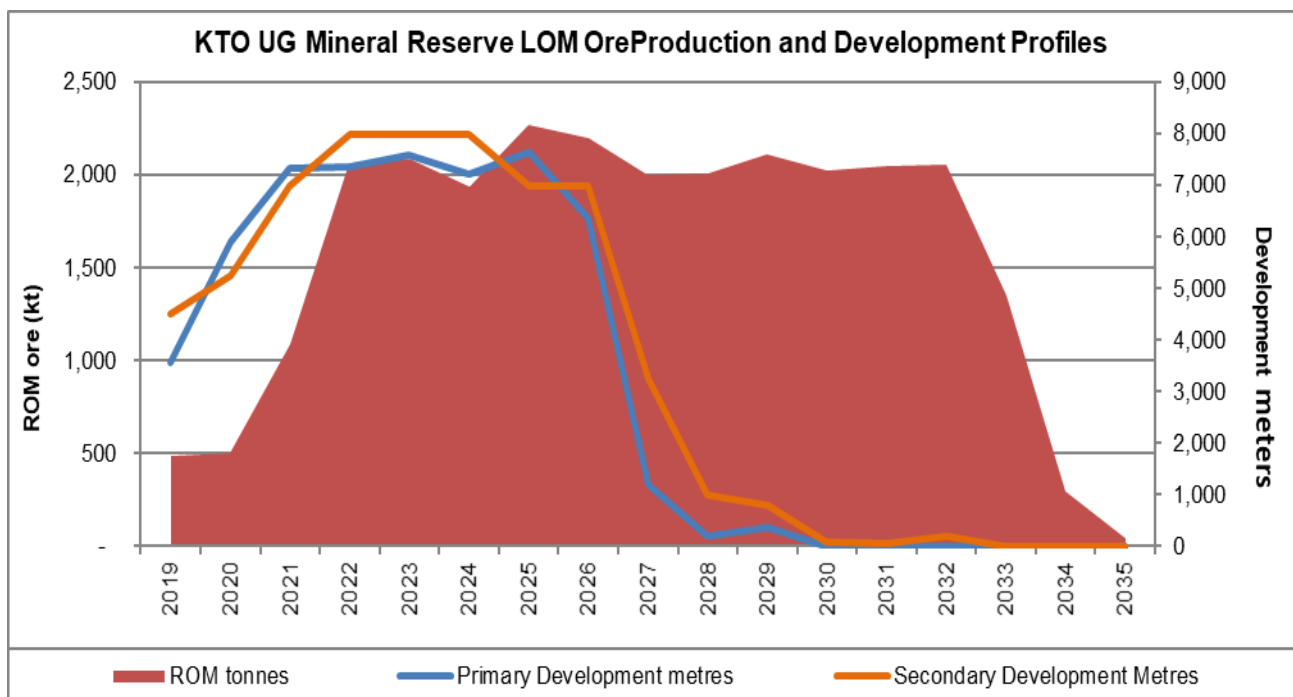


Figure 95: KTO UG Mineral Reserve LOM ore production and development profiles

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16.6.1.5 KTO UG Mining and Support Equipment

Mine production and support equipment for development and ore stoping is provided in Table 94.

Table 94: KTO UG mining and support equipment

KTO UG equipment	Unit	2019	2020	2021	2022-2033
Atlas Copco Boomer 282	#	6	6	11	12
Atlas Copco Simba S7D	#	2	2	2	7
Atlas Copco XAS 186 DD	#	3	3	4	11
Atlas Copco XAS 47	#	1	1	1	1
Getman 2-500L	#	2	2	1	2
AARD UV80	#	1	1	1	1
Atlas Copco ST1530	#	0	0	0	0
CAT R2900G	#	2	2	4	8
Atlas Copco MT5020	#	2	2	1	1
CAT AD45B	#	2	2	5	14
CAT Skid Steer Loader 226B	#	2	2	2	2
Fermel Boom Basket	#	2	2	2	2
Scissor Lift	#	6	6	6	6
Scissor Lift	#	1	1	1	1
Fermel Mediator	#	1	1	1	1
Piccinini Mixer Truck 3500	#	1	1	1	1
CAT Wheel Loader 950H	#	2	2	2	2
Atlas Copco ST1030	#	1	1	0	0
CAT 1600	#	3	3	3	3
Atlas Copco MT436B	#	3	3	3	0
CAT AD30	#	0	0	3	3
CAT 422E Backhoe loader	#	3	3	3	3
CAT Bulldozer D7	#	2	2	1	0
CAT Track Dozer D6R	#	2	2	2	3
CAT Grader 140H	#	2	2	2	2
CAT Compacter CS533D	#	1	1	1	1
Renault 420	#	2	2	2	2
Renault 330	#	2	2	2	2
Putzmeister Wetkret	#	1	1	1	1
Atlas Copco Scaletec	#	3	3	2	2
Atlas Copco Cabletec	#	2	2	2	2
AARD UV80	#	3	3	2	3

16.6.1.6 KTO Mineral Reserve LOM Plan Reconciliation

The previously stated Mineral Reserve estimate was published as at 31 December 2017. The variance between the Mineral Reserve estimates at 31 December 2017 and 31 December 2018 is listed below in Table 95. There is a decrease of 0.3 Mt in the KTO UG Mineral Reserve estimate.

Table 95: KTO UG LOM Mineral Reserve reconciliation

KTO UG	December 31, 2018 Mineral Reserve	December 31, 2017 Mineral Reserve	Variance
ROM (Mt)	26.6	26.9	-0.3
Recovered Cu (kt)	688.6	696.7	-8.1

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KTO UG	December 31, 2018 Mineral Reserve	December 31, 2017 Mineral Reserve	Variance
Recovered Co (kt)	93.4	94.8	-1.4
ROM %TCu grade	3.40	3.40	-
ROM %TCo grade	0.54	0.54	-

16.6.2 T17 UG

16.6.2.1 Background

The T17 OP had been mined to its ultimate final pit shell by 2013. Further pit optimizations indicated that the north-eastern portion of the pit could be extended but it would require a river diversion and additional measures to protect the GCM crusher adjacent to the final pit, as shown in Figure 96. A strategic decision was made in 2013 not to go ahead with the extension but to rather incorporate the remaining OP Mineral Reserve into the UG mine design, as shown in Figure 97, to avoid the risks associated with the OP extension. The pit was dewatered in 2013 after being dormant for two years, in preparation for the T17 pit-bottom portal, and 0.1 Mt of OP tonnes were mined to create the appropriate pit-bottom elevation in the vicinity of the portal.

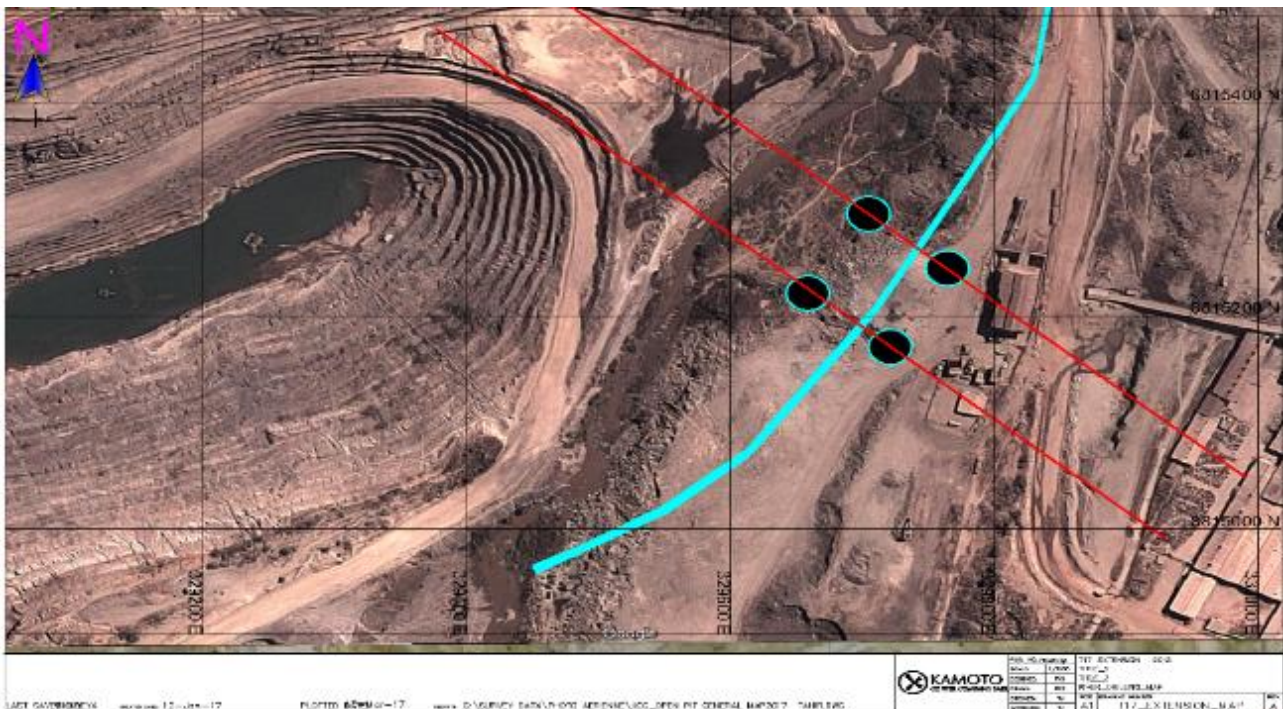


Figure 96: T17 OP extension with crusher protection and road and river diversions as planned in 2012

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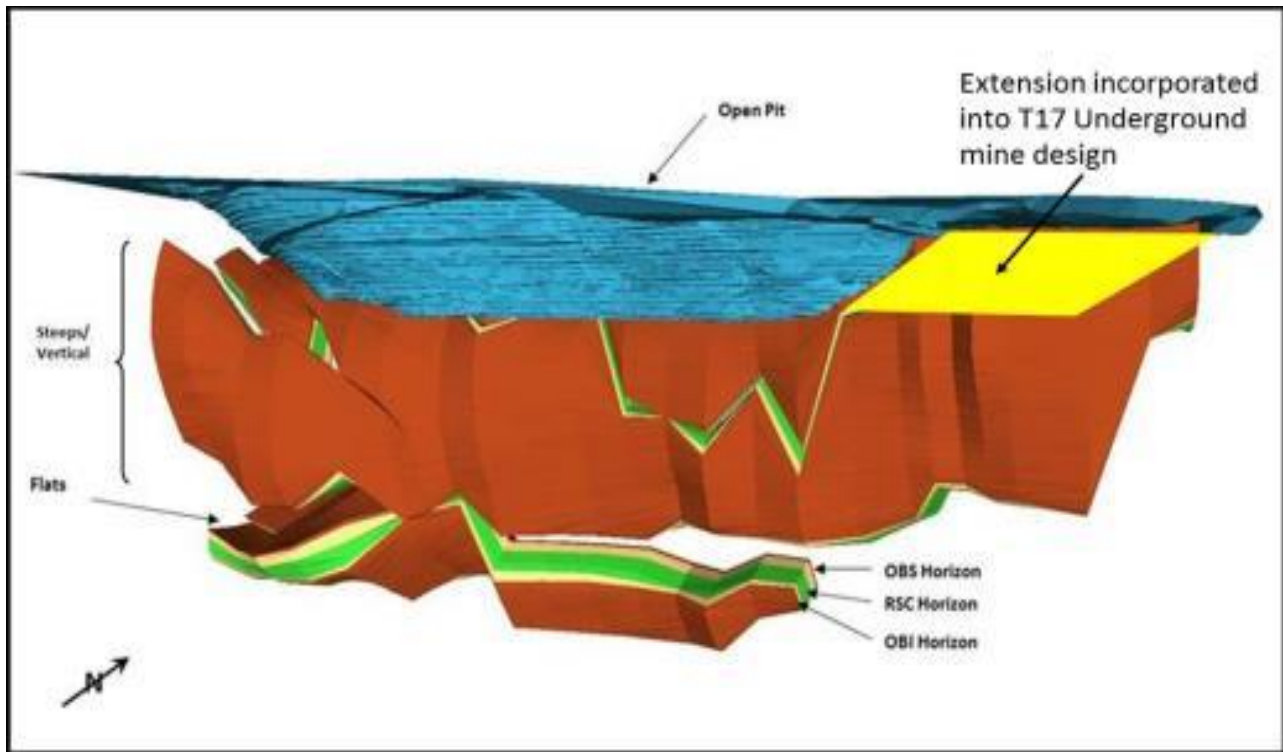


Figure 97: The portion of open pit extension that has been added to the underground mine design

16.6.2.2 T17 UG Modifying Factors

The modifying factors that were used for the T17 UG design in the mining model for conversion to Mineral Reserves include:

- The stopes are designed at a cut-off of 1.2% TCu;
- Cut-off grade is calculated using an assumed long term price of \$6,500 per tonne of copper and long term price of \$60,000 per tonne of cobalt;
- An 8% combined geological loss and mining loss was applied to all longitudinal and transverse stopes;
- An allowance was made for 7.8% dilution on all longitudinal and transverse stopes at 0.5%TCu;
- The longitudinal stoping design consists of 48 m stopes with an 8 m wide pillar (12% rib pillar loss); and
- The transverse stoping design consists of 20 m stopes with an 8 m wide pillar (35% rib pillar loss).

Price assumptions listed above are different than the current long term price assumptions. However, the variances are not material for the Mineral Reserves.

16.6.2.3 Mining Strategy as Defined in the PFS

The T17 UG orebody is near vertical, lending itself to LLHR and Backfill as being executed at the KTO UG zone Etang North (see Figure 92), there are however areas where the OBS, RSC and OBI are mined together in a transverse stoping configuration. The mine design for the T17 UG deposit is shown in Figure 98 and the schematic in Figure 100. The design includes a 30 m crown pillar between the pit bottom and the first levels that will be extracted at the end of the T17 UG LOM. Access to the underground workings is through two portals. A portal at the bottom of the pit facilitates early access and the top portal to facilitate hauling of the ore when mine reaches the full capacity. Re-shaping of the existing benches in the open pit within the immediate vicinity of the proposed portals took place in 2013 and 2014 to accommodate the establishment of the portals. KCC management approved the project and construction of the bottom portal was started in 2013 and can be

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seen in Figure 99. Other connections to the surface will include three ventilation return holes and one raise bored intake hole.

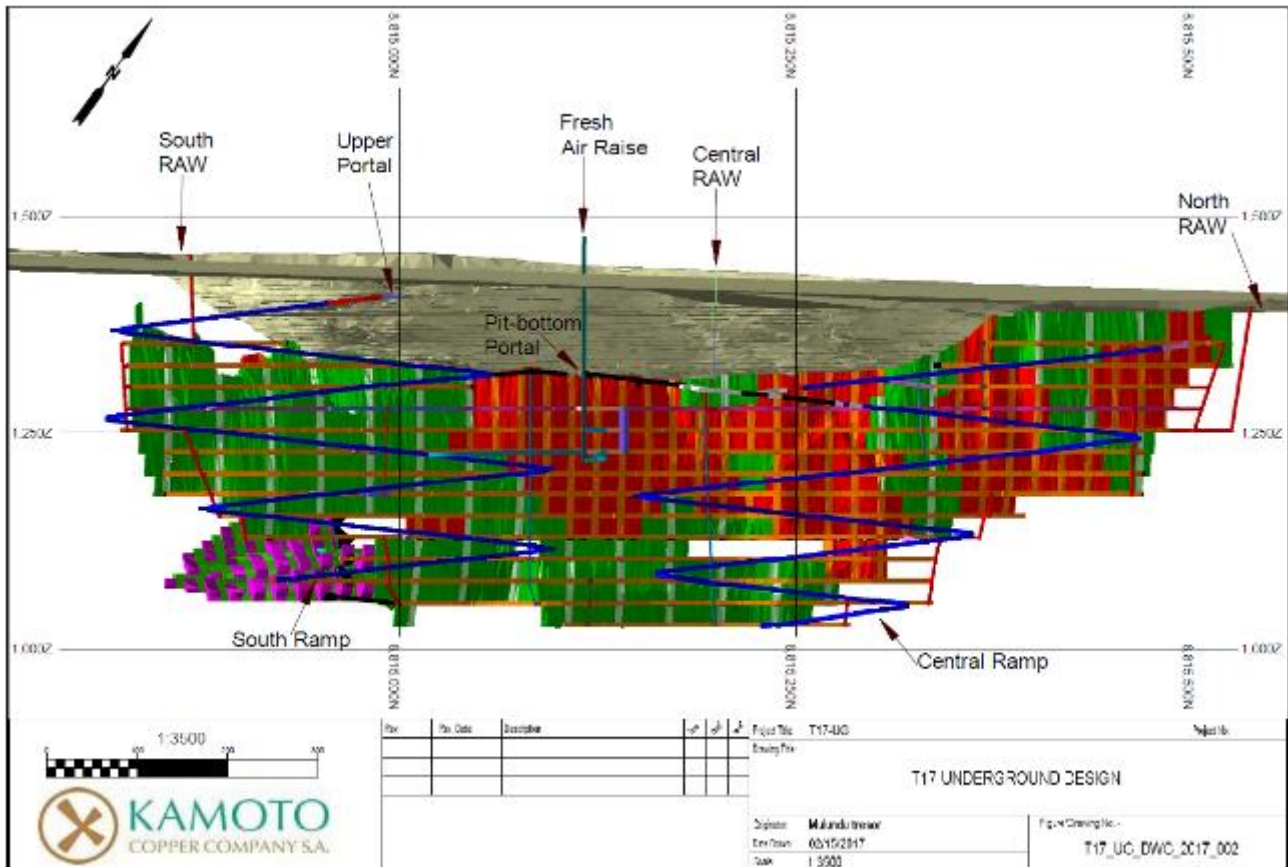


Figure 98: T17 UG LOM design



Figure 99: Left - T17 UG bottom portal construction after project approval. Right – T17 UG Portal construction and development completed

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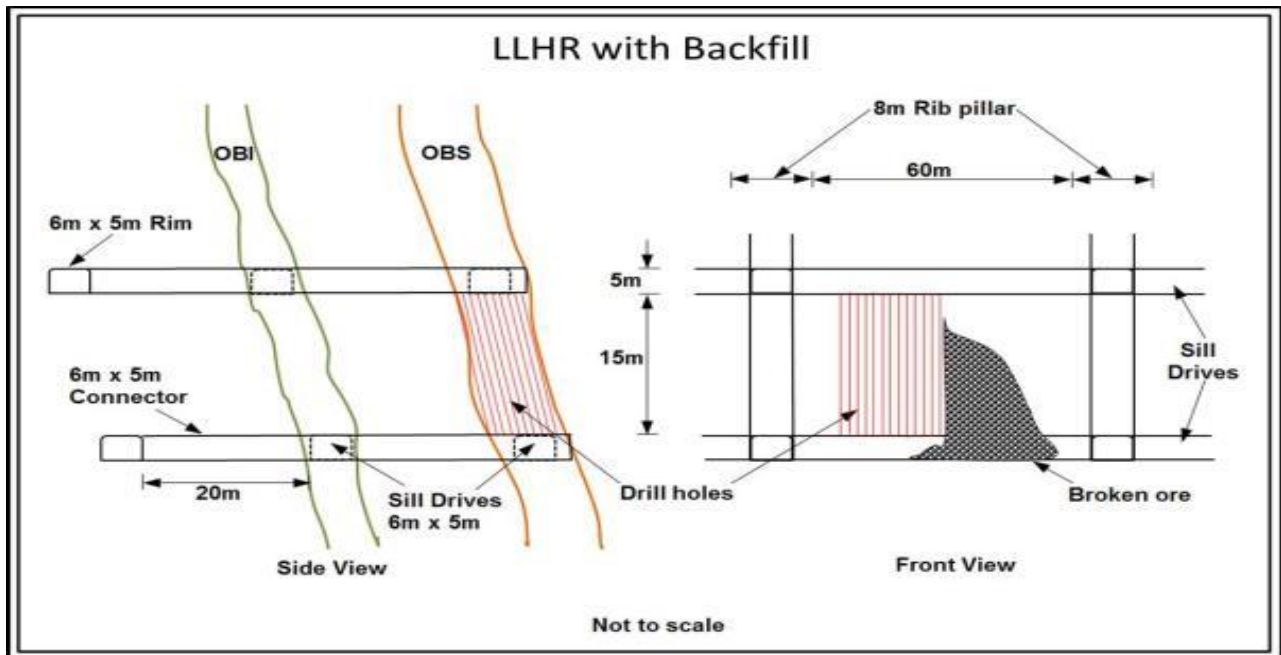


Figure 100: Schematic of LLHR with backfill mining method

16.6.2.4 T17 UG Mineral Reserve LOM Production Schedule

The Mineral Reserve LOM plan schedule for T17 UG was completed in Deswik mine design and scheduling software. The scheduling outputs for the Mineral Reserve estimate are shown in Table 96. Steady state production is reached in 2027.

Table 96: T17 UG Mineral Reserve LOM schedule

T17 UG Mineral Reserve	Unit	2019	2020	2021	2022	2023	2024	2025	2026	2027
Mineral Reserve ore	Mt	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.8	1.3
ROM Cu grade	%	0.00	0.00	0.00	0.00	0.00	0.00	4.48	4.14	3.87
ROM Co grade	%	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.66	0.86
Sec. Dev.	km	0.0	0.0	0.0	0.0	0.0	0.0	1.4	2.9	4.0
Prim. Dev.	km	0.0	0.0	0.0	0.0	0.0	0.9	3.4	5.9	6.6
Recovered Cu	kt	0.0	0.0	0.0	0.0	0.0	0.0	7.5	29.0	42.4
Recovered Co	kt	0.0	0.0	0.0	0.0	0.0	0.0	0.6	3.5	7.2
T17 UG Mineral Reserve	Unit	2028	2029	2030	2031	2032	2033	2034	2035	Total
Mineral Reserve ore	Mt	1.5	1.5	1.5	1.5	1.4	1.0	0.6	0.0	11.3
ROM Cu grade	%	3.67	3.39	3.56	3.57	3.66	3.51	3.46	0.00	3.7
ROM Co grade	%	0.66	0.48	0.51	0.62	0.68	0.64	0.53	0.00	0.6
Sec. Dev.	km	4.2	3.0	3.0	3.0	2.5	0.5	0.0	0.0	24.5
Prim. Dev.	km	5.0	2.6	0.0	0.0	0.0	0.0	0.0	0.0	24.4
Recovered Cu	kt	46.1	43.0	45.2	45.2	44.9	29.9	16.3	0.0	349.7
Recovered Co	kt	6.3	4.6	4.9	6.0	6.3	4.1	1.9	0.0	45.6

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The Mineral Reserve LOM production profile with ROM head grade and development requirements are shown in Figure 101 and Figure 102 below.

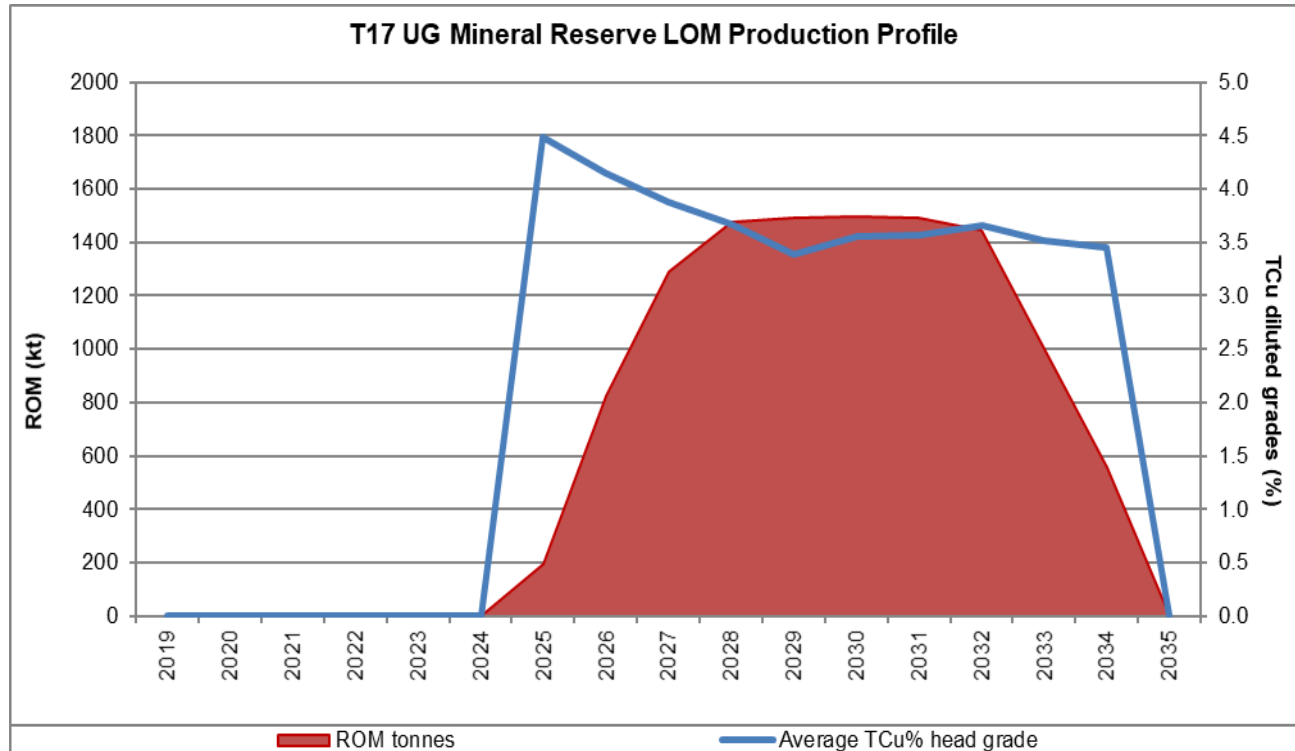


Figure 101: T17 UG Mineral Reserve LOM production profile

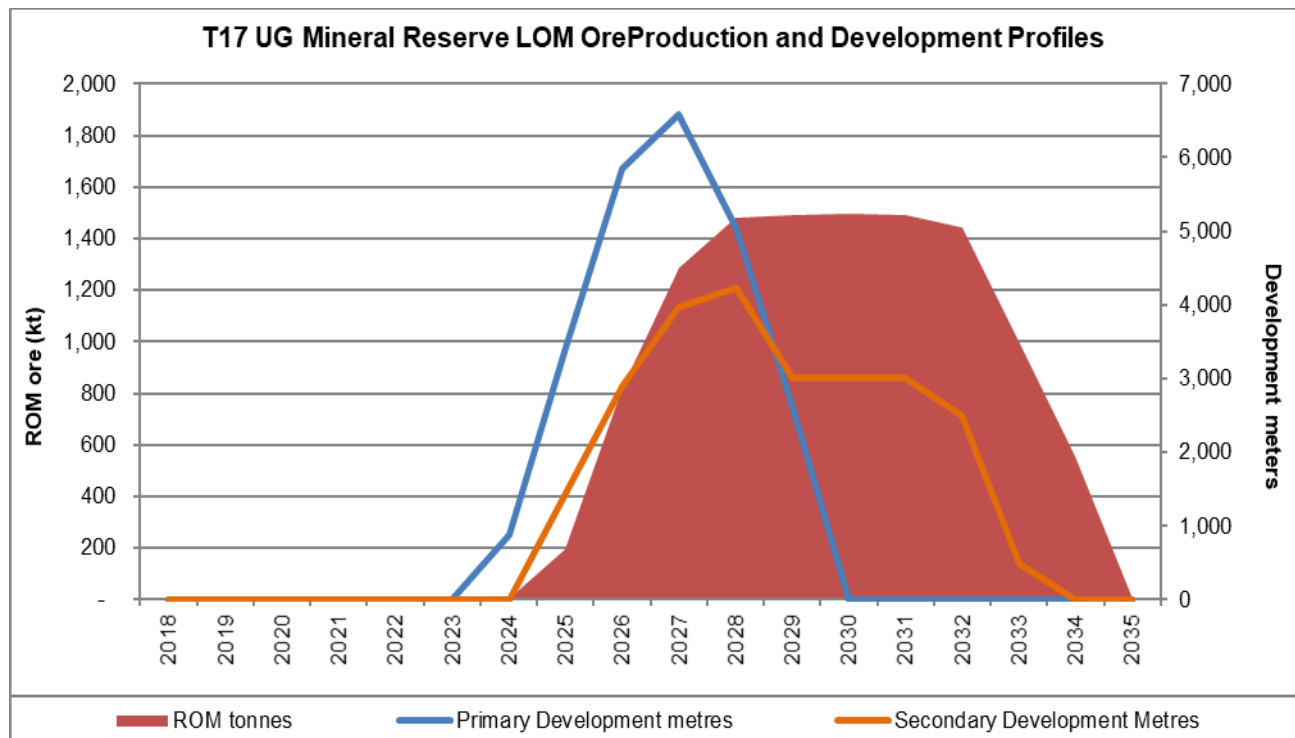


Figure 102: T17 UG Mineral Reserve LOM production and development profiles

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16.6.2.5 T17 UG Mining and Support Equipment

Mine production equipment for secondary development and stoping is provided in Table 97.

Table 97: T17 UG mining and support equipment

T17 UG Equipment	Unit	2024	2025	2026	2027- 2034
Emulsion Car LC	#	1	4	4	4
AARD UV80	#	1	1	1	1
Atlas Copco Simba S7D	#	0	3	3	3
Atlas Copco XAS 186 DD	#	3	3	3	3
CAT AD45B	#	2	7	7	7
CAT R2900G	#	2	5	5	5
CAT Skid Steer Loader 226B	#	1	1	1	1
Scissor Lift	#	1	1	1	1
Piccinini Mixer Truck 3500	#	1	1	1	1
Putzmeister Wetkret	#	1	1	1	1
Fermel Mediator	#	3	8	8	8
LDV	#	4	10	10	10
Atlas Copco Boomer 282	#	2	3	3	3
CAT Grader 140H	#	1	1	1	1
CAT 422E Backhoe loader	#	1	2	2	2
Utility Truck	#	1	4	4	4

Primary development will be developed by contractors. The cost of primary development is included in capital cost based on contractor rates.

16.6.2.6 T17 UG Mineral Reserve LOM Plan Reconciliation

The previously stated Mineral Reserve estimate was published as at 31 December 2016. The variance between the Mineral Reserve estimates at 31 December 2016 and 31 December 2017 is listed below in Table 98. There is no variance in the T17 UG Mineral Reserve estimates.

Table 98: T17 UG Mineral Reserve reconciliation

T17 UG	December 31, 2018 Mineral Reserve	December 31, 2017 Mineral Reserve	Variance
ROM production (Mt)	11.3	11.3	-
Recovered Cu (kt)	349.7	349.7	-
Recovered Co (kt)	45.6	45.6	-
ROM %TCu grade	3.65	3.65	-
ROM %TCo grade	0.62	0.62	-

16.6.3 Underground LOM Mineral Reserve Estimate

The updated LOM Mineral Reserve estimate is based on the mining methods described in the section above and is compliant with the CIM Code. The entire Mineral Resource area was sub-divided into smaller portions per zone to aid the mine design and process schedule. The basis of conversion from a Mineral Resource to Mineral Reserve is that the Measured Mineral Resources convert to Proven Mineral Reserves and the Indicated Mineral Resources to Probable Mineral Reserves. The Mineral Reserve estimate is shown in Table 99 below.

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Table 99: Underground LOM Mineral Reserves as at December 31, 2018¹⁻⁸

Mining operation	Proven			Probable			Total		
	Mt	%TCu	%TCo	Mt	%TCu	%TCo	Mt	%TCu	%TCo
KTO UG	6.5	3.60	0.56	20.1	3.34	0.53	26.6	3.40	0.54
T17 UG	2.2	3.42	0.54	9.1	3.71	0.64	11.3	3.65	0.62
Total	8.7	3.56	0.55	29.2	3.45	0.57	37.9	3.48	0.56

- 1) The Mineral Reserve estimates have been prepared in accordance with the classification criteria of the NI 43-101;
- 2) Grade measurements reported as percent (%), tonnage measurements are in dry metric units;
- 3) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 4) The projects included were developed to a minimum PFS level of accuracy;
- 5) The LOM plans of the existing operations were developed to a minimum PFS level of accuracy;
- 6) Approximately 11.8% of the KTO UG Mineral Reserve are pillars that are recoverable;
- 7) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 8) The Mineral Reserve estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

16.7 OP Mining

Three surface operations are included in the LOM plan, namely KOV OP, Mashamba East OP and the KITD Tailings project. The KOV OP and Mashamba East OP LOM plan and Mineral Reserve are based on conventional load and haul open pit operations. The KITD Tailings LOM plan and Mineral Reserve is based on hydraulic mining. The first section below outlines the process followed in developing these LOM plans.

16.7.1 LOM Planning Process Overview

The LOM Mineral Reserve production from the surface mining operations is based on the available pit space and to maintain the production requirement. An overall production profile for the total KCC operation of 300,000 tonnes of recovered copper per annum is targeted. The LOM planning process for surface mining operations can be summarised in the following steps:

- Selective Mining Unit (SMU) selection;
- Pit optimization and pit selection;
- Pushback analyses and selection;
- Pit pushback and dump design;
- Scheduling unit selection and design; and
- Production planning.

A brief description of each step is given in this section. These apply to KOV OP and Mashamba East OP.

16.7.1.1 Mining Model

The selective mining unit (SMU) is defined as the smallest unit that can be mined selectively as a complete unit. SMUs applied to the various Mineral Resource block models are shown in Table 100 below.

Table 100: Selected SMU dimensions per pit

Mining Operation	Unit	SMU
Mashamba East OP	m	10 x 10 x 5
KOV OP	m	10 x 10 x 10

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16.7.1.2 Dilution

Dilution is defined as the waste material intentionally added during the mining process to the defined in-situ Mineral Resources, in order to make it practically mineable. The methodology applied to estimate the dilution is as follows:

- On the ore contacts (where the in-situ Mineral Resource block consists of a percentage ore material and a percentage waste material) the tonnage and grade of the SMU is defined as the weighted average tonnage and grade of the ore and waste materials contained in the original unmodified blocks; and
- In cases where the total in-situ Mineral Resource block is ore, the corresponding SMU is defined as a 100% ROM block with the same grade attributes as the in-situ blocks.

16.7.1.3 Mining Loss

Mining loss is a percentage estimate of in-situ ore not recovered after mining has taken place. It is an economic mineralization modelled as ore but mined as waste or left in the pit during the mining activities due to structural and mining losses, or does not exist due to artisanal mining in the past. Estimation of mining losses requires an understanding of the Mineral Resource estimation, mine geology, blasting and mining equipment. The dip, strike, width and length of the ore zones within the deposit are the most significant consideration for mining recovery and mining dilution. The variability in geometry of the deposit also has a significant influence on the efficiency of the ore mining. The loading capabilities of the selected equipment must match the blasted rock pile profile and dig-ability. Finally, the equipment bucket size of the selected equipment and the direction of mining relative to the deposit's geometry and blast displacement also influences the mining loss and dilution.

16.7.1.4 Pit Optimization

One of the outputs of the pit optimization process is to determine the position and extent of the final pit boundary. The Datamine™ pit optimization software Net Present Value Scheduler (NPVS) was employed for this purpose.

NPVS uses the Lerchs-Grossmann algorithm to determine the optimal shape for a pit in three dimensions. The method is applied to a block model of the orebody, and progressively constructs lists of the related blocks that should, or should not, be mined. The final list defines a pit outline that has the highest total relative value, subject to the required pit slopes. This outline includes every block that "adds value" when waste stripping is taken into account and excludes every block that "destroys value". It takes into account all revenues and costs as well as mining and processing parameters.

Although a detailed description of the NPVS methodology is beyond the scope of this report, the following provides a brief summary. Pit optimization can be divided into two processes:

- Creation of a range of nested pit shells of increasing sizes. This is done by varying the product price and generating a pit shell at each price point;
- Selection of the optimal pit shell is achieved by generating various production schedules for each pit shell and calculating the net present value for each schedule. The output of this process is a series of "pit-versus-value" curves; and
- Selection of pushbacks for the required production rates and ore blend.

16.7.1.5 Pit Design

The mining method applied is conventional open pit mining, consisting of drilling, blasting, loading and hauling. As part of the mine planning process, a pit design is undertaken once an optimal pit shell has been selected. The pit design process considers:

- Safe operations;
- Continuous access to the individual blocks and the working benches;
- Equipment units and movement requirements;

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- Geotechnical recommendations;
- Water handling;
- Backfill opportunities; and
- The phasing of operations or pre-stripping.

Design work was performed in Hexagon's MineSight general mine planning software. The selected optimum pit shell was used as the design limits. All the input parameters were incorporated to create a three dimensional pit design. The pit design was used to evaluate the tonnages and grades of the various ore types. Pit designs were created based on the current mining methodology which includes mining at 10 m bench. Ramp and pit access designs considered the largest expected hauler dimension, ensuring safe and practical execution. All pit designs adhere to the current geotechnical requirements.

16.7.1.6 Scheduling Units

Block designs were conducted based on a typical blast block or practical bench and production block dimension. Ramps were designed and scheduled separately at appropriate rates. The block designs simulated the scheduling units. Each block could contain a range of material types that could be selectively hauled to separate locations (ROM stockpile, various stockpiles or waste dumps).

16.7.1.7 Production Scheduling

Schedules were produced in Hexagon's MineSight Schedule Optimizer software. Operating slope angles for each pit and pushback were maintained and increased to the final pit slopes as the ultimate pit is reached. Schedules considered the available pit space, number and size of excavators required, and the practical constraints of each pit.

16.7.2 KOV OP

16.7.2.1 Background

The KOV OP LOM plan includes Mineral Reserve of 60.5 Mt of ore at a head grade of 3.70% TCu to the year 2028.

16.7.2.2 Modifying Factors

The KOV OP Mineral Reserve estimate was based on a revised pit optimization process and mine design incorporating the revised geotechnical parameters. The pit optimization process was based on the 2016 updated Mineral Resource model and diluted mining model or SMU model. The SMU block size was adjusted to reflect the current mining practices that considered planned equipment sizes and high production rates. The SMU model was adjusted from a 10 m x 10 m x 5 m block size to a 10 m x 10 m x 10 m block size that resulted in modelled dilution increasing from 8% to 20%.

The Inferred Mineral Resource portion was considered as waste for the purposes of the Mineral Reserve production schedule.

A total mining dilution of 20% was applied on a SMU basis, and a cut-off grade of 0.65% TCu was applied at the KOV OP. The basis of the cut-off grade estimate was to determine the breakeven cost based on selling, processing and royalty costs. The cut-off grade considered revenues generated from copper and cobalt with the appropriate processing recoveries applied.

A total of 5% mining losses was applied as a tonnage loss that does not impact the ROM head grade.

The costs, revenues and recoveries used are shown in Table 101.

16.7.2.3 Pit Optimization

A pit optimization process for KOV OP was completed during the 2017 planning cycle due to the changes in:

- Economic parameters;
- Mineral Resource model;

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- Geotechnical parameters;
- Infrastructure relocation strategy; and
- Waste handling strategy.

Pit optimization parameters are set out in Table 101 below. Mining and processing costs are inclusive of sustaining capital costs. Price assumptions listed in Table 101 and used in 2017 optimization are different than the current long term price assumptions. However, these variances are not material for the Mineral Reserves.

Table 101: KOV OP optimization parameters

Optimization Parameters	Unit	KOV OP Parameter
Reference mining cost	US\$/t	2.61
Processing cost	US\$/t	67.77
Selling cost – Cu	US\$/t Cu	749
Selling cost – Co	US\$/t Co	4,904
Discount rate	%	10.7
Cu recovery – oxide	%	85
Co recovery – oxide	%	65
Cu recovery – sulphide	%	76.14
Co recovery – sulphide	%	65
Cu price	US\$/t	6,500
Co price	US\$/t	55,000

16.7.2.4 Pit Design

No updated design was undertaken. The ultimate pit and pushbacks previously defined were applied. The face positions as at the end of 31 December 2018 are presented in Figure 103.

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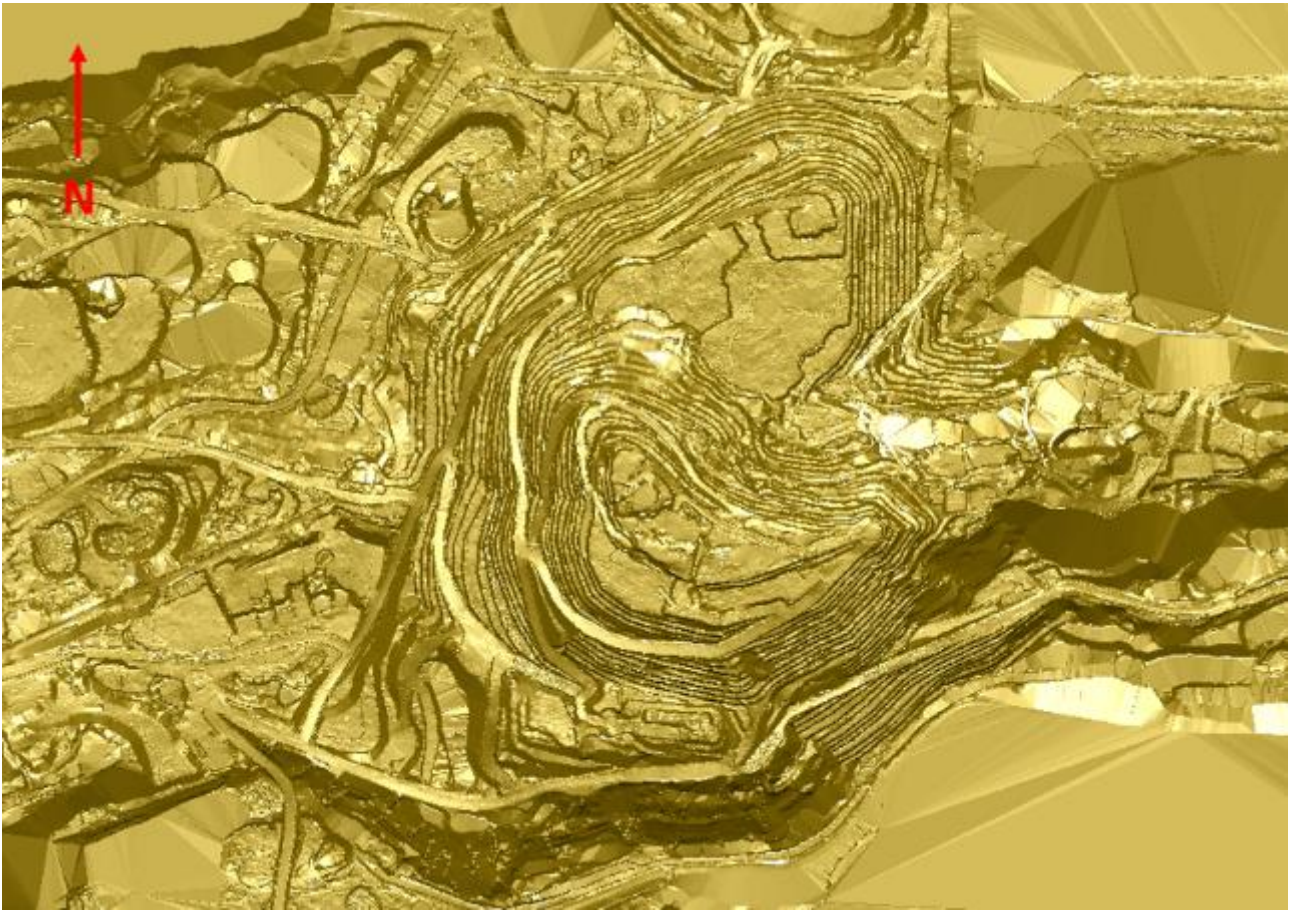


Figure 103: KOV OP survey as at 31 December 2018

Pit floor and high wall dewatering are critical aspects of successfully operating the KOV OP. High wall dewatering infrastructure to minimize the risk of slope failures is operational and will be expanded upon. The current pit layout and dewatering infrastructure are shown in Figure 104 below.

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Figure 104: KOV OP showing pit bottom dewatering

Pit design criteria as applied to KOV OP are shown below in Table 102.

Table 102: KOV OP design criteria

Pit Design Criteria	Unit	KOV
Bench height	m	10.0
Berm width	m	var
Batter angle	degrees	65.0
Ramp width	m	35.0
Ramp gradient	degrees	5.7 (1 in 10)

16.7.2.5 KOV OP LOM Mineral Reserve Production Schedule

The LOM Mineral Reserve production from the KOV OP is planned at a maximum of 8.0 Mtpa. The LOM plan profile is shown in in Table 103 and does not include the Inferred Mineral Resources.

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Table 103: KOV OP Mineral Reserve production profile

KOV OP LOM Mineral Reserve plan	Unit		2019	2020	2021	2022	2023	2024	2025	
Mineral Reserve ore	Mt		3.6	4.6	7.1	3.7	3.9	4.1	4.7	
Average Mineral Reserve %TCu	%		3.79	3.81	4.07	4.65	4.42	4.87	4.36	
Average Mineral Reserve %TCu	%		0.52	0.52	0.66	0.50	0.48	0.62	0.54	
Recovered Cu	kt		116.4	150.1	244.4	145.3	144.7	167.9	173.3	
Recovered Co	kt		12.2	15.7	30.3	11.9	12.0	16.3	16.4	
Waste	Mt		46.5	46.7	35.1	24.9	49.2	71.2	60.5	
KOV OP LOM Mineral Reserve plan	Unit	2026	2027	2028	2029	2030	2031	2032	2033	Total
Mineral Reserve ore	Mt	8.0	6.7	6.5	4.8	0.0	0.0	0.0	0.0	57.5
Average Mineral Reserve %TCu	%	3.17	3.45	3.43	1.91	0.00	0.00	0.00	0.00	3.72
Average Mineral Reserve %TCu	%	0.49	0.42	0.34	0.17	0.00	0.00	0.00	0.00	0.48
Recovered Cu	kt	214.3	195.4	190.9	77.8	0.0	0.0	0.0	0.0	1820.5
Recovered Co	kt	25.3	18.2	14.5	5.4	0.0	0.0	0.0	0.0	178.3
Waste	Mt	57.9	6.0	3.3	1.7	0.0	0.0	0.0	0.0	403.0

16.7.2.6 KOV OP Mining and Support Equipment

Mine production equipment for ore production and waste stripping, as well as support equipment for Mashamba East, is provided in Table 104.

Table 104: KOV OP mining and support equipment

KOV Equipment	Unit	2019	2020	2021	2022	2023-2030
CAT 793D Truck	#	22	26	23	23	23
CAT 777D Truck	#	5	5	5	5	5
Terex 340H Shovel	#	0	0	0	0	0
CAT 6060FS Shovel	#	2	2	2	2	2
CAT 6060BH Shovel	#	1	1	1	1	1
CAT 6030FS Shovel	#	1	1	1	1	1
CAT 385B Excavator	#	2	2	2	2	2
CAT 390D Excavator	#	1	1	4	4	2
Hitachi 470	#	1	1	1	1	1
Hitachi 870	#	2	4	4	4	4
CAT 336D	#	4	4	4	4	

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KOV Equipment	Unit	2019	2020	2021	2022	2023-2030
CAT 992K Wheel Loader	#	2	2	2	2	2
CAT 980	#	1	1	1	1	1
CAT 777D Water Truck	#	2	2	2	2	2
CAT D10T Dozer	#	9	9	9	9	9
CAT D11T Dozer	#	4	4	4	4	4
CAT 834H Dozer	#	1	1	1	1	1
CAT 18M Grader	#	2	2	2	2	2
CAT 24M Grader	#	3	3	3	3	3
CAT CS76XT Compactor	#	1	1	1	1	1
CAT 740 Truck	#	3	3	3	3	3
CAT 745 Truck	#	2	2	6	6	6
CAT MD5125 Track Drill	#	1	1	1	1	1
CAT MD 6240 Rotary Drill	#	1	1	1	1	1
Epiroc D65	#	3	3	3	3	3

16.7.2.7 KOV OP LOM Mineral Reserve Plan Reconciliation

The previously stated Mineral Reserve estimate was published as at 31 December, 2017. The variance between the Mineral Reserve estimates at 31 December 2017 and 31 December, 2018 is shown in Table 105. There is a decrease of 3.0 Mt in the KOV OP Mineral Reserves due to mining activities in 2018.

Table 105: KOV OP Mineral Reserve reconciliation

KOV OP	December 31, 2018 Mineral Reserve	December 31, 2017 Mineral Reserve	Variance
Ore tonnes (Mt)	57.5	60.5	-3.0
Recovered Cu tonnes (kt)	1,820.5	1901.9	-81.4
Recovered Co tonnes (kt)	178.3	187.2	-8.9
Waste tonnes (Mt)	403	434.5	-31.6
%TCu grade	3.70	3.70	-
%TCo grade	0.48	0.48	-

16.7.3 Mashamba East OP

16.7.3.1 Background

The Mashamba East OP is now an operating mine with waste pre-stripping having commenced in January 2017. KCC completed a PFS in November 2016 to update the Mineral Reserve estimate after receiving positive results from the exploration drilling and updated Mineral Resource model. The PFS yielded the positive NPV. The Mashamba East OP LOM plan as at December 31, 2018 includes Mineral Reserve of 31.6 Mt of ore at a head grade of 2.15% TCu to the year 2029.

16.7.3.2 Modifying Factors

A total of 5% mining losses was applied as a tonnage loss that does not impact the ROM head grade.

A cut-off grade of 0.65% TCu was applied at the Mashamba East OP. The basis of the cut-off grade estimation is to determine the break-even cost based on selling, processing and royalty cost. The cut-off grade considered

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revenues generated from copper and cobalt with the appropriate processing recoveries applied. The costs, revenues and recoveries allowed for are shown in Table 106. An average of 7.9% dilution is applied on SMU basis.

16.7.3.3 Pit Optimization

The pit optimization completed for the 2016 PFS is still considered current, and as a result no further pit optimization work was done for the Mashamba East OP during the subsequent planning cycles. The parameters used during the 2016 pit optimization are listed in Table 106. The overall final pit slope angles of 25° and 40° were applied in the Southern and Northern portions of the pit. Mining and processing costs are inclusive of sustaining capital and are based on the 2016 assumptions used in PFS at that time.

Price assumptions listed in Table 106 and used in 2016 optimization are different than the current long term price assumptions. However, the variances are not material for the Mineral Reserves.

Table 106: Mashamba East OP optimization parameters used in 2016 PFS

Optimization Parameters	Unit	Mashamba East OP
Reference mining cost	USD/t	3.50
Processing cost	USD/t	66.90
Selling cost – Cu	USD/t Cu	740
Selling cost – Co	USD/t Co	4,904
Discount rate	%	10.7
Cu recovery – oxide	%	85
Co recovery – oxide	%	65
Cu recovery – sulphide	%	76.14
Co recovery – sulphide	%	65
Cu price	USD/t	6,750
Co price	USD/t	30,000

16.7.3.4 Pit Design

A graphical representation of the final pit design with current topography is shown in Figure 105 and Figure 106.

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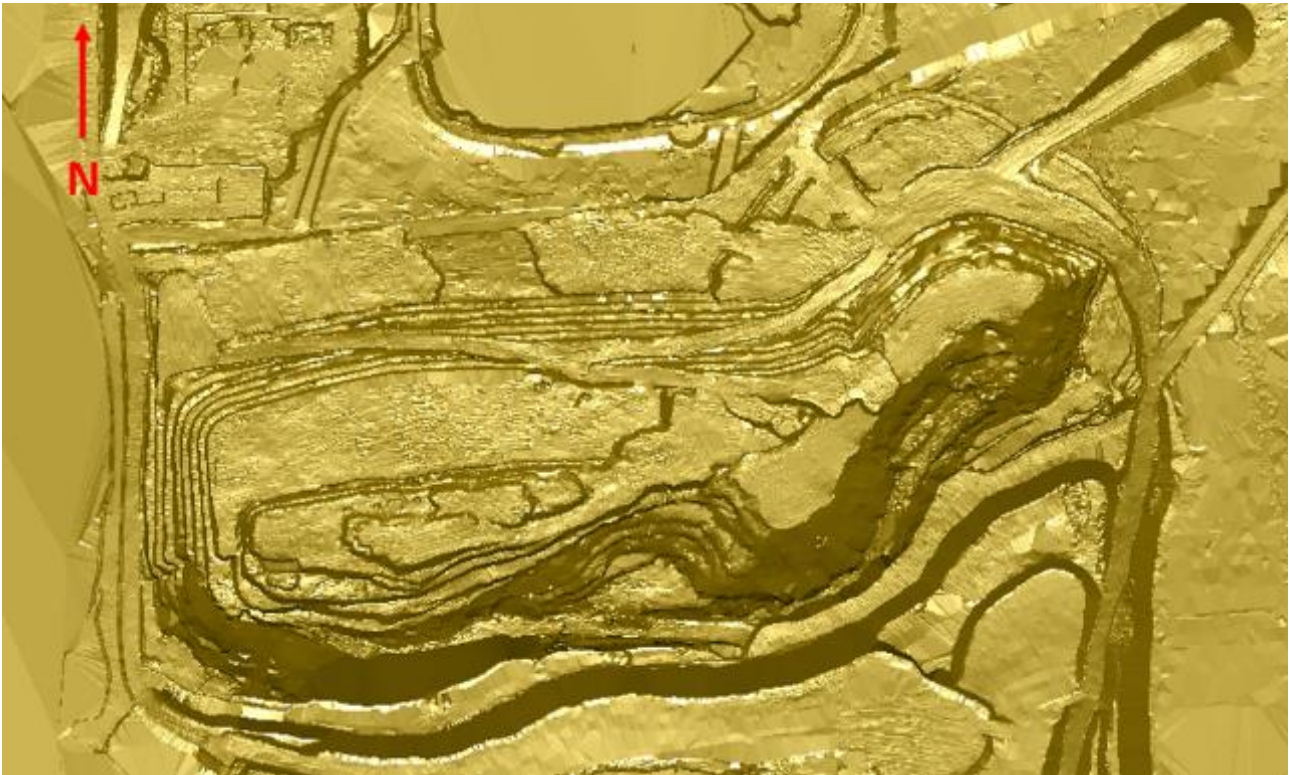


Figure 105: Mashamba East current topography

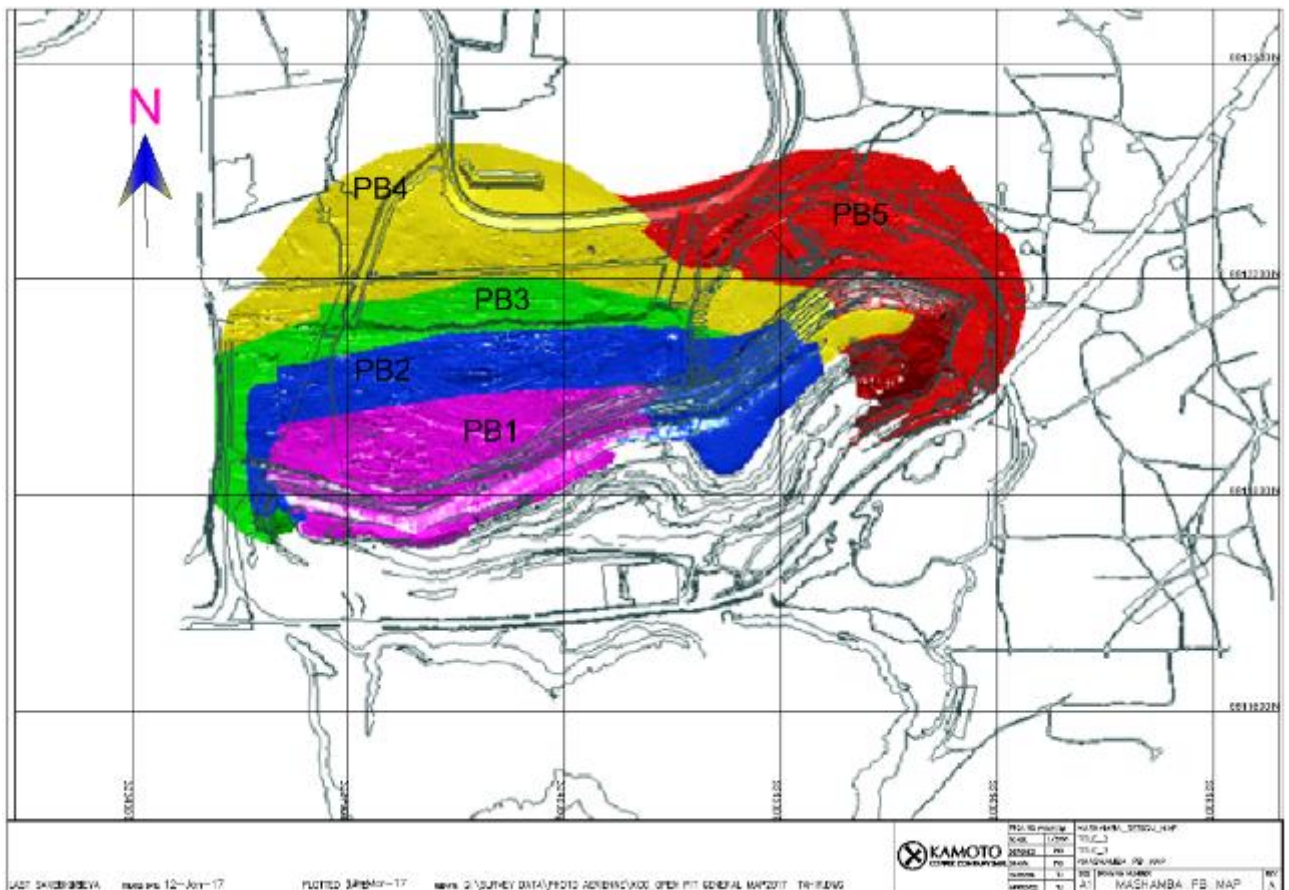


Figure 106: Mashamba East pit design

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Mashamba East pit design parameters are shown in Table 107 below.

Table 107: Mashamba East OP design criteria

Pit Design Criteria	Unit	Mashamba East OP
Bench height	m	10.0
Berm width	m	6.8
Batter angle	degrees	65.0
Ramp width	m	35.0
Ramp gradient	degrees	5.7 (1 in 10)

16.7.3.5 Mashamba East OP LOM Mineral Reserve Production Schedule

LOM Mineral Reserve Production from the Mashamba East OP project is planned at a maximum of 5.2 Mtpa. The LOM plan profile is shown in Table 108 below.

Table 108: Mashamba East OP project LOM Mineral Reserve production profile

Mashamba East OP LOM Mineral Reserve plan	Unit		2019	2020	2021	2022	2023	2024	2025	
Mineral Reserve ore	Mt		3.6	4.2	1.3	5.2	5.2	5.0	4.0	
Average Mineral Reserve %TCu	%		2.91	2.57	1.75	2.35	2.37	1.87	1.74	
Average Mineral Reserve %TCo	%		0.50	0.46	0.63	0.67	0.68	0.67	0.69	
Recovered Cu	kt		87.9	92.8	19.4	103.9	104.8	80.0	59.3	
Recovered Co	kt		11.6	12.7	5.3	22.7	23.0	21.9	18.0	
Waste	Mt		22.4	22.6	19.3	21.6	21.4	26.7	24.0	
Mashamba East OP LOM Mineral Reserve plan	Unit	2026	2027	2028	2029	2030	2031	2032	2033	Total
Mineral Reserve ore	Mt	0.4	1.4	1.0	0.2	0.0	0.0	0.0	0.0	31.6
Average Mineral Reserve %TCu	%	1.25	0.99	1.17	1.94	0.00	0.00	0.00	0.00	2.15
Average Mineral Reserve %TCo	%	0.44	0.40	0.35	0.09	0.00	0.00	0.00	0.00	0.60
Recovered Cu	kt	4.3	12.0	9.5	3.6	0.0	0.0	0.0	0.0	577.5
Recovered Co	kt	1.2	3.7	2.2	0.1	0.0	0.0	0.0	0.0	122.3
Waste	Mt	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	194.7

16.7.3.6 Mashamba East OP Mining and Support Equipment

Mine production equipment is provided, with all support equipment allocated within the KOV equipment, in Table 109.

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Table 109: Mashamba East OP mining and support equipment

Mashamba East Open Pit Equipment	Unit	2019	2020	2021	2022	2023-2030
CAT 793D Truck	#	13	9	12	12	12
RH-340 Shovel	#	2	2	2	2	2

16.7.3.7 Mashamba East OP LOM Mineral Reserve Plan Reconciliation

There is no change in the Mineral Reserve estimate as shown in Table 110 below.

Table 110: Mashamba East OP Mineral Reserve reconciliation

Description	December 31, 2018 Mineral Reserve	December 31, 2017 Mineral Reserve	Variance
Ore tonnes (Mt)	31.6	32.1	-0.5
Contained Cu tonnes (Kt)	577.52	684.6	-4.4
Contained Co tonnes (Kt)	122.31	191.6	-2.2
Waste tonnes (Mt)	195	231.0	-36.3
%TCu grade	2.2	2.13	-
%TCu grade	0.6	0.60	-

16.7.4 KITD Hydraulic Mining

16.7.4.1 Background

The KITD project area consists of a tailings dam which formed as a result of the discharge of fine tailings from KTC between 2008 and 2013. The KITD was constructed by conventional upstream discharge methods and is a U-shaped tailings dam, constrained to the west, east and south flank by civil engineered designs (SRK Consulting, 2008) containment walls whilst the northern flank is formed by a steep natural rising ground.

A portion of the KITD is situated on PE8841, belonging to GCM. As per the provisions of the AJVA with GCM, KCC has the legal right to mine these tailings as they were discharged from its KTC operation.

The KTC processed ore from T17 OP, KOV OP and KTO UG to produce the concentrate that was piped to Luilu Plant for solvent extraction / electro-winning.

The recoveries obtained from the conventional concentrating techniques of the sulphide and oxide ores left valuable amounts of copper and cobalt to discharge into KITD. In June/July 2013, the dam was nearing full capacity, and tails were instead redirected to the Mupine OP.

KCC began re-mining and reprocessing the KITD in January 2017 to assess the processing viability of the KITD material which is still on going. With positive results achieved, KCC decided to reclaim all of the KITD and add it to its assets by completing a PFS. Figure 81 shows the relative location of the KITD project.

KCC commissioned an internal team of professionals to execute a PFS started in July 2017 to update the Mineral Reserve estimate after receiving the positive results from the recent exploration drilling and the updated Mineral Resource model. The PFS on the KITD was required to advance this project to the next phase. The PFS was completed in October 2017 and yielded a positive NPV.

As a result of the PFS, the KITD contains 7.4 Mt of ore in the Probable Mineral Reserve category at an average TCu of 1.49% and TCo of 0.16% as at December 31, 2017. This tonnage is reported after depletion due to mining in 2017. The KITD updated Mineral Reserves as at December 31, 2018 are 5.7 Mt of ore at a head grade of 1.44% TCu to the year 2021

16.7.4.2 Modifying Factors

Mining related modifying factors applied to the KITD are based on first principals:

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- A cut-off grade of 0.20% TCu and 0.07% TCo;
- A mining dilution of 0.66%; and
- Mining losses of 5%.

Other inputs used for estimation of the Mineral Reserves are set out in Table 111. Price assumptions listed in Table 111 and used in 2017 PFS are different than the current long term price assumptions. However, the variances are not material for the Mineral Reserves.

Table 111: KITD inputs for the Mineral Reserve estimation

Parameters	Unit	KITD Parameter
Hydraulic mining cost	USD/t ore	1.20
Processing cost	USD/t ore	18.40
Selling cost – Cu	USD/t ore	894.2
Selling cost – Co	USD/t ore	Included in Cu selling cost
Discount rate	%	11.8
Cu recovery – overall	%	48.0
Co recovery – overall	%	19.5
Cu price	USD/t	6,500
Co price	USD/t	55,000

16.7.4.3 Mine Design

16.7.4.3.1 Introduction

The Mineral Reserve estimate for the KITD is based on hydraulic mining which is quite common for tailings retreatment projects. Hydraulic mining was selected due to the proven technology, cost effectiveness, as well as technical and operational ease thereof. The system consists of:

- Fixed pumps and piping equipment for high pressure pumping of clean process water to re-pulp the dry ROM tailings with mobile monitoring units;
- Fixed screening equipment to remove debris and oversize material on site; and
- Fixed slurry pumps and piping equipment to pump the slurry to KTC for re-floatation and concentration.

16.7.4.3.2 Hydraulic Mining Method

The hydraulic mining system simplifies the remaining operation in three ways. First, the hydraulic system eliminates the need for excavating and hauling equipment normally associated with the conventional mining. Second, the energy produced by the water jets serves to break down clay aggregates allowing more efficient separation, and finally, the hydraulic action provides the appropriate slurry necessary for the subsequent hydrometallurgical beneficiation.

KCC is currently operating three barges with two guns each (water cannons). Water is sprayed at 3.58 m³/minute at a pressure between 16 and 25 bar at the tailings pile in order to cut the upstream face.

The monitors wash tailings down the face of the pile into a sump as shown in Figure 108. Warman, a type of sump pump, pumps the slurry to a vibrating screen that rejects material such as scraps of wood and other debris in the tailings. The undersized material passes through the screens into a tank from where it is pumped to the processing plant.

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The monitors, sumps and sump pumps advance as the face of the tailings pile recedes. The equipment is advanced frequently so that the slurry paths are kept short, which in turn, prevents the solution from migrating through the pile and into the groundwater.

In gravity deposition testing of the KITD dry material on a restricted surface, it is formed into a cone-shaped heap and the slope angle of the cone was repeatedly given an average value of 45° . For safety reasons, the actual cutting face at the KITD hydraulic mining is between 30° and 45° . This is achieved by maintaining a distance of 5 m between the cutting lines. Figure 107 shows the cutting angle of the face at the KITD.



Figure 107: KITD hydraulic mining cutting angle

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Figure 108: Monitor cutting face top-down at the KITD

16.7.4.3.3 Mining Blocks

The KITD mining blocks (Figure 109) were designed in Deswik mine planning software. In selecting mining blocks, consideration was also given to optimising copper feed grade to KTC, and the potential for expansion of the hydraulic mining activities. The KITD mine plan includes 10 mining blocks which will be mined in sequential ascending order. Each block will be mined as a single face due to thickness of the tailings which averages around 10 m.

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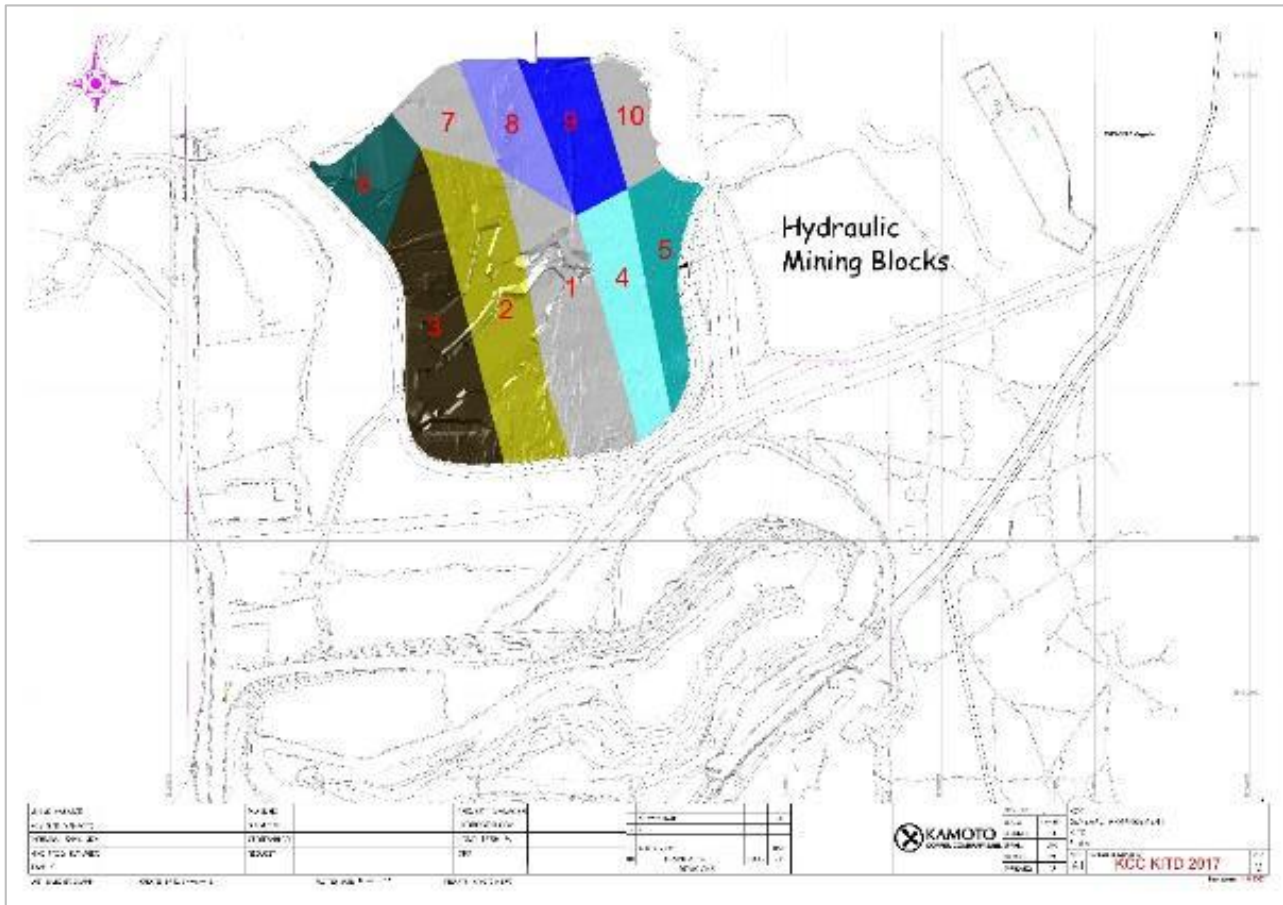


Figure 109: KITD mining blocks and sequence

16.7.4.4 KITD LOM Mineral Reserve Production Schedule

The KITD updated Mineral Reserve LOM is 3 years, with the LOM schedule finishing in 2021. The current KITD Mineral Reserve as at December 31, 2018 is shown in Table 112 and Figure 109. Schedules were produced in Deswik mine planning software between 200 and 500 TPH in 2019 to 2021 with an effective utilization of 78% in wet season and 93% in dry season.

Table 112: KITD LOM production schedule

KITD LOM Mineral Reserve	Unit	2019	2020	2021	Total
Probable Mineral Reserve	t	2,572,467	2,081,808	1,062,073	5,716,349
Average Mineral Reserve %TCu	%	1.44	1.44	1.44	1.44
Average Mineral Reserve %TCu	%	0.17	0.17	0.17	0.17

The Mineral Reserve estimate for the KITD situated within the KCC lease area (PE525) was estimated at 2.8 Mt at 1.42% TCu and 0.16% TCo, and classified as Probable Mineral Reserve. The Mineral Reserve estimate for the KITD situated on block 14 of PE8841 was estimated at 2.9 Mt at 1.45% TCu and 0.17% TCo, and classified as Probable Mineral Reserve.

16.7.5 Open Pit LOM Mineral Reserve Production Scheduling

The open pit consolidated Mineral Reserve LOM plan schedules for KOV OP and Mashamba East OP were produced in MineSight Schedule Optimizer software. The KITD production schedule was produced in

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DeswikCAD schedule software. The schedule considers the mining width, number and size of excavators required and the practical constraints of each pit.

The ex-pit ore and waste tonnes generated throughout the LOM from the open pit operations peaks at 90 Mtpa as seen in Figure 110 below.

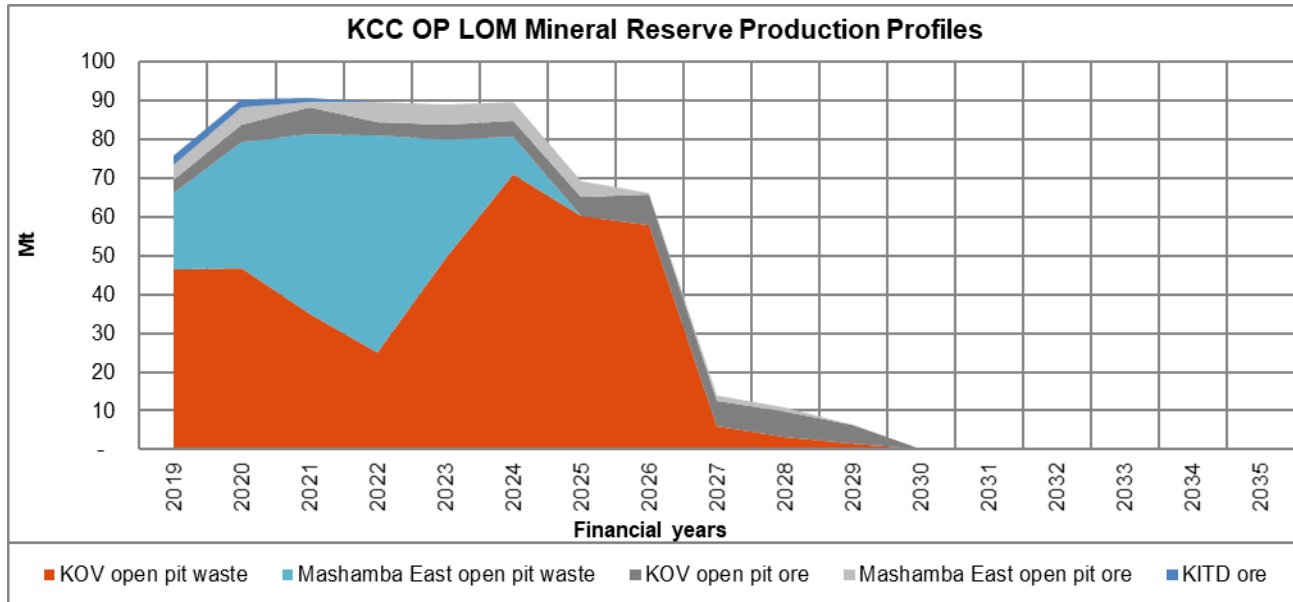


Figure 110: KCC open pit production profiles

The LOM ore production profile as shown in Figure 110 is based only on the Indicated Mineral Resource above the economical cut-off grade within the optimized pit design after applying appropriate level of mining loss and dilution. The selected optimized pit shell was used as guideline to create the final production schedule.

16.7.6 Open Pit LOM Mineral Reserve Estimate

The LOM Mineral Reserve of the surface mining operations is listed in Table 113 below.

Table 113: Surface mining operations Mineral Reserve table as at December 31, 2018¹⁻¹⁰

Surface Mining Operation	Proven			Probable			Total		
	Mt	%TCu	%TCo	Mt	%TCu	%TCo	Mt	%TCu	%TCo
KOV OP	-	-	-	57.5	3.72	0.48	57.5	3.72	0.48
Mashamba East OP	-	-	-	31.6	2.15	0.60	31.6	2.15	0.60
KITD (PE525)	-	-	-	2.8	1.42	0.16	2.8	1.42	0.16
KITD (PE8841)	-	-	-	2.9	1.45	0.17	2.9	1.45	0.17
Total	-	-	-	94.8	3.06	0.50	94.8	3.06	0.50

- 1) The Mineral Reserve estimates have been prepared in accordance with the classification criteria of the NI 43-101;
- 2) The Mineral Reserves are current as at December 31, 2018.
- 3) The KITD Probable Mineral Reserve within KCC lease PE525 is reported at 2.8Mt and 2.9 Mt within GCM lease PE8841.
- 4) The KITD portion situated on PE8841 belong to GCM for which KCC has the legal right to mine the tailings discharged from its KTC operation as per the provisions of the AJVA concluded with GCM;
- 5) Grade measurements reported as percent (%), tonnage measurements are in metric units;
- 6) Tonnages are reported as Mt rounded to one decimal place; grades are rounded to two decimal places;
- 7) Projects included were developed to a minimum PFS level of accuracy;

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- 8) LOM plans of existing operations were developed to a minimum PFS level of accuracy;
- 9) Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content; and
- 10) The Mineral Reserve estimates are for KCC's entire interest, whereas the Company owns 75% of KCC.

16.8 Hydrogeology and Dewatering

16.8.1 Rainfall and Recharge

Rainfall is highly seasonal, exhibiting a well-defined wet season between November and March, and a dry season between May and September. October and April are transition months between the wet and dry seasons. The mean annual rainfall for KOV OP based on 9 years of data (2006-2015) is 1051 mm/yr. Figure 111 shows the monthly average rainfall for KOV OP (2006-2015), Kolwezi Airport (2004-2012) and Solwezi Airport (1961-2004). At KCC the maximum rainfall month is December with 204 mm. Daily rainfall data has been collected from a manual rain gauge in the KOV pit bottom since 2006, and over this period the maximum recorded daily rainfall (24 hour) was 76 mm on 1st January 2017.

A storm rainfall analysis was carried out by Golder (September 2015), to determine the 24 hour rainfall depth for various return periods. The 24 hour storm rainfall depths at KOV OP for the 20 year and 100 year events were calculated to be 74 mm and 80 mm respectively.

The legacy mining activity over the previous decades has resulted in terrain which has sparse vegetation over large areas such as the abandoned tailings dams and waste dumps. The absence of vegetation combined with the intensity of wet season rainfall results in high runoff volumes on slopes, and low potential for evapotranspiration. Shallower slopes and disturbed surfaces (e.g. waste dumps) have higher infiltration capacities and may receive higher recharge. Enhanced groundwater recharge can also occur in areas of poor drainage and water ponding such as the tailings dam facilities and within the flooded pits (e.g. Mupine and GH).

Groundwater recharge is estimated by SRK (2009) to be 5-13% of mean annual rainfall (MAR) based on their knowledge of similar environments. The best estimates of recharge are derived from groundwater hydrograph analysis and groundwater model calibration. A 3D groundwater model (MODFLOW) was completed in 2018, and areas of much higher recharge (>26% MAR) are required over the Roan group geology to achieve the correct water balance for KOV OP.

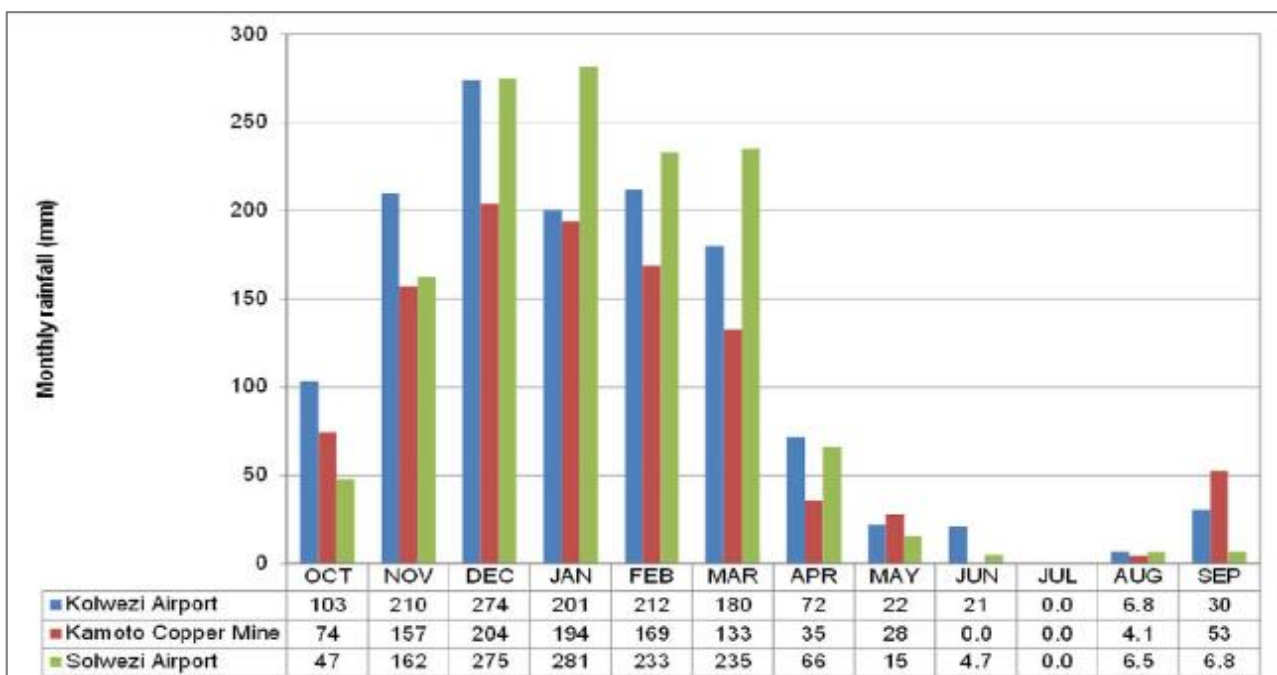


Figure 111: Mean monthly rainfall data from nearby rain gauges (Golder, Sep 2015)

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16.8.2 Hydrostratigraphy and Structure

16.8.2.1 General Hydrostratigraphy

Groundwater occurrence in the mine series aquifers (Roan Series of the Katangan System) forming the ore bodies and surrounding strata is substantial, and groundwater inflows to the KOV OP require 3,500-4,000 m³/hr of dewatering to maintain a stable pit floor water level.

Regionally, groundwater storage is present within the weathered saprolitic profile and at the interface between the saprolite and partially weathered saprock. The properties of the saprolite depend on the source rock, but generally the highly weathered saprolite has high primary porosity but often poor permeability, and typically results in a zone of perched groundwater. Beyond the weathered zone, secondary porosity and permeability exists within an interconnected network of cross-cutting fractures, however the majority of the permeability and groundwater movement is strataform, within highly permeable leached calcitic and silicified dolomites containing bedding-parallel dissolution voids and fissures. Within the Kolwezi Nappe, these lithologies are present within faulted fragments often separated by the lower permeability RAT lithology, hence it is the orientation, structural arrangement and extent of these fragments, which control how the groundwater moves towards and into the open pits and the underground mine.

The Roan Series lithologies are shown in Table 114, with the conceptual hydrostratigraphy. The 2017 hydrogeology drilling and testing program has improved the current understanding by providing tested estimates of the hydraulic conductivity for each lithology (Further detail is provided in Section: 16.9.2.6).

Table 114: Stratigraphy of the orebody fragments and conceptual hydraulic characteristics

Lithology	Formation	Thickness (m)	Aquifer potential	Nature of Permeability
RGS	R3	100+	Aquitard	Often present as highly weathered saprolite of low permeability but moderate primary porosity.
CMN	R2	50-120	Major Aquifer	Typically Calcite dolomites which are highly altered and prone to cellular (dissolution) weathering leaving permeable voids and fissures.
SDS		50-80	Minor Aquifer	Finely bedded carbonaceous shales with minor fracture permeability
BOMZ		1-5	Minor Aquifer	Well bedded dolomitic shales with minor fracture permeability
SDB		10-15	Minor Aquifer	Lower permeability well bedded dolomitic shales with minor fracture permeability
RSC		10-25	Major Aquifer	Silicified dolomite with high permeability from extreme cellular (dissolution) weathering, Often semi-confined by overlying strata,
RSF		5-15	Major Aquifer	Finely laminated silicified dolomite with high permeability from cellular (dissolution) weathering
DSTRAT		3-5	Minor Aquifer	Stratified dolomite with minor fracture permeability
RAT GRISE		3-5	Aquitard	Low permeability massive argillic dolomite
BR RAT		1-3	Aquitard	Low permeability brecciated Rat Lilas and Rat Gris
RAT LILAS	R1	5-150	Aquitard	Low permeability massive dolomite with localised permeable ologist.

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In the Roan Series, the most permeable lithologies are found within two discrete intervals of the R2 formation; the CMN in the upper R2 formation, and the RSC-RSF around 60-100 m below.

The CMN, RSC and RSF are highly leached and altered Dolomites. Dissolution of dolomites through groundwater circulation can lead to high permeability fissuring over time as dissolution is focused along the preferential flow pathways. The CMN, RSC and RSF exhibit such dissolution fissures and extreme cellular weathering is common. In the RSC and RSF, silicification may help preserve the void apertures by strengthening the rock mass in these altered zones.

The stratigraphy separating the CMN and RSC-RSF is a sequence of lower permeability carbonaceous and dolomitic shales (SDS, BOMZ and SDB). Whilst fracturing of these units imparts a degree of permeability, the shales act as an aquitard and a semi-confining layer on top of the RSF and RSC lithologies.

At the bottom of the Roan Series are the basal RAT lithologies (RAT Grise, Br RAT and RAT Lilas). The RAT are massive argillaceous dolomites, with relatively low fracture density and permeability. The RAT Lilas is present as a considerable thickness (up to 150 m) and is a widespread basal unit marking the lower limit for groundwater storage and flow. RAT also separates adjacent structural fragments, reducing hydraulic continuity between them.

As the major permeable units are strataform, the structural arrangement and dip orientation of the fragments are important factors in controlling the flow of groundwater into the pit. Where bedding dips into the pit and is intersected in the pit slope, there is a direct 'pathway' for groundwater flow and discharge. In steeply dipping stratigraphy striking parallel to the pit slope, however, lower permeability aquitard units can present a barrier to flow.

16.8.2.2 KOV OP

The Oliveira, Variante and FNSR fragments in the north of KOV OP are oriented favourably for groundwater inflow to the pit, and the Variante and Oliveira (Cut 2) particularly, are highly leached and fractured resulting in high bulk permeability in the fragments.

The Oliveira fragment is an elongate syncline with an average strike of northwest-southeast orientation. The northern limb of the syncline dips towards the pit and an artesian dewatering well (KCC6) indicates a confined aquifer at the pit bottom, within the Oliveira fragment. The high flow rate (estimated to be 300-400 m³/hr) from the well indicates both high pressure and substantial transmissivity most likely in the RSC-RSF lithologies.

The Variante fragment is considered to be the greatest source of groundwater inflow to the KOV OP. The fragment is not only highly leached and fractured imparting high permeability, it also dips steeply to the north-east exposing permeable lithologies at the topographic surface to the north of the pit, and therefore presents a direct flow pathway down-dip into the pit. The Variante fragment extends away from the KOV OP towards the northeast and it has been proposed that this extension creates a hydraulic connection with potential sources of recharge from the Musonoie River and the Kingamyambo Tailings Dam. Whilst it is possible that these sources contribute some elevated recharge to the Variante, they are unlikely to be sufficient to sustain the inflow rates experienced in the KOV OP. Instead, it is more plausible that the Variante receives groundwater recharge and inflows from a number of regional sources including inflows from the surrounding regional groundwater system (i.e. fractured rock and saprolite aquifers). The regional northeast-southwest oriented fault structures may present permeable sources of groundwater inflow to the Variante, through fault breccias or more heavily fractured rock oriented parallel to the faults.

On the southern part of KOV OP are the Virgule and KTE fragments. The Virgule fragment dips to the southwest and the KTE fragment is steeply dipping to the north. The two are separated by a large scale fault but join together to give the appearance of a synform structure. The dip orientation of these fragments, combined with the presence of lower permeability RAT to the south of KTE (and forming a large part of the southern pit slope), explain the absence of widespread groundwater seepage in the southern pit wall. The KTE fragment is also dewatered by the underground mine in this area.

16.8.2.3 KTO UG

With the exception of a hydraulic connection to KTE UG, KTO UG mine is hydraulically isolated from the KOV OP, as the underground Mineral Resource fragments are separated by lower permeability RAT and RGS.

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Major inflow zones in the underground are associated with the orebody's RSC unit. In some locations, short term inflows are experienced when mining intersects highly porous haematite bands in the RAT, however these pockets of groundwater storage are not connected to a wider permeable fracture network.

16.8.2.4 T17 OP

In T17 OP, the permeable orebody units have a vertical configuration due to the folding present in the T17 fragment. Groundwater flow to the T17 OP is mainly from the south along the orebody. The fault present on the southeast of the T17 OP is considered to have permeability to a depth of around 200 m. Below 200 m depth the permeability along the fault is thought to decrease (Golder, 2013). The Musonoi River is routed directly adjacent to the T17 OP on its eastern side. It is highly likely that infiltration of water from the Musonoi River is occurring and entering the pit groundwater system. However the presence of lower permeability RAT between the river and the T17 OP limits the magnitude of this interaction.

16.8.2.5 Mashamba East OP

The Mashamba East fragment, like the Variante, contains highly leached orebody units, which are likely to have similar high permeability. The orebody strikes northwest in the western part of the pit, and rotates round to strike east in the central part of the pit. The orebody dips towards the north at 30-45°. To the west of Mashamba East OP, the orebody continues into the neighbouring Mashamba West OP which is operated by La Sino-Congolaise des Mines SARL (Sicomines). There is a strong hydraulic connection between the two concessions, presumably due to the lateral continuation of the RSC aquifer. This hydraulic connection is evidenced in the draining of the Mashamba East OP due to dewatering activities in the Mashamba West OP. As a result of pumping from the Mashamba West OP, the pit lake in the Mashamba East OP drained at a rate of approximately 3 m/month and reached the pit floor elevation of 1,350 mamsl in July 2015. The pit is currently dry.

16.8.2.6 Hydraulic properties

Packer testing is a standard methodology for obtaining the permeability estimates from individual boreholes. The procedure involves isolating an interval of a diamond-drilled hole with a packer, and injecting water into the open hole at a certain pressure. The injection flow rate and pressure are used to calculate the permeability. Packer tests were conducted as part of the 2017 drilling program. The packer test results from the site investigation ranged from 10^{-9} m/s (no water uptake by formation) to 10^{-4} m/s. The packer test hydraulic conductivity estimates are shown in Table 115, lumped by the lithological groups as follows:

- Above orebody (AOB) consisting of RGS, CMN, and SDS
- Ore body superior (OBS) consisting of BOMZ and SDB
- The single lithology RSC
- Orebody inferior (OBI) consisting of RSF, DSTRAT, RAT GRISE, and BR RAT, and
- Below orebody (BOB) consisting of RAT LILAS and BR HET.

Table 115: Hydraulic conductivity estimates derived from the packer testing

Lithology Group	Lithology	Hydraulic conductivity estimate			
		Min	Max	Geometric Average	Number of tests completed (n)
		(m/s)	(m/s)	(m/s)	
AOB	RGS, CMN, SDS	7×10^{-8}	1×10^{-4}	2×10^{-6}	16
OBS	BOMZ, SDB	7×10^{-8}	2×10^{-6}	3×10^{-7}	3
RSC	RSC	8×10^{-8}	7×10^{-7}	2×10^{-7}	3
OBI	RSF, DSTRAT, RAT GRISE, BR RAT	8×10^{-8}	7×10^{-7}	2×10^{-7}	3
BOB	RAT LILAS, BR HET	1×10^{-9}	6×10^{-6}	1×10^{-7}	26

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Packer test estimates of hydraulic conductivity often show bias towards more stable and competent rock, as the packer itself requires such rock to safely inflate and create a seal for testing. This bias is illustrated well in the pumping test analysis carried out on the data collected during the commissioning of dewatering well KCC-14A. The analysis indicates transmissivity of between $2.8 \times 10^{-3} \text{ m}^2/\text{day}$ and $9.9 \times 10^{-3} \text{ m}^2/\text{day}$. This is an order of magnitude higher than the packer test results for similar AOB lithologies.

16.8.3 Groundwater Levels

16.8.3.1 KOV OP

Around the KOV OP, groundwater levels are controlled by the magnitude of groundwater flow towards the pit, the permeability of the pit geology, and the elevation of the pit bottom. For the KOV OP, there is currently a network of 20 vibrating wire piezometers and 14 standpipe piezometers used to collect groundwater level data. Groundwater levels are measured manually, three times a week. Data loggers are currently installed at 29 piezometers around KOV OP to increase the frequency and accuracy of the groundwater level measurements. These data loggers record the groundwater level every hour. The level of water in the pit floor is also regularly surveyed. Table 116 shows the groundwater monitoring network, and example monitoring data for November 2, 2017. A map of existing and planned locations of the piezometers and wells is shown on Figure 112. The groundwater level monitoring network has been well established for KOV OP as 13 of the 20 vibrating wire piezometers were installed in 2017.

Table 116: Groundwater level data as of November 1, 2018

Piezometer ID	Collar Elevation (mamsl)	Depth to Water (mamsl)	Groundwater Level Elevation (mamsl)
SP10	1,308	93	1,215
SP1	1,335	63	1,272
SP5	1,448	133	1,315
WC-01	1,453	30	1,423
WC-02	1,454	50	1404
DCP-6P	1,399	101	1,294
SP2	1,417	90	1,327
DCP-1D	1,411	113	1,298
VWP1	1,281	101	1,180
KCC14	1,317	60	1,257
KCC22	1,309	95	1,214
SP6	1,362	63	1,299
GH02	1,412	54	1,358
MUP-01	1,448	67	1,381
MUP-14	1,442	31	1,411
CHO-01B	1,436	147	1,289
CHO-01C	1,436	147	1,289
CHO-05	1,454	182	1,272
CHO-06	1,468	427	1,041
CHO-07	1,444	165	1,279
CHO-08	1,301	62	1,239
CHO-09	1,345	37	1,308

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Piezometer ID	Collar Elevation (mamsl)	Depth to Water (mamsl)	Groundwater Level Elevation (mamsl)
CHO-10	1,415	129	1,286
WC-01	1,453	30	1,423
WC-02	1454	Piezo dry	-
KOV-0830	1,452	165	1,287
KOV-0836	1,449	68	1,381
KOV-0840	1,420	134	1,286
KOV-0841	1,411	96	1,315
KOV-0846	1,280	81	1,199
KOV-0848	1,345	152	1,193
KOV-0849	1,362	34	1,328
KOV-0857	1,351	88	1,263
KOV-0858	1,360	68	1,292
KOV-0859	1,341	56	1,285
KOV-0872	1,292	101	1,191

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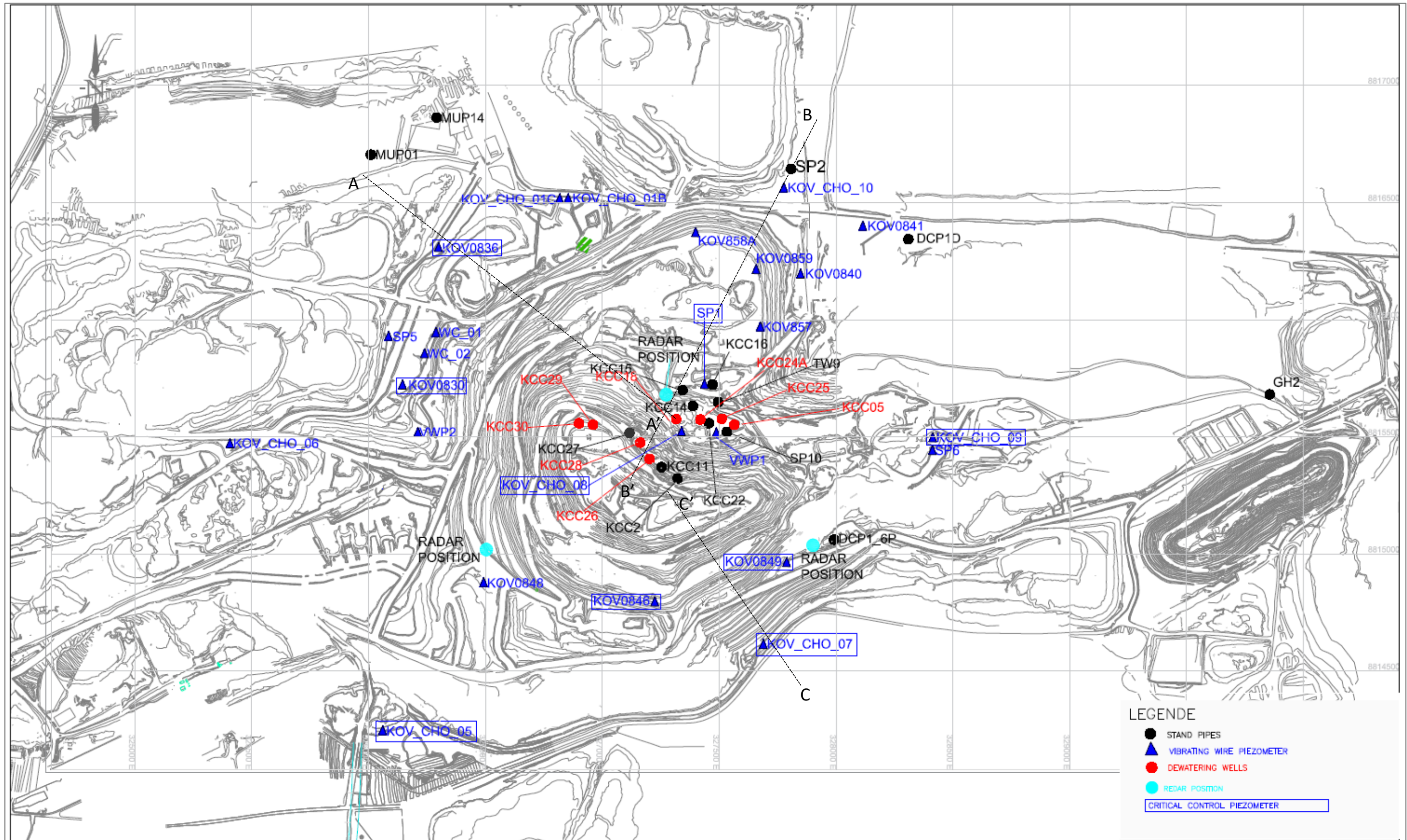


Figure 112: Map of KOV OP dewatering wells and groundwater monitoring network

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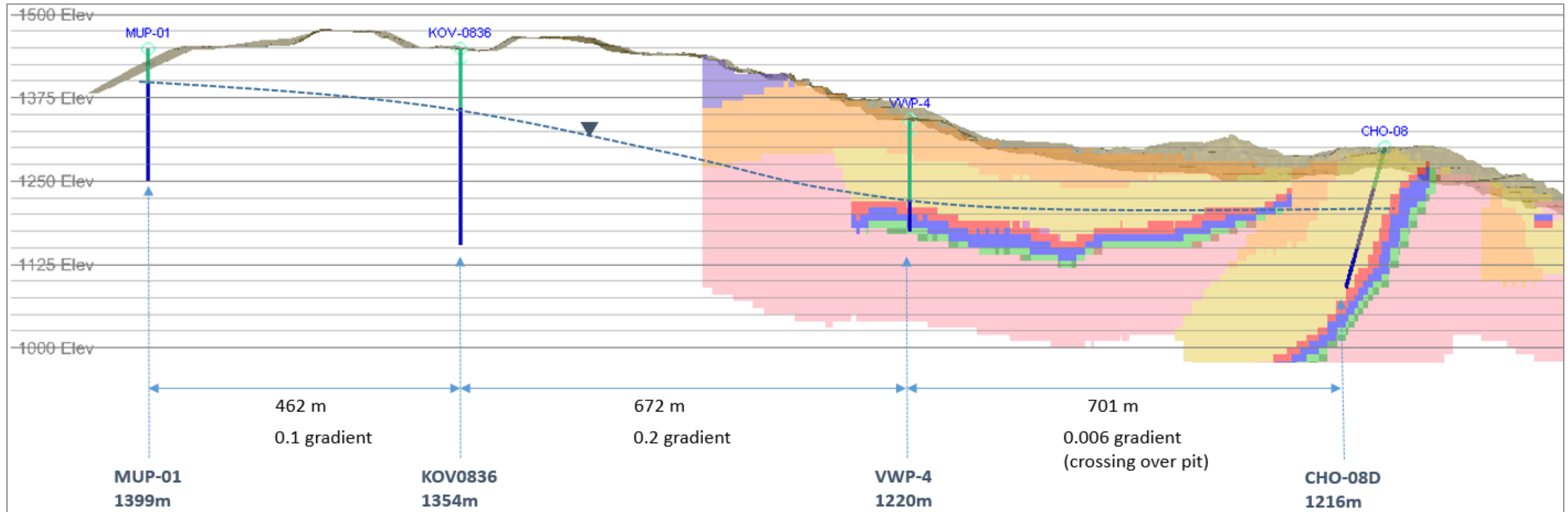


Figure 113: Hydrogeology section showing piezometers and interpolated groundwater levels in to the west of KOV OP

The diagram is a geological cross-section showing a subsurface profile. The vertical axis represents elevation in meters, ranging from 800 to 1500. The horizontal axis represents distance in meters, ranging from 0 to 1080. The profile shows a series of geological units in different colors: light blue, yellow, orange, red, green, and brown. A dashed line indicates a 0.05 gradient. Several boreholes are shown as vertical lines with labels: SP-2, CHO-10, KOV-0840, KOSR8, KOC15, TW09, KOC14, SP-10, MNP-1, CHO-08, and KOSR0234. Key points are marked: CHO-10 at 1300m and CHO-08 at 1245m.

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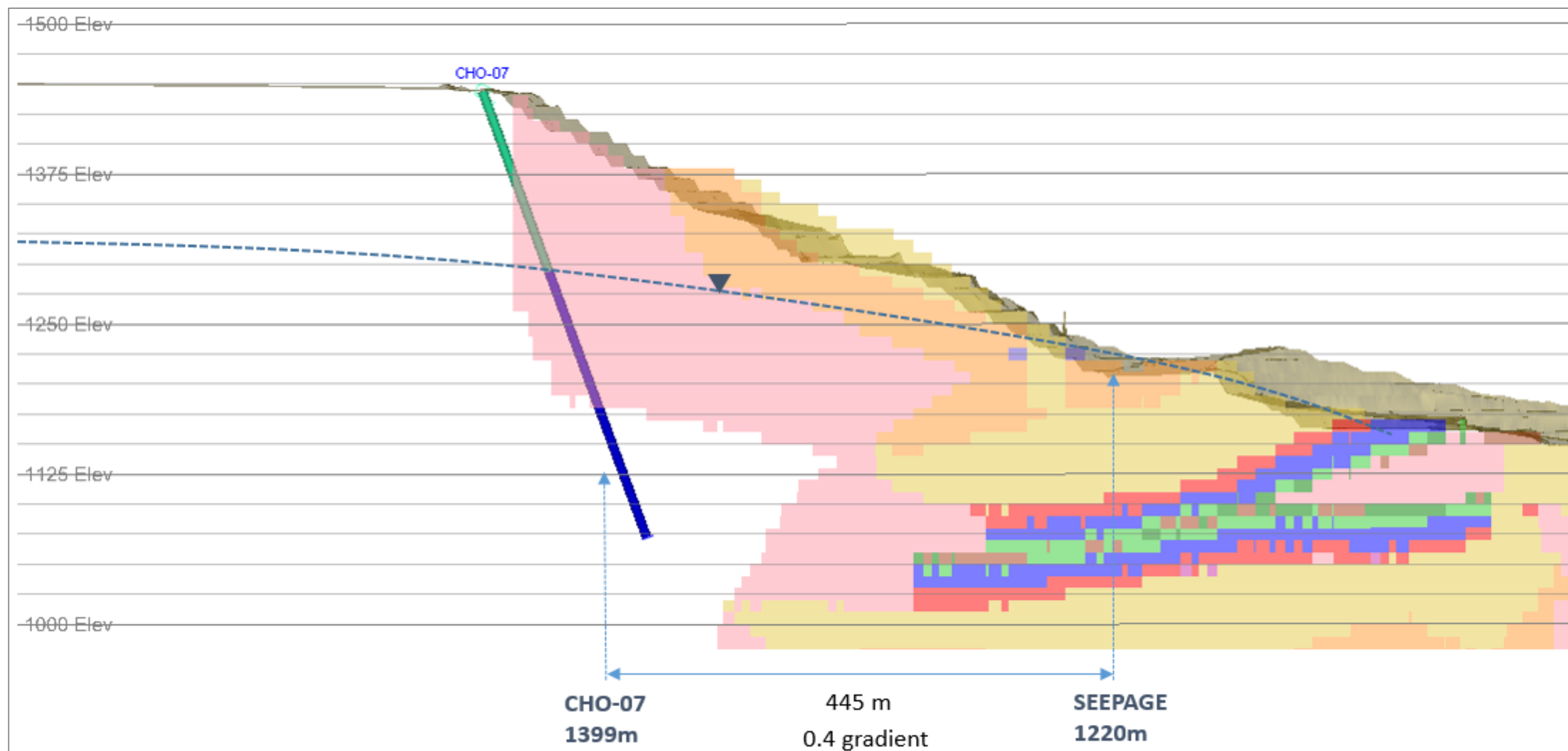


Figure 115: Hydrogeology section C-C' showing piezometers and interpolated groundwater levels to the southeast of KOV OP

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Dewatering of the KOV OP has caused depressed groundwater levels immediately around the pit, and a hydraulic gradient towards the pit over a wider area. In-pit groundwater levels are up to 200 m lower than ambient groundwater levels 1 km away from the pit floor.

Figure 113, Figure 114 and Figure 115 show groundwater levels plotted on cross sections of the KOV OP for the northwest, northeast, and southwest pit walls respectively. The section lines are shown on the map in Figure 126.

The cross-sections in Figure 113, Figure 114 and Figure 115 show approximate hydraulic gradients towards the pit of 0.2 from the northwest of KOV OP, 0.5 from the northeast, and 0.4 from the southeast. Assuming similar rates of recharge in these areas, the shallower gradient to the northeast is indicative of higher permeability here. Groundwater levels to the southwest of KOV OP are heavily influenced by dewatering from the KTO UG and the pit slopes are fully dewatered there. To the southeast, the steeper hydraulic gradient of 0.4 suggests lower relative permeability, which is due to the predominance of the RAT lithology in this part of the pit.

The historical data shows the impact of pit dewatering on the surrounding groundwater levels. There are very few piezometers which show data over an extended historical period, as many original piezometers have become dry as the groundwater levels have fallen. Figure 116 is a plot of the groundwater hydrographs for piezometers which best reflect the historical and recent trends. The data shows that the groundwater levels were relatively stable up to the beginning of 2011, after which groundwater levels begin to fall in response to further dewatering of the KOV OP. Piezometers within, or close to the pit experience this effect to a much greater degree (e.g. KCC22 and SP10).

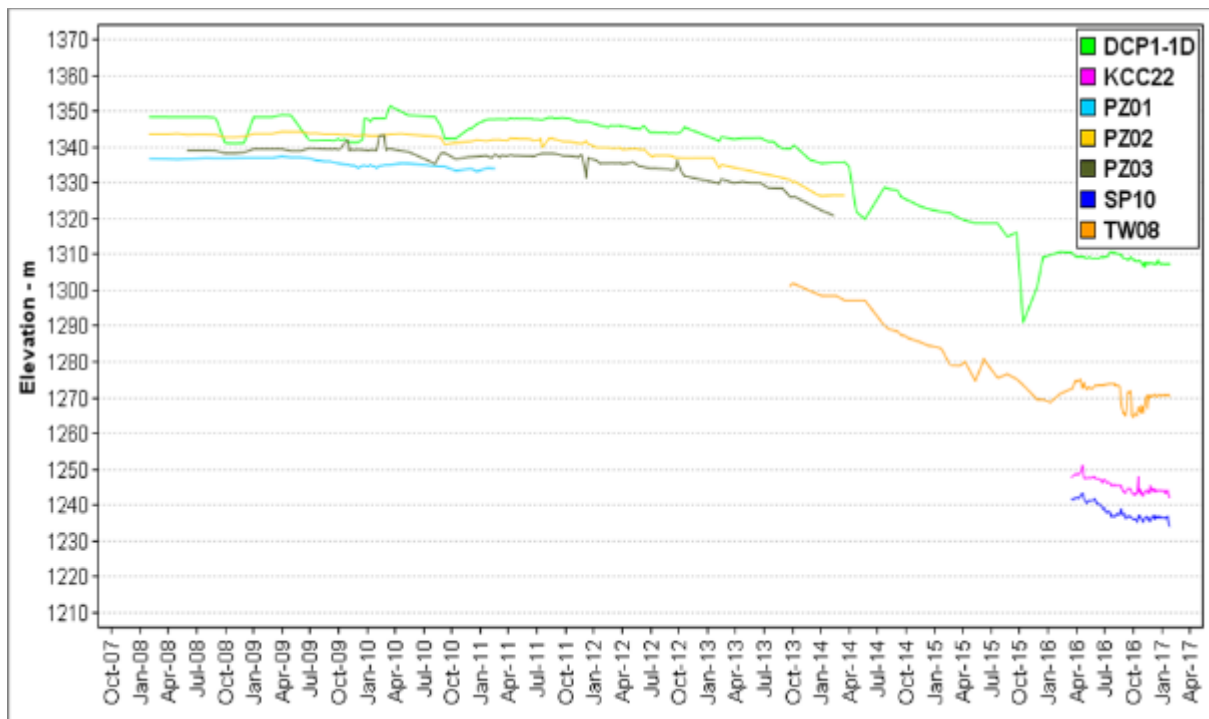


Figure 116: Historical groundwater levels for KOV OP

16.8.3.2 T17 OP

Groundwater levels around the T17 OP were measured in a single piezometer T17-01, which became dry in December 2012. Prior to this, the measured groundwater levels in T17-01 were reasonably stable at 50 m to 60 m below the ground surface.

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16.8.3.3 Mashamba East OP

Prior to 2017, there was no groundwater monitoring network at the Mashamba East OP. Between January and April 2017, 6 vibrating wire piezometers were installed around the perimeter of the pit. The piezometer locations are shown in Figure 117, and the details in Table 117.

The historical data indicate a pre-mining groundwater level of 1,415 mamsl. Dewatering of Mashamba East OP was carried out between 1984 and 1989. When production at Mashamba East OP ceased in 1988, due to the lack of funds and increasing costs, the pit was allowed to flood. The ensuing pit lake water level was approximately equivalent to the pre-mining groundwater level. Both the Mashamba East and Mashamba West pits were flooded until dewatering activities commenced in Mashamba West OP in late 2013. As a result of pumping from Mashamba West OP, the pit lake in the Mashamba East OP drained at a rate of approximately 3 m per month and reached the pit floor elevation of 1,350 mamsl in July 2015. The pit is currently dry.

Table 117: Vibrating wire piezometers at Mashamba East OP

Piezometer ID	Collar Elevation (mamsl)	Depth to Water (mamsl)	Groundwater Level Elevation (mamsl)
ME-CHO-01	1,435	126	1,309
ME-CGO-02	1,433	123	1,310
ME-CGO-03	1,426	113	1,313
ME-CHO-02	1,458	147	1,311
ME-CHO-04	1,445	121	1,324
ME-CHO-05	1,454	138	1,316

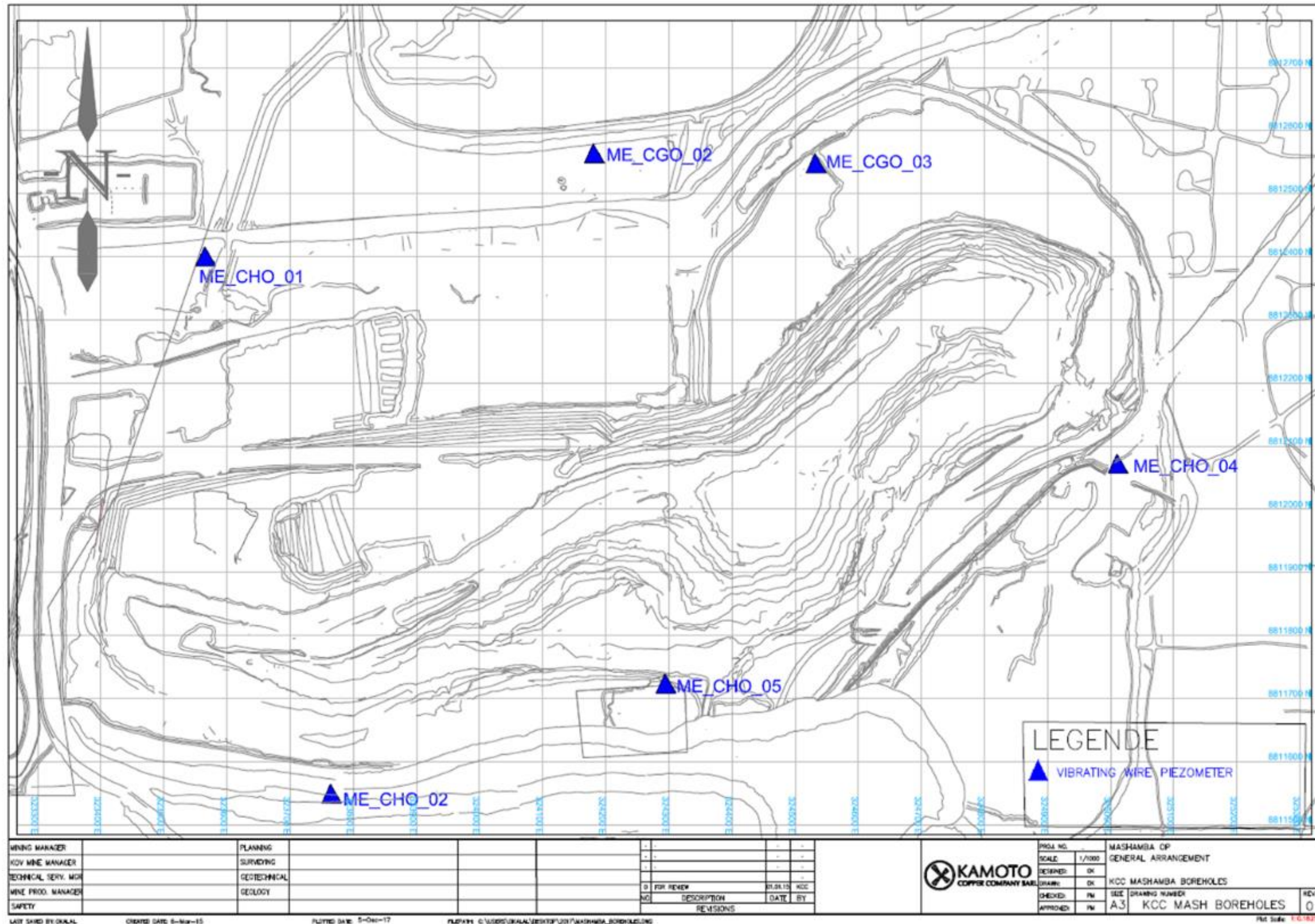


Figure 117: Map of the piezometer locations at Mashamba East OP

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16.8.4 Dewatering

16.8.4.1 Dewatering Strategy

Currently KOV OP, T17 OP and KTO UG are being actively dewatered. The objectives of dewatering these mines is to lower the groundwater table to below the active mining level to ensure dry working conditions for optimal operational efficiency. In addition, lowering groundwater levels can improve the slope stability in weak pit slopes by increasing the effective stress.

The role of water in slope stability is a commonly misunderstood concept, and the terms depressurisation and dewatering are often used interchangeably but require very different approaches in terms of water management. Groundwater will naturally flow towards the lowest point of discharge, typically towards the pit floor. In permeable lithologies inflows to the pit can be large and require a significant dewatering effort, however the pressure gradients are typically shallow, and pore pressure does not build up behind the pit slope. Conversely, in lower permeability geology the inflows may be very low, but can result in a significant pressure build-up behind the slope. The zones of high groundwater inflow therefore do not necessarily indicate high pore pressure. Typically the greatest need for depressurisation is in the lower permeability slopes which do not passively drain into the pit.

The need for slope depressurisation measures requires targeted and discrete pore pressure data from behind the pit slopes. A total of 19 vibrating wire piezometers were installed in KOV OP and Mashamba East OP in 2017 to collect this information. The analysis of the pore pressure data is ongoing to determine which parts of the pit, if any, require depressurisation in the form of horizontal drains.

In addition to expansion of the groundwater monitoring network, a program of dewatering well drilling in 2017-18 has increased the number of KOV dewatering wells to 8, with a combined pumping capacity of around 1,800 m³/hr.

16.8.4.2 KOV OP

In the KOV OP, dewatering is achieved with a combination of in-pit sump pumping, which accounts for 2,800 m³/hr to 3,200 m³/hr and pumping from the dewatering wells in the Variante and at pit bottom at a rate of 1,300 – 1,800 m³/hr.

Figure 118 shows the historical dewatering rates at the KOV OP, from 1978 up to December 2016. Figure 118 also shows the progression of dewatering over this period from in-pit water levels of 1,370 m in September 1983, to 1,145 m just prior to the slope failure of the north wall in March 2016. After GCM operations ceased in the year 2000, the KOV OP was allowed to partially flood, and between 2004 and 2009 the pit water level stabilised at 1,270 m. The KCC pumping infrastructure was commissioned in November 2011, and since this time the pit water levels have fallen steeply through pit deepening and dewatering. Groundwater inflows have increased as the pit has expanded and the dewatering rate has increased from 2,000 m³/hr in 1995 to the current average in-pit pumping rate of 4,100 m³/hr. This pumping rate is sufficient to hold the pit bottom water level steady at 1,175 m. Current pumping capacity is 10,000 m³/hr, which is able to manage more than twice this rate of inflow if required.

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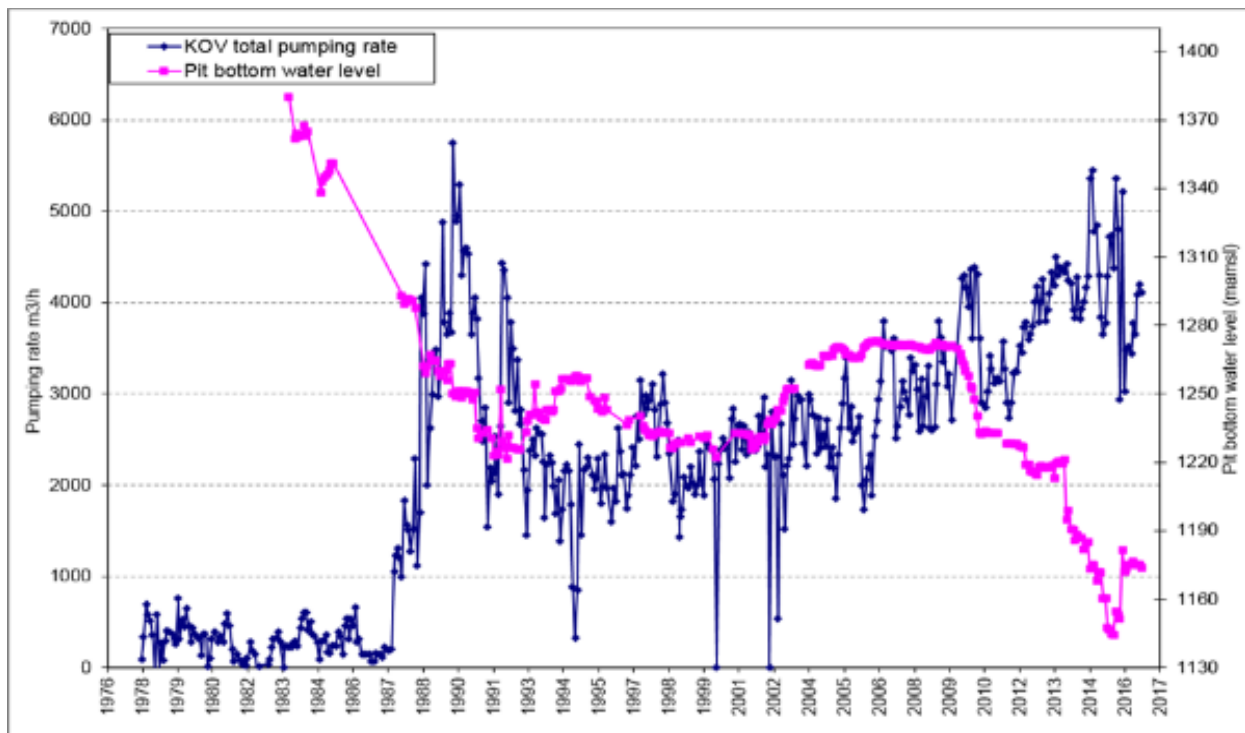


Figure 118: Plot of total dewatering rates from the KOV OP and the pit bottom water level

16.8.4.3 KTO UG

In the KTO UG, groundwater inflows are highest in the KOV aquifer close to KTE in the east, and the Kabulungu aquifer to the south. The dewatering rates from the KTO UG have steadily decreased over time from around 2,300 m³/hr in 2008 to their current level of 1,300 m³/hr, as the regional groundwater levels have decreased and equilibrated to dewatering of the adjacent KOV OP. Figure 119 shows dewatering rates from the KTO UG. Depressurisation drill holes assist with lowering the groundwater levels in the KOV aquifer adjacent to KTE. Initial flow rate from these drains of more than 1,000 m³/hr has decreased significantly to an average of 500 m³/hr in 2016. Most groundwater inflows in the KTO UG are uncontrolled seepages which are pumped from ponding areas to sumps at secondary pumping stations, where the water is pumped to the principal pumping system at level 369m. The two pumping rooms at 369 m have a combined pumping capacity of 6,700 m³/hr, and discharge water directly to the surface.

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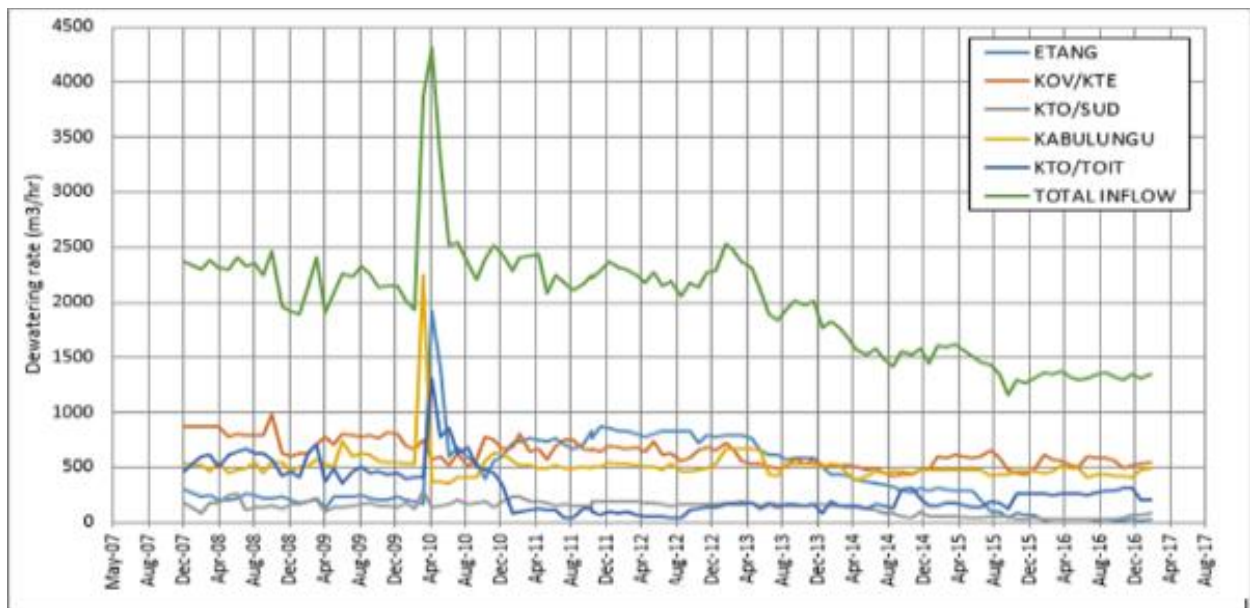


Figure 119: Dewatering rates for KTO UG

16.8.4.4 T17 OP

T17 OP is dewatered through pumping of the pit bottom sump. Despite the proximity of this pit to the Musonoi River, groundwater inflow to the T17 OP is relatively low in comparison to the KOV OP. The average pumping rate from the T17 OP was 393 m³/hr between 2014 and 2016. The T17 fragment, and the permeable units within, are spatially limited to the pit extent and surrounded by RAT on the eastern and the southern side of the pit, and SDS/CMN on the western side.

16.8.4.5 Mashamba East OP

At the Mashamba East OP, groundwater levels are currently below the pit floor, and dewatering of the Mashamba West OP reduces the immediate need for dewatering. There is a risk that the dewatering from the Mashamba West OP is terminated in advance of the completion of mining at the Mashamba East OP. There is a need to establish, and maintain control over, a dewatering system of sufficient capacity to effectively dewater the Mashamba East OP independently of the neighbouring pit.

The drilling and commissioning of new groundwater pumping wells will be required to build the needed capacity at the Mashamba East OP. In absence of a hydrogeological dataset, it is assumed that the historical dewatering system of six wells will be sufficient to bring the Mashamba East OP into production. In absence of pumping from the Mashamba West OP, perimeter wells of at least 250 m depth may be required to maintain the groundwater levels below the pit floor elevation.

16.8.5 Storm Water Management

16.8.5.1 Design Principles

The high rainfall and very erodible soils make design, installation and maintenance of the storm water management plan (SWMP) an ongoing activity throughout the year. The SWMP at KCC is extensive and is designed to:

- Minimise water inflow to the pits by diverting drainage around the pit perimeters, or by intercepting drainage in sumps;
- Limit erosion and sediment mobilisation; and
- Limit the transport of sediment into the pits and deposition of sediment at the pit bottoms.

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The SWMP requires intensive remediation in the dry season, primarily to restore the sump storage capacities and reopen the drains where sedimentation has occurred.

16.8.5.2 KOV OP

The SWMP is challenging in the KOV OP, due to the large contributing catchment areas, the steep in-pit topography, and the highly erodible and poorly consolidated mine surfaces, and the waste dumps which make up the pit catchment area. The SWMP consists of a number of typical design features and purpose built facilities to manage the runoff water, such as:

- Crowned and sheeted roads and ramps to efficiently shed runoff and minimise the erosion;
- Road-side drains with water bars to divert the drainage where needed; and
- Sumps to collect the drainage in the topographic depressions and at the intermediate levels within the pit to minimise the drainage flow to the pit floor.

Apart from the pit floor, from where the majority of the groundwater inflows are pumped, there are 7 sumps to manage water in the KOV OP. The most important of these is the KTE sump, which has a combined catchment area of 195 ha. The KTE sump intercepts a significant volume of storm water drainage which would otherwise report to the pit floor. Similarly, KTE sump allows for the interception of transported sediment and reduces the haulage distance on excavated sediments that re-enter the pit when the sump is re-excavated in the dry season.

Other important features of the SWMP at KOV OP are the Oliveira 4 and 2A sumps to the north of the pit. This area is a natural depression, and these sumps collect the drainage water which would otherwise flow over the pit crest.

16.8.5.3 T17 OP

T17 OP has favourable topography for diverting drainage around the pit. The south-eastern side of the pit has the highest elevation, and drainage is routed around the pit perimeter with a berm and a drain. The water is discharged on the north-eastern side into the Musonoi River.

Apart from the pit bottom, there are no in-pit sumps at the T17 OP.

16.8.5.4 Mashamba East OP

At the Mashamba East OP, water inflows to the pit are minimised through the diversion of surface water runoff around the pit perimeter. However, storm runoff falling within the pit perimeter will report to the pit floor. Currently all in-pit drainage flows to the lowest pit bottom elevation towards the eastern side of the pit, and infiltrates to the deep groundwater level beneath.

Should a sump pumping system be needed in future to clear these inflows, the pump capacity required to pump the 100 year, 24 hour storm rainfall event is 2,845 m³/hr for a 48 hour pumping period. These estimates are based on a conservative runoff model (i.e. equivalent runoff coefficient of 0.9).

The Mashamba East OP is located in an area of relatively level topography, which naturally drains to the east. Immediately to the south of the pit is the Mashamba waste rock dump. Surface drainage from the dump drains north down the ramp towards the southern rim of the pit. A drainage trench diverts this water towards the east where it discharges into a natural drainage system. A raised berm at the crest restricts the surface drainage from entering the pit at other points on the pit perimeter.

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17.0 Recovery Methods: Plant and Processing

17.1 Introduction

The KCC process plant is comprised of KTC and Luilu. KTC is located approximately 4 km from Kolwezi and the Luilu Plant is 6 km from KTC and approximately 9 km from Kolwezi.

Historically KTC treated a mixed ore (predominantly oxide with disseminated sulphide mineralization) from KOV and a sulphide ore (almost exclusively sulphide) from KTO through two separate mill and float circuits, to produce both an oxide concentrate and a sulphide concentrate from the mixed ore, and a sulphide concentrate from the sulphide ore.

In the new KTC circuit the mixed ore from KOV OP and Mashamba East OP produce a sulphide pre-float concentrate, and oxide form tailings, which are pumped to Luilu for further processing. The sulphide ore from KTO continues to be treated in a separate sulphide mill and float circuit. The two sulphide concentrates are pumped to the roasters at Luilu for further processing.

The WOL Project includes the design of optimized copper and cobalt circuits to produce up to 300 ktpa of copper cathode metal and 30 ktpa of cobalt as a cobalt hydroxide cake. The design utilizes as much of the existing KCC facilities as possible.

Further work has started on expansion projects to increase the cobalt production capacity up to 40 ktpa of dried cobalt hydroxide salt and to supply the full sulphuric acid requirements of the WOL plant by the construction of a sulphur burning acid plant.

KML suspended cobalt sales on November 6th 2018 as a result of the uranium content in its cobalt hydroxide. The level exceeded the acceptable limit allowed for export of the product through main African ports to customers. A study was completed by SENET that identified two methods of removing the uranium; Ion exchange (IX) and Precipitation using the phosphate ion. At the time of writing this TR KCC is evaluating both solutions.

17.2 WOL Plant

Several requirements were identified in order to convert the current KCC operation to a WOL facility. The WOL Process Block Flow Diagrams are shown in Figure 120 and Figure 121 below.

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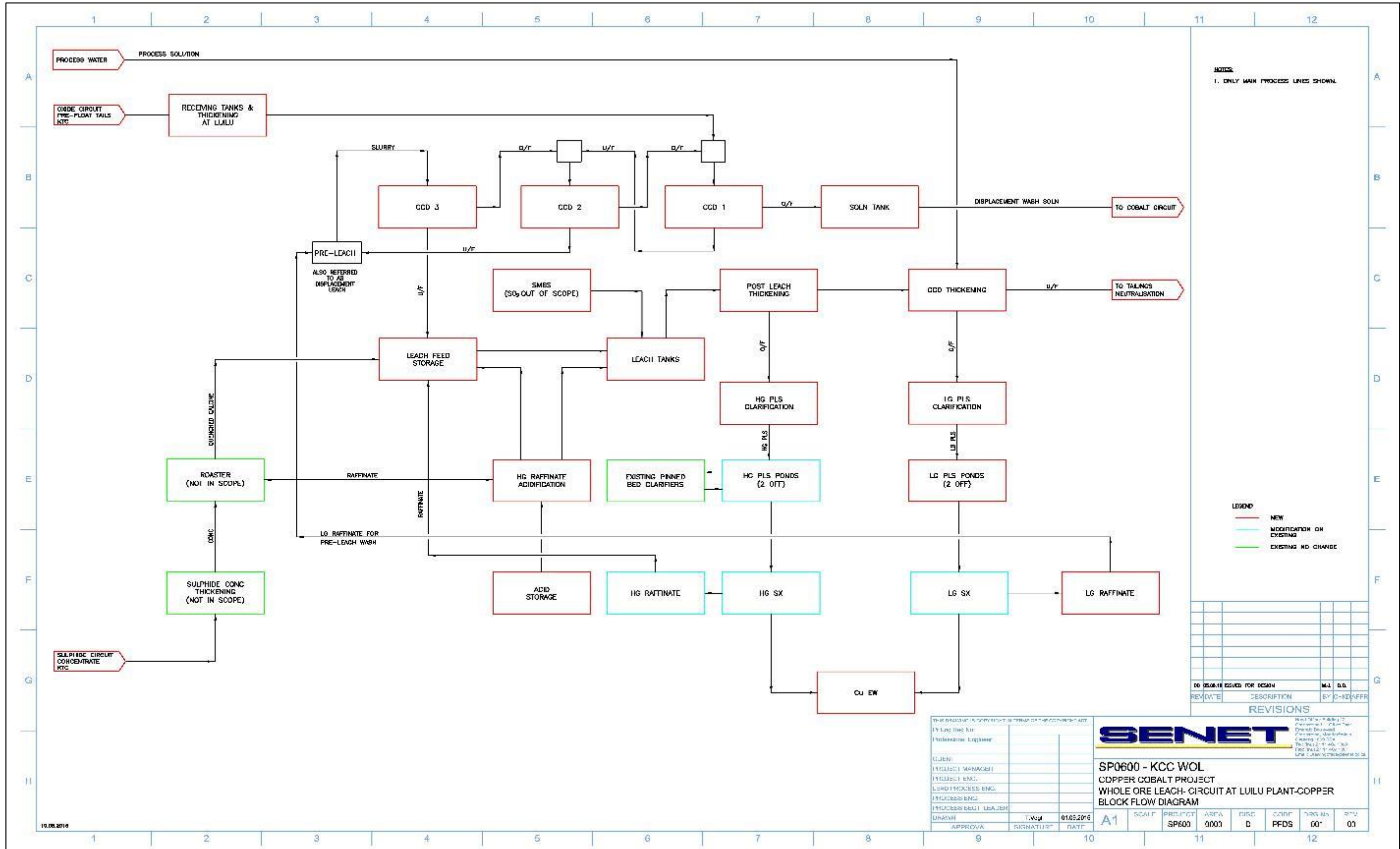
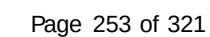


Figure 120: Block flow diagram – leach and CCD circuits

re 121: Block flow diagram – cobalt plant upgrade circuit



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17.2.1 KTC Upgrade

KCC is currently operating the WOL hydrometallurgical processing at their existing Luilu operation. The KCC WOL project is designed to process the mixed ore flotation tails from the KTC plant, as well as the roasted sulphide concentrate, to produce 300 ktpa of copper metal and 30 ktpa of cobalt contained in cobalt hydroxide. Further expansion is planned to achieve cobalt production capacity of 40 ktpa.

The process plant design is based on the following:

- Historical operating data and historical testwork;
- METSIM Simulation Model based on assumed inputs;
- Supplementary grind and flotation testwork by KTC which is to be conducted on site;
- Leach testwork;
- Settling testwork;
- Filtration testwork;
- Solution clarity testwork;
- Reagent consumption testwork; and
- Ore characterization.

The comminution simulation conducted by New Concept Projects (NCP), for which the final report was released in August 2016.

17.2.1.1 Process Plant Description

17.2.1.1.1 Milling

Due to WOL plant requirements, milling is required to grind the ore to 80 % passing 150 μm , the size most suitable for both the sulphide mineral extraction and the leaching process. Ore is transferred from the crushed ore storage stockpile dedicated to each mill circuit. The cyclone overflow slurry product from each circuit will be fed to the flotation plant via the one new 500 m³ surge tanks.

The mixed ore milling consist of CM1 and BM1 (or CM2 and BM2), CM5, CM6 and BM6, CM7 and BM7 milling circuits.

The CM1/BM1 (or CM2/BM2) mills are operated in a closed circuit with a common cyclone clusters. CM5, CM6/BM6 and CM7/BM7 also have the same arrangement. All circuits will produce a grind of 80% -150 μm from a feed with an F₈₀ of 150 mm. BM1/BM2, CM5, BM6 and BM7 operate with a ball load of up to 35%.

The feed to each mill circuit is measured and recorded using a weightometer installed on each AG or SAG mill feed conveyor. Process water, together with the crushed ore, is added to each AG or SAG mill to achieve a slurry solids content of 75% by mass within the mill. Scats conveyors are installed at the discharge of CM5, CM6 and CM7 to remove material from the milling area to a dedicated stockpile outside the building. Table 118 presents the mills required on duty for each level of throughput.

Table 118: Comminution milling circuits

Throughput	23,000 tpd	28,000 tpd	33,000 tpd
Mills on duty	CM5 CM6/7	CM1/2 CM5 CM6/7	CM1/2 CM5 CM6 CM7

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17.2.1.2 Flotation

The flotation circuit consists of two distinct parallel flotation circuits followed by a common cleaner circuit.

From the surge tank slurry is distributed to the two flotation circuits via a dedicated pumping system.

17.2.1.2.1 Rougher/Scavenger Flotation Circuit 1

Rougher/Scavenger Flotation Circuit 1 consists of $23 \times 50 \text{ m}^3$ tank cells. Conditioned slurry will be pumped from the new surge tanks through a slurry splitter to four of the tank cell banks. Each of the first three banks has $6 \times 50 \text{ m}^3$ tank cells and the last bank has $5 \times 50 \text{ m}^3$ tank cells.

The concentrate from the rougher cells will gravitate to Tank TK 035, and then be pumped to Tank TK 059. The concentrate from the respective scavenger cells will gravitate to Tank TK 036, and also be pumped to Tank TK 059. The concentrate from Tank TK 059 will be pumped to the cleaner flotation cells. The tails from the rougher cells will gravitate to the scavenger cells. The tails from the scavenger cells will gravitate to Tank TK 032, and then be pumped to the final tails distribution tanks.

Table 119 presents the flotation circuit 1 capacities for different throughputs.

Table 119: Rougher/Scavenger flotation circuit 1

Throughput	23,000 tpd	28,000 tpd	33,000 tpd
Number of cells	$18 \times 50 \text{ m}^3$	$23 \times 50 \text{ m}^3$	$23 \times 50 \text{ m}^3$

17.2.1.2.2 Rougher Flotation Circuit 2

Rougher/Scavenger Flotation Circuit 1 consists of $112 \times 8.5 \text{ m}^3$ Wemco cells. Conditioned slurry will be pumped from the new surge tanks through a slurry splitter to the eight Wemco cells banks. Each of the banks has $14 \times 8.5 \text{ m}^3$ Wemco cells.

The concentrate from the rougher/scavenger cells will gravitate to Tank TK 085, and then be pumped to the cleaner circuit through a splitter. The tails from the rougher cells will gravitate to the scavenger cells. The tails from the scavenger cells will gravitate to Tank TK 075, and then be pumped to the final tails distribution tanks.

Table 120 presents the flotation circuit 2 capacities for different throughputs.

Table 120: Rougher/Scavenger flotation circuit 2

Throughput	23,000 tpd	28,000 tpd	33,000 tpd
Number of cells	$84 \times 8.5 \text{ m}^3$	$112 \times 8.5 \text{ m}^3$	$112 \times 8.5 \text{ m}^3$

Note that for 33,000 t/d the excess capacity will report to Rougher/Scavenger Circuit 1, as the larger cells will be able to handle the additional capacity.

17.2.1.2.3 Concentrate Cleaning

A concentrate cleaner circuit exists to upgrade the copper and sulphur content of the flotation concentrate. This cleaner circuit is common to both Circuit 1 and Circuit 2.

The concentrate from all flotation banks will gravitate to Tank TK 085, and then be pumped to the cleaner circuit. Feed to the cleaner circuit will be pumped through a splitter to the two cleaner banks. The splitter will also receive the concentrates produced in the Circuit 1 Tank cells. Each cleaner bank has $12 \times 8.5 \text{ m}^3$ tank cells. The concentrate upgraded in the cleaner banks will report to either Tank TK 081 or Tank TK 086, and then be pumped to the concentrate transfer section. The tails from the cleaner banks will be pumped to Tank TK 090, and then be pumped back to the new surge tanks for recycle through flotation.

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17.2.1.3 Concentrate and Tailings Transfer

17.2.1.3.1 Flotation Tailings Transfer

The final tails will be pumped to the two existing 500 m³ tails tanks, from which the tails will be pumped to the Luilu plant as WOL feed. An additional four transfer tanks are planned for installation to facilitate consistent feed volumes to the Luilu plant.

After reworking the KITD material the tails will be pumped to the tailings facility or utilized as KTO UG backfill material. After the KITD is depleted, low grade copper ore from KOV and Mashamba East will be processed through this circuit to continue feeding oxide and sulphide concentrate to the Luilu plant, as well as providing UG backfill material.

17.2.1.3.2 Concentrate Transfer

The final concentrate will be pumped to the existing 300 m³ concentrate tank, and will then be pumped to the Luilu plant.

17.2.1.4 Water Services

The existing water reticulation system will be used.

17.2.1.5 Reagents

The existing reagent plant will be used.

17.2.1.6 Sampling and Metal Accounting

New metallurgical samplers will be installed on the float feed and tails in 2019.

17.2.1.7 Process Recommendations

Supplementary grind and flotation testwork for mill simulations and flotation was completed on site in 2016. The flotation flowsheet development is planned to be completed in 2019 and will include:

- Flotation testwork at a grind of P₈₀ 150 µm to confirm kinetics;
- Installation of a scats recycle system on CM5, CM6 and CM7 circuits to cope with the additional scats tonnage at the higher feed rates;
- Installation of one new surge tank and linear sahead of the float plant;
- Installation of metallurgical samplers on flotation feed and flotation tail streams for metal accounting purposes;
- Delta water balance report review;
- NCP mill simulation report review and METSIM balance updating;
- Blower air distribution review;
- Commissioning of the existing automated ball loading system on CM5; and
- Current plant operating philosophy review.

17.2.2 Luilu Plant Upgrade

The KCC WOL process plant is an agitated leach, SX and EW copper and cobalt hydrometallurgical processing facility. The KCC WOL hydrometallurgical processing facility is designed to process approximately 9.3 Mtpa of sulphide pre-float tailings from KTC as well as 200 Ktpa of roasted sulphide concentrate processed through the Luilu Roaster Plants. The plant is designed to produce a maximum of 300 Ktpa of copper metal and 30 Ktpa of cobalt contained in cobalt hydroxide. Further expansion is planned in 2018 and 2019 to achieve cobalt production capacity of 40 Ktpa.

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The KCC WOL processing facility is designed to integrate some existing processing facilities already in use at Luilu, and, where required, additional capacity is included.

The pre leach, leach and the post leach circuits are all new installations. The plant is designed to run as two parallel streams of pre leach, leach and post leach, with common tailings handling, reagents and utilities.

The facility is designed to utilise the existing Luilu Solvent Extraction plant, with some minor process modifications to the SX configuration. The current SX at the Luilu processing facility has three solvent extraction trains, all operating with a single high grade PLS feed. Part of the changes to the SX processing facility included the conversion of the SX plant to two high grade (HG) streams and a single low grade (LG) stream operating in optimised series parallel configuration to produce a lean low grade raffinate for the cobalt circuit. Additional PLS and raffinate ponds are included into the design to improve the solution handling capabilities of the facility.

The KCC WOL processing facility is designed to use the existing Luilu copper Electrowinning plant. A new tank house, TH03, with the capacity of 40 Ktpa was built to meet the maximum planned copper production targets.

The KCC WOL processing facility is designed to produce cobalt only in a hydroxide salt form. The cobalt salt will be produced from additional equipment included in the design (SENET, 2016). KCC may evaluate the production of cobalt metal in the future.

17.2.2.1 Copper Circuit

The new copper circuit consists out of the following:

- Pre-Leach Dewatering (Receiving & Pre-Leach CCD – 2 trains);
- Leaching – 2 Trains;
- Post-Leach Thickening – 2 Trains;
- Counter-Current Decantation – 2 Trains;
- Neutralisation and Neutralised Tailings Transportation;
- Solvent Extraction;
- Utilities;
- Electrowinning;
- Reagents;
- Compressed Air; and
- Water.

17.2.2.1.1 Pre-leach Dewatering

Receiving

The pre-leach dewatering process is meant to remove the maximum possible solution from the slurry feed to the leach circuit.

The pre-float tailings from the KTC is discharged into a new feed receiving box. The discharge from this box is sampled then discharged into two feed receiving tanks and then pumped to the two 38 m receiving thickeners. The feed to the two thickeners is diluted by internal dilution in a feed well with a resulting solids density of 5% to 10% m/m. The dilute thickener feed is flocculated along the thickener feed pipe as well as in the feed well to achieve the required flocculation rate.

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The underflow slurry from the two receiving thickeners, at 55% to 60% m/m solids, is transferred into pre-leach thickener mixer tanks. Underflow pumps on each receiving thickener are equipped with a bypass to pump to the opposite pre-leach CCD train or directly to the leach feed tank.

The overflow solution from the two receiving thickeners is routed into the receiving thickener overflow tank, from where it is pumped to the existing Process Water Return Holding Thickener. The receiving thickener overflow is then returned to the KTC process water reticulation.

Pre-leach CCD

The pre leach CCD consists of two trains of thickeners (each train has three 38 m diameter thickeners with associated mixer tanks). The pre-leach is designed to reduce the acid consumption by consuming residual acid in the low grade raffinate bleed, with an associated reduction in the lime consumption in the iron/manganese precipitation circuit.

In this circuit, slurry flow is counter-current to the low grade raffinate copper solution used for washing of the slurry.

A receiving tank receives the feed from KTC, the slurry is then fed from the receiving tank to the receiving thickeners. Thickened underflow slurry from the two receiving thickeners is transferred into the pre-leach thickener 1 mixing tanks as shown in shown in Figure 120 above.

In the mixing tanks, the slurry is thoroughly mixed with acid-depleted low grade raffinate overflow from the second pre-leach thickener and fed into the first pre-leach thickener of each train. The underflow from the first thickener is fed to the second thickener mixing tank where it is vigorously contacted with low grade raffinate from the third thickener. This mixed slurry is fed into the second thickener. The overflow solution from the first thickener is pumped via a transfer tank to the Iron/Manganese (Fe/Mn) circuit.

The underflow from the second thickener is pumped into the pre-leach tank where it is mixed with LG Raffinate. The slurry from the pre-leach tank is pumped to the mixing tank of the third pre leach thickener where it is vigorously mixed with fresh low grade raffinate from the Low Grade Raffinate pond.

The partially leached slurry is pumped from pre-leach thickener three to the Leach Feed Tank or directly to leach trains 1 and 2 in the event of the Leach Feed Tank being offline.

The ability to bypass one pre-leach CCD thickener has been included in the design of the thickeners' underflow and overflow transfer.

17.2.2.1.2 Leaching

The leach circuit will be operated in two parallel trains. Each of the parallel leach trains consists of six leach tanks fed from one common leach feed tank. The six tanks in one leach train each have a volume of 1,100 m³.

The partially leached slurry from the underflow of the third pre-leach thickener is pumped into the leach storage tank, where it is contacted with re-pulped calcine and acidified HG raffinate. Acidified raffinate (majority raffinate) is added to the leach storage tank from the acid dilution tank for density control. pH is not controlled in the leach storage tank, as it serves only as a mixing/storage tank. The level in the acid dilution tank is controlled by bypassing HG raffinate directly to the leach storage tank.

Leach slurry is transferred from the leach storage tank to the two parallel leach circuits using the leach feed pumps.

Each of the leach tanks is equipped with a de-sanding pump at the bottom. The de-sanding pumps circulate the slurry from the bottom of the tank up the tank discharge and prevent collection of coarse material in the leach tanks.

SMBS/SO₂ gas is dispersed into the leach tanks via pipes/spargers to provide the reducing environment required for cobalt leaching

The leach storage and leach tanks are all closed tanks, with provision for fume extraction. The fumes extracted from the leach circuit are processed through the SO₂ gas scrubbers.

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SMBS (or SO₂ gas in the future) is added to facilitate leaching of the Co+3 by converting it to the soluble Co+2. Monitoring of reduction of Co+3 to soluble Co+2 is achieved by maintaining an ORP (Oxidation/Reduction potential) of between 350 and 400 mV. The circuit is designed to allow SO₂ gas to be used in the future.

The pH profile of the leach train is controlled by the addition of acidified raffinate to each of the leach tanks. The acidified raffinate is made by contacting 98% H₂SO₄ with HG raffinate solution in the acid dilution tanks and is thoroughly mixed by the tank agitator to achieve a well-acidified HG raffinate for acidic dilution and pH control.

Leach tailings from the last two tanks are pumped to each train's respective post leach thickener feed tanks.

The leach tails pumps are configured with a bypass to pump to either the Post Leach thickener or the associated CCD number 1.

17.2.2.1.3 Post Leach Thickening

Leached slurry exiting the last leach tank of each train is pumped into the respective post-leach thickener. The two post leach thickeners are built with internal forced-dilution systems which enables the feed to the thickener to be diluted to 5% to 10% m/m solids. Flocculant is contacted with slurry in the thickener feed tank and along the thickener feed pipe as the slurry gravitates into the thickener feedwell.

The post-leach thickener removes excess solution from the leach tailings slurry to produce a post-leach thickener underflow with a solids content of 55-60% m/m and thickener overflow solution rich in copper and cobalt, the HG PLS.

Post-leach thickener underflow is pumped to the CCD circuit where the slurry is washed with effluent from the cobalt recovery circuit. If there is insufficient effluent from the cobalt recovery circuit, then process water and/or low grade raffinate from the low grade SX can be utilised.

The overflow solution from the post leach thickeners is pumped to the HG clarifier for each train respectively.

HG Clarifier underflow is returned back to its respective post-leach thickener or to the first CCD thickener in each respective train, while HG clarifier overflow is pumped to the existing Dirty HG PLS pond.

A provision is available to add coagulant solution to the overflow tank of the post-leach thickener. In addition, the clarifiers have a provision available to add flocculant solution to aid settling.

17.2.2.1.4 Counter Current Decantation

Leached slurry from the post-leach thickening circuit is transferred to the CCD circuit to wash the valuable soluble metal content from the leach slurry. The KCC plant consists of two parallel CCD circuits, each with seven CCD wash stages.

Underflow slurry from the two post leach thickeners is pumped to the respective first CCDs' mixing tanks, where it is diluted with internal thickener overflow product. From these mixing tanks, diluted slurry is gravitated to the respective thickener feedwell.

Each of the CCD thickeners has built in internal dilution system that allows slurry to be diluted to between 5% and 10% for feed into the thickener. Flocculant is added to the thickener feed tank and along the thickener feed pipe as the slurry gravitates into the thickener feed well. A provision is available to add flocculant solution to the feed well of each of the CCD thickeners.

Each of the CCD thickeners produces underflow slurry of between 55% and 60% m/m solids concentration. Slurry washing is achieved by counter-current washing whereby the leach tailings slurry is washed with a combination of cobalt effluent (preferred wash solution), barren process solution and low grade raffinate solution in a counter-current flow. A provision is made to acidify the CCD wash solution to maintain a low pH of the wash solution if required to prevent possible precipitation of the contained valuable metal with excessively higher pH. Low grade raffinate wash solution can be added to the 5th CCD in line if required. Low grade raffinate is relatively low in soluble copper, but high in soluble cobalt, hence cobalt effluent or process water is used to wash the back end of the CCD circuit.

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Slurry flows inter-stage downstream through the CCD circuit from the first CCD thickener (CCD 1) to the last CCD thickener (CCD 7). Wash solution flows inter-stage upstream through the CCD circuit CCD 7 to CCD 1. Underflow slurry from the last CCD in line is pumped to the tailings transfer tank.

Mixing thickener feed in thickeners $1 < n < 7$ are achieved by receiving underflow from thickener $n-1$ underflow pumps, and overflow solution from thickener $n+1$ needle tank pumps. These feed streams are mixed in an agitated mixing tank, which gravitates towards the thickener feedwell. Internal dilution from each thickener utilizes internal overflow solution pumped from its overflow to ensure feedwell feed is diluted to 5% – 10% m/m solids.

Overflow from the first CCD thickener is pumped into the respective LG Clarifier. The clarifiers overflow is the LG PLS and is gravitated to the Dirty LG PLS pond. The clarifier underflow slurry at a solids content of 6% by mass is pumped to the respective first CCD thickener.

A provision is available to add coagulant solution to the overflow tank of the first CCD thickener. In addition, the clarifiers have a provision available to add flocculant solution to aid settling.

All CCD thickeners have the possibility to bypass a maximum of one thickener.

17.2.2.1.5 Tailings Neutralisation

Tailings from the CCD is transferred from the Tailings Transfer Tank to the neutralisation circuit using the transfer pumps. The tailings are neutralised by the addition of milk of lime. The neutralised tailings slurry is transferred to the Tailings Storage Facility (TSF) by the Neutralised Tails Transfer Pumps.

17.2.2.1.6 Solvent Extraction

The existing SX circuit consists of three SX trains made up of three extraction stages and two strip stages per train.

The first two SX trains are configured to receive HG PLS. The third train will be converted to operate on an optimised series parallel configuration on LG PLS with series mixer settler configuration being the first and the third extraction stage and the parallel being the second extraction stage.

Two raffinate solutions will be produced from the SX plant namely:

- HG raffinate; and
- LG raffinate.

High Grade Solvent Extraction

HG PLS is pumped from the existing Dirty HG PLS Pond. Two thirds is pumped through the existing pinned bed clarifiers to the HG PLS Pond while the other third is pumped directly to the HG PLS Pond.

Clarified HG PLS is pumped by two independent pumps (one per train with a common standby) from the HG PLS Pond through two independent lines to the first extraction stage of the HG SX Trains. A common priming tank is used for the pumping from the PLS Pond and an additional pump will be required. The aqueous solution (HG PLS to start with) is then transferred from E1 to E2, from E2 to E3 and from E3 to the raffinate after-settler. The raffinate is then finally pumped from the raffinate after-settler to the new HG Raffinate Pond.

The ability to bypass all of the SX stages (aqueous and organic) has been included in the design. This will give the ability to bypass individual settlers, minimizing loss of production.

The HG Raffinate is pumped from the HG Raffinate Pond to the Leach Circuit and a portion of the HG Raffinate is distributed to the roasters for calcine repulp.

Low Grade Solvent Extraction

Dirty LG PLS overflows from the Dirty LG PLS Pond to the LG PLS Pond so that some clarification takes place in the Dirty LG PLS Pond.

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The LG PLS is then pumped from the LG PLS Pond via priming tanks to the LG SX Train. The LG SX Train is configured in an optimised series/parallel configuration. Half of the LG PLS is pumped to E1 then transferred to E3 and finally to the LG Raffinate After-Settler. The other half of the LG PLS is pumped to E2 then transferred to the LG Raffinate After-Settler. The LG Raffinate is pumped from the LG Raffinate After-Settler to the existing LG Raffinate Pond.

The LG Raffinate is pumped via a priming tank from the LG Raffinate Ponds to the Pre-Leach Circuit. The ability to bypass the pre-leach circuit and pump the LG raffinate directly to the FAM Precipitation Circuit or the CCD wash water tank has been included in the design. This will give the ability to bypass individual settlers, minimizing loss of production.

17.2.2.1.7 Electrowinning

There are three Electrowinning facilities at KCC, EW2, EW3 and EW4. A description of EW3 is included below and similar applies to EW2 and EW4.

EW3 has three existing tank houses, Tank house 01 (TH001), Tank house 02 (TH002) and Tank house (TH003). This description is for TH003.

The advance electrolyte exiting the multimedia filters in SX is transferred into the existing EW3 Advance Electrolyte Tanks then pumped through a recovery heat exchanger to the polishing cells. The recovery heat exchanger is used to recover heat from the hot spent electrolyte being pumped to the solvent extraction circuit.

The cleaned heated advance electrolyte is circulated to polishing cells; TH003 has 28 of the total of 136 cells, at a feed rate of 338 m³/h. These polishing cells help to remove the last traces of organic from the advance electrolyte and protect the other commercial cells from organic breaking through the multimedia filters.

Polishing cell discharge flows to the circulating electrolyte tank where it is joined by the recycled spent electrolyte. The circulating electrolyte is pumped by the circulating electrolyte pump to the commercial cells for deposition of the copper. Overflow from the commercial cells is spent electrolyte.

Spent electrolyte discharging from the commercial cells is returned to the spent electrolyte tank. Excess spent electrolyte is recycled to the circulating electrolyte tank for further copper deposition.

Each EW cell contains 48 stainless steel cathodes and 49 lead anodes. Copper is deposited on the cathodes electrolytically.

The copper growth cycle is generally between six and eight days with an average of seven days. Operation at high current density will give a shorter growth cycle and an increased probability of quality problems while operating at low current density will give a longer growth cycle and a lower probability of quality problems.

Copper is stripped from the cathodes at the end of the growth cycle. One third of the cathodes (every third cathode) are removed from a cell and washed with hot water (60 – 70° C) to remove any electrolyte from the cathode. The wash water is heated by inline heaters. The cathodes are then placed in a stripping machine where the copper cathode is removed from the stainless steel mother plates. After stripping the stainless steel cathodes are returned to the electrowinning cells. This process is then repeated twice more in order to harvest a full cell.

Spent electrolyte is pumped by pumps to the heat recovery heat exchanger to heat the incoming electrolyte. The spent electrolyte is further cooled (to increase the efficiency of the SX stripping process) by circulating cooling water in heat exchangers. The cooled spent electrolyte is delivered into the spent electrolyte return tank in the SX strip circuit.

Cooling water is circulated through the cooling heat exchanger cooling water pumps. Cooling water heated by the spent electrolyte is cooled in cooling tower.

Concentrated sulphuric acid (H₂SO₄) is added to the Advance Electrolyte to maintain an acid level of 180 g/l to compensate for the acid loss caused by the impurity bleed in the backwash circuit of the multimedia filters in SX, losses by acid mist and dilution from cathode wash water.

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17.2.2.2 Cobalt Circuit

The cobalt circuit is designed to produce 30 Ktpa of cobalt as a cobalt hydroxide cake. Further expansion is planned in 2019 to achieve the cobalt production capacity up to 40 Ktpa. The plant will consist of a combination of modified existing equipment and new equipment for additional processing capacity.

The cobalt plant consists of the following unit operations:

- Iron, aluminium and manganese (FAM) precipitation;
- Iron, aluminium and manganese thickening;
- Copper removal (precipitation);
- Copper removal Thickening;
- Cobalt precipitation with MgO (stage 1);
- Cobalt stage 1 thickening and filtration;
- Cobalt precipitation with lime (stage 2) (Deferred);
- Cobalt stage 2 thickening (Deferred);
- Wet cobalt bagging and cobalt drying and bagging;
- Reagent plants for:
 - a) Flocculants
 - b) Lime
 - c) Magnesia make up
 - d) Coagulant
 - e) Phosphoric acid
- Utilities.
 - a) Fe/Al/Mn precipitation air
 - b) Gland service water
 - c) Plant and instrument air
 - d) Potable water (safety showers/eye wash)
 - e) Process water
 - f) Raw water
 - g) SO₂ gas (deferred)

17.2.2.2.1 Iron, Aluminium and Manganese Precipitation

The iron, aluminium and manganese (FAM) removal step will use two (or one) of the three available old concentrate leach trains. Each of these trains consists of three mechanically agitated tanks with an average volume of approximately 565 m³. These tanks are closed, with fume extraction covers.

The overflow solution from the third pre-Leach CCD thickeners is pumped into the Fe/Mn feed distribution tank. Pre-leach bypass solution (LG raffinate), reclaimed cobalt hydroxide slurry and lime are also fed into this tank.

From the Fe/Mn feed distribution tank, the solution is pumped to mechanically agitated tanks in parallel arrangement with pumps. The overflows from these two tanks are discharged into the second set of mechanically agitated tanks and the overflow from these two tanks discharges into the last set of tanks. The product from the last two tanks, which are in parallel, is pumped to the Fe/Mn thickener.

Lime, air and sulphur dioxide gas (future) is fed to each of the six precipitation tanks to maintain a pH of 3.5 and Eh of 560 mV.

The second train of Fe/Mn precipitation tanks are used as a common standby for Fe/Mn precipitation and the copper precipitation.

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When air/SO₂ gas is used, the off-gas from the Fe/Mn and copper precipitation reactors is treated by the SO₂ off-gas scrubber (future) to remove excess SO₂ gas and entrained liquor and then discharged to atmosphere. The effluent solution from the scrubber is returned to the pre-leach circuit.

17.2.2.2.2 Iron and Manganese Thickening

The Fe/Mn thickener mixing tank receives precipitation product from both tanks as well as internal dilution from the thickener overflow pump. From the mixing tank, diluted slurry gravitates into the thickener feedwell. The existing thickener number Z4 will be used for this duty. Flocculant is added to the thickener feed tank and along the thickener feed pipe as the slurry gravitates into the thickener feed well. The underflow slurry is pumped into a distribution tank by the underflow pumps. Part of the thickener underflow is pumped from the distribution tank back to the Fe/Mn Feed distribution Tank to serve as Fe/Mn re-seed slurry, while the rest of the thickener underflow is pumped to the 3rd Cu CCD thickeners. The option of pumping the FAM thickener underflow to the tailings neutralisation section is also included.

The second Fe/Mn thickener and associated equipment is a common standby for the Fe/Mn and copper thickening circuit. The existing thickener number Z5 will be used for this duty.

17.2.2.2.3 Copper Precipitation

The copper precipitation step will utilize the third (or common second) available old concentrate leach train. This train consists of three mechanically agitated tanks with an average volume of approximately 565 m³. These tanks are closed, with fume extraction covers.

Overflow from Fe/Mn Thickener is pumped directly into the first of three mechanically agitated precipitation tanks, where it is contacted with lime to form a copper rich precipitate. A pH of 4.8 to 6.0 is maintained in these tanks by the addition of lime. Phosphoric acid (25-35 g/L) is added to the second copper precipitation tank to precipitate uranium from the solution as calcium uranyl phosphate. Residual iron and aluminium in solution are also precipitated as phosphates. Transfers between tanks are by gravity, while product from the last tank is pumped to the copper removal thickener mixing tank. Air and sulphur dioxide gas (future) can be fed as an option to each of the three precipitation tanks to maintain an Eh of 560mV for further Mn precipitation as required.

17.2.2.2.4 Copper Removal Thickening

The copper thickener mixing tank receives feed from the copper precipitation tank and dilution liquor from the thickener overflow pump. From the mixing tank, diluted slurry gravitates to the thickener feedwell. Flocculant is added to the thickener feed tank and along the thickener feed pipe as the slurry gravitates into the thickener feed-well.

Thickener underflow is pumped to the distribution tank. Two thirds is recycled to the copper precipitation reactors as seed material and one third is sent back to the copper leach tanks for redissolution.

Thickener overflow is pumped from the overflow tank to the cobalt Stage 1 precipitation reactors.

17.2.2.2.5 Cobalt Precipitation with MgO (Stage 1)

The cobalt precipitation with MgO stage 1 will utilize 4 new mechanically agitated tanks in series, each with a volume of approximately 1,100 m³. Cobalt rich liquor is pumped from the Overflow Tank to the first tank of the cobalt stage 1 Precipitation Reactors, where it is contacted with MgO that is added at 115% of the stoichiometric requirement. Product from the final precipitate tank is pumped to the cobalt stage 1 thickener.

17.2.2.2.6 Cobalt Stage 1 Thickening and Filtration

Thickening

The cobalt stage 1 thickening will utilize an existing thickener, where it will produce a cobalt rich precipitate in the underflow and cobalt depleted overflow liquor.

The cobalt thickener receives product from cobalt stage 1 precipitation tank 4 via pump, and internal dilution from thickener overflow pumps ensures a diluted slurry enters the feedwell. Flocculant is added to the thickener feed tank and along the thickener feed pipe as the slurry gravitates into the thickener feed-well.

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Thickener underflow is pumped to the cobalt hydroxide pressure filter feed tank, while the overflow is pumped to the CCD circuit or the future cobalt precipitation stage 2 (with lime). The second cobalt thickener and associated equipment is a common standby for the cobalt stage 1 circuit and the future cobalt stage 2 circuit.

Filtration

Cobalt hydroxide from cobalt stage 1 thickener underflow is pumped into an agitated filter feed tank. Part of the thickener underflow is pumped back to the cobalt stage 1 precipitation reactors to serve as cobalt re-seed slurry, while the rest is pumped to the pressure filters. Filter cake is discharged onto discharge conveyors and will be sent for wet bagging or drying and bagging.

Filtrate from filters is collected in a filtrate tank, from where it is pumped to the cobalt precipitation stage 1 Thickener.

17.2.2.2.7 Cobalt Precipitation with Lime (Stage 2) - Deferred

The cobalt precipitation with lime stage 2 will utilize 4 new mechanically agitated tanks in series, each with a volume of approximately 1,100 m³. The overflow of the cobalt stage 1 thickener will feed the stage 2 precipitation tanks.

Liquor is pumped from cobalt stage 1 thickener overflow tank to the first precipitation tank, where it is contacted with lime at a controlled pH of approximately 8.5. Product from the final precipitate tank is pumped to the cobalt stage 2 thickener.

17.2.2.2.8 Cobalt Stage 2 Thickening - Deferred

The cobalt stage 2 thickening will utilize an existing thickener, where it will produce a cobalt rich precipitate in underflow and cobalt depleted overflow liquor.

The cobalt stage 2 thickener receives product from the cobalt stage 2 precipitation tank 5 via pump, as well as internal dilution from thickener overflow pumps to ensure a diluted slurry enters the feedwell.

Thickener underflow is pumped to the underflow tank. Part of the thickener underflow is pumped back to the cobalt stage 2 precipitation reactors to serve as cobalt re-seed slurry, while the rest is pumped to the Pre-Leach CCD Section. The ability to transfer the thickener underflow to the FAM circuit is also included if the pH is low enough in the FAM feed.

17.2.2.2.9 Cobalt Cake Drying and Packaging

Filter cake from the cobalt pressure filters is conveyed to a wet cobalt filter cake stockpile. The wet cake is transferred by a front-end loader to the wet cake bagging plant or to the cobalt hydroxide cake dryer. Dried cobalt hydroxide is bagged by the dry cobalt hydroxide bagging plant. The anticipated moisture is expected to be reduced from approximately 65% to 15%.

17.3 Planned WOL Copper Recovery

17.3.1 Sulphide Copper Recovery

The overall sulphide recovery is calculated as 82.80%. This overall recovery applies only to the ore fed as "pure" sulphide.

The overall recovery was obtained by taking into account the following recoveries/factors:

- Flotation: 92% recovery; and
- Roaster conversion factor: 90% recovery.

Therefore overall sulphide recovery = $0.92 \times 0.90 = 82.80\%$

17.3.2 Mixed Ore Copper Recovery

The overall mixed ore recovery is based on two separate process routes, namely the sulphide fraction of the mixed ore, and the oxide fraction of the mixed ore. This recovery is therefore a function of the Ox-Sul fractions

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in the ore. The sulphide fraction undergoes a flotation recovery, followed by a roaster conversion/recovery and finally a leach/SX/EW recovery, whereas the oxide fraction undergoes only a leach/SX/EW recovery. The overall mixed ore recovery is a weighted combined recovery of these two fractions.

17.3.2.1 Sulphide Recovery within the Mixed Ore

As an example of the recovery for sulphides in the mixed ore, the following factors would be applied:

- Sulphide ratio in mixed ore: 15%
- Flotation recovery: 50%
- Roaster conversion: 90%

Therefore the sulphide fractional recovery = $0.15 \times 0.50 \times 0.90 = 7\%$

17.3.2.2 Oxide Recovery within the Mixed Ore

For oxide in the mixed ore, the following factors are applied:

- Oxide ratio in mixed ore: 85%
- Flotation recovery: 100%
- Leach/SX/EW recovery: 92%

Therefore the oxide fractional recovery = $0.85 \times 0.92 = 78\%$

The overall mixed ore recovery is the cumulative recovery of $7\% + 78\% = 85\%$

17.4 Planned WOL Cobalt Recovery

The overall cobalt recovery for sulphide or mixed ore is planned at 65%.

17.5 Planned WOL Projects and Upgrades

KCC is currently planning two major WOL Project upgrades. The first is the cobalt debottlenecking project, which will increase cobalt capacity from 30 ktpa to 40 ktpa. The second project is the acid plant, which will allow KCC to produce both sulphuric acid and liquid SO_2 .

17.5.1.1 Cobalt Debottlenecking Project

Based on a more detailed analysis of the mine plan it became evident that cobalt production for certain years may potentially be in excess of 30 ktpa, being the current cobalt production capability of the WOL project. Further, wet cobalt hydroxide material contains in excess of 60% moisture. The objective of the Cobalt Debottlenecking Project was to upgrade the existing cobalt plant design in order to increase production capacity up to 40 ktpa of cobalt, in form of a dry cobalt hydroxide salt. The Cobalt Debottlenecking Project is only upgrading the design of the cobalt plant to increase its production capacity from 30 ktpa to 40 ktpa by continuous process improvement and necessary equipment from the cobalt Stage 1 precipitation, the rest of the facility is not affected by the increase in cobalt capacity.

There is no change in the Cobalt Process Block Flow Diagram is shown in Figure 121 presented in the previous Section 17.2. The only areas that are modified by the Cobalt Debottlenecking Project are:

- Cobalt Stage 1 Precipitation;
- Cobalt Stage 1 Thickening and Filtration; and
- Cobalt Drying and Bagging.

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17.5.1.1.1 Cobalt Precipitation with MgO (Stage 1)

The residence time in the cobalt precipitation section is not affected by the increased cobalt production so the existing tanks are adequate and will not be modified. It is, however necessary to upgrade the MgO addition to the precipitation tanks and it is necessary to build a new MgO make-up facility.

17.5.1.1.2 Cobalt Stage 1 Thickening and Filtration

Thickening

The cobalt stage 1 thickener is adequate to process the increased cobalt throughput and the only modifications are to the underflow pumping arrangement. It is necessary to upgrade the underflow pumping arrangement by the installation of an additional pump, and the system will utilise two lines operating in parallel to the pressure filtration feed tank.

Filtration

In order to process the increased cobalt throughput, it is necessary to increase the cobalt hydroxide filtration capacity by the addition of three filter presses. Each filter press discharges onto a dedicated discharge conveyor which all discharge onto a common transfer conveyor. The transfer conveyor discharges onto a wet cobalt hydroxide stockpile.

In addition, it is necessary to upgrade the filtrate system by the installation of a new pump and return line to the Cobalt Stage 1 Thickener.

The balance of the equipment in the filtration area remains as per the WOL process design.

17.5.1.1.3 Cobalt Cake Drying and Packaging

Wet filter cake is stored on the cobalt filter cake stockpile and can be wet bagged or it can be sent to the dryer before bagging.

Wet Bagging

Wet cobalt hydroxide filter cake is transferred from the cobalt filter cake stockpile by a front-end loader and loaded via a chute into bags. The bags are stored before sale.

Cobalt Dryer

Two flash dryers operating in parallel have been included in order to dry the cobalt hydroxide filter cake.

Wet cobalt hydroxide filter cake is transferred from the cobalt filter cake stockpile by a front-end loader, and loaded via a chute onto one of the dryer feed conveyors. The wet cobalt filter cake is transferred by conveyors into the dryers. The cobalt filter cake is dried to 15% moisture in the dryers and then bagged for sale.

A diesel fired burner is used to generate hot gasses which are used to dry the cobalt hydroxide in the flash dryers.

17.5.1.1.4 Cobalt Debottlenecking Project Update

The cobalt debottlenecking projects (the "Cobalt Projects") are expected to be completed and in full production during Q3 2020. Commissioning of two filter presses was completed earlier in the year, while the third filter press was commissioned during Q3 2019. The commissioning of the MgO plant was also completed earlier in the year. During Q3 2019 phase 1 of the MgO plant has been undergoing a rectification program to correct certain aspects of the project which is expected to be complete Q1 2020. While commissioning the cobalt dryers there was additional work required on the units due to a mechanical failure of the burners. Dryer one is undergoing a temporary fix which should be complete by the end of October 2019 whilst dryer two is undergoing a long term rectification plan and is expected to be online by the end of Q4 2019. Dryer one is expected to be operating at full capacity during Q1 2020 while dryer two is expected to be operating at full capacity in Q2 2020.

The objective of the Cobalt Projects is to upgrade the existing cobalt plant design in order to reduce bottlenecks by modification of the precipitation, thickening & filtration and drying & bagging processes. This will align the

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design of the cobalt plant with the average life-of-mine cobalt production plan of 30,000 tonnes per annum. These improvements integrate with the existing WOL processing facilities at Luilu.

17.5.1.2 Acid Plant

The increased production of copper (and cobalt) at Luilu requires substantially more sulphuric acid. This sulphuric acid was initially going to be supplied from Glencore operations in Zambia, but increased acid requirements by other users has resulted in a strain being placed on the acid supply in the DRC. KCC has therefore decided to construct a sulphur burning acid plant at the Luilu refinery.

In addition to the acid, a reducing agent is required to improve the recovery of cobalt in the leach and an oxidizing agent is required in the iron/aluminium/manganese precipitation. The same reagent, sulphur dioxide, can be used for both of these. It is usually generated by a sulphur burner. The original design utilized Sodium Metabisulphite (SMBS) instead of sulphur dioxide gas, despite SMBS being far more expensive. As a result, generating sulphur dioxide gas as a side-stream from a sulphur burning acid plant provides a major cost saving.

The objective of the Acid Plant Project is to construct a facility for the production of sulphuric acid and liquid SO₂, starting from solid sulphur, and based on double conversion double absorption (DCDA) process. The plant capacity, expressed as 100% H₂SO₄ and based on a stream factor of 333 days per year, is:

- Sulphuric Acid (as 100% H₂SO₄): 2,200 tpd; or
- Sulphuric Acid (as 100% H₂SO₄): 1,900 tpd and Liquid SO₂ (as 100% H₂SO₄): 200 tpd.

Solid sulphur is delivered to the facility in bags and stored under cover. The bags are manually loaded and the sulphur is transferred to the sulphur melting area. The sulphur is melted and filtered then transferred to the molten sulphur storage tanks. The sulphur is then burnt in air to produce SO₂ gas and the SO₂ gas is reacted with oxygen to produce SO₃. The SO₃ is then absorbed by concentrated (98%) sulphuric acid. The concentrated sulphuric acid is finally diluted back to 98% and the excess sulphuric acid is removed from the system. The concentrated sulphuric acid produced is stored before transfer to the existing storage facility.

A portion of the SO₂ bearing gas from the sulphur burner is fed to the liquid SO₂ plant. The SO₂ plant uses partial condensation at low temperature in order to produce pure liquid SO₂. The off-gas leaving this unit after condensation still holds a residual amount of SO₂, which is removed before release to the atmosphere. SO₂ removal is completed by sending back the off-gas to the first stage of the catalytic converter and therefore producing more sulphuric acid.

As part of the sulphuric acid generation process, excess heat can be used to drive a Steam Turbine Generator (STG). The KCC acid plant design includes a 32 MW STG which will be able to utilize this excess heat to provide power back to KCC's operation, and is expected to be commissioned during Q3 2020. The STG will be able to provide power to facilitate the acid plant's own production as well as send excess power to the Luilu plant under the following specifications and limits:

- At 100% capacity (2200 tpd of acid) an average of 10.5 MW is required by the Acid Plant, with a max requirement of 11.0 MW;
- At 100% capacity (1900 tpd of acid and 200 tpd of SO₂) an average of 10.5 MW is required by the Acid Plant, with a max requirement of 11.0 MW;
- At minimum capacity (approximately 60% - 1200 tpd of acid) an average of 9.0 MW is required by the Acid Plant, with a max requirement of 9.5 MW; and
- At minimum capacity (approximately 60% - 1200 tpd of acid and 200 tpd of SO₂) an average of 9.0 MW is required by the Acid Plant, with a max requirement of 9.5 MW.

Although the capacity of the STG is designed at 32 MW the generated power will be based on the steam energy received from the Acid Plant process. Depending on the operating condition of the Acid Plant the following power generation output is expected:

- At 100% production (2200 tpd of acid only) – 26 MW peak; and

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- At 100% production (1900 tpd of acid and 200 tpd of SO₂) – 24.9 MW peak.

At minimum capacity the STG will produce 11 MW, therefore sustaining only the power required to operate the Acid Plant with extremely little power available for cogeneration and distribution into the rest of the Luilu plant.

Based on the design estimates, a direct relationship (linear) on the amount of cogeneration power available can be assumed if the Acid plant operates at a level between minimum and 100% capacity.

17.5.1.3 Uranium Removal

Although uranium occurs naturally in KCC's feed ore previous test work conducted by other experts made no mention of this element. Therefore, no allowance was made in the WOL project for the process to remove uranium from the final products. Moreover, historically KCC produced cobalt metal, and as uranium is not plated on the cobalt cathode it was always disposed of via the spent electrolyte and eventually through to the TSF.

On November 6th 2018 KML announced the suspension of cobalt sales as a result of the uranium content in its cobalt hydroxide. The level exceeded the acceptable limit allowed for export of the product through main African ports to customers. An investigation was carried out to determine:

1. The levels of uranium that could report in plant feed from the two operating pits, KITD and the underground mine;
2. How the uranium would deport across the plant; and
3. What options were available for uranium removal.

The outcomes at this current stage of investigation are below.

1. Average uranium values in different mining areas are as follows:
 - KOV open pit – 22.2 g/Tonne
 - Mashamba open pit – 8.5 g/Tonne
 - Kamoto underground – 53.6 g/Tonne
 - KITD – 17.3 g/Tonne

These values have been obtained within 825 blasthole samples inside of confirmed ore zones for KOV and Mashamba, 84 samples within KTO UG from diamond drill holes, and 38 samples within KITD diamond drill holes. Sampling for uranium began in late 2018 and is ongoing.

2. Dependant on the oxidation state of the uranium between 30 and 50% of the uranium leaches in the leaching circuit. The leached uranium all reports to the low grade raffinate which is treated through the Cobalt plant. About 30% of the contained uranium is precipitated in the iron removal step and another 50% is precipitated in the copper removal step. The remainder precipitates with the cobalt hydroxide and results in uranium levels that exceeds the saleable limits.
3. Two methods of removing the uranium were identified
 - Ion exchange (IX) using an anionic exchange resin that would target the uranyl anionic species. This technology is commonly implemented in the extraction of primary uranium.
 - Precipitation using the phosphate ion (PO₄³⁻). This method has already been implemented, and KCC has demonstrated consistent success at producing saleable cobalt utilizing the phosphate ion. Literature indicated that some uranyl phosphate minerals are extremely insoluble at pH levels greater than 3.5

The investigation determined that it would be best to carry out test work on both options and to advance both in parallel until such time as a choice could be made on the basis that uranium contamination in the cobalt hydroxide would always be less than the accepted level of 75 ppm.

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The test work, carried out by CM solutions in Johannesburg, demonstrated that both technologies should achieve the required levels of uranium removal.

The investigation also identified that implementation of the ion exchange technology would involve the installation of significant new processing equipment including large adsorption columns and resin elution vessels. SENET was commissioned to provide a feasibility study that would estimate the capital and operating costs as well as an implementation schedule.

The phosphate removal technique on the other hand did not require any significant capital expenditure and could be implemented quickly on a trial basis. There was however doubt that the method would prove to be 100% effective.

At the time of writing this TR both options are still being developed in parallel.

17.5.1.3.1 IX Plant (Future Installation)

Solution from the copper precipitation thickeners' overflow will feed to the IX Plant. The copper thickener overflow will discharge into the uranium ion exchange feed tank.

The solution from the uranium IX feed tank will be pumped upwards through the uranium adsorption columns and barren solution exits the top of the columns while the uranium in the feed loads onto the resin in the column. The barren solution with a low uranium concentration will discharge into the barren solution tank. The barren solution will be pumped via a linear screen to the cobalt precipitation section.

When the resin in one stage of the column is fully loaded with uranium ions, the feed to the column will be stopped and the loaded resin transferred into the transfer-elution vessel.

The feed solution entrained with the resin will be washed out of the resin with water before the uranium is eluted from the resin in the transfer-elution vessel by pumping eluant (110 g/l H_2SO_4) downwards through the vessel and the resin bed. The solution exiting the bottom of the transfer-elution vessel is termed eluate. In order to reduce the acid consumption and improve the water balance the elution is done in two phases:

- Recycled eluant, termed intermediate eluant from the intermediate eluant storage tank is pumped through the elution-transfer vessel and the eluate transferred to the eluate storage tank
- Fresh eluant is pumped from the eluant storage tank through the elution-transfer vessel and the eluate transferred to the intermediate eluate solution storage tank

The final eluate will be discharged to the leach feed tank or the tailings neutralisation tanks, but the leach feed tank will be the normal discharge location.

The fresh eluant makeup will be by mixing concentrated sulphuric acid (98%) with raw water in an inline mixer.

Allowance has been made for the emergency storage of resin in a conical tank when the columns are offline for maintenance. The resin will be transferred back into the columns by means of an eductor. An eductor operates with pressure energy within a high-pressure motive liquid to pump, a secondary low-pressure fluid.

Fresh resin will be loaded by a resin loading tank and transferred into the adsorption columns by an eductor.

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18.0 Project Infrastructure

The current KCC infrastructure as listed in different Chapters (1.5, 1.7, 5.0, 16.0, and 17.0) of the TR is sufficient to handle surface and underground production, however process plant upgrades related to the WOL and electrification of the Mashamba East OP are required.

18.1 Process Plant Upgrade

The KCC process plant comprises both KTC and Luilu. The major changes in the infrastructure are related to the Luilu plant upgrades to enable the treatment of the mixed ore. The associated additions to the plant infrastructure completed to date are follows:

- Overland piping between KTC and Luilu to transport the slurry from the mixed ore pre-float circuit;
- Two receiving thickeners installed at Luilu to dewater the mixed ore pre-float tails ahead of the new leach facility;
- Two pre-leach CCD thickeners; each train with three units;
- Two leach trains; each train with six tanks for leaching of the mixed ore pre-float tailings at Luilu;
- Two trains of thickeners and CCDs; each train with eight units to recover the PLS ahead of solvent extraction;
- SX/EW plants;
- Lime plant upgrade to match milk of lime preparation capacity with 300 ktpa of copper production
- Cobalt plant upgrade;
- Modifications to the existing leach and CCD facilities to accommodate cobalt hydroxide salt production; and
- Sulphuric acid storage tank farm for the acid used in the new leach facility.

The associated additions to the plant infrastructure under commission are follows:

- Dryers for the cobalt hydroxide salt;

The associated additions to the plant infrastructure under engineering and construction are as follows:

- A 2,200 ton per day sulphuric acid plant to produce acid and sulphur dioxide in accordance with the WOL requirements;
- Sulphuric acid storage tank farm 2 and 3;
- IX plant;
- Installation of Z7, Z8, and Z11 cobalt thickeners;
- Development and refurbishment of the power supply infrastructure;
- Installation of the power factor correction equipment;
- Installation of three new substations; and
- Upgrade of several outdated switchgear boards with new ABB VD4 breakers.

18.2 Mining and KTC Infrastructure Upgrade

The major changes in the mining and KTC infrastructure currently in engineering and construction phase are the following:

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- An upgrade and relocation of the KOV OP dewatering facilities to ensure facilities are not in an area that will be mined, and to ensure a constant dewatering capacity of 10,000 m³/h from the pit bottom sump to the desilting ponds or GH pit;
- Installation of surge tanks equipped with a linear screen on the feed to eliminate coarse material and flow fluctuations from feed to floatation section;
- Installation of a pebble crusher to crush and recycle 20 tph of scats discharged by CM5, CM6 and CM7 utilising trains of new conveyors for the transfer of scats;
- Installation of a static grizzly on crusher B4 feed bin as well as a rock breaker to control the size of ore fed to the crusher;
- Engineering work for the Near West (Luilu) Tailings Storage Facility in order to commence phase 1 construction in 2020;
- Construction of the 1450 lift on the Mupine Tailings Dam during 2019 and 2020
- Installation of power factor correction equipment; and
- Upgrade of several outdated switchgear boards with new ABB VD4 breakers.

18.3 Electrification for Mashamba East OP

18.3.1 Introduction

Figure 122 shows the electrical plan layout for the Mashamba East OP. A new cable and overhead line are being constructed to provide power to the Mashamba East OP with the capacity to power two RH-340 electric shovels and any future dewatering pumps. The line will operate on 15 kilovolt (kV) and will span a distance of 2,500 m with a total overhead conductor requirement of 7,500 m for three sets and a total cable requirement of 3,400 m.

The line will be aluminium ACSR Panther, which is suitable for a maximum of 12 Mega-volt ampere (MVA), with the possibility of short term overloading.

The line will be fed from the R1 switchboard at the Kadi substation. Two containerized substations will be required at the Mashamba East OP. One will be for 6.6 kV and the other will be for 15 kV for the receiving and distribution of power to various end users. Two local minisubs will be required for the local lighting supply and the power requirements for any future dewatering pumps.

18.3.2 Overhead Lines

18.3.2.1 Scope of Work

The following scope of work will be required for the Mashamba East OP electrification:

- Cleaning of 900 m long ducts from the Kadi substation to the B3 substation;
- Replacing of existing 70 mm² copper cable with 185 mm² cable to feed the B3 substation;
- Constructing a trench from the B3 substation to the first pole of the overhead line passing through the warehouse at the new training center area;
- Installation of two new 185 mm² size cables from the existing switchboard at the Kadi substation;
- Installation of 15 m high poles to support the 7,000 m long powerline;
- Installation of three conductor phases and the ground wires;
- Installation of two 630 ampere (amp) disconnects at the first end of the powerline;
- Connecting two 185 mm² cables and installing surge arresters;

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- Installation of one 630 amp disconnect at the end of the powerline;
- Connecting cable and installing the surge arresters at the end of the powerlines;
- Painting and marking of poles as per KCC standards; and
- Performing of open line tests (continuity and insulation tests).

18.3.2.2 Materials Estimate

Table 121 shows estimated required materials for the Mashamba East OP electrification.

Table 121: Material estimate for the Mashamba East OP electrification

Item	Description	Unit	Quantity
01	Surge Arrester 10 kiloamperes (kA) -17.5 kV	pces	6
02	Outdoor disconnect 630 A -17.5 kV (complete unit)	pces	3
03	Poles HEB 200 of 15 metre height	pces	30
04	20 – 30 kV Insulators for overhead line	pces	90
05	Cable 3 x 185 mm ² /copper – XLPE	m	3,400
06	Aluminum power cable ACSR - 261 mm ² (Panther)	m	7,500
07	Steel ground wire 35 mm ²	m	2,500
08	Indoor and outdoor termination Kits for 3 x 185 mm ² /Cu	pces	lot
09	All accessories for overhead line	pces	lot
10	Materials/accessories for earthing	Lot	lot
11	Paint (rust proof + Aluminum) for poles and black for markings	Liters	lot

18.3.3 Excavation

A limited amount of excavation is required to bury the proposed cables from overhead powerlines in the ground. This is on-going work, which is usually done during normal pit operation and covered under the mine operating cost.

18.3.4 Switchgear at KADI Substation

The following would be required at the KADI substation in order to install the switchgear.

- Move the feeder for Reyrolle switchgear from the double bus board to a single bus board;
- Install new protection relays; and
- Install the new metering system.

18.3.5 Mini Substation (400 V)

A new 15 kV to 400 volts (V) 250 kVA mini substation with the following specifications will be required:

- Standalone mini substations and transformers will be required. Each mini substation and transformer will be provided with an earth rod at two diagonally opposite corners. The earth rods will be connected to the mini substation earth bar or equipment earthing points, as the case may be, with 70 mm² PVC insulated copper earth wire. The resistance to earth of the system shall not exceed 1.0 ohm;
- The transformers will be designed for 50 hertz (Hz) operating frequency and will be double wound, oil immersed, natural air cooled;

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- A 17.5 kV, SF6 or vacuum ring main unit comprising of 2 x 630 on load isolators will be installed. Switch disconnectors will be suitable for fault making and load breaking duty;
- Two 100 amp circuit breakers equipped with a self-powered transformer protection relay will be required. The transformer protection will also include neutral earth fault protection. The circuit breaker will be connected to the transformer;
- Switches will be independent manually operated type, and will be possible to lock the same in each position (on/off/earth); and
- 400 V compartment containing:
 - a) 1 X 400 amp three pole incomer air circuit breaker; and
 - b) 4 X 100 amp three pole feeder air circuit breakers, each complete with 30 milliamp earth leakage units.

18.3.6 Mini Substation (550v)

A new 15 kV to 550 v 1000 kVA mini substation with the following specifications will be required:

- Standalone mini substations and transformers will be required. Each miniature substation and transformer will be provided with an earth rod at two diagonally opposite corners. The earth rods will be connected to the mini substation earth bar or equipment earthing points, as the case may be, with 70 mm² PVC insulated copper earth wire. The resistance to earth of the system will not exceed 1.0 ohm;
- The transformers shall be designed for 50 Hz operating frequency and shall be double wound, oil immersed, natural air-cooled;
- 17.5 kV, SF6 or vacuum ring main unit;
- 2 x 630 amp on load isolators (Alternative 1 x on load switch-disconnectors) Switch disconnectors shall be suitable for fault making and load breaking duty;
- 2 x 250 amp circuit breakers equipped with a self-powered transformer protection relay. The transformer protection shall include neutral earth fault protection. The circuit breaker shall be connected to the transformer;
- Switches shall be independent manually operated type, and shall be possible to lock the same in each position (on/off/earth); and
- 550 V compartment containing:
 - a) 1 x 800 amp three pole incomer air circuit breaker; and
 - b) 4 x 250 amp three pole feeder air circuit breakers each complete with 250 milliamp earth leakage units.

18.3.7 Distribution Transformers

Two new 4 MVA 15 kV to 6.6 kV transformers have been included in the cost estimate.

18.3.8 Substation (15 kV)

A containerized MV substation has been included in the cost estimate to receive the incoming 15 kV power and distribute it to various end users.

18.3.9 Substation (6.6 kV)

This containerized substation will be used to distribute power to the electric shovels.

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18.3.10 Cost Estimate

Table 122 shows cost estimate required for the Mashamba East OP electrification.

Table 122: Cost estimate for the Mashamba East OP electrification

Description	Cost (USD) '000'
Overhead line	840
Cables and termination	510
Excavation	25
Switchgear at KADI substation	100
400 V 250 kVA mini substation	60
550 V 1,000 kVA mini substation	80
550 V pump starters	120
Lighting	50
Distribution transformers	120
15 kV substation	540
6.6 kV substation	490
Fencing	20
Civil Work	50
Contingency (15%)	450
Total	3,445

Figure 122: Mashamba East OP reticulation layout



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19.0 Market Studies and Contracts

19.1 Markets

19.1.1 Copper

Copper is a major industrial metal (ranking third after iron and aluminium by consumption) because it is highly conductive (electrically and thermally), highly ductile and malleable, and resistant to corrosion. Electrical applications of copper include power transmission and generation; building wiring; motors; transformers; telecommunications; electronics and electronic products; and renewable energy production systems. Copper and brass (an alloy of copper) are the primary metal used in plumbing pipes, taps, valves and fittings. Further applications of copper include decorative features; roofing; marine applications; heat exchangers; and in alloys used for gears, bearings and turbine blades.

Global copper mine production was 20.6 Mt in 2018, with Chile the largest producer accounting for 28% of world copper mine production at 5.8 Mt of copper. In 2018, China accounted for almost 45% of world copper smelter production, followed by Japan, Chile and the Russian Federation with 8%, 6% and 5% share respectively. Global refined copper production increased to 24.1 Mt in 2018, which included 4 Mt of secondary refined production. Global refined copper usage reached 24.5 Mt in 2018 as shown in Table 123.

Table 123: Global refined copper market balance

Production	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Mine production (Mt)	15.9	16	16	16.7	18.2	18.4	19.1	20.4	20.1	20.6
Refined production (Mt)	18.2	19	19.6	20.2	21.1	22.5	22.8	23.4	23.5	24.1
Consumption	17.9	19.1	19.7	20.5	21.4	22.9	23.1	23.5	23.7	24.5
Copper price (USD/t)	5,199	7,558	8,826	7,953	7,349	6,830	5,493	4,870	6,200	6,645

(Source: ICSG - Monthly Press Release 23 September 2019 and LME)

As shown above in Table 123 and below in Figure 123, the copper price has demonstrated significant volatility over the course of the last ten years. After reaching a peak of \$10,160/t in February 2011, the price trended negative until the end of 2016, at which point it steadily increased to over \$7,300/t in 2018. The London Metal Exchange (LME) 3 month price for copper over the first nine months of 2019 averaged USD 6,056/t, as shown in Figure 123.



Figure 123: The LME copper price between January 2009 and September 2019 (Source: LME)

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Given the volatility of the copper price, a single copper price forecast of \$6,500/t is used in the economic evaluation, as shown in Table 124. The long-term real price assumption is seen to be appropriate, when considering the upper and lower bounds of the nominal copper price over the last ten years, the average nominal copper price over the last two years of \$6,400/t and consensus pricing in August 2019, represented by a basket of global banks ranging from \$6,200 to 7,500/t, with a median long term real price estimate of \$6,600/t.

The price disclosures in Table 124, reflect the historic pricing guidelines in line with the guidance provided for compiling this technical report.

Table 124: LME Copper price forecasts

Copper Price (USD/tonne)	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
KML – Macroeconomic Assumptions	6,047	6,500	6,500	6,500	6,500	6,500	6,500	6,500	6,500	6,500

19.1.2 Cobalt

Cobalt has many commercial, industrial and military applications. The leading use of cobalt is for rechargeable battery electrodes. The temperature stability and heat- and corrosion-resistance of cobalt-based superalloys makes them suitable for use in turbine blades for jet turbines and gas turbine engines. Other uses of cobalt include vehicle airbags; catalysts for the petroleum and chemical industries; cemented carbides and diamond cutting and abrasion tools; drying agents for paints, varnishes, and inks; dyes and pigments; ground coats for porcelain enamels; high-speed steels; magnetic recording media; magnets; and steel-belted radial tyres.

Global refined production of cobalt was 116,937 Mt in 2017, with half of the production coming from the DRC, the largest producer. Table 125 shows the historical global refined cobalt market production.

Table 125: Global refined cobalt market balance

Production	2009	2010	2011	2012	2013	2014	2015	2016	2017
Total world refined production (t)	62,228	79,254	82,247	77,189	84,904	91,754	98,113	93,889	116,937
Cobalt price (USD/t avg)	35,031	41,337	30,005	28,783	27,029	30,866	29,233	26,433	61,288
Cobalt price (USD/lb avg)	15.89	18.75	13.61	13.06	12.26	14.00	13.26	11.99	27.80

(Source Cobalt News April 2018, Published by the Cobalt Development Institute)

The cobalt price reached a record of USD 48.63/lb in March 2008, falling in line with other commodities to a 5-year low of USD 11/lb in December 2008. The price subsequently recovered and was relatively stable until a rally commenced in January 2017 which culminated in a peak of USD 43.45/lb in April 2018. A sharp price decline resulted in a low approaching \$12/lb in July 2019, while prices recovered thereafter and surpassed USD 14.69/lb in August 2019. The 3-year average on this date is USD 16.69/lb. Figure 124 shows cobalt price from January 2016 to December 2018.

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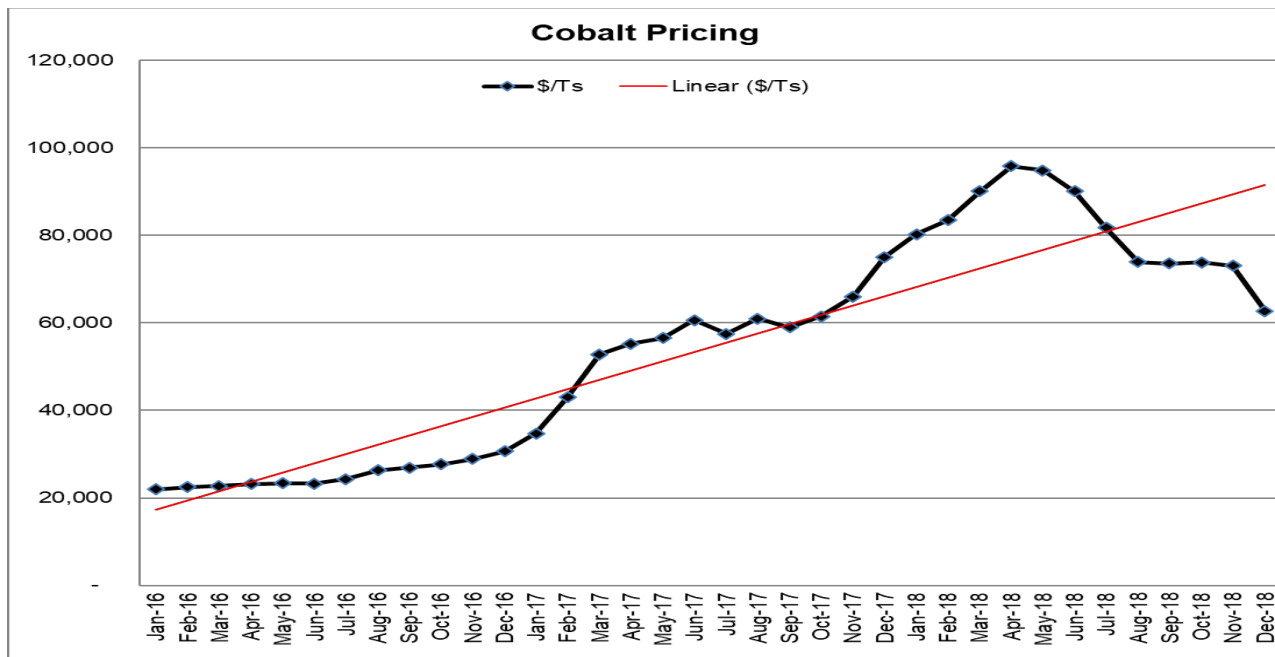


Figure 124: The MB cobalt price between January 2016 and December 2018 (Source: Fastmarkets MB)

The cobalt market is smaller and less liquid than other bulk commodities and base metals, and has experienced high price volatility. The emerging electric vehicle revolution is a significant potential source of cobalt demand, growing at a strong rate albeit off a low base. Meanwhile supply side additions and suspensions impact the overall market balance. The combination of these factors, together with cobalt metal market fundamentals which are separate to cobalt intermediate (hydroxide) fundamentals, has led to volatility in the cobalt price and a wider range of analyst expectations compared to base metals. The long-term real terms assumption of \$27/lb is within a consensus range (in August 2019) representing 10 global banks (e.g. Barclays, Deutsche Bank, Macquarie) ranging \$13.50-\$30/lb, with the average being \$21.90/lb

The price disclosures in Table 126, reflect the historic pricing guidelines in line with the guidance provided for compiling this technical report.

Table 126: Cobalt price forecast

Cobalt price (USD/lb)	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
KML – Macroeconomic Assumptions	12.7	20.0	23.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0

19.2 Contracts

KML has entered into offtake agreements with Glencore International AG, pursuant to which Glencore will buy 100% of the quantities of copper and cobalt produced by KCC over the life of the Project. The offtake agreements are negotiated at arm's length at standard market terms and approved by the Company's independent Corporate Governance committee.

20.0 Environmental Studies, Permitting and Social Community Impact

These sections highlight key points with regards to the environmental and social aspects of KCC operations.

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20.1 Environmental Studies and Potential Impacts

This section summarises the outcomes of the environmental assessment and potential environmental aspects that could materially impact and/or impair the issuer's ability to extract the Mineral Resources or Mineral Reserves.

20.1.1 Summary of Environmental Studies

KCC has commissioned the following Environmental Impact Studies since 2011 as part of various permitting applications, which cover current and future mining operations for the proposed LOM:

- The original Environmental Impact Study (EIS) was undertaken by DRC EMEC Green and approved by the Direction de la Protection de l'Environnement Minier (DPEM) in 2011;
- The 2011 EIS was updated by SRK Consulting in 2014 and subsequently approved by the DPEM in August 2014;
- The 2014 EIS was updated by SRK Consulting in 2017 to assess changes in KCC operations, namely the WOL project. It was submitted to the DPEM for review in 2017 and resubmitted in April 2018 to address additional information requested by the authorities. DPEM approved this EIS in June 2018; and
- The April 2018 EIS was updated by SRK consulting to assess additional changes in KCC operations, namely the sulphuric acid plant and ion exchange plant. It was submitted to the DPEM in December 2018 and resubmitted in May 2019 to address additional information requested by the authorities. DPEM approved this EIS in August 2019.

The latest version of the EIS (2019) is aligned with the requirements of the 2018 Mining Code and Mining Regulation (2018) and it includes a description of the Project, legal framework, environmental and social baseline, assessment of the environmental and social impacts, proposed mitigation and management measures within an environmental management plan, a sustainable development plan, and a rehabilitation and closure plan. The EIS covers all active mining permits, namely PE525, PE11602 and PE4961.

The baseline environmental conditions, as described in the EIS, are summarised as follows (SRK, 2019):

KCC operations lie north of the plateau of Manika, in a zone of rolling hills known as the Kibarian belt. The plateau of Manika is made up of sand dunes of the Kalahari formation and is at an altitude of between 1 400 to 1 500 mamsl and covers a surface of about 150 km². The Luilu River to the west and the Musonoi River to the East, both originate in the high lying areas of the Manika plateau, south of Kolwezi. The topography within the mine area has been significantly altered by previous mining operations and the site is characterised by industrial buildings, workshops, processing plants, waste rock dumps, tailings storage facilities and associated mine infrastructure.

The dominant soils in and around the mine site are those that are commonly found in Katanga Province, consisting mainly of ferrasols, plinthols and gleysols, which are not suitable for crop production. Land use in and around the mine site comprises mining and mineral processing/industrial use, housing, subsistence agriculture of cassava fields and corn, abandoned farmland and original and secondary vegetation.

The average annual rainfall at Kolwezi Airport is 1 323 mm. The majority of rainfall is received from November to March. There is on average low rainfall from May to August. The average temperatures range between 17 and 23°C with higher temperatures experienced between October and December and lower temperatures occurring between June and July. The prevailing wind directions are relatively consistent throughout the year, and are characterised as being from north-west and east to south-east. Air quality varies with the season due to the variations in temperature, humidity and rainfall. During the dry season dust that is suspended in the air layers close to the ground may cause frequent, but not significant, hazes which reduces visibility. However, during the wet season, rainfall removes dust from the atmosphere and improves the air quality. The dry season is characteristic of very dry and dusty conditions within the mining area and Kolwezi Town.

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The Luilu and Musonoi rivers flow through the mine site and meet downstream of the mine site. The surface water is used for domestic use (water and food supply, washing, etc.) and agricultural users use streams for both irrigation and the watering of small livestock. However, as a result of previous mining activities, rivers have undergone serious environmental degradation. Almost all the villages around the mine site rely on groundwater for their daily domestic use. Groundwater usage in the surrounding villages is from water supply boreholes and shallow hand dug wells. KCC undertakes surface and groundwater monitoring in line with DRC regulatory requirements and good international industry practice..

Kolwezi and surrounds are characterised by mining and related industries interspersed with towns and villages that are closely associated with these industries, e.g. Luilu Settlement with Luilu Plant. Noise measurements taken from settlement areas indicate that the ambient noise intensity approaches and sometimes exceeds the limit of the DRC guidelines (55 dBA during day and 50 dBA at night). The primary activities producing noise are drilling and blasting dump trucks loading, wastes and ores transport, ore crushing and waste handling at the KOV and Mashamba East operations. The noise emissions from these industries are therefore generally accepted as part of the normal ambient noise climate as long as they are not particularly disturbing.

The mine site falls within the Central Zambian Miombo Woodland ecoregion in the southern part of the country, consisting mainly of flat plains or rolling hills covered by woodland vegetation that contain seasonal wetlands, or dambos, occupying shallow depressions. Due to anthropogenic activities this forest has undergone significant deterioration and is almost non-existent in the mining area. The flora is characterised by: shrub thickets including species such as *Uapaca robynsii*; disturbed savannah habitat supporting ruderal (pioneer) species; and vegetation on termite mounds. As a result of habitat disturbance from historic mining activity, slash and burn agriculture and local hunting practices, the mine site and surrounding area is characterised by low faunal diversity and richness.

In addition to the recent environmental impact study (2019), as per the requirements of the 2018 Mining Code, KCC also supports the following actions:

- Annual environmental reports are prepared by KCC and submitted to the DRC regulatory authorities;
- Periodic third party environmental audits are undertaken at KCC; and
- Regular environmental inspections are undertaken at KCC by the DRC regulatory authorities.

20.1.2 Environmental Issues

The natural environment in and around the mine site has been significantly impacted and transformed by human activities such as land clearing, settlement and various mining operations, which have taken place over many decades. The latest environmental impact study (SRK, 2019) identified the following potential environmental and social impacts due to KCC mining operations:

- Impact P-1: Potential loss of soil resources due to changes in soil characteristics, productivity and a change in land use.
Significance after mitigation = Low
- Impact P-2: Potential contamination of soils which affects land capability.
Significance after mitigation = Low
- Impact P-3: Potential increase in ambient dust concentrations.
Significance after mitigation = Low
- Impact P-4: Potential increase in gaseous emissions.
Significance after mitigation = Medium
- Impact P-5: Potential reduction in groundwater quantity.

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Significance after mitigation = Low

- Impact P-6: Potential change in surface water hydrology.

Significance after mitigation = Low

- Impact P-7: Potential contamination of surface water and groundwater.

Significance after mitigation = Medium

- Impact B-1: Potential modification and loss of indigenous fauna and flora.

Significance after mitigation = Low

SRK (2019) concluded that the majority of the identified potential negative impacts arising from KCC mining operations can be managed to remain within acceptable environmental limits so long as the measures set out in the Environmental Management Plan for the Project (EMP) are implemented and KCC continues to strengthen its environmental management system. Positive impacts can be enhanced by implementing community development programs set out in the Sustainable Development Plan.

KCC is committed to implementing the mitigation and management measures within the EMP and undertakes environmental monitoring in accordance with DRC regulatory requirements. KCC is developing an improved Environmental Management System which will be implemented as part of KCC's on-going efforts of continuous environmental improvement. The management system will contain plans and procedures to help manage environmental aspects and impacts, and help ensure legal compliance.

20.2 Waste, Tailings, Monitoring and Water Management

This Section provides the requirements and plans for waste and tailings management, site monitoring, and water management both during operations and post mine closure.

20.2.1 Waste Management

The question of the application of the 2018 Mining Code to KCC also impacts the approval process of some of the future waste dumps to be developed for the continuity of its operations. Article 279 of the 2018 Mining Code provides for an approval procedure for certain installations such as tailings and dumps when located within a certain distance from residential dwellings. KCC is currently considering several options to increase its tailings and dumping capacity for the life of mine. While KCC chose to comply voluntarily with this requirement, KCC has clarified that it was under no obligation to do so and reserved entirely its rights pursuant to the Stability Guarantee. See also Section 4.6.1.1.

20.2.2 Tailings Management and Disposal

20.2.2.1 Tailings Sources

Tailings from the KCC operations are produced by two plants, namely KTC and the Luilu plant.

20.2.2.2 Historical Disposal

Tailings were historically disposed of directly into Luilu River from various mining companies located in Kolwezi.

Prior to the authorisation by DPEM for KCC to utilize the disused Mupine Pit for tailings disposal, tailings from the Luilu plant were pumped through pipelines to KTC to be discharged into the KITD.

The KITD capacity was increased in 2010 to meet the increasing demand for tailings storage capacity. The crest/contour of the existing wall was raised by 2 m to provide additional capacity. This increased the life of this tailings dam by one year. No tailings from KTC have been deposited into the KITD since May 2013 and it is currently being re-mined via a hydromining methodology. The target end date for mining is 2021.

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20.2.2.3 Current Disposal

Currently, tailings from Luilu and KTC are discharged into the Mupine Pit. The Mupine Pit is located about 3.5 km northeast of KTC and 2 km southeast of Luilu plant. The final tailings comprise a mixture of the tailings from the oxide and sulphide processing circuits. A portion of tailings (20%) is utilised in KTO for backfilling the UG workings, with the remainder deposited into the Mupine Pit.

Currently, the Luilu plant treats both oxide and sulphide concentrates to produce copper and cobalt cathodes for export. Aluminium and iron are precipitated to obviate process interferences.

Tailings from Luilu are neutralised before being discharged to Mupine TSF.

According to the EIS (SRK, 2019), from the preliminary work (toxicity characterizing leaching procedure (TCLP)) and acid base accounting (ABA), the tailings produced by KTC are classified as low risk in terms of the 2018 Mining Code. Whereas the tailings produced by Luilu plant are classified as high risk due to their acid generating potential and concentrations of copper and cobalt. The mixture of the two tailings stream transforms into a composite flux of low risk due to the KTC tailings' high neutralization potential.

20.2.2.4 Tailings Deficiency and Future Disposal

KCC has been in discussions with third parties for land use rights for future tailings facilities to meet the life of mine tailings deposition requirements. KCC is expecting to resolve and build the tailings facilities for life of mine in near future. If KCC and third parties (including GCM) are not able to reach an agreement on the tailings deficiency, this may trigger adverse consequences for KCC's operations, up to the shutdown of the operations by the end of Q1 2021. Three options are available for future tailings disposal:

- Near West Tailings Storage Facility – this is the preferred location situated immediately south of the Luilu plant. Surface rights negotiations with GCM are underway and DPEM approval has been granted;
- Mupine OP Tailings Storage Facility – feasibility studies have been completed for Mupine OP to contain additional tailings; the first 2 stages of earthworks has been completed. Mupine TSF has a final capacity of 37 Mt; and
- Far West Tailings Facility - the location is a greenfield site approximately 4 km southwest of the Luilu plant and 5 km northwest of KTC. The site covers a maximum surface area of about 420 ha. The envisaged maximum capacity is 83 Mm³ or 99 Mt at an assumed in-situ dry density of 1.2t/m³. This facility has not yet been subject to an environmental impact study.

20.2.3 Site Environmental Monitoring

Environmental monitoring at KCC includes the following.

Surface and groundwater monitoring:

Water samples are analysed for pH, Total Suspended Solids and metals listed in the 2018 Mining Code. Sampling is undertaken as follows:

- Three times a week at discharge points;
- Weekly in drainage channels and rivers; and
- Monthly in groundwater monitoring boreholes.

Air and noise monitoring:

- Monthly dust fallout monitoring through dust buckets;
- Continuous monitoring of SO₂, NO₂, and Particulate Matter; and
- Noise monitoring in operational areas and closest affected communities.

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Post-closure monitoring is important for KCC to determine whether the mitigation and rehabilitation measures are effective. Monitoring for air, water and soil quality will be undertaken during the LOM and will be compared to DRC Standards and relevant Guidelines to inform remediation strategies during the LOM and post closure, if required. Should monitoring results indicate that post-closure monitoring is required, relevant post-closure monitoring plan(s) will be developed in consultation with DP EM and other relevant authorities and stakeholders (SRK, 2019).

20.2.4 Water Management

20.2.4.1 Surface Water

According to the EIS (SRK, 2019) the Luilu and Musonoi rivers that form the western and eastern boundaries of the KCC mining area have been the recipients of mining-related discharges for several decades. Both rivers are heavily silted with tailings from historical mining and the water quality is degraded to the extent that in-stream and riparian habitats cannot sustain diverse aquatic ecosystems. In addition, due to its poor quality the river water is also not fit for use by the communities immediately around the mining area and most other purposes.

The dewatering water from KOV and T17 OPs is used as makeup water in the ore processing circuits. Surplus water is discharged to the Musonoi River. Routine monitoring is undertaken at all discharge points. Future plans exist to use GH pit (previously mined by GCM) as a settling basin to reduce suspended solids prior to discharge of dewatering water. Surplus water is also discharged from Mupine TSF provided that the necessary water quality conditions are met.

The measures implemented by KCC to address potential adverse water quality effects are as follows:

- Treatment by means of neutralisation of the tailings from Luilu prior to being sent to the TSF;
- Separation of the potentially contaminated areas from non-contaminated by means of diversion measures;
- In the event of an environmental incident occurring that could have a negative impact on water quality, the incident will be reported and addressed promptly;
- The GH pit will be used as a settling pond;
- Normal processing plant operations are undertaken via closed circuit;
- TSF discharge is checked daily, various settling and emergency ponds exist downstream of KCC operations and more are planned in the future;
- KITD is being mined out (reducing associated risks to water quality);
- Geochemical analysis was undertaken to assess the potential for acid generation from waste rock at KCC operations. Recent analysis determined that Mashamba East waste rock is unlikely to generate acidity (SRK, 2017); and
- Implementation of waste management procedures for example, waste items to be stored away from watercourses and covered during the rainy season.

20.3 Project Permitting

The section below presents the status of existing permits and any known requirements to post performance or reclamation bonds.

20.3.1 Project Permits

As per the applicable DRC mining law and regulations, mining licences are mandatory before carrying out any mine activity. KCC was issued with the following mining permits: PE 525, PE 11601, PE 11602, PE 4960, PE 4961 and PE 4963.

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As part of their legal obligations KCC was also issued with the following approvals from the DPEM:

- Updated EIS and EMP prepared by SRK (2014) covering: PE 525, PE 11601, PE 11602, PE 4960, PE 4961, and PE 4963 as per article 455 of the DRC Mining Regulations. This was approved in August 2014;
- Updated EIS and EMP prepared by SRK (2017) and resubmitted in April 2018 covering: PE525, PE11602, and PE4961 This was approved in June 2018; and
- Updated EIS and EMP prepared by SRK (2018) and resubmitted in May 2019 covering: PE525, PE11602, and PE4961): This was approved by DPEM in August 2019 and CAMI in September 2019.

20.3.2 Financial Guarantee

In terms of the financial guarantee required by DRC Law, the EIS (SRK, 2019) estimated the closure costs. The total closure costs for all mining license areas and surface rights, including the latest infrastructure upgrades, were \$132,005,375.

In terms of the annual financial guarantee payments as required by the 2018 Mining Code, KCC has undertaken the following and is up to date with the required annual payments.

PE 4960, 4961 and 4963 surety bonds issued to the DPEM:

- USD 76,250 issued on October 7, 2011;
- USD 238,260 issued on August 6, 2012;
- USD 390,750 issued on August 26, 2013;
- USD 552,770 issued on August 18, 2014; and
- USD 705,257 issued on August 28, 2015.

PE 11601, 11602 and 525 surety bonds issued to the DPEM:

- USD 121,760 issued on April 15, 2014; and
- USD 380,495 issued on 28 August 2015.

Surety bonds issued to the DPEM for combined permits PE525, PE11602, PE4960, PE4961 and PE4963:

- USD 1,491,287 issued on 28 August 2016;
- USD 1,902,511 issued on 28 August 2017; and
- USD 6,961,734 issued on 13 November 2018

20.4 Social and Community Related Requirements and Plans

This section presents potential social or community related requirements and plans for the project and the status of any negotiations and/or agreements with local communities.

The following contributions to local health, safety and community are noted:

- All KCC employees and contractors continue to receive introductory and refresher training for the SafeWork Program which incorporates 12 fatal hazard protocols and associated lifesaving behaviours. The rolling 12 month Lost Time Injury Frequency Rate (LTIFR, based on 1 M man hours) was 0.17 in December 2018;
- KCC collaborates with local health zones to assess the overall health conditions and define a common strategy to address the major community health risks and concerns. KCC is active in the rehabilitation of existing health facilities, training of medical staff of reference hospitals and local health centres,

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awareness initiatives in communities, periodic drugs donation and assistance to vaccination campaigns in urban and rural areas;

- In addition, KCC works in synergy with the local representatives of PNMLS (Plan National Multisectoriel de Lutte au SIDA), the national program for the fight against HIV, to raise awareness in communities through sensitization initiatives and peer educators; and
- KCC social investments are aligned to the national 5 pillars (Education, health, infrastructure, agriculture and governance) program and are implemented through participative stakeholder engagement and open community dialogue.

The main domains of intervention and key projects are infrastructure, agriculture, and education.

20.4.1 Environmental Management Plan

As part of the EIS update an Environmental Management Plan (EMP) has been developed (SRK, 2019). The EMP contains the environmental, social and safety management and monitoring commitments that KCC will implement to manage the negative impacts and enhance the positive impacts identified in the EIS and associated with KCC operational activities.

The EMP is intended as a living document that will be regularly reviewed and updated by KCC in response to any changes to the operations, changes in the KCC organisational structure and changes in the relevant DRC legislation, as well as other guidelines and practices that KCC subscribes to.

The objectives of the EMP are to:

- Ensure that the Project operates in compliance with the DRC legal requirements and international good practice pertaining to environmental, social and safety management and rehabilitation and closure;
- Ensure that the Project operates in accordance with KCC's environmental, social and safety procedures;
- Ensure that the identified potential impacts are managed; and
- Provide a reference against which future monitoring and auditing can be undertaken

KCC is committed to continual improvement of its operations.

The approach considered for developing management measures includes:

- Meeting DRC legal requirements, international good practice and KCC environmental, social and safety procedures;
- Considering principles of environmental sustainability;
- Favouring avoidance and prevention over minimization and mitigation in dealing with negative impacts (where practical to do so);
- Where avoidance is not possible, reducing, restoring, or offsetting the impact;
- Enhancing positive impacts; and
- Being accountable and accepting responsibility for those impacts.

20.4.2 Social Requirements

As per the 2018 Mining Code, the proponent /developer must present a sustainable development plan (SDP) as part of its EIS. In this context, KCC reiterated that it and its shareholders reserve entirely the right to the continuing application of the provisions of the 2002 Mining Code for a period of 10 years pursuant to the stability guarantee provided at Article 276 of the 2002 Mining Code.

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The SDP presented in the EIS (SRK, 2019) and approved by DPEM aims to improve the economic, cultural and social well-being of local communities affected by the Project during operations and post closure. This will be applied for the following:

- Enhancement of local capacities and human capital;
- Development of income generating activities (including development of 65 cooperatives in terms of general services, agriculture, fish farming and livestock);
- Improvement of civil infrastructure such as schools, hospitals, etc.;
- Strengthening capacities of vulnerable groups;
- Improving governance that includes support for local development and participation in community development initiatives; and
- Improving environmental sustainability.

From 2014 to 2018, KCC has spent USD 28 M on various social, agricultural, educational and infrastructural programs in the DRC.

Inspections by DPEM are conducted regularly to confirm KCC's compliance with SDP and EMP.

20.4.3 Infrastructure

The following contributions to local infrastructure are noted:

- Local initiatives have included the rehabilitation of the airport of Kolwezi, the rehabilitation and maintenance of roads in and around Kolwezi, participation to the construction of Lualaba bridge;
- In collaboration with Regie de distribution d'eau (REGIDESO) and SNEL, the national agencies for water and electricity, KCC provides financial and technical assistance for the improvement of existing infrastructure and facilitates communities' access to water and electricity supply through the construction of new facilities and distribution systems;
- KCC has supported SNEL with the donation of cables and transformers and the punctual rehabilitation of undamaged facilities. With REGIDESO, KCC collaborated on the project for the rehabilitation of the filtering plant and Luilu project for drilling and equipment of two water wells in the community of Luilu; and
- At national level, in partnership with the government, SNEL and other mining companies, KCC is participating in the INGA project to increase electricity production and supply.

20.4.4 Education

The contributions to local education are noted:

- KCC is committed in supporting the improvement of education at all levels and ensuring the development of a skilled workforce;
- In collaboration with the provincial government, KCC engaged in the construction of a new building for the University of Kolwezi. The facility features an auditorium, 8 classrooms with a capacity of 150 students each, administrative offices, toilets and is fully supplied with electricity and water;
- KCC provides assistance to local technical schools. KCC assisted ISTA (Institute Supérieur des Techniques Appliquées) with the donation of a library, computers, and a fully-equipped chemistry laboratory, the rehabilitation of the auditorium and the purchase of a nearby property for future extensions. KCC also supported the full rehabilitation of ITIMA (Institute of Industrial Techniques of Manika), including the auditorium, three workshops, toilet and fencing wall; and

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- In regard to primary education, KCC builds and rehabilitates education facilities and provides teachers with various educational materials such as books.

KCC also organizes summer camps for children, living in communities affected by artisanal mining. KCC provides educational and recreational material and also one meal per day for each child. Approximately this program concerns 5,400 children. Via this initiative, KCC has the opportunity to raise awareness in children and their mothers on the risks and dangers linked to artisanal mining and possible alternatives.

20.4.5 Agriculture

The contributions to local agriculture are noted:

- Through KCC's community development program, it assists local cooperatives with the aim of improving their skills and long-term productivity and decreasing the dependence on the mining sector;
- KCC cooperatives are involved in the cultivation of vegetables, livestock, beekeeping and sewing;
- KCC provides different types of support: all cooperatives take part in continuous training schemes in order to reinforce their organizational structure and decision-making process, improve the management of their financial means and deepen their knowledge of agriculture. KCC assists with inputs donation, improvement of irrigation systems and the shift from purely-subsistence rural livelihoods to transformation, conservation and commercialization of the products; and
- KCC also supports the development of small and medium local enterprises via in-kind donations and professional training.

20.4.6 Additional Social Plans and Commitments

Over and above the regulatory requirements in terms of the SDP, KCC has provided additional funding for the period 2014 to 2019 for various community investments. Additional social reporting undertaken by KCC includes:

- Input to the Glencore sustainability report (www.glencore.com/sustainability);
- Development of Social Key Performance Indicators (KPIs) for its corporate social responsibility initiatives;
- Community Engagement Plan – which outlines KCCs Stakeholder Engagement process inclusive of grievance recording and procedures as well as local media coverage; and
- Socio-economic Baseline surveys and reporting to enable KCC to monitor and track its Social KPIs.

20.4.7 Social/Community Issues

Based on the risk assessment approach outlined in Section 20.1, there are no social or community issues that have a moderate or high risk of material impact on KCCs ability to extract the Mineral Reserves or Mineral Resources.

Issues that may have a low risk of impacting KCC operations (e.g. unlikely to affect Mineral Resource or Mineral Reserve extraction and would only result in disruption for minor or short periods of time) are:

- Artisanal Miners – There were a number of illegal artisanal miners observed at KCC as well as adjacent mining concessions that are not under the control of KCC. The vast area, large number and organised nature of the artisanal groups poses a challenge;
- Limited economic and employment opportunities for young people; high dependency on the mining sector exposes the economic context to several uncertainties and possible shocks linked to the fluctuations of the commodity market; and
- Migrations of job seekers from different provinces; potential disruption of community structures. This phenomenon is mainly due to attractiveness of the mining sector and the concentration of several

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artisanal and industrial operations in the area. Through its community development program, KCC promotes activities and projects for economic diversification and professional skills development for small-businesses (restaurants, welding, carpentry, etc.)

20.5 Mine Closure

A discussion of mine closure (remediation and reclamation) requirements and costs is contained in Section 24.2.

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21.0 Capital and Operating Costs

21.1 Capital Cost Estimates

Capital expenditure is required by KCC to develop new and existing mining areas; acquire and replace mining equipment; develop new and repair existing processing facilities; develop and repair general infrastructure, buildings, facilities, ponds and tailings facilities; and expenditure of a general nature to expand and maintain the operations of KCC. Table 127 shows the nature of the capital expenditure required for each operational area.

Table 127: Nature of capital expenditure by operational area

Operational Area	Nature of Capital Expenditure Requirement
KTO UG	Purchase of UG mining equipment to meet LOM plans and maintain production targets. Development costs to access mining areas. Ventilation and dewatering infrastructure. A major ventilation raise is required near Etang North. Primary development is regarded as capital expenditure and secondary development as operating costs.
KOV OP	Purchase of mining equipment required to meet LOM plans, waste stripping to access the orebody and dewatering. Pre-stripping on all OPs is treated as capital expenditure.
T17 UG	The initial development of a portal from the surface, a ventilation raise and dewatering infrastructure. Thereafter, purchase of mining equipment required to meet LOM plans, development to access mining areas, ventilation and dewatering infrastructure.
KITD	The current infrastructure is sufficient to support the requirements of the KITD production
Mashamba East OP	Purchase of mining equipment required to meet LOM plans.
Processing plants	Development of a whole ore leach process from 2015 – 2020, consisting of a leach CCD circuit, SX and EW plant upgrades and a cobalt plant upgrade. A sulphuric acid and SO ₂ plant with a storage facility are also forecasted to be built in 2019 – 2020.
Effluent ponds and tailings	Development of the ponds and tailings facilities, and raising the earth containment embankment. Potential purchase of additional land of surface rights necessary for the construction of appropriate tailings storage facilities in line with the life of mine of KCC.
Power supply	Development and refurbishment of power supply infrastructure.
Environmental and social	Tarring of roads (to reduce dust and road safety hazards); dust monitoring equipment; equipment for sulphur dioxide emission reductions/monitoring; surface water management (containment and management); general and hazardous waste management (trenches and buildings); ad hoc equipment for ground water; water settlement facilities for suspended solids radiation monitoring and survey equipment; and emergency response equipment and vehicles. As part of corporate social responsibility programs, investment will be required in DRC schools, medical facilities, and local development initiatives such as farming programs.
General	Unallocated infrastructure of a general nature required to sustain the operations of KCC. This includes stay in business capital for maintenance, engineering, finance, supply chain, IT and general management.

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Table 128 presents the expected capital expenditure annually from 2019 – 20221, the total from 2023 – 2034, and the total over the Mineral Reserve LOM. These costs represent the current best estimate based upon the past experience from actual costs incurred, tenders received and current tax legislation.

Table 128: Capital expenditure by year

Operational Capital (USD million)	2 019	2 020	2 021	2 022	2023-2034	LOM Total
Mining						
KOV open pit	25.75	83.28	34.83	35.36	368.64	547.86
KTO and T17 underground	3.42	18.34	12.78	4.19	20.85	59.57
KTO and T17 underground primary development	0.35	3.05	18.40	19.93	106.56	148.29
Mine Technical Services	1.80	2.89	-	-	-	4.69
Total	31.32	107.56	66.01	59.48	496.05	760.42
Processing Plants						
KTC	2.69	20.02	2.18	-	-	24.89
Luilu	18.69	44.82	20.01	26.30	20.00	129.82
Processing Technical Services	0.25	3.37	0.26	0.10	-	3.98
Total	21.63	68.21	22.45	26.40	20.00	158.69
Service Departments						
Engineering	2.35	13.17	-	-	-	15.52
Environment	0.17	2.20	-	-	-	2.37
HR	1.67	4.49	-	-	-	6.16
HSE	0.22	1.11	-	-	-	1.33
IT	0.63	2.76	0.86	0.85	-	5.09
Medical	0.09	2.29	0.71	0.40	-	3.49
Other	-	5.00	-	-	-	5.00
Closure costs					132.01	132.01
SCM	0.69	2.90	0.88	2.06	-	6.53
Security	2.17	4.95	-	-	-	7.12
Total	7.99	38.87	2.45	3.31	132.01	184.62
Total Operational Capital	60.94	214.63	90.91	89.19	648.06	1 103.73
Expansion Projects						
Mining						
Open pit deferred stripping	104.00	-	-	9.01	1 015.45	1 128.46
Other	-	44.00	64.99	64.91	453.74	627.64
Total	104.00	44.00	64.99	73.92	1 469.19	1 756.10

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Operational Capital (USD million)	2 019	2 020	2 021	2 022	2023-2034	LOM Total
Processing Plants						
Whole ore leach project	10.30	4.95	-	-	-	15.25
Tailings facilities	9.12	37.50	44.62	36.02	-	127.26
IX Plant	11.51	-	30.69	-	-	42.20
Cobalt driers / EW	13.01	0.99	-	-	-	14.00
Acid Plant	204.51	57.52	-	-	-	262.03
Other	4.08	0.42	34.08	1.00	-	39.58
Total	252.54	101.37	109.39	37.02	-	500.32
Total Expansion Projects	356.54	145.37	174.38	110.94	1 469.19	2 256.42
Total Capital, Operational and Projects	417.48	360.00	265.29	200.13	2 117.25	3 360.14

21.2 Operating Cost Estimates

The operating cost estimates related to the project are shown in Table 129.

Table 129: Operating costs by area

Area	Description/Basis	Cost Applied
KTO UG	Based on current and forecast costs as an owner operation.	Weighted average cost over LOM: USD 59.80/t ore mined.
KOV OP	Based on current and forecast costs as an owner operation.	Weighted average cost over LOM: USD 2.92/t mined excluding sustaining capital.
T17 UG	Based on estimated costs as an owner operation, using KTO as a benchmark.	Weighted average cost over LOM: USD 59.80/t ore mined.
Mashamba East OP	Based on estimated costs as an owner operation, using KOV as a benchmark.	Weighted average cost over LOM: USD 2.92t mined excluding sustaining capital.
KTC and Luilu	Plant costs for reagents, consumables and electricity.	Sulphide Ore combined processing cost USD 71.71/t mined excluding sustaining capital. Oxide Ore combined processing cost USD 70.36/t mined excluding sustaining capital.
Mining Technical Services	Mining technical services costs.	Included in the OP and UG costs per tonne
Metallurgy Technical Services	Metallurgy technical services costs.	USD 27.97/t finished Cu

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Area	Description/Basis	Cost Applied
Engineering	Maintenance/construction and other centralised costs.	USD 172.5/t finished Cu
Administration	Administration costs not allocated.	USD 599.36/t finished Cu

The actual cost of major operating items is detailed on an annual basis in Table 130 and Figure 125.

Table 130: Major operational expenditure

Operating Costs (USD million)	2019	2020	2021	2022	2023 - 2034	LOM Total
Mining	199.81	264.51	230.68	237.57	3,858.03	4,790.77
KTC and Luilu	688.12	720.44	727.47	688.37	6,858.48	9,682.89
Technical Services	23.07	39.04	42.64	42.64	497.99	645.38
Engineering	77.00	59.34	50.36	50.95	609.29	846.94
General and Administration	268.00	146.01	152.34	149.09	2,263.59	2,979.02

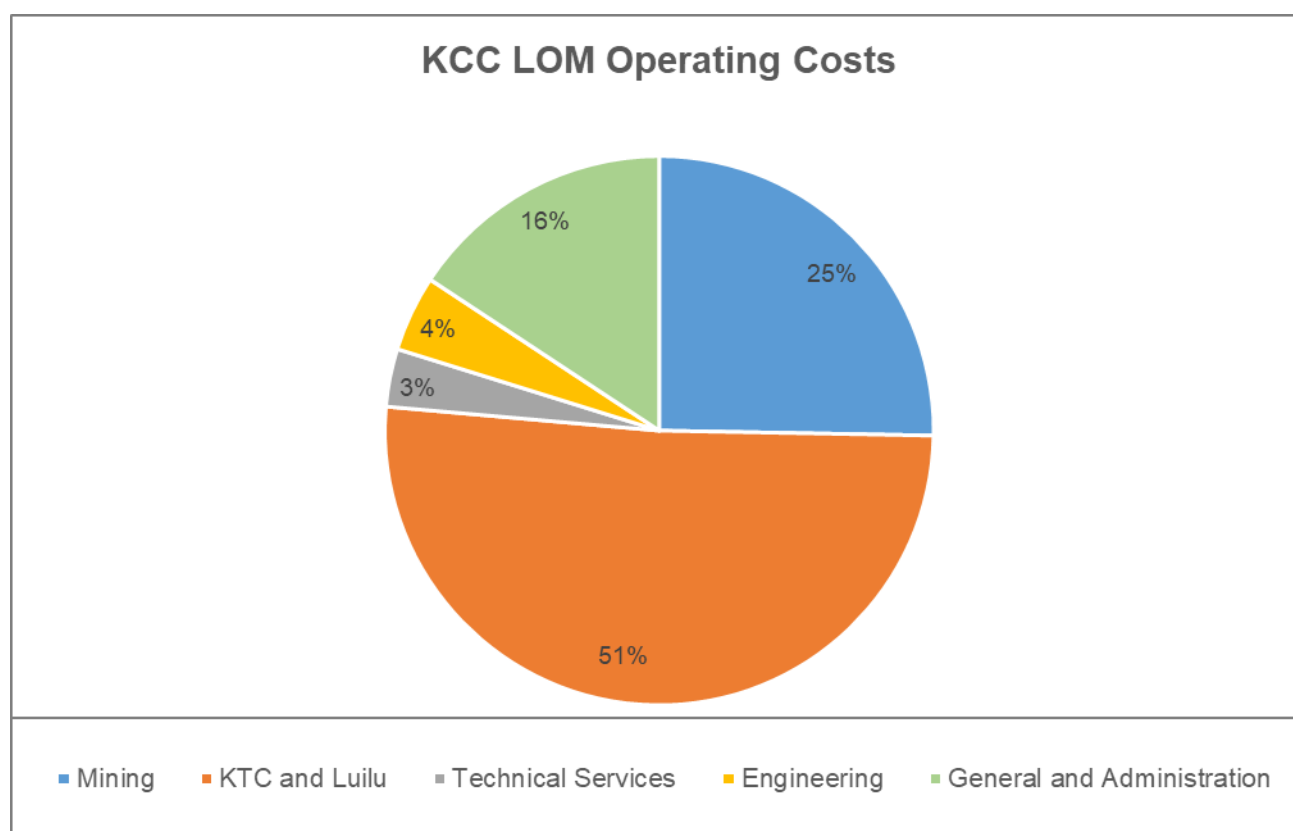


Figure 125: Major operational expenditure

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22.0 Economic Analysis

The economic assessment has been excluded as per instruction 1 of Item 22 of Form 43-101F1.

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23.0 Adjacent Properties

The following companies are operating in the neighbourhood of KCC:

- Gécamines holds the majority of the surface and mining rights surrounding the properties of KCC and has been entering into various operating or lease agreements with third parties, the details of which agreement are unknown to the Company. Some of the agreements entered into by Gécamines with third parties may have an impact on the future operations of the Company, especially in terms of future tailings storage facilities;
- Sicominis is currently operating the Mashamba West OP, which is adjacent to the Mashamba East OP. It does not have any known existing, potential, or reasonable future material impact on the Mashamba East OP;
- METALKOL tailing reclamation project is located north of T17 and east of KOV OP. This project is currently operational and does not have any known existing, potential, or reasonable future material impact on T17 or KOV OP; and
- Comus is currently operating an open pit mine in the proximity of T17. It does not have any known existing, potential, or reasonable future material impact on the T17.

There is no other information from these properties, which is applicable to the KML Material Assets for disclosure in this TR.

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24.0 Other Relevant Data and Information

24.1 Tailings

24.1.1 Introduction

This section covers the existing tailings deposition practices related to the Mupine Pit and KITD, giving a description of current operations and recommendations for improvement. It also provides for an overview of the Near West Tailings Dam project, which is critical for the continuity of the operations of KCC and alignment of the tailings capacity with the life of mine. Figure 127 shows the current tailing sites.

24.1.2 Tailings Management Overview

The key points related tailings management are noted below:

- There is a satisfactory tailings management structure in place. Personnel have been assigned to manage tailings deposition. Good reporting structures were also evident. The monitoring of the volume of material deposited into the pit is being well monitored;
- Tailings from KTC and Luilu have been and will be deposited into the Mupine OP from the southern flank of the pit. The tailings will be deposited from the top perimeter of the pit and allowed to cascade down the inner side slope to the floor of the pit. Permitting for this operation is in place; and
- The estimated remaining life of the Mupine Pit deposition has been extended to Q1 2021. KCC conducts regular bathymetric surveys of the pit floor to ascertain the tailings profile and depth below the water surface. This is used to assess the remaining life of the pit deposition; (the expansion project is explained in the Figure 126 below).

Mupine Tailings Storage Facility projects

- Embankment to be built in 4 phases between 2017 and 2020 to enable 4 years tailings deposition
- All 2017 planned works completed under budget and within schedule
- Some 2018 works commenced

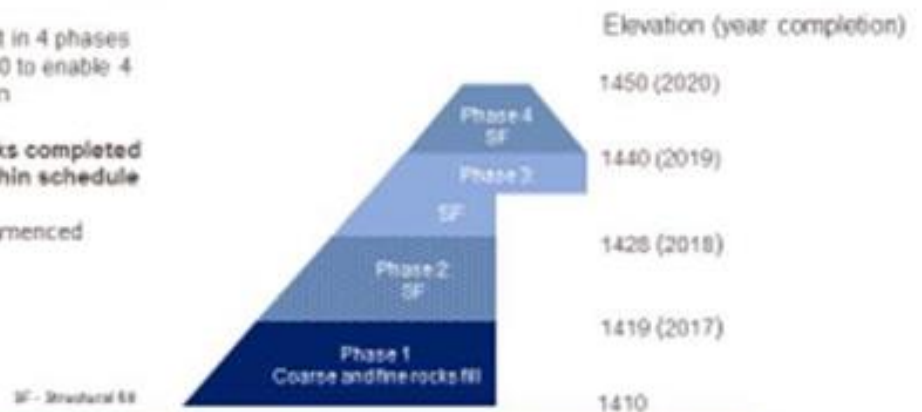


Figure 126: Mupine tailing storage facility project

Mupine Pit

KTC Plant

Kamoto Interim Tailings Dam (KITD)

Mashamba Tailings Storage Facility

0 500 1000 2000

FILE NAME	KOC_TAILING_LAYOUT
DATE	2010/05/07
DESIGNED BY	YH
CHECKED BY	YH
APPROVED BY	YH
SCALE	1:1000
PROJECT	KOC_TAILING_LAYOUT_MAP
DATE	2010/05/07
DESIGNED BY	YH
CHECKED BY	YH
APPROVED BY	YH
SCALE	1:1000
PROJECT	KOC_TAILING_LAYOUT_MAP

LAST SAVED BY: CREATED BY: YH PLOTTED BY: YH

FILE D:\SURVEY DATA\PHOTO AERIAL\KOC_OPEN PIT GENERAL MAP\2010 TAILING

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- No tailings from KTC have been deposited into the KITD since May 2013 and it is presently being re-mined, with the target end date for mining being 2021;
- The emergency tailings dam located on the eastern flank of the KITD is no longer used to deposit tailings and this tailing facility will also be mined out by the end of 2021;
- Water is pumped back to KTC from the Mupine Pit for process water used in the extraction process;
- Downstream of the KITD there has been considerable historical tailings spillage prior to KML taking ownership. The spillage extends up to the mining lease area of KCC. Spillage from KTC is routed to this “informal” tailings disposal area. The paddock has a bund wall for containment and it is estimated that it will also be completely cleared during the re-mining process that is in progress; and
- KCC is assessing numerous alternatives for incremental tailings capacity, including a potential Near West site and a Mupine Mega pit. As KCC currently does not have all the rights it requires in relation to these areas, discussions are ongoing with the relevant third parties. The areas being considered are not active mining zones and would provide sufficient space for life of mine tailings capacity. If KCC and third parties (including GCM) are not able to reach an agreement on the tailings deficiency, this may trigger adverse consequences for KCC’s operations, up to the shutdown of the operations by the end of Q1 2021. The Near West TSF site is as shown in Figure 128 and Figure 129.



Figure 128: Proposed Near West TSF expansion facility site

24.1.3 KCC Tailings

KTC receives ore from the open pit and underground mining operations. The oxide ore and the sulphide concentrates are routed to the Luilu plant. The tailings stream is split via hydro cyclones as follows; 15% - 20% coarse underflow to underground backfill operations and the remaining 80% overflow to the Mupine Pit. The mass of tailings deposited into the Mupine Pit from the Luilu Plant is 100% of the mass of the ore processed minus the mass of the metal removed.

24.1.4 Mupine Pit

The Mupine TSF has +/- 1 and half years’ life at current deposition rates and planned expansions. During 2017 early works were completed, and expansions to the 1,428 elevation was completed in 2018. The 1,441 elevation work will be completed in 2019, and the final 1450 elevation raise will be completed in 2020. The

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current estimated cost of the project from 2019 to 2020 is \$ 19.0 M. The tailings delivery pipes from both WOL and KTC enter the TSF on the southern flank. Tailings are deposited from the top perimeter and allowed to cascade down to the floor pit.

Daily observations of the tailings lines have yielded no evidence of spillages. The OMS protocols have been implemented and well managed since deposition in November 2017.

Water is decanted from the TSF to KTC, via a floating barge and pump arrangement. This operation is well managed in terms of deposition, health and safety, and environmental protocols.

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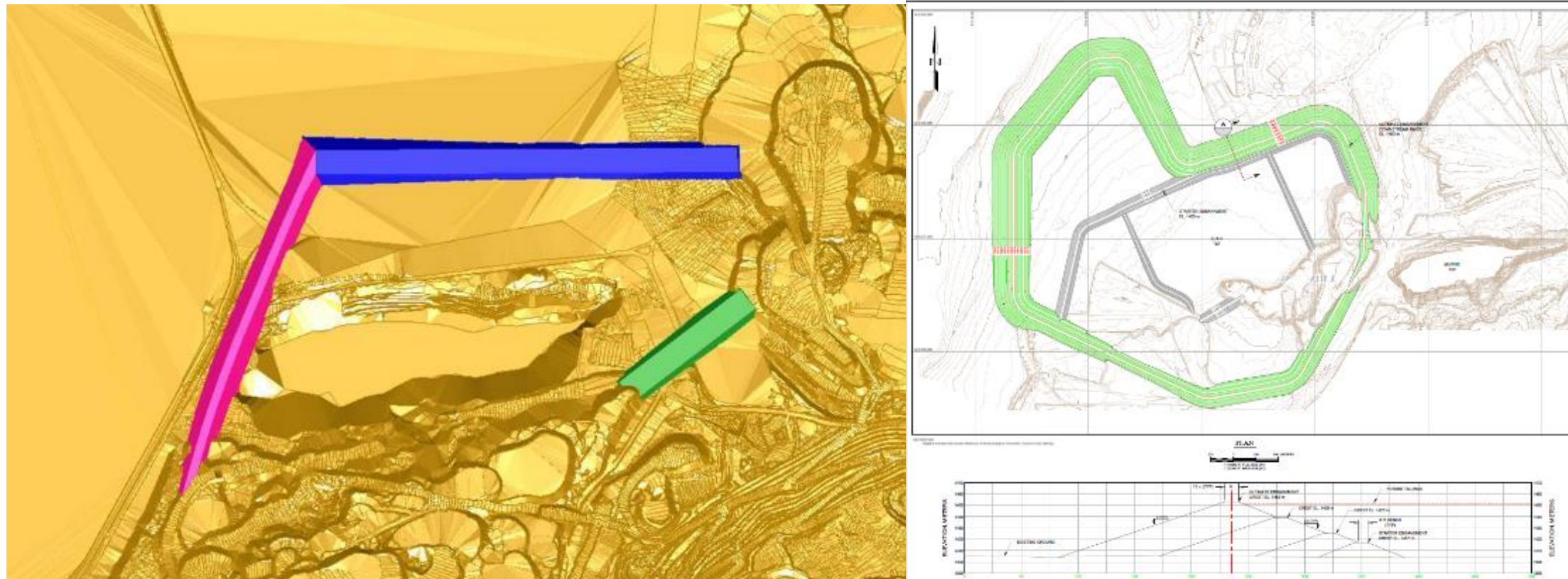


Figure 129: Mupine Mega Dam and Near West TSF layout

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24.1.5 Opportunities for Improvement

In general the tailings deposition into the Mupine Pit is being well managed. Management and reporting structures are all in place.

An opportunity for improvement that has been identified, but is not considered critical, is that the delivery pipelines used for tailings deposition into the pit should be extended down to just above the current water level in the pit. These pipes can then be cut off as the water level rises. The delivery pipe should be kept above the water visually inspect deposition taking place. This will eliminate segregation as the tailings falls from the top perimeter of the pit. Erosion of the inner side slopes of the pit will also be eliminated.

Separately, KCC is assessing numerous alternatives for incremental tailings capacity, including a potential Near West site and a Mupine Mega pit. As KCC currently does not have all the rights it requires in relation to these areas, discussions are ongoing with the relevant third parties. The areas being considered are not active mining zones and would provide sufficient space for life of mine tailings capacity. If KCC and third parties (including GCM) are not able to reach an agreement on the tailings deficiency, this may trigger adverse consequences for KCC's operations, up to the shutdown of the operations by the end of Q1 2021.

24.1.6 Near West Tailings Facility

Knight Piesold has been commissioned to provide designs for stage 1 and 2 of the proposed new KCC Luilu TSF to be located at the "Near West" site immediately South of the Luilu Plant and west of the Mupine TSF. A full design report by Amec Foster Wheeler was reviewed and used as the basis of Knight Piesold's new design of Near West. Knight Piesold 2018 report presents the design, analysis, and construction considerations for the stage 1 and 2 Near West TSF. The Near West TSF will initially be constructed in two stages using the centerline construction method. Stage 1 has been designed to store tailings for two years and stage 2 has been designed to provide an additional 3 years of tailings storage capacity. The tailings streams reporting to the Near West TSF will come from two different sources: the KTC plant and the Luilu plant. Combined, the tailings production rate reporting to the Near West TSF is 25,500 tonnes per day (tpd) with an additional approximate 10,000 tpd tailings going to UG mine backfill according to the design criteria. Future expansions to the Near West TSF are anticipated to be generally to the west and north of the stage 2 facility. Based on conceptual layouts of the ultimate facility presented by Morrison Geotechnical Solutions, Inc. (MGS, 2014), the ultimate site could provide tailings storage for more than 20 years of tailings production.

The stage 1 and 2 Near West TSF is formed by two perimeter embankments, herein designated as the "main embankment" and the "south embankment", that abut existing waste dumps to form an enclosed basin for tailings storage. The main embankment measures approximately 3,900 m long and forms the northern and western perimeter of the TSF. The main embankment will be keyed into waste dumps which form the majority of the southern and south-eastern portions of the TSF basin. The south embankment measures approximately 450 m long and will be constructed to bridge the gap between the waste dumps on the south side of the TSF. Embankment fill materials will be sourced from designated portions of the waste dumps, from embankment foundation excavations, and excavation of the proposed spillway pond.

The TSF embankments have been designed as zoned earth fill embankments and are composed predominantly with structural fill to be sourced from nearby waste dumps. Additionally, an upstream zone of low permeability fill, sourced from native clayey soils, and internal drains are included in the embankment design. Analytical and geochemical testing on a sample of WOL tailings indicate the tailings classify as "highly hazardous" according to the DRC (2003) regulation, and would require "Level B" water tightness measures. A linear low-density polyethylene (LLDPE) geomembrane liner has been included in the design on the embankment faces to reduce seepage through the embankments.

A water balance was developed for the stage 1 and 2 Near West TSF as part of this scope of work. The stage 1 Near West TSF has been designed to have storage capacity to contain the 95th percentile free water volume as determined by the water balance plus direct precipitation and runoff from the Probable Maximum Precipitation (PMP), while maintaining 1.5 m freeboard. An auxiliary spillway is included in the design for the stage 2 expansion, signifying that the facility is only required to store 50% of the PMP according to DRC (2003) regulation. The spillway will be located over the existing waste dump on the south side of the TSF and will discharge into a surge pond (spillway pond) south of the south embankment.

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It is understood that the tailings water chemistry may not be suitable for reuse in the process plants without some dilution with freshwater. Due to uncertainty of the percent dilution required, the water balance included a sensitivity analysis to evaluate freshwater dilution of tailings reclaim water equal to 20%, 35% and 50% of the required process water, meaning that rates of tailings water reclaim from the TSF would be reduced accordingly. The water balance results demonstrate that 50% dilution results in accumulation of water at the TSF, which is not a sustainable water management practice. As such, additional water management practices are anticipated to be necessary during operations of the Near West TSF. These practices may need to include treatment of TSF water (followed by release to the environment or reuse in the process), construction of a separate return water pond, and/or enhanced evaporation at the TSF.

24.1.7 Mupine Mega Dam

The Mupine Mega Dam is currently in the pre-feasibility stages. The capacity in the footprint will allow for an estimated additional 65-70Mt of tailings capacity. Dam wall design will be centerline construction with possible liner placed on the upstream wall. The land use of the facility is needed to be shared by the respective parties and / or leased from GCM by KCC.

24.2 Mine Closure

24.2.1 Background

Based on available information at the time, indicative closure costs were estimated by SRK in May 2019 as part of the recent update to the EIS.

It is noted that in some instances the responsibility for mine rehabilitation and closure is shared with GCM which historically conducted extensive mining over notable portions of the KCC concession areas. Some mining is still being conducted by GCM in proximity of the KCC concession areas with selective infrastructure being shared by the respective parties and / or leased from GCM by KCC.

The apportionment of costs for shared responsibilities remained challenging and all effort was made to identify the infrastructure at Luilu plant area that is solely the responsibility of KCC. In the case of mining related features and disturbances, this delineation of responsibility was also not as straight-forward. The waste rock dumps at KOV OP are a typical example where subsequent waste rock deposition by KCC has been conducted over the historical GCM dumps. To resolve this matter, the new defined waste rock dumps for KOV and Mashamba East OPs will be the full responsibility of KCC, whilst the host of other onsite waste rock dumps remain the responsibility of GCM.

The “relationship” between KCC and GCM is directed by the AJVA for the period from 2009 to 2024, with two 10 year renewable options in place. Sicominas, a third party, is currently developing and mining the concession to the west of the PE 525 KCC concession.

24.2.2 Acid Rock Drainage

The primary sources of acid generation are from waste rock at KCC operations due to oxidation of sulphide minerals, dissolution of acidic-sulphate minerals and hydrolysis of metals from sulphide and acidic-sulphate minerals. Sulphide sulphur in the waste rock is below detection limit (< 0.1). The waste rock is unlikely to generate acidity in the long term. Acidic drainage will only occur when the rate of acid generation exceeds the rate of acid neutralisation. Based on the 2018 Mining Code requirements and international criterion, the ABA results indicate that Mashamba East waste rock is unlikely to generate acidity and contains excess Neutralization Potential (NP) in the form of carbonates and aluminosilicates. The Net Acid Generation (NAG) tests indicate that the samples tested all have a low potential to generate acidity with the NAG values all being less than 1 kg/t H₂SO₄ (SRK, 2019).

KTC tailings are neutral prior to deposition (neutral pH), while tailings from Luilu Plant are anticipated to be pH=10. Tailings analytical and geochemical characterisation conducted by M&L Laboratory Services (M&L) in January 2016 indicate that the WOL tailings classify as highly hazardous based on total chromium and lead content. Geochemical testing on the WOL tailings indicated a sulphide content of 1.6 % and a Net Neutralization Potential (NNP) of -38.7 kg CaCO₃ /tonne, which indicates that the tailings are likely to generate acid (SRK, 2017). Tailings generated by Luilu processing operations are neutralised before being sent to the Mupine TSF.

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24.2.3 Approach and Limitations to Costing

24.2.3.1 Approach

The 2019 closure costs were updated from:

- The 2015 closure costing undertaken in the “Review and Update of Scheduled and Unscheduled Closure Costs for Katanga Copper Company for Independent Technical Report as at 2015” (Golder, June 2015);
- The budget and financial plan prepared for the 2014 EIS and EMPP which was based on the information provided in the 2011 Estimate (PE525, PE1161 and PE1162) and 2007 Estimate (PE4960, PE4961 and PE4963); and
- The budget and financial plan prepared for the April 2018 EIS and EMP, which was based on the 2014 estimate.

24.2.3.2 Limitations

- The closure costs were estimated based on engineering drawings and measurements from aerial photography. No detailed inventories were assessed for the closure cost calculations;
- Potential revenue from equipment or scrap has not been included in the estimate;
- The costs are based on the identified closure concepts to be implemented. No level of engineering has yet been undertaken around any of the proposed closure measures;
- At the time of compiling the closure estimate for the Ion Exchange plant there was no design or Bill of Materials was available. The closure cost was estimated based on the size of the plant, therefore the estimate has a very high uncertainty factor. The closure estimate of the Ion Exchange Plant must be updated in future EIS updates/revisions when more information is available;
- SRK has assumed that sufficient material will be available locally for use as covers where appropriate. No provision has been made to commercially acquire these materials;
- The extent of soil contamination remediation has not been determined. An assumption that a portion of the metallurgical plants will require remediation has been included in the assessment;
- The assessment has included the biophysical closure costs. The assessment therefore does not include the internal and external social costs or costs related to the community, including any constructive obligations that may exist;
- The assessment does not include any company overheads that may be incurred during closure;
- The assessment does not include security costs which may be incurred during closure;
- The mine is considering a pit backfilling option, which will be dependent on the geochemical suitability of waste rock material as well as other engineering considerations. Once the mine has further investigated backfilling options, and determined feasibility, the liability for this activity will be included in the periodic updates of the liability;
- No allowance has been made for the revegetation the WRDs (Waste Rock Dump) as trials are required to determine the appropriate technology to adopt for this activity. This will require field trials as well as community engagement and agreement with DPEM;
- Pit lake investigations are not yet available and are required to inform further measures for pit lake formation and management. The current assumption is that the pits will remain a hydraulic sink, with water quality that meets the relinquishment criteria and that no further intervention is required to manage water post closure;
- Buildings that were handed back to Gecamines and no longer used by KCC since 2015 have been removed from the closure costs;

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- Once the final detailed design of the Near West TSF is available and the closure plan for the structure has been completed, the liability for this facility must be included in the site wide closure cost estimate; and
- Closure provisioning for KITD TSF has been removed as the area is earmarked for the new Mashamba WRD once hydraulic mining of the TSF has been completed.

24.2.4 Costing-related Assumptions and Qualifications

24.2.4.1 General

The closure of the Kamoto Mine Complex is likely to be challenging due to the interspersed mining activities undertaken by KCC and GCM. KCC operations are governed by the AJVA, which defines the partnership terms with GCM. The associated liabilities between KCC and GCM are directed by the AJVA for a period from 2009 to 2024, with two 10-year renewable options in place. Sicomin, a third party, is currently developing and mining the concession to the west of the PE 525 KCC concession and it is likely that new concessions may be granted to other mining companies in future.

KCC will periodically review their commitments to closure and rehabilitation to ensure that the measures are feasible, relevant and implementable. The measures as provided in the ESMP aim to mitigate potential environmental and social impacts upon closure, where possible, and reduce potential residual impacts to acceptable levels, in accordance with accepted good practice, legal requirements and KCC's requirements.

24.2.4.2 Site Specific

24.2.4.2.1 Infrastructural Areas

The buildings that existed at the time of KCC commencing mining operations will be handed back to GCM. This includes improvements and additions made to the buildings. Only newly constructed buildings will require demolition, if GCM and the community have no further use for such buildings, which can only be determined closer to closure.

KCC will be responsible for the removal of all equipment that can be removed irrespective of whether it has a value.

The following measures will be taken for the infrastructure and equipment not leased by KCC from GCM:

- Prior to removal from site, all equipment posing a risk to the environment and health and safety of people, will be decontaminated and then certified as contaminant free;
- Where possible, third party uses for equipment will be identified and materials sold or donated to these parties;
- Equipment that cannot be fully used in its entirety will be broken down into different metal types and sent to scrap for metal recycling;
- All equipment that cannot be sold /donated /recycled as scrap will be placed in a suitable pit bottom and covered with a suitable layer of waste rock to deter people from scouring the area for sellable materials;
- Foundations of buildings not handed back to GCM will be broken in situ to limit ponding and covered with suitable material;
- Following demolition and removal of rubble and other waste, including contaminated soils, the area will be scarified to remove any residual material;
- A soil layer to a depth suitable to meet the post-closure land use, will be placed onto the area and reshaped to approximate surrounding topography, taking requirements for erosion control into cognisance; and
- All concrete will be disposed of at pit bottom and covered with waste rock.

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24.2.4.3 Mining Areas

24.2.4.3.1 Stockpiles and Waste Rock Dumps

All useable stockpiles of raw and/or saleable material will be processed and removed off-site at closure and as a result there will be no reclamation requirement.

At scheduled closure the KOV ROM stockpile footprint area would be rehabilitated by shaping and ripping of the footprint area to prepare for vegetation establishment. Placement of a growth medium has not been provided for as it is understood that acceptable levels of natural re-vegetation of disturbed areas occurs with time. The remaining waste rock and/or over burden dumps will be shaped and vegetated.

To achieve long-term stability of waste rock dumps, the following activities will be undertaken:

- Where practical and space permits, KCC will cut back the slopes to a safer slope angle. Where possible, cutting back will be maximised on the slopes that are adjacent to the pit rims, so that cut back material can fall into the pits, thereby minimising the increase in footprint that occurs as result of cut and fill activities. This will however, only be possible if pit slope stability allows the dumps to be cut back;
- Where it is not possible to cut back slopes due to space constraints, appropriate erosion control measures will be implemented, for example revegetation;
- A suitable growth media will be placed on the WRDs to facilitate the establishment of a vegetative cover, consisting of indigenous species;
- As erosion is still expected from the dumps, a series of toe paddocks will be established around the toes of the facilities, where the accumulation of sediment is likely to impact on the receiving environment; and
- Given the locations and geometries of the end of life WRDs, it is likely that the toes of the WRDs will be bordering on concession boundaries and communities that will restrict battering down of the outer slopes. KCC will however consult with affected communities and the relevant authorities to obtain permission to potentially batter-down the outer slopes.

24.2.4.3.2 Open Pits

Areas where excavations remain following the demolition of infrastructure will be reshaped and covered with a suitable growth media and the area revegetated. This approach will also be undertaken, as far as practical to do so, for ad hoc excavations across the site. Where there is a deficit of material in the immediate vicinity to backfill, excavations will be re-contoured to minimise erosion and make the area safe and stable.

Warning signs will be placed in strategic locations around excavations and open pits. The final voids of the T-17, KOV and Mashamba East open pits, will remain at closure. Due to the geometry of the open pits, these will be deep voids with benched "high walls" along all edges. It is likely that pit lakes will have developed in those pits not actively dewatered during the latter stages of operations (i.e. due to regional ground water levels and relatively high rainfall). Two high wall portals will be present at the T-17 pit.

The following measures will be implemented upon the closure of the open pits:

- All equipment will be pulled out of the pits at the end of their individual life of mine;
- All ramps will be rendered inaccessible by either placing rock barriers across the ramp or by blasting the ramp to prevent both human and vehicle access;
- A berm will be provided along pit rims that are not protected by mine associated infrastructure, such as waste dumps; and
- The open pits will be left to fill naturally with ground and rain water to form a pit lake.

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24.2.4.4 TSFs

It has been assumed that for scheduled closure, the Mupine Pit tailings will be slightly overfilled compared to the surrounding surface topography requiring vegetation establishment directly onto the tailings. If this is not the case and deposition ceased below the in-pit water level, the measures will be implemented for the unscheduled situation. For unscheduled closure, a bund will be constructed around the extent of the pit lip. Further closure measures will be similar to those proposed for the Near West TSF as outlined below.

The measures required to stabilise the long-term TSFs (including Mupine) and ensure that they are not a source of dust and /or contaminated runoff include:

- Treatment and management of overflows from tailings ponds;
- Levelling of the surface coverage of the tailings ponds to maintain positive drainage and reduce the accumulation of runoff on the cover;
- Construction of a durable, chemically stable, protective cover covered with a layer of topsoil to facilitate revegetation to reduce wind erosion;
- Placement of rocks on the embankment to eliminate erosion;
- The demolition and burial of all infrastructure (pipes, pumps, plinths, etc.) not required post closure;
- During the last year of operations, tailings depositing will be undertaken to create a beach draining to a low point on the facility, acting as an evaporation facility for accumulated rainfall;
- A spillway will be constructed to allow water accumulation during the periods of excessive rainfall to spill and return to designated approved water storage; and
- During operation, the impact of the tailing dams on the groundwater will be monitored through appropriate sampling.

24.2.4.5 Waste

All waste management areas in the mine complex area will be appropriately closed at the end of the LoM. Any waste management facilities developed for the mine complex area will be closed by constructing an engineered cap to reduce water ingress into the waste body. The cap will be revegetated with grass species to minimise the potential for deep rooted plants compromising the landfill engineered cap. Storm water drainage around the facilities will be retained.

During operation, KCC will investigate and implement waste reduction strategies throughout the LOM. This includes waste reduction, reuse and recycling as far as is possible, given the constraints of supporting local infrastructure and regulations. Suitable waste disposal options will be investigated and implemented and will consider the risks and impacts to the environment and communities.

Waste sites (non-hazardous and hazardous) will be closed with a suitable cover. The groundwater below the waste sites will be sampled to determine whether any hazardous substances have infiltrated the groundwater. If groundwater contamination is identified, appropriate remediation action will be taken and post-closure monitoring will continue until the risk is deemed acceptable.

Any stockpiled recyclable material will be removed from site, subject to agreement with the regulatory authorities, by appropriate recycling companies. Economically salvageable materials will be disassembled and sold, where practical.

24.2.4.6 General Surface Rehabilitation

Areas affected by the Project will be revegetated, where practical to do so, using local plant species where possible and other species identified where appropriate.

This will include areas occupied by the processing plants, waste rock dumps, tailings storage facilities, and other infrastructure. This will occur after remediation of the sites (if necessary).

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The main goals for revegetation of the site at its closure is to:

- Stabilise disturbed areas against erosion and sedimentation;
- Help control dust emissions associated with exposed areas; and
- Restore vegetation with appropriate species.

Previously disturbed and compacted areas where infrastructure have been demolished, will be ripped to allow for the natural establishment of vegetation. The ripped areas will be covered with a suitable growth media (up to 300 mm), if required, to facilitate vegetation establishment. Suitable indigenous plant species will be identified for use where vegetation does not establish naturally and to stabilise slopes and larger mine structures such as WRDs and TSFs.

Not all operational areas will require rehabilitation and revegetation, as some facilities will be handed back to GCM post-closure (as indicated in the Joint Venture Agreement). The revegetation of these areas will be the responsibility of GCM, in line with its own mine closure plan.

Monitoring will take place for a period of 6 years post closure or until the mine's relinquishment criteria have been achieved, as required by Article 96, DRC Mining Regulations.

24.2.5 Closure Cost Determination

24.2.5.1 Summary of Closure Costs

A summary of the updated closure costs situations is reflected in Table 131. These costs are same as estimated in May 2019.

Table 131: Scheduled closure costs for KCC as at May 2019

*Refer to limitations in section 24.2.3.2 for items excluded from cost estimate.

No.	Item	Total Cost (USD)
1	Dismantling of processing plants and related structures (including overland conveyors, pipelines and power lines) and demolition of buildings and structures	30,473,804
1a	Sulfuric Acid Plant	1,423,036
1b	Ion Exchange Plant	1,047,619
2	Rehabilitation of access and haul roads	1,354,000
3	Opencast reclamation including final voids and ramps	16,976,500
4	Rehabilitation of waste rock dumps	17,359,893
5	Rehabilitation of tailings storage facilities and ponds (basic)	10,868,240
6	General surface rehabilitation	13,772,307
7	Post-closure monitoring	1,704,393
8	Aftercare and maintenance	5,916,105
	Sub Total 1	101,142,962
9	Preliminary and general at 18%	16,831,044
10	Contingencies at 15%	14,028,370
	Sub Total 2	30,862,413
	Grand Total (Sub Total 1 + Sub Total 2)	132,005,375

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25.0 Interpretations and Conclusions

25.1 Mining and Exploration

25.1.1 Mining History

Mining in the Kolwezi district has a long history, stretching back to the early 1900s. At its peak, in the late 1980's, the district accounted for roughly 7% of the world's copper production and 62% of the cobalt. In the mid-to late 1990's, production declined to a virtual standstill as a result of various political and technical factors. In 2002, a new Mining Code was established and since 2008, after the merger of KCC and DCP, significant growth in mining production has occurred. Because of a slump in metal prices in 2015, operations were put on care and maintenance. During this time, construction of the WOL facility was undertaken to achieve planned recoveries and reduce operating costs. Production recommenced, with first copper being produced in December 2017.

On January 27, 2018, the DRC parliament adopted the 2018 Mining Code, changing several provisions relating to mining permits awards and renewals, royalties and taxation, regulatory controls and national preference for subcontracting. Further detail on the 2018 Mining Code is included in Sections 4.6.1 and 25.3.7.3.

25.1.2 Exploration History and Future Potential

The earliest exploration drill hole on record was drilled in 1942. Numerous exploration drill holes were drilled on a systematic grid in the 1950's and 1960's. In the 1980's and early 1990's, a drilling campaign was undertaken on section lines spaced approximately 100 m apart. Since 2008, extensive drilling focused on defining the KOV OP pushbacks, KTO UG developments, and upgrading the Mineral Resource classification has been conducted. Geological mapping in the open pits and underground developments is on-going. Exploration potential exists inside the KCC concession for both the extension of known ore bodies and the discovery of new mineralized zones.

25.1.3 Database Management and Data Verification

Data from the drilling programs are stored in a Datamine™ GDMS Fusion8 database. A QA/QC program was initiated on the data collected since 2009, and improvements are notable year on year. A twin drilling program was carried out in 2007 to confirm the GCM drilling database. The results of these programs are considered to be acceptable for Mineral Resource estimation purposes.

25.1.4 Geological Modeling, Mineral Estimation Methods, and Mineral Resource Classification

Geological modelling, exploratory statistics, and geostatistics were carried out using Datamine™ Studio RM, and are considered acceptable for Mineral Resource estimation purposes. Ordinary kriging was utilized for the Mineral Resource estimation and was considered acceptable. However, if insufficient geostatistical results were available, IDW was utilized. The Mineral Resource classification is based on several factors, including: number of samples, number of drill holes, distance to a sample, drill hole spacing, and time period of drilling. Both the Mineral Resource estimation and the classification methods are considered acceptable.

25.1.5 Metallurgical Testwork

Metallurgical testwork has been conducted on KCC core since 2006, and the WOL Project testwork was initiated in 2015. This work included: comminution, flotation, and hydrometallurgical studies. The results of this testwork confirm the process design parameters

25.2 Conclusions

This report has taken into consideration the following:

- Depleted Mineral Resources and Mineral Reserves as at December 31, 2018;
- Mining activities continued during 2019 and updated depleted Mineral Resources and Mineral Reserves for 2019 will be released as a part of the 2019 year end Mineral Resources and Mineral Reserves update;

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- Mineral Resources and Mineral Reserves changed for KTO UG, KOV OP, Mashamba East OP and KITD due to mining activities in 2018;
- There is no change in Mineral Resources and Mineral Reserves for T17 UG;
- Cobalt debottlenecking – upgrades to the MgO make up facility, thickening, filtration and cobalt dryers will allow KCC to process at the design capacity of up to 40Ktpa of Cobalt contained as Cobalt Hydroxide salt;
- Acid plant – construction of the acid plant, which is capable of producing 2,200 tpd of sulphuric acid or 1,900 tpd of sulphuric acid and 200 tpd of liquid SO₂, will be completed during 2020;
- Process improvements to the plant and processing infrastructure, including the WOL project, are intended to deliver 300 ktpa of saleable copper; and
- Uranium removal from cobalt hydroxide product by either precipitation using the phosphate ion and/or ion exchange using an anionic exchange resin.

KCC is an economically viable operation and this TR presents a reasonable overview of KCC's potential value. As such it is not without inherent risks. KCC believes that these risks can be mitigated to various levels by management actions.

25.3 Risk Analysis

A summary of the risks involved in the technical analysis of the Material Assets of the project include:

25.3.1 General Risks

The KCC site operates adjacent to historical and community settlements, that include artisanal miners and artisanal ore buying counters, which may impact the operations of KCC. Due to the vicinity of these surrounding settlements, KCC is subject to intrusions which can significantly impact the security and safety of operations.

Due to the extended surface of the KCC mining permits which are not all adjacent to each other, the concession is crossed by regional and public access roads which are open to third parties vehicles. This increases the risk of traffic accidents for KCC employees, contractors and deliveries who access via these roads within the KCC Concession.

There is a risk of fatalities as a result of employees and contractors not being able to identify hazards, implement effective controls or being non-compliant with these controls. General acceptance of risk within the country requires the company to continually train and educate employees, supervisors and management, to assess compliance and verify the effectiveness of the Safety Management Systems.

There is a risk of threat to physical security of the operations and personnel due to political unrest or civil wars. Operating in the DRC also exposes employees and contractors to epidemic / pandemic diseases such as Ebola, tuberculosis, cholera, yellow fever and malaria, which may impact operations and project delivery. Although the Company has put in place emergency response and evacuation plans, as well as medical services, there is a risk that these cannot be implemented to the necessary standard in a timely manner if events such as civil war, civil riots or large epidemic / pandemic diseases spreading quickly.

25.3.2 Mining Risks

25.3.2.1 Underground Risks

The major underground risks that could have a negative impact on the overall KCC planned production profile include:

- The quality of backfill placement in the KTO UG to guarantee the stability of the mining areas remains a risk. Improvements have been implemented by KCC as mitigation measures since 2012. Quality placement is required to ensure that extraction ratios are achieved. Key areas are where backfill is

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required to extract pillars (or the secondary stopes), where old rehabilitated areas are mined in Zone 3, and in the bottom of Etang North below existing stopes; and

- The presence of a collapsed zone in the central plateau portion of the KTO UG orebody may influence stress distribution in adjacent mining areas.

25.3.2.2 Surface Risks

The major risks from the open pits that could have a material effect on the overall KCC planned production profile includes:

- A high production rate across all open pit areas, including KOV OP and Mashamba East OP, of up to a combined 90 Mtpa of total material requires sufficient working areas, face length and ramps for the production fleet to operate optimally. The open pits space risk is being managed through the simultaneous mining of multiple pushbacks. The model shows that establishing two to three pushbacks reduces this special risk, and enables the mining fleets to operate efficiently. Mining from a single pushback would generally result in lower efficiencies and lower productions rates;
- Continuous dewatering of the KOV OP is required. Limiting the dewatering potentially introduces risks associated with the pit stability and the production schedule. This risk is being mitigated by a comprehensive borehole drilling program to intercept water sources and divert them to controllable areas. This will result in better control of inflow of water and effective manage of its discharge from the pit;
- Small slope failures could have a negative impact on access systems to the mining benches which may jeopardize production and put at risk employees and contractors working in these areas. Detailed planning, independent reviews and ongoing assessments and real time monitoring is installed;
- One of the highest environmental impact risk can occur during the dry season when either the background dust levels or dust generated from the operations may be higher than the limits imposed by the regulator. Continual monitoring and response plans are the primary controls.
- In addition, given that KCC operates an old mining site where open pit mining waste rehabilitation has not been completed to current standards, KCC faces an elevated risk of erosion and sediment movement from historic operations. During the rainy season, excessive rainfall events may inundate control structures releasing sediments in excess of the prescribed limits. High rainfall events also present an inundation risk to some of the mine operating areas and significantly disrupt the operations.

25.3.3 Processing Risks

The KCC operations involve an ore concentrator, sulphide roaster, WOL, solvent extraction (SX), electro-winning (EW) and cobalt pressure filters and driers. These processes involve the utilization and manipulation of various forms of hazardous products and energy sources such as chemicals, electricity, heat, high pressure and industrial scale equipment (or a combinations of these) that can cause harm the employees, contractors and surrounding communities if not controlled efficiently.

KCC is also exposed to a risk of severe injuries or fatalities, should a fire occur at the SX facility. Control in place specifically require limited and highly controlled personnel access, together with emergency response teams and systems to mitigate this risk.

Finally, the deliveries of material and consumables, as well as the dispatch of final products (copper cathodes and cobalt hydroxide) is via road transport through the operating site to the ports of delivery. This increases the risk of traffic accidents, theft and damage of the material, consumable and product shipped from and to the operating site in DRC.

25.3.3.1 Unavailability and Quality of Key Reagents for Metallurgical Processing

KCC faces a potential risk that critical process reagents may not be available in the region in the required quantities or quality, leading to reduced production of copper and cobalt. Although there have been recent improvements in supply chain management and inventory control, the border controls at Kasumbalesa border post increased the transit time from 4 to 7 weeks for trucks from South Africa and from 2 to 3 weeks for trucks

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from Zambia. This caused a truck shortage in South Africa and Zambia, which affected supply of lime and other materials recently. These issues could impact production of copper and cobalt.

25.3.3.2 Power Availability and Supply Fluctuations

Power requirements to operate at the scheduled production profile are approximately 150-180MW and there are risks that this power may not be available through the national grid and may lead to power disruptions or supply fluctuation. KCC has entered into an agreement with the state utility, SNEL, to refurbish the DC link between Kinshasa and Kolwezi SCK/RO stations and to refurbish additional power infrastructure within the DRC to increase the available power supply from the Inga hydroelectricity facility to 450MW to the Katanga Province. The work with SNEL is progressing well and the power supply constraints seem less of a risk.

25.3.4 Tailings

KCC has been in discussions with third parties for land use rights for future tailings facilities to meet the life of mine tailings deposition requirements. Failure to obtain the required land use rights could impact planned metal production levels and Mineral Reserves. For a complete discussion, refer to Section 20.2.2.

The KCC site has been in operation for decades previously as a Gecamines operation. Disused tailings disposal facilities are located around the operating site. Generally, these are located near or below surrounding land areas.

Current tailings disposal facility (Mupine) is being elevated to above natural ground and present a risk of dam failure.

25.3.5 Capital Risks

25.3.5.1 Escalation of Costs

Capital cost overruns may require more funding and reduce project returns.

Access to the appropriate skilled and trained contractors can have an impact on Project delivery timetable.

25.3.6 Operating Risks

25.3.6.1 Underdeveloped In-country Institutional Infrastructure and Capacity

The DRC's institutional infrastructures have limited capacity and may not have the ability to deliver on the approvals' requirements of the project, such as environmental or feasibility studies, reducing the project feasibility or causing delays to achieve construction in due time. This risk is rated as high.

25.3.6.2 Senior Management and Technical Expertise

Recruiting and retaining senior management and critical technical expertise to manage and operate the mines and processing plants is an ongoing issue. It potentially affects the ability of the project to run optimally and comply with legislation. Mitigating measures include reviewing KMLs and KCCs employment strategy, recruitment and retention plan.

25.3.6.3 Artisanal / Illegal Miners

Artisanal mining activities are not permitted within KCC's mining concession. When artisanal miners gain entry, there is a risk of severe injuries or fatalities to either themselves, security personnel or employees during either pursuit, detainment or aggressive interactions. There have been multiple fatalities of illegal miners working in generally inaccessible benches in the open pit where their activities have caused collapses.

Illegal miners also create situations where production is disrupted and a loss of copper and cobalt ore from the reserves.

Perimeter security is maintained by employees, contractor and DRC Police forces (Mines Police). The protection provided by Mines Police at KCC site is governed by a Memorandum of Understanding.

In 2019, the KOV OP was overrun by large groups of artisanal miners (approximately 3000 individuals), interrupting the operations in some areas of the concession. The DRC government decided to dispatch the DRC army (FARDC) in the areas surrounding KCC's concession. There is a risk that KCC suffers reputational

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damage due to any accusations (proven or not) that the FARDC, located in proximity to the KCC operations, are involved in human rights violations.

Due to the activities and the generally ineffective application of regulatory controls of artisanal mining, in the DRC, there is a risk that cobalt production may be restricted into some international markets due to a general perception that DRC cobalt may be associated with, human rights abuses, corruption and activities involving women and child labor. For KCC, this increases the reputational risk and increases the requirements to demonstrate traceability for open market access.

25.3.7 Sovereign Risk

The DRC continues to be at risk of being affected by varying degrees of political and economic instability in the future, which is outside of KCC's control. Changes to the DRC legal system (such as the 2018 Mining Code) may expose KCC's operations to increased operational and financial risks and associated compliance costs which were not foreseen at the inception of KCC's investments and projects.

25.3.7.1 Regulatory and Corporate Law Risk

As a result of the factors and risks specific to operating in the DRC, there is a risk that the Company may face investigation, fines, penalties or sanctions resulting from future regulatory action in Canada, the United States or other jurisdictions in which the Company operates, which may have an adverse impact on the Company's business, reputation, prospects or financial condition.

25.3.7.2 Jurisdictional Risk

The Company's cash flow is derived from and dependent on material mining operations located in the DRC. The stability of revenue from the Company's mining operations and the carrying value of its investment in its 75% DRC operating subsidiary KCC is directly impacted by certain risks specific to operating in that jurisdiction, which risks can create financial, legal and economic uncertainty. Such risks include, but are not limited to: political unrest; risk of corruption in interaction with state and non-state actors, including violations under applicable foreign corrupt practices statutes; arbitrary changes in laws regulations, policies, taxation, price controls and exchange controls; limitations on the repatriation of earnings; and limitations on mineral exports. These risks may limit or disrupt the Company's operations and projects, restrict the movement of funds or result in the deprivation of contractual rights or the taking of property by nationalization, expropriation or other means without fair compensation.

In December 2017, the United States government designated Mr. Gertler and several of his affiliated companies as SDNs by way of Executive Order 13818 ("the "Order"). The purpose of the Order is to block and prohibit all transactions in all property and interests in property of the designated parties and prohibit U.S. persons from dealing with such parties, or any entity owned or controlled by, or acting on behalf of, the designated parties. The Order applies to all property and interests in property located in the U.S., property that comes within the U.S. or property that comes within the possession or control of a U.S. person. In addition, entities not designated as SDNs but that are owned, at 50% or greater interest (directly or indirectly, individually or in the aggregate), by one or more SDNs also have to be treated as being subject to the same restrictions imposed by the Order. In addition to restrictions applicable to U.S. persons or activities in the United States, non-U.S. persons who are determined by the U.S. government to "have materially assisted, sponsored, or provided financial, material, or technological support for, or goods or services to or in support of" a party designated under the Order could themselves be designated as an SDN. As described in Section 4.6.3, the Company has pre-existing contractual obligations in DRC to make certain payments to Ventora, a company also designated as an SDN and owned by Mr. Gertler, which obligations pre-date Ventora's designation as an SDN and which arose when AHIL acquired such rights from GCM (which rights were subsequently assigned to Ventora). The Company believes that payment in non-US dollars of royalties to Ventora without the involvement of US persons would appropriately address all applicable sanctions obligations.

Independent surveys and assessments place the DRC in the top 10 highest risk countries to industry nationalization. It is also classified high on the list of Corruption and medium in terms travel risk.

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25.3.7.3 2018 Mining Code

On January 27, 2018, the DRC parliament adopted the 2018 Mining Code, replacing the previous 2002 Mining Code, which provided for a 10-year stability clause by which existing operations would, after introduction of any new regulations amending the 2002 Mining Code, continue to benefit under the terms of the 2002 Mining Code for 10 years.

On March 7, 2018, representatives of the Company, along with fellow international miners with operations in the DRC, met with the President of the DRC and outlined concerns, the cornerstone being the need to preserve the stability arrangement provided in the 2002 Mining Code. Additional concerns included the increase of mining royalties, conditions regarding renewal of mining permits, tax on "super-profits", the increase of import duties, repatriation of funds and strategic substances qualification. It was agreed that the miners' concerns would be addressed and a working group hosted by the Ministry of Mines, which would include the miners and civil society was to have commenced a 30-day review of the concerns. Despite repeated attempts, no such formal discussions have taken place.

The 2018 Mining Code was published in the Official Gazette on March 28, 2018. The position of the DRC government and the application of the new measures provided by the 2018 Mining Code and subsequent mining regulations published on June 8, 2018 as well as an instruction issued by the Secretary General of the Ministry of Mines on 16 June 2018, confirmed that the 2018 Mining Code is intended to supersede the 10-year stability clause provided by the 2002 Mining Code and that all new obligations will be immediately applicable.

The 2018 Mining Code substantially amended the 2002 Mining Code, introducing a number of new obligations, taxes and charges, and increasing existing ones. KML and KCC consider that they are entitled to the continuing application of all relevant provisions of the 2002 Mining Code, including, but not limited to, the tax, custom and exchange rate regimes existing thereunder, for a further ten years starting from the date of entry into force of the 2018 Mining Code, pursuant to Article 276, paragraph 2 of the 2002 Mining Code (the "2002 Stability Clause"). KML and KCC have consistently contested the applicability of the 2018 Mining Code and reserved all rights in this regard. Where in this section there is reference to agreement or compliance with the 2018 Mining Code, KCC has stated that it was under no obligation to do so and reserved entirely its rights pursuant to the 2002 Stability Clause. Further detail on the 2018 Mining Code is included in Sections 4.6.1 and 25.3.7.37.1.

The royalties on copper and cobalt under the 2018 Mining Code were increased from 2% of the net revenue basis to 3.5% of the gross revenue calculated on the basis of the average price of the concerned commodity on international markets. Additionally, the DRC customs authorities have established cobalt tables to predetermine the quality of cobalt products for exports. The tables consist of multiple quality brackets, within which the cobalt product for export is automatically deemed to be the top end of the bracket for the purpose of the royalty calculation. The new rates and basis have been implemented in June 2018.

As allowed by the 2018 Mining Code, on November 24, 2018, the Prime Minister of DRC published a decree qualifying cobalt as a strategic mineral substances, resulting in a royalty payable of 10% instead of 3.5% on KCC contained cobalt sales. The effect of such a decree on KCC operations to date is currently relatively limited due to the fact that cobalt quantities exported since the issuance of the decree have been impacted by the presence of uranium in the cobalt produced. The 2018 Mining Code also provides for the possible adoption by the Order of the Ministers responsible for Mines and Finance of an entirely different regulatory regime for "strategic mineral substances" including cobalt, concerning the conditions for access, research, exploitation and marketing thereof. The non-deductibility of freight costs lead KCC to add back freight costs related to the lots exported after June 8, 2018. As at September 30, 2019, the loss to be carried forward indefinitely for tax purposes has been reduced by USD 107.4 million.

The 2018 Mining Code creates a 50% tax on 'exceptional' profits, also called the "Super Profit Tax" or "SPT", made on a given mining project when the prices of the metals and minerals produced exceed by more than 25% the prices used in the bankable feasibility study ("BFS") for that project. Specifically, the 2018 Mining Code introduced a new SPT equating to 50% of the portion of the effective EBITDA higher than 125% of the EBITDA presented by the BFS, when the concerned commodity price exceeds 125% of the price used in the bankable feasibility study. In July 2019, KCC settled its 2018 SPT liability with the authorities. As part of a

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memorandum of understanding entered into in connection with the settlement of its 2018 SPT liability, KCC agreed to produce a BFS for future SPT calculation by the end of 2019.

The copper and cobalt price assumptions used in the BFS are key judgments applied in estimating the impact resulting from the changes in the 2018 Mining Code on Katanga's estimated fair value. The 2018 Mining Code provides for the application of substantive rates of imports duties (essentially 5% or 10% instead of 3%) when the mining license has been issued for more than 6 years. KCC has been importing under substantive regime since July 2018, resulting in an additional monthly cost of \$3 million (duties and taxes collected by DGDA).

The non-deductibility of interest payable to group companies (such as interest payable on intragroup loans) remains unclear and will depend on the interpretation of the new regulations by the DRC central bank. Following the enactment and entry into force of the 2018 Mining Code and Ministerial Decree n°020/CAB/MIN/EDD/AAN/TNT/SAA/2019, the Ministry of Environment has updated the list of classified installations to be declared by KCC to serve as basis for the calculation of various environmental taxes. KCC indicated that it will proceed with making the requisite declarations on the basis of the updated schedule.

The application of the 2018 Mining Code to KCC also impacts the approval process of some of the future waste dumps to be developed for the continuity of its operations. Article 279 of the 2018 Mining Code provides for an approval procedure for certain installations such as tailings and dumps when located within a certain distance from residential dwellings. KCC chose to comply with this requirement.

As part of the approval procedures for the Ion Exchange Plant (the "IX Plant"), the Ministry of Mines has requested that KCC submit a complete BFS for the entire KCC project, pursuant to the 2018 mining regulations made under the 2018 Mining Code. Given the effectiveness of the interim solutions using phosphoric acid over the past two quarters, KCC undertook instead to provide a BFS for the entire KCC project by 2019 year-end which will cover the IX Plant long term option.

Renewal of the concession permits also have been amended by the 2018 Mining Code adding several requirements which have increased the risk of non-renewal of the permits held by KCC. In particular, the renewal of the permits are subject to the submission of a bankable feasibility study and the assignment of 5% of the shares to the DRC State. The renewal of the permits is also subject to the submission of a new social plan (which needs to be complied with) and the uninterrupted operations of the mining permits. Although KCC's mining permits were not issued under the 2018 Mining Code and some conditions appear not be applicable, in virtue of the Stability Clause, the 2018 Mining Code additional requirements to renew the permits are considered as a high risk for KCC.

The rules governing foreign exchange under the 2018 Mining Code could also have a significant impact on the Company. Specifically, the 2018 Mining Code provides that KCC is required to repatriate to the DRC at least 60% of the proceeds from exports of mineral products, which proceeds may only be used for domestic expenses of KCC. The remaining 40% of export proceeds may be used for foreign expenses, including payment of foreign suppliers, servicing of foreign debt and the payment of dividends to non-DRC resident shareholders. Under the 2002 Mining Code, only 40% of export proceeds were required to be repatriated. This restriction may affect the ability of KCC to fully maintain its supply relationships with certain key non-DRC suppliers, service the Residual Debt and pay dividends to KMFL and ultimately to the Company.

KCC is considering certain commercial and structural options to mitigate the negative impact of such restrictions.

25.3.8 Economic and Market Risks

25.3.8.1 Commodity Prices

Copper and cobalt market prices are significant drivers of the profitability for KCC and the value of KML's interest in KCC. These prices are subject to wide fluctuations beyond the control of KCC or KML due to factors such as: demand for the commodities caused by global economic conditions and prospects, supply from various sources, currency and interest rate changes, and speculative activities. Sustained commodity prices below the costs of production may cause the curtailment or suspension of operations. There is some scope to manage market risk through hedging, but this may lead to loss of upside during periods of high commodity prices.

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25.3.8.2 Operating Costs

Project operating costs also affect the profitability of KML and the value of KCC projects. Operating costs are subject to a wide range of pressures such as energy prices, oil prices, chemical prices, labour costs and inflation. KCC is currently ramping up production in accordance with the project delivery schedule. Delay in the delivery of these projects can effect output and operating costs accordingly.

25.3.8.3 Currency Risk

Project revenues are in USD; however, input costs may be in other currencies, specifically the South African Rand (ZAR). Variations in currency exchange rates can affect production costs and overall project profitability.

25.3.9 Mine Closure Risk

The closure of the Material Assets may be challenging due to intermingled mining activities undertaken by KCC and GCM. The Amended Joint Venture Agreement, Reinforced, Consolidated and Modified, defines the partnership terms with GCM for a period from 2009 to 2024, with renewable options in place. The associated liabilities between KCC and GCM are governed by the Amended Joint Venture Agreement.

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26.0 Recommendations

The following work programs are recommended:

- Acquire and construction of a tailings facility with sufficient capacity for the life-of-mine;
- Upgrading of Inferred Mineral Resources to Indicated Mineral Resources, specifically for KOV OP, KTE UG, KOV UG, Mashamba East OP and KTO UG; and
- Upgrading of Mineral Reserves for KTE UG and KOV UG after upgrading Inferred Mineral Resource to Indicated Mineral Resource.

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KCC 2019 NI 43-101 TECHNICAL REPORT

28.0 Date and Signatures Pages

This TR on the Material Assets of KML located in Kolwezi, Democratic Republic of Congo is submitted to KML and is effective as of November 7, 2019. A list of QPs is shown in Table 132.

Table 132: List of QPs

QPs	Role	Company Name	Qualification	Responsible for Chapters	Date Signed	Signature
Tahir Saleem Usmani	Mine Technical Services Manager	KCC	BSc. Mining Engineering M. Eng. Mine optimization. P.Eng. (APEGA, Alberta Canada) Registration No. 93038	1, 2, 3, 4, 5, 6, 15, 16, 18.2, 18.3, 19, 20, 21, 22, 23, 24.2, 25, 26, 27 and 28		
Christiano Santos Goncalves	Principal Resource Geologist	Golder (UK)	BSc. Geology CP (Geo), MAusIMM (MAusIMM) Registration No. 306079	7, 8, 9, 10, 11, 12 and 14		
Nicholas Dempers	Principal Process Engineer	SENET	BSc, MSc Chemical Engineering Pr.Eng and FSAIMM (ECSA and SAIMM) Registration No. 20150196	1.7.2, 13, 17, 18.1, 18.2, 21 and 24.1		

APPENDIX A

Certificates of QPs

Certificate of Qualified Person – Nicholas Dempers

I, Nicholas Dempers, do hereby certify that

- 1) I am a Principal Process Engineer at SENET (Pty) Ltd, Building 12, Greenstone Hill Office Park, Emerald Boulevard, Greenstone Hill, Greenstone 1609, Modderfontein, Gauteng, South Africa, and have been employed in this position since 2015.
- 2) This Certification applies to the report titled "NI 43-101 TECHNICAL REPORT ON THE MATERIAL ASSETS OF KATANGA MINING LIMITED, LUALABA PROVINCE DEMOCRATIC REPUBLIC OF CONGO dated November 7, 2019 (Technical Report)."
- 3) I am a graduate with a BSc Chemical Engineering from the University of Cape Town in 1998. In addition, I have obtained a MSc Chemical Engineering Degree from the University of Cape Town in 2000 and BCom Degree from the University of South Africa in 2007.
- 4) I am a practising Process Engineer/Metallurgist and have practised my profession continuously since 2001. I have over 18 years' experience in the minerals industry. I have been involved in the process operation (production) and plant design, from conceptualization to complete project execution, of more than 10 mineral process projects, as well as more than five process plant studies for major commodities including cobalt, copper, gold, uranium and platinum group metals (PGMs).
- 5) I am a Professional Engineer with the Engineering Council of South Africa (Reg. No. 20150196), and I am fellow of the Southern African Institute of Mining and Metallurgy.
- 6) I have read the definition of "qualified person" set out in National Instrument 43-101 (NI 43-101) and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfil the requirements to be a "qualified person" for the purposes of NI 43-101.
- 7) I visited the KTC Site in December 2017.
- 8) I am responsible as Qualified Person for Chapters 1.7.2, 13, 17, 18.1, 18.2, 21 and 24.1 of the Technical Report.
- 9) I am independent of the issuer as described in section 1.5 of NI 43-101.
- 10) I do not own or expect to receive any interest (direct, indirect or contingent) in the property described herein. I am independent of Katanga Mining Limited.
- 11) I have had any prior involvement with Katanga Mining Limited operations on processing side.
- 12) I have read NI 43-101 and form 43-101F1 and the parts of the Technical Report for which I am responsible (Chapters 1.7.2, 13, 17, 18.1, 18.2, 21 and 24.1) have been prepared in compliance with that instrument and form.
- 13) At the effective date of the Technical Report, to the best of my knowledge, information and belief, the parts of the Technical Report for which I am responsible for compiling information (Chapters 1.7.2, 13, 17, 18.1, 18.2, 21 and 24.1), contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated - November 07, 2019

(signed) Nicholas Dempers

Name: Nicholas Dempers

Qualification: MSc Eng (Chem), BSc Eng (Chem), BCom (Man), Pr.Eng (RSA), Reg.No 20150196, FSAIMM (RSA) SENET (Pty) Ltd.
Principal Process Engineer

Certificate of Qualified Person – Christiano Santos Goncalves

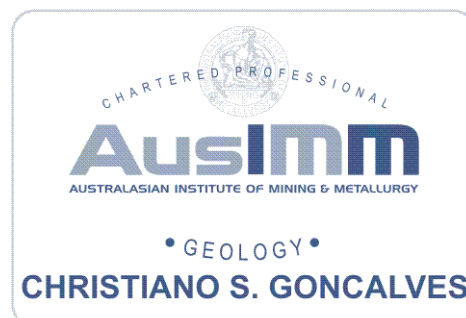
I, Christiano Santos Goncalves, AusIMM CP (Geology), do hereby certify that:

- 1) I am employed as a Principal Resource Geologist
at: Golder Associates UK Ltd.
Cavendish House, Bourne End Business Park, Cores End Road, Bourne End, Buckinghamshire, SL8 5AS, UK
Telephone: +44 1628 851867; Fax: +44 1628 851852; Email: cgsantos@golder.com
- 2) This Certification applies to the report titled “NI 43-101 TECHNICAL REPORT ON THE MATERIAL ASSETS OF KATANGA MINING LIMITED, LUALABA PROVINCE DEMOCRATIC REPUBLIC OF CONGO dated November 7, 2019 (Technical Report).”
- 3) I graduated with a Bachelor’s degree in Geology from UNESP University of Rio Claro-SP, Brazil in 1998.
- 4) I am a Chartered Professional (Geology) and member in good standing of the Australasian Institute of Mining and Metallurgy (#306079).
- 5) I have practiced my profession continuously since graduation. My relevant background with respect to this project is over fifteen years of experience in mine geology and mineral resource evaluation of mineral projects nationally and internationally in a variety of commodities.
- 6) I have read the definition of “Qualified Person” set out in National Instrument 43-101 (NI 43-101) and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a “Qualified Person” for the purpose of NI 43-101.
- 7) I have personally completed a site visit of the Material Asset that is the subject of the Technical Report in December 2018.
- 8) I am responsible as Qualified Person for Chapters 7, 8, 9, 10, 11, 12 and 14 of the Technical Report.
- 9) I am independent of the issuer as described in section 1.5 of NI 43-101.
- 10) I have had prior involvement with the Katanga Mining Limited operations on previous audits and have provided technical advice on the exploration program.
- 11) I have read NI 43-101 and form 43-101F1, and the parts of the Technical Report for which I am responsible (Chapters 7, 8, 9, 10, 11, 12 and 14) have been prepared in compliance with that instrument and form.
- 12) At the effective date of the Technical Report, to the best of my knowledge, information and belief, the parts of the Technical Report for which I am responsible for compiling information (Chapters 7, 8, 9, 10, 11, 12 and 14), contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated – November 07, 2019

(signed) Christiano Santos Goncalves

Name: Christiano Santos Goncalves,
Qualification: BSc Geology
AusIMM CP (Geology)



Certificate of Qualified Person – Tahir Usmani

I, Tahir S Usmani, PEng_QP, do hereby certify that:

- 1) I am employed as a Chief Long Term Mine Planning Engineer at:
Kamoto Copper Company S.A. (KCC)
Usines de Luilu, Commune de Dilala, Kolwezi, Lualaba Province, Democratic Republic of Congo
Telephone: + 243 976025233; Email: tusmani@katangamining.com
- 2) This Certification applies to the report titled “NI 43-101 TECHNICAL REPORT ON THE MATERIAL ASSETS OF KATANGA MINING LIMITED, LUALABA PROVINCE DEMOCRATIC REPUBLIC OF CONGO dated November 7, 2019 (Technical Report).”
- 3) I graduated with Masters of Engineering (MEng) in Mine Optimization from University of Alberta Edmonton, Alberta, Canada in 2012 and Bachelors of Science degree in Mining Engineering from University of Engineering and Technology Lahore, Pakistan in 1995.
- 4) I am a Registered Professional Engineer in good standing of The Association of Professional Engineers and Geoscientists of Alberta (APEGA) (#93038) in Canada.
- 5) I have practiced my profession continuously since graduation. My relevant background with respect to this project is over twenty years of experience in technical mining engineering, mine operations, mine planning, project management, evaluation of mineral projects and the preparation of Mineral Reserve estimates for open pit mines, underground mines and greenfield projects on a range of commodities that include, iron ore, gold, silver, copper, cobalt, lead, zinc, diamonds, ruby, heavy mineral sands and coal nationally and internationally. More than five years of this experience is relevant to the style of mineralisation, mining methods and subject matter of this report.
- 6) I have read the definition of “Qualified Person” set out in National Instrument 43-101 (NI 43 101) and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a “Qualified Person” for the purpose of NI 43-101.
- 7) I have personally completed several visits of the Material Asset that is the subject of the Technical Report several times during the course of this study in 2019 and prior.
- 8) I am responsible as Qualified Person for review and compiling of the Technical Report and specifically for the Chapters 1, 2, 3, 4, 5, 6, 15, 16, 18.2, 18.3, 19, 20, 21, 22, 23, 24.2, 25, 26, 27 and 28.
- 9) I am not independent of the issuer as described in section 1.5 of NI 43-101.
- 10) I am an employee of KCC and have involvement with the KCC operations on the technical side.
- 11) I do not own or expect to receive any interest (direct, indirect or contingent) in the property described herein; however, I receive annual salary, discretionary bonus and medical benefits from KCC.
- 12) I have read NI 43-101 and form 43-101F1 and the parts of the Technical Report for which I am responsible (Chapters 1, 2, 3, 4, 5, 6, 15, 16, 18.2, 18.3, 19, 20, 21, 22, 23, 24.2, 25, 26, 27 and 28) have been prepared in compliance with that instrument and form.
- 13) At the effective date of the Technical Report, to the best of my knowledge, information and belief, the parts of the Technical Report for which I am responsible for compiling information (Chapters 1, 2, 3, 4, 5, 6, 15, 16, 18.2, 18.3, 19, 20, 21, 22, 23, 24.2, 25, 26, 27 and 28), contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated – November 07, 2019

(signed) Tahir S Usmani

Name: Tahir S Usmani,
Qualification: MEng, BSc Mining Engineering
PEng, APEGA Canada

